Undergraduate Texts in Mathematics



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*(continued after index)*

Larry Smith

# Linear Algebra

Third Edition

With 23 Illustrations

Springer

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# Preface

This text was originally written for a one semester course in linear algebra at the (U.S.) sophomore undergraduate level, preferably directly following a one variable calculus course, so that linear algebra could be used in a course on multidimensional calculus and/or differential equations. Students at this level generally have had little contact with complex numbers or abstract mathematics, so the book deals almost exclusively with real finite-dimensional vector spaces, but in a setting and formulation that permits easy generalization to abstract vector spaces. The parallel complex theory is developed in part in the exercises.

The goal of the first two editions was the principal axis theorem for real symmetric linear transformations. Twenty years of teaching in Germany, where linear algebra is a one year course taken in the first year of study at the university, has modified that goal. The principal axis theorem becomes the first of two goals, and to be achieved as originally planned in one semester, a more or less direct path is followed to its proof. As a consequence there are many subjects that are not developed, and this is intentional: this is only an introduction to linear algebra. As compensation, a wide selection of examples of vector spaces and linear transformations is presented, to serve as a testing ground for the theory. Students with a need to learn more linear algebra can do so in a course in abstract algebra, which is the appropriate setting. Through this book they will be taken on an excursion to the algebraic/analytic zoo, and introduced to some of the animals for the first time. Further excursions can teach them more about the curious habits of some of these remarkable creatures.

In the second edition of the book I added, among other things, a safari into the wilderness of canonical forms, where the hardy student could pursue the Jordan form, which has become the second goal of this book, with the tools developed in the preceding chapters. In this edition I have added the tip of the iceberg of invariant theory to show that linear algebra alone is not capable of solving these canonical forms problems, even in the simplest case of $2 \times 2$ complex matrices.

Göttingen, Germany, February 1998 *Larry Smith*

# Contents

# Chapter 1
# Vectors in the Plane and in Space

In physics certain quantities such as *force, displacement, velocity,* and *acceleration* possess both a magnitude and a direction and they are most usually represented geometrically by drawing an arrow with the magnitude and direction of the quantity in question. Physicists refer to the arrow as a *vector*, and call the quantity so represented a *vector quantity*. In the study of the calculus the student has no doubt also encountered what are called vectors, particularly in connection with the study of lines and planes and the differential geometry of space curves. We begin by reviewing these ideas and codifying the algebra of vectors.

## 1.1 First Steps

Vectors as they appear in physics and the study of curves and surfaces can be described as **ordered pairs** of points $(\mathbf{P}, \mathbf{Q})$ which we call the **vector from P to Q** and often denote by $\overrightarrow{\mathbf{PQ}}$. This is substantially the same as the physics definition, since all it amounts to is a technical description of the word "arrow". **P** is called the **initial point** and **Q** the **terminal point**.

For our purposes it will be convenient to regard two vectors as being equal if they have the same magnitude and the same direction. In other words, we will regard $\overrightarrow{\mathbf{PQ}}$ and $\overrightarrow{\mathbf{RS}}$ as determining the *same vector* if $\overrightarrow{\mathbf{RS}}$ results by moving $\overrightarrow{\mathbf{PQ}}$ parallel to itself.

N.B. Vectors that conform to this definition are called *free vectors*, since we are "free to pick" their initial point. Not all "vectors" that occur in nature conform to this convention. If the vector quantity depends not only on its direction and magnitude but as well on its initial point it is called a *bound vector*. For Example, torque is a bound vector. In the force vector diagram represented by Figure 1.1.1 $\overrightarrow{\mathbf{PQ}}$ does not have the same effect as $\overrightarrow{\mathbf{RS}}$ in pivoting a bar. In this book we will consider only free vectors.

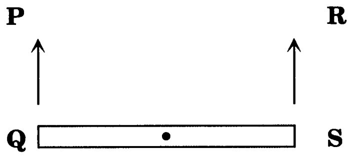


Figure 1.1.1

With this convention of equality of vectors in mind it is clear that if we *fix a point* **O** *in space* called the **origin**, then we may regard all our vectors as having their initial point at **O**. The vector $\overrightarrow{\mathbf{OP}}$ will very often be abbreviated to $\vec{\mathbf{P}}$ if the point **O** which serves as the origin of all vectors is clear from context. The vector $\vec{\mathbf{P}}$ is called the **position vector** of the point **P** relative to the origin **O**.

In physics vector quantities such as force vectors are often added together to obtain a resultant force vector. This process may be described as follows. Suppose an origin **O** has been fixed. Given vectors $\vec{\mathbf{P}}$ and $\vec{\mathbf{Q}}$ their sum is defined by the Figure 1.1.2. That is, draw the parallelogram determined by the three points **P**, **O** and **Q**. Let **R** be the fourth vertex and set $\vec{\mathbf{P}} + \vec{\mathbf{Q}} = \vec{\mathbf{R}}$.

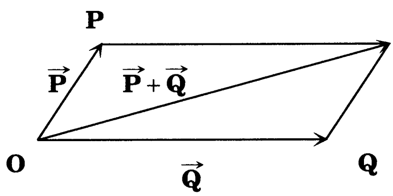


Figure 1.1.2

The following basic rules of vector algebra may be easily verified by elementary Euclidean geometry.

(1) $\vec{\mathbf{P}} + \vec{\mathbf{Q}} = \vec{\mathbf{Q}} + \vec{\mathbf{P}}$.
(2) $(\vec{\mathbf{P}} + \vec{\mathbf{Q}}) + \vec{\mathbf{R}} = \vec{\mathbf{P}} + (\vec{\mathbf{Q}} + \vec{\mathbf{R}})$.
(3) $\vec{\mathbf{P}} + \vec{\mathbf{O}} = \vec{\mathbf{P}} = \vec{\mathbf{O}} + \vec{\mathbf{P}}$.

It is also possible to define the operation of multiplying a vector by a number. Suppose we are given a vector $\vec{\mathbf{P}}$ and a number $a$. If $a > 0$, we let $a\vec{\mathbf{P}}$ be the vector with the same direction as $\vec{\mathbf{P}}$ only $a$ times as long (see Figure 1.1.3).

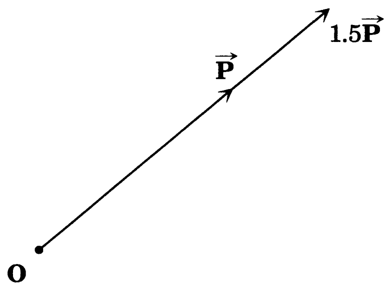


Figure 1.1.3

If $a < 0$, we set $a\vec{\mathbf{P}}$ equal to the vector of magnitude $a$ times the magnitude of $\vec{\mathbf{P}}$ but having direction *opposite* to that of $\vec{\mathbf{P}}$ (see Figure 1.1.4).

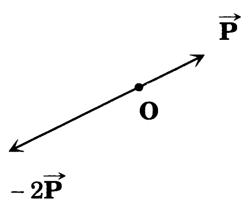


Figure 1.1.4

If $a = 0$ we set $a\vec{\mathbf{P}}$ equal to $\vec{\mathbf{O}}$. It is then easy to show that vector algebra satisfies the following additional rules.

(4) $\vec{\mathbf{P}} + (-1\vec{\mathbf{P}}) = \vec{\mathbf{O}}$.
(5) $a(\vec{\mathbf{P}} + \vec{\mathbf{Q}}) = a\vec{\mathbf{P}} + a\vec{\mathbf{Q}}$.
(6) $(a + b)\vec{\mathbf{P}} = a\vec{\mathbf{P}} + b\vec{\mathbf{P}}$.
(7) $(ab)\vec{\mathbf{P}} = a(b\vec{\mathbf{P}})$.
(8) $0\vec{\mathbf{P}} = \vec{\mathbf{O}}$, $1\vec{\mathbf{P}} = \vec{\mathbf{P}}$.

Note that Rule 6 involves two types of addition, namely addition of numbers and addition of vectors.

Vectors are particularly useful in studying lines and planes in space. Suppose that an origin **O** has been fixed and $L$ is the line through the two points **P** and **Q** as in Figure 1.1.5. Suppose that **R** is any other point on $L$. Consider the position vector $\vec{\mathbf{R}}$. Since the two points **P**, **Q** completely determine the line $L$, it is quite reasonable to look for some relation among the vectors $\vec{\mathbf{P}}$, $\vec{\mathbf{Q}}$, and $\vec{\mathbf{R}}$.

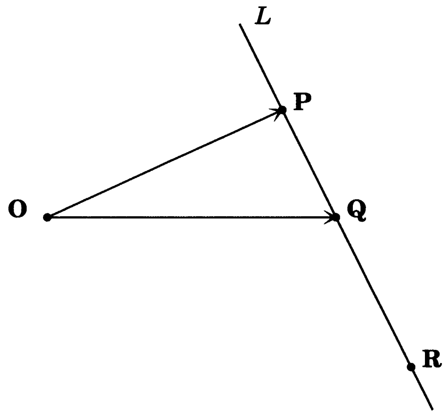


Figure 1.1.5

One such relation is provided by Figure 1.1.6.

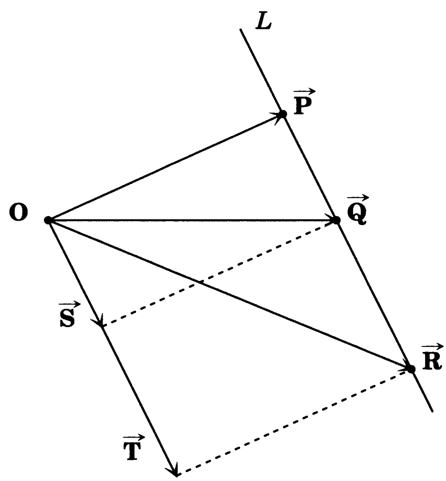


Figure 1.1.6

Observe that

$$\vec{\mathbf{S}} + \vec{\mathbf{P}} = \vec{\mathbf{Q}}.$$

Therefore, if we write $-\vec{\mathbf{P}}$ for $(-1)\vec{\mathbf{P}}$, we see that

$$\vec{\mathbf{S}} = \vec{\mathbf{Q}} - \vec{\mathbf{P}}.$$

Notice that there is a number $t$ such that

$$\vec{\mathbf{T}} = t\vec{\mathbf{S}}.$$

Moreover,

$$\vec{\mathbf{R}} = \vec{\mathbf{P}} + \vec{\mathbf{T}},$$

and hence we find

$$(*) \qquad \vec{\mathbf{R}} = \vec{\mathbf{P}} + t(\vec{\mathbf{Q}} - \vec{\mathbf{P}}).$$

Equation (*) is called the **vector equation** of the line $L$. To make practical computations with this equation it is convenient to introduce in addition to the origin **O** a Cartesian coordinate system as in Figure 1.1.7. Every point **P** then has coordinates $(x, y, z)$, and if we have two points **P** and **Q** with coordinates $(x_{\mathbf{P}}, y_{\mathbf{P}}, z_{\mathbf{P}})$ and $(x_{\mathbf{Q}}, y_{\mathbf{Q}}, z_{\mathbf{Q}})$ then it is quite easy to check that the vector $\vec{\mathbf{P}} + \vec{\mathbf{Q}}$ is the position vector of the point with components $(x_{\mathbf{P}} + x_{\mathbf{Q}}, y_{\mathbf{P}} + y_{\mathbf{Q}}, z_{\mathbf{P}} + z_{\mathbf{Q}})$.

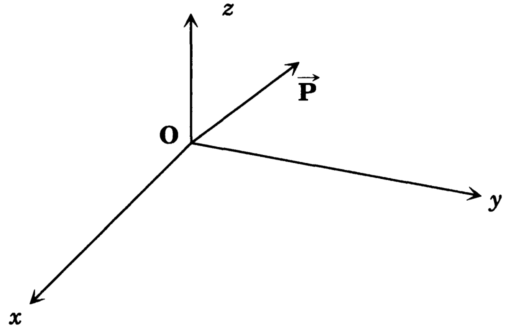


Figure 1.1.7

Likewise, for a number $a$ the vector $a\vec{\mathbf{P}}$ is the position vector of the point with coordinates $(ax_{\mathbf{P}}, ay_{\mathbf{P}}, az_{\mathbf{P}})$. Thus we find by considering the coordinates of the points represented equation (*) that $(x, y, z)$ lies on the line $L$ through **P**, **Q** if and only if there is a number $t$ such that

$$(**) \qquad \begin{aligned} x &= x_{\mathbf{P}} + t(x_{\mathbf{Q}} - x_{\mathbf{P}}), \\ y &= y_{\mathbf{P}} + t(y_{\mathbf{Q}} - y_{\mathbf{P}}), \\ z &= z_{\mathbf{P}} + t(z_{\mathbf{Q}} - z_{\mathbf{P}}). \end{aligned}$$

**EXAMPLE 1:** Does the point $(1, 2, 3)$ lie on the line passing through the points $(4, 4, 4)$ and $(1, 0, 1)$?

**SOLUTION:** Let $L$ be the line through $\mathbf{P} = (4, 4, 4)$ and $\mathbf{Q} = (1, 0, 1)$. Then the points of $L$ must satisfy the equations

$$x = 4 + t(1-4) = 4 - 3t,$$
$$y = 4 + t(0-4) = 4 - 4t,$$
$$z = 4 + t(1-4) = 4 - 3t,$$

where $t$ is a number. Let us check whether this is possible: namely, does there exist a number $t$ such that

$$1 = 4 - 3t,$$
$$2 = 4 - 4t,$$
$$3 = 4 - 3t.$$

The first equation gives

$$-3 = -3t \qquad t = 1.$$

Putting this in the last equation gives

$$3 = 4 - 3 = 1,$$

which is impossible. Therefore $(1, 2, 3)$ does not lie on the line through $(4, 4, 4)$ and $(1, 0, 1)$.

**EXAMPLE 2:** Let $L_1$ be the line through the points $(1, 0, 1)$ and $(1, 1, 1)$. Let $L_2$ be the line through the points $(0, 1, 0)$ and $(1, 2, 1)$. Determine whether the lines $L_1$ and $L_2$ intersect. If so find their point of intersection.

**SOLUTION:** The equations of $L_1$ are

$$x = 1 + t_1(1-1) = 1,$$
$$y = 0 + t_1(1-0) = t_1,$$
$$z = 1 + t_1(1-1) = 1.$$

The equations of $L_2$ are

$$x = 0 + (1-0)t_2 = t_2,$$
$$y = 1 + (2-1)t_2 = 1 + t_2,$$
$$z = 0 + (1-0)t_2 = t_2.$$

If a point lies on both of these lines we must have

$$1 = t_2,$$
$$t_1 = 1 + t_2,$$
$$1 = t_2.$$

Therefore $t_2 = 1$ and $t_1 = 2$. Hence $(1, 2, 1)$ is the only point these lines have in common.

**EXAMPLE 3:** Determine whether the lines $L_1$ and $L_2$ with equations

$$L_1 \quad \begin{cases} x = 1 - 3t, \\ y = 1 + 3t, \\ z = t, \end{cases}$$

$$L_2 \quad \begin{cases} x = -2 - 3t, \\ y = 4 + 3t, \\ z = 1 + t, \end{cases}$$

have a point in common.

**SOLUTION:** If a point $(x, y, z)$ lies on both lines, it must satisfy both sets of equations, so there is a number $t_1$ such that

$$\begin{aligned} x &= 1 - 3t_1, \\ y &= 1 + 3t_1, \\ z &= t_1, \end{aligned}$$

and a number $t_2$ with

$$\begin{aligned} x &= -2 - 3t_2, \\ y &= 4 + 3t_2, \\ z &= 1 + t_2, \end{aligned}$$

and the answer to the problem is reduced to determining whether in fact two such numbers can be found, that is if the simultaneous equations

$$(*) \qquad \begin{aligned} 1 - 3t_1 &= -2 - 3t_2, \\ 1 + 3t_1 &= 4 + 3t_2, \\ t_1 &= 1 + t_2, \end{aligned}$$

have any solutions. Writing these equations in the more usual form they become

$$\begin{aligned} 3 &= 3t_1 - 3t_2, \\ -3 &= -3t_1 + 3t_2, \\ -1 &= -t_1 + t_2. \end{aligned}$$

By dividing the first equation by 3, the second by $-3$, and multiplying the third by $-1$ we get

$$\begin{aligned} 1 &= t_1 - t_2, \\ 1 &= t_1 - t_2, \\ 1 &= t_1 - t_2, \end{aligned}$$

giving as the only requirement on $t_1$ and $t_2$ that

$$t_1 = 1 + t_2.$$

What does this mean? It means that no matter what value of $t_2$ we choose, there is a value of $t_1$, namely $t_1 = 1 + t_2$, that satisfies equations (✠). By varying the values of $t_2$ we get all the points on the line $L_2$. For each such value of $t_2$ the fact that there is a (corresponding) value of $t_1$ solving equations (✠) shows that every point of the line $L_2$ lies on the line $L_1$. Therefore these lines must be the same!

The lesson to be learned from this Example is that the equations of a line are not unique. This should be geometrically clear since we only used two points of the line to determine the equations, and there are many such possible pairs of points.

**EXAMPLE 4:** Determine if the lines $L_1$ and $L_2$ with equations

$$L_1 \quad \begin{cases} x = 1 + t, \\ y = 1 + t, \\ z = 1 - t, \end{cases}$$

$$L_2 \quad \begin{cases} x = 2 + t, \\ y = 2 - t, \\ z = 2 - t, \end{cases}$$

have a point in common.

**SOLUTION:** As in Example 3, our task is to determine whether the set of simultaneous equations

$$(✠✠) \qquad \begin{aligned} 1 + t_1 &= 2 + t_2, \\ 1 + t_1 &= 2 - t_2, \\ 1 - t_1 &= 2 - t_2, \end{aligned}$$

has any solutions. In more usual form these equations become

$$\begin{aligned} -1 &= -t_1 + t_2, \\ -1 &= -t_1 - t_2, \\ -1 &= t_1 - t_2. \end{aligned}$$

Adding the first two equations gives

$$-2 = -2t_1,$$

so $t_1$ must equal 1. Putting this into the last equation, we get

$$-1 = 1 - t_2,$$

so $t_2$ must equal 2. But substituting these values of $t_1$ and $t_2$ into either of the first two equations leads to a contradiction, namely

$$-1 = -1 + 2 = 1,$$
$$-1 = -1 - 2 = -3.$$

Therefore, no values of $t_1$ and $t_2$ can simultaneously satisfy equations (✠✠), so the lines have no point in common.

In Chapter 13 we will take up the study of solving simultaneous linear equations in detail. There we will explain various techniques and "tests" that will make the problems encountered in Examples 3 and 4 routine.

Suppose that **P**, **Q**, and **R** are three noncollinear points, i.e., the points **P**, **Q** and **R** do not all lie on the same line. Then they determine a unique plane Π. If we introduce a fixed origin **O** then it is possible to derive a vector equation satisfied by the points of Π. Considering Figure 1.1.8 shows that

$$\vec{\mathbf{A}} - \vec{\mathbf{Q}} = s(\vec{\mathbf{P}} - \vec{\mathbf{Q}}) + t(\vec{\mathbf{R}} - \vec{\mathbf{Q}});$$

that is,

$$(✠✠) \qquad \vec{\mathbf{A}} = s(\vec{\mathbf{P}} - \vec{\mathbf{Q}}) + t(\vec{\mathbf{R}} - \vec{\mathbf{Q}}) + \vec{\mathbf{Q}}.$$

Equation (✠✠) is called the **vector equation** of the plane Π. Compare it to the vector equation of a line. Note the presence of the two parameters $s$ and $t$ instead of the single parameter $t$.

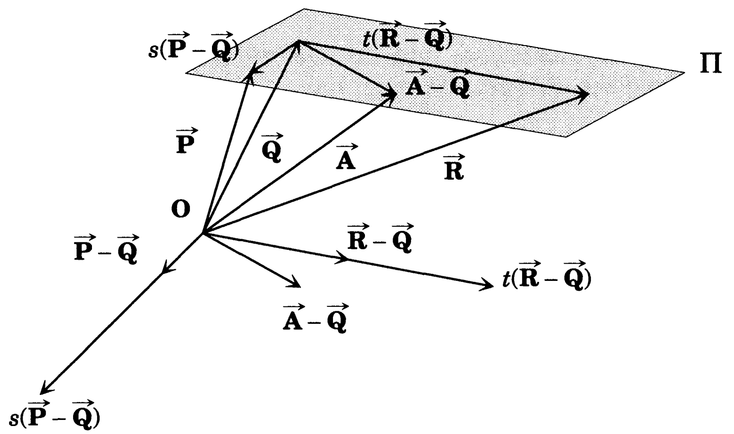


Figure 1.1.8

If we now introduce a coordinate system and pass to components in equation (✠✠), we obtain

$$
(✣) \qquad \begin{aligned}
x &= s(x_{\mathbf{P}} - x_{\mathbf{Q}}) + t(x_{\mathbf{R}} - x_{\mathbf{Q}}) + x_{\mathbf{Q}},\\
y &= s(y_{\mathbf{P}} - y_{\mathbf{Q}}) + t(y_{\mathbf{R}} - y_{\mathbf{Q}}) + y_{\mathbf{Q}},\\
z &= s(z_{\mathbf{P}} - z_{\mathbf{Q}}) + t(z_{\mathbf{R}} - z_{\mathbf{Q}}) + z_{\mathbf{Q}}.
\end{aligned}
$$

We may regard equation (✣) as the equation of the plane $\Pi$ or we may regard it as a system of three equations in the two unknowns $s$, $t$ which we may formally eliminate and obtain the more familiar equation of a plane

$$
(✣✣) \qquad ax + by + cz + d = 0,
$$

where we may take (or twice these values, or –7 times, etc.)

$$
\begin{aligned}
a &= (y_{\mathbf{R}} - y_{\mathbf{Q}})(z_{\mathbf{P}} - z_{\mathbf{Q}}) - (z_{\mathbf{R}} - z_{\mathbf{Q}})(y_{\mathbf{P}} - y_{\mathbf{Q}}),\\
b &= (z_{\mathbf{R}} - z_{\mathbf{Q}})(x_{\mathbf{P}} - x_{\mathbf{Q}}) - (x_{\mathbf{R}} - x_{\mathbf{Q}})(z_{\mathbf{P}} - z_{\mathbf{Q}}),\\
c &= (x_{\mathbf{R}} - x_{\mathbf{Q}})(y_{\mathbf{P}} - y_{\mathbf{Q}}) - (y_{\mathbf{R}} - y_{\mathbf{Q}})(x_{\mathbf{P}} - x_{\mathbf{Q}}),\\
d &= -(ax_{\mathbf{Q}} + by_{\mathbf{Q}} + cz_{\mathbf{Q}}).
\end{aligned}
$$

Equation (✣✣) is also called the equation of the plane $\Pi$.

**EXAMPLE 5:** Find the equation of the plane through the points

$$(1, 0, 1), \qquad (0, 1, 0), \qquad (1, 1, 1).$$

Determine whether the point $(0, 0, 0)$ lies in this plane.

**SOLUTION:** We know that the equation has the form

$$ax + by + cz + d = 0,$$

and all we must do is find suitable values for $a, b, c, d$. (Remember they are not unique). We must have

$$
\begin{aligned}
a + c + d &= 0,\\
b + d &= 0,\\
a + b + c + d &= 0,
\end{aligned}
$$

since the points $(1, 0, 1)$, $(0, 1, 0)$, and $(1, 1, 1)$ lie in this plane. Thus

$$a + c = 0, \qquad d = 0, \qquad b = 0, \qquad a = -c.$$

So the plane has the equation

$$x - z = 0,$$

and $(0, 0, 0)$ lies in it.

**EXAMPLE 6:** Determine the equation of the line of intersection of the planes

$$x - z = 0,$$
$$x + y + z + 1 = 0.$$

**SOLUTION:** The line in question has an equation of the form

$$x = a + ut,$$
$$y = b + vt,$$
$$z = c + wt,$$

for suitable numbers $a, b, c, u, v, w$. Since such points must lie in both planes we have

$$a + ut - (c + wt) = 0,$$
$$a + ut + b + vt + c + wt + 1 = 0,$$

for all values of $t$. Put $t = 0$. Then

$$a - c = 0,$$
$$a + b + c + 1 = 0.$$

The first equation yields $a = c$. Combining this with the second equation and setting $b = 1$ yields $2a + 2 = 0$. Hence $a = -1 = c$. Next put $t = 1$. Then

$$0 = a + ut - (c + wt) = -1 + u + 1 - w,$$
$$0 = a + ut + b + vt + c + wt + 1$$
$$= -1 + u + 1 + v - 1 + w + 1.$$

The first equation yields $u = w$. Combining this with the second equation and setting $u = 1$ yields $w = u = 1$ and $v = -2$. Then

$$x = -1 + t,$$
$$y = 1 - 2t,$$
$$z = -1 + t,$$

are the equations of a line containing the two points $(-1, 1, -1)$ and $(0, -1, 0)$ that lies in both planes and hence must be the line of intersection.

## 1.2 Exercises

**1.** On the dedication page of this book are the names, and where they worked, of mathematicians who are connected with the history of linear algebra and whose names appear elsewhere in this book. Find out who they were: this means that you may have to go to the library and learn how to look up subjects from mathematics and history. Do it.

**2.** Suppose that an origin **O** and a coordinate system have been fixed. Let **P** be a point. Define vectors $\overrightarrow{\mathbf{E}_1}, \overrightarrow{\mathbf{E}_2}$, and $\overrightarrow{\mathbf{E}_3}$ by requiring that they be the position vectors of the points $(1, 0, 0), (0, 1, 0)$, and $(0, 0, 1)$, respectively. Let the coordinates of **P** be $(x_\mathbf{P}, y_\mathbf{P}, z_\mathbf{P})$. Show that

$$\overrightarrow{\mathbf{P}} = x_\mathbf{P}\overrightarrow{\mathbf{E}_1} + y_\mathbf{P}\overrightarrow{\mathbf{E}_2} + z_\mathbf{P}\overrightarrow{\mathbf{E}_3}.$$

The vectors

$$x_\mathbf{P}\overrightarrow{\mathbf{E}_1}, \qquad y_\mathbf{P}\overrightarrow{\mathbf{E}_2}, \qquad z_\mathbf{P}\overrightarrow{\mathbf{E}_3}$$

are called the **component vectors** of $\overrightarrow{\mathbf{P}}$ relative to the given coordinate system.

**3.** Find the equation of the line through the points $(1, 0, -1), (2, 3, -1)$. Does the point $(0, 1, -1)$ lie on this line?

**4.** Does the point $(1, 1, 1)$ lie in the plane containing the points $(1, 1, 0)$, $(0, 1, 1), (1, 0, 1)$?

**5.** Does the line through the points $(1, 1, 1), (1, -1, 1)$ lie in the plane containing the points $(1, -1, 0), (1, 0, -1), (-1, 1, 1)$?

**6.** Show that the point $(1, -2, 1)$ lies on the line through the two points $(0, 1, -1)$ and $(2, -5, 3)$.

**7.** Find the equation of the plane $\Pi$ containing the three points $(1, 1, 1)$, $(1, 2, 3)$, and $(2, -1, 0)$.

(1) Find an equation for a line $L_1$ with exactly one point in common with $\Pi$.

(2) Find an equation for a line $L_2$ with no points in common with $\Pi$ (i.e., $L_2$ should be *parallel* to $\Pi$).

(3) Find an equation for a line $L_3$ lying in the plane $\Pi$.

**8.** Let $\overrightarrow{\mathbf{P}} = (x_1, y_1, z_1)$, $\overrightarrow{\mathbf{Q}} = (x_2, y_2, z_2)$ be two points. Show that the midpoint of the line segment joining them has coordinates

$$\left(\frac{x_1 + x_2}{2}, \quad \frac{y_1 + y_2}{2}, \quad \frac{z_1 + z_2}{2}\right).$$

**9.** Show that the midpoints of the edges of a quadrangle[1] are the vertices of a parallelogram.

[1] A **quadrangle** is a closed figure in a plane with four straight sides: Examples

**10.** Find the equation of the line through the origin bisecting the angle formed by **AOB**, where $\mathbf{A} = (1, 0, 0)$, $\mathbf{B} = (0, 0, 1)$.

**11.** Verify that vectors $\overrightarrow{\mathbf{PQ}}$ and $\overrightarrow{\mathbf{RS}}$ represent the same vector $\vec{\mathbf{T}}$, where $\mathbf{P} = (0, 1, 1)$, $\mathbf{Q} = (1, 3, 4)$, $\mathbf{R} = (1, 0, -1)$, $\mathbf{S} = (2, 2, 2)$. Find the coordinates of $\vec{\mathbf{T}}$.

**12.** Find the sum of the vectors $\overrightarrow{\mathbf{PQ}}$ and $\overrightarrow{\mathbf{RS}}$, where $\mathbf{P} = (0, 1, 1)$, $\mathbf{Q} = (1, 0, 0)$, $\mathbf{R} = (1, 0, -1)$, $\mathbf{S} = (2, 2, 2)$.

**13.** Let $\mathbf{P} = (1, 1)$, $\mathbf{Q} = (2, 3)$, $\mathbf{R} = (-2, 3)$, $\mathbf{S} = (1, -1)$. Find $\overrightarrow{\mathbf{PQ}} - \overrightarrow{\mathbf{RS}}$, $\overrightarrow{\mathbf{PQ}} + \overrightarrow{\mathbf{RS}}$.

**14.** Show that the points **A, B, C, D** with the following coordinates form a parallelogram in a plane: $\mathbf{A} = (1, 1)$, $\mathbf{B} = (3, 2)$, $\mathbf{C} = (2, 3)$, $\mathbf{D} = (0, 2)$.

**15.** Let $\mathbf{P} = (1, 0, 1)$, $\mathbf{Q} = (1, 1, 1)$, and $\mathbf{R} = (-1, 1, -1)$. Find the coordinates of $\vec{\mathbf{T}}$, where

(1) $\vec{\mathbf{T}} = 2\vec{\mathbf{P}} - \vec{\mathbf{Q}}$,
(2) $\vec{\mathbf{T}} = \overrightarrow{\mathbf{PQ}}$,
(3) $\vec{\mathbf{T}} = 2\vec{\mathbf{R}}$,
(4) $\vec{\mathbf{T}} = -\vec{\mathbf{R}}$,
(5) $\vec{\mathbf{T}} = \overrightarrow{\mathbf{PQ}} + \overrightarrow{\mathbf{PR}}$,
(6) $\vec{\mathbf{T}} = a\vec{\mathbf{P}} + b\vec{\mathbf{Q}} + c\vec{\mathbf{R}}$, where $a$, $b$, $c$ are given constants.

**16.** In each of (1)–(7) below find a vector equation of the line satisfying the given conditions:

(1) Passing through the point $\mathbf{P} = (-2, 1)$ and having slope $\frac{1}{2}$.
(2) Passing through the point $(0, 3)$ and parallel to the $x$-axis.
(3) The tangent line to $y = x^2$ at $(2, 4)$.
(4) The line parallel to the line of (c) passing through the origin.
(5) The line passing through points $(1, 0, 1)$ and $(1, 1, 1)$.
(6) The line passing through the origin and the midpoint of the line segment $\overrightarrow{\mathbf{PQ}}$, where $\mathbf{P} = (\frac{1}{2}, \frac{1}{2}, 0)$, $\mathbf{Q} = (0, 0, 1)$.
(7) The line in the $xy$-plane passing through $(1, 1, 0)$ and $(0, 1, 0)$.

**17.** In each of (1)–(7) determine a vector equation of the plane satisfying the given conditions:

(1) The plane determined by $(0, 0)$, $(1, 0)$ and $(1, 1)$.
(2) The plane determined by $(0, 0, 1)$, $(1, 0, 1)$, and $(1, 1, 1)$.

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are squares and rectangles, but *any* four sided figure with straight sides is a quadrangle; the angles do not have to be right angles, and the sides do not all have to have the same length.

(3) The plane determined by $(1, 0, 0)$, $(0, 1, 0)$, and $(1, 1, 1)$. Does the origin lie on this plane?
(4) The plane parallel to the $xy$-plane and containing the point $(1, 1, 1)$.
(5) The plane through the origin and containing the points $\mathbf{P} = (1, 0, 0)$, $\mathbf{Q} = (0, 1, 0)$
(6) The plane through three points **A**, **B**, **C**, where $\mathbf{A} = (1, 0, 1)$, $\mathbf{B} = (-1, 2, 3)$, and $\mathbf{C} = (2, 6, 1)$. Does the origin lie on this plane?
(7) The plane parallel to $yz$-plane passing through the point $(1, 1, 1)$.

**18.** Let $a$ be a real number. Determine the equation of a line $L$ in the plane, passing through the origin, such that the area of the figure bounded by $L$ and the two lines

$$x + y - a = 0, \quad x = 0$$

is $a^2$.

# Chapter 2
# Vector Spaces

In the previous chapter we reviewed the intuitive notions connected with vectors in space and their algebra. We derived vector equations for lines and planes and saw how once a coordinate system was chosen these vector equations lead to the familiar equations of analytic geometry. However, particularly in application to physics, it is often very important to know the relation between the equations for the same plane (or line) in different coordinate systems. This leads us to the notion of a *coordinate transformation*. The appropriate domain in which to study such transformations is that of the abstract vector spaces to be introduced in this chapter.

## 2.1 Axioms for Vector Spaces

The following definition lays the basis for our study of vector spaces.

DEFINITION : *A* **vector space** *is a set* $\mathcal{V}$*, whose elements are called vectors, together with two operations. The first operation, called* **vector addition**, *assigns to each pair of vectors* **A** *and* **B** *a vector denoted by* $\mathbf{A} + \mathbf{B}$*, called their* **sum**. *The second operation, called* **scalar multiplication**, *assigns to each vector* **A** *and each number*[1] $r$ *a vector denoted by* $r\mathbf{A}$*. The two operations are required to have the following properties:*

AXIOM 1. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ *for each pair of vectors* **A** *and* **B** *(commutative law of vector addition).*

AXIOM 2. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$ *for each triple of vectors* **A**, **B**, *and* **C** *(associative law of vector addition).*

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[1] Unless stated to the contrary the word ***number*** will mean a real number.

**Axiom 3.** *There is a unique vector* $\mathbf{0}$*, called the* **zero vector***, such that* $\mathbf{A} + \mathbf{0} = \mathbf{A}$ *for every vector* $\mathbf{A}$.

**Axiom 4.** *For each vector* $\mathbf{A}$ *there corresponds a unique vector* $-\mathbf{A}$ *such that* $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$.

**Axiom 5.** $r(\mathbf{A} + \mathbf{B}) = r\mathbf{A} + r\mathbf{B}$ *for each real number* $r$ *and each pair of vectors* $\mathbf{A}$ *and* $\mathbf{B}$.

**Axiom 6.** $(r + s)\mathbf{A} = r\mathbf{A} + s\mathbf{A}$ *for each pair of real numbers* $r$ *and* $s$ *and each vector* $\mathbf{A}$.

**Axiom 7.** $(rs)\mathbf{A} = r(s\mathbf{A})$ *for each pair* $r$, $s$ *of real numbers and each vector* $\mathbf{A}$.

**Axiom 8.** *For each vector* $\mathbf{A}$, $1\mathbf{A} = \mathbf{A}$.

The development of linear algebra in this book uses the *axiomatic method*. That is, ***vector***, ***vector addition***, and ***scalar multiplication*** constitute the basic terms of the theory. They are not defined, but rather our study of linear algebra will be based on the properties of these terms as specified by the preceding eight axioms. In the axiomatic treatment, what vectors, vector addition, and scalar multiplication are is immaterial; rather what is important is the properties that these quantities have as consequences of the axioms. Thus in our development of the theory we may not use properties of vectors that are not stated in, or are consequences of, the preceding axioms. We may use any properties of vectors, etc. that are stated in the axioms: for example, that the vector $\mathbf{0}$ is unique, or that $\mathbf{A} = 1\mathbf{A}$ for any vector $\mathbf{A}$. On the other hand, we may ***not*** use that a vector is an arrow with a specified head and tail.

The advantage of the axiomatic approach is that results so obtained will apply to any special case or example that we wish to consider. The converse is definitely false, i.e., results obtained in a special case need not apply to all. Presently we will see a large number of examples of vector spaces. Let us begin with some elementary consequences of the axioms.

**Proposition 2.1.1:** $0\mathbf{A} = \mathbf{0}$.

**Proof:** We have

$$\mathbf{A} = 1\mathbf{A} = (1 + 0)\mathbf{A} = 1\mathbf{A} + 0\mathbf{A} = \mathbf{A} + 0\mathbf{A}$$

by using Axioms 8, 6, and 8 again. To the equation

$$\mathbf{A} = \mathbf{A} + 0\mathbf{A}$$

we apply Axiom 1, getting

$$\mathbf{A} = 0\mathbf{A} + \mathbf{A}.$$

If we apply Axiom 4, we obtain by Axiom 2,

$$\begin{aligned}\mathbf{0} &= \mathbf{A} + (-\mathbf{A}) = (0\mathbf{A} + \mathbf{A}) + (-\mathbf{A}) = 0\mathbf{A} + (\mathbf{A} + (-\mathbf{A}))\\ &= 0\mathbf{A} + \mathbf{0} = 0\mathbf{A}\end{aligned}$$

where the last equality is by Axiom 3. That is ,

$$\mathbf{0} = 0\mathbf{A},$$

which is the desired conclusion. □

NOTATION : *We will reserve boldface capital letters for vectors and small italic letters for numbers (which are also called* **scalars***).* [2]

The proof of Proposition 2.1.1 was given in considerable detail to illustrate how results are deduced by the axiomatic method. In the sequel we will not be so detailed in our proofs, leaving to the reader the task of providing as much detail as is felt necessary.

**PROPOSITION 2.1.2:** $(-1)\mathbf{A} = -\mathbf{A}$.

**PROOF :** By Proposition 2.1.1 we have

$$\mathbf{0} = 0\mathbf{A} = (1-1)\mathbf{A} = 1\mathbf{A} + (-1)\mathbf{A} = \mathbf{A} + (-1)\mathbf{A}.$$

Add $-\mathbf{A}$ to both sides, giving

$$\begin{aligned}-\mathbf{A} &= -\mathbf{A} + (\mathbf{A} + (-1)\mathbf{A}) = (-\mathbf{A} + \mathbf{A}) + (-1)\mathbf{A}\\ &= (\mathbf{A} - \mathbf{A}) + (-1)\mathbf{A} = \mathbf{0} + (-1)\mathbf{A} = (-1)\mathbf{A} + \mathbf{0}\\ &= (-1)\mathbf{A},\end{aligned}$$

as required. □

**PROPOSITION 2.1.3:** $\mathbf{0} + \mathbf{A} = \mathbf{A}$.

**PROOF :** Exercise. □

These formal deductions may seem like a sterile intellectual exercise—an indication of the absurdity of too much reliance on abstraction and formalism. Quite to the contrary, however, they illustrate the advantages of the abstract formulation of a mathematical theory. For if the basic terms are not defined, the possibility is opened of assigning to them content in new and unforeseen ways. If in this way the axioms become true statements about the meanings assigned to the basic

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[2] Actually the font used for numbers is called `\mit` in $\mathcal{LS}$TEX and not `\it`. See the list of fonts used at the end of the book.

terms *vector, vector addition*, and *scalar multiplication*, then we have constructed a **model** for the abstract theory: and in this model all the theorems, propositions, etc. that we deduced from the axioms are also true statements about the assigned meanings of *vector, vector addition*, and *scalar multiplication*.

## 2.2 Cartesian (or Euclidean) Spaces

One of the standard models of a vector space is the generalization of the familiar Euclidean plane usually encountered in a calculus course. Before plunging into a discussion of these examples, we pause to make some comments on the axioms. Note that Axioms 3 and 4 are of a "different character" than the others. Apart from Axioms 3 and 4, verifying that an axiom holds in an example usually may be reduced to some property of real numbers. Axioms 3 and 4 are different in that they assert the existence of things and their uniqueness. In other words, one will have to specify which vector in the example under discussion is the zero vector, and then verify that it has the property stated in the axiom and that it is the only vector in the example with this property.

Here is one standard model of a vector space.

DEFINITION: *Let $k$ be a positive integer. The* **Cartesian $k$-space,** *denoted by $\mathbb{R}^k$, is the set of all $k$-tuples $(a_1, a_2, ,\ldots, a_k)$ of $k$ real numbers, together with the two operations (vector addition)*

$$(a_1, a_2, \ldots, a_k) + (b_1, b_2, \ldots, b_k) = (a_1 + b_1, a_2 + b_2, \ldots, a_k + b_k)$$

*and (scalar multiplication)*

$$r(a_1, \ldots, a_k) = (ra_1, ra_2, \ldots, ra_k).$$

If $(a_1, \ldots, a_k)$ is a vector in $\mathbb{R}^k$, then the number $a_i$ is called the $i$-th **component** of the vector $(a_1, \ldots, a_k)$. We can think of $\mathbb{R}^1$ as being just $\mathbb{R}$ by with its usual addition and multiplication by ignoring the parentheses ( ) around the numbers.

THEOREM 2.2.1: *For each positive integer $k$, $\mathbb{R}^k$ is a vector space.*

Before beginning the proof of Theorem 2.2.1, let us consider exactly what it is that we are trying to prove. We are going to assign meanings to the three basic terms of the axioms for a vector space. Namely, by a vector we will mean a $k$-tuple $(a_1, \ldots, a_k)$ of real numbers $a_1, \ldots, a_k$. For vectors $\mathbf{A} = (a_1, \ldots, a_k)$ and $\mathbf{B} = (b_1, \ldots, b_k)$, the equality of the two vectors $\mathbf{A} = \mathbf{B}$ means that $a_1 = b_1, a_2 = b_2, \ldots, a_k = b_k$. By addition of **A** and **B**, we shall mean the vector $(a_1 + b_1, \ldots, a_k + b_k)$; that is, we define

$$\mathbf{A} + \mathbf{B} = (a_1 + b_1, \ldots, a_k + b_k).$$

Likewise, for a real number $r$ and a $k$-tuple **A** we define

$$r\mathbf{A} = (ra_1, \ldots, ra_k).$$

Axioms 1–8 for a vector space then become statements about $k$-tuples, and *we must verify that they are true statements.*

**PROOF OF 2.2.1:** We will verify the axioms in turn.

**AXIOM 1.** Let $\mathbf{A} = (a_1, \ldots, a_k)$, $\mathbf{B} = (b_1, \ldots, b_k)$. Then

$$\begin{aligned} \mathbf{A} + \mathbf{B} &= (a_1 + b_1, \ldots, a_k + b_k) \\ &= (b_1 + a_1, \ldots, b_k + a_k) = \mathbf{B} + \mathbf{A}, \end{aligned}$$

so Axiom 1 is true, since addition of real numbers is commutative.

**AXIOM 2.** Let $\mathbf{A} = (a_1, \ldots, a_k)$, $\mathbf{B} = (b_1, \ldots, b_k)$ and $\mathbf{C} = (c_1, \ldots, c_k)$. Then by the associative law for addition of real numbers,

$$\begin{aligned} (\mathbf{A} + \mathbf{B}) + \mathbf{C} &= (a_1 + b_1, \ldots, a_k + b_k) + (c_1, \ldots, c_k) \\ &= (a_1 + b_1 + c_1, \ldots, a_k + b_k + c_k) \\ &= (a_1, \ldots, a_k) + (b_1 + c_1, \ldots, b_k + c_k) \\ &= \mathbf{A} + (\mathbf{B} + \mathbf{C}), \end{aligned}$$

so Axiom 2 holds.

**AXIOM 3.** We let $\mathbf{0} = (0, \ldots, 0)$. Then for any $\mathbf{A} = (a_1, \ldots, a_k)$ we will have

$$\mathbf{A} + \mathbf{0} = (a_1 + 0, \ldots, a_k + 0) = (a_1, \ldots, a_k) = \mathbf{A}.$$

Moreover if $\mathbf{B} = (b_1, \ldots, b_k)$ is any vector such that

$$\mathbf{A} + \mathbf{B} = \mathbf{A},$$

then

$$(a_1 + b_1, \ldots, a_k + b_k) = (a_1, \ldots, a_k),$$

and therefore

$$\begin{array}{rclcrcl} a_1 + b_1 & = & a_1 & \Rightarrow & b_1 & = & 0, \\ a_2 + b_2 & = & a_2 & \Rightarrow & b_2 & = & 0, \\ & \vdots & & & & \vdots & \\ a_k + b_k & = & a_k & \Rightarrow & b_k & = & 0; \end{array}$$

i.e., $\mathbf{B} = \mathbf{0}$. Thus $\mathbf{0}$ is the unique vector with the property that $\mathbf{A} + \mathbf{0} = \mathbf{A}$, and Axiom 3 holds.

**AXIOM 4.** Let $\mathbf{A} = (a_1, \ldots, a_k)$ and set $-\mathbf{A} = (-a_1, \ldots, -a_k)$. Then

$$\begin{aligned}\mathbf{A} + (-\mathbf{A}) &= (a_1, \ldots, a_k) + (-a_1, \ldots, -a_k)\\ &= (a_1 - a_1, \ldots, a_k - a_k) = (0, \ldots, 0) = \mathbf{0}.\end{aligned}$$

Moreover, if $\mathbf{C} = (c_1, \ldots, c_k)$ is any vector such that

$$\mathbf{A} + \mathbf{C} = \mathbf{0},$$

then

$$(a_1 + c_1, \ldots, a_k + c_k) = (0, \ldots, 0),$$

and therefore

$$\begin{array}{rcccrcl} a_1 + c_1 &=& 0 &\Rightarrow& c_1 &=& -a_1,\\ a_2 + c_2 &=& 0 &\Rightarrow& c_2 &=& -a_2,\\ &\vdots& & & &\vdots& \\ a_k + c_k &=& 0 &\Rightarrow& c_k &=& -a_k;\end{array}$$

i.e., $\mathbf{C} = -\mathbf{A}$. Thus $-\mathbf{A}$ is the unique vector with the property that $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$ and Axiom 4 holds.

**AXIOM 5.** Let $r$ be a real number and let $\mathbf{A} = (a_1, \ldots, a_k)$ and $\mathbf{B} = (b_1, \ldots, b_k)$ be vectors. Then

$$\begin{aligned}r(\mathbf{A} + \mathbf{B}) &= r(a_1 + b_1, \ldots, a_k + b_k) = (r(a_1 + b_1), \ldots, r(a_k + b_k))\\ &= (ra_1 + rb_1, \ldots, ra_k + rb_k) = (ra_1, \ldots, ra_k) + (rb_1, \ldots, rb_k)\\ &= r(a_1, \ldots, a_k) + r(b_1, \ldots, b_k) = r\mathbf{A} + r\mathbf{B},\end{aligned}$$

so that Axiom 5 is satisfied.

**AXIOM 6.** Let $r, s$ be numbers and $\mathbf{A} = (a_1, \ldots, a_k)$. Then

$$\begin{aligned}(r + s)\mathbf{A} &= ((r + s)a_1, \ldots, (r + s)a_k)\\ &= (ra_1 + sa_1, \ldots, ra_k + sa_k)\\ &= (ra_1, \ldots, ra_k) + (sa_1, \ldots, sa_k)\\ &= r(a_1, \ldots, a_k) + s(a_1, \ldots, a_k)\\ &= r\mathbf{A} + s\mathbf{A},\end{aligned}$$

so Axiom 6 holds.

**AXIOM 7.** Let $r, s$ be numbers and $\mathbf{A} = (a_1, \ldots, a_k)$. Then

$$\begin{aligned}(rs)\mathbf{A} &= (rsa_1, \ldots, rsa_k) = r(sa_1, \ldots, sa_k)\\ &= r(s(a_1, \ldots, a_k)) = r(s(\mathbf{A})),\end{aligned}$$

so Axiom 7 holds.

**AXIOM 8.** Instant.

Therefore, $\mathbb{R}^k$ is a vector space. $\square$

## 2.3 Some Rules for Vector Algebra

In the chapters that follow we will introduce many additional examples of vector spaces. To close this chapter let us indicate some further elementary consequences of the axioms: the generalized associative and commutative laws, which allow us to drop parentheses from formulas when they serve no useful purpose (such as grouping terms together to emphasize the result of a computation not shown in the text).

**PROPOSITION 2.3.1** (Generalized Associative Law): *Let $n \geq 3$ be an integer. Then any two ways of associating a sum*

$$\mathbf{A}_1 + \cdots + \mathbf{A}_n$$

*of $n$ vectors give the same vector. Consequently, sums may be written without parentheses.*

The proof of this proposition is elementary and may be carried out by induction on $n$. Similarly, we have:

**PROPOSITION 2.3.2** (Generalized Commutative Law): *Let $n$ be any integer $\geq 2$. Then the sum of any $n$ vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ is independent of the order in which the sum is taken.*

It will be convenient to introduce some notations for describing the relationships between vectors and vector spaces, or more generally between sets and their elements.

**NOTATION:** *We will use the symbol $\in$ as an abbreviation for "is an element of." Thus $x \in S$ should be read as, $x$ is an element of the set $S$. The symbol $\subseteq$ is an abbreviation for "is contained in." Thus $S \subseteq T$ should be read as, the set $S$ is contained in the set $T$. If $S \subseteq T$ and $S \neq T$ then we use the symbol $\subset$ or $\subsetneqq$ to indicate this, and write for example $S \subsetneqq T$ in this case.*[3]

*If $S$ and $T$, are sets then the collection of elements contained in either set is denoted by $S \cup T$. Thus $x \in S \cup T$ is equivalent to $x \in S$ or $x \in T$. The collection of all elements common to both sets is denoted by $S \cap T$. Thus $x \in S \cap T$ is equivalent to $x \in S$ and $x \in T$. The set $S \cup T$ is called the* **union** *of $S$ and $T$, and $S \cap T$ is the* **intersection** *of $S$ and $T$. We denote by $\varnothing$ the empty set, i.e., the set with no elements.*

The axioms for a vector space that we have given are the axioms for a **real** vector space, that is, a vector space whose scalars are the *real numbers*, which we have denoted by $\mathbb{R}$. It is also possible, and often important, to study vector spaces whose scalars are the *complex*

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[3] There are many other variations of these symbols, such as; e.g., $\subsetneqq$, $\not\subseteq$, etc..

*numbers*, which we denote by $\mathbb{C}$. A vector space with complex scalars is called a **complex vector space**. The axioms for a complex vector space are exactly the same as for a real vector space except that the numbers (= scalars) are to be complex. The generic example of a complex vector space is the complex Cartesian space $\mathbb{C}^k$ of $k$-tuples $\mathbf{A} = (a_1, \ldots, a_k)$ of *complex* numbers, where for vectors $\mathbf{A} = (a_1, \ldots, a_k)$ and $\mathbf{B} = (b_1, \ldots, b_k)$ and for scalars $r \in \mathbb{C}$, vector addition and scalar multiplication are given by

$$\mathbf{A} + \mathbf{B} = (a_1 + b_1, \ldots, a_k + b_k)$$

and

$$r\mathbf{A} = (ra_1, \ldots, ra_k).$$

In the next few chapters we will study only real vector spaces, indicating where necessary the modifications required in the complex case.

## 2.4 Exercises

**1.** Let $A = \left\{(2a, a) \mid a \in \mathbb{R}\right\}$, $B = \left\{(b, b) \mid b \in \mathbb{R}\right\}$. Find $A \cup B$ and $A \cap B$.

**2.** Denote by $\mathbb{Z}$ the set of all integers and let $A = \left\{(2n, n) \mid n \in \mathbb{Z}\right\}$, $B = \left\{(k + 1, k) \mid k \in \mathbb{Z}\right\}$. Find $A \cup B, A \cap B$.

**3.** If $A \subseteq B, B \subseteq C$, then $A \subseteq C$.

**4.** If $A \subseteq B, A \subseteq C$, then $A \subseteq B \cap C$.

**5.** If $A \supseteq B, A \supseteq C$, then $A \supseteq B \cup C$.

**6.** Show that $A \cap (B \cap C) = (A \cap B) \cap (A \cap C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, where $A, B, C$ are sets.

**7.** Let $A = \left\{x \in \mathbb{R} \mid |x| > 1\right\}$, $B = \left\{x \in \mathbb{R} \mid -2 < x < 3\right\}$. Find $A \cup B$ and $A \cap B$.

**8.** Let $\mathcal{V}$ be a vector space. Prove each of the following statements:
- (a) If $\mathbf{A} \in \mathcal{V}$ and $a$ is a number, then $a\mathbf{A} = \mathbf{0}$ if and only if $a = 0$ or $\mathbf{A} = \mathbf{0}$, or both.
- (b) If $\mathbf{A} \in \mathcal{V}$ and $a$ is a number, then $a\mathbf{A} = \mathbf{A}$ if and only if $a = 1$ or $\mathbf{A} = \mathbf{0}$, or both.

**9.** Let $\mathcal{V}$ be the set of all ordered pairs of real numbers $(a, b)$. Define an addition for the elements of $\mathcal{V}$ by the rule

$$(a, b) \oplus (c, d) = (a + c, b + d),$$

and a multiplication of elements of $\mathcal{V}$ by numbers by the rule

$$r \cdot (c, d) = (rc, 0).$$

Is $\mathcal{V}$ with these two operations a vector space? Justify your answer.

**10.** Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces. Denote by $\mathcal{V} \oplus \mathcal{W}$ the set of all ordered pairs $(\mathbf{A}, \mathbf{B})$, where $\mathbf{A} \in \mathcal{V}$ and $\mathbf{B} \in \mathcal{W}$. Define an addition for the elements of $\mathcal{V} \oplus \mathcal{W}$ by the rule

$$(\mathbf{A}, \mathbf{B}) + (\mathbf{C}, \mathbf{D}) = (\mathbf{A} + \mathbf{C}, \mathbf{B} + \mathbf{D}),$$

and a multiplication of elements of $\mathcal{V}$ by numbers by the rule

$$r \cdot (\mathbf{A}, \mathbf{B}) = (r\mathbf{A}, r\mathbf{B}).$$

Is $\mathcal{V} \oplus \mathcal{W}$ with these two operations a vector space? Justify your answer. (Beware of the fact that $+$ and $\cdot$ are used with multiple meanings in the preceding equations. $\mathcal{V} \oplus \mathcal{W}$ is called the **direct sum** of $\mathcal{V}$ and $\mathcal{W}$.)

**11.** A **translation** of the plane $\Pi$ is a function $T : \Pi \longrightarrow \Pi$ with the following two properties:

① There is a constant $k$, called the **length** of the translation, such that for every point $p \in \Pi$ the distance from $p$ to $T(p)$ is equal to $k$.

② For any two points $p, q \in \Pi$ the distance from $p$ to $q$ is the same as the distance from $T(p)$ to $T(q)$.

Introduce Cartesian coordinates in $\Pi$ and show:

(a) If $T$ is a translation, then for every $p = (x, y) \in \Pi$, $T(p) = (x + l_1, x + l_2)$, where $T(0,0) = (l_1, l_2)$. $T$ is said to be the **translation by** $l = (l_1, l_2)$.

(b) If $l = (l_1, l_2) \in \Pi$, show that

$$T(p) = (x + l_1, y + l_2), \qquad p = (x, y) \in \Pi$$

defines a translation of $\Pi$.

If $T, S$ are translations, define their sum $T \oplus S$ by

$$(T \oplus S)(p) = T(S(p)).$$

(c) Show that $T \oplus S$ is again a translation.

If $T$ is the translation by $l = (l_1, l_2)$ and $a$ is a number, define $a \cdot T$ to be the translation by $(al_1, al_2)$.

(d) Show that the set of translations with the addition $\oplus$ and scalar multiplication $\cdot$ is a vector space.

**12.** Show that if a vector space contains two elements, then it contains infinitely many.

**13.** Let $\mathbf{A}, \mathbf{B}$ be elements of the vector space $\mathcal{V}$. Show that there exists a unique element $\mathbf{X}$ of $\mathcal{V}$ such that $\mathbf{A} + \mathbf{X} = \mathbf{B}$.

# Chapter 3
# Examples of Vector Spaces

Before embarking on our study of the elementary properties of vector spaces and their linear subspaces in the succeeding chapters, let us collect a list of examples of vector spaces. Of basic importance are the three examples $\mathbb{R}^k$, $\mathcal{P}_n(\mathbb{R})$, and Fun$(S)$ described in Section 3.1. They illustrate three of the viewpoints that have contributed to the development of linear algebra and its applications: geometry, analysis, and combinatorics.

## 3.1 Three Basic Examples

We have already encountered the Cartesian $k$-space $\mathbb{R}^k$ in Section 2.2, and so for the sake of completeness let us begin by listing this example:

EXAMPLE 1: $\mathbb{R}^k$.

The first new example in this chapter is primarily designed to destroy the belief that a vector is a quantity with both direction and magnitude and to give meaning to the phrase in our comments on axiomatics in Chapter 2 that *"the possibility is opened of assigning to them (the axioms of a vector space) content in new and unforeseen ways."*

EXAMPLE 2: $\mathcal{P}_n(\mathbb{R})$.

The vectors in $\mathcal{P}_n(\mathbb{R})$ are polynomials

$$p(x) = a_0 + a_1x + \cdots + a_mx^m$$

of degree less than or equal to $n$, that is, $m \leq n$. Addition of vectors is to be ordinary addition of polynomials, and multiplication of a polynomial by a number will be the ordinary product of a polynomial by a number. With these interpretations of the basic terms, namely

$$\begin{aligned} \text{vector} &\leftrightarrow \text{polynomial of degree} \le n, \\ \text{vector additon} &\leftrightarrow \text{addition of polynomials}, \\ \text{scalar multiplication} &\leftrightarrow \text{multiplication of a polynomial} \\ &\phantom{\leftrightarrow} \text{ by a number}, \end{aligned}$$

we obtain an example of a vector space. To verify that $\mathcal{P}_n(\mathbb{R})$ is indeed a vector space we must check that the eight declarative sentences obtained from these interpretations of the basic terms *vector, vector addition, scalar multiplication*, are true sentences. This is a straightforward deduction from the (assumed) properties of real numbers following the pattern of Proposition 2.2.1 and will be left to the diligent reader.

Note that in this example it is very difficult to say what direction or length a vector has.

**EXAMPLE 3:** Let $S$ be a nonempty set and Fun($S$) the set of all functions $f : S \longrightarrow \mathbb{R}$. If $f, g \in \text{Fun}(S)$, define their **sum**

$$f + g : S \longrightarrow \mathbb{R}$$

by

$$(f + g)(s) = f(s) + g(s),$$

and for a real number $r$ define

$$rf : S \longrightarrow \mathbb{R}$$

by

$$(rf)(s) = r(f(s)),$$

for all $s \in S$.

Equipped with these definitions of vector addition and scalar multiplication the set Fun($S$) becomes a vector space. The zero vector of Fun($S$) is the function

$$0 : S \longrightarrow \mathbb{R}$$

defined by

$$0(s) = 0$$

for all $s \in S$; that is, 0 is the constant function that takes the value 0 for all $s \in S$. The negative of $f \in \text{Fun}(S)$ is the function

$$-f : S \longrightarrow \mathbb{R}$$

defined by

$$(-f)(s) = -f(s).$$

It is routine to verify that the axioms of a real vector space are satisfied for Fun($S$).

This completes the list of basic examples. The exercises at the end of the chapter introduce further examples and develop some of the properties of $\mathbb{R}^k$, $\mathcal{P}_n(\mathbb{R})$, and Fun($S$).

## 3.2 Further Examples of Vector Spaces

We collect in this section some further examples of vector spaces that indicate only a small portion of the vector spaces encountered in practice.

**Example 1:** $\mathbb{C}$ as a real vector space.

Recall that $\mathbb{C}$ denotes the **complex numbers**. A complex number[1] looks like

$$c = a + b\boldsymbol{i},$$

where $a, b$ are real numbers, and $\boldsymbol{i}$ is a number such that $\boldsymbol{i}^2 = -1$. The number $a$ is called the **real part** of $c$ and the number $b$ the **imaginary part** of $c$. Addition of complex numbers is componentwise, i.e.,

$$c' + c'' = (a' + a'') + (b' + b'')\boldsymbol{i}$$

for complex numbers $c' = a' + b'\boldsymbol{i}$ and $c'' = a'' + b''\boldsymbol{i}$. Multiplication is determined by the rule

$$c'c'' = (a'a'' - b'b'') + (a'b'' + a''b')\boldsymbol{i}.$$

The vectors in our vector space will be complex numbers. Addition of vectors is to be the ordinary addition of complex numbers, and scalar multiplication the familiar process of multiplying a complex number by a real number. With these interpretations of the basic terms,

$$\begin{aligned} \text{vector} &\leftrightarrow \text{complex number} \\ \text{vector addition} &\leftrightarrow \text{addition of complex numbers} \\ \text{scalar multiplication} &\leftrightarrow \text{multiplication of a complex number by a} \\ &\phantom{\leftrightarrow}\ \text{real number,} \end{aligned}$$

we obtain an example of a vector space. The verifications are again routine.

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[1] We are not concerned here with the logical construction of the complex numbers, but take their basic arithmetic properties as known. For a construction of complex numbers using ideas developed later in this book see Appendix B, so you may want to come back to this example after further study of this book.

Note that in this example we are not using all of the structure that we have, for, we may in effect, multiply two complex numbers, that is, in this example we may multiply two vectors, something it is not always possible to do in a vector space. This is a possibility[2] worthy of further study, and we will do just that when we study spaces of linear transformations and linear algebras.

**REMARK:** The product of two polynomials of degree at most $n$ will have degree at most $2n$, but in general will not be of degree $n$, so you cannot multiply elements of $\mathcal{P}_n(\mathbb{R})$ in any obvious way.

Examples 2 and 3 in Section 3.1 are very important, and they are prototypes for many others of the same type. These examples are characterized by the fact that their "vectors" are actually functions of some type or other. Here are some related examples.

**EXAMPLE 2:** $\mathrm{Fun}_{\mathbb{C}}(S)$.

We can make the set of all **complex-valued functions** on a nonempty set $S$, i.e., $\mathrm{Fun}_{\mathbb{C}}(S) = \{f : S \longrightarrow \mathbb{C}\}$, into a complex vector space by setting

$$(f+g)(s) = f(s) + g(s),$$
$$(cf)(s) = c(f(s))$$

for all $f, g \in \mathrm{Fun}_{\mathbb{C}}(S)$, $s \in S$, and $c \in \mathbb{C}$.

**EXAMPLE 3:** $\mathcal{P}(\mathbb{R})$.

The simplest way to obtain a space akin to but different from $\mathcal{P}_n(\mathbb{R})$ is simply to remove the restriction that the polynomials have degree at most $n$. In this way we obtain the vector space $\mathcal{P}(\mathbb{R})$ whose vectors are the polynomials

$$p(x) = a_0 + a_1 x + \cdots + a_m x^m$$

with no bound on $m$. The interpretation of the basic terms we propose in this example is:

$$\begin{aligned} \text{vector} &\leftrightarrow \text{polynomial,} \\ \text{vector addition} &\leftrightarrow \text{addition of polynomials,} \\ \text{scalar multiplication} &\leftrightarrow \text{multiplication of a polynomial} \\ &\phantom{\leftrightarrow}\ \text{by a number.} \end{aligned}$$

---

[2] The multiplication of complex numbers enjoys several important properties, among which are:

(1) the commutative law for multiplication $c'c'' = c''c'$ holds;
(2) for every nonzero complex number $c$ there is an inverse $c^{-1}$ such that $cc^{-1} = 1$, and hence
(3) the cancellation law if $c'c'' = 0$ and $c' \neq 0$ then $c'' = 0$ holds.

The only **finite-dimensional** (a term we will define and study in Chapter 6) real vector space with these properties is in fact the complex numbers.

It is again a routine verification that the vector space axioms are satisfied.

**Remark:** In $\mathcal{P}(\mathbb{R})$ it is possible to multiply two vectors and again receive a vector in $\mathcal{P}(\mathbb{R})$. However, with this product not every element of $\mathcal{P}(\mathbb{R})$ has an inverse.

**Example 4:** Let $S$ be a nonempty set and $T \subseteq S$. We denote by $\mathrm{Fun}(S, T)$ the set of all functions

$$f : S \longrightarrow \mathbb{R}$$

such that[3]

$$f(s) = 0 \qquad \forall\ s \in T.$$

The set $\mathrm{Fun}(S, T) \subseteq \mathrm{Fun}(S)$ is a vector space in its own right. For if $f, g \in \mathrm{Fun}(S, T)$ and $s \in T$, then

$$(f + g)(s) = f(s) + g(s) = 0 + 0 = 0,$$

and if $r \in \mathbb{R}$, then

$$(rf)(s) = rf(s) = r \cdot 0 = 0,$$

so that $f + g \in \mathrm{Fun}(S, T)$, $rf \in \mathrm{Fun}(S, T)$, and therefore Axioms 1,2, and 5–8, which hold in $\mathrm{Fun}(S)$, also hold in $\mathrm{Fun}(S, T)$. Finally, note that $0 \in \mathrm{Fun}(S, T)$ so Axiom 3 holds, and since $(-1)f = -f$ is also in $\mathrm{Fun}(S, T)$ whenever $f \in \mathrm{Fun}(S, T)$, we see that Axiom 4 holds as well.

**Example 5:** $\mathcal{C}(a, b)$.

This is a very fancy example and is included only to indicate the wealth of possible examples of vector spaces.

Let $a$ and $b$ be real numbers with $a < b$. The vectors of $\mathcal{C}(a, b)$ are the continuous functions defined for $a \leq x \leq b$. Addition of vectors is to be addition of functions. That is, if $f$ and $g$ are functions defined and continuous for $a \leq x \leq b$ then $f + g$ is the function defined by

$$(f + g)(x) = f(x) + g(x)$$

for all $a \leq x \leq b$. It is an important theorem of the calculus that $f + g$ is again a continuous function for $a \leq x \leq b$. Scalar multiplication is to be defined as the ordinary product of a function by a number. If the function is continuous then its product by a number is also continuous.

[3] The symbol $\forall$ is an abbreviation for "for all", e.g., $\forall\ s \in T$ means for all elements $s$ in the set $T$. If the set $T$ is empty, there are no such $s$, and hence no condition.

The basic terms of a vector space are to be interpreted as follows in this example:

$$\begin{aligned} \text{vector} &\leftrightarrow \text{continuous function on } a \le x \le b, \\ \text{vector addition} &\leftrightarrow \text{addition of functions}, \\ \text{scalar multiplication} &\leftrightarrow \text{multiplication of a function by a number.} \end{aligned}$$

Again the vector space axioms are easily verified.

Finally we will close this introduction to examples of vector spaces by describing a rather artificial example.

**Example 6:** Let $\mathcal{V}$ be the set of all *positive* real numbers and define for $A, B \in \mathcal{V}$ a **vector sum** by

$$A + B = A \cdot B,$$

where the product on the right is the usual product of numbers. If $a$ is a number and $A \in \mathcal{V}$ define

$$a \cdot A = A^a$$

that is the number $A$ raised to the $a$-th power. Note that since $A > 0$ the so is $A^a$. For example, with these definitions

$$2 + 3 = 6,$$

$$2 \cdot 3 = 9.$$

We claim that with these definitions of vector, vector addition and scalar multiplication $\mathcal{V}$ becomes a vector space. The details of verification are left to you.

The preceding list only barely scratches the surface of the enormous variety of examples of vector spaces. More examples will appear as we progress through the book and will by no means exhaust the possibilities.

## 3.3 Exercises

**1.** Verify that Fun($S$) is indeed a vector space.

**2.** Let $\mathcal{E}$ be the subset of $\mathcal{P}_k(\mathbb{R})$ defined by

$$\mathcal{E} = \left\{ p(x) \mid p(x) \in \mathcal{P}_k(\mathbb{R}) \text{ and } p(-x) = p(x) \right\}.$$

Show that $\mathcal{E}$ becomes a vector space if we define the vector operations (vector addition and scalar multiplication) to be those of $\mathcal{P}_k(\mathbb{R})$.

**3.** The set of all continuous functions $y = f(x)$, $-\infty < x < \infty$ satisfying the differential equation

$$y'' - y' - 2y = 0$$

is a vector space. (In fact, any solution of this differential equation is a linear combination of $y_1 = e^{-x}$ and $y_2 = e^{2x}$.)

**4.** The set of all solutions of a linear differential equation

$$a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_{n-1}(x)y' + a_n(x)y = 0,$$

where $a_0(x)$ is not zero on $[a, b]$ and $a_i(x)$ are continuous on $[a, b]$, $i = 1, 2, \ldots, n$, is a vector space.

**5.** Let $\mathcal{D} = \left\{f \in C(a, b) \mid f \text{ is differentiable on } (a, b)\right\}$. Show that $\mathcal{D}$ with the operations induced from $C(a, b)$ is a vector space.

**6.** Let $\mathcal{P}(a, b)$ denote the set of all functions on the interval $[a, b]$ which are polynomials. Then $\mathcal{P}(a, b) \subset \mathcal{D} \subset C(a, b)$. Show $\mathcal{P}(a, b)$ is a vector space with the operations induced from either $\mathcal{D}$ or $C(a, b)$. Is this the same vector space as $\mathcal{P}(\mathbb{R})$?

**7.** Consider the set $\mathcal{D}^{(n)}$ of all continuous functions on the closed interval $[a, b]$ that are $n$-times differentiable functions on the open interval $(a, b)$. $\mathcal{D}^{(n)}$ is also a vector space. Is $\mathcal{P}(a, b) \subset \mathcal{D}^{(n)}$?

**8.** Let $S$ be a nonempty set and Fun($S$) the vector space of real valued functions on $S$. If $A, B \subseteq S$ show that

$$\text{Fun}(S, A) \cap \text{Fun}(S, B) = \text{Fun}(S, A \cup B).$$

If $A \subseteq B$, show that Fun($S, A$) contains Fun($S, B$) and the vector operations coincide.

**9.** Let $\mathcal{V}$ be a vector space and

$$\varphi : \mathcal{V} \longrightarrow \mathbb{R}$$

such that

(LF 1) $\varphi(\mathbf{A} + \mathbf{B}) = \varphi(\mathbf{A}) + \varphi(\mathbf{B})$, $\forall\ \mathbf{A}, \mathbf{B} \in \mathcal{V}$,
(LF 2) $\varphi(r\mathbf{A}) = r\varphi(\mathbf{A})$, $\forall\ A \in \mathcal{V}$, $\forall\ r \in \mathbb{R}$.

Show that $\mathcal{N} = \left\{\mathbf{A} \in \mathcal{V} \mid \varphi(\mathbf{A}) = 0\right\}$ is a vector space with respect to the operation of vector addition and scalar multiplication as defined in $\mathcal{V}$.

**10.** Let $\mathcal{V}$ be a vector space and $S$ a nonempty set. Define Fun($S$, $\mathcal{V}$) to be the set of all functions from $S$ to $\mathcal{V}$. If $f, h \in$ Fun($S$, $\mathcal{V}$) and $r$ is a number, define:

(FUN 1) $(f+h)(s) = f(s) + h(s)\ \forall\ s \in S$,
(FUN 2) $(rf)(s) = r \cdot f(s)\ \forall\ s \in S$.

Show that with these operations as vector addition and scalar multiplication $\mathrm{Fun}(S, \mathcal{V})$ is a vector space. (Note that in this notation $\mathrm{Fun}(S) = \mathrm{Fun}(S, \mathbb{R})$.)

**11.** Let $\mathcal{V}$ be a vector space and $S$ a nonempty set with $s \in S$ a fixed element. Define

$$e_s : \mathrm{Fun}(S, \mathcal{V}) \longrightarrow \mathcal{V}$$

by

$$e_s(f) = f(s).$$

Show that

$$\mathcal{K} = \left\{ f \in \mathrm{Fun}(S, \mathcal{V}) \mid e_s(f) = \mathbf{0} \right\}$$

is a vector space with respect to the operations of vector addition and scalar multiplication defined in $\mathrm{Fun}(S, \mathcal{V})$ (see the previous exercise).

**12.** Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces. Show that the Cartesian product $\mathcal{V} \times \mathcal{W}$ consisting of all the ordered pairs $(\mathbf{A}, \mathbf{B})$, $\mathbf{A} \in \mathcal{V}$, $\mathbf{B} \in \mathcal{W}$, becomes a vector space if we define vector addition and scalar multiplication componentwise, that is,

$$(\mathbf{A}', \mathbf{B}') + (\mathbf{A}'', \mathbf{B}'') = (\mathbf{A}' + \mathbf{A}'', \mathbf{B}' + \mathbf{B}''),$$
$$r(\mathbf{A}, \mathbf{B}) = (r\mathbf{A}, r\mathbf{B}).$$

**13.** Let $\mathcal{V}$ be a vector space, $S$ a nonempty set, and $s, t \in S$. Define

$$\varphi : \mathrm{Fun}(S, \mathcal{V}) \longrightarrow \mathcal{V} \times \mathcal{V}$$

by

$$\varphi(f) = (f(s), f(t)).$$

Show that

$$\mathcal{N} = \left\{ f \in \mathrm{Fun}(S, \mathcal{V}) \mid \varphi(f) = \mathbf{0} \right\}$$

is a vector subspace of $\mathrm{Fun}(S, \mathcal{V})$.

**14.** Show that the complex numbers $\mathbb{C}$ have the following properties (which are familiar for the real numbers):

(1) Addition and multiplication of complex numbers are commutative.
(2) If $a, b, c \in \mathbb{C}$ then $a(b+c) = ab + ac$.
(3) If $c', c'' \in \mathbb{C}$ and $c'c'' = 0$, then either $c' = 0$ or $c'' = 0$.
(4) if $a$, $b$ and $c$ are complex numbers, and $a \neq 0$ then the equation

$$az + b = c$$

has a unique solution $z \in \mathbb{C}$.

What properties do the complex numbers have that the real numbers do not have?

**15.** Let $\ell_0$ be the set of all sequences of real numbers. Show that the usual rules for adding sequences and multiplying them by a real number makes $\ell_0$ into a vector space.

**16.** Let

$$\ell_\infty = \left\{(a_0, a_1, \ldots, a_n, \ldots) \mid \text{the sequence } a_0, a_1, \ldots \text{ is bounded}\right\}$$

(recall that a sequence of real numbers $(a_0, a_1, \ldots)$ is bounded if there is some real number $M$ such that $|a_i| < M \; \forall\, i$). Show that the usual rules for adding sequences and multiplying them by real numbers makes $\ell_\infty$ into a vector space.

# Chapter 4
# Subspaces

Not only the vectors of a vector space $\mathcal{V}$ are interesting, but also those subsets of the vector space that are themselves vector spaces when the opertaions of $\mathcal{V}$ are restricted to them. Such subsets are called **vector subspaces** or **linear subspaces.**

## 4.1 Basic Properties of Vector Subspaces

The following definition captures the essential idea that a subset of a vector space be a "linear" subset.

DEFINITION : *A nonempty subset $\mathcal{U}$ of a vector space $\mathcal{V}$ is called a* **linear subspace,** *or* **vector subspace,** *of $\mathcal{V}$ if and only if the following two conditions are satisfied:*

(1) *if $\mathbf{A} \in \mathcal{U}$ and $\mathbf{B} \in \mathcal{U}$ then $\mathbf{A} + \mathbf{B} \in \mathcal{U}$, and*

(2) *if $\mathbf{A} \in \mathcal{U}$ and $r \in \mathbb{R}$ then $r\mathbf{A} \in \mathcal{U}$.*

These two conditions assert that applying the two basic vector operations to elements of the collection $\mathcal{U}$ give again elements of the collection $\mathcal{U}$. If the vector space $\mathcal{V}$ is complex then condition (2) should be replaced by

(2$\mathbb{C}$) if $\mathbf{A} \in \mathcal{U}$ and $c \in \mathbb{C}$ then $c\mathbf{A} \in \mathcal{U}$,

and likewise in the sequel. Note that $\mathcal{V}$ is a linear subspace of itself, and we may refer to $\mathcal{V}$ as being a **linear space**.

PROPOSITION **4.1.1**: *If $\mathcal{U}$ is a linear subspace of the vector space $\mathcal{V}$ then $\mathcal{U}$ is itself a vector space if we define vector addition and scalar multiplication as in $\mathcal{V}$.*

**PROOF:** Notice that conditions (1), (2) in the definition of subspace assure us that we have operations on $\mathcal{U}$, i.e., if $\mathbf{A}$ and $\mathbf{B}$ belong to $\mathcal{U}$, so do $\mathbf{A}+\mathbf{B}$, and if $r$ belongs to $\mathbb{R}$, $r\mathbf{A}$ also belongs to $\mathcal{U}$. The properties expressed by Axioms 1, 2, 5, 6, 7, and 8 are valid for vectors in $\mathcal{V}$ and hence for vectors in the smaller set $\mathcal{U}$. To verify Axiom 3, we first show that $\mathbf{0} \in \mathcal{U}$. Since $\mathcal{U}$ is nonempty, there exists at least one vector $\mathbf{A} \in \mathcal{U}$. By (2.1) and condition (2) for a subspace, $\mathbf{0} = 0\mathbf{A} \in \mathcal{U}$. Axiom 3 is now immediate, since it holds in $\mathcal{V}$. To verify Axiom 4, suppose that $\mathbf{A} \in \mathcal{U}$. Then $(-1)\mathbf{A} \in \mathcal{U}$, but by Proposition 2.1.2, $(-1)\mathbf{A} = -\mathbf{A}$, and therefore $-\mathbf{A} \in \mathcal{U}$, and Axiom 4 holds. □

There are some trivial examples of subspaces, to wit: $\mathcal{V}$ is always a subspace of $\mathcal{V}$ and the set consisting of the zero vector alone $\{\mathbf{0}\}$ is always a subspace of $\mathcal{V}$. We often abuse notation and write $\mathbf{0}$ for this subspace. There is a also trivial example that is *not* a subspace, namely the empty set $\varnothing$, which the definition of subspace explicitly excludes.

**EXAMPLE 1:** Here is a less trivial example of a linear subspace. We consider in $\mathbb{R}^n$ the subset $\mathcal{U}$ consisting of all vectors $\mathbf{A} = (a_1, \ldots, a_n)$ with the property that $a_1 + \cdots + a_n = 0$. If $\mathbf{A} = (a_1, \ldots, a_n)$, $\mathbf{B} = (b_1, \ldots, b_n) \in \mathcal{U}$ and $c$ is a number, then

$$\mathbf{A}+\mathbf{B} = (a_1 + b_1, \ldots, a_n + b_n),$$
$$c \cdot \mathbf{A} = (ca_1, \ldots, ca_n)$$

and

$$a_1 + b_1 + \cdots + a_n + b_n = (a_1 + \cdots + a_n) + (b_1 + \cdots + b_n) = 0 + 0 = 0,$$
$$ca_1 + \cdots ca_n = c(a_1 + \cdots a_n) = c(0) = 0,$$

so $\mathcal{U}$ is a linear subspace of $\mathbb{R}^n$.

**EXAMPLE 2:** The solution space to a linear homogeneous equation is a generalization of the preceding example. A linear homogeneous equation in the variables $x_1, \ldots, x_n$ is an equation of the form

$$(*) \qquad a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0.$$

A solution to this equation is a sequence of $n$ numbers $(s_1, \ldots, s_n)$ such that

$$a_1s_1 + \cdots + a_ns_n = 0.$$

If $\mathbf{A} = (s_1, \ldots, s_n)$ and $\mathbf{B} = (t_1, \ldots, t_n)$ are solutions to $(*)$, define their sum by

$$\mathbf{A}+\mathbf{B} = (s_1 + t_1, \ldots, s_n + t_n).$$

We claim that $\mathbf{A} + \mathbf{B}$ is again a solution to (*). For we have

$$\begin{aligned}
&a_1(s_1 + t_1) + a_2(s_2 + t_2) + \cdots + a_n(s_n + t_n) \\
&= a_1 s_1 + a_1 t_1 + a_2 s_2 + a_2 t_2 + \cdots + a_n s_n + a_n t_n \\
&= a_1 s_1 + a_2 s_2 + \cdots + a_n s_n + a_1 t_1 + a_2 t_2 + \cdots + a_n t_n \\
&= 0 + 0 = 0,
\end{aligned}$$

as we claimed. Next, define $a\mathbf{A}$ for a number $a$ to be

$$a\mathbf{A} = (as_1, \ldots, as_n).$$

Simple manipulation shows that

$$\begin{aligned}
a_1(as_1) + \cdots + a_n(as_n) &= aa_1 s_1 + \cdots + aa_n s_n \\
&= a(a_1 s_1 + \cdots + a_n s_n) \\
&= a(0) = 0,
\end{aligned}$$

so $a\mathbf{A}$ is again a solution to (*). We define a vector space $\mathcal{V}$ by the interpretation

$$\begin{aligned}
\text{vector} &\leftrightarrow \text{solution to } (*), \\
\text{vector addition} &\leftrightarrow \text{as defined above}, \\
\text{scalar multiplication} &\leftrightarrow \text{as defined above}.
\end{aligned}$$

To show that $\mathcal{V}$ is a vector space we will show that it is actually a linear subspace of $\mathbb{R}^n$. For by definition the vectors of $\mathcal{V}$ are sequences $(s_1, \ldots, s_n)$ of numbers and hence are vectors in $\mathbb{R}^n$. The process of adding solutions and multiplying solutions by scalars is exactly the process of adding vectors in $\mathbb{R}^n$ and multiplying a vector of $\mathbb{R}^n$ by a number. In our preceding discussion we checked the following:

(1) If $\mathbf{A}, \mathbf{B} \in \mathcal{V}$, then $\mathbf{A} + \mathbf{B} \in \mathcal{V}$.
(2) If $\mathbf{A} \in \mathcal{V}$, then $a\mathbf{A} \in \mathcal{V}$ for any number $a$.

Thus we may apply Proposition 4.1.1 to conclude that $\mathcal{V}$ is a vector space. But wait! In order to apply 4.1.1 to $\mathcal{V}$, we must know that $\mathcal{V}$ is nonempty, that is, that (*) has at least one solution. Happily this is a simple point, because $(0, \ldots, 0)$ is a solution to (*), as one easily sees, since

$$a_1 0 + a_2 0 + \cdots + a_n 0 = 0 + \cdots + 0 = 0.$$

Thus $\mathcal{V}$ is a linear subspace of $\mathbb{R}^n$.

The preceding example may be extended from one equation to many, but this is a topic for further study. (See chapter 13.)

**DEFINITION:** *If* $\mathbf{A}_1, \ldots, \mathbf{A}_n$ *are vectors of* $\mathcal{V}$, *then a* **linear combination** *of* $\mathbf{A}_1, \ldots, \mathbf{A}_n$ *is a vector of the form*

$$\mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n,$$

*where* $a_1, \ldots, a_n$ *are numbers.*

**DEFINITION:** *If the vectors* $\mathbf{A}_1, \ldots, \mathbf{A}_n$ *are fixed, the* **linear span** *of* $\mathbf{A}_1, \ldots, \mathbf{A}_n$, *denoted by* $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$, *or* Span $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$, *is the set of all vectors of* $\mathcal{V}$ *that are linear combinations of* $\mathbf{A}_1, \ldots, \mathbf{A}_n$.

**PROPOSITION 4.1.2:** *Suppose that* $\mathbf{A}_1, \ldots, \mathbf{A}_n$ *are vectors of* $\mathcal{V}$. *Then* $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$ *is a linear subspace of* $\mathcal{V}$.

**PROOF:** We must verify that the two conditions of the definition of a linear subspace are satisfied by the set of all linear combinations of $\mathbf{A}_1, \ldots, \mathbf{A}_n$. So suppose

$$\mathbf{A}', \mathbf{A}'' \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n).$$

Then

$$\begin{aligned} \mathbf{A}' &= a_1'\mathbf{A}_1 + \cdots + a_n'\mathbf{A}_n, \\ \mathbf{A}'' &= a_1''\mathbf{A}_1 + \cdots + a_n''\mathbf{A}_n, \end{aligned}$$

for suitable numbers $a_1', \ldots, a_n', a_1'', \ldots, a_n''$. Then using the generalized associative and commutative laws, we find

$$\begin{aligned} \mathbf{A}' + \mathbf{A}'' &= a_1'\mathbf{A}_1 + \cdots + a_n'\mathbf{A}_n + a_1''\mathbf{A}_1 + \ldots + a_n''\mathbf{A}_n \\ &= a_1'\mathbf{A}_1 + a_1''\mathbf{A}_1 + \cdots + a_n'\mathbf{A}_n + a_n''\mathbf{A}_n \\ &= (a_1' + a_1'')\mathbf{A}_1 + \cdots + (a_n' + a_n'')\mathbf{A}_n, \end{aligned}$$

which shows that $\mathbf{A}' + \mathbf{A}''$ is again a linear combination of $\mathbf{A}_1, \ldots, \mathbf{A}_n$, that is, $\mathbf{A}' + \mathbf{A}'' \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$. Similarly if $\mathbf{A} \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$ and $r \in \mathbb{R}$, then

$$\mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n$$

for suitable numbers $a_1, \ldots, a_n$, so

$$\begin{aligned} r\mathbf{A} &= r(a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n) \\ &= ra_1\mathbf{A}_1 + \cdots + ra_n\mathbf{A}_n \end{aligned}$$

showing that $r\mathbf{A} \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$. Therefore $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$ is a subspace of $\mathcal{V}$. $\square$

The idea of the linear span is not restricted to finite sets of vectors, but may be extended to arbitrary sets of vectors as follows.

**DEFINITION:** *Let $\mathcal{V}$ be a vector space and $E \subseteq \mathcal{V}$; that is, $E$ is a collection of vectors in $\mathcal{V}$. A linear combination of vectors in $E$ is a vector in $\mathcal{V}$ of the form*

$$a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_n\mathbf{A}_n,$$

*where $\mathbf{A}_1, \ldots, \mathbf{A}_n \in E$. The* **linear span** *of $E$ denoted by $\mathcal{L}(E)$, or* Span $\{E\}$, *is the set of all vectors that are linear combinations of vectors of $E$.*

**PROPOSITION 4.1.3:** *Let $\mathcal{V}$ be a vector space and $E \subseteq \mathcal{V}$. Then $\mathcal{L}(E)$ is a linear subspace of $\mathcal{V}$.*

The proof of Proposition 4.1.3 follows closely the proof of Proposition 4.1.2 and will be left to the diligent student. Note that the linear span allows us to assign to each sub*set* $E$ of a vector space $\mathcal{V}$ a sub*space* $\mathcal{L}(E)$ of $\mathcal{V}$, and that $E \subseteq \mathcal{L}(E)$.

**PROPOSITION 4.1.4:** *Let $\mathcal{V}$ be a vector space and $E \subseteq \mathcal{V}$. Then $E = \mathcal{L}(E)$ if and only if $E$ is a linear subspace of $\mathcal{V}$.*

**PROOF:** Suppose that $E$ is a linear subspace of $\mathcal{V}$. If the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ belong to $E$ and $a_1, \ldots, a_n$ are numbers, then the vector $a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n$ belongs to $E$ because $E$ is closed under the operations of scalar multiplication and vector addition. Therefore, $\mathcal{L}(E) \subseteq E$. Since $E \subseteq \mathcal{L}(E)$, we conclude that $E = \mathcal{L}(E)$.

Conversely, suppose that $E = \mathcal{L}(E)$. If $\mathbf{A}, \mathbf{B} \in E$, then $\mathbf{A} + \mathbf{B}$ is certainly a linear combination of vectors in $E$ and hence $\mathbf{A} + \mathbf{B}$ belongs to $\mathcal{L}(E)$, which, since $\mathcal{L}(E) = E$, leads us to conclude $\mathbf{A} + \mathbf{B} \in E$. Likewise, $a\mathbf{A}$ is a linear combination of vectors of $E$ and hence belongs to $\mathcal{L}(E) = E$. Therefore, $E$ is closed under vector addition and scalar multiplication, and hence $E$ is a linear subspace of $\mathcal{V}$. $\square$

The preceding propositions show that in general a vector space has an abundance of subspaces.

**EXAMPLE 3:** In $\mathbb{R}^3$ consider the subspace spanned by the two vectors $\mathbf{A} = (-1, 0, 1)$ and $\mathbf{B} = (0, 1, 0)$, see Figure 4.1.1. Note that this is just the plane through the origin with equation $x + z = 0$. That is, the vectors in $\mathcal{L}(\mathbf{A}, \mathbf{B})$ are those vectors $(x, y, z) \in \mathbb{R}^3$ whose coordinates satisfy the equation $x - z = 0$.

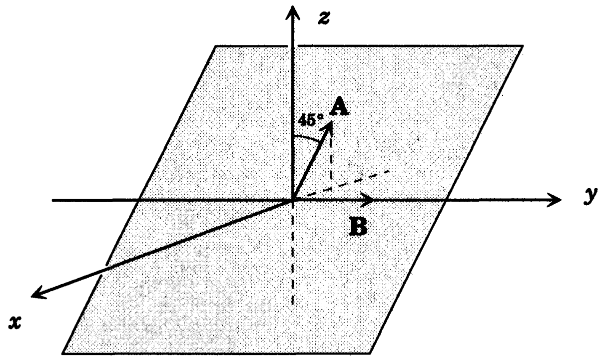


Figure 4.1.1

**PROPOSITION 4.1.5:** *Let $\mathcal{S}$ and $\mathcal{T}$ be subspaces of $\mathcal{V}$. Then $\mathcal{S} \cap \mathcal{T}$ is also a subspace of $\mathcal{V}$.*

**PROOF:** $\mathbf{0} \in \mathcal{S}$ and $\mathbf{0} \in \mathcal{T}$ since $\mathcal{S}$ and $\mathcal{T}$ are subspaces. Therefore $\mathbf{0} \in \mathcal{S} \cap \mathcal{T}$.

Suppose that $\mathbf{A} \in \mathcal{S} \cap \mathcal{T}$ and $\mathbf{B} \in \mathcal{S} \cap \mathcal{T}$. Then $\mathbf{A} \in \mathcal{S}$ and $\mathbf{B} \in \mathcal{S}$. Since $\mathcal{S}$ is a subspace, $\mathbf{A} + \mathbf{B} \in \mathcal{S}$. Likewise, $\mathbf{A} \in \mathcal{T}$ and $\mathbf{B} \in \mathcal{T}$, and since $\mathcal{T}$ is a subspace, $\mathbf{A} + \mathbf{B} \in \mathcal{T}$. Therefore, $\mathbf{A} + \mathbf{B} \in \mathcal{S} \cap \mathcal{T}$.

If $r$ is a number, then since $\mathcal{S}$ and $\mathcal{T}$ are subspaces, $r\mathbf{A} \in \mathcal{S}$ and $r\mathbf{A} \in \mathcal{T}$, so $r\mathbf{A} \in \mathcal{S} \cap \mathcal{T}$, showing that $\mathcal{S} \cap \mathcal{T}$ is again a subspace of $\mathcal{V}$. □

**DEFINITION:** *If $\mathcal{S}$ and $\mathcal{T}$ are subspaces of $\mathcal{V}$, their* **sum**, *denoted by $\mathcal{S} + \mathcal{T}$, is defined to be the set of all vectors $\mathbf{C}$ in $\mathcal{V}$ of the form*

$$\mathbf{C} = \mathbf{A} + \mathbf{B},$$

*where $\mathbf{A} \in \mathcal{S}$ and $\mathbf{B} \in \mathcal{T}$.*

**PROPOSITION 4.1.6:** *If $\mathcal{S}$ and $\mathcal{T}$ are subspaces of the vector space $\mathcal{V}$, then so is $\mathcal{S} + \mathcal{T}$.*

**PROOF:** Suppose that $\mathbf{C}_1, \mathbf{C}_2 \in \mathcal{S} + \mathcal{T}$. Write

$$\mathbf{C}_1 = \mathbf{A}_1 + \mathbf{B}_1, \qquad \mathbf{A}_1 \in \mathcal{S}, \mathbf{B}_1 \in \mathcal{T},$$

$$\mathbf{C}_2 = \mathbf{A}_2 + \mathbf{B}_2, \qquad \mathbf{A}_2 \in \mathcal{S}, \mathbf{B}_2 \in \mathcal{T}.$$

Then

$$\mathbf{C}_1 + \mathbf{C}_2 = \mathbf{A}_1 + \mathbf{B}_1 + \mathbf{A}_2 + \mathbf{B}_2 = (\mathbf{A}_1 + \mathbf{A}_2) + (\mathbf{B}_1 + \mathbf{B}_2).$$

Let $\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2$, $\mathbf{B} = \mathbf{B}_1 + \mathbf{B}_2$. Since $\mathcal{S}$ and $\mathcal{T}$ are subspaces, $\mathbf{A} \in \mathcal{S}$ and $\mathbf{B} \in \mathcal{T}$. Since

$$\mathbf{C}_1 + \mathbf{C}_2 = \mathbf{A} + \mathbf{B},$$

it follows that $\mathbf{C}_1 + \mathbf{C}_2 \in \mathcal{S} + \mathcal{T}$.

Next suppose that $\mathbf{C} \in \mathcal{S} + \mathcal{T}$ and $r \in \mathbb{R}$. Then we may write

$$\mathbf{C} = \mathbf{A} + \mathbf{B}, \qquad \mathbf{A} \in \mathcal{S}, \mathbf{B} \in \mathcal{T},$$

and hence

$$r\mathbf{C} = r(\mathbf{A} + \mathbf{B}) = (r\mathbf{A}) + (r\mathbf{B}).$$

Since $\mathcal{S}$ and $\mathcal{T}$ are subspaces, $r\mathbf{A} \in \mathcal{S}$, $r\mathbf{B} \in \mathcal{T}$ and hence $r\mathbf{C} \in \mathcal{S} + \mathcal{T}$. Finally $\mathbf{0} = \mathbf{0} + \mathbf{0} \in \mathcal{S} + \mathcal{T}$ shows $\mathcal{S} + \mathcal{T}$ contains the zero vector. $\square$

**PROBLEM :** Suppose that $\mathcal{S}$, $\mathcal{T}$ are subspaces of $\mathcal{V}$. When is $\mathcal{S} \cup \mathcal{T}$ again a subspace of $\mathcal{V}$?

**SOLUTION :** If and only if $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$.

## 4.2 Examples of Subspaces

Examples of subspaces will appear naturally in the course of our study. We collect here three examples related to our basic vector spaces $\mathbb{R}^n$, $\mathcal{P}_k(\mathbb{R})$, and Fun($S$).

**EXAMPLE 1 :** Consider the subset $\mathcal{U}$ of $\mathbb{R}^n$ consisting of vectors $\mathbf{A} = (a_1, \ldots, a_n)$ satisfying the condition

$$(a_1, \ldots, a_{n-1}, a_n) = (a_2, \ldots, a_n, a_1).$$

It is straightforward to verify that $\mathcal{U} \neq \emptyset$, and the two conditions of a linear subspace are satisfied by the vectors of $\mathcal{U}$. Indeed, a vector belongs to $\mathcal{U}$ if and only if all its coordinates are equal. For example, the vector $(1, \ldots, 1)$ belongs to $\mathcal{U}$, and in fact, $\mathcal{U} = \operatorname{Span}\{(1, \ldots, 1)\}$. For if all the coordinates of $\mathbf{A}$ are equal, and $a = a_1$ say, then $\mathbf{A} = a(1, \ldots, 1) \in \operatorname{Span}\{(1, \ldots, 1)\}$.

**EXAMPLE 2 :** Consider the subset $\mathcal{Z}$ of $\mathcal{P}_k(\mathbb{R})$ consisting of polynomials whose constant term is zero, i.e.,

$$p(x) \in \mathcal{Z} \iff p(x) = a_1 x + \cdots + a_k x^k.$$

Note that a polynomial $p(x)$ belongs to $\mathcal{Z}$ if and only if $p(x) = x\bar{p}(x)$ for a unique polynomial $\bar{p}(x) \in \mathcal{P}_{k-1}(\mathbb{R})$. Therefore, if $p(x), q(x) \in \mathcal{Z}$ and $r \in \mathbb{R}$, we have

$$p(x) + q(x) = x\bar{p}(x) + x\bar{q}(x) = x(\bar{p}(x) + \bar{q}(x)),$$

$$rp(x) = rx\bar{p}(x) = x(r\bar{p}(x)),$$

so $rp(x), p(x) + q(x) \in \mathcal{Z}$, and hence $\mathcal{Z}$ is a subspace of $\mathcal{P}_k(\mathbb{R})$.

**Example 3:** Let $S$ be the set consisting of the three elements $a, b, c$ and consider all the functions $f$ in Fun($S$) with the property that

$$f(a) + f(b) + f(c) = 0.$$

Call this set of functions $\mathcal{N}$. If $f, g \in \mathcal{N}$ and $r \in \mathbb{R}$, then by the definition of vector addition and scalar multiplication in Fun($S$) we find that

$$\begin{aligned}(f+g)(a) + (f+g)(b) + (f+g)(c) \\ &= f(a) + g(a) + f(b) + g(b) + f(c) + g(c) \\ &= f(a) + f(b) + f(c) + g(a) + g(b) + g(c) = 0, \\ (rf)(a) + (rf)(b) + (rf)(c) &= r \cdot f(a) + r \cdot f(b) + r \cdot f(c) \\ &= r\left(f(a) + f(b) + f(c)\right) = r \cdot 0 = 0,\end{aligned}$$

and therefore $\mathcal{N}$ is a linear subspace of Fun($S$).

You might want to think through what happens in Examples 2 and 3 if the condition that something be zero is replaced by the same thing being, say, $\pi$.

## 4.3 Exercises

**1.** Which of the following collections of vectors in $\mathbb{R}^3$ are linear subspaces?

(1) $\mathcal{U} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 = 0\}$.
(2) $\mathcal{U} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_2 = 0\}$.
(3) $\mathcal{U} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 = 0\}$.
(4) $\mathcal{U} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 = 1\}$.
(5) $\mathcal{U} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 \geq 0\}$.

**2.** Determine the subspace of $\mathbb{R}^3$ that is the linear span of the three vectors $(1, 0, 1), (0, 1, 0), (0, 1, 1)$.

**3.** Repeat Exercise 2 for $(1, 0, 0), (0, 1, 0), (1, 1, 1)$.

**4.** Let $\mathcal{U} = \left\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = a\right\}$ for $a \in \mathbb{R}$ fixed. Show that $\mathcal{U}$ is a vector subspace if and only if $a = 0$.

**5.** Show that $\mathcal{P}_s(\mathbb{R})$ is a linear subspace of $\mathcal{P}_r(\mathbb{R})$ whenever $s \leq r$.

**6.** Show that $\mathcal{P}_k(\mathbb{R})$ is always a subspace of $\mathcal{P}(\mathbb{R})$.

**7.** What is the span of $\{1 + x, 1 - x\}$ in $\mathcal{P}(\mathbb{R})$?

**8.** What is the span of $\left\{1, x^2, x^4\right\}$ in $\mathcal{P}_4(\mathbb{R})$?

**9.** Find a vector that spans the subspace $2x - 3y = 0$ of $\mathbb{R}^2$.

**10.** Find a pair of vectors that span the subspace $x + y - 2z = 0$ of $\mathbb{R}^3$.

**11.** Suppose $\mathcal{S}, \mathcal{T}$ are subspaces of $\mathcal{V}$ and $\mathcal{S} \cap \mathcal{T} = \mathbf{0}$. Show that every vector in $\mathcal{S} + \mathcal{T}$ can be written *uniquely* in the form $\mathbf{A} + \mathbf{B}$, with $\mathbf{A} \in \mathcal{S}, \mathbf{B} \in \mathcal{T}$. Construct an example to show that this is false if $\mathcal{S} \cap \mathcal{T} \neq \mathbf{0}$.

**12.** Show that any nonzero vector spans $\mathbb{R}^1$.

**13.** Show that the two sets of vectors

$$\{\mathbf{A} = (1, 1, 0), \mathbf{B} = (0, 0, 1)\}$$

and

$$\{\mathbf{C} = (1, 1, 1), \mathbf{D} = (-1, -1, 1)\}$$

span the same subspace of $\mathbb{R}^3$.

**14.** Let $\mathcal{V}$ be the set of pairs of numbers $\mathbf{A} = (a_1, a_2)$. If $\mathbf{A}, \mathbf{B} \in \mathcal{V}$ define

$$\mathbf{A} + \mathbf{B} = (a_1 + b_1, a_2 + b_2),$$

where $\mathbf{B} = (b_1, b_2)$. If $a$ is a number define

$$a\mathbf{A} = (aa_1, 0).$$

Is $\mathcal{V}$ a vector space? Why?

**15.** Let $\ell_\infty$ be the vector space of bounded sequences (see Chapter 3, Exercise 16). Which of the following collections of vectors in $\ell_\infty$ are linear subspaces?

(1) $\left\{(a_0, a_1, \ldots, a_n, \ldots) \mid |a_i| < 5 \ \forall\, i \in \mathbb{N}_0\right\}$.

(2) $\left\{(a_0, a_1, \ldots, a_n, \ldots) \mid a_i = 0 \ \forall\, i > i_0,\ i_0 \in \mathbb{N}_0 \text{ fixed}\right\}$.

(3) $\left\{(a_0, a_1, \ldots, a_n, \ldots) \mid a_i = a_j \text{ for all } i \text{ and } j\right\}$.

(4) $\left\{(a_0, a_1, \ldots, a_n, \ldots) \mid a_i \in \mathbb{Z} \ \forall\, i \in \mathbb{N}_0\right\}$.

**16.** Suppose $\mathcal{V}$ is a vector space and $E, F$ are subsets of $\mathcal{V}$. Show

(1) $E \subseteq F \Rightarrow \mathcal{L}(E) \subseteq \mathcal{L}(F)$.

(2) $\mathcal{L}(E \cup F) = \mathcal{L}(E) + \mathcal{L}(F)$.

(3) $\mathcal{L}(E \cap F) \subseteq \mathcal{L}(E) \cap \mathcal{L}(F)$.

**17.** Let $\mathcal{V}$ be a vector space and $E, F \subseteq \mathcal{V}$. Suppose $\mathcal{L}(E) \subseteq \mathcal{L}(F)$. Is it true that $E \subseteq F$?

**18.** Suppose that $\mathcal{V}$ is a vector space and $E \subseteq \mathcal{V}$. If $\mathcal{U}$ is a subspace containing $E$, show that $\mathcal{U}$ contains $\mathcal{L}(E)$.

**19.** Suppose $\mathcal{V}$ is a vector space and $E \subseteq \mathcal{V}$. Show that $\mathcal{L}(E) = \cap\{\mathcal{U} \mid \mathcal{U}$ is a subspace of $\mathcal{V}$ and $\mathcal{U}$ contains $E\}$.

**20.** For any subspace $\mathcal{U}$ of $\mathcal{V}$ show that $\mathcal{U} + \mathcal{U} = \mathcal{U}$.

**21.** Let $\mathcal{S}$ and $\mathcal{T}$ be linear subspaces of the vector space $\mathcal{V}$. Show that $\mathcal{S} \cup \mathcal{T} = \mathcal{V}$ if and only if $\mathcal{S}$ or $\mathcal{T}$, or both, is equal to $\mathcal{V}$.

**22.** Find all the linear subspaces of $\mathbb{R}^2$.

**23.** Find all the linear subspaces of $\mathbb{R}^3$.

**24.** Show that a subset $E$ of a vector space $\mathcal{V}$ that does not contain $\mathbf{0}$ is not a subspace of $\mathcal{V}$.

**25.** Let $E$ be the subset of $\mathbb{R}^2$ defined by $E = \left\{(x, y) \mid x \geq 0,\ y \in \mathbb{R}\right\}$. Is $E$ a subspace of $\mathbb{R}^2$?

**26.** Let $E = \left\{(x, 2x + 1) \mid x \in \mathbb{R}\right\}$, so $E$ is a subset of $\mathbb{R}^2$. Is $E$ a subspace of $\mathbb{R}^2$?

**27.** (1) Let $E = \left\{(2a, a) \mid a \in \mathbb{R}\right\}$. Is $E$ a subspace of $\mathbb{R}^2$?
(2) Let $B = \left\{(b, b) \mid b \in \mathbb{R}\right\}$. Is $B$ a subspace of $\mathbb{R}^2$?
(3) What is $E \cap B$?
(4) Is $E \cup B$ a subspace of $\mathbb{R}^2$?
(5) What is $E + B$?

**28.** Let $\mathcal{S}$ and $\mathcal{T}$ be subspaces of $\mathcal{V}$. Prove:
(1) $\mathcal{S} + \mathcal{T} = \mathcal{L}(S \cup T)$.
(2) $\mathcal{S} \cap (\mathcal{S} + \mathcal{T}) = \mathcal{S}$.
(3) $\mathcal{S} + \mathcal{T} = \mathcal{T} + \mathcal{S}$.
(4) If $\mathcal{S} \subseteq \mathcal{T}$, then $\mathcal{S} + \mathcal{T} = \mathcal{T}$.

**29.** Let $\mathcal{V}$ be a vector space, $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathcal{V}$. Suppose $\mathbf{A} + \mathbf{B} + \mathbf{C} = \mathbf{0}$. Show that $\mathcal{L}(\mathbf{A}, \mathbf{B}) = \mathcal{L}(\mathbf{B}, \mathbf{C})$.

**30.** Let $\mathcal{V}$ be a vector space and $\mathcal{W}$ a subspace of $\mathcal{V}$. Show that

$$\left\{\mathbf{A} \in \mathcal{V} \mid \mathbf{A} \notin \mathcal{W}\right\}$$

is not a vector subspace of $\mathcal{V}$.

**31.** Let $\mathcal{V}$ be a vector space, $\mathcal{W}$ a subspace of $\mathcal{V}$, and $\mathbf{A}, \mathbf{B} \in \mathcal{V}$. Assume that

$$\mathbf{A} \notin \mathcal{W} \text{ but } \mathbf{A} \in \mathcal{L}(\mathcal{W} \cup \{\mathbf{B}\}).$$

Show that $\mathbf{B} \in \mathcal{L}(\mathcal{W} \cup \{\mathbf{A}\})$.

**32.** Let $\mathcal{S}$, $\mathcal{T}$, $\mathcal{U}$ be subspaces of a vector space $\mathcal{V}$. Is it always true that

$$\mathcal{S} \cap (\mathcal{T} + \mathcal{U}) = (\mathcal{S} \cap \mathcal{T}) + (\mathcal{S} \cap \mathcal{U})?$$

**33.** Can it happen for two subspaces $\mathcal{S}$ and $\mathcal{T}$ of $\mathcal{V}$ that $\mathcal{S} \cap \mathcal{T} = \varnothing$?

**34.** Can you find two vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$ such that $\mathcal{L}(\mathbf{A}, \mathbf{B}) = \mathbb{R}^3$?

**35.** Let $\mathcal{V}$ be a vector space, $\mathcal{W}$ a subspace of $\mathcal{V}$, $S$ a set, and $s \in S$. Show that

$$\mathcal{U} = \left\{ f \in \text{Fun}(S, \mathcal{V}) \mid f(s) \in \mathcal{W} \right\}$$

is a subspace of Fun$(S, \mathcal{V})$.

**36.** Let $S$ be a set and $T \subseteq S$. Is it true that

$$\text{Fun}(S) = \text{Fun}(S, T) + \text{Fun}(S, S - T)?$$

**37.** Let $\mathcal{V}$ be a vector space and $S$ a set. Let Fun$(S, \mathcal{V})$ be defined as in Chapter 3 Exercise 10

(1) If $\mathcal{V}$ is a subspace of $\mathcal{W}$, show that Fun$(S, \mathcal{V})$ is a subspace of Fun$(S, \mathcal{W})$.

(2) If $\mathcal{S}$ and $\mathcal{T}$ are subspaces of $\mathcal{V}$, show that

$$\text{Fun}(S, \mathcal{S}) + \text{Fun}(S, \mathcal{T}) = \text{Fun}(S, \mathcal{S} + \mathcal{T}).$$

# Chapter 5
# Linear Independence and Dependence

A nontrivial vector space[1] always contains infinitely many elements. However, it is often possible to describe the elements by means of only finitely many vectors, and in such a way that the description of each individual vector is unique.

The essential idea involved is quite simple. If $\mathbf{A}_1, \ldots, \mathbf{A}_k$ are vectors in $\mathcal{V}$ then a **linear combination of** $\mathbf{A}_1, \ldots, \mathbf{A}_k$ is any vector $\mathbf{A}$ in $\mathcal{V}$ of the form

$$a_1\mathbf{A}_1 + \cdots + a_k\mathbf{A}_k \in \mathcal{V},$$

where $a_1, \ldots, a_k$ are real numbers. If at least one of the numbers $a_1, \ldots, a_k$ is nonzero, then the linear combination is called **nontrivial**. The linear combination

$$0\mathbf{A}_1 + \cdots + 0\mathbf{A}_k$$

is called the **trivial linear combination** of $\mathbf{A}_1, \ldots, \mathbf{A}_k$ and is equal to the zero vector $\mathbf{0}$ of $\mathcal{V}$. The study of properties of linear combinations lies at the heart of linear algebra.

## 5.1 Basic Definitions and Examples

The following definition is basic for what follows.

DEFINITION: *A set of vectors $E$ is said to be* **linearly dependent** *if there exist distinct vectors $\mathbf{A}_1, \ldots, \mathbf{A}_k$ in $E$ and numbers $a_1, \ldots, a_k$ not all zero such that*

$$(*) \qquad a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_k\mathbf{A}_k = \mathbf{0}.$$

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[1] With real or complex scalars.

*Equation* (*) *is called a* **linear relation** *between* $\mathbf{A}_1, \ldots, \mathbf{A}_k$.

In other words, a set of vectors $E$ is linearly dependent if there is a nontrivial linear combination of distinct vectors chosen from $E$ that is equal to the zero vector.

**EXAMPLE 1:** Let $E$ be the set of vectors

$$E = \{(0, 1, 0), (0, 1, 1), (0, 0, 1)\}$$

in $\mathbb{R}^3$. Then $E$ is a linearly dependent set of vectors because

$$1(0, 1, 0) + 1(0, 0, 1) + (-1)(0, 1, 1) = (0, 0, 0).$$

**EXAMPLE 2:** Let $E$ be the set of vectors

$$E = \left\{p(x) \mid \text{the degree of } p(x) \text{ is at most } 1\right\}$$

in $\mathcal{P}_3(\mathbb{R})$. Then $E$ is a linearly dependent set of vectors because the vectors $1$, $x$, $1 + x$ belong to $E$ and

$$1 \cdot (1 + x) + (-1) \cdot x + (-1) \cdot (1) = 0.$$

**DEFINITION:** *A set of vectors $E$ that is not linearly dependent is said to be* **linearly independent**.

**EXAMPLE 3:** Let $E$ be the vectors

$$\{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$$

in $\mathbb{R}^3$. Then $E$ is a linearly independent set of vectors. In order to prove this we suppose some linear combination of these vectors is the zero vector, i.e., that there are numbers $a_1$, $a_2$, $a_3$, such that

$$a_1(1, 1, 1) + a_2(0, 1, 1) + a_3(0, 0, 1) = (0, 0, 0).$$

A little vector algebra gives

$$a_1(1, 1, 1) + a_2(0, 1, 1) + a_3(0, 0, 1) = (a_1, a_1 + a_2, a_1 + a_2 + a_3),$$

so we must have

$$(0, 0, 0) = (a_1, a_1 + a_2, a_1 + a_2 + a_3),$$

which means that

$$a_1 = 0, \qquad a_1 + a_2 = 0, \qquad a_1 + a_2 + a_3 = 0,$$

so of necessity

$$a_1 = 0, \qquad a_2 = 0, \qquad a_3 = 0.$$

Thus the only linear combination of the original vectors giving the zero vector must be the one where $a_1$, $a_2$, and $a_3$ are zero. Therefore, the set $E$ cannot be linearly dependent and hence must be linearly independent.

**EXAMPLE 4:** Regard the complex numbers $\mathbb{C}$ as a real vector space. Let $E$ be the set of vectors $\{1,\ \boldsymbol{i}\}$ in $\mathbb{C}$. Then $E$ is linearly independent. For suppose that

$$a_1(1) + a_2(\boldsymbol{i}) = 0.$$

Let $a_1 - a_2\boldsymbol{i}$ be the conjugate complex number. Then $0 = a_1 + a_2\boldsymbol{i}$ implies

$$0 = (a_1 + a_2\boldsymbol{i})(a_1 - a_2\boldsymbol{i}) = a_1^2 + a_2^2,$$

so $a_1 = 0 = a_2$. Therefore $E$ must be linearly independent.

**REMARK:** The proof above uses the conjugate of a complex number. There is an alternative proof avoiding this which goes as follows: Since we consider $\mathbb{C}$ only as a real vector spaces, if there are two real numbers $a_1,\ a_2$ such that

$$a_1 1 + a_2\boldsymbol{i} = 0,$$

then

$$(*) \qquad a_1 = -a_2\boldsymbol{i}.$$

Since $a_1$ is real from assumption, as is $a_2$, $(*)$ says that $a_1$ is also purely imaginary. The only way to avoid a contradiction is for $a_1 = 0$ and $a_2 = 0$. Thus $\{1,\ \boldsymbol{i}\}$ is a set of linearly independent vectors.

**EXAMPLE 5:** Let $E$ be the set of vectors $\{1 + x,\ 1 - x\}$ in $\mathcal{P}_1(\mathbb{R})$. Then $E$ is linearly independent. For suppose to the contrary that $\{1 + x,\ 1 - x\}$ is linearly dependent. Then there would exist numbers $a_1$, $a_2$, not both zero, such that

$$a_1(1 + x) + a_2(1 - x) = 0.$$

Then we would have

$$\begin{aligned} 0 &= a_1(1 + x) + a_2(1 - x) = a_1 + a_1 x + a_2 - a_2 x \\ &= (a_1 + a_2) + (a_1 - a_2)x. \end{aligned}$$

Remember that a polynomial is identically zero if and only if all its coefficients are zero. Therefore, we have

$$a_1 + a_2 = 0, \qquad a_1 - a_2 = 0.$$

Solving these equations, we find

$$a_1 = 0, \qquad a_2 = 0$$

which is a contradiction to the assumption that $a_1$ and $a_2$ are not both zero. Therefore, $\{1 + x,\ 1 - x\}$ is a linearly independent set of vectors.

**EXAMPLE 6:** Let $S$ be a set. For each $s \in S$ the **characteristic function of** $s$ is the function

$$\chi_s : S \longrightarrow \mathbb{R}$$

defined by

$$\chi_s(t) = \begin{cases} 1 & \text{if } t = s, \\ 0 & \text{if } t \neq s. \end{cases}$$

If $s_1, \ldots, s_k \in S$ are distinct, then their characteristic functions $\chi_{s_1}, \ldots, \chi_{s_k} \in \mathrm{Fun}(S)$ are linearly independent. To see this, suppose that

$$a_1\chi_{s_1} + \cdots + a_k\chi_{s_k} = \mathbf{0}$$

is a linear relation between $\chi_{s_1}, \ldots, \chi_{s_k}$. Then

$$\begin{aligned} \mathbf{0} &= (a_1\chi_{s_1} + \cdots + a_k\chi_{s_k})(s_i) \\ &= a_1\chi_{s_1}(s_i) + \cdots + a_k\chi_{s_k}(s_i) \\ &= a_1 0 + \cdots + a_{i-1}0 + a_i 1 + a_{i+1}0 + \cdots + a_k 0 = a_i, \end{aligned}$$

so setting $i$ successively equal to $1, \ldots, k$ we see that $a_1 = 0$, $a_2 = 0, \ldots, a_k = 0$ and therefore $\chi_{s_1}, \ldots, \chi_{s_k}$ are linearly independent.

## 5.2 Properties of Independent and Dependent Sets

A very quick test for a linearly dependent set is the following:

**PROPOSITION 5.2.1:** *If a set of vectors $E$ contains the vector $\mathbf{0}$, it is linearly dependent.*

**PROOF:** Clearly

$$1 \cdot \mathbf{0} = \mathbf{0},$$

so letting $\mathbf{A}_1 = \mathbf{0} \in E$, $a_1 = 1 \in \mathbb{R}$, and $k = 1$, we satisfy the condition of linear dependence. □

**COROLLARY 5.2.2:** *If $E$ is a linear subspace of $\mathcal{V}$, then $E$ is a linearly dependent set of vectors.*

**PROOF:** A linear subspace always contains $\mathbf{0}$. Apply Proposition 5.2.1. □

**DEFINITION:** *A vector* $\mathbf{A}$ *is said to be* **linearly dependent on a set of vectors** $E$ *if* $\mathbf{A} \in \mathcal{L}(E)$.

**PROPOSITION 5.2.3:** *A set of vectors $E$ is linearly dependent if and only if there is a vector $\mathbf{A}$ in $E$ linearly dependent on the remaining vectors of $E$, i.e., $\mathbf{A}$ is linearly dependent on the set $E \setminus \{\mathbf{A}\}$.*

**PROOF:** Suppose that $E$ is linearly dependent. Then we may find distinct vectors $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_k$ in $E$, and numbers $a_1, \ldots, a_k$, not all zero, such that

$$a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_k\mathbf{A}_k = \mathbf{0}.$$

Since not all the numbers $a_1, \ldots, a_k$ are zero, we can by changing the order assume that $a_1 \neq 0$. Then we have

$$a_1\mathbf{A}_1 = -a_2\mathbf{A}_2 - a_3\mathbf{A}_3 - \cdots - a_k\mathbf{A}_k,$$

and since $a_1 \neq 0$,

$$\mathbf{A}_1 = \frac{-a_2}{a_1}\mathbf{A}_2 + \frac{-a_3}{a_1}\mathbf{A}_3 + \cdots + \frac{-a_k}{a_1}\mathbf{A}_k,$$

and hence $\mathbf{A}_1 \in \mathcal{L}(\mathbf{A}_2, \ldots, \mathbf{A}_k)$, which shows (since $\mathbf{A}_1, \ldots, \mathbf{A}_k$ are distinct) that $\mathbf{A}_1$ is linearly dependent on the remaining vectors of $E$.

Conversely, if there is a vector $\mathbf{A}$ in $E$ that is linearly dependent on the remaining vectors of $E$, we may find distinct vectors $\mathbf{A}_2, \ldots, \mathbf{A}_k$, different from $\mathbf{A}$, and numbers $a_2, \ldots, a_k$ such that

$$\mathbf{A} = a_2\mathbf{A}_2 + \cdots + a_k\mathbf{A}_k.$$

Then

$$\mathbf{0} = (-1)\mathbf{A} + a_2\mathbf{A}_2 + \cdots + a_k\mathbf{A}_k$$

is a nontrivial (the coefficient of $\mathbf{A}$ is $-1 \neq 0$) linear relation among $\mathbf{A} = \mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_k$, showing that $E$ is a linearly dependent set. □

**THEOREM 5.2.4:** *If $E$ is a finite set of vectors spanning the linear subspace $\mathcal{U}$ of $\mathcal{V}$, that is, $\mathcal{L}(E) = \mathcal{U}$, then there exists a subset $F$ of $E$ such that $F$ is a linearly independent set of vectors and $\mathcal{L}(F) = \mathcal{U} = \mathcal{L}(E)$.*

**PROOF:** If $E$ is linearly independent, there is nothing to prove. So suppose that $E$ is a linearly dependent set of vectors. By Proposition 5.2.3 there exists a vector $\mathbf{A}$ that is linearly dependent on the remaining vectors of $E$. Denote the set of remaining vectors by $E'$. Thus $\mathbf{A} \in \mathcal{L}(E')$. Therefore, $\mathcal{L}(E') = \mathcal{L}(E)$ because $\mathcal{L}(E') \subseteq \mathcal{L}(E)$ and (see Chapter 4, Exercises 16 and 20)

$$\begin{aligned}\mathcal{L}(E) = \mathcal{L}(\{\mathbf{A}\} \cup E') &\subseteq \mathcal{L}(\mathbf{A}) + \mathcal{L}(E') \subseteq \mathcal{L}(E') + \mathcal{L}(E') \\ &= \mathcal{L}(E').\end{aligned}$$

Therefore, we can repeat our argument on $E'$. That is, either $E'$ is linearly independent, in which case we are done, or it is linearly dependent and we can use the preceding argument to reduce the size of $E'$ by one vector. Since the set $E$ of vectors that we began with is finite the theorem will follow by repeating the argument a finite number of times. □

**EXAMPLE 1:** Let $E = \{(1, 0, 0),\ (0, 1, 0),\ (0, 0, 1),\ (1, 1, 1)\}$ be vectors in $\mathbb{R}^3$. Find a linearly independent set $F$ that is a subset of $E$ such that $\mathcal{L}(F) = \mathcal{L}(E)$.

**SOLUTION:** The proof of Theorem 5.2.4 suggests that we look for a vector in $E$ linearly dependent on the remaining vectors of $E$ and throw it away. If that doesn't lead to a linearly independent set with the same span, do it again, etc. Observe that

$$(1, 1, 1) = 1(1, 0, 0) + 1(0, 1, 0) + 1(0, 0, 1),$$

so that setting $F = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ we have $\mathcal{L}(F) = \mathcal{L}(E)$. The set $F$, however, is a linearly independent set of vectors and so we are done.

**ALTERNATIVE SOLUTION:** Proceeding as in the previous solution, we note that $(1, 0, 0) = 1(1, 1, 1) + (-1)(0, 1, 0) + (-1)(0, 0, 1)$, and setting $H = \{(1, 1, 1),\ (0, 1, 0),\ (0, 0, 1)\}$ we have, $\mathcal{L}(H) = \mathcal{L}(E)$. The set $H$, however, is a linearly independent set of vectors and so we done.

**MORAL:** If $E$ is a finite set of vectors in $\mathcal{V}$, then there need not exist a *unique* subset $F \subseteq E$ with $F$ linearly independent and $\mathcal{L}(F) = \mathcal{L}(E)$.

**EXAMPLE 2:** Let $S$ be a *finite* set and $X = \{\chi_s \mid s \in S\} \subseteq \mathrm{Fun}(S)$. These characteristic functions span $\mathrm{Fun}(S)$ because for any $f \in \mathrm{Fun}(S)$ we have the equality

$$f = \sum_{s \in S} f(s)\chi_s.$$

This is seen by evaluating the right hand side at an arbitrary element $t \in S$, giving

$$\left[\sum_{s \in S} f(s)\chi_s\right](t) = \sum_{s \in S} f(s) \cdot \chi_s(t) = \left[\sum_{s \neq t} f(s) \cdot 0\right] + f(t) = f(t)$$

as required.

**REMARK:** Note that none of the functions $\chi_s$ for $s \in S$ can be left out and still leave us with a set of functions that span $\mathrm{Fun}(S)$, for the only way to write $\chi_s$ as a linear combination of characteristic functions is as

$$\chi_s = \sum_{t \in S} \delta_{s,t}\chi_t = 1 \cdot \chi_s.$$

**REMARK:** Example 2 cannot be extended to infinite sets. For if $S$ is an infinite set then the function

$$f : S \longrightarrow \mathbb{R}$$

defined by $f(s) = 1$, $s \in S$ is not a *(finite)* linear combination of characteristic functions.

## 5.3 Exercises

**1.** Which of the following sets of vectors in $\mathbb{R}^3$ are linearly dependent and which are linearly independent?

$E = \{(1, 1, 1),\ (0, 1, 0),\ (1, 0, 1)\}$.
$F = \{(1, 1, 1),\ (1, 1, 0),\ (1, 0, 0)\}$.
$G = \{(1, 1, 1),\ (1, 1, 0),\ (1, 0, 1)\}$.
$H = \{(1, 0, 0),\ (0, 1, 0),\ (1, 1, 1)\}$.
$K = \{(1, 1, 1),\ (0, 1, 0),\ (0, 0, 1)\}$.

**2.** Which of the following sets of vectors in $\mathcal{P}_2(\mathbb{R})$ are linearly dependent and which are linearly independent?

$E = \left\{1,\ x,\ x^2\right\}$.
$F = \left\{1 + x,\ 1 - x, x^2,\ 1\right\}$.
$G = \left\{x^2 - 1,\ x + 1,\ x^2 - x,\ x^2 + x\right\}$.
$H = \left\{x - x^2,\ x^2 - x\right\}$.
$K = \left\{1,\ 1 - x,\ 1 - x^2\right\}$.

**3.** Show that the set of vectors in $\mathbb{R}^3$

$$E = \{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$$

is linearly dependent but that any set of three of them is linearly independent.

**4.** Let $\ell_\infty$ be the vector space of bounded sequences of numbers (see Chapter 3 Exercise 16). Which of the following collections of vectors in $\ell_\infty$ are linearly independent?

(1) $\{(1, 1, \ldots, 1, \ldots),\ (2, 1, 1, \ldots, 1, \ldots),\ (0, 1, 1, \ldots, 1, \ldots)\}$.
(2) $\{(1, 0, \ldots, 0, \ldots),\ (0, 1, 0, , \ldots, 0, \ldots),\ (0, 0, \ldots, 0, 1, 0 \ldots), \ldots\}$.
(3) $\{(1, , 1, 1, 0, \ldots, 0, \ldots), (1, 1, 0, \ldots, 0, \ldots),\ (1, 0, 1, 0, \ldots, 0, \ldots)\}$.
(4) $\{(1, 0, 0, \ldots, 0, \ldots),\ (1, 1, 0, , \ldots, 0, \ldots),\ (1, 1, \ldots, 1, 0, \ldots, 0), \ldots, \}$.
(5) $\{(1, 1/2, \ldots, 1/n, \ldots), (0, 1/2, 1/3, \ldots, 1/n, \ldots),$
$(0, 0, 1/3, \ldots, 1/n, \ldots), \ldots, (0, , \ldots, 0, 1/n, \ldots), \ldots, \}$.

**5.** Let $\mathcal{U}$ be the subspace of $\mathbb{R}^5$ spanned by the vectors

$$E = \{(1, 1, 0, 0, 1),\ (1, 1, 0, 1, 1),\ (0, 1, 1, 1, 1),\ (2, 1, -1, 0, 1)\}.$$

Find a linearly independent subset $F$ of $E$ with $\mathcal{L}(F) = \mathcal{U}$.

**6.** Let $\mathcal{U}$ be the subspace of $\mathcal{P}_3(\mathbb{R})$ spanned by

$$E = \left\{x^3,\ x^3 - x^2,\ x^3 + x^2,\ x^3 - 1\right\},$$

find a linearly independent subset $F$ of $E$ spanning $\mathcal{U}$. (**HINT:** Try to use Theorem 5.2.4.)

**7.** For each of the linearly dependent sets $S$ in Exercise 4 use Theorem 5.2.4 to find a linearly independent subset of vectors with the same span as $S$.

**8.** Under what conditions on the numbers $a$ and $b$ are the vectors $(1, a)$, $(1, b)$ linearly independent in $\mathbb{R}^2$.

**9.** Suppose that $E$ and $F$ are sets of vectors in $\mathcal{V}$ with $E \subseteq F$. Prove that if $E$ is linearly dependent, then so is $F$.

**10.** Suppose that $E$ and $F$ are sets of vectors in $\mathcal{V}$ with $E \subseteq F$. Prove that if $F$ is linearly independent then so is $E$.

**11.** Show that functions $e^x$ and $e^{2x}$ form a set of linearly independent vectors in $\mathcal{C}(-\infty, \infty)$.

**12.** Show that

$$\left\{\cos x,\ \sin x,\ \sin\left(x + \frac{\pi}{2}\right)\right\}$$

is a set of linearly dependent vectors in $\mathcal{C}(-\infty, \infty)$.

**13.** Is the pair of complex numbers $\alpha + \beta\mathbf{i}$, $\alpha - \beta\mathbf{i}$ a set of linearly independent vectors in $\mathbb{C}$, where $\alpha, \beta$ are any nonzero real numbers?

**14.** Show that the set of polynomials $E = \left\{x^2, 1 + x^2\right\}$ is a set of linearly independent vectors in $\mathcal{P}_2(\mathbb{R})$. What is the space spanned by $E$? Is $\mathcal{L}(E) = \mathcal{P}_2(\mathbb{R})$? If not, find a vector in $\mathcal{P}_2(\mathbb{R})$ that does not belong to $\mathcal{L}(E)$, and show that together with $E$ they form a set of linearly independent vectors in $\mathcal{P}_2(\mathbb{R})$.

**15.** Let $E$ be a set of vectors in the vector space $\mathcal{V}$. Show that $E$ is linearly independent if and only if for every proper subset $E'$ of $E$, $\mathcal{L}(E') \neq \mathcal{L}(E)$. Equivalently, show that $E$ is linearly dependent if and only if there exists a proper subset $E'$ of $E$ such that $\mathcal{L}(E') = \mathcal{L}(E)$.

**16.** Let $S = \{x_1, x_2, x_3\}$ and consider the functions $f, g, h \in \text{Fun}(S)$ defined by

$$\begin{aligned} f(x_1) &= 0, \quad f(x_2) = 1, \quad f(x_3) = 1, \\ g(x_1) &= 1, \quad g(x_2) = 0, \quad g(x_3) = 1, \\ h(x_1) &= 1, \quad h(x_2) = 1, \quad h(x_3) = 0. \end{aligned}$$

Does $\{f, g, h\}$ form a linearly independent set?

**17.** Let $E', E''$ be linearly independent sets of vectors in $\mathcal{V}$. Show that:
(1) $E' \cap E''$ is linearly independent.
(2) $E' \cup E''$ is linearly independent if and only if $\mathcal{L}(E') \cap \mathcal{L}(E'') = \{0\}$.

**18.** Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathcal{V}$ be linearly independent vectors. Show that the vectors $\mathbf{A}+\mathbf{B}, \mathbf{B}+\mathbf{C}, \mathbf{C}+\mathbf{A}$ are linearly independent.

**19.** Let $\mathcal{V}$ be a vector space and $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D} \in \mathcal{V}$. Show that the vectors

$$\mathbf{A}+\mathbf{B}+\mathbf{C}+\mathbf{D}, \quad 2\mathbf{A}+2\mathbf{B}+\mathbf{C}-\mathbf{D}, \quad \mathbf{A}-\mathbf{B}+\mathbf{C}, \quad \mathbf{A}-\mathbf{C}+\mathbf{D}$$

are linearly dependent. Find a maximal linearly independent subset of these four vectors.

**20.** Let $\mathcal{S}, \mathcal{T}$ be subspaces of a vector space $\mathcal{V}$, and $\mathbf{S} \in \mathcal{S}, \mathbf{T} \in \mathcal{T}, \mathbf{S} \neq \mathbf{0}, \mathbf{T} \neq \mathbf{0}$. Assume $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$. Show that $\{\mathbf{S}, \mathbf{T}\}$ is a linearly independent set of vectors.

**21.** Show that any four vectors in $\mathbb{R}^3$ must be linearly dependent.

**22.** Let $\mathcal{S}, \mathcal{T}$ be subspaces of $\mathcal{V}$ and $S$ a set. Suppose $f \in \text{Fun}(S, \mathcal{S})$, $g \in \text{Fun}(S, \mathcal{T})$, and $\mathcal{S} \cap \mathcal{T} = \mathbf{0}$. Show that regarded as vectors in $\text{Fun}(S, \mathcal{V})$, the vectors $f$ and $g$ are linearly independent.

**23.** Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathcal{V}$. If $\mathbf{C} \notin \mathcal{L}(\mathbf{A}, \mathbf{B})$, show that $\mathbf{A}$ and $\mathbf{B}$ are linearly independent if and only if $\mathbf{A}+\mathbf{C}$ and $\mathbf{B}+\mathbf{C}$ are linearly independent.

**24.** Let $f_1, \ldots, f_k \in \mathcal{P}(\mathbb{R})$ be polynomials of positive degree. Suppose $f_1, \ldots, f_k$ are linearly independent. Does it follow that their derivatives $df_1/dx, \ldots, df_k/dx$ are linearly independent?

**25.** Let $S$ be a set. A **permutation** $\sigma$ of $S$ is a bijective (i.e., a one to one and onto) mapping $\sigma : S \longrightarrow S$. For a permutation $\sigma$ of $S$ and $f \in \text{Fun}(S)$ define

$$f_\sigma : S \longrightarrow \mathbb{R} \text{ by } f_\sigma(s) = f(\sigma(s)).$$

Suppose $f_1, \ldots, f_n \in \text{Fun}(S)$ are linearly independent. Show for any permutation $\sigma$ that $(f_1)_\sigma, \ldots, (f_n)_\sigma$ are also linearly independent.

**26.** If $f_1, \ldots, f_n \in \mathcal{P}_m(\mathbb{R})$ are linearly dependent then so are their derivatives $df_1/dx, \ldots, df_n/dx$.

**27.** Let $\mathcal{V}$ be the set of all polynomials $p(x) \in \mathcal{P}_m(\mathbb{R})$, $m \geq 2$, such that

$$p(0) = 0 \text{ and } \left.\frac{d}{dx}p(x)\right|_{x=0} = 0.$$

Show that $\mathcal{V}$ is a subspace of $\mathcal{P}_m(\mathbb{R})$ and find a linearly independent set of polynomials that span this subspace.

# Chapter 6 Finite-Dimensional Vector Spaces and Bases

If a vector space $\mathcal{V}$ contains a finite number of vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ that span $\mathcal{V}$ then we say that $\mathcal{V}$ is **finite-dimensional**. Finite-dimensional vector spaces have particularly nice properties, and they provide a suitable background for the study of many linear phenomena.

If $\mathbf{A}_1, \ldots, \mathbf{A}_n$ span $\mathcal{V}$ and $\mathbf{A} \in \mathcal{V}$, then we can always find numbers $a_1, \ldots, a_n$ (depending on $\mathbf{A}$, of course) such that

$$(*) \qquad \mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n .$$

However, this way of writing $\mathbf{A}$ need not be unique, i.e., if

$$\mathbf{A} = \bar{a}_1\mathbf{A}_1 + \cdots + \bar{a}_n\mathbf{A}_n ,$$

then we cannot in general conclude that

$$a_i = \bar{a}_i \text{ for } i = 1, \ldots, n .$$

Spanning sets where the representation $(*)$ is unique are very special. In this chapter we will investigate their properties.

## 6.1 Finite-Dimensional Vector Spaces

We begin with a definition.

**DEFINITION:** *A vector space $\mathcal{V}$ is said to be* **finite-dimensional** *if there exists a **finite** set of vectors $E$ with $\mathcal{L}(E) = \mathcal{V}$.*

Our three basic examples $\mathbb{R}^k$, $\mathcal{P}_n(\mathbb{R})$, and $\mathrm{Fun}(S)$ when $S$ is a finite set are all finite-dimensional.

**Example 1:** $\mathbb{R}^k$ is finite-dimensional.

To see this we introduce the vectors

$$\mathbf{E}_i = (0, 0, \ldots, 1, 0, \ldots, , 0), \qquad i = 1, \ldots, k.$$

For example, if $k = 4$, there are four such vectors, namely

$$\mathbf{E}_1 = (1, 0, 0, 0),$$
$$\mathbf{E}_2 = (0, 1, 0, 0),$$
$$\mathbf{E}_3 = (0, 0, 1, 0),$$
$$\mathbf{E}_4 = (0, 0, 0, 1).$$

If $\mathbf{A}$ is any vector in $\mathbb{R}^k$, then $\mathbf{A} = (a_1, , \ldots, , a_k)$, and hence we may write $\mathbf{A}$ in the form of a linear combination of $\mathbf{E}_1, \ldots, \mathbf{E}_k$:

$$\mathbf{A} = a_1\mathbf{E}_1 + \cdots + a_k\mathbf{E}_k.$$

For example, in $\mathbb{R}^4$ we have $(1, 2, 3, 4) = 1\mathbf{E}_1 + 2\mathbf{E}_2 + 3\mathbf{E}_3 + 4\mathbf{E}_4$. Thus $\mathbb{R}^k = \mathcal{L}(\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_k)$, and since $E = \{\mathbf{E}_1, \ldots, \mathbf{E}_k\}$ is a finite set, $\mathbb{R}^k$ is finite-dimensional.

**Example 2:** $\mathcal{P}_n(\mathbb{R})$ is finite-dimensional.

The finite set of polynomials $E = \left\{1, x, x^2, \ldots, x^n\right\}$ spans the vector space of polynomials $\mathcal{P}_n(\mathbb{R})$. To see this, suppose that $p(x)$ is any vector in $\mathcal{P}_n(\mathbb{R})$. Then

$$p(x) = a_0 \cdot 1 + a_1 \cdot x + \cdots + a_n \cdot x^n,$$

and hence $\mathcal{P}_n(\mathbb{R}) = \mathcal{L}(1, x, x^2, \ldots, x^n)$ and therefore $\mathcal{P}_n(\mathbb{R})$ is finite-dimensional.

**Example 3:** Let $S$ be a *finite* set. Then $\mathrm{Fun}(S)$ is finite-dimensional because the characteristic functions $X = \left\{\chi_s \mid s \in S\right\}$ span $\mathrm{Fun}(S)$ (see Chapter 5 Example 6).

**Theorem 6.1.1:** *Let $\mathcal{V}$ be a finite-dimensional vector space. Then there exists a finite set of linearly independent vectors $B$ spanning $\mathcal{V}$; that is,*

(1) $\mathrm{Span}(B) = \mathcal{V}$ *and;*
(2) *$B$ is a linearly independent set of vectors in $\mathcal{V}$.*

**Proof:** Since $\mathcal{V}$ is finite-dimensional, there is a finite set of vectors $E$ with $\mathrm{Span}(E) = \mathcal{V}$. By Theorem 5.2.4 we may find a linearly independent set $B \subseteq E$ with $\mathrm{Span}(B) = \mathrm{Span}(E)$. The set $B$ is as required. $\square$

The property that a set of vectors be both linearly independent and span $\mathcal{V}$ is most important and fundamental to further developments. We therefore introduce:

**DEFINITION:** *A set of vectors $E$ in a vector space $\mathcal{V}$ is called a* **basis** *for $\mathcal{V}$ if $E$ is linearly independent and $\mathcal{L}(E) = \mathcal{V}$.*

**EXAMPLE 4:** The vectors $(1, 1)$, $(1, 0)$ are a basis for $\mathbb{R}^2$.

To show this we must check that $(1, 1), (1, 0)$ are linearly independent and span $\mathbb{R}^2$. We check first that they span $\mathbb{R}^2$. If $(x, y) \in \mathbb{R}^2$, we wish to find numbers $a, b$ such that

$$(x, y) = a(1, 1) + b(1, 0) = (a + b, a).$$

Therefore,

$$y = a, \qquad x = a + b,$$

so solving for $a$ and $b$ we get

$$a = y, \qquad b = x - y,$$

and therefore

$$(x, y) = y(1, 1) + (x - y)(1, 0),$$

showing that $(x, y) \in \mathcal{L}\{(1, 1), (1, 0)\}$, and therefore that $\{(1, 1), (1, 0)\}$ spans $\mathbb{R}^2$. To show that $(1, 1), (1, 0)$ are linearly independent, we suppose that

$$a(1, 1) + b(1, 0) = (0, 0)$$

is a linear relation between them. As above, we then find

$$0 = a, \qquad 0 = a + b,$$

whence $a = 0$ and $b = 0$ as required.

**EXAMPLE 5:** The vectors

$$1, x - 1, (x - 2)(x - 1)$$

form a basis for $\mathcal{P}_2(\mathbb{R})$.

We must again check linear independence and spanning. To check linear independence, suppose

$$0 = a_1 \cdot 1 + a_2(x - 1) + a_3(x - 2)(x - 1)$$

is a linear relation. Multiplying out gives

$$\begin{aligned} 0 &= a_1 + a_2(x - 1) + a_3(x^2 - 3x + 2) \\ &= a_1 + a_2x - a_2 + a_3x^2 - 3a_3x + 2a_3 \\ &= (a_1 - a_2 + 2a_3) + (a_2 - 3a_3)x + a_3x^2, \end{aligned}$$

so we must have (n.b. a polynomial is zero if and only if all its coefficients are zero)

$$0 = a_1 - a_2 + 2a_3,$$
$$0 = a_2 - 3a_3,$$
$$0 = a_3,$$

which, solving in reverse order, gives

$$a_3 = 0, \qquad a_2 = 0, \qquad a_1 = 0,$$

as required.

To show that $\{1, x-1, (x-2)(x-1)\}$ is a spanning set, let $p(x) = a_0 + a_1x + a_2x^2 \in \mathcal{P}_2(\mathbb{R})$. We are looking for numbers $b_0, b_1, b_2$ such that

$$a_0 + a_1x + a_2x^2 = b_0 + b_1(x-1) + b_2(x-2)(x-1).$$

Multiplying out and equating coefficients, we obtain the simultaneous equations

$$a_0 = b_0 - b_1 + 2b_2,$$
$$a_1 = b_1 - 3b_2,$$
$$a_2 = b_2,$$

which may be solved to yield

$$b_2 = a_2,$$
$$b_1 = a_1 + 3a_2,$$
$$b_0 = a_0 + a_1 + a_2,$$

so

$$p(x) = (a_0 + a_1 + a_2)1 + (a_1 + 3a_2)(x-1) + a_2(x-2)(x-1),$$

and therefore $\{1, x-1, (x-2)(x-1)\}$ spans $\mathcal{P}_2(\mathbb{R})$.

**EXAMPLE 6:** In $\mathbb{R}^3$ the vectors $(1, -1, 0), (0, 1, -1)$ are a basis for the subspace

$$\mathcal{V} = \left\{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\right\}.$$

To see this, we first show that $(1, -1, 0), (0, 1, -1)$ span $\mathcal{V}$. So let $(x, y, z) \in \mathcal{V}$. Then

$$(*) \qquad x + y + z = 0.$$

We wish to write

$$(x, y, z) = a(1, -1, 0) + b(0, 1, -1),$$
$$= (a, b-a, -b)$$

so we must have

$$x = a, \qquad y = b - a, \qquad z = -b.$$

To solve for $a$ and $b$ use the first and last equations giving,

$$a = x, \qquad b = -z.$$

But is this consistent with the middle equation? Substituting gives

$$y = -z - x,$$

and recalling $(*)$, we see that this is valid precisely for the vectors of $\mathcal{V}$. Thus if $(x, y, z) \in \mathcal{V}$, then

$$(x, y, z) = x(1, -1, 0) - z(0, 1, -1),$$

and therefore the vectors $(1, -1, 0), (0, 1, -1)$ span $\mathcal{V}$. The check for linear independence of $\{(1, -1, 0), (0, 1, -1)\}$ is easy, and we omit it.

Example 6 is illustrated in Figure 6.1.1.

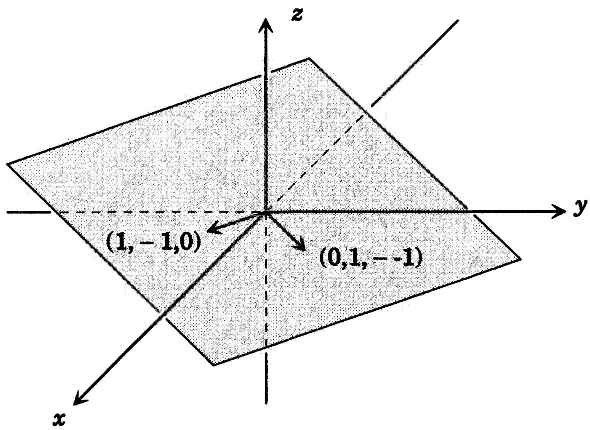


Figure 6.1.1

## 6.2 Properties of Bases

Theorem 6.1.1 shows that every finite-dimensional vector space has a basis. Actually, more is true, namely, any two bases for a finite-dimensional vector space contain the same number of vectors. This is an important result, since the number of vectors in a basis, being independent of the basis, allows us to define the **dimension** of a vector space as the number of vectors in any (and hence all) bases. To prove this we will need a few preliminary steps, the first of which is:

**PROPOSITION 6.2.1:** *Let $E = \{\mathbf{A}_1, \ldots, \mathbf{A}_k\}$ be a finite set of vectors. Then $E$ is linearly dependent if and only if*

$$\mathbf{A}_m \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_{m-1})$$

*for some $m \leq k$.*

**PROOF:** If for some $m \leq k$ we have $\mathbf{A}_m \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_{m-1})$ then $\mathbf{A}_m$, is linearly dependent on the remaining vectors of $E$ and hence by Proposition 5.2.4 $E$, is a linearly dependent set of vectors.

On the other hand, suppose that $E$ is a linearly dependent set of vectors. Then there exists a linear relation

$$a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_k\mathbf{A}_k = \mathbf{0}.$$

Choose $m$ to be the largest integer between 1 and $k$ for which $a_m \neq 0$. Then since $a_{m+1} = \cdots = a_n = 0$, we may write our linear relation as

$$a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_m\mathbf{A}_m = \mathbf{0},$$

and since $a_m \neq 0$, we may solve for $\mathbf{A}_m$, obtaining

$$\mathbf{A}_m = \frac{-a_1}{a_m}\mathbf{A}_1 + \frac{-a_2}{a_m}\mathbf{A}_2 + \cdots + \frac{-a_{m-1}}{a_m}\mathbf{A}_{m-1}$$

which shows that $\mathbf{A}_m \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_{m-1})$. □

**PROPOSITION 6.2.2:** *Let $E = \{\mathbf{A}_1, \ldots, \mathbf{A}_k\}$ be a finite set of vectors in $\mathcal{V}$. If $F$ is any linearly independent set of vectors in $\mathcal{L}(E)$, then $F$ is finite and the number of elements in $F$ is at most $k$.*

**PROOF:** Suppose that $H$ is a finite subset of $F$. Let the vectors of $H$ be $\mathbf{B}_1, \ldots, \mathbf{B}_s$. Note that $\mathbf{B}_1, \ldots, \mathbf{B}_s$ are linearly independent. We must show that $s \leq k$. To do this we consider the set of vectors

$$G_1 = \{\mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_k\}.$$

Note that this set is linearly dependent by Proposition 5.2.3, since

$$\mathbf{B}_s \in \mathcal{L}(E) = \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k).$$

Therefore, we may apply Proposition 6.2.1, to conclude that $G_1$ is linearly dependent in another way; namely, there exists a vector in the set $G_1$ linearly dependent on the ***preceding*** vectors. This vector cannot be $\mathbf{B}_s$ because $\mathbf{B}_s \neq \mathbf{0}$. Therefore, it must be an $\mathbf{A}_i$, and by rearranging terms we may assume that it is $\mathbf{A}_k$. That is,

$$\mathbf{A}_k \in \mathcal{L}(\mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_{k-1}).$$

Set $\mathbf{E}_1 = \{\mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_{k-1}\}$ and note that $\mathcal{L}(\mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_{k-1}) = \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k)$. Reasoning as before, we see that some vector in the set

$$G_2 = \{\mathbf{B}_{s-1}, \mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_{k-1}\}$$

must be linearly dependent on the vectors that precede it. It cannot be $\mathbf{B}_{s-1}$, since $\mathbf{B}_{s-1} \neq 0$. Nor can it be $\mathbf{B}_s$, since $\{\mathbf{B}_{s-1}, \mathbf{B}_s\}$ are linearly independent. Therefore, it must be an $\mathbf{A}_j$, and again by shuffling we may assume that it is $\mathbf{A}_{k-1}$. Thus

$$\mathbf{A}_{k-1} \in \mathcal{L}(\mathbf{B}_{s-1}, \mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_{k-2}),$$

and hence

$$\begin{aligned} \mathcal{L}(\mathbf{B}_{s-1}, \mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_{k-2}) &= \mathcal{L}(\mathbf{B}_s, \mathbf{A}_1, \ldots, \mathbf{A}_{k-1}) \\ &= \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k). \end{aligned}$$

Suppose next that $s > k$. Then we may repeat the above argument $k$ times and conclude that

$$\mathcal{L}(\mathbf{B}_{s-k+1}, \mathbf{B}_{s-k+2}, \ldots, \mathbf{B}_s) = \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k).$$

But then

$$\mathbf{B}_{s-k} \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k) = \mathcal{L}(\mathbf{B}_{s-k+1}, \ldots, \mathbf{B}_s),$$

and hence $\{\mathbf{B}_1, \ldots, \mathbf{B}_s\}$ is a linearly dependent set by Proposition 5.2.3, contrary to our hypothesis. Therefore $s \leq k$, as required. $\square$

**THEOREM 6.2.3:** *A finite-dimensional vector space $\mathcal{V}$ has a basis. Any two bases contain the same number of elements.*

**PROOF:** We proved the first statement in Theorem 6.1.1. To prove the second statement, suppose that

$$\begin{aligned} E &= \{\mathbf{A}_1, \ldots, \mathbf{A}_n\}, \\ F &= \{\mathbf{B}_1, \ldots, \mathbf{B}_m\} \end{aligned}$$

are bases for $\mathcal{V}$. Then $F$ is a linearly independent set in $\mathcal{L}(E) = \mathcal{V}$, so that $m \leq n$ by Proposition 6.2.2. Likewise, reversing the roles of $E$ and $F$, $E$ is a linearly independent set in $\mathcal{V} = \mathcal{L}(F)$, so $n \leq m$, therefore $m = n$. $\square$

**EXAMPLE 1:** $\mathcal{P}(\mathbb{R})$ is not finite-dimensional.

To see this consider the set $F = \left\{1, x, x^2, \ldots, x^n, \ldots\right\}$ of vectors in $\mathcal{P}(\mathbb{R})$. This set is linearly independent. If $\mathcal{P}(\mathbb{R})$ were finite-dimensional, say

$\mathcal{P}(\mathbb{R}) = \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k)$, then applying Proposition 6.2.2, we would conclude that $F$ is a finite set, which it is not. Therefore, $\mathcal{P}(\mathbb{R})$ is not finite-dimensional.

If a vector space $\mathcal{V}$ is not finite-dimensional, then we say it is **infinite-dimensional**. It is possible to combine Theorem 6.2.3 with the ideas of the preceding example to characterize finite and infinite-dimensional vector spaces in terms of the sets of linearly independent vectors that they contain.[1]

**THEOREM 6.2.4:** *A vector space $\mathcal{V}$ is finite-dimensional if and only if every linearly independent set of vectors in $\mathcal{V}$ is finite. A vector space $\mathcal{W}$ is infinite-dimensional if and only if it contains an infinite linearly independent set of vectors.*

**PROOF:** If $\mathcal{V}$ is finite-dimensional, then proposition 6.2.2 says that every linearly independent set of vectors in $\mathcal{V}$ is finite. On the other hand, suppose that every linearly independent set in $\mathcal{V}$ is finite, but the vector space $\mathcal{V}$ is not finite-dimensional. (Remember that this means that $\mathcal{V}$ is not spanned by any finite set of vectors in $\mathcal{V}$.) Let $\mathbf{A}_1 \neq \mathbf{0} \in \mathcal{V}$. Then $\{\mathbf{A}_1\}$ is a linearly independent set of vectors. Since $\mathcal{V}$ is not finite-dimensional, $\mathcal{L}(\mathbf{A}_1) \neq \mathcal{V}$. Therefore, we may select a vector $\mathbf{A}_2$ in $\mathcal{V}$ that is not in $\mathcal{L}(\mathbf{A}_1)$. From Proposition 6.2.1 it follows that $\{\mathbf{A}_1, \mathbf{A}_2\}$ is a linearly independent set. Let us repeat this process. In this way we obtain vectors $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3, \ldots$ such that $\mathbf{A}_{i+1} \notin \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_i)$, and hence by Proposition 6.2.1 the infinite set $\{\mathbf{A}_1, \mathbf{A}_2, \ldots\}$ of vectors in $\mathcal{V}$ is linearly independent. This is a contradiction of the fact that every linearly independent set of vectors in $\mathcal{V}$ is finite. Therefore, the assumption that $\mathcal{V}$ is not finite-dimensional must be false, so $\mathcal{V}$ is finite-dimensional. $\square$

**EXAMPLE 2:** If $S$ is an infinite set, then $\mathrm{Fun}(S)$ is infinite-dimensional.

To see this, notice that in Example 6 of Chapter 5 it was shown that the set $X = \left\{\chi_s \mid s \in S\right\}$ of characteristic functions is a linearly independent set of vectors in $\mathrm{Fun}(S)$. If $S$ is infinite, this set is infinite.

Theorem 6.2.4 says that the concept of finite-dimensionality, which we defined in terms of spanning properties of vectors, is equivalent to certain linear independence properties of vectors. This is one reason

[1] The definition of finite-dimensional and infinite-dimensional vector spaces depends on spanning properties of vectors; thus the following results provide an unexpected connection between these two concepts.

that bases are so important: They combine the spanning and linear independence concepts.

**DEFINITION:** *Let $\mathcal{V}$ be a finite-dimensional vector space. Then the number of vectors in any basis for $\mathcal{V}$ is called the* **dimension** *of $\mathcal{V}$ and is denoted by* $\dim(\mathcal{V})$.

**EXAMPLE 3:** $\dim(\mathbb{R}^k) = k$.

We have already seen that the set $E = \{\mathbf{E}_1, \ldots, \mathbf{E}_k\}$ is a basis for $\mathbb{R}^k$.

**EXAMPLE 4:** $\dim(\mathcal{P}_n)(\mathbb{R}) = n + 1$.

The set $E = \left\{1, x, x^2, \ldots, x^n\right\}$ is a basis for $\mathcal{P}_n(\mathbb{R})$ and contains $n + 1$ elements.

**NOTATION:** *If $S$ is any set, we write $|S|$ for the number of elements of $S$ when $S$ is finite and let $|S| = \infty$ when $S$ is not finite.*

By combining Examples 3 and 2 we obtain

**EXAMPLE 5:** $\dim(\mathrm{Fun}(S)) = |S|$.

## 6.3 Using Bases

If $\mathcal{V}$ is a finite-dimensional vector space and $\mathbf{A}_1, \ldots, \mathbf{A}_n$ a fixed basis for $\mathcal{V}$, then using this basis we can represent the vectors of $\mathcal{V}$ as $n$-tuples of numbers, just as for $\mathbb{R}^n$. But be careful: The representation of a vector in one basis may have nothing obvious to do with its representation with respect to some other basis. For this reason the choice of the *best* basis for a computation can often only be made after certain maneuvers.

**THEOREM 6.3.1:** *Let $\mathcal{V}$ be a finite-dimensional vector space with basis $\mathbf{A}_1, \ldots, \mathbf{A}_n$. Then any vector $\mathbf{A} \in \mathcal{V}$ may be written uniquely as a linear combination*

$$\mathbf{A} = a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_n\mathbf{A}_n.$$

*The numbers in the sequence $(a_1, \ldots, a_n)$ are called the* **coordinates** *(or* **components***) of $\mathbf{A}$ relative to the ordered basis $\mathbf{A}_1, \ldots, \mathbf{A}_n$.*

**PROOF:** Since $\mathbf{A}_1, \ldots, \mathbf{A}_n$ is a basis for $\mathcal{V}$, they span $\mathcal{V}$, and hence there is at least one way to write $\mathbf{A}$ as a linear combination

$$\mathbf{A} = a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_n\mathbf{A}_n.$$

Suppose that

$$\mathbf{A} = b_1\mathbf{A}_1 + b_2\mathbf{A}_2 + \cdots + b_n\mathbf{A}_n$$

were another way to write $\mathbf{A}$ as a linear combination of $\mathbf{A}_1, \ldots, \mathbf{A}_n$. Then

$$\mathbf{0} = \mathbf{A} - \mathbf{A} = (a_1 - b_1)\mathbf{A}_1 + (a_2 - b_2)\mathbf{A}_2 + \cdots + (a_n - b_n)\mathbf{A}_n.$$

Since $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is a basis, it is linearly independent, and hence all of the coefficients in the preceding equation must be zero. That is,

$$a_1 - b_1 = 0, \qquad a_2 - b_2 = 0, \qquad \ldots \qquad a_n - b_n = 0,$$

and hence

$$a_1 = b_1, \qquad a_2 = b_2, \qquad \ldots \qquad a_n = b_n,$$

which establishes uniqueness. $\square$

**EXAMPLE 1:** Find the coordinates of $2 - x$ relative to the basis $\{(1 - x), (1 + x)\}$ for $\mathcal{P}_1(\mathbb{R})$.

**SOLUTION:** We have by Theorem 6.3.1

$$2 - x = a(1 - x) + b(1 + x)$$

for unique numbers $a$ and $b$. Multiplying out gives

$$2 - x = a + b + (b - a)x,$$

and equating coefficients gives

$$2 = a + b, \qquad -1 = -a + b.$$

Therefore, $b = \frac{1}{2}$ and $a = \frac{3}{2}$. So the answer is $\left(\frac{3}{2}, \frac{1}{2}\right)$.

**EXAMPLE 2:** Find the coordinates of $(1, 1, 1)$ relative to the basis $\{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ for $\mathbb{R}^3$.

**SOLUTION:** $(1, 0, 0)$. Think about this one!

**EXAMPLE 3:** Find the coordinates of $1 + x + x^2$ relative to the basis $\{1, x - 1, (x - 2)(x - 1)\}$ for $\mathcal{P}_2(\mathbb{R})$.

**SOLUTION:** We require three numbers $a, b, c$ such that

$$1 + x + x^2 = a + b(x - 1) + c(x - 2)(x - 1).$$

Multiplying out gives

$$\begin{aligned} 1 + x + x^2 &= a + b(x - 1) + c(x^2 - 3x + 2) \\ &= a + bx - b + cx^2 - 3cx + 2c \\ &= (a - b + 2c) + (b - 3c)x + cx^2, \end{aligned}$$

and so equating coefficients gives

$$1 = a - b + 2c,$$
$$1 = b - 3c,$$
$$1 = c,$$

which may be solved to give

$$c = 1, \qquad b = 4, \qquad a = 3,$$

whence

$$1 + x + x^2 = 3 \cdot 1 + 4(x - 1) + (x - 2)(x - 1),$$

so that $(3, 4, 1)$ are the coordinates of $1 + x + x^2$ relative to the basis $\{1, x - 1, (x - 2)(x - 1)\}$.

**THEOREM 6.3.2** (Basis Extension Theorem): *Let $\mathcal{V}$ be a finite-dimensional vector space and $\mathbf{A}_1, \ldots, \mathbf{A}_m$ linearly independent vectors in $\mathcal{V}$. Then there exist $n = \dim(\mathcal{V}) - m$ vectors $\mathbf{B}_1, \ldots, \mathbf{B}_n$ in $\mathcal{V}$ such that the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1, \ldots, \mathbf{B}_n\}$ is a basis for $\mathcal{V}$.*

**PROOF:** If $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m) = \mathcal{V}$, then there is nothing to prove. So we may suppose that $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m) \neq \mathcal{V}$. Let $\mathbf{B}_1$ be a vector in $\mathcal{V}$ that is not in $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m)$. Then the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1\}$ is linearly independent. For if it were linearly dependent, then by Proposition 6.2.1 some vector would be linearly dependent on the preceding ones. The vector cannot be an $\mathbf{A}_i$ because $\mathbf{A}_1, \ldots, \mathbf{A}_m$ are linearly independent. It cannot be $\mathbf{B}_1$ because we chose $\mathbf{B}_1$ such that $\mathbf{B}_1 \notin \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m)$. Therefore,

$$\{\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1\}$$

is a linearly independent set.

If $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1) = \mathcal{V}$, we are done. If not, we may repeat the preceding argument starting with the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1$. In this way we obtain a set

$$\{\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1, \mathbf{B}_2\}$$

of linearly independent vectors in $\mathcal{V}$. Again, if $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1, \mathbf{B}_2) = \mathcal{V}$, we are done, otherwise, we continue to repeat the preceding argument as often as we can. By Theorem 6.2.2 this process must stop with a set

$$\{\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1, \ldots, \mathbf{B}_k\}$$

when $m + k = \dim(\mathcal{V})$, in which case $k = n$ and

$$\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1, \ldots, \mathbf{B}_n) = \mathcal{V}$$

as required. □

**THEOREM 6.3.3 :** *Let $\mathcal{V}$ be a finite-dimensional vector space and $\mathcal{U}$ a linear subspace of $\mathcal{V}$. Then $\mathcal{U}$ is finite-dimensional and* $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$.

**PROOF :** Suppose that $\mathcal{U}$ is not finite-dimensional. Then by Theorem 6.2.4 there is an infinite linearly independent set $E$ of vectors in $\mathcal{U}$. But then $E$ is certainly an infinite linearly independent set of vectors in $\mathcal{V}$, and hence again by Theorem 6.2.4 $\mathcal{V}$ is infinite-dimensional, which is a contradiction. Therefore, $\mathcal{U}$ is finite-dimensional. Let $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ be a basis for $\mathcal{U}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ a basis for $\mathcal{V}$. Then by Theorem 6.2.2 $m = \dim(\mathcal{U}) \leq n = \dim(\mathcal{V})$. □

Very often in dealing with problems of a concrete nature, we will be working in one of the finite-dimensional vector spaces $\mathbb{R}^n$, $\mathcal{P}_n(\mathbb{R})$, or $\text{Fun}(S)$ whose dimension we already know. If we wish to check that some set $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ of vectors is a basis in one of these spaces, it turns out that we must check only *one* of the two characteristic properties, i.e., *linear independence, and spanning,* of a basis. More precisely we have:

**THEOREM 6.3.4 :** *Let $\mathcal{V}$ be a finite-dimensional vector space of dimension $n$. If the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ in $\mathcal{V}$ are linearly independent, then they are a basis.*

**PROOF :** By the basis extension theorem (Theorem 6.3.2) if $\mathbf{A}_1, \ldots, \mathbf{A}_n$ is not a basis we may find vectors $\mathbf{B}_1, \ldots, \mathbf{B}_m$ such that $\mathbf{A}_1, \ldots, \mathbf{A}_n, \mathbf{B}_1, \ldots, \mathbf{B}_m$ is a basis for $\mathcal{V}$. But then $\dim(\mathcal{V}) = m + n \neq n$, a contradiction. □

**THEOREM 6.3.5 :** *Let $\mathcal{V}$ be a finite-dimensional vector space of dimension $n$. If the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ span $\mathcal{V}$, they are a basis.*

**PROOF :** Since the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ span $\mathcal{V}$, some subset of $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is a basis for $\mathcal{V}$ by Theorem 6.1.1. By reordering the vectors we may assume that $\{\mathbf{A}_1, \ldots, \mathbf{A}_m\}$ is a basis for $\mathcal{V}$, where $m \leq n$. Then $m = \dim(\mathcal{V}) = n$ by Theorem 6.3.3, which shows that $m = n$ and hence that $\mathbf{A}_1, \ldots, \mathbf{A}_n$ is a basis for $\mathcal{V}$. □

**COROLLARY 6.3.6 :** *Let $\mathcal{V}$ be a finite-dimensional vector space of dimension $n$. Suppose that $\mathcal{U}$ is a linear subspace of $\mathcal{V}$ and* $\dim(\mathcal{U}) = n$. *Then $\mathcal{U} = \mathcal{V}$.*

**PROOF :** Since $\dim(\mathcal{U}) = n$, there is a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ for $\mathcal{U}$. By Theorem 6.3.4, $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ must be a basis for $\mathcal{V}$. Thus

$$\mathcal{U} = \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n) = \mathcal{V},$$

as required. □

**EXAMPLE 4:** The vectors $(1, 1, 1), (1, 1, 0), (1, 0, 0)$ are a basis for $\mathbb{R}^3$.

There are three vectors and $\mathbb{R}^3$ has dimension 3, so by theorem 6.3.4 we need only check that the three vectors are linearly independent. If

$$a_1(1, 1, 1) + a_2(1, 1, 0) + a_3(1, 0, 0) = (0, 0, 0),$$

then

$$(a_1 + a_2 + a_3, a_1 + a_2, a_1) = (0, 0, 0),$$

so

$$\begin{aligned} a_1 + a_2 + a_3 &= 0, \\ a_1 + a_2 &= 0, \\ a_1 &= 0, \end{aligned}$$

and $a_1 = a_2 = a_3 = 0$. So they are linearly independent.

**EXAMPLE 5:** Are the vectors $(1, 1, -2), (0, 3, -3)$ a basis for the subspace

$$\mathcal{V} = \left\{(x, y, z) \,|\, x + y + z = 0\right\}$$

of $\mathbb{R}^3$?

**SOLUTION:** We already know, by Example 6 of Chapter 5 Section 5.1 that $\dim(\mathcal{V}) = 2$. So it will suffice to check

(1). $(1, 1, -2)$ and $(0, 3, -3)$ belong to $\mathcal{V}$

(2) $(1, 1, -2)$ and $(0, 3, -3)$ are linearly independent.

Condition (1) is easy, since

$$1 + 1 - 2 = 0 \quad \text{and} \quad 0 + 3 - 3 = 0.$$

To check (2), suppose

$$a(1, 1, -2) + b(0, 3, -3) = 0$$

is a linear relation. Then

$$\begin{aligned} (0, 0, 0) &= (a, a, -2a) + (0, 3b, -3b) \\ &= (a, a + 3b, -2a - 3b), \end{aligned}$$

so equating coefficients gives

$$\begin{aligned} 0 &= a, \\ 0 &= a + 3b, \\ 0 &= -2a - 3b, \end{aligned}$$

and $a = 0, b = 0$, as required.

**EXAMPLE 6:** For what values of $r$ are the vectors $(r, 1, 1), (1, r, 1), (1, 1, r)$ a basis for $\mathbb{R}^3$?

**SOLUTION:** Since $\dim(\mathbb{R}^3) = 3$, it will suffice to determine when the vectors in question are linearly independent. So suppose that

$$a(r, 1, 1) + b(1, r, 1) + c(1, 1, r) = 0$$

is a linear relation. Then

$$(ar + b + c, a + br + c, a + b + cr) = \mathbf{0} = (0, 0, 0),$$

so taking coordinates gives

$$ar + b + c = 0,$$
$$a + br + c = 0,$$
$$a + b + cr = 0.$$

From the last equation we get

$$(*) \qquad cr = -(a + b).$$

Multiplying the second equation by $r$ and substituting gives

$$\begin{aligned} 0 &= ar + br^2 - (a + b) = a(r - 1) + b(r^2 - 1) \\ &= (r - 1)[a + b(r + 1)]. \end{aligned}$$

So if $r - 1 \neq 0$, we may divide by $r - 1$ and solve for $a$ to get

$$(**) \qquad a = -(r + 1)b.$$

Multiplying the first equation by $r$ and substituting in $(*)$ and $(**)$, we obtain

$$\begin{aligned} 0 &= [-(r + 1)b]r^2 + br - (a + b) \\ &= -r^2(r + 1)b + rb - [-(r + 1)b + b] \\ &= [-r^2(r + 1) + r + (r + 1) - 1]b \\ &= (-r^3 - r^2 + r + r)b = -(r^3 + r^2 - 2r)b \\ &= -r(r^2 + r - 2)b = -r(r + 2)(r - 1)b. \end{aligned}$$

So assuming $r \neq 0, -2, 1$ we may divide by $-r(r + 2)(r - 1)$ and conclude that $b = 0$, whence $a = 0$ by $(**)$ and $c = 0$ by $(*)$. Therefore, the vectors are independent when $r \neq 0, 1, -2$. For $r = 0$, the original system simplifies to

$$0 = b + c,$$
$$0 = a + c,$$
$$0 = a + b.$$

Substituting the first equation into the second gives

$$0 = a - b$$

and adding to the third gives

$$0 = 2a,$$

giving $a = 0$ and $b = 0$, $c = 0$, so again the vectors are independent. For $r = 1$, the three vectors are the same vector three times, so they cannot be a basis for $\mathbb{R}^3$. And finally, for $r = -2$, there is the linear relation

$$(-2, 1, 1) + (1, -2, 1) + (1, 1, -2) = (0, 0, 0),$$

so the vectors are not linearly dependent. In conclusion, the vectors $(r, 1, 1), (1, r, 1), (1, 1, r)$ are a basis for $\mathbb{R}^3$ if and only if $r \neq 1, -2$.

## 6.4 Exercises

**1.** Which of the following sets of vectors are bases for $\mathbb{R}^3$?

$$E = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}.$$
$$F = \{(1, 0, 1), (0, 1, 0), (1, 1, 1)\}.$$
$$G = \{(-1, 1, 1), (1, -1, 1), (1, 1, -1)\}.$$
$$H = \{(1, -1, 0), (1, 0, -1), (-1, 0, 1)\}.$$

**2.** Which of the following sets of vectors are bases for $\mathcal{P}_2(\mathbb{R})$?

$$E = \left\{1, 1 + x, 1 + x + x^2\right\}.$$
$$F = \{1, (x - 1), (x - 1)(x - 2)\}.$$
$$G = \left\{x^2 + x + 1, x^2 - x + 1, x^2 - x - 1\right\}.$$
$$H = \left\{x^2 + x + x^2 + 1, x + 1\right\}.$$

**3.** Find a basis for each of the following subspaces of $\mathbb{R}^3$.

$$\mathcal{U} = \left\{(x, y, z) \mid x - z = 0\right\}.$$
$$\mathcal{S} = \left\{(x, y, z) \mid x + y + z = 0\right\}.$$
$$\mathcal{T} = \left\{(x, y, z) \mid x - y + z = 0\right\}.$$
$$\mathcal{W} = \left\{(x, y, z) \mid x = 0 \text{ and } y + z = 0\right\}.$$

**4.** Find a basis for each of the folowing the subspaces of $\mathcal{P}_3(\mathbb{R})$:

$$\mathcal{V} = \left\{p(x) \mid p(0) = 0\right\}.$$

$$\mathcal{S} = \left\{p(x) \mid \frac{d}{dx}p(x) = 0\right\}.$$

$$\mathcal{T} = \left\{p(x) \mid p(x) = a_0 + a_1x + a_3x^3\right\}.$$

$$\mathcal{W} = \left\{p(x) \mid p(-x) = p(x)\right\}.$$

**5.** Calculate the dimension of each of the subspaces in the preceding two exercises.

**6.** Let $\ell_\infty$ denote the vector space of bounded sequences (see Chapter 3 Exercise 16). Which of the following sets of vectors are bases for $\ell_\infty$?

$$E = \left\{(1, 0, \ldots, 0, \ldots), (0, 1, 0, \ldots, 0, \ldots), \ldots, (0, , \ldots, 0, 1, 0, \ldots), \ldots\right\}.$$

$$F = \left\{(1, 1, , \ldots, 1, \ldots), (0, 1, \ldots, 1, \ldots), \ldots, (0, 0, \ldots, 0, 1, 1, \ldots), \ldots\right\}.$$

$$F = \left\{(1, 1, \ldots, 1, \ldots), (2, 1, \ldots, 1, \ldots), \ldots, (n, 1, \ldots, 1, \ldots), \ldots\right\}.$$

$$G = \left\{(1, 0, \ldots, 0, \ldots), (0, 1/2, 0, \ldots, 0, \ldots), \ldots, (0, \ldots, 1/n, 0, \ldots), \ldots\right\}.$$

**7.** For each of the following subsets of $\ell_\infty$ show that the subset is a linear subspace and find a basis.

$\mathcal{A} = \left\{(a_0, a_1, \ldots, a_n, \ldots) \mid a_i = a_j \ \forall\ i, j \in \mathbb{N}_0\right\}$,

$\mathcal{S}_{n_0} = \left\{(a_0, a_1, \ldots, a_n, \ldots) \mid a_i = 0 \ \forall\ i > n_0\right\}$, where $n_0 \in \mathbb{N}_0$ is a fixed nonnegative integer.

**8.** Suppose that $\mathcal{V}$ is a finite-dimensional vector space and $\mathcal{S}$ is a linear subspace of $\mathcal{V}$. Show that there exists a linear subspace $\mathcal{T}$ of $\mathcal{V}$ such that $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$ and $\mathcal{S} + \mathcal{T} = \mathcal{V}$. (Hint: Study the proof of the basis extension theorem.)

**9.** Suppose that $\mathcal{S}$ and $\mathcal{T}$ are subspaces of a finite-dimensional vector space $\mathcal{V}$, and $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$. Show that

$$\dim(\mathcal{S} + \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$$

**10.** Suppose that $\mathcal{S}$ and $\mathcal{T}$ are subspaces of a finite-dimensional vector space $\mathcal{V}$. Show that

$$\dim(\mathcal{S} + \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T}).$$

(**HINT:** Choose a basis for $\mathcal{S} \cap \mathcal{T}$, extend it to a basis for $\mathcal{S}$, and extend it to a basis for $\mathcal{T}$. Show that the result spans $\mathcal{V}$ and count the number of vectors that occur twice.)

**11.** Suppose that $\mathcal{S}_1, \ldots, \mathcal{S}_k$ are subspaces of a finite-dimensional vector space $\mathcal{V}$. Find a formula for the dimension of $\mathcal{S}_1 + \cdots + \mathcal{S}_k$. (**HINT:** Try the case $k = 3$.)

**12.** What is the dimension of $\mathbb{C}$ as a vector space over $\mathbb{R}$? What is the dimension of $\mathbb{C}$ as a vector space over $\mathbb{C}$?

**13.** Let $\mathcal{S} = \{(a, 0, 0)\}$ and $\mathcal{T} = \{(0, b, b)\}$, where $a$ and $b$ are real numbers. Show $\mathcal{S}$ and $\mathcal{T}$ are subspaces of $\mathbb{R}^2$. Find a basis for $\mathcal{S} + \mathcal{T}$.

**14.** Let $\mathcal{S}$ be the subspace of $\mathbb{R}^3$ given by

$$\mathcal{S} = \left\{(x, y, z) \mid y - z = 0\right\}.$$

Find a subspace $\mathcal{T}$ of $\mathbb{R}^3$ such that $\mathcal{S} \cap \mathcal{T} = \left\{\mathbf{0}\right\}$ and $\mathcal{S} + \mathcal{T} = \mathbb{R}^3$.

**15.** Under what conditions on the number $a$ will the vectors

$$(a, 1, 0), (1, a, 1), (0, 1, a)$$

be a basis for $\mathbb{R}^3$?

**16.** Let $\mathbf{A}_1, \ldots, \mathbf{A}_n$ be a basis for $\mathbb{R}^n$. Set

$$\mathcal{S} = \left\{ \sum_{i=1}^{n} a_i \mathbf{A}_i \mid a_1, \ldots, a_n \in \mathbb{R} \text{ and } a_2 = \cdots = a_n \right\},$$

$$\mathcal{T} = \left\{ \sum_{i=1}^{n} a_i \mathbf{A}_i \mid a_1, \ldots, a_n \in \mathbb{R} \text{ and } \sum_{i=1}^{n} a_i = 0 \right\}.$$

Show that $\mathcal{S}$ and $\mathcal{T}$ are vector subspaces and find bases for $\mathcal{S}$, $\mathcal{T}$, $\mathcal{S} \cap \mathcal{T}$, and $\mathcal{S} + \mathcal{T}$.

**17.** Let $\mathbf{A}_1, \ldots, \mathbf{A}_n$ be vectors in $\mathcal{V}$. Suppose that $n = \dim(\mathcal{V})$. Show that $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are linearly independent if and only if $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$ has dimension $n$.

**18.** The equation $y = 3x$ defines a straight line in the plane. Show that if $\mathbf{A}, \mathbf{B}$ are points on this line then $\vec{\mathbf{A}}, \vec{\mathbf{B}}$ are linearly dependent vectors.

**19.** Let $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ be four distinct points on a plane $\Pi$. Show that the vectors $\overrightarrow{\mathbf{AB}}, \overrightarrow{\mathbf{AC}}, \overrightarrow{\mathbf{AD}}$ form a set of linearly dependent vectors.

**20.** Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be three points in a space. Show that $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are not colinear (i.e., they all lie on the same line) if and only if $\left\{\overrightarrow{\mathbf{AB}}, \overrightarrow{\mathbf{AC}}\right\}$ is a set of linearly independent vectors.

**21.** Let $S = \{u, v, w\}$. Show that the functions $f, g, h \in \text{Fun}(S)$ defined by

$$\begin{array}{lll} f(u) = 1, & g(u) = 1, & h(u) = 1, \\ f(v) = 1, & g(v) = 1, & h(v) = 0, \\ f(w) = 1, & g(w) = 0, & h(w) = 0, \end{array}$$

are a basis for Fun($S$). Find the coordinates of the characteristic functions $\chi_u, \chi_v, \chi_w$ relative to this basis.

**22.** Let $S$ be a finite set and $\mathcal{V}$ a finite-dimensional vector space. Show that Fun($S$, $\mathcal{V}$) is finite-dimensional and find its dimension.

**23.** Let $S$ be a finite set and $T \subseteq S$. Find the dimension of Fun($S$, $T$).

**24.** Let $\mathcal{V}$ be a finite-dimensional vector space over $\mathbb{C}$. By forgetting a part of the structure, $\mathcal{V}$ becomes a real vector space. Show that $\dim_{\mathbb{R}}(\mathcal{V}) = 2\dim_{\mathbb{C}}(\mathcal{V})$, where $\dim_{\mathbb{C}}(\mathcal{V})$ is the dimension of $\mathcal{V}$ as a complex vector space, and $\dim_{\mathbb{R}}(\mathcal{V})$ is its dimension regarded as a real vector space.

**25.** Let $n \in \mathbb{N}$ be an integer and $t_0, \ldots, t_n \in \mathbb{R}$ real numbers that are pairwise distinct. Show that the polynomials

$$\Phi_j(x) = \prod_{\substack{i=0 \\ i \neq j}}^{n} \frac{x - t_i}{t_j - t_i} \qquad j = 0, 1, \ldots, n$$

are a basis for $\mathcal{P}_n(\mathbb{R})$. (The polynomials $\Phi_0(x), \ldots, \Phi_n(x)$ are called **Lagrange interpolation polynomials** in approximation theory and the points $a_0, \ldots, a_n$ are called **support points**.) Show that $\Phi_i(t_i) = 1$ for $i = 0, \ldots, n$.

**26.** Let $\mathcal{V}$ be an infinite-dimensional vector space. Show that there exist vectors $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n, \ldots \in \mathcal{V}$ such that any finite subset of the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_n, \ldots\}$ is linearly independent.

# Chapter 7
# The Elements of Vector Spaces: A Summing Up

The purpose of this chapter is twofold: to work out a number of numerical examples to illustrate and illuminate the portion of the theory of vector spaces we have developed so far, and to extend the theory in several different directions.

## 7.1 Numerical Examples

This section is devoted to numerical examples.

EXAMPLE 1: Determine whether or not the vector $\mathbf{A} = (1, -2, 0, 3) \in \mathbb{R}^4$ is a linear combination of the vectors

$$\mathbf{B}_1 = (3, 9, -4, -2), \qquad \mathbf{B}_2 = (2, 3, 0, -1), \qquad \mathbf{B}_3 = (2, -1, 2, 1).$$

That is, does $\mathbf{A}$ belong to the linear span of $\mathbf{B}_1, \mathbf{B}_2$, and $\mathbf{B}_3$, or, in symbols, is $\mathbf{A} \in \mathcal{L}(\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3)$?

SOLUTION: $\mathbf{A} \in \mathcal{L}(\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3)$, if and only if there are numbers $b_1, b_2, b_3$ such that

$$\mathbf{A} = b_1\mathbf{B}_1 + b_2\mathbf{B}_2 + b_3\mathbf{B}_3.$$

So we check if such numbers can exist, i.e., if there are numbers $b_1, b_2, b_3$ such that

$$\begin{aligned}(1, -2, 0, 3) &= b_1(3, 9, -4, -2) + b_2(2, 3, 0, -1) + b_3(2, -1, 2, 1)\\ &= (3b_1 + 2b_2 + 2b_3, 9b_1 + 3b_2 - b_3, -4b_1 + 2b_3, -2b_1 - b_2 + b_3).\end{aligned}$$

Therefore, equating components gives

$$\begin{array}{lrcl}
(1) & 1 & = & 3b_1 + 2b_2 + 2b_3, \\
(2) & -2 & = & 9b_1 + 3b_2 - b_3, \\
(3) & 0 & = & -4b_1 \phantom{+ 2b_2} + 2b_3, \\
(4) & 3 & = & -2b_1 - b_2 + b_3.
\end{array}$$

Add (4) to (2) to get

(5) $$1 = 7b_1 + 2b_2.$$

Equation (5) yields

$$2b_2 = 1 - 7b_1,$$

(6) $$b_2 = \frac{1 - 7b_1}{2}.$$

Equation (3) gives

(7) $$\begin{aligned} 2b_3 &= 4b_1, \\ b_3 &= 2b_1. \end{aligned}$$

Putting (6) and (7) into (4) gives in succession the equations

$$\begin{aligned}
3 &= -2b_1 + \frac{7b_1 - 1}{2} + 2b_1, \\
6 &= -4b_1 + 7b_1 - 1 + 4b_1, \\
7 &= 7b_1, \\
1 &= b_1.
\end{aligned}$$

So

$$b_1 = 1, \qquad b_2 = -3, \qquad b_3 = 2,$$

and hence

$$\mathbf{A} = \mathbf{B}_1 - 3\mathbf{B}_2 + 2\mathbf{B}_3.$$

So $\mathbf{A}$ is a linear combination of $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3$ and hence $\mathbf{A}$ belongs to $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3)$.

**EXAMPLE 2:** Determine whether or not the vector

$$\mathbf{A} = 1 + x - 2x^2 + 4x^3$$

belongs to the subspace of $\mathcal{P}_3(\mathbb{R})$ spanned by the vectors

$$\mathbf{B}_1 = 1 - x, \qquad \mathbf{B}_2 = 1 - x^2, \qquad \mathbf{B}_3 = 1 - x^3.$$

**SOLUTION:** Note that $\mathbf{A}$ belongs to $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3)$ if and only if there are numbers $b_1$, $b_2$, $b_3$ such that

$$\mathbf{A} = b_1\mathbf{B}_1 + b_2\mathbf{B}_2 + b_3\mathbf{B}_3,$$

that is ,

$$\begin{aligned} 1 + x - 2x^2 + 4x^3 &= b_1(1-x) + b_2(1-x^2) + b_3(1-x^3) \\ &= b_1 + b_2 + b_3 - b_1x - b_2x^2 - b_3x^3, \end{aligned}$$

and hence

$$\begin{aligned} 1 &= b_1 + b_2 + b_3, \\ 1 &= -b_1, \\ -2 &= -b_2, \\ 4 &= -b_3. \end{aligned}$$

Therefore the only possibility is

$$b_1 = -1, \qquad b_2 = 2, \qquad b_3 = -4.$$

But then we receive the impossible equation

$$1 = -1 + 2 - 4 = -3.$$

Therefore, there are no numbers $b_1$, $b_2$, $b_3$ such that

$$\mathbf{A} = b_1\mathbf{B}_1 + b_2\mathbf{B}_2 + b_3\mathbf{B}_3,$$

and hence $\mathbf{A}$ does not belong to the subspace spanned by $\mathbf{B}_1$, $\mathbf{B}_2$, and $\mathbf{B}_3$.

**EXAMPLE 3:** In $\mathbb{R}^4$ let $\mathcal{S}$ and $\mathcal{T}$ be the subspaces defined by

$$\mathcal{S} = \left\{(a_1, a_2, a_3, a_4) \mid a_1 - a_2 + a_3 - a_4 = 0\right\},$$

$$\mathcal{T} = \left\{(a_1, a_2, a_3, a_4) \mid a_1 + a_2 + a_3 + a_4 = 0\right\}.$$

Find a basis for $\mathcal{S} \cap \mathcal{T}$.

**SOLUTION:** A vector $\mathbf{A} = (a_1, a_2, a_3, a_4)$ in $\mathbb{R}^4$ belongs to $\mathcal{S} \cap \mathcal{T}$ if and only if

$$\begin{aligned} a_1 - a_2 + a_3 - a_4 &= 0, \\ a_1 + a_2 + a_3 + a_4 &= 0, \end{aligned}$$

or

$$\begin{aligned} a_1 + a_3 &= 0, \\ a_2 + a_4 &= 0, \end{aligned}$$

or

$$a_1 = -a_3,$$
$$a_2 = -a_4.$$

Let

$$\mathbf{B}_1 = (1, 0, -1, 0),$$
$$\mathbf{B}_2 = (0, -1, 0, 1).$$

Note that $\mathbf{B}_1$ is a vector in $\mathcal{S} \cap \mathcal{T}$ because

$$1 = -(-1),$$
$$0 = 0.$$

and $\mathbf{B}_2$ is a vector in $\mathcal{S} \cap \mathcal{T}$ because

$$0 = -0,$$
$$-1 = -(1).$$

We claim that $\{\mathbf{B}_1, \mathbf{B}_2\}$ is a basis for $\mathcal{S} \cap \mathcal{T}$. To prove this we must demonstrate two facts: first; that $\{\mathbf{B}_1, \mathbf{B}_2\}$ is a linearly independent set of vectors, and second; that $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2) = \mathcal{S} \cap \mathcal{T}$.

$\mathbf{B}_1, \mathbf{B}_2$ *are linearly independent*: Suppose to the contrary that $\{\mathbf{B}_1, \mathbf{B}_2\}$ is linearly dependent. Then there are numbers $b_1, b_2$, not both zero, such that

$$\mathbf{0} = b_1\mathbf{B}_1 + b_2\mathbf{B}_2,$$

or equivalently,

$$\begin{aligned}(0, 0, 0, 0) &= b_1(1, 0, -1, 0) + b_2(0, -1, 0, 1) \\ &= (b_1, -b_2, -b_1, b_2),\end{aligned}$$

and hence

$$0 = b_1, \qquad 0 = -b_2,$$

so $b_1 = 0 = b_2$, a contradiction. Hence $\mathbf{B}_1, \mathbf{B}_2$ must be linearly independent.

$\mathbf{B}_1, \mathbf{B}_2$ *span* $\mathcal{S} \cap \mathcal{T}$: Suppose that $\mathbf{A} \in \mathcal{S} \cap \mathcal{T}$. Then as we have seen,

$$\mathbf{A} = (a_1, a_2, -a_1, -a_2).$$

Therefore,

$$\mathbf{A} = a_1(1, 0, -1, 0) - a_2(0, -1, 0, 1),$$

and hence

$$\mathbf{A} = a_1\mathbf{B}_1 - a_2\mathbf{B}_2,$$

so $\mathbf{A} \in \mathcal{L}(\mathbf{B}_1, \mathbf{B}_2)$. Therefore, $\mathcal{S} \cap \mathcal{T}$ is contained in $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2)$. But since $\mathbf{B}_1, \mathbf{B}_2$ belong to $\mathcal{S} \cap \mathcal{T}$, $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2)$ is contained in $\mathcal{S} \cap \mathcal{T}$. The only conclusion possible is therefore that $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2) = \mathcal{S} \cap \mathcal{T}$.

Hence we have found that the vectors

$$\mathbf{B}_1 = (1, 0, -1, 0), \qquad \mathbf{B}_2 = (0, -1, 0, 1)$$

are a basis for $\mathcal{S} \cap \mathcal{T}$. Note that $\mathcal{S} \cap \mathcal{T}$ therefore has dimension 2.

**EXAMPLE 4:** Show that the vectors

$$\mathbf{A}_1 = (1, -1, -1, 1), \qquad \mathbf{A}_2 = (1, -2, -2, 1), \qquad \mathbf{A}_3 = (0, 1, 1, 0)$$

and

$$\mathbf{B}_1 = (1, 0, 0, 1), \qquad \mathbf{B}_2 = (0, -1, -1, 0)$$

span the same linear subspace of $\mathbb{R}^4$. Find a basis for this subspace.

**SOLUTION:** If you think about it for a moment, you will see that we must prove the following facts:

$$\begin{aligned}
\mathbf{A}_1 &\in \mathcal{L}(\mathbf{B}_1, \mathbf{B}_2),\\
\mathbf{A}_2 &\in \mathcal{L}(\mathbf{B}_1, \mathbf{B}_2),\\
\mathbf{A}_3 &\in \mathcal{L}(\mathbf{B}_1, \mathbf{B}_2),\\
\mathbf{B}_1 &\in \mathcal{L}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3),\\
\mathbf{B}_2 &\in \mathcal{L}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3).
\end{aligned}$$

To do this, we note the following

$$\begin{aligned}
\mathbf{A}_1 &= \mathbf{B}_1 + \mathbf{B}_2,\\
\mathbf{A}_2 &= \mathbf{B}_1 + 2\mathbf{B}_2,\\
\mathbf{A}_3 &= -\mathbf{B}_2,\\
\mathbf{B}_1 &= \mathbf{A}_1 + \mathbf{A}_3,\\
\mathbf{B}_2 &= -\mathbf{A}_3.
\end{aligned}$$

These equations are obtained as in Example 1 in this section.

If we next note that $\{\mathbf{B}_1, \mathbf{B}_2\}$ is linearly independent, we see that $\{\mathbf{B}_1, \mathbf{B}_2\}$ is a basis for $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2) = \mathcal{L}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3)$, and hence we have shown
$2 = \dim(\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2)) = \dim(\mathcal{L}(\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3))$.

**EXAMPLE 5:** Let $\mathcal{S}$ and $\mathcal{T}$ be the subspaces of $\mathbb{R}^3$ defined by

$$\mathcal{S} = \left\{(x, y, z) \mid x = y = z\right\},$$
$$\mathcal{T} = \left\{(x, y, z) \mid x = 0\right\}.$$

Show that $\mathcal{S} + \mathcal{T} = \mathbb{R}^3$.

**Solution:** We must show that any vector $\mathbf{A}$ in $\mathbb{R}^3$ may be written in the form $\mathbf{A} = \mathbf{S} + \mathbf{T}$, where $\mathbf{S}$ is a vector in $\mathcal{S}$ and $\mathbf{T}$ is a vector in $\mathcal{T}$. So let $\mathbf{A} = (a_1, a_2, a_3)$ and note that

$$\mathbf{A} = (a_1, a_1, a_1) + (0, a_2 - a_1, a_3 - a_1).$$

The vector $(a_1, a_1, a_1)$ belongs to $\mathcal{S}$ and the vector $(0, a_2 - a_1, a_3 - a_1)$ belongs to $\mathcal{T}$. Therefore, $\mathbf{A} = \mathbf{S} + \mathbf{T}$, as required.

**Example 6:** Show that the polynomials

$$\mathbf{A}_1 = 1, \qquad \mathbf{A}_2 = t - 1, \qquad \mathbf{A}_3 = (t-1)^2 = t^2 - 2t + 1$$

form a basis for $\mathcal{P}_2(\mathbb{R})$. Find the coordinates of the vector

$$\mathbf{B} = 2t^2 - 5t + 6$$

relative to this ordered basis.

**Solution:** To show that $\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3\}$ is a basis, we first apply Theorem 6.3.4 to conclude that it suffices to show that $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ are linearly independent. This is easy. For if

$$\mathbf{0} = a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + a_3\mathbf{A}_3 = (a_1 - a_2 + a_3) + (a_2 - 2a_3)t + a_3t^2,$$

then

$$\begin{aligned} a_3 &= 0, \\ a_2 - 2a_3 &= 0, \\ a_1 - a_2 + a_3 &= 0, \end{aligned}$$

or

$$a_3 = 0, \qquad a_2 = 0, \qquad a_1 = 0,$$

so $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ cannot be linearly dependent. Hence they form a basis for $\mathcal{P}_2(\mathbb{R})$ by Theorem 6.3.4 as $\mathcal{P}_2(\mathbb{R})$ is 3-dimensional.

To find the coordinates of $\mathbf{B}$ we write

$$\mathbf{B} = b_1\mathbf{A}_1 + b_2\mathbf{A}_2 + b_3\mathbf{A}_3,$$

or equivalently,

$$2t^2 - 5t + 6 = b_1 + b_2(t-1) + b_3(t-1)^2.$$

Rearranging terms we get

$$6 - 5t + 2t^2 = b_1 - b_2 + b_3 + (b_2 - 2b_3)t + b_3t^2,$$

so

$$\begin{aligned} 6 &= b_1 - b_2 + b_3, \\ -5 &= b_2 - 2b_3, \\ 2 &= b_3. \end{aligned}$$

Therefore,

$$b_3 = 2,$$
$$b_2 = -5 + 2b_3 = -1,$$
$$b_1 = 6 + b_2 - b_3 = 6 - 1 - 2 = 3.$$

Therefore, the coordinates of $6 - 5t + 2t^2$ relative to the ordered basis $\left\{1, (1-t), (1-t)^2\right\}$ are $(3, -1, 2)$.

**EXAMPLE 7:** Find the dimension of the subspace of $\mathbb{R}^3$ given by

$$\mathcal{V} = \left\{(x, y, z) \mid x + 2y + z = 0, x + y + 2z = 0, 2x + y + z = 0\right\}.$$

**SOLUTION:** A vector $\mathbf{A} = (x, y, z)$ belongs to $\mathcal{V}$ if and only if

$$\begin{aligned} x + 2y + z &= 0, \\ x + y + 2z &= 0, \\ 2x + y + z &= 0, \end{aligned}$$

or

$$\begin{aligned} -x + y \quad &= 0, \\ -x + \quad z &= 0, \\ 2x + y - z &= 0, \end{aligned}$$

or

$$y = x, \qquad z = x, \qquad 2x + x - x = 0,$$

or

$$y = x, \qquad z = x, \qquad 2x = 0,$$

or

$$y = 0, \qquad z = 0, \qquad x = 0,$$

so $\mathcal{V} = \{\mathbf{0}\}$ and $\dim(\mathcal{V}) = 0$.

**EXAMPLE 8:** Consider the set $S = \{u, v, w\}$ and let

$$\mathcal{V} = \left\{f \in \mathrm{Fun}(S) \mid f(u) = f(w)\right\}.$$

Show that $\mathcal{V}$ is a subspace of Fun($S$) and find its dimension.

**SOLUTION:** To check that $\mathcal{V}$ is a subspace note first of all that it is nonempty, since $\mathbf{0} \in \mathcal{V}$. If $f, g \in \mathcal{V}$ then

$$(f + g)(u) = f(u) + g(u) = f(w) + g(w) = (f + g)(w),$$

so $f + g \in \mathcal{V}$. Likewise for a number $r$

$$(rf)(u) = rf(u) = rf(w) = (rf)(w),$$

so $rf \in \mathcal{V}$, and therefore $\mathcal{V}$ is a subspace of Fun($S$).

To find the dimension of $\mathcal{V}$, recall from Section 6.2 Example 5 that $\dim(\text{Fun}(S)) = 3$. Moreover, $\mathcal{V} \neq \text{Fun}(S)$ because $\chi_u \notin \mathcal{V}$. Therefore, $\dim(\mathcal{V}) < 3$; i.e., $\dim(\mathcal{V}) = 0, 1$, or $2$. A little experimentation shows that the two vectors $\chi_u - \chi_w, \chi_v$ belong to $\mathcal{V}$. They are also linearly independent, for if

$$a(\chi_u - \chi_w) + b\chi_v = \mathbf{0}$$

is a linear relation, then

$$a\chi_u + b\chi_v - a\chi_w = \mathbf{0},$$

and evaluating at $u$ and $v$ gives

$$0 = a\chi_u(u) + b\chi_v(u) - a\chi_w(u) = a,$$

$$0 = a\chi_u(v) + b\chi_v(v) - a\chi_w(v) = b,$$

so $a = 0$, $b = 0$, as required. Therefore, $\dim(\mathcal{V}) \geq 2$, which since $\dim(\mathcal{V}) \leq 2$, shows that $\dim(\mathcal{V}) = 2$.

## 7.2 Exercises

**1.** Let $\mathbf{A} = (1, 0, 1)$, $\mathbf{B} = (-1, 1, 0)$, $\mathbf{C} = (0, 1, 1)$ be three vectors of $\mathbb{R}^3$. Show that $\mathcal{L}(\mathbf{A}, \mathbf{B}) = \mathcal{L}(\mathbf{B}, \mathbf{C})$.

**2.** $\{\mathbf{B}_1, \mathbf{B}_2\}$ is a basis for $\mathcal{L}(\mathbf{B}_1, \mathbf{B}_2)$ in Eexample 4 of this chapter. Find the coordinates (in components) of $\mathbf{A}_1$ relative to this basis $\{\mathbf{B}_1, \mathbf{B}_2\}$. Do the same problem for $\mathbf{A}_2$ and $\mathbf{A}_3$.

**3.** Let $\mathbf{P} = (a, b)$ be a vector in $\mathbb{R}^2$.
  (1) Show that $a, b$ are the coordinates of $\mathbf{P}$ relative to the basis $\mathbf{E}_1 = (1, 0)$, $\mathbf{E}_2 = (0, 1)$.
  (2) Find the coordinates of $\mathbf{P} = (a, b)$ relative to the basis $\mathbf{E}_1 = (1, 0)$ and $\mathbf{F}_2 = (0, 2)$.
  (3) Find the coordinates of $\mathbf{P}$ relative to the basis $\mathbf{G} = (1, 1)$ and $\mathbf{H} = (-1, 2)$.

**4.** Are the polynomials $\left\{x + x^3, 1 + x^2\right\}$ a set of linearly independent vectors of $\mathcal{P}_3(\mathbb{R})$? If so, is $\left\{x + x^3, 1 + x^2\right\}$ a basis for $\mathcal{P}_3(\mathbb{R})$? What is the dimension of $\mathcal{L}(x + x^3, 1 + x^2)$?

**5.** The solution space $\mathcal{S}$ of $x - 2y + 3z = 0$ is a subspace of $\mathbb{R}^3$. Show that $\dim(\mathcal{S}) = 2$. Find a basis for $\mathcal{S}$.

**6.** Let $S$ be a set. A map

$$\tau : S \longrightarrow S$$

is called an **involution** if

$$\tau(\tau(s)) = s, \forall\ s \in S.$$

Let $\tau$ be an involution on the set $S$ and define

$$\text{Fun}_+(S) = \left\{f \in \text{Fun}(S) \mid f(\tau(s)) = f(s), \forall\ s \in S\right\},$$

$$\text{Fun}_-(S) = \left\{f \in \text{Fun}(S) \mid f(\tau(s)) = -f(s), \forall\ s \in S\right\}.$$

(1) Show that $\text{Fun}_+(S)$ and $\text{Fun}_-(S)$ are subspaces of $\text{Fun}(S)$.
(2) Show that $\text{Fun}_+(S) \cap \text{Fun}_-(S) = \{\mathbf{0}\}$.
(3) Let $f \in \text{Fun}(S)$ and define

$$f_+ : S \longrightarrow \mathbb{R} \text{ by } f_+(s) = f(s) + f(\tau(s)),$$

$$f_- : S \longrightarrow \mathbb{R} \text{ by } f_-(s) = f(s) - f(\tau(s)).$$

Show that $f_+ \in \text{Fun}_+(S)$ and $f_- \in \text{Fun}_-(S)$.
(4) Show that $\text{Fun}(S) = \text{Fun}_+(S) + \text{Fun}_-(S)$.
(5) If $S$ is finite show that

$$\dim(\text{Fun}_+(S)) + \dim(\text{Fun}_-(S)) = |S|.$$

**7.** Let $\mathcal{V}$ be a vector space, and $\mathcal{S}$, $\mathcal{T}$ finite-dimensional subspaces of $\mathcal{V}$. If $\mathcal{S}$ and $\mathcal{T}$ have the same dimension and $\mathcal{S} \subseteq \mathcal{T}$, show that $\mathcal{S} = \mathcal{T}$.

**8.** Let $S = \{s, t, u\}$ and let $\tau : S \longrightarrow S$ be the involution obtained by interchanging $s$ and $u$ and leaving $t$ alone, that is

$$\tau(s) = u, \qquad \tau(t) = t, \qquad \tau(u) = s.$$

Find bases for $\text{Fun}_+(S)$ and $\text{Fun}_-(S)$ (see Exercise 6).

**9.** Find a basis for the subspaces of $\mathcal{P}_4(\mathbb{R})$ spanned by the polynomials $f \in \mathcal{P}_4(\mathbb{R})$ that satisfy the equation

$$f(z) = z^4 f\left(\frac{1}{z}\right).$$

**10.** Let $\mathcal{S}$, $\mathcal{T}$, $\mathcal{U}$ be vector subspaces of the vector space $\mathcal{V}$. Form the nine subspaces

$$\begin{array}{lll} \mathcal{S}, & \mathcal{S} \cap \mathcal{T}, & \mathcal{S} \cap \mathcal{U}, \\ \mathcal{S} + \mathcal{T}, & \mathcal{S} \cap (\mathcal{T} + \mathcal{U}), & \mathcal{S} + (\mathcal{T} \cap \mathcal{U}), \\ \mathcal{S} + \mathcal{T} + \mathcal{U}, & (\mathcal{S} \cap \mathcal{T}) + (\mathcal{S} \cap \mathcal{U}), & (\mathcal{S} + \mathcal{T}) \cap (\mathcal{S} + \mathcal{U}). \end{array}$$

Show that for $\mathcal{X}, \mathcal{Y}$ any two of these nine subspaces, either $\mathcal{X} \subseteq \mathcal{Y}$ or $\mathcal{Y} \subseteq \mathcal{X}$. Find an example with the maximum number of distinct subspaces among the nine.

# Chapter 8
# Linear Transformations

Vector spaces and the algebra of vectors are interesting and useful mathematical tools, but the justification for their importance is that they serve as the arena for the study of linear transformations. Linear transformations are the fundamental objects of study in a basic linear algebra course. In this central chapter we introduce linear transformations and study their most basic properties, postponing detailed numerical examples and their more intensive study to the following chapters.

In physics, when vectors are used by fixing a coordinate system and employing components, it is very important to understand how the components will change if the coordinate system is changed. A change of coordinates is best expressed in terms of transformations of vectors. The simplest such transformations are the linear ones.

In the calculus when the study of curves in the plane and space are introduced the student encounters *vectorvalued functions of a scalar*. The simplest such functions are the linear ones. Linear transformations are the generalization of these linear functions to higher dimensions.

## 8.1 Definition of Linear Transformations

Let us begin with an example. A typical linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^3$ is given by a system of linear equations

$$\begin{aligned} y_1 &= a_{1,1}x_1 + a_{1,2}x_2 \\ y_2 &= a_{2,1}x_1 + a_{2,2}x_2 \\ y_3 &= a_{3,1}x_1 + a_{3,2}x_2 \end{aligned}$$

where $\mathbf{X} = (x_1, x_2)$ are the components of a vector $\mathbf{X}$ of $\mathbb{R}^2$, and $\mathbf{Y} = (y_1, y_2, y_3)$ are the components of the vector $\mathbf{Y}$ of $\mathbb{R}^3$ that corresponds to $\mathbf{X}$ under the particular transformation. The numbers $a_{1,1}, \ldots, a_{3,2}$ are fixed and determine the transformation. The rectangular array of numbers

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{bmatrix}$$

is called the **matrix** of the transformation (relative to the usual bases for $\mathbb{R}^2$ and $\mathbb{R}^3$). It is cumbersome and inconvenient to deal with linear transformations in terms of a system of linear equations. Instead, we shall define them as functions between vector spaces with two simple properties.

**DEFINITION:** *Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces. A* **linear transformation $\mathsf{T}$ from $\mathcal{V}$ to $\mathcal{W}$**, *written $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$, is a function that assigns to each vector $\mathbf{A}$ in $\mathcal{V}$ a vector $\mathsf{T}(\mathbf{A})$ in $\mathcal{W}$ such that the following two properties hold:*

(1) $\mathsf{T}(\mathbf{A} + \mathbf{B}) = \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{B})$ *for all* $\mathbf{A}, \mathbf{B} \in \mathcal{V}$.

(2) $\mathsf{T}(a\mathbf{A}) = a\mathsf{T}(\mathbf{A})$ *for all numbers $a$ and all* $\mathbf{A} \in \mathcal{V}$.

We might paraphrase this definition by saying that a linear transformation from $\mathcal{V}$ to $\mathcal{W}$ is a function that preserves vector addition and scalar multiplication.

It is to be emphasized that as soon as the dimensions of $\mathcal{V}$ and $\mathcal{W}$ exceed 1 there is an enormous number of linear transformations from $\mathcal{V}$ to $\mathcal{W}$, and it is their variety makes the subject of linear algebra really interesting. The classification of linear transformations is what will concern us throughout the remainder of this book, although we will only scratch the surface of classification theory!

Let us begin by examining one linear transformation in each of our basic examples of vector spaces.

**EXAMPLE 1:** Consider the function $\mathsf{R} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$\mathsf{R}(x,\ y) = (x \cos\vartheta - y \sin\vartheta, x \sin\vartheta + y \cos\vartheta),$$

where $\vartheta$ is a fixed number satisfying $0 \leq \vartheta \leq 2\pi$. We claim that $\mathsf{R}$ is a linear transformation. To prove this we must verify that $\mathsf{R}$ possesses the two properties characteristic of linear transformations.

We suppose that $\mathbf{A} = (a_1, a_2)$, $\mathbf{B} = (b_1, b_2)$ are vectors in $\mathbb{R}^2$. Then

$$\begin{aligned}
\mathsf{R}(\mathbf{A}+\mathbf{B}) &= \mathsf{R}(a_1+b_1, a_2+b_2)\\
&= ((a_1+b_1)\cos\vartheta - (a_2+b_2)\sin\vartheta, (a_1+b_1)\sin\vartheta + (a_2+b_2)\cos\vartheta)\\
&= (a_1\cos\vartheta - a_2\sin\vartheta + b_1\cos\vartheta - b_2\sin\vartheta, a_1\sin\vartheta + a_2\cos\vartheta + b_1\sin\vartheta + b_2\cos\vartheta)\\
&= (a_1\cos\vartheta - a_2\sin\vartheta, a_1\sin\vartheta + a_2\cos\vartheta) + (b_1\cos\vartheta - b_2\sin\vartheta, b_1\sin\vartheta + b_2\cos\vartheta)\\
&= \mathsf{R}(\mathbf{A}) + \mathsf{R}(\mathbf{B}),
\end{aligned}$$

and hence the first characteristic property of a linear transformation is satisfied. To verify the second property, we suppose that $a$ is a number and compute

$$\begin{aligned}
\mathsf{R}(a\mathbf{A}) &= \mathsf{R}(aa_1, aa_2)\\
&= (aa_1\cos\vartheta - aa_2\sin\vartheta, aa_1\sin\vartheta + aa_2\cos\vartheta)\\
&= (a(a_1\cos\vartheta - a_2\sin\vartheta), a(a_1\sin\vartheta + a_2\cos\vartheta))\\
&= a(a_1\cos\vartheta - a_2\sin\vartheta, a_1\sin\vartheta + a_2\cos\vartheta)\\
&= a\mathsf{R}(\mathbf{A}),
\end{aligned}$$

so that the second characteristic property of a linear transformation holds for $\mathsf{R}$ also. Therefore $\mathsf{R} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ is a linear transformation.

Geometrically, $\mathsf{R}$ has a very simple interpretation. It is a **rotation** of the plane $\mathbb{R}^2$ through the angle $\vartheta$ (in radian measure) in the counter-clockwise direction as shown in Figure 8.1.1.

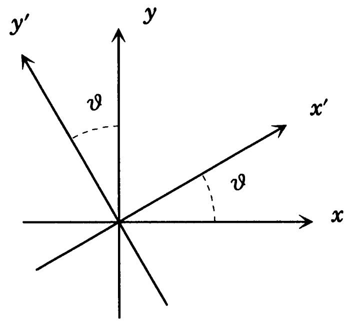


Figure 8.1.1

(In the figure, $x' = \mathsf{R}(x)$ and $y' = \mathsf{R}(y)$.) In studying linear transformations we will often seek such simple geometric interpretations.

**EXAMPLE 2:** Consider the vector space of all polynomials $\mathcal{P}(\mathbb{R})$ and define a mapping $\mathsf{D}$ of this space into itself by

$$\mathsf{D}(p(x)) = \frac{d^3p(x)}{dx^3} + 4\frac{d^2p(x)}{dx^2} + \frac{dp(x)}{dx} - 6p(x).$$

For example, if $p(x) = x^3 + 2x^2 + 3x + 4$, then

$$\begin{aligned}\mathsf{D}(p(x)) &= 3 \cdot 2 + 4(6x + 4) + (3x^2 + 4x + 3) - 6(x^3 + 2x^2 + 3x + 4)\\ &= -6x^3 - 9x^2 + 10x + 1.\end{aligned}$$

The mapping $\mathsf{D}$ defines a linear transformation $\mathsf{D} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R})$. Indeed, it defines a linear transformation $\mathsf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R})$ for any integer $n$. This follows easily from the fact that differentiation of polynomials preserves sums and the derivative of a constant times a polynomial is the constant times the derivative of the polynomial, and finally that differentiation lowers the degree of a polynomial.

The linear transformation $\mathsf{D} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R})$ is the prototype of a **linear differential operator**; see, e.g., Chapter 18.

**EXAMPLE 3** (See also Chapter 5, Exercise 25): Let $S$ be a finite set. A **permutation** of $S$ is a bijective mapping $\sigma : S \longrightarrow S$. For example if $S = \{a, b, c\}$, then the mapping

$$\sigma(a) = b, \quad \sigma(b) = c, \quad \sigma(c) = a$$

is a permutation of $S$, whereas

$$\sigma(a) = b, \quad \sigma(b) = c, \quad \sigma(c) = c$$

is not.

If $\sigma : S \longrightarrow S$ is a permutation of the finite set $S$, then we may define a function (think about *change of variables* in the calculus and see Exercise 25 in Chapter 5)

$$\sigma^* : \mathrm{Fun}(S) \longrightarrow \mathrm{Fun}(S)$$

by the rule

$$\sigma^*(f)(s) = f(\sigma(s)) \qquad \forall\ s \in S.$$

We then find for $f, h \in \mathrm{Fun}(S)$ and $a, b \in \mathbb{R}$ that

$$\begin{aligned}\sigma^*(af + bh)(s) &= (af + bh)(\sigma(s))\\ &= af(\sigma(s)) + bh(\sigma(s)) = a\sigma^*(f(s)) + b\sigma^*(h(s)),\end{aligned}$$

which satisfies both properties of a linear transformation. Hence the mapping $\sigma^* : \mathrm{Fun}(S) \longrightarrow \mathrm{Fun}(S)$ is a linear transformation. We say that the linear transformation $\sigma^* : \mathrm{Fun}(S) \longrightarrow \mathrm{Fun}(S)$ is **induced** by $\sigma$.

## 8.2 Examples of Linear Transformations

Let us continue by examining some of the boundless examples of linear transformations. Example 1 in the previous section has many variants. Here are but a few.

**EXAMPLE 1:** Consider the function from $\mathbb{R}^3$ to $\mathbb{R}^2$ given by

$$\mathsf{P}(x, y, z) = (x, y).$$

We claim that $\mathsf{P}$ is a linear transformation. To prove this we must verify that $\mathsf{P}$ has the two characteristic properties of a linear transformation. This is done in the following computations: We set $\mathbf{A} = (a_1, a_2, a_3)$, $\mathbf{B} = (b_1, b_2, b_3)$, and let $a$ be a number. For property (1) we have

$$\begin{aligned}\mathsf{P}(\mathbf{A}+\mathbf{B}) &= \mathsf{P}(a_1 + b_1, a_2 + b_2, a_3 + b_3)\\ &= (a_1 + b_1, a_2 + b_2) = (a_1, a_2) + (b_1, b_2)\\ &= \mathsf{P}(\mathbf{A}) + \mathsf{P}(\mathbf{B}),\end{aligned}$$

and for property (2),

$$\mathsf{P}(a\mathbf{A}) = \mathsf{P}(aa_1, aa_2, aa_3) = (aa_1, aa_2) = a(a_1, a_2) = a\mathsf{P}(\mathbf{A}).$$

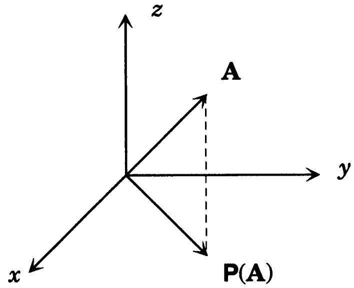


Figure 8.2.1

**EXAMPLE 2:** A slight variation of Example 1 is given by the function $\mathsf{P} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ given by

$$\mathsf{P}(x, y, z) = (x, y, 0).$$

It is routine to check that $\mathsf{P}$ is a linear transformation. Geometrically, $\mathsf{P}$ is the **projection** onto the $xy$-plane, as in figure 8.2.1. It is possible to combine this linear transformation with the rotation of Example 1 to obtain more complicated linear transformations. Many of the linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$ are of this composite form.

Example 1 in Section 8.1 is an example of a **rotation**: The following is an example of a **reflection**.

**EXAMPLE 3:** Let $\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be defined by

$$\mathbf{S}(x, y, z) = \frac{1}{3}(x - 2y - 2z, -2x + y - 2z, -2x - 2y + z).$$

The verification that $\mathbf{S}$ is a linear transformation is routine and left to the reader. This linear transformation may seem somewhat mysterious at first, but one of the basic goals of this book is to provide an *explanation* for the discussion that follows.

The first step in analyzing this linear transformation is to note that the vectors in the plane $\Pi = \left\{(x, y, z) \mid x + y + z = 0\right\}$ are left **fixed** by $\mathbf{S}$; i.e., if $\mathbf{A} \in \Pi$, then $\mathbf{S}(\mathbf{A}) = \mathbf{A}$. This is verified as follows: If $\mathbf{A} = (a_1, a_2, a_3)$ and $\mathbf{A} \in \Pi$ then

$$\mathbf{S}(a_1, a_2, a_3) = \frac{1}{3}(a_1 - 2a_2 - 2a_3, -2a_1 + a_2 - 2a_3, -2a_1 - 2a_2 + a_3)$$

$$= \tfrac{1}{3}(3a_1 - 2(a_1 + a_2 + a_3), 3a_2 - 2(a_1 + a_2 + a_3), 3a_3 - 2(a_1 + a_2 + a_3))$$

$$= \frac{1}{3}(3a_1, 3a_2, 3a_3) = (a_1, a_2, a_3),$$

since $a_1 + a_2 + a_3 = 0$, because $\mathbf{A} \in \Pi$.

The next step is to note that $\mathbf{S}(1, 1, 1) = (-1, -1, -1)$, since

$$\mathbf{S}(1, 1, 1) = \frac{1}{3}(1 - 4, 1 - 4, 1 - 4) = (-1, -1, -1) = -(1, 1, 1).$$

So the vector $(1, 1, 1)$ is reversed by $\mathbf{S}$, or, what is the same thing, reflected in the plane $\Pi$, i.e., sent by $\mathbf{S}$ to its mirror image in the plane $\Pi$. The vector $(1, 1, 1)$ is perpendicular to the plane $\Pi$ and it is not too hard to verify that $\mathbf{S}$ preserves lengths (in the usual sense of the Euclidean length of a vector $\mathbf{A}$ defined by the familiar equation $||\mathbf{A}|| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ from analytic geometry) and angles (see Chapter 15), and therefore $\mathbf{S}$ is an **orthogonal reflection** with **reflecting plane** $\Pi$.

Rotations, reflections, and projections are some of the basic building blocks from which all linear transformations may be constructed.

**EXAMPLE 4:** Define a function $\mathbf{T}$ from $\mathbb{R}^3$ to $\mathcal{P}_2(\mathbb{R})$ by

$$\mathbf{T}(a_1, a_2, a_3) = a_1 + a_2x + a_3x^2.$$

It is routine to verify that $\mathbf{T}$ is a linear transformation. Presently, we will be able to use it to show that $\mathbb{R}^3$ and $\mathcal{P}_2(\mathbb{R})$ are in some sense the *same* vector space.

**EXAMPLE 5:** Let $S$ be a set and $s_0 \in S$ a fixed element. The **evaluation map** at $s_0$ is defined by

$$e_{s_0}(f) = f(s_0) \qquad \forall f \in \mathrm{Fun}(S)$$

and defines a mapping

$$e_{s_0} : \mathrm{Fun}(S) \longrightarrow \mathbb{R}.$$

If $f, h \in \mathrm{Fun}(S)$ and $a, b \in \mathbb{R}$, then

$$e_{s_0}(af + bh) = (af + bh)(s_0) = af(s_0) + bh(s_0) = ae_{s_0}(f) + be_{s_0}(h),$$

so $e_{s_0} : \mathrm{Fun}(S) \longrightarrow \mathbb{R}$ is a linear transformation.

## 8.3 Properties of Linear Transformations

In this section we start investigating some of the basic properties of linear transformations.

**PROPOSITION 8.3.1:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation. Then $\mathsf{T}(\mathbf{0}) = \mathbf{0}$. (Note that we have used the same symbol $\mathbf{0}$ to denote the zero vector of $\mathcal{V}$ and of $\mathcal{W}$.)*

**PROOF:** We have

$$\mathsf{T}(\mathbf{0}) = \mathsf{T}(0\mathbf{0}) = 0\mathsf{T}(\mathbf{0}) = \mathbf{0}$$

by Proposition 2.1.1. $\square$

**PROPOSITION 8.3.2:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation. If $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are vectors in $\mathcal{V}$ and $a_1, \ldots, a_n$ are numbers, then*

$$\mathsf{T}(a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_n\mathbf{A}_n) = a_1\mathsf{T}(\mathbf{A}_1) + a_2\mathsf{T}(\mathbf{A}_2) + \cdots + a_n\mathsf{T}(\mathbf{A}_n).$$

**PROOF:** We apply the fact that $\mathsf{T}$ preserves vector sums to write

$$\mathsf{T}(a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n) = \mathsf{T}(a_1\mathbf{A}_1) + \cdots + \mathsf{T}(a_n\mathbf{A}_n),$$

and now we apply the fact that $\mathsf{T}$ preserves scalar products to write

$$\mathsf{T}(a_1\mathbf{A}_1) + \cdots + \mathsf{T}(a_n\mathbf{A}_n) = a_1\mathsf{T}(\mathbf{A}_1) + a_2\mathsf{T}(\mathbf{A}_2) + \cdots + a_n\mathsf{T}(\mathbf{A}_n),$$

which is the desired conclusion. $\square$

**PROPOSITION 8.3.3:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation. Suppose that $\mathcal{U}$ is a linear subspace of $\mathcal{V}$, and set*

$$\mathsf{T}(\mathcal{U}) = \left\{ \mathsf{T}(\mathbf{A}) \in \mathcal{W} \mid \mathbf{A} \in \mathcal{U} \right\}$$

*(that is, $\mathsf{T}(\mathcal{U})$ is the set of all vectors in $\mathcal{W}$ that are of the form $\mathsf{T}(\mathbf{A})$ for some vector $\mathbf{A}$ in $\mathcal{V}$). Then $\mathsf{T}(\mathcal{U})$ is a linear subspace of $\mathcal{W}$.*

**PROOF:** Suppose that $\mathbf{A}', \mathbf{B}'$ are vectors in $\mathsf{T}(\mathcal{U})$. Then there are vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{U}$ such that

$$\mathbf{A}' = \mathsf{T}(\mathbf{A}) \text{ and } \mathbf{B}' = \mathsf{T}(\mathbf{B}).$$

Then $\mathbf{A} + \mathbf{B}$ is in $\mathcal{U}$, since $\mathcal{U}$ is a linear subspace of $\mathcal{V}$. Therefore, $\mathbf{C} = \mathsf{T}(\mathbf{A} + \mathbf{B})$ is in $\mathsf{T}(\mathcal{U})$. But since $\mathsf{T}$ is a linear transformation,

$$\mathbf{C} = \mathsf{T}(\mathbf{A} + \mathbf{B}) = \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{B}) = \mathbf{A}' + \mathbf{B}',$$

so we see that $\mathbf{A}' + \mathbf{B}'$ is in $\mathsf{T}(\mathcal{U})$.

Next we suppose that $a$ is a number. Since $\mathcal{U}$ is a linear subspace of $\mathcal{V}$, $a\mathbf{A} \in \mathcal{U}$. Therefore $\mathbf{C} = \mathsf{T}(a\mathbf{A}) \in \mathsf{T}(\mathcal{U})$. But since $\mathsf{T}$ is a linear transformation $\mathbf{C} = \mathsf{T}(a\mathbf{A}) = a\mathsf{T}(\mathbf{A}) = a\mathbf{A}'$ so that $a\mathbf{A}'$ is in $\mathsf{T}(\mathcal{U})$ also. $\square$

**PROPOSITION 8.3.4:** *Suppose that* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *is a linear transformation and* $E$ *is a set of vectors in* $\mathcal{V}$. *Then*

$$\mathsf{T}(\mathcal{L}(E)) = \mathcal{L}(\mathsf{T}(E)),$$

*where* $\mathsf{T}(E)$ *denotes the set of vectors in* $\mathcal{W}$ *of the form* $\mathsf{T}(\mathbf{A})$ *for some vector* $\mathbf{A}$ *in* $E$.

**PROOF:** The proof is similar to that of Proposition 8.3.3. We must show that each vector in $\mathsf{T}(\mathcal{L}(E))$ is in $\mathcal{L}(\mathsf{T}(E))$ and conversely.

A vector in $\mathsf{T}(\mathcal{L}(E))$ is a vector $\mathsf{T}(\mathbf{A})$ for some $\mathbf{A}$ in $\mathcal{L}(E)$, so there are vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ in $E$ and numbers $a_1, \ldots, a_n$ with

$$\mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n.$$

Therefore, by Proposition 8.3.2,

$$\mathsf{T}(\mathbf{A}) = a_1\mathsf{T}(\mathbf{A}_1) + \cdots + a_n\mathsf{T}(\mathbf{A}_n),$$

and since the righthand side belongs to $\mathcal{L}(\mathsf{T}(E))$, we see that $\mathsf{T}(\mathcal{L}(E)) \subseteq \mathcal{L}(\mathsf{T}(E))$.

To prove the converse, suppose that $\mathbf{A}'$ belongs to $\mathcal{L}(\mathsf{T}(E))$. Then there are vectors $\mathbf{A}'_1, \ldots, \mathbf{A}'_n$ in $\mathsf{T}(E)$ and numbers $a_1, \ldots, a_n$ such that

$$\mathbf{A}' = a_1\mathbf{A}'_1 + \cdots + a_n\mathbf{A}'_n.$$

There are also vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ in $E$ with

$$\mathbf{A}'_1 = \mathsf{T}(\mathbf{A}_1), \mathbf{A}'_2 = \mathsf{T}(\mathbf{A}_2), \ldots, \mathbf{A}'_n = \mathsf{T}(\mathbf{A}_n).$$

Let

$$\mathbf{A} = a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_n\mathbf{A}_n.$$

Then $\mathbf{A}$ is a vector in $\mathcal{L}(E)$, and by Proposition 8.3.2

$$\begin{aligned}\mathsf{T}(\mathbf{A}) &= \mathsf{T}(a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n)\\ &= a_1\mathsf{T}(\mathbf{A}_1) + \cdots + a_n\mathsf{T}(\mathbf{A}_n)\\ &= a_1\mathbf{A}_1' + \cdots + a_n\mathbf{A}_n' = \mathbf{A}',\end{aligned}$$

so that $\mathbf{A}'$ belongs to $\mathsf{T}(\mathcal{L}(E))$; that is, $\mathcal{L}(\mathsf{T}(E)) \subseteq \mathsf{T}(\mathcal{L}(E))$ as was to be shown. □

**PROPOSITION 8.3.5:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation. Then*

$$\left\{\mathbf{A} \in \mathcal{V} \mid \mathsf{T}(\mathbf{A}) = \mathbf{0}\right\}$$

*is a linear subspace of $\mathcal{V}$.*

**PROOF:** Let $\mathbf{A}, \mathbf{B} \in \mathcal{V}$ and $a, b \in \mathbb{R}$. Suppose that $\mathsf{T}(\mathbf{A}) = \mathbf{0} = \mathsf{T}(\mathbf{B})$. Then

$$\mathsf{T}(a\mathbf{A} + b\mathbf{B}) = a\mathsf{T}(\mathbf{A}) + b\mathsf{T}(\mathbf{B}) = a\mathbf{0} + b\mathbf{0} = \mathbf{0},$$

so $\left\{\mathbf{A} \in \mathcal{V} \mid \mathsf{T}(\mathbf{A}) = \mathbf{0}\right\}$ is a linear subspace of $\mathcal{V}$. □

**PROPOSITION 8.3.6:** *Suppose that $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ and $\mathsf{S} : \mathcal{W} \longrightarrow \mathcal{U}$ are linear transformations. Then the composite function*

$$\mathsf{S} \cdot \mathsf{T} : \mathcal{V} \xrightarrow{\mathsf{T}} \mathcal{W} \xrightarrow{\mathsf{S}} \mathcal{U}$$

*is a linear transformation.*

**PROOF:** We must verify that $\mathsf{S} \cdot \mathsf{T}$ has the two characteristic properties of a linear transformation. So suppose that $\mathbf{A}, \mathbf{B}$ belong to $\mathcal{V}$ and $a, b$ are numbers. Then using the facts that $\mathsf{S}$ and $\mathsf{T}$ are linear transformations, we find

$$\begin{aligned}\mathsf{S} \cdot \mathsf{T}(a\mathbf{A} + b\mathbf{B}) &= \mathsf{S}(\mathsf{T}(a\mathbf{A} + b\mathbf{B}))\\ &= \mathsf{S}(\mathsf{T}(a\mathbf{A}) + \mathsf{T}(b\mathbf{B}))\\ &= \mathsf{S}(a\mathsf{T}(\mathbf{A})) + \mathsf{S}(b\mathsf{T}(\mathbf{B}))\\ &= a\mathsf{S} \cdot \mathsf{T}(\mathbf{A}) + b\mathsf{S} \cdot \mathsf{T}(\mathbf{B}),\end{aligned}$$

and hence by Proposition 8.3.2, $\mathsf{S} \cdot \mathsf{T}$ is a linear transformation. □

**EXAMPLE 1:** Consider the differentiation operator $\mathsf{D} = \dfrac{d}{dx}$ as a mapping on a polynomial space, i.e.,

$$\mathsf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}), \qquad \mathsf{D}(p(x)) = \frac{d}{dx}(p(x)).$$

The derivative of a polynomial of degree $n$ is at most of degree $n-1$, and the derivative of a polynomial of degree 0 is the zero polynomial. Therefore, $\underbrace{\mathsf{D}\cdots\mathsf{D}}_{\leftarrow n+1\rightarrow} = \mathbf{0}$, since the $(n+1)$-st derivative of a polynomial of degree at most $n$ is the zero polynomial. Therefore, the composition of the operator $\mathsf{D}$ with itself $n+1$ times is the zero linear transformation from $\mathcal{P}_n(\mathbb{R})$ to $\mathcal{P}_n(\mathbb{R})$.

The linear transformation $\mathsf{D} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R})$ is what is called **a nilpotent transformation.** The nilpotent transformations play a central role in the more advanced theory of linear algebra, and we will have more to do with them in later chapters.

## 8.4 Images and Kernels of Linear Transformations

Associated to a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ are two important linear subspaces, the **kernel** and the **image**[1] of $\mathsf{T}$. These are defined with the help of Propositions 8.3.5 and 8.3.3 as follows:

**DEFINITION:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation. The* **kernel** *of $\mathsf{T}$ is defined to be the set*

$$\ker(\mathsf{T}) = \left\{\mathbf{A} \in \mathcal{V} \mid \mathsf{T}(\mathbf{A}) = \mathbf{0}\right\}.$$

*The* **image** *of $\mathsf{T}$ is defined to be the set*

$$\operatorname{Im}(\mathsf{T}) = \left\{\mathbf{B} \in \mathcal{W} \mid \exists\, \mathbf{A} \in \mathcal{W} \textit{ with } \mathsf{T}(\mathbf{A}) = \mathbf{B}\right\}.$$

**PROPOSITION 8.4.1:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation. Then*

*(i)* $\ker(\mathsf{T})$ *is a linear subspace of $\mathcal{V}$,*
*(ii)* $\operatorname{Im}(\mathsf{T})$ *is a linear subspace of $\mathcal{W}$.*

**PROOF:** Apply Propositions 8.3.5 and 8.3.3. $\square$

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[1] The image of a linear transformation $\mathsf{T}$ is also referred to as the **range** of $\mathsf{T}$.

**EXAMPLE 1:** Consider the rotation $\mathsf{R} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$\mathsf{R}(x,\ y) = (x \cos \vartheta - y \sin \vartheta, x \sin \vartheta + y \cos \vartheta),$$

where $\vartheta$ is a fixed number satisfying $0 \leq \vartheta \leq 2\pi$. Recall (see Section 8.1 Example 1) that $\mathsf{R}$ is a rotation through the angle with radian measure $\vartheta$ in the counterclockwise direction. Then the only vector that $\mathsf{R}$ *rotates* to $\mathbf{0}$ is the zero vector $\mathbf{0}$ itself. Every vector $\mathbf{A} \in \mathbb{R}^2$ is the image under $\mathsf{R}$ of the vector $\mathbf{B}$ obtained by rotating $\mathbf{A}$ through an angle $\vartheta$ in the *clockwise* direction. Therefore,

$$\ker(\mathsf{R}) = \{\mathbf{0}\},$$
$$\operatorname{Im}(\mathsf{R}) = \mathbb{R}^2.$$

**EXAMPLE 2:** Consider the linear transformation $\mathsf{F} : \mathbb{R}^3 \longrightarrow \mathbb{R}^1$ defined by

$$\mathsf{F}(x, y, z) = x + y + z \qquad \forall\ (x, y, z) \in \mathbb{R}^3.$$

A vector $(x, y, z) \in \mathbb{R}^3$ lies in $\ker(\mathsf{F})$ if and only if $x + y + z = 0$. The set of all such vectors forms the 2-dimensional subspace of $\mathbb{R}^3$ spanned by the vectors $(1, -1\ 0)$, $(0, 1, -1)$. (N.B. These are, of course, not the only two vectors that span $\ker(\mathsf{F})$.) The vector $1 \in \mathbb{R}^1$ lies in the image of $\mathsf{F}$, since, for example, $\mathsf{F}(1, 0, 0) = 1$. Therefore, $\operatorname{Im}(\mathsf{F}) = \mathbb{R}^1$ , since it is a linear subspace and contains a basis for $\mathbb{R}^1 = \mathbb{R}$.

**EXAMPLE 3:** Consider the linear transformation $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ defined by

$$\mathsf{T}(u, v) = (u, v - u, v) \qquad \forall\ (u, v) \in \mathbb{R}^2.$$

Then the image of $\mathsf{T}$ is spanned by the vectors

$$(1, -1, 0) = \mathsf{T}(1, 0),$$
$$(0, 1, -1) = \mathsf{T}(0, 1),$$

so $\operatorname{Im}(\mathsf{T})$ is the plane in $\mathbb{R}^3$ that occurs as the kernel of the linear transformation $\mathsf{F}$ in the preceding example. The kernel of $\mathsf{T}$ is the trivial subspace $\{\mathbf{0}\}$ of $\mathbb{R}^2$.

**EXAMPLE 4:** Consider the differentiation operator

$$\mathsf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R})$$

given by

$$\mathsf{D}(p(x)) = \frac{d}{dx} p(x).$$

We may therefore compose $\mathbf{D}$ with itself repeatedly to obtain the linear transformations

$$\begin{aligned}
\mathbf{D}^2 &= \mathbf{D} \cdot \mathbf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}), \\
\mathbf{D}^3 &= \mathbf{D} \cdot \mathbf{D}^2 : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}), \\
&\vdots \\
\mathbf{D}^m &= \mathbf{D} \cdot \mathbf{D}^{m-1} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}).
\end{aligned}$$

The kernel of $\mathbf{D}$ consists of all those polynomials with $\frac{d}{dx}(p(x)) = 0$. The only such polynomials are the constant polynomials. Thus

$$\ker(\mathbf{D}) = \mathcal{P}_0(\mathbb{R}).$$

Since differentiation of polynomials lowers the degree of the polynomial, the image of $\mathbf{D}$ lies in $\mathcal{P}_{n-1}(\mathbb{R})$. By the fundamental theorem of the calculus,

$$p(x) = \frac{d}{dt}\left[\int_0^x p(t)dt\right],$$

and hence any polynomial of degree at most $n-1$ lies in the image of $\mathbf{D}$. (N.B. Since the indefinite integral of a polynomial of degree $n$ is of degree $n+1$, it does not lie in $\mathcal{P}_n(\mathbb{R})$, so we cannot use integration to prove that it lies in the image of $\mathbf{D}$.) Therefore,

$$\operatorname{Im}(\mathbf{D}) = \mathcal{P}_{n-1}(\mathbb{R}).$$

Likewise, the kernel of $\mathbf{D}^2$ consists of all those polynomials with $\frac{d^2}{dx^2}(p(x)) = 0$. The only such polynomials are the linear polynomials. Thus

$$\ker(\mathbf{D}^2) = \mathcal{P}_1(\mathbb{R}),$$

and by integrating twice we see that

$$\operatorname{Im}(\mathbf{D}^2) = \mathcal{P}_{n-2}(\mathbb{R}).$$

Continuing in this way, we obtain

$$\begin{aligned}
\ker(\mathbf{D}) &= \mathcal{P}_0(\mathbb{R}), \\
\operatorname{Im}(\mathbf{D}) &= \mathcal{P}_{n-1}(\mathbb{R}), \\
\ker(\mathbf{D}^2) &= \mathcal{P}_1(\mathbb{R}), \\
\operatorname{Im}(\mathbf{D}^2) &= \mathcal{P}_{n-2}(\mathbb{R}),
\end{aligned}$$

and, more generally,

$$\ker(\mathbf{D}^m) = \mathcal{P}_{m-1}(\mathbb{R}),$$
$$\operatorname{Im}(\mathbf{D}^m) = \mathcal{P}_{n-m}(\mathbb{R}).$$

In particular,

$$\ker(\mathbf{D}^{n+1}) = \mathcal{P}_n(\mathbb{R}),$$
$$\operatorname{Im}(\mathbf{D}^{n+1}) = \{\mathbf{0}\}.$$

That is,

$$\mathbf{D}^{n+1} = \mathbf{0} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R}).$$

The following proposition is central to the study of linear equations. Its proof is important because it indicates one process whereby one *solves* a system of linear homogeneous equations.

**THEOREM 8.4.2** (Dimension Formula): *Let* $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *be a linear transformation. If* $\mathcal{V}$ *is finite-dimensional, then* $\ker(\mathbf{T})$ *and* $\operatorname{Im}(\mathbf{T})$ *are finite-dimensional and*

$$\dim(\mathcal{V}) = \dim(\operatorname{Im}(\mathbf{T})) + \dim(\ker(\mathbf{T})).$$

**PROOF:** Since $\mathcal{V}$ is finite-dimensional and $\ker(\mathbf{T})$ is a subspace of $\mathcal{V}$, it too is finite dimensional by Theorem 6.3.3. Choose a basis $\mathbf{A}_1, \dots, \mathbf{A}_s$ for $\ker(\mathbf{T})$. By the basis extension theorem (Theorem 6.3.2) we may find vectors $\mathbf{B}_1, \dots, \mathbf{B}_t$ such that $\mathbf{A}_1, \dots, \mathbf{A}_s, \mathbf{B}_1, \dots, \mathbf{B}_t$ is a basis for $\mathcal{V}$. By Proposition 8.3.4, we have

$$\begin{aligned}
\operatorname{Im}(\mathbf{T}) = \mathbf{T}(\mathcal{V}) &= \mathbf{T}(\mathcal{L}(\mathbf{A}_1, \dots, \mathbf{A}_s, \mathbf{B}_1, \dots, \mathbf{B}_t)) \\
&= \mathcal{L}(\mathbf{T}(\mathbf{A}_1), \dots, \mathbf{T}(\mathbf{A}_s), \mathbf{T}(\mathbf{B}_1), \dots, \mathbf{T}(\mathbf{B}_t)) \\
&= \mathcal{L}(\mathbf{0}, \dots, \mathbf{0}, \mathbf{T}(\mathbf{B}_1), \dots, \mathbf{T}(\mathbf{B}_t)) \\
&= \mathcal{L}(\mathbf{T}(\mathbf{B}_1), \dots, \mathbf{T}(\mathbf{B}_t)),
\end{aligned}$$

and hence $\operatorname{Im}(\mathbf{T})$ is finite-dimensional.

We claim that the vectors $\mathbf{T}(\mathbf{B}_1), \dots, \mathbf{T}(\mathbf{B}_t)$ are a basis for $\operatorname{Im}(\mathbf{T})$. The preceding equation shows that $\mathbf{T}(\mathbf{B}_1), \dots, \mathbf{T}(\mathbf{B}_t)$ span $\operatorname{Im}(\mathbf{T})$, and so we must show that they are linearly independent. To this end, suppose that there are numbers $b_1, \dots, b_t$ such that

$$b_1\mathbf{T}(\mathbf{B}_1) + b_2\mathbf{T}(\mathbf{B}_2) + \cdots + b_t\mathbf{T}(\mathbf{B}_t) = \mathbf{0}.$$

Let $\mathbf{B} = b_1\mathbf{B}_1 + \cdots + b_t\mathbf{B}_t$. Then $\mathbf{B}$ is a vector in $\mathcal{V}$. By Proposition 8.3.2,

$$\mathbf{T}(\mathbf{B}) = b_1\mathbf{T}(\mathbf{B}_1) + \cdots + b_1\mathbf{T}(\mathbf{B}_t) = \mathbf{0},$$

so $\mathbf{B} \in \ker(\mathbf{T})$. Since the vectors $\mathbf{A}_1, \dots, \mathbf{A}_s$ are a basis for $\ker(\mathbf{T})$, we may find numbers $a_1, \dots, a_s$ such that

$$\mathbf{B} = a_1\mathbf{A}_1 + \cdots + a_s\mathbf{A}_s,$$

or

$$b_1\mathbf{B}_1 + \cdots + b_t\mathbf{B}_t = a_1\mathbf{A}_1 + \cdots + a_s\mathbf{A}_s,$$

which may be written

$$\mathbf{0} = a_1\mathbf{A}_1 + \cdots + a_s\mathbf{A}_s - b_1\mathbf{B}_1 - \cdots - b_t\mathbf{B}_t,$$

which is a linear relation among the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_s, \mathbf{B}_1, \ldots, \mathbf{B}_t$. The vectors $\mathbf{A}_1, \ldots, \mathbf{A}_s, \mathbf{B}_1, \ldots, \mathbf{B}_t$ are a basis for $\mathcal{V}$. Hence we must have

$$a_1 = \cdots = a_s = -b_1 = \cdots = -b_t = 0,$$

so in particular, $b_1 = \cdots = b_t = 0$, and therefore $\mathsf{T}(\mathbf{B}_1), \ldots, \mathsf{T}(\mathbf{B}_t)$ are linearly independent and hence a basis for $\text{Im}(\mathsf{T})$.

Thus

$$\dim(\ker(\mathsf{T})) = s,$$

$$\dim(\text{Im}(\mathsf{T})) = t,$$

$$\dim(\mathcal{V}) = s + t,$$

as was to be shown. □

## 8.5 Some Fundamental Constructions

We begin this section by describing a very general process for constructing linear transformations, called the **linear extension construction**.

Suppose we are given the following data:

(1) $\mathcal{V}, \mathcal{W}$ vector spaces.

(2) $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ an ordered basis for $\mathcal{V}$.

(3) $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ an ordered $n$-tuple of vectors in $\mathcal{W}$. They need not be a basis; in fact, *they need not even be distinct.*

We define a function $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ as follows. Let $\mathbf{A}$ be a vector in $\mathcal{V}$. By Theorem 6.3.1 there are unique numbers $a_1, \ldots, a_n$ (the coordinates of $\mathbf{A}$ relative to $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$) such that

$$\mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n.$$

Define

$$\mathsf{T}(\mathbf{A}) = a_1\mathbf{B}_1 + \cdots + a_n\mathbf{B}_n.$$

Note that

$$\mathsf{T}(\mathbf{A}_i) = \mathbf{B}_i, \qquad i = 1, 2, \ldots, n.$$

**REMARK:** The reason that $\mathsf{T}$ is welldefined is that the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are a basis for $\mathcal{V}$: In order to define $\mathsf{T}$, we first must write

$$\mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n,$$

and we are using the fact that $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ spans $\mathcal{V}$. The fact that $\mathsf{T}(\mathbf{A})$ is unambiguously defined in $\mathcal{W}$ uses the fact that the coordinates of $\mathbf{A}$ are unique, which is true because $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are linearly independent.

Note also that the linear transformation $\mathsf{T}$ depends not only on the sets of vectors $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$, but also on their order. To change the order of the vectors $\mathbf{A}_i$ and/or $\mathbf{B}_j$ will in general change the transformation $\mathsf{T}$.

We call the function $\mathsf{T}$ the **linear extension** of

$$\mathsf{T}(\mathbf{A}_i) = \mathbf{B}_i, \qquad i = 1, \ldots, n.$$

The adjective ***linear*** is justified by:

**THEOREM 8.5.1** (Linear Extension Theorem): *Suppose that $\mathcal{V}$ and $\mathcal{W}$ are vector spaces, $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ a basis for $\mathcal{V}$, and $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ vectors in $\mathcal{W}$. Then the linear extension of*

$$\mathsf{T}(\mathbf{A}_i) = \mathbf{B}_i, \qquad i = 1, \ldots, n,$$

*is a linear transformation*

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}.$$

**PROOF:** We must show that

$$\mathsf{T}(\mathbf{A} + \mathbf{C}) = \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{C}),$$
$$\mathsf{T}(r\mathbf{A}) = r\mathsf{T}(\mathbf{A}),$$

for all vectors $\mathbf{A}$ and $\mathbf{C}$ in $\mathcal{V}$ and all numbers $r$. So suppose that

$$\mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n,$$
$$\mathbf{C} = c_1\mathbf{A}_1 + \cdots + c_n\mathbf{A}_n.$$

Then, of course,

$$\mathbf{A} + \mathbf{C} = (a_1 + c_1)\mathbf{A}_1 + (a_2 + c_2)\mathbf{A}_2 + \cdots + (a_n + c_n)\mathbf{A}_n$$

by the usual sort of argument involving shuffling vectors around. Thus according to our definition of $\mathsf{T}$, we have

$$\mathsf{T}(\mathbf{A} + \mathbf{C}) = (a_1 + c_1)\mathbf{B}_1 + \cdots + (a_n + c_n)\mathbf{B}_n;$$

hence

$$\begin{aligned}\mathsf{T}(\mathbf{A}+\mathbf{C}) &= a_1\mathbf{B}_1 + c_1\mathbf{B}_1 + \cdots + a_n\mathbf{B}_n + c_n\mathbf{B}_n \\ &= (a_1\mathbf{B}_1 + a_2\mathbf{B}_2 + \cdots + a_n\mathbf{B}_n) + (c_1\mathbf{B}_1 + c_2\mathbf{B}_2 + \cdots + c_n\mathbf{B}_n) \\ &= \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{C}).\end{aligned}$$

In a similar manner we may show that

$$\mathsf{T}(r\mathbf{A}) = r\mathsf{T}(\mathbf{A})$$

for all vectors $\mathbf{A}$ in $\mathcal{V}$ and numbers $r$. Therefore, $\mathsf{T}$ is a linear transformation. □

The linear extension construction has a number of surprising properties. Here are some of them (see also Theorem 8.6.3):

**PROPOSITION 8.5.2:** *Let $\mathbf{A}_1, \ldots, \mathbf{A}_n$ be vectors in the vector space $\mathcal{V}$. Let $\mathbf{E}_1, \ldots, \mathbf{E}_n \in \mathbb{R}^n$ be the standard basis and*

$$\mathsf{T} : \mathbb{R}^n \longrightarrow \mathcal{V}$$

*the linear extension of*

$$\mathsf{T}(\mathbf{E}_i) = \mathbf{A}_i \qquad i = 1, 2, \ldots, n.$$

*Then $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are linearly independent if and only if* $\ker(\mathsf{T}) = \{\mathbf{0}\}$.

**PROOF:** Suppose that $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are linearly independent. If $\mathbf{B} = (b_1, \ldots, b_n) \in \mathbb{R}^n$, then

$$\mathsf{T}(\mathbf{B}) = \mathsf{T}(b_1, \ldots, b_n) = b_1\mathbf{A}_1 + b_2\mathbf{A}_2 + \cdots + b_n\mathbf{A}_n.$$

Suppose that

$$\mathsf{T}(b_1, \ldots, b_n) = \mathbf{0}.$$

Then

$$b_1\mathbf{A}_1 + \cdots + b_n\mathbf{A}_n = \mathbf{0},$$

which, since $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are linearly independent, implies

$$(b_1, \ldots, b_n) = (0, \ldots, 0).$$

That is, $\ker(\mathsf{T}) = \{\mathbf{0}\}$.

On the other hand, suppose that $\ker(\mathsf{T}) = \{\mathbf{0}\}$. If

$$a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n = \mathbf{0}$$

is a linear relation among $\mathbf{A}_1, \ldots, \mathbf{A}_n$, then

$$\mathsf{T}(a_1, \ldots, a_n) = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n = \mathbf{0}.$$

So $(a_1, \ldots, a_n) = (0, \ldots, 0)$, since $\ker(\mathsf{T}) = \{\mathbf{0}\}$. Therefore, the linear relation is the trivial one, so $\mathbf{A}_1, \ldots, \mathbf{A}_n$ are linearly independent. □

Similarly, we have:

**PROPOSITION 8.5.3:** *Let* $\mathbf{A}_1, \ldots, \mathbf{A}_n$ *be vectors in the vector space* $\mathcal{V}$. *Let* $\mathbf{E}_1, \ldots, \mathbf{E}_n \in \mathbb{R}^n$ *be the standard basis for* $\mathbb{R}^n$ *and*

$$\mathsf{T} : \mathbb{R}^n \longrightarrow \mathcal{V}$$

*the linear extension of*

$$\mathsf{T}(\mathbf{E}_i) = \mathbf{A}_i \qquad i = 1, 2, \ldots, n.$$

*Then* $\mathbf{A}_1, \ldots, \mathbf{A}_n$ *span* $\mathcal{V}$ *if and only if* $\mathrm{Im}(\mathsf{T}) = \mathcal{V}$.

**PROOF:** Exercise. □

We postpone a discussion of examples of the linear extension construction to the next chapter.

The set of all linear transformations between two vector spaces is more than just a set: It carries a natural structure as a vector space in its own right, as we show next.

**DEFINITION:** *Suppose that* $\mathsf{S}, \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *are linear transformations. Let*

$$\mathsf{S} + \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$$

*be the function defined by*

$$(\mathsf{S} + \mathsf{T})(\mathbf{A}) = \mathsf{S}(\mathbf{A}) + \mathsf{T}(\mathbf{A})$$

*for all* $\mathbf{A} \in \mathcal{V}$.

**PROPOSITION 8.5.4:** *If* $\mathsf{S}, \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *are linear transformations, then so is* $\mathsf{S} + \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$.

**PROOF:** We suppose that $\mathbf{A}, \mathbf{B}$ belong to $\mathcal{V}$, and we compute as follows:

$$\begin{aligned}(\mathsf{S} + \mathsf{T})(\mathbf{A} + \mathbf{B}) &= \mathsf{S}(\mathbf{A} + \mathbf{B}) + \mathsf{T}(\mathbf{A} + \mathbf{B}) \\ &= \mathsf{S}(\mathbf{A}) + \mathsf{S}(\mathbf{B}) + \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{B}) \\ &= \mathsf{S}(\mathbf{A}) + \mathsf{T}(\mathbf{A}) + \mathsf{S}(\mathbf{B}) + \mathsf{T}(\mathbf{B}) \\ &= (\mathsf{S} + \mathsf{T})(\mathbf{A}) + (\mathsf{S} + \mathsf{T})(\mathbf{B}).\end{aligned}$$

For any number $a$ we have

$$\begin{aligned}(\mathsf{S} + \mathsf{T})(a\mathbf{A}) &= \mathsf{S}(a\mathbf{A}) + \mathsf{T}(a\mathbf{A}) \\ &= a\mathsf{S}(\mathbf{A}) + a\mathsf{T}(\mathbf{A}) \\ &= a(\mathsf{S}(\mathbf{A}) + \mathsf{T}(\mathbf{A})) \\ &= a((\mathsf{S} + \mathsf{T})(\mathbf{A})).\end{aligned}$$

Hence the function $\mathsf{S} + \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ possesses the two characteristic properties of a linear transformation. □

**PROPOSITION 8.5.5:** *If* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *is a linear transformation and* $a$ *is a number, then* $a\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *is also a linear transformation.*

**PROOF:** The proof is similar to that of the preceding proposition and is left to the reader. $\square$

**DEFINITION:** *If* $\mathcal{V}$ *and* $\mathcal{W}$ *are vector spaces, the* **zero transformation**, *denoted by* $\mathbf{0} : \mathcal{V} \longrightarrow \mathcal{W}$, *is defined by* $\mathbf{0}(\mathbf{A}) = \mathbf{0}$ *for all* $\mathbf{A}$ *in* $\mathcal{V}$. *If* $\mathcal{V} = \mathcal{W}$, *the* **identity transformation**, *denoted by* $\mathsf{I} : \mathcal{V} \longrightarrow \mathcal{V}$, *is defined by* $\mathsf{I}(\mathbf{A}) = \mathbf{A}$ *for all* $\mathbf{A} \in \mathcal{V}$.

Note that taken together with the preceding propositions we are on our way to showing that the set of all linear transformations from $\mathcal{V}$ to $\mathcal{W}$ ***is again a vector space***. This is indeed true, and a very important theorem.

**THEOREM 8.5.6:** *Let* $\mathcal{V}$ *and* $\mathcal{W}$ *be vector spaces and* $\mathcal{L}(\mathcal{V}, \mathcal{W})$ *the set of all linear transformations from* $\mathcal{V}$ *to* $\mathcal{W}$. *Then* $\mathcal{L}(\mathcal{V}, \mathcal{W})$ *is a vector space.*

**PROOF:** We have defined an addition for two linear transformations and an operation of multiplying a linear transformation by a number. As zero vector for $\mathcal{L}(\mathcal{V}, \mathcal{W})$ we choose the zero transformation $\mathbf{0}$, and we must verify that the axioms of a vector space given in Chapter 2 are satisfied. The negative of a transformation $\mathsf{T}$ in Axiom 4 is to be $(-1)\mathsf{T}$. The verifications are routine and left to the reader. $\square$

Note that Proposition 8.5.6 should finally lay to rest the idea that a vector is a quantity with direction and magnitude. The vector spaces $\mathcal{L}(\mathcal{V}, \mathcal{W})$ are enormously important and we will see more and more of them. The special case $\mathcal{W} = \mathbb{R} = \mathbb{R}^1$ occurs so often that it is given a special name, **the dual space of** $\mathcal{V}$, and the special notation $\mathcal{V}^*$.

There should be no confusion between the notation $\mathcal{L}(\mathcal{V}, \mathcal{W})$ for the **space of all linear transformations from** $\mathcal{V}$ **to** $\mathcal{W}$ and $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k)$ for the linear span $\mathrm{Span}(\mathbf{A}_1, \ldots, \mathbf{A}_k)$ of the vectors $\mathbf{A}_1, \ldots, \mathbf{A}_k \in \mathcal{V}$.

## 8.6 Isomorphism of Vector Spaces

The vector spaces $\mathbb{R}^3$, $\mathcal{P}_2(\mathbb{R})$, and $\mathrm{Fun}(\{a, b, c\})$ are all 3-dimensional vector spaces. In what sense are they the same, and in what sense are they different? They have the same dimension, but they do not *look* the same. The following definition allows us to compare such non look alikes in a convenient way.

**DEFINITION:** *A linear transformation* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *is said to be an* **isomorphism** *if there exists a linear transformation* $\mathbf{S} : \mathcal{W} \longrightarrow \mathcal{V}$ *such* *that*

$$\mathbf{S} \cdot \mathbf{T}(\mathbf{A}) = \mathbf{A} \quad \forall\, \mathbf{A} \in \mathcal{V},$$

$$\mathbf{T} \cdot \mathbf{S}(\mathbf{B}) = \mathbf{B} \quad \forall\, \mathbf{B} \in \mathcal{W}.$$

*The linear transformations* **S** *and* **T** *are said to be* **inverse isomorphisms,** *and* **T** *is called the* **inverse** *of* **S** *and conversely. If there is a linear transformation* $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *that is an isomorphism, then we say that* $\mathcal{V}$ *and* $\mathcal{W}$ *are* **isomorphic.**

If $\mathcal{V}$ and $\mathcal{W}$ are isomorphic vector spaces, then in some sense they are the same. More precisely, an isomorphism **T** will translate true theorems in $\mathcal{V}$ into true theorems in $\mathcal{W}$, and conversely.

**EXAMPLE 1:** The vector spaces $\mathbb{R}^{n+1}$ and $\mathcal{P}_n(\mathbb{R})$ are isomorphic for all nonnegative integers $n$.

To see this, we define linear transformations

$$\mathbf{T} : \mathbb{R}^{n+1} \longrightarrow \mathcal{P}_n(\mathbb{R})$$

and

$$\mathbf{S} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathbb{R}^{n+1}$$

as follows:

$$\mathbf{T}(a_1, \ldots, a_{n+1}) = a_1 + a_2 x + a_3 x^2 + \cdots + a_{n+1} x^n$$

and

$$\mathbf{S}(b_0 + b_1 x + \cdots + b_n x^n) = (b_0, b_1, \ldots, b_n).$$

It is routine to verify that **S, T** are linear transformations and that

$$\mathbf{S} \cdot \mathbf{T}(a_1, \ldots, a_{n+1})) = (a_1, \ldots, a_{n+1}) \quad \forall\, (a_1, \ldots, a_{n+1}) \in \mathbb{R}^{n+1},$$

$$\mathbf{T} \cdot \mathbf{S}(p(x)) = p(x) \qquad \forall\, p(x) \in \mathcal{P}_n(\mathbb{R}).$$

Hence **S** and **T** are isomorphisms.

In view of this example and the theorems to follow, the reader might be puzzled why, if the two vector spaces $\mathbb{R}^{n+1}$ and $\mathcal{P}_n(\mathbb{R})$ are isomorphic, we bothered to introduce both of them rather than stick to good old $\mathbb{R}^{n+1}$. The answer is not simple and involves in part the "style" of the mathematics of the second half of this century. However, to be more specific, one reason to introduce both examples is that each suggests certain natural phenomena. For example, $\mathbb{R}^{n+1}$ lends itself quite nicely to the rotations, reflections and projections (see Section 8.1 Example 1, Section 8.2 Examples 12.1, and 3) that we have discussed in this chapter. Such linear transformations make little geometric sense in $\mathcal{P}_n(\mathbb{R})$ directly. On the other hand, $\mathcal{P}_n(\mathbb{R})$ suggests the linear transformation **D**, of Example 1 of Section 8.3 which in the context of $\mathbb{R}^{n+1}$ is more than a little forced and artificial. As remarked previously, the nilpotent transformations, of which **D** is the most natural example, are destined to play a central role in the further study of linear transformations.

Finally, there is no *natural* way to construct an isomorphism from $\mathbb{R}^{n+1}$ to $\mathcal{P}_n(\mathbb{R})$. For example, we leave to the reader the verification that each of the linear transformations

$$\mathsf{T}_1(a_1, \ldots, a_{n+1}) = a_2 + a_3x + \cdots + a_{n+1}x^{n-1} + a_1x^n,$$

$$\mathsf{T}_2(a_1, \ldots, a_{n+1}) = a_1 + (a_1 + a_2)x + \cdots + (a_1 + a_2 + \cdots + a_{n+1})x^n,$$

is an isomorphisms of $\mathbb{R}^{n+1}$ to $\mathcal{P}_n(\mathbb{R})$. The ***choice*** of a particular isomorphism may simplify or complicate a problem.

For example, in the theory of approximation one considers the following problem: Given a function $f$ defined on the interval $[a, b]$ of real numbers, and the values of $f$ at the $n+1$ points $c_0, \ldots, c_n$, find a polynomial of degree $n$ taking the value $f(c_i)$ at the point $c_i$ for $i = 0, \ldots, n$ that is a good approximation to $f$. If one chooses for $\mathcal{P}_n(R)$ the basis of Lagrange interpolation polynomials (see Exercise 25 in Chapter 6), then it is a rather easy matter to show that this problem has a unique solution, and one is therefore able to estimate the deviation of $f$ from the approximating polynomial quite well.

**PROPOSITION 8.6.1:** *A linear transformation* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *is an isomorphism if and only if* $\ker(\mathsf{T}) = \{\mathbf{0}\}$ *and* $\operatorname{Im}(\mathsf{T}) = \mathcal{W}$.

**NOTE:** We are required to construct a linear transformation

$$\mathsf{S} : \mathcal{W} \longrightarrow \mathcal{V}$$

such that

$$\mathsf{S} \cdot \mathsf{T}(\mathbf{A}) = \mathbf{A} \quad \forall\, \mathbf{A} \in \mathcal{V},$$

$$\mathsf{T} \cdot \mathsf{S}(\mathbf{B}) = \mathbf{B} \quad \forall\, \mathbf{B} \in \mathcal{W}.$$

In order to do this, we will first show that the linear transformation $\mathsf{T}$ has a special property, namely for each vector $\mathbf{B}$ in $\mathcal{W}$ there is exactly one vector $\mathbf{A}$ in $\mathcal{V}$ with $\mathsf{T}(\mathbf{A}) = \mathbf{B}$. This result is important enough to be of separate interest.

**PROPOSITION 8.6.2:** *Suppose that* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *is a linear transformation with* $\ker(\mathsf{T}) = \{\mathbf{0}\}$. *Then for each vector* $\mathbf{B}$ *in* $\operatorname{Im}(\mathsf{T})$ *there is exactly one vector* $\mathbf{A}$ *in* $\mathcal{V}$ *such that* $\mathsf{T}(\mathbf{A}) = \mathbf{B}$.

**PROOF:** Since $\mathbf{B}$ is a vector in $\operatorname{Im}(\mathsf{T})$, there is certainly *at least* one vector $\mathbf{A}$ such that $\mathsf{T}(\mathbf{A}) = \mathbf{B}$. If there were another vector $\mathbf{C}$ with $\mathsf{T}(\mathbf{C}) = \mathbf{B}$, then we would have

$$\mathsf{T}(\mathbf{A} - \mathbf{C}) = \mathsf{T}(\mathbf{A}) - \mathsf{T}(\mathbf{C}) = \mathbf{B} - \mathbf{B} = \mathbf{0}.$$

Therefore $\mathbf{A} - \mathbf{C} \in \ker(\mathsf{T}) = \{\mathbf{0}\}$, and hence $\mathbf{A} - \mathbf{C} = \mathbf{0}$ which implies $\mathbf{A} = \mathbf{C}$ so $\mathbf{C}$ could not have been different from $\mathbf{A}$. □

**EXAMPLE 2 :** Consider the linear transformation of Section 8.3 Example 12.1

$$\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

defined by

$$\mathsf{T}(x, y, z) = (x, y, 0).$$

The vector $(1, 1, 0)$ belongs to $\mathrm{Im}(\mathsf{T})$, and

$$(1, 1, 0) = \mathsf{T}(1, 1, 1),$$

$$(1, 1, 0) = \mathsf{T}(1, 1, -1),$$

so more than one vector is mapped by $\mathsf{T}$ to $(1, 1, 0)$. Of course, $\ker(\mathsf{T}) = \left\{(0, 0, z) \mid z \in \mathbb{R}\right\} \neq \{\mathbf{0}\}$.

Proposition 8.6.2 is in a sense quite surprising. Of course, if $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ has the property that for each $\mathbf{B} \in \mathrm{Im}\mathsf{T}$ there is exactly one $\mathbf{A}$ in $\mathcal{V}$ with $\mathsf{T}(\mathbf{A}) = \mathbf{B}$, then $\ker(\mathsf{T}) = \{\mathbf{0}\}$. For $\mathbf{0} \in \mathrm{Im}(\mathsf{T})$, since $\mathrm{Im}(\mathsf{T})$ is a subspace of $\mathcal{W}$ and $\mathsf{T}(\mathbf{0}) = \mathbf{0}$ since $\mathsf{T}$ is a linear transformation. Therefore, if $\mathbf{A} \in \mathcal{V}$ with $\mathsf{T}(\mathbf{A}) = \mathbf{0}$ we must have $\mathbf{A} = \mathbf{0}$. What is surprising is that the converse holds, and this is the content of Proposition 8.6.2. It is a reflection of the *homogeneity* of linear transformations.

**PROOF OF PROPOSITION 8.6.1:** Since $\ker(\mathsf{T}) = \mathbf{0}$ and $\mathrm{Im}(\mathsf{T}) = \mathcal{W}$, there is for each vector $\mathbf{B}$ in $\mathcal{W}$ exactly one vector $\mathbf{A}$ in $\mathcal{V}$ such that $\mathsf{T}(\mathbf{A}) = \mathbf{B}$. Define a function

$$\mathsf{S} : \mathcal{W} \longrightarrow \mathcal{V}$$

by setting $\mathsf{S}(\mathbf{B}) = \mathbf{A}$, where $\mathbf{A} \in \mathcal{V}$ is the unique vector such that $\mathsf{T}(\mathbf{A}) = \mathbf{B}$. Note that since there is exactly one $\mathbf{A}$ in $\mathcal{V}$ with $\mathsf{T}(\mathbf{A}) = \mathbf{B}$, this definition yields a welldefined function $\mathsf{S} : \mathcal{W} \longrightarrow \mathcal{V}$.

We claim that $\mathsf{S}$ is a linear transformation. To prove this, we suppose that $\mathbf{B}, \mathbf{C}$ are vectors in $\mathcal{W}$. Let $\mathbf{A}$ and $\mathbf{D}$ be the unique vectors of $\mathcal{V}$ with

$$\mathsf{T}(\mathbf{A}) = \mathbf{B} \text{ and } \mathsf{T}(\mathbf{D}) = \mathbf{C}.$$

Since $\mathsf{T}$ is a linear transformation,

$$\mathsf{T}(\mathbf{A} + \mathbf{D}) = \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{D}) = \mathbf{B} + \mathbf{C}.$$

Therefore, $\mathbf{A} + \mathbf{D}$ is the unique vector of $\mathcal{V}$ with

$$\mathsf{T}(\mathbf{A} + \mathbf{D}) = \mathbf{B} + \mathbf{C}.$$

In terms of our function $\mathsf{S}$, this says that

$$\mathsf{S}(\mathbf{B}+\mathbf{C}) = \mathbf{A}+\mathbf{D} = \mathsf{S}(\mathbf{B}) + \mathsf{S}(\mathbf{C}).$$

If $r$ is a number, then since $\mathsf{T}$ is a linear transformation,

$$\mathsf{T}(r\mathbf{A}) = r\mathsf{T}(\mathbf{A}) = r\mathbf{B}.$$

Therefore, by the definition of $\mathsf{S}$,

$$\mathsf{S}(r\mathbf{B}) = r\mathbf{A} = r\mathsf{S}(\mathbf{A}).$$

Thus we have shown that the function

$$\mathsf{S} : \mathcal{W} \longrightarrow \mathcal{V}$$

is a linear transformation. It is immediate from the definitions that

$$\mathsf{S}(\mathsf{T}(\mathbf{A})) = \mathbf{A} \qquad \forall\, \mathbf{A} \in \mathcal{V},$$
$$\mathsf{T}(\mathsf{S}(\mathbf{B})) = \mathbf{B} \qquad \forall\, \mathbf{B} \in \mathcal{W}.$$

Therefore, $\mathsf{T}$ is an isomorphism.

To prove the converse direction, that is, if $\mathsf{T}$ is an isomorphism then $\ker(\mathsf{T}) = \{\mathbf{0}\}$, goes as follows: Suppose $\mathsf{T}(\mathbf{A}) = \mathbf{0}$. Let $\mathsf{S}$ be an inverse isomorphism to $\mathsf{T}$. Then $\mathbf{A} = \mathsf{S} \cdot \mathsf{T}(\mathbf{A}) = \mathsf{S}(\mathbf{0}) = \mathbf{0}$. □

Let us look at some examples:

**EXAMPLE 3:** Let $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ be the linear transformation

$$\mathsf{T}(a_1, a_2, a_3, a_4) = (a_1, a_1 + a_2, a_1 + a_2 + a_3, a_1 + a_2 + a_3 + a_4).$$

We claim that $\mathsf{T}$ is an isomorphism. To see this we take advantage of Theorem 8.6.1 and *do not* have to construct a linear transformation

$$\mathsf{S} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$$

such that

$$\mathsf{S} \cdot \mathsf{T}(\mathbf{A}) = \mathbf{A} \qquad \forall\, \mathbf{A} \in \mathbb{R}^4,$$
$$\mathsf{T} \cdot \mathsf{S}(\mathbf{B}) = \mathbf{B} \qquad \forall\, \mathbf{B} \in \mathbb{R}^4,$$

although such an $\mathsf{S}$ will certainly exist.

First notice that $\ker(\mathsf{T}) = \{\mathbf{0}\}$. For if

$$\mathsf{T}(a_1, a_2, a_3, a_4) = \mathbf{0},$$

then

$$(a_1, a_1 + a_2, a_1 + a_2 + a_3, a_1 + a_2 + a_3 + a_4) = (0, 0, 0, 0),$$

so

$$\begin{aligned} a_1 \qquad\qquad\qquad &= 0 \\ a_1 + a_2 \qquad\qquad &= 0 \\ a_1 + a_2 + a_3 \qquad &= 0 \\ a_1 + a_2 + a_3 + a_4 &= 0 \end{aligned}$$

or

$$a_1 = 0, \qquad a_2 = 0, \qquad a_3 = 0, \qquad a_4 = 0.$$

Next note that by the dimension formula (Theorem 8.4.2) we have

$$4 = \dim(\mathbb{R}^4) = \dim(\mathrm{Im}\mathbf{T})) + \dim(\ker(\mathbf{T})) = \dim(\mathrm{Im}(\mathbf{T})) + 0.$$

Therefore, $\dim(\mathrm{Im}(\mathbf{T})) = 4$. Thus by Corollary 6.3.6, $\mathrm{Im}(\mathbf{T}) = \mathbb{R}^4$. To summarize,

$$\mathrm{Im}(\mathbf{T}) = \mathbb{R}^4,$$

$$\ker(\mathbf{T}) = \{\mathbf{0}\},$$

so $\mathbf{T}$ is an isomorphism by Proposition 8.6.1.

Here is a nice application of the linear extension construction (see Theorem 8.5.1).

**THEOREM 8.6.3:** *Suppose that $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces. If $\dim(\mathcal{V}) = \dim(\mathcal{W})$, then $\mathcal{V}$ and $\mathcal{W}$ are isomorphic.*

**PROOF:** Let $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ be a basis for $\mathcal{V}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ a basis for $\mathcal{W}$. Note that the two bases have the same number of elements. Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be the linear transformation that is the linear extension of

$$\mathbf{T}(\mathbf{A}_i) = \mathbf{B}_i \qquad i = 1, 2, \ldots, n.$$

Since $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ is a basis for $\mathcal{W}$, it follows from Proposition 8.5.3 that $\mathrm{Im}(\mathbf{T}) = \mathcal{W}$. Therefore, by the dimension formula (Theorem 8.4.2)

$$\begin{aligned} n = \dim(\mathcal{V}) &= \dim(\mathrm{Im}(\mathbf{T})) + \dim(\ker(\mathbf{T})) \\ &= \dim(\mathcal{W}) + \dim(\ker(\mathbf{T})) \\ &= n + \dim(\ker(\mathbf{T})), \end{aligned}$$

and hence $\dim(\ker(\mathbf{T})) = n - n = 0$. So $\ker(\mathbf{T}) = \{\mathbf{0}\}$, and hence $\mathbf{T}$ is an isomorphism by Theorem 8.6.1. $\square$

Notice that the converse of Theorem 8.6.3 is also true. That is, if $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces that are isomorphic, then they must have the same dimension. To see that this is so, suppose

$$\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$$

is a linear transformation that is an isomorphism. Let $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ be a basis for $\mathcal{V}$, so that $\dim(\mathcal{V}) = n$. Then

$$\begin{aligned}\mathcal{W} &= \mathrm{Im}(\mathsf{T}) = \mathsf{T}(\mathcal{V}) = \mathsf{T}(\mathrm{Span}(\mathbf{A}_1, \ldots, \mathbf{A}_n)) \\ &= \mathrm{Span}(\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_n))\end{aligned}$$

so that the vectors $\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_n)$ span $\mathcal{W}$ and hence satisfy half of the basis condition. It remains to show that $\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_n)$ are linearly independent, for then they are a basis, and hence $\dim(\mathcal{W}) = n$ also. So suppose

$$a_1\mathsf{T}(\mathbf{A}_1) + a_2\mathsf{T}(\mathbf{A}_2) + \cdots + a_n\mathsf{T}(\mathbf{A}_n) = \mathbf{0}$$

is a linear relation among $\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_n)$. Then

$$\begin{aligned}\mathbf{0} &= a_1\mathsf{T}(\mathbf{A}_1) + a_2\mathsf{T}(\mathbf{A}_2) + \cdots + a_n\mathsf{T}(\mathbf{A}_n) \\ &= \mathsf{T}(a_1\mathbf{A}_1) + \mathsf{T}(a_2\mathbf{A}_2) + \cdots + \mathsf{T}(a_n\mathbf{A}_n) \\ &= \mathsf{T}(a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_n\mathbf{A}_n),\end{aligned}$$

and hence $a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n \in \ker(\mathsf{T})$. By Theorem 8.6.1 $\ker(\mathsf{T}) = \{\mathbf{0}\}$, so

$$a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n = \mathbf{0}.$$

Since $\mathbf{A}_1, \ldots, \mathbf{A}_n$ is a basis for $\mathcal{V}$, this says that

$$a_1 = \cdots = a_n = 0.$$

Hence the only linear relation among the vectors $\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_n)$ is the trivial relation, and hence they are linearly independent.

To summarize:

**THEOREM 8.6.4:** *Two finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$ are isomorphic if and only if they have the same dimension.* □

**COROLLARY 8.6.5:** *If a vector space $\mathcal{V}$ has finite dimension $n$, then $\mathcal{V}$ is isomorphic to $\mathbb{R}^n$.* □

## 8.7 Exercises

**1.** Show that each of the following is a linear transformation:
- (1) $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by $\mathsf{T}(x, y) = (2x - y, x)$.
- (2) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ defined by $\mathsf{T}(x, y, z) = (z, x + y)$.
- (3) $\mathsf{T} : \mathbb{R} \longrightarrow \mathbb{R}^2$ defined by $\mathsf{T}(x) = (2x, -x)$.
- (4) $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ defined by $\mathsf{T}(x, y) = (x + y, y, x)$.

**2.** Show that each of the following is *not* a linear transformation:
(1) $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by $\mathsf{T}(x, y) = (x^2, y^2)$.
(2) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ defined by $\mathsf{T}(x, y, z) = (x + y + z, 1)$.
(3) $\mathsf{T} : \mathbb{R} \longrightarrow \mathbb{R}^2$ defined by $\mathsf{T}(x) = (1, -1)$.
(4) $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ defined by $\mathsf{T}(x, y) = (xy, y, x)$.

**3.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}$ be defined by $\mathsf{T}(x, y, z) = x - 3y + 2z$. Show that $\mathsf{T}$ is linear. Find a basis for the kernel of $\mathsf{T}$. What are $\dim(\ker(\mathsf{T}))$ and $\dim(\mathrm{Im}(\mathsf{T}))$?

**4.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be defined by $\mathsf{T}(x, y, z) = (y, 0, z)$. Show that $\mathsf{T}$ is linear. Find bases for $\ker(\mathsf{T})$ and $\mathrm{Im}(\mathsf{T})$. What are their dimensions? Let $\mathcal{U} \subset \mathbb{R}^3$ be the linear subspace consisting of all vectors with $y = 0$ (that is the $xz$-plane). Find a basis for $\mathsf{T}(\mathcal{U})$.

**5.** Let $\mathsf{T} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_2(\mathbb{R})$ be the linear transformation defined by

$$\mathsf{T}(a_0 + a_1x + a_2x^2 + a_3x^3) = a_1 + a_2x + a_3x^2.$$

Find a basis for $\ker(\mathsf{T})$ and $\mathrm{Im}(\mathsf{T})$. What are their dimensions?

**6.** Let $\mathsf{T} : \mathbb{R} \longrightarrow \mathbb{R}$ be a linear transformation. Show that there exists a number $t$, depending only on $\mathsf{T}$, such that $\mathsf{T}(x) = tx$ for all $x \in \mathbb{R}$.

**7.** Suppose that $\mathsf{T} : \mathbb{R} \longrightarrow \mathbb{R}$ is a linear transformation such that $\mathsf{T}(3) = -4$. Calculate $\mathsf{T}(-7)$. (**HINT:** See the previous exercise.)

**8.** Linear transformations from $\mathbb{R}^2$ to $\mathbb{R}$ are particularly simple. To see this prove: Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be a linear transformation. Show that there exist two numbers $a$ and $b$, depending only on $\mathsf{T}$, such that $\mathsf{T}(x, y) = ax + by$ for all $(x, y)$ in $\mathbb{R}^2$.

**9.** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be a linear transformation. Suppose $\mathsf{T}(1, 1) = 3, \mathsf{T}(1, 0) = 4$. Calculate $\mathsf{T}(2, 1)$. (**HINT:** See the previous exercise.)

**10.** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be a linear transformation. Show that there are numbers $a, b \in \mathbb{R}$ such that $\mathsf{T} \cdot \mathsf{T} + a \cdot \mathsf{T} + b \cdot \mathsf{I} = \mathbf{0}$.

**11.** Let $\mathsf{S} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be the **shift operator**, which is defined by the formula $\mathsf{S}(a_1, \ldots, a_n) = (0, a_1, a_2, \ldots, a_{n-1})$. Compute the dimensions of the kernel and image of $\mathsf{S}$. Do the same for $\mathsf{S}^k = \underbrace{\mathsf{S} \cdots \mathsf{S}}_{\longleftarrow k \longrightarrow}$. Show that $\mathsf{S}^n = \mathbf{0}$.

**12.** A linear transformation $\mathsf{M} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R})$ is defined by

$$\mathsf{M}(p(x)) = xp(x),$$

that is ,

$$\mathsf{M}(a_0 + a_1x + \cdots + a_nx^n) = a_0x + a_1x^2 + \cdots + a_nx^{n+1}.$$

Find ker($\mathbf{M}$) and Im($\mathbf{M}$). Does $\mathbf{M}^n = \mathbf{0}$ for any $n$?

**13.** A linear transformation $\mathbf{T} : \mathcal{V} \longrightarrow W$ is said to be **injective** if for each pair of vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$ with $\mathbf{A} \neq \mathbf{B}$ we have $\mathbf{T}(\mathbf{A}) \neq \mathbf{T}(\mathbf{B})$. Show that $\mathbf{T}$ is injective if and only if $\ker(\mathbf{T}) = \{\mathbf{0}\}$.

**14.** Let $\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^4, \mathbf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^2$ be the linear transformations defined by

$$\mathbf{S}(a_1, a_2, a_3) = (a_1 + a_2, a_1 + a_3, a_2 + a_3, a_1 + a_2 + a_3),$$

$$\mathbf{T}(b_1, b_2, b_3, b_4) = (b_1 + b_2, b_3 + b_4).$$

Compute $\mathbf{T} \cdot \mathbf{S}(1, 1, 1)$.

**15.** Let $\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ and $\mathbf{P} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformations defined by

$$\mathbf{S}(a_1, a_2, a_3) = (0, a_1, a_2),$$

$$\mathbf{P}(a_1, a_2, a_3) = (a_1, a_2, 0).$$

Calculate $\dim(\mathrm{Im}(\mathbf{S}))$, $\dim(\ker(\mathbf{S}))$, $\dim(\mathrm{Im}(\mathbf{P}))$, $\dim(\ker(\mathbf{P}))$. Find $\mathbf{S} \cdot \mathbf{P}(1, 0, 1)$, $\mathbf{P} \cdot \mathbf{S}(1, 0, 1)$, $(\mathbf{S} + \mathbf{P})(1, 0, 1)$, and $(\mathbf{P} - 2\mathbf{S})(1, 0, 1)$.

**16.** Suppose that $\mathcal{V}$ is a finite-dimensional vector space and $\mathbf{S}, \mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ are linear transformations. Let $\mathbf{A}_1, \ldots, \mathbf{A}_n$ be a basis for $\mathcal{V}$. Suppose further that

$$\mathbf{S}(\mathbf{A}_i) = \mathbf{T}(\mathbf{A}_i), \qquad i = 1, 2, \ldots, n.$$

Show that $\mathbf{S} = \mathbf{T}$.

**17.** Suppose that $\mathcal{V}$ is a finite-dimensional vector space and $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ a linear transformation. Show that the following are equivalent:

(1) $\mathbf{T}$ is an isomorphism.
(2) $\ker(\mathbf{T}) = \{\mathbf{0}\}$ (i.e. $\mathbf{T}$ is injective).
(3) $\mathrm{Im}(\mathbf{T}) = \mathcal{V}$ (i.e. $\mathbf{T}$ is surjective).

**18.** Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with the same dimension and $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ a linear transformation. Show that $\mathbf{T}$ is injective if and only if it is surjective.

**19.** Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation and $\mathbf{A}_1, \ldots, \mathbf{A}_n$ a basis for $\mathcal{V}$. Show that $\mathbf{T}$ is injective if and only if $\mathbf{T}(\mathbf{A}_1), \ldots, \mathbf{T}(\mathbf{A}_n)$ are linearly independent.

**20.** Let $\mathcal{V}$ be a finite dimensional vector space and $\mathcal{W}', \mathcal{W}''$ vector subspaces of $\mathcal{V}$. Suppose that

$$\mathcal{W}' \cap \mathcal{W}'' = \{\mathbf{0}\},$$

$$\mathcal{W}' + \mathcal{W}'' = \mathcal{V}.$$

Show that $\mathcal{V}$ is isomorphic to $\mathcal{W}' \oplus \mathcal{W}''$. (See Chapter 2 Exercise 10 for the definition of $\mathcal{W}' \oplus \mathcal{W}''$.)

**21.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation. Show that $\mathsf{T} \cdot \mathsf{T} = \mathsf{I}$ if and only if $\mathcal{V} = \ker(\mathsf{T} - \mathsf{I}) \oplus \ker(\mathsf{T} + \mathsf{I})$. (See Chapter 2 Exercise 10 for the definition of $\mathcal{V}' \oplus \mathcal{V}''$, where $\mathcal{V}'$ and $\mathcal{V}''$ are vector spaces.)

**22.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear extension of $\mathsf{T}(\mathbf{E}_i) = \mathbf{A}_i$, $i = 1, 2, 3$, where $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ is the standard basis for $\mathbb{R}^3$ and

$$\mathbf{A}_1 = (0, 1, 1), \qquad \mathbf{A}_2 = (-1, 0, 1), \qquad \mathbf{A}_3 = (0, 1, 2).$$

Find $\mathrm{Im}(\mathsf{T})$ and $\ker(\mathsf{T})$. Also find $\mathsf{T}(1, 2, 3)$.

**23.** Let $\mathsf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be the linear extension of

$$\begin{aligned} \mathsf{T}(\mathbf{E}_1) &= (0, 0, \ldots, 0), \\ \mathsf{T}(\mathbf{E}_2) &= (1, 0, \ldots, 0), \\ \mathsf{T}(\mathbf{E}_3) &= (0, 2, 0, \ldots, 0), \\ &\vdots \\ \mathsf{T}(\mathbf{E}_n) &= (0, 0, \ldots, n-1, 0), \end{aligned}$$

where $\mathbf{E}_1, \ldots, \mathbf{E}_n \in \mathbb{R}^n$ is the standard basis. Show that $\mathsf{T}$ is a nilpotent operator and more specifically that $\mathsf{T}^n = \mathbf{0}$. Find $\mathrm{Im}(\mathsf{T})$ and $\ker(\mathsf{T})$.

**24.** Let $S$ and $T$ be finite sets. Show that $\mathrm{Fun}(S)$ and $\mathrm{Fun}(T)$ are isomorphic if and only if $S$ and $T$ have the same number of elements.

**25.** Let $S = \{a, b, c\}$ and let $\mathsf{T} : \mathrm{Fun}(S) \longrightarrow \mathbb{R}^2$ be the linear extension of

$$\begin{aligned} \mathsf{T}(\chi_a) &= (0, 1), \\ \mathsf{T}(\chi_b) &= (1, 1), \\ \mathsf{T}(\chi_c) &= (1, 0). \end{aligned}$$

Show that the kernel of $\mathsf{T}$ is the subspace spanned by the single function $f : S \longrightarrow \mathbb{R}$ defined by $f(a) = 1, f(b) = -1, f(c) = 1$.

**26.** Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces and let $S \subset \mathcal{V}$ be a subset. Define

$$\mathsf{L} : \mathrm{Fun}(S, \mathcal{W}) \longrightarrow \mathcal{L}(\mathrm{Span}(S), \mathcal{W})$$

by

$$\mathsf{L}(f)(a_1\mathbf{S}_1 + \cdots + a_n\mathbf{S}_n) = a_1 f(\mathbf{S}_1) + \cdots + a_n f(\mathbf{S}_n)$$
$$\forall\, \mathbf{S}_1, \ldots, \mathbf{S}_n \in S, a_1, \ldots, a_n \in \mathbb{R}, n \in \mathbb{N}.$$

Show that $\mathsf{L}$ is an isomorphism of vector spaces.

**27.** Let $S$ be a set and define

$$\varphi : \mathcal{L}(\mathrm{Fun}(S), \mathbb{R}) \longrightarrow \mathrm{Fun}(S)$$

by $\varphi(\mathsf{T})(s) = \mathsf{T}(\chi_s)$ for all $\mathsf{T} \in \mathcal{L}(\mathrm{Fun}(S), \mathbb{R})$, where $\chi_s$ is the characteristic function of $s$. Show that $\varphi$ is a linear transformation. Show that it is an isomorphism if $S$ is finite. What happens when $S$ is infinite?

**28.** Let $S, T$ be sets and $\varphi : S \longrightarrow T$ a function. Define

$$\varphi^* : \mathrm{Fun}(T) \longrightarrow \mathrm{Fun}(S)$$

by $\varphi^*(f)(s) = f(\varphi(s))$ for $s \in S$ and $f \in \mathrm{Fun}(T)$.

(1) Show that $\varphi^*$ is a linear transformation.
(2) Show that when $\varphi$ is surjective, $\varphi^*$ is injective.
(3) Show that when $\varphi$ is injective, $\varphi^*$ is surjective.
(4) In general, show that $\ker(\varphi^*) = \mathrm{Fun}(T, T - \varphi(S))$.

**29.** Let $\mathcal{V}', \mathcal{V}''$ be vector subspaces of the finite-dimensional vector space $\mathcal{V}$. Show that there is an isomorphism $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ such that $\mathsf{T}$ maps $\mathcal{V}'$ isomorphically to $\mathcal{V}''$ if and only if $\dim(\mathcal{V}') = \dim(\mathcal{V}'')$.

**30.** Let $\mathcal{V}$ be a finite-dimensional vector space. Consider pairs of subspaces $(\mathcal{V}_1, \mathcal{V}_2)$ in $\mathcal{V}$. We say that two such pairs $(\mathcal{V}_1', \mathcal{V}_2'')$ and $(\mathcal{V}_1', \mathcal{V}_2'')$ are **equivalent** if and only if there is an isomorphism $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ such that $\mathsf{T}$ maps $\mathcal{V}_i'$ isomorphically onto $\mathcal{V}_i$ for $i = 1, 2$. Try to find necessary and sufficient conditions for two pairs of subspaces to be equivalent. (**Hint:** Think about the formula

$$\dim(\mathcal{V}_1 + \mathcal{V}_2) = \dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - \dim(\mathcal{V}_1 \cap \mathcal{V}_2)$$

and how it is proved. Consider how many different dimensions you can associate to a pair of vector spaces $\mathcal{V}_1, \mathcal{V}_2$, and do not forget the first hint).

**31.** Let $\ell_2$ denote the set of all sequences $a_0, a_1, \ldots, a_n, \ldots$ of real numbers that are **square summable**, i.e., such that $\sum_{i=0}^{\infty}(a_i)^2 < \infty$, and let $\ell_0$ be the set of all sequences of real numbers $a_0, a_1, \ldots, a_n, \ldots$. Show that the usual rules for adding sequences and multiplying a sequence by a real number make $\ell_2$ and $\ell_0$ into vector spaces. Are they isomorphic?

# Chapter 9
# Linear Transformations: Examples and Applications

In this chapter we present some numerical examples to illustrate the discussion of linear transformations in Chapter 8. We also show how linear transformations can be applied to solve some concrete problems in linear algebra.

## 9.1 Numerical Examples

Our objective in this section is to present several numerical examples to illustrate the ideas introduced in Chapter 8 such as the kernel and image of a linear transformation, the linear extension construction, the dimension formula, etc.

**EXAMPLE 1:** Which of the following functions are linear transformations $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ and which are not?

(1) $\mathsf{T}(x, y, z) = (x + y + z, 0, 0)$.
(2) $\mathsf{T}(x, y, z) = (y, z, x)$.
(3) $\mathsf{T}(x, y, z) = (x + y + z, 1, -1)$.
(4) $\mathsf{T}(x, y, z) = (xyz, 0, 0)$.

**SOLUTION:** The first and second of these are easily seen to be linear transformations. For example, we will check (2).

Let $\mathbf{A} = (a_1, a_2, a_3)$, $\mathbf{B} = (b_1, b_2, b_3)$. Then we have

$$\begin{aligned}\mathsf{T}(\mathbf{A} + \mathbf{B}) &= \mathsf{T}(a_1 + b_1, a_2 + b_2, a_3 + b_3)\\ &= (a_2 + b_2, a_3 + b_3, a_1 + b_1)\\ &= (a_2, a_3, a_1) + (b_2, b_3, b_1)\\ &= \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{B}),\end{aligned}$$

and

$$\begin{aligned}\mathsf{T}(r\mathbf{A}) &= \mathsf{T}(r\ a_1, r\ a_2, r\ a_3)\\ &= (r\ a_2, r\ a_3, r\ a_1)\\ &= r(a_2, a_3, a_1)\\ &= r\mathsf{T}(\mathbf{A}).\end{aligned}$$

The first function be checked similarly.

Let us try to check the third transformation by this method. We get

$$\begin{aligned}\mathsf{T}(\mathbf{A}+\mathbf{B}) &= \mathsf{T}(a_1+b_1, a_2+b_2, a_3+b_3)\\ &= (a_1+b_1+a_2+b_2+a_3+b_3, 1, -1)\\ &= (a_1+a_2+a_3+b_1+b_2+b_3, 1, -1),\end{aligned}$$

while

$$\begin{aligned}\mathsf{T}(\mathbf{A})+\mathsf{T}(\mathbf{B}) &= (a_1+a_2+a_3, 1, -1)+(b_1+b_2+b_3, 1, -1),\\ &= (a_1+a_2+a_3+b_1+b_2+b_3, 2, -2)\end{aligned}$$

and we note that

$$(a_1+a_2+a_3+b_1+b_2+b_3, 1, -1) \neq (a_1+a_2+a_3+b_1+b_2+b_3, 2, -2),$$

because $1 \neq 2$. Therefore, $\mathsf{T}(\mathbf{A}+\mathbf{B}) \neq \mathsf{T}(\mathbf{A})+\mathsf{T}(\mathbf{B})$, and hence $\mathsf{T}$ is ***not*** a linear transformation.

Actually, with a little more familiarity with linear transformations it should be immediately obvious that $\mathsf{T}$ is not a linear transformation. All that is needed is a ***reason*** why it isn't. Here is one simple reason:

$$\mathsf{T}(0,0,0) = (0,1,-1) \neq (0,0,0),$$

and according to Proposition 8.3.1,

$$\mathsf{T}(0,0,0) = (0,0,0)$$

if $\mathsf{T}$ is linear. Therefore, $\mathsf{T}$ is not linear.

Let us look at the last example. If it is a linear transformation, then we must have

$$\mathsf{T}(r\mathbf{A}) = r\mathsf{T}(\mathbf{A})$$

for all numbers $r$ and vectors $\mathbf{A} = (a_1, a_2, a_3)$. Let us therefore calculate both sides of the above equation.

$$\begin{aligned}\mathsf{T}(r\mathbf{A}) &= \mathsf{T}(ra_1, ra_2, ra_3)\\ &= (ra_1ra_2ra_3, 0, 0)\\ &= (r^3a_1a_2a_3, 0, 0)\\ &= r^3(a_1a_2a_3, 0, 0),\end{aligned}$$

while

$$r\mathbf{T}(\mathbf{A}) = r(a_1a_2a_3, 0, 0).$$

So we must ask ourselves whether it is true that

$$r^3(a_1a_2a_3, 0, 0) = r(a_1a_2a_3, 0, 0)$$

for *all* numbers $r$ and vectors $\mathbf{A} = (a_1, a_2, a_3)$. The answer is clearly ***no***. For if we set $a_1 = a_2 = a_3 = 1$ and $r = 2$, then

$$r^3(a_1a_2a_3, 0, 0) = 8(1, 0, 0) = (8, 0, 0),$$
$$r(a_1a_2a_3, 0, 0) = 2(1, 0, 0) = (2, 0, 0),$$

and these vectors are clearly not equal, so $\mathbf{T}$ cannot be a linear transformation in this case.

**EXAMPLE 2 :** Let $\mathbf{T}, \mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformations given by the formulas

$$\mathbf{T}(x, y, z) = (x + y + z, 0, 0),$$
$$\mathbf{S}(x, y, z) = (y, z, x).$$

Calculate

| | | | |
|---|---|---|---|
| $(a)$ | $(\mathbf{T} \cdot \mathbf{S})(1, 0, 1)$. | $(d)$ | $(\mathbf{S} - \mathbf{T})(1, 0, 1)$. |
| $(b)$ | $(\mathbf{S} \cdot \mathbf{T})(1, 0, 1)$. | $(e)$ | $(\mathbf{S} \cdot (\mathbf{S} + \mathbf{T}) \cdot \mathbf{T})(1, 0, 1)$. |
| $(c)$ | $(\mathbf{S} + \mathbf{T})(1, 0, 1)$. | $(f)$ | $(\mathbf{T} \cdot \mathbf{S}^2 \cdot \mathbf{T})(1, 0, 1)$. |

**SOLUTION :** This requires nothing but determination:

(a)
$$\begin{aligned}(\mathbf{T} \cdot \mathbf{S})(1, 0, 1) &= \mathbf{T}(\mathbf{S}(1, 0, 1)) && \text{(by definition of composition)}\\ &= \mathbf{T}(0, 1, 1) && \text{(by definition of } \mathbf{S})\\ &= (2, 0, 0). && \text{(by definition of } \mathbf{T})\end{aligned}$$

(b)
$$\begin{aligned}(\mathbf{S} \cdot \mathbf{T})(1, 0, 1) &= \mathbf{S}(\mathbf{T}(1, 0, 1))\\ &= \mathbf{S}(2, 0, 0)\\ &= (0, 0, 2).\end{aligned}$$

(c)
$$\begin{aligned}(\mathbf{S} + \mathbf{T})(1, 0, 1) &= \mathbf{S}(1, 0, 1) + \mathbf{T}(1, 0, 1)\\ &= (0, 1, 1) + (2, 0, 0)\\ &= (2, 1, 1).\end{aligned}$$

(d)
$$\begin{aligned}(\mathbf{S} - \mathbf{T})(1, 0, 1) &= \mathbf{S}(1, 0, 1) - \mathbf{T}(1, 0, 1)\\ &= (0, 1, 1) - (2, 0, 0)\\ &= (-2, 1, 1).\end{aligned}$$

(e)
$$\begin{aligned}(\mathbf{S}(\mathbf{S} + \mathbf{T})\mathbf{T})(1, 0, 1) &= (\mathbf{S}(\mathbf{S} + \mathbf{T}))(\mathbf{T}(1, 0, 1)\\ &= (\mathbf{S}(\mathbf{S} + \mathbf{T}))(2, 0, 0)\\ &= \mathbf{S}((\mathbf{S} + \mathbf{T})(2, 0, 0))\end{aligned}$$

$$\begin{aligned} &= \mathbf{S}(\mathbf{S}(2,0,0) + \mathbf{T}(2,0,0)) \\ &= \mathbf{S}((0,0,2) + (2,0,0)) \\ &= \mathbf{S}(2,0,2) = (0,2,2). \end{aligned}$$

(f)
$$\begin{aligned} \mathbf{TS}^2\mathbf{T}(1,0,0) &= \mathbf{TS}^2(1,0,0) = \mathbf{T}(0,1,0) \\ &= (1,0,0). \end{aligned}$$

**EXAMPLE 3:** Let $\mathbf{T} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathbb{R}^2$ be the function defined by

$$\mathbf{T}(p(x)) = \left(p(0), p'(0)\right).$$

($p'(x)$ is the derivative of $p(x)$ and $p'(0)$ the value of $p'(x)$ at $x = 0$.) Is $\mathbf{T}$ a linear transformation?

**SOLUTION:** Recall that the vector operations in $\mathcal{P}_n(\mathbb{R})$ are defined by the rules

$$(p+q)(x) = p(x) + q(x) \ \forall\ x \in \mathbb{R},$$
$$r \cdot p(x) = rp(x) \ \forall\ x \in \mathbb{R}.$$

Therefore, if $p(x), q(x) \in \mathcal{P}_n(\mathbb{R})$ are polynomials and we put $x = 0$ in the rule for vector addition, we find

$$(p+q)(0) = p(0) + q(0).$$

Passing from a polynomial to its derivative is also a linear operation, i.e.,

$$\frac{d}{dx}(p(x) + q(x)) = \frac{d}{dx}(p(x)) + \frac{d}{dx}(q(x)),$$

and therefore

$$(p+q)'(0) = (p' + q')(0) = p'(0) + q'(0).$$

So from the definition of $\mathbf{T}$ we find

$$\begin{aligned} \mathbf{T}(p(x) + q(x)) &= (p(0) + q(0), p'(0) + q'(0)) \\ &= (p(0), p'(0)) + (q(0), q'(0)) = \mathbf{T}(p(x)) + \mathbf{T}(q(x)), \end{aligned}$$

and hence $\mathbf{T}$ is a linear transformation

**EXAMPLE 4:** Let $\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation[1] given by the formula

$$\mathbf{S}(x, y, z)) = (0, x, y).$$

Find the bases for the kernel and image of $\mathbf{S}$, $\mathbf{S}^2$, and $\mathbf{S}^3$.

---

[1] This transformation is called a **shift operator** (see Exercise 11 in Chapter 8).

**SOLUTION:** Let us work with $\mathbf{S}$ first. The kernel of $\mathbf{S}$ consists of all the vectors $\mathbf{A} = (a_1, a_2, a_3)$ such that $\mathbf{S}(\mathbf{A}) = \mathbf{0}$. This means $\mathbf{A} \in \ker(\mathbf{S})$ if and only if

$$(0, 0, 0) = \mathbf{S}(\mathbf{A}) = (0, a_1, a_2),$$

that is,

$$0 = 0, \qquad 0 = a_1, \qquad 0 = a_2,$$

and there is *no restriction* on $a_3$. Thus

$$\ker(\mathbf{S}) = \left\{(0, 0, z) \in \mathbb{R}^3\right\},$$

or in geometric terms the kernel of $\mathbf{S}$ is the $z$-axis. Therefore a basis for the kernel of $\mathbf{S}$ consists of the single vector $\{(0, 0, 1)\}$. (There are many other bases, for example, $\{(0, 0, -99/100)\}$ is also a basis for $\ker(\mathbf{S})$.)

To find a basis for the image of $\mathbf{S}$ we note that $\mathbf{B} \in \mathrm{Im}(\mathbf{S})$ if and only if $\mathbf{B} = \mathbf{S}(\mathbf{A})$ for some vector $\mathbf{A} \in \mathbb{R}^3$. Therefore, $\mathbf{B} \in \mathrm{Im}(\mathbf{S})$ if and only if

$$(b_1, b_2, b_3) = \mathbf{B} = \mathbf{S}(\mathbf{A}) = (0, a_1, a_2),$$

that is,[2]

$$b_1 = 0, \qquad b_2 = a_1, \qquad b_3 = a_2,$$

which means that

$$\mathrm{Im}(\mathbf{S}) = \{(0, b_2, b_3)\},$$

or put geometrically, $\mathrm{Im}(\mathbf{S})$ is the $yz$-plane. Therefore, a basis for $\mathrm{Im}(\mathbf{S})$ is the set $\{(0, 1, 0), (0, 0, 1)\}$ and $\{(0, 1, -1), (0, 1, 0)\}$ is another basis.

As a partial check on our work, we note that by the dimension formula (Theorem 8.4.2), that

$$\dim(\ker(\mathbf{S})) + \dim(\mathrm{Im}\mathbf{S}) = 1 + 2 = 3 = \dim(\mathbb{R}^3),$$

just as the dimension formula predicted.

We must apply the same analysis to the linear transformation $\mathbf{S}^2$. First let us find a "formula" for $\mathbf{S}^2$. We have

$$\begin{aligned}\mathbf{S}^2(x, y, z) &= \mathbf{S}(\mathbf{S}(x, y, z)) = \mathbf{S}(0, x, y)\\ &= (0, 0, x).\end{aligned}$$

(Remember that $\mathbf{S}$ shifts the coordinates of a vector to the right one unit, inserts a 0 in the first place and strikes off the third coordinate.) So a vector $\mathbf{A}$ belongs to the kernel of $\mathbf{S}^2$ if and only if

$$(0, 0, 0) = \mathbf{S}^2(x, y, z) = (0, 0, a_1),$$

---

[2] For example, $(0, 2, 3) = \mathbf{S}(2, 3, 0)$, and $(0, 2, 3) = \mathbf{S}(2, 3, 1)$, etc.

that is,

$$0 = 0, \qquad 0 = 0, \qquad 0 = a_1,$$

so that $\mathbf{A} \in \ker(\mathbf{S}^2)$ if and only if $a_1 = 0$, while $a_2$ and $a_3$ are arbitrary. Thus

$$\ker(\mathbf{S}^2) = \left\{(0, a_2, a_3) \in \mathbb{R}^3\right\}$$

or geometrically, $\ker(\mathbf{S}^2)$ is the $yz$-plane, and $\{(0, 1, 0), (0, 0, 1)\}$ is a basis for $\ker(\mathbf{S}^2)$.

To find a basis for the image of $\mathbf{S}^2$ we note that $\mathbf{B} \in \mathrm{Im}(\mathbf{S}^2)$ if and only if $\mathbf{B} = (\mathbf{S}^2)(\mathbf{A})$ for some vector $\mathbf{A}$ in $\mathbb{R}^3$. That is,

$$(b_1, b_2, b_3) = \mathbf{B} = \mathbf{S}^2(\mathbf{A}) = (0, 0, a_1),$$

or

$$b_1 = 0, \qquad b_2 = 0, \qquad b_3 = a_1.$$

Therefore, we find that

$$\mathrm{Im}(\mathbf{S}^2) = \left\{(0, 0, z) \in \mathbb{R}^3\right\},$$

or $\mathrm{Im}(\mathbf{S}^2)$ is the $z$-axis, which has basis $\{(0, 0, 1)\}$.

Again we may check our work with the dimension formula:

$$\dim(\ker(\mathbf{S}^2)) + \dim(\mathrm{Im}(\mathbf{S}^2)) = 2 + 1 = 3 = \dim(\mathbb{R}^3),$$

just as required.

Finally, we note that

$$\mathbf{S}^3(x, y, z) = (0, 0, 0),$$

so that $\mathrm{Im}(\mathbf{S}^3) = \{(0, 0, 0)\}$ and $\ker(\mathbf{S}^3) = \mathbb{R}^3$. A basis for the kernel of $\mathbf{S}^3$ is thus any basis for $\mathbb{R}^3$, such as $\{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$, and a basis for the image of $\mathbf{S}^3$ is $\varnothing$, the empty set.

**EXAMPLE 5:** Let $\mathbf{S}, \mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by be the linear transformations

$$\mathbf{S}(x, y, z) = (y, z, x),$$
$$\mathbf{T}(x, y, z) = (x + y + z, 0, 0).$$

Find a basis for the kernel of $\mathbf{S} + \mathbf{T}$.

**SOLUTION :** Let us find a formula for $\mathsf{S}+\mathsf{T}$. Notice that

$$\begin{aligned}(\mathsf{S}+\mathsf{T})(x, y, z) &= \mathsf{S}(x, y, z)+\mathsf{T}(x, y, z)\\ &= (y, z, x)+(x+y+z, 0, 0)\\ &= (x+2y+z, z, x).\end{aligned}$$

That is,

$$(\mathsf{S}+\mathsf{T})(x, y, z) = (x+2y+z, z, x).$$

Therefore, $\mathbf{A} \in \ker(\mathsf{S}+\mathsf{T})$ if and only if

$$(0,0,0) = (\mathsf{S}+\mathsf{T})(\mathbf{A}) = (a_1+2a_2+a_3, a_3, a_1),$$

or

$$0 = a_1+2a_2+a_3, \qquad 0 = a_3, \qquad 0 = a_1,$$

which is true if and only if

$$0 = a_2, \qquad 0 = a_3, \qquad 0 = a_1,$$

therefore, $\ker(\mathsf{S}+\mathsf{T}) = \{(0,0,0)\}$ so the empty set is a basis for $\ker(\mathsf{S}+\mathsf{T})$.

Notice that from the dimension formula Theorem 8.4.2 it follows that $\dim(\mathrm{Im}(\mathsf{S}+\mathsf{T})) = 3$ and therefore that $\mathrm{Im}(\mathsf{S}+\mathsf{T}) = \mathbb{R}^3$. You might want to verify this by showing, for example, that the three standard basis vectors $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ for $\mathbb{R}^3$ belong to $\mathrm{Im}(\mathsf{S}+\mathsf{T})$: This is an unpleasant computation.

**EXAMPLE 6 :** Which of the following linear transformations are isomorphisms

(a) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathcal{P}_2(\mathbb{R})$ defined by
$$\mathsf{T}(a_1, a_2, a_3) = a_1+(a_1+a_2)x+(a_1+a_2+a_3)x^2.$$
(b) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathcal{P}_2(\mathbb{R})$ defined by $\mathsf{T}(a_1, a_2, a_3) = a_1+(a_1+a_2)x$.

(c) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ defined by
$$\mathsf{T}(a_1, a_2, a_3) = (a_1+a_2, a_2+a_3, a_3+a_1, a_1+a_2+a_3).$$
(d) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by
$$\mathsf{T}(a_1, a_2, a_3) = (a_1+a_2+a_3, a_1+a_2, a_1+a_2).$$

**SOLUTION :** Remember, to check that $\mathsf{T}$ is an isomorphism you do ***not*** want to use the definition of an isomorphism if you can avoid it, because that requires a construction of another linear transformation $\mathsf{S}$, and that may be quite hard to do. Instead, we will try to use properties of linear transformations such as Proposition 8.6.1, Theorem 8.6.3.

(a) Note that $\ker(\mathsf{T}) = \{\mathbf{0}\}$ because

$$\mathbf{0} = \mathsf{T}(\mathbf{A}) = \mathsf{T}(a_1, a_2, a_3) = a_1+(a_1+a_2)x+(a_1+a_2+a_3)x^2$$

if and only if

$$0 = a_1, \qquad 0 = a_1 + a_2, \qquad 0 = a_1 + a_2 + a_3,$$

or

$$a_1 = 0, \qquad a_2 = 0, \qquad a_3 = 0.$$

Next we apply the dimension formula (Theorem 8.4.2), which gives

$$3 = \dim(\mathbb{R}^3) = \dim(\ker(\mathsf{T})) + \dim(\operatorname{Im}(\mathsf{T})) = 0 + \dim(\operatorname{Im}(\mathsf{T})),$$

so

$$\dim(\operatorname{Im}(\mathsf{T})) = 3.$$

Recall that $\dim(\mathcal{P}_2(\mathbb{R})) = 3$. Therefore, $\operatorname{Im}(\mathsf{T})$, which is a linear subspace of $\mathcal{P}_2(\mathbb{R})$ by Proposition 8.3.3, has the same dimension as $\mathcal{P}_2(\mathbb{R})$ and hence $\operatorname{Im}(\mathsf{T}) = \mathcal{P}_2(\mathbb{R})$. Therefore, $\mathsf{T}$ is an isomorphism by Proposition 8.6.1.

For those who are interested, the linear transformation

$$\mathsf{S} : \mathcal{P}_2(\mathbb{R}) \longrightarrow \mathbb{R}^3$$

given by the formula

$$\mathsf{S}(b_0 + b_1x + b_2x^2) = (b_0, b_1 - b_0, b_2 - b_1 - b_0)$$

is actually the transformation required by the definition, and whose existence is assured by Proposition 8.6.1.

(b) This transformation is *not* an isomorphism. We may see this by applying Proposition 8.6.1, because

$$\mathsf{T}(0, 0, 1) = \mathbf{0},$$

and hence $\ker(\mathsf{T}) \neq \{\mathbf{0}\}$.

(c) This is *not* an isomorphism, because if it were, then by Theorem 8.6.4 we would have $3 = \dim(\mathbb{R}^3) = \dim(\mathbb{R}^4) = 4$, which is impossible.

(d) This transformation is also *not* an isomorphism. To see this we note that $\mathbf{B} \in \operatorname{Im}(\mathsf{T})$ if and only if there is an $\mathbf{A}$ with $\mathbf{B} = \mathsf{T}(\mathbf{A})$, so in this case

$$(b_1, b_2, b_3) = \mathbf{B} = \mathsf{T}(\mathbf{A}) = (a_1 + a_2 + a_3, a_1 + a_2, a_1 + a_2),$$

and therefore

$$\begin{aligned} b_1 &= a_1 + a_2 + a_3, \\ b_2 &= a_1 + a_2, \\ b_3 &= a_1 + a_2, \end{aligned}$$

and in particular, $b_2 = b_3$. Therefore, the vector $(0, 1, 2)$ does not belong to $\operatorname{Im}(\mathsf{T})$, so by Proposition 8.6.1 $\mathsf{T}$ cannot be an isomorphism, since $\operatorname{Im}(\mathsf{T}) \neq \mathbb{R}^3$.

**EXAMPLE 7:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathcal{P}_5(\mathbb{R})$ be the linear transformation that is the linear extension of

$$\mathsf{T}(1, 1, 1) = x^2 + x^4,$$
$$\mathsf{T}(1, 1, 0) = x + x^3 + x^5,$$
$$\mathsf{T}(1, 0, 0) = 1.$$

Calculate $\mathsf{T}(0, 0, 1)$.

**SOLUTION:** First of all we ought to note that the vectors $\{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ are a basis for $\mathbb{R}^3$, so that transformation $\mathsf{T}$ is welldefined by the linear extension construction. To calculate $\mathsf{T}(0, 0, 1)$ we must first write

$$(0, 0, 1) = a_1(1, 1, 1) + a_2(1, 1, 0) + a_3(1, 0, 0),$$

that is, we must find the coordinates of $(0, 0, 1)$ relative to the basis (*ordered* basis) $(1, 1, 1), (1, 1, 0), (1, 0, 0)$. This is just some elementary algebra. Suppose

$$(0, 0, 1) = a_1(1, 1, 1) + a_2(1, 1, 0) + a_3(1, 0, 0).$$

Then

$$\begin{aligned}(0, 0, 1) &= (a_1, a_1, a_1) + (a_2, a_2, 0) + (a_3, 0, 0)\\ &= (a_1 + a_2 + a_3, a_1 + a_2, a_1),\end{aligned}$$

so

$$0 = a_1 + a_2 + a_3, \qquad 0 = a_1 + a_2, \qquad 1 = a_1,$$

or

$$a_1 = 1, \qquad a_2 = -1, \qquad a_3 = 0.$$

So

$$(0, 0, 1) = 1(1, 1, 1) - 1(1, 1, 0) + 0(1, 0, 0),$$

and therefore

$$\begin{aligned}\mathsf{T}(0, 0, 1) &= 1(x^2 + x^4) - 1(x + x^3 + x^5) + 0 \cdot 1\\ &= -x + x^2 - x^3 + x^4 - x^5.\end{aligned}$$

Notice that since the set of vectors

$$\left\{x^2 + x^4, x + x^3 + x^5, 1\right\}$$

is linearly independent in $\mathcal{P}_5(\mathbb{R})$, the kernel of the transformation $\mathsf{T}$ of the preceding example is $\{\mathbf{0}\}$. (Why? Because of the dimension formula 8.4.2. See also Exercise 21 in Chapter 8.)

Here is another example involving linear extensions.

**EXAMPLE 8:** Let $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^2$ be the linear transformation that is the linear extension of

$$\begin{aligned}
\mathsf{T}(1, 0, 0, 0) &= (1, 1),\\
\mathsf{T}(1, 1, 0, 0) &= (0, 1),\\
\mathsf{T}(1, 1, 1, 0) &= (1, 0),\\
\mathsf{T}(1, 1, 1, 1) &= (-1, -1).
\end{aligned}$$

Calculate $\mathsf{T}(4, 3, 2, 1)$.

**SOLUTION:** Again, as in Example 7 in this section, the first step will be to find coordinates of $(4, 3, 2, 1)$ relative to the (***ordered***) basis

$$\{(1, 0, 0, 0), (1, 1, 0, 0), (1, 1, 1, 0), (1, 1, 1, 1)\}$$

for $\mathbb{R}^4$. So write

$$(4, 3, 2, 1) = a_1(1, 0, 0, 0) + a_2(1, 1, 0, 0) + a_3(1, 1, 1, 0) + a_4(1, 1, 1, 1)$$

and use the method of equating coefficients to compute $a_1$, $a_2$, $a_3$, $a_4$.

$$\begin{aligned}
(4, 3, 2, 1) &= (a_1, 0, 0, 0) + (a_2, a_2, 0, 0) + (a_3, a_3, a_3, 0) + (a_4, a_4, a_4, a_4)\\
&= (a_1 + a_2 + a_3 + a_4, a_2 + a_3 + a_4, a_3 + a_4, a_4),
\end{aligned}$$

so

$$\begin{array}{rcrcrcrcr}
4 &=& a_1 &+& a_2 &+& a_3 &+& a_4,\\
3 &=& & & a_2 &+& a_3 &+& a_4,\\
2 &=& & & & & a_3 &+& a_4,\\
1 &=& & & & & & & a_4,
\end{array}$$

which implies

$$a_1 = 1, \qquad a_2 = 1, \qquad a_3 = 1, \qquad a_4 = 1.$$

Thus

$$(4, 3, 2, 1) = 1(1, 0, 0, 0) + 1(1, 1, 0, 0) + 1(1, 1, 1, 0) + 1(1, 1, 1, 1),$$

so

$$\begin{aligned}
\mathsf{T}(4, 3, 2, 1) &= (1, 1) + (0, 1) + (1, 0) + (-1, -1)\\
&= (1 + 0 + 1 - 1, 1 + 1 + 0 - 1)\\
&= (1, 1).
\end{aligned}$$

## 9.2 Some Applications

In approximation theory one often studies the problem of approximating a function by simpler functions, subject to the constraint that the function and its approximation agree at some fixed set of points. To be specific, let us suppose that $f$ is a realvalued function and $t_0 < t_1 < \cdots < t_n \in \mathbb{R}$ are fixed points. We seek a polynomial of degree $n$ that agrees with $f$ at $t_0, t_1, \ldots, t_n$. In fact, we will show that there is a unique such polynomial, and we will develop a formula for it.

To this end we introduce[3] the **Lagrange interpolation polynomials** of degree $n$, $\Phi_0, \Phi_1, \ldots, \Phi_n$, which are defined as follows:

$$\Phi_j(x) = \prod_{i \neq j} \frac{x - t_i}{t_j - t_i}, \qquad j = 0, 1, \ldots, n.$$

**Proposition 9.2.1:** *For each integer $n$, the $n + 1$ Lagrange interpolation polynomials $\Phi_0, \Phi_1, \ldots, \Phi_n$ are a basis for $\mathcal{P}_n(\mathbb{R})$.*

**Proof:** Note that

$$\Phi_j(t_i) = \delta_{i,j} = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j, \end{cases}$$

and therefore the linear transformation

$$\mathsf{T} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathbb{R}^{n+1}$$

defined by

$$\mathsf{T}(p) = (p(t_0), p(t_1), \ldots, p(t_n))$$

maps $\Phi_j$ to the $(j + 1)$-st standard basis vector of $\mathbb{R}^n$, namely, $\mathbf{E}_{j+1} = (0, \ldots, 0, 1, 0, \ldots, 0)$, where naturally the 1 appears in the $(j+1)$-st coordinate $(j = 0, 1, \ldots, n)$. Therefore, the image of $\mathsf{T}$ is all of $\mathbb{R}^{n+1}$. Hence, since $\mathcal{P}_n(\mathbb{R})$ and $\mathbb{R}^{n+1}$ have the same dimension, $\mathsf{T}$ must be an isomorphism, and therefore $\Phi_0, \Phi_1, \ldots, \Phi_n$ is a basis for $\mathcal{P}_n(\mathbb{R})$. $\square$

**Corollary 9.2.2:** *Let $f$ be a realvalued function of one real variable and $t_0 < t_1 < \cdots < t_n \in \mathbb{R}$. Then there exists a unique polynomial $P_f(t)$ of degree $n$, depending of course on $f$, such that*

$$f(t_i) = P_f(t_i), \qquad i = 0, 1, \ldots, n.$$

---

[3] The diligent reader will already have encountered these in Exercise 25 of Chapter 6.

**PROOF :** The polynomial

$$P_f(t) = \sum_{i=0}^{n} f(t_i)\Phi_i(t)$$

has degree $n$ and agrees with $f(t)$ at the points $t_0, t_1, \ldots, t_n$. If $Q_f(t)$ is any other polynomial of degree $n$ with this property, then, since $\Phi_0, \Phi_1, \ldots, \Phi_n$ is a basis for $\mathcal{P}(\mathbb{R})$ (Proposition 9.2.1), it follows that

$$Q_f = \sum_{j=0}^{n} a_j\Phi_j$$

for certain uniquely defined numbers $a_0, a_1, \ldots, a_n$. Evaluating this equation at $t_i$ yields

$$f(t_i) = Q_f(t_i) = \sum_{j=0}^{n} a_j\Phi_j(t_i) = a_i,$$

and hence $Q_f = P_f$, as required. $\square$

## 9.3 Exercises

**1.** Let $k(x)$ be a *fixed* polynomial. Define functions

$$\mathbf{M}_{k(x)} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R}),$$
$$\mathbf{L}_{k(x)} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R}),$$

by

$$\mathbf{M}_{k(x)}(p(x)) = k(x)p(x),$$
$$\mathbf{L}_{k(x)}(p(x)) = \int_0^x p(t)k(t)dt.$$

If $k(x) = 1$ for all $x \in \mathbb{R}$, we drop the subscript and write $\mathbf{M}$ and $\mathbf{L}$.

(1) Show that $\mathbf{M}_{k(x)}$ and $\mathbf{L}_{k(x)}$ are linear transformations for all polynomials $k(x)$. Calculate

$$\mathbf{M}_{x^2+2x^3}(x + x^4) \quad \text{and} \quad \mathbf{L}_{x+x^2}(1 + x^3).$$

(2) Calculate $\mathbf{D} \cdot \mathbf{M}_x - \mathbf{M}_x \cdot \mathbf{D} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R})$, where $\mathbf{D}$ is defined in Example 2 of Section 8.1. That is find a *formula* for

$$(\mathbf{D} \cdot \mathbf{M}_x - \mathbf{M}_x \cdot \mathbf{D})(p(x)).$$

(3) Show that $\mathbf{L} \cdot \mathbf{M}_{k(x)} = \mathbf{L}_{k(x)} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R})$.

**2.** Show that $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$\mathsf{T}(x, y) = (ax + by, cx + dy),$$

$a, b, c, d$ fixed real numbers, is a linear transformation and that $\mathsf{T}$ is an isomorphism if and only if $ad \neq bc$.

**3.** Show that $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by

$$\mathsf{T}(x, y, z) = (a_1x + a_2y + a_3z, b_1x + b_2y + b_3z, c_1x + c_2y + c_3z)$$

is a linear transformation. Any linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$ takes this form, where $a_i, b_j, c_k$ are scalars.

**4.** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be defined by $\mathsf{T}(x, y) = (x - y, 2x + y)$ and let $\mathsf{S} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be defined by $\mathsf{S}(x, y) = (y - 2x, x + y)$.

(1) Find $\mathsf{T} \cdot \mathsf{S}(1, 0)$, $(\mathsf{T} + \mathsf{S})(1, 0)$, $\mathsf{S} \cdot \mathsf{T}(1, 0)$, $(2\mathsf{T} - \mathsf{S})(1, 0)$, $\mathsf{S}^2(1, 0)$, and $\mathsf{T}^2(1, 0)$.

(2) Find $\mathsf{T} \cdot \mathsf{S}(x, y)$ and $\mathsf{S} \cdot \mathsf{T}(x, y)$.

(3) What are the vectors $(x, y)$ satisfying $\mathsf{T}(x, y) = (1, 0)$?

**5.** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be defined by $\mathsf{T}(x, y) = (x, x)$. What are the kernel of $\mathsf{T}$ and the image of $\mathsf{T}$?

**6.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be defined by $\mathsf{T}(x, y, z) = (x + y, y + z, z + x)$. Is $\mathsf{T}$ an isomorphism? Find $\mathsf{T}(1, 0, 0)$, $\mathsf{T}(0, 1, 0)$ and $\mathsf{T}(0, 0, 1)$.

**7.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation satisfying the condition

$$\mathsf{T}(x, y, z) = (0, 0, 0) \quad \text{whenever} \quad 2x - y + z = 0.$$

Let $\mathsf{T}(0, 0, 1) = (1, 2, 3)$. Find $\mathsf{T}(1, 0, 0)$ and $\mathsf{T}(0, 1, 0)$. What is $\dim(\mathrm{Im}(\mathsf{T}))$?

**8.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation given by the formula

$$\mathsf{T}(x, y, z) = (y + z, x + z, x + y).$$

Show that $\mathsf{T}$ is an isomorphism and find an inverse for $\mathsf{T}$.

**9.** Let $\mathcal{V}$ be a finite-dimensional vector space over $\mathbb{R}$, and $\mathbf{E}_1, \ldots, \mathbf{E}_n$ a basis for $\mathcal{V}$. Define for each $i$, $1 \leq i \leq n$, a function

$$\mathsf{E}_i^* : \mathcal{V} \longrightarrow \mathbb{R}$$

by the formula

$$\mathsf{E}_i^*(\mathbf{A}) = a_i, \qquad 1 \leq i \leq n,$$

where

$$\mathbf{A} = a_1\mathbf{E}_1 + \cdots + a_n\mathbf{E}_n.$$

(1) Show that $\mathsf{E}_1^*, \ldots, \mathsf{E}_n^*$ are linear transformations.

(2) Find a basis for $\ker(\mathsf{E}_i^*)$.

(3) Let $\mathbf{S} : \mathcal{V} \longrightarrow \mathcal{L}(V, \mathbb{R})$ be the linear extension of the map

$$\mathbf{E}_i \longrightarrow \mathbf{E}_i^*, \qquad i = 1, \ldots, n.$$

Show that $\mathbf{S}$ is an isomorphism.

(4) Compute $\dim_{\mathbb{R}} \mathcal{L}(\mathcal{V}, \mathbb{R})$.

**10.** Let $\mathcal{V}$ be a vector space and define $\mathcal{V}^* = \mathcal{L}(\mathcal{V}, \mathbb{R})$. Define a linear transformation

$$\mathbf{B} : \mathcal{V} \longrightarrow \mathcal{V}^{**} \qquad (= \mathcal{L}(\mathcal{V}^*, \mathbb{R}) = \mathcal{L}(\mathcal{L}(\mathcal{V}, \mathbb{R}), \mathbb{R}))$$

by

$$\mathbf{B}(\mathbf{A})(f) = f(\mathbf{A}).$$

(1) Show that $\mathbf{B}$ is a linear transformation.
(2) Show that $\ker(\mathbf{B}) = \{\mathbf{0}\}$.
(3) Show that $\mathbf{B}$ is an isomorphism if $\mathcal{V}$ is finite-dimensional.
(4) Show by an example that $\mathrm{Im}(\mathbf{B}) \neq \mathcal{V}^{**}$ when $\mathcal{V}$ is not finite-dimensional.

**11.** Let $E = \{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ be a basis for the finite-dimensional vector space $\mathcal{V}$. Show that the linear extension construction gives an isomorphism

$$\mathbf{L} : \mathrm{Fun}(E) \longrightarrow \mathcal{V}$$

by

$$\mathbf{L}(f) = f(\mathbf{E}_1) \cdot \mathbf{E}_1 + \cdots + f(\mathbf{E}_n) \cdot \mathbf{E}_n.$$

**12.** Let $E = \{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ be a basis for the finite-dimensional vector space $\mathcal{V}$. Show that the map (remember, $\mathcal{V}^* = \mathcal{L}(\mathcal{V}, \mathbb{R})$ and the remarks following Theorem 8.5.6)

$$\mathbf{C} : \mathcal{V}^* \longrightarrow \mathrm{Fun}(E)$$

defined by

$$\mathbf{C}(\mathbf{T})(\mathbf{E}_i) = \mathbf{T}(\mathbf{E}_i), \qquad i = 1, \ldots, n,$$

is a vectorspace isomorphism.

**13.** Define a function

$$\mathbf{S} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R})$$

by

$$\mathbf{S}(p(x)) = p(x + 1).$$

Determine whether $\mathbf{S}$ is a linear transformation.

**14.** Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces. Suppose $E = \{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ is a basis for $\mathcal{V}$ and $\{\mathbf{F}_1, \ldots, \mathbf{F}_m\}$ a basis for $\mathcal{W}$. For each ordered pair $(i, j)$ show that the function $\mathsf{M}_{i,j} : \mathcal{V} \longrightarrow \mathcal{W}$, defined by the formula

$$\mathsf{M}_{i,j}(\mathbf{A}) = a_j \mathbf{F}_i, \qquad \mathbf{A} = a_1\mathbf{E}_1 + \cdots + a_n\mathbf{E}_n,$$

is a linear transformation.

(1) Show that $\left\{\mathsf{M}_{i,j} \mid 1 \leq i \leq n, 1 \leq j \leq\right\}$ is a basis for $\mathcal{L}(V, W)$.

(2) Show that $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = nm$.

**15.** Let $\mathcal{V}$ be a vector space and suppose that $\mathsf{S}, \mathsf{T} \in \mathcal{V}^* = \mathcal{L}(\mathcal{V}, \mathbb{R})$. Define

$$\mathsf{L} : \mathcal{V} \longrightarrow \mathbb{R}^2$$

by

$$\mathsf{L}(\mathbf{A}) = (\mathsf{S}(\mathbf{A}), \mathsf{T}(\mathbf{A})).$$

(1) Show that $\mathsf{L}$ is a linear transformation.
(2) Show that $\ker(\mathsf{L}) = \ker(\mathsf{S}) \cap \ker(\mathsf{T})$.

**16.** Let $\mathcal{V}$ be a vector space and define (see Theorem 8.5.6)

$$\mathsf{C}_i : \mathcal{L}(V, \mathbb{R}) \longrightarrow \mathcal{L}(V, \mathbb{R}^2), \qquad i = 1, 2,$$

by

$$\begin{aligned} \mathsf{C}_1(\mathsf{T})(\mathbf{A}) &= (\mathsf{T}(\mathbf{A}), 0), \\ \mathsf{C}_2(\mathsf{T})(\mathbf{A}) &= (0, \mathsf{T}(\mathbf{A})), \end{aligned} \qquad \forall\, \mathbf{A} \in \mathcal{V}, \mathsf{T} \in \mathcal{L}(\mathcal{V}, \mathbb{R}).$$

(1) Show that $\mathsf{C}_i$ is a linear transformation for $i = 1, 2$.
(2) Show that $\ker(\mathsf{C}_i) = \{\mathbf{0}\}$, $i = 1, 2$.
(3) Show that $\operatorname{Im}(\mathsf{C}_1) \cap \operatorname{Im}(\mathsf{C}_2) = \{\mathbf{0}\}$.
(4) Let $\mathcal{L}_i = \operatorname{Im}(\mathsf{C}_i)$, $i = 1, 2$ and show that

$$\mathcal{L}(V, \mathbb{R}^2) = \mathcal{L}_1 + \mathcal{L}_2.$$

**17.** Let $\mathcal{V}$ be a vector space over $\mathbb{R}$. A **complex structure** on $\mathcal{V}$ is a linear transformation

$$\mathsf{J} : \mathcal{V} \longrightarrow V$$

such that for all $\mathbf{A}$ in $\mathcal{V}$,

$$\mathsf{J}^2(\mathbf{A}) = -\mathbf{A}.$$

(1) Show that

$$\mathsf{J}(x, y) = (-y, x)$$

is a complex structure on $\mathbb{R}^2$.

(2) Show that

$$\mathsf{J}(x, y) = (y, -x)$$

is a different complex structure on $\mathbb{R}^2$.

**18.** For a real vector space $\mathcal{V}$ with complex structure $\mathbf{J}$ define a multiplication of vectors in $\mathcal{V}$ by **complex** scalars by the formula

$$(a + b\mathbf{i}) \cdot \mathbf{A} = a\mathbf{A} + b\mathbf{J}(\mathbf{A}).$$

(1) Show that this definition of complex scalar multiplication makes $\mathcal{V}$ into a complex vector space.
(2) Show that on $\mathbb{R}^3$ there is no complex structure.

**19.** A linear transformation $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is called **invertible** if it is an isomorphism.
(1) If $\mathcal{V}$ is finite-dimensional, show that $\mathbf{T}$ is invertible if and only if $\ker\mathbf{T} = \{\mathbf{0}\}$.
(2) If $\mathbf{S}, \mathbf{T}$ are invertible, show that $\mathbf{S} \cdot \mathbf{T}$ is also. What is $(\mathbf{S} \cdot \mathbf{T})^{-1}$?

**20.** A sequence of real numbers $a_0, a_1, \ldots, a_n, \ldots$ is said to be **eventually zero** if there is an integer $k$ such that $a_i = 0$ for all $i > k$. Define $\mathbb{R}^\infty$ to be the set of all sequences $(a_0, a_1, \ldots, a_n, \ldots)$ that are eventually zero. Show that the usual addition of sequences and multiplication of sequences by a real number makes $\mathbb{R}^\infty$ into a vector space.
(1) Prove that $\mathcal{P}(\mathbb{R})$ and $\mathbb{R}^\infty$ are isomorphic.
(2) Show that $\mathcal{P}(\mathbb{R})$ is not isomorphic to $\ell_0$.
(3) Conclude that Theorem 8.6.3 need not be true in the infinite-dimensional case.

# Chapter 10
# Linear Transformations and Matrices

In this chapter we will begin the study of matrices, which are the analogue for linear transformations of coordinates for vectors. Our approach will be to consider first the case of a linear transformation

$$\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

in some detail and then abstract the salient features to the general case. This leads to an agebra for matrices, which we then pursue further.

## 10.1 Linear Transformations and Matrices in $\mathbb{R}^3$

Let us suppose that we have been given a fixed linear transformation $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$. As usual, we will denote by $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ the standard basis vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ in $\mathbb{R}^3$. Let

$$\mathsf{T}(\mathbf{E}_1) = (a_{1,1}, a_{2,1}, a_{3,1}),$$

$$\mathsf{T}(\mathbf{E}_2) = (a_{1,2}, a_{2,2}, a_{3,2}),$$

$$\mathsf{T}(\mathbf{E}_3) = (a_{1,3}, a_{2,3}, a_{3,3}).$$

Notice that the subscript $i, j$ on an $a$ means that it is the $i$-th coordinate of $\mathsf{T}(\mathbf{E}_j)$ and that $i, j$ assume the values $1, 2, 3$. Of course, using the vectors $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ we can write these equations as

$$\mathsf{T}(\mathbf{E}_1) = a_{1,1}\mathbf{E}_1 + a_{2,1}\mathbf{E}_2 + a_{3,1}\mathbf{E}_3,$$

$$\mathsf{T}(\mathbf{E}_2) = a_{1,2}\mathbf{E}_1 + a_{2,2}\mathbf{E}_2 + a_{3,2}\mathbf{E}_3,$$

$$\mathsf{T}(\mathbf{E}_3) = a_{1,3}\mathbf{E}_1 + a_{2,3}\mathbf{E}_2 + a_{3,3}\mathbf{E}_3.$$

If we use the $\sum$-notation for sums from the calculus, we see that we may replace these three equations by the single equation

$$\mathsf{T}(\mathbf{E}_j) = \sum_{i=1}^{3} a_{i,j}\mathbf{E}_i, \qquad j = 1, 2, 3.$$

Suppose that $\mathbf{X} = (x_1, x_2, x_3)$ is any vector in $\mathbb{R}^3$. Then of course ,

$$\mathbf{X} = x_1\mathbf{E}_1 + x_2\mathbf{E}_2 + x_3\mathbf{E}_3.$$

Therefore,

$$\begin{aligned}\mathsf{T}(\mathbf{X}) &= \mathsf{T}(x_1\mathbf{E}_1 + x_2\mathbf{E}_2 + x_3\mathbf{E}_3)\\ &= x_1\mathsf{T}(\mathbf{E}_1) + x_2\mathsf{T}(\mathbf{E}_2) + x_3\mathsf{T}(\mathbf{E}_3)\end{aligned}$$

because $\mathsf{T}$ is a linear transformation. Referring to the preceding list of equations for $\mathsf{T}(\mathbf{E}_j)$, $j = 1, 2, 3$, we find

$$\begin{aligned}\mathsf{T}(\mathbf{X}) = {}& x_1(a_{1,1}\mathbf{E}_1 + a_{2,1}\mathbf{E}_2 + a_{3,1}\mathbf{E}_3)\\ &+ x_2(a_{1,2}\mathbf{E}_1 + a_{2,2}\mathbf{E}_2 + a_{3,2}\mathbf{E}_3) + x_3(a_{1,3}\mathbf{E}_1 + a_{2,3}\mathbf{E}_2 + a_{3,3}\mathbf{E}_3),\end{aligned}$$

or after some algebraic manipulations, that

$$\begin{aligned}\mathsf{T}(\mathbf{X}) ={}&(a_{1,1}x_1 + a_{1,2}x_2 + a_{1,3}x_3)\mathbf{E}_1 + (a_{2,1}x_1 + a_{2,2}x_2 + a_{2,3}x_3)\mathbf{E}_2\\ &+ (a_{3,1}x_1 + a_{3,2}x_2 + a_{3,3}x_3)\mathbf{E}_3,\end{aligned}$$

which in the more compact $\sum$-notation may be written

$$\mathsf{T}(\mathbf{X}) = \sum_{i=1}^{3}\sum_{j=1}^{3} a_{i,j}x_j\mathbf{E}_i.$$

Thus we see that using the array of 9 numbers

$$(a_{i,j})_{\substack{i=1,2,3\\ j=1,2,3}}$$

we may calculate the value of the linear transformation $\mathsf{T}$ on any vector $\mathbf{X}$ whatsoever. Notice carefully that it is not just the collection of 9 numbers that matters, but the pattern (which is $3 \times 3$) that they fit into. Notice also that we have tacitly agreed to ***fix the ordered basis*** $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ for $\mathbb{R}^3$ and calculate all coordinates with respect to this ordered basis. Let us formalize these ideas with a definition.

**DEFINITION:** *Let* $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ *be a linear transformation. Suppose that*

$$\mathsf{T}(\mathbf{E}_j) = \sum_{i=1}^{3} a_{i,j}\mathbf{E}_i, \qquad j = 1, 2, 3,$$

*for the unique numbers* $(a_{i,j})_{i,j=1,2,3}$. *The* **matrix of** $\mathsf{T}$ *relative to the ordered basis* $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ *is the* $3 \times 3$ *array of numbers*

$$\mathsf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}.$$

Notice that the number $a_{i,j}$ occurs as the entry in the $i$-th row and $j$-th column of the matrix of $\mathsf{T}$. In order to write down the matrix of $\mathsf{T}$ we must calculate the coordinates of $\mathsf{T}(\mathbf{E}_j)$ relative to the ordered basis $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ and write the resulting numbers as the $j$-th ***column*** of the matrix of $\mathsf{T}$.

**EXAMPLE 1:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation given by

$$\mathsf{T}(x, y, z) = (x + 2z, y - x, z + y).$$

Calculate the matrix of $\mathsf{T}$ relative to the standard basis $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ of $\mathbb{R}^3$.

**SOLUTION:** We must first calculate the coordinates of $\mathsf{T}(\mathbf{E}_1)$, $\mathsf{T}(\mathbf{E}_2)$, $\mathsf{T}(\mathbf{E}_3)$ and write them down according to the rule explained above. We have

$$\mathsf{T}(\mathbf{E}_1) = \mathsf{T}(1, 0, 0) = (1, -1, 0) = (1)\mathbf{E}_1 + (-1)\mathbf{E}_2 + (0)\mathbf{E}_3,$$

$$\mathsf{T}(\mathbf{E}_2) = \mathsf{T}(0, 1, 0) = (0, 1, 1) = (0)\mathbf{E}_1 + (1)\mathbf{E}_2 + (1)\mathbf{E}_3,$$

$$\mathsf{T}(\mathbf{E}_3) = \mathsf{T}(0, 0, 1) = (2, 0, 1) = (2)\mathbf{E}_1 + (0)\mathbf{E}_2 + (1)\mathbf{E}_3.$$

Therefore, the matrix of $\mathsf{T}$ relative to the basis $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ is

$$\begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

**EXAMPLE 2:** Let $\mathsf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the shift operator (see Section 9.1 Example 4) given by

$$\mathsf{S}(x, y, z) = (0, x, y).$$

Calculate the matrix of $\mathsf{S}$ relative to the standard basis $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ of $\mathbb{R}^3$.

**SOLUTION:** We have

$$\mathsf{S}(\mathbf{E}_1) = \mathsf{S}(1,0,0) = (0,1,0) = 0\mathbf{E}_1 + 1\mathbf{E}_2 + 0\mathbf{E}_3,$$

$$\mathsf{S}(\mathbf{E}_2) = \mathsf{S}(0,1,0) = (0,0,1) = 0\mathbf{E}_1 + 0\mathbf{E}_2 + 1\mathbf{E}_3,$$

$$\mathsf{S}(\mathbf{E}_3) = \mathsf{S}(0,0,1) = (0,0,0) = 0\mathbf{E}_1 + 0\mathbf{E}_2 + 0\mathbf{E}_3,$$

so the matrix we seek is

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

In order for the matrix of a linear transformation $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ to be of any real use to us it should be possible to answer questions about $\mathsf{T}$ by using just its matrix, and for each operation on $\mathsf{T}$ to have a corresponding operation of the matrix of $\mathsf{T}$. To illustrate this point let us suppose we are given two linear transformations

$$\mathsf{T}, \mathsf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

that have matrices

$$\mathsf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}, \qquad \mathsf{B} = \begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} \\ b_{2,1} & b_{2,2} & b_{2,3} \\ b_{3,1} & b_{3,2} & b_{3,3} \end{bmatrix}$$

relative to the standard basis $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ of $\mathbb{R}^3$. We can form the new linear transformation

$$\mathsf{S} \cdot \mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3,$$

which is the composition of $\mathsf{S}$ and $\mathsf{T}$. The question we pose is, What is the matrix of $\mathsf{S} \cdot \mathsf{T}$ relative to the standard basis of $\mathbb{R}^3$? To answer this question we simply follow the procedure of the two preceding examples, taking care not to become too entangled in the notations. Let us therefore calculate $\mathsf{S} \cdot \mathsf{T}(\mathbf{E}_1)$, $\mathsf{S} \cdot \mathsf{T}(\mathbf{E}_2)$, and $\mathsf{S} \cdot \mathsf{T}(\mathbf{E}_3)$ in terms of what has been given to us, namely the matrices $\mathsf{A}$ and $\mathsf{B}$. We have

$$\begin{aligned}
\mathsf{S} \cdot \mathsf{T}(\mathbf{E}_1) &= \mathsf{S}(\mathsf{T}(\mathbf{E}_1)) \\
&= \mathsf{S}(a_{1,1}\mathbf{E}_1 + a_{2,1}\mathbf{E}_2 + a_{3,1}\mathbf{E}_3) \\
&= a_{1,1}\mathsf{S}(\mathbf{E}_1) + a_{2,1}\mathsf{S}(\mathbf{E}_2) + a_{3,1}\mathsf{S}(\mathbf{E}_3) \\
&= a_{1,1}(b_{1,1}\mathbf{E}_1 + b_{2,1}\mathbf{E}_2 + b_{3,1}\mathbf{E}_3) \\
&+ a_{2,1}(b_{1,2}\mathbf{E}_1 + b_{2,2}\mathbf{E}_2 + b_{3,2}\mathbf{E}_3) \\
&+ a_{3,1}(b_{1,3}\mathbf{E}_1 + b_{2,3}\mathbf{E}_2 + b_{3,3}\mathbf{E}_3),
\end{aligned}$$

which after a little algebraic manipulation gives

$$\begin{aligned}\mathsf{S}\cdot\mathsf{T}(\mathbf{E}_1) &= (b_{1,1}a_{1,1} + b_{1,2}a_{2,1} + b_{1,3}a_{3,1})\mathbf{E}_1\\ &+ (b_{2,1}a_{1,1} + b_{2,2}a_{2,1} + b_{2,3}a_{3,1})\mathbf{E}_2\\ &+ (b_{3,1}a_{1,1} + b_{3,2}a_{2,1} + b_{3,3}a_{3,1})\mathbf{E}_3.\end{aligned}$$

Thus the first column of the matrix we are seeking is

$$\begin{bmatrix} b_{1,1}a_{1,1} + b_{1,2}a_{2,1} + b_{1,3}a_{3,1}\\ b_{2,1}a_{1,1} + b_{2,2}a_{2,1} + b_{2,3}a_{3,1}\\ b_{3,1}a_{1,1} + b_{3,2}a_{2,1} + b_{3,3}a_{3,1}\end{bmatrix}.$$

In a similar manner we compute $\mathsf{S}\cdot\mathsf{T}(\mathbf{E}_2)$, $\mathsf{S}\cdot\mathsf{T}(\mathbf{E}_3)$, and we find that

$$\begin{aligned}\mathsf{S}\cdot\mathsf{T}(\mathbf{E}_2) &= (b_{1,1}a_{1,2} + b_{1,2}a_{2,2} + b_{1,3}a_{3,2})\mathbf{E}_1\\ &+ (b_{2,1}a_{1,2} + b_{2,2}a_{2,2} + b_{2,3}a_{3,2})\mathbf{E}_2\\ &+ (b_{3,1}a_{1,2} + b_{3,2}a_{2,2} + b_{3,3}a_{3,2})\mathbf{E}_3,\end{aligned}$$

and

$$\begin{aligned}\mathsf{S}\cdot\mathsf{T}(\mathbf{E}_3) &= (b_{1,1}a_{1,3} + b_{1,2}a_{2,3} + b_{1,3}a_{3,3})\mathbf{E}_1\\ &+ (b_{2,1}a_{1,3} + b_{2,2}a_{2,3} + b_{2,3}a_{3,3})\mathbf{E}_2\\ &+ (b_{3,1}a_{1,3} + b_{3,2}a_{2,3} + b_{3,3}a_{3,3})\mathbf{E}_3.\end{aligned}$$

Therefore, the matrix of $\mathsf{S}\cdot\mathsf{T}$ relative to the standard basis of $\mathbb{R}^3$ is given by

$$\begin{bmatrix} b_{1,1}a_{1,1} + b_{1,2}a_{2,1} + b_{1,3}a_{3,1} & b_{1,1}a_{1,2} + b_{1,2}a_{2,2} + b_{1,3}a_{3,2} & b_{1,1}a_{1,3} + b_{1,2}a_{2,3} + b_{1,3}a_{3,3}\\ b_{2,1}a_{1,1} + b_{2,2}a_{2,1} + b_{2,3}a_{3,1} & b_{2,1}a_{1,2} + b_{2,2}a_{2,2} + b_{2,3}a_{3,2} & b_{2,1}a_{1,3} + b_{2,2}a_{2,3} + b_{2,3}a_{3,3}\\ b_{3,1}a_{1,1} + b_{3,2}a_{2,1} + b_{3,3}a_{3,1} & b_{3,1}a_{1,2} + b_{3,2}a_{2,2} + b_{3,3}a_{3,2} & b_{3,1}a_{1,3} + b_{3,2}a_{2,3} + b_{3,3}a_{3,3}\end{bmatrix}.$$

While the above formulas are quite formidable to contemplate, and hard to read because of the multiple subscripts and small typeface, there is really a very simple rule underlying the computation of the matrix of $\mathsf{S}\cdot\mathsf{T}$. To explain this we write $\mathsf{C}$ for this matrix, so

$$\mathsf{C} = \begin{bmatrix} c_{1,1} & c_{1,2} & c_{1,3}\\ c_{2,1} & c_{2,2} & c_{2,3}\\ c_{3,1} & c_{3,2} & c_{3,3}\end{bmatrix}.$$

The entry $c_{i,j}$ is calculated by taking the $i$-th ***row*** of $\mathsf{B}$,

$$[\, b_{i,1}, b_{i,2}, b_{i,3}\,],$$

and the $j$-th ***column*** of $\mathsf{A}$,

$$\begin{bmatrix} a_{1,j}\\ a_{2,j}\\ a_{3,j}\end{bmatrix},$$

multiplying the elements in the corresponding positions, and adding up the resulting products; that is,

$$c_{i,j} = b_{i,1}a_{1,j} + b_{i,2}a_{2,j} + b_{i,3}a_{3,j} = \sum_{k=1}^{3} b_{i,k}a_{k,j}.$$

This row-by-column method of multiplication for matrices, which perhaps some of you have seen before, is not artificial, but a natural consequence of the attempt to solve the problem we posed, namely, to calculate the matrix of $\mathbf{S} \cdot \mathbf{T}$ from $\mathbf{A}$ and $\mathbf{B}$.

## 10.2 Some Numerical Examples

Here are some concrete, numerical examples, to help clarify the preceding discussion.

EXAMPLE 1: Let $\mathbf{S}$ and $\mathbf{T}$ be the linear transformations introduced in Examples 1 and 2 of the preceding section. Calculate the matrix of the linear transformation $\mathbf{S} \cdot \mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ relative to the standard basis $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ of $\mathbb{R}^3$.

SOLUTION: Recall that we found the matrices

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

to be the matrices of $\mathbf{S}$ and $\mathbf{T}$ respectively. Therefore, using the row-by-column multiplication procedure described in the preceding section, we find that the required matrix is

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}.$$

EXAMPLE 2: With $\mathbf{S}$ and $\mathbf{T}$ as in the preceding example, calculate the matrix of $\mathbf{T} \cdot \mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ relative to the standard basis of $\mathbb{R}^3$.

SOLUTION: We have to employ row-by-column multiplication in the opposite order from the preceding example. We write

$$\begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

and so the required matrix is

$$\begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix},$$

which is quite a bit different from what we found in the preceding example.

There is an important fact to be drawn from this, namely, multiplication of matrices by the row-by-column rule is ***not*** in general commutative. Usually, the product of two, or more matrices, depends on the order in which they are multiplied.

**EXAMPLE 3:** With $\mathsf{S}$ and $\mathsf{T}$ as in the preceding example calculate the matrix of $\mathsf{T}+\mathsf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ relative to the standard basis of $\mathbb{R}^3$.

**SOLUTION:** We calculate the coordinates of $(\mathsf{T}+\mathsf{S})(\mathbf{E}_1), (\mathsf{T}+\mathsf{S})(\mathbf{E}_2), (\mathsf{T}+\mathsf{S})(\mathbf{E}_3)$ and write them into the required matrix using the rules already explained. We have

$$\begin{aligned}(\mathsf{T}+\mathsf{S})(\mathbf{E}_1) &= \mathsf{T}(1,0,0)+\mathsf{S}(1,0,0) = (1,-1,0)+(0,1,0)\\ &= (1,0,0) = (1)\mathbf{E}_1+(0)\mathbf{E}_2+(0)\mathbf{E}_3,\end{aligned}$$

$$\begin{aligned}(\mathsf{T}+\mathsf{S})(\mathbf{E}_2) &= \mathsf{T}(0,1,0)+\mathsf{S}(1,0,0) = (0,1,1)+(0,0,1)\\ &= (0,1,2) = (0)\mathbf{E}_1+(1)\mathbf{E}_2+(2)\mathbf{E}_3,\end{aligned}$$

$$\begin{aligned}(\mathsf{T}+\mathsf{S})(\mathbf{E}_3) &= \mathsf{T}(0,0,1)+\mathsf{S}(0,0,1) = (2,0,1)+(0,0,0)\\ &= (2,0,1) = (2)\mathbf{E}_1+(0)\mathbf{E}_2+(1)\mathbf{E}_3.\end{aligned}$$

So the matrix of $\mathsf{T}+\mathsf{S}$ relative to the standard basis of $\mathbb{R}^3$ is

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}.$$

It is interesting to compare this to the matrices for $\mathsf{T}$ and $\mathsf{S}$, which are

$$\begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Carefull examination shows that the entries in the matrix for $\mathsf{T}+\mathsf{S}$ are entry for entry the sums of the entries in the matrices for $\mathsf{T}$ and $\mathsf{S}$.

These examples suggest that we should be able to develop an **algebra of matrices** that mirrors the operations we have for linear transformations. This is indeed possible, and it is the subject of the next chapter.

The matrix of a linear transformation $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ gives a very compact and convenient way to specify the transformation. This is particularly so if the matrix has a large number of zero entries, or a particularly simple form. However, this may not be the case for a particular $\mathsf{T}$ if we only allow ourselves to use the standard basis $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ for $\mathbb{R}^3$. This suggests that we introduce a matrix for $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ relative to

***any*** ordered basis $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ for $\mathbb{R}^3$, or more general still, relative to a ***pair*** of ordered bases $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ and $\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3$ for $\mathbb{R}^3$. Of course, there is nothing sacred about $\mathbb{R}^3$, and so we might as well consider the general case of a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$, where $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces. We will take up this subject in Chapter 11 after introducing some elementary matrix notions.

## 10.3 Matrices and Their Algebra

In Section 10.1 we saw that a linear transformation $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ can be represented (that is, is completely determined) by 9 numbers arranged in a $3 \times 3$ array. In this section we will study such arrays, which are called **matrices**. We will return to the connection between matrices and linear transformations in the next chapter.

DEFINITION: *A rectangular array of numbers composed of $m$ rows and $n$ columns*

$$\mathsf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

*is called an $m \times n$* **matrix** *(read "$m$ by $n$ matrix").*[1]

The elements $a_{i,1}, a_{i,2}, \ldots, a_{i,n}$ form the $i$-th **row of $\mathsf{A}$**, and the elements

$$\begin{matrix} a_{1,j} \\ a_{2,j} \\ \vdots \\ a_{m,j} \end{matrix}$$

form the $j$-th **column** of $\mathsf{A}$. We will often write

$$\mathsf{A} = (a_{i,j})_{\substack{1 \le i \le m \\ 1 \le j \le n}}$$

for $\mathsf{A}$, or simply

$$\mathsf{A} = (a_{i,j})$$

when $m$ and $n$ are understood from context. Note that the order of the subscripts is important; the first subscript denotes the row, and the second subscript the column, to which an entry belongs.

Just as with vectors in $\mathbb{R}^n$, two matrices are equal if and only if they have the same entries. That is:

---

[1] We also say that the matrix $\mathsf{A}$ is of, or has, **size** $m \times n$.

**DEFINITION:** *If* $\mathbf{A} = (a_{i,j})$ *and* $\mathbf{B} = (b_{i,j})$ *are* $m \times n$ *matrices, then* $\mathbf{A} = \mathbf{B}$ *if and only if* $a_{i,j} = b_{i,j}$ *for* $i = 1, 2, \ldots, m$ *and* $j = 1, \ldots, n$.

Our study of linear transformations suggests the following definitions.

**DEFINITION:** *If* $\mathbf{A} = (a_{i,j})$ *and* $\mathbf{B} = (b_{i,j})$ *are* $m \times n$ *matrices, their* **sum,** $\mathbf{A} + \mathbf{B}$*, is the matrix* $\mathbf{C} = (c_{i,j})$*, where* $c_{i,j} = a_{i,j} + b_{i,j}$, $i = 1, 2, \ldots, m$ *and* $j = 1, 2, \ldots, n$.

**DEFINITION:** *If* $\mathbf{A} = (a_{i,j})$ *is an* $m \times n$ *matrix, and* $r$ *is a number then* $r\mathbf{A}$*, the* **scalar multiple of A by** $r$ *is the matrix* $\mathbf{C} = (c_{i,j})$*, where* $c_{i,j} = ra_{i,j}$*, and* $i = 1, \ldots, m$ *and* $j = 1, \ldots, n$.

The following result is a routine verification of definitions:

**PROPOSITION 10.3.1:** *The matrices of size* $m \times n$ *form a vector space under the operations of matrix addition and scalar multiplication. We denote this vector space by* $\mathrm{Mat}_{m,n}$. □

The dimension of the vector space $\mathrm{Mat}_{m,n}$ is not hard to compute. We take our lead from the method we used to show that $\dim(\mathbb{R}^n) = n$. Introduce the $m \times n$ matrix $\mathbf{E}(r, s) = (e_{i,j})$ by the requirement

$$e_{i,j} = \begin{cases} 1 & i = r, j = s, \\ 0 & \text{otherwise.} \end{cases}$$

For example, the $6 \times 4$ matrix $\mathbf{E}(3, 2)$ is

$$\mathbf{E}(3, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The matrices $\mathbf{E}(r, s)$ are called **elementary matrices.** It is then a routine verification to prove:

**PROPOSITION 10.3.2:** *The vectors*

$$\left\{\mathbf{E}(r, s) \mid r = 1, 2, \ldots, m; s = 1, 2, \ldots, n\right\}$$

*form a basis for* $\mathrm{Mat}_{m,n}$*. Therefore,* $\dim(\mathrm{Mat}_{m,n}) = mn$. □

Recall that for vector spaces $\mathcal{V}$ and $\mathcal{W}$ we introduced (see Theorem 8.5.6) the vector space $\mathcal{L}(\mathcal{V}, \mathcal{W})$ of linear transformations from $\mathcal{V}$ to $\mathcal{W}$. If $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional, then we will see that the innocent looking Proposition 10.3.2 implies the useful fact that $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = \dim(\mathcal{V}) \cdot \dim(\mathcal{W})$. We record this, since the symmetry it implies is a priori not at all clear.

**COROLLARY 10.3.3 :** *If $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces then* $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = \dim(\mathcal{V}) \cdot \dim(\mathcal{W}) = \dim(\mathcal{L}(\mathcal{W}, \mathcal{V}))$. □

**EXAMPLE 1:** Here are some examples of matrix addition and scalar multiplication

$$\begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & -1 \\ -2 & -1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 3 \\ 2 & -9 & 4 \\ 7 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 3+0 & 1-1 & 4+3 \\ 2+2 & 0-9 & -1+4 \\ -2+7 & -1+6 & 0+1 \end{bmatrix}$$
$$= \begin{bmatrix} 3 & 0 & 7 \\ 4 & -9 & 3 \\ 5 & 5 & 1 \end{bmatrix},$$

$$4\begin{bmatrix} 3 & 1 \\ 7 & 4 \\ 6 & -4 \end{bmatrix} = \begin{bmatrix} 12 & 4 \\ 28 & 16 \\ 24 & -16 \end{bmatrix}.$$

The discussion of the matrix of the composition of two linear transformations in $\mathbb{R}^3$ from Section 10.1 suggests that the following might be a useful definition.

**DEFINITION:** *If* $\mathbf{A} = (a_{i,j})$ *is an* $m \times n$ *matrix and* $\mathbf{B} = (b_{i,j})$ *is an* $n \times p$ *matrix their* **matrix product**[2] $\mathbf{A} \cdot \mathbf{B}$ *is the* $m \times p$ *matrix* $\mathbf{A} \cdot \mathbf{B} = (c_{i,j})$ *where*

$$c_{i,j} = \sum_{k=1}^{n} a_{i,k} b_{k,j}$$

*for* $i = 1, \ldots, m; j = 1, \ldots, p$.

Thus the entry of the $i$-th row and $j$-th column of the product $\mathbf{A} \cdot \mathbf{B}$ is obtained by taking the $i$-th row

$$\left[ a_{i,1}, a_{i,2}, \ldots, a_{i,n} \right]$$

of the matrix $\mathbf{A}$ and the $j$-th column of the matrix $\mathbf{B}$

$$\begin{bmatrix} b_{1,j} \\ b_{2,j} \\ \vdots \\ b_{n,j} \end{bmatrix},$$

[2] The matrix product is also denoted by $\mathbf{AB}$, i.e., without the $\cdot$ if no confusion can arise.

multiplying the corresponding entries together, and adding the resulting products, i.e.,

$$a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,k}b_{k,j} + \ldots + a_{i,n}b_{n,j},$$

$i = 1, 2, \ldots, m; j = 1, 2, \ldots, p$.

***Note that for the product of* A *and* B *to be defined, the number of columns of* A *must be equal to the number of rows of* B.** Thus the ***order*** in which the product of **A** and **B** is taken is very important, for **A** · **B** can be defined without the product **B** · **A** being defined at all, much less being equal to **A** · **B**.

**EXAMPLE 2:** Compute the matrix product

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

**SOLUTION:** Note the answer must be a $1 \times 1$ matrix.

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 \end{bmatrix} = \begin{bmatrix} 32 \end{bmatrix}.$$

The product in the reverse order,

$$\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 12 \\ 5 & 10 & 15 \\ 6 & 12 & 18 \end{bmatrix},$$

does not even have the same size: It is $3 \times 3$ and not $1 \times 1$. So the products in different orders are certainly not equal.

**EXAMPLE 3:** Compute the matrix product

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

**SOLUTION:**

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

**EXAMPLE 4:** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{bmatrix}.$$

Calculate the product **A** · **B**.

**SOLUTION:** We have

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+2+3 & 2+4+6 & 3+6+9 & 4+8+12 \\ 4+5+6 & 8+10+12 & 12+15+18 & 16+20+24 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 12 & 18 & 24 \\ 15 & 30 & 45 & 60 \end{bmatrix}.$$

Note that the product $\mathbf{B} \cdot \mathbf{A}$ is ***not*** defined.

**DEFINITION:** *A matrix* $\mathbf{A}$ *is said to be* **square of size** $n$ *if it has* $n$ *rows and* $n$ *columns (that is, the number of rows equals the number of columns equals* $n$*).*

If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of size $n$, then the products $\mathbf{AB}$ and $\mathbf{BA}$ are both defined and are square matrices of size $n$. However, they may not be equal.

**EXAMPLE 5:** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}.$$

Compute the matrix products $\mathbf{AB}$ and $\mathbf{BA}$.

**SOLUTION:** We have

$$\mathbf{AB} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 6 & 3 \end{bmatrix}, \quad \mathbf{BA} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 2 & 3 \end{bmatrix},$$

and we see that $\mathbf{AB} \neq \mathbf{BA}$.

As the preceding example shows, even if $\mathbf{AB}$ and $\mathbf{BA}$ are defined, we should not expect that $\mathbf{AB} = \mathbf{BA}$.

**NOTE:** If $\mathbf{A}$ is a square matrix, then $\mathbf{AA}$ is defined and is denoted by $\mathbf{A}^2$. Similarly,

$$\underset{\longleftarrow n \longrightarrow}{\mathbf{A} \cdots \mathbf{A}}$$

is defined and denoted $\mathbf{A}^n$.

**EXAMPLE 6:** Let

$$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Calculate $\mathbf{A}^2$.

**SOLUTION:** We have

$$\mathbf{A}^2 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus not only does matrix multiplication behave strangely in that it is not commutative, it is also possible for the square of a matrix with nonzero entries to have only zero entries.

Here is a summary of the basic rules of matrix algebra. (Assume that the indicated operations are defined, that is, that the sizes are correct for the operations to make sense. $\mathbf{0}$ is the matrix[3] all of whose entries are 0.)

$$\begin{aligned}
\mathbf{A} + \mathbf{B} &= \mathbf{B} + \mathbf{A}, \\
\mathbf{A} + (\mathbf{B} + \mathbf{C}) &= (\mathbf{A} + \mathbf{B}) + \mathbf{C}, \\
r(\mathbf{A} + \mathbf{B}) &= r\mathbf{A} + r\mathbf{B}, \\
(r + s)\mathbf{A} &= r\mathbf{A} + s\mathbf{A}, \\
0\mathbf{A} &= \mathbf{0}, \\
\mathbf{A} + (-1)\mathbf{A} &= \mathbf{0}, \\
\mathbf{A} + \mathbf{0} &= \mathbf{A}, \\
(\mathbf{A} + \mathbf{B}) \cdot \mathbf{C} &= \mathbf{A} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{C}, \\
\mathbf{C} \cdot (\mathbf{A} + \mathbf{B}) &= \mathbf{C} \cdot \mathbf{A} + \mathbf{C} \cdot \mathbf{B}, \\
\mathbf{0} \cdot \mathbf{A} &= \mathbf{0} = \mathbf{A} \cdot \mathbf{0}, \\
\mathbf{A} \cdot (\mathbf{B} \cdot \mathbf{C}) &= (\mathbf{A} \cdot \mathbf{B}) \cdot \mathbf{C}.
\end{aligned}$$

Of course, the first few of these rules were needed to prove Proposition 10.3.1.

## 10.4 Special Types of Matrices

In discussing matrices, it is convenient to distinguish certain special types of matrices.

***The identity matrix.*** The **identity matrix of size $n$** is the square $n \times n$ matrix

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & & \cdots & 0 \\ 0 & 1 & & \cdots & 0 \\ 0 & 0 & \ddots & \cdots & 0 \\ \vdots & & & & \cdots \\ 0 & 0 & & \cdots & 1 \end{bmatrix} = (i_{i,j}), \quad \text{where } i_{i,j} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

[3] $\mathbf{0}$ is called the **zero matrix**.

For example, the identity matrices of sizes 1, 2, 3, and 4 are

$$[1], \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The following important facts are easily verified:

$$\begin{aligned} \mathbf{I} \cdot \mathbf{B} &= \mathbf{B} \quad \text{for any } n \times p \text{ matrix } \mathbf{B}, \\ \mathbf{A} \cdot \mathbf{I} &= \mathbf{A} \quad \text{for any } m \times n \text{ matrix } \mathbf{A}. \end{aligned}$$

***Scalar matrices.*** A square matrix $\mathbf{A} = (a_{i,j})$ is called a **scalar matrix** if $\mathbf{A} = r\mathbf{I}$ for some number $r$. For example,

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is a scalar matrix, but

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

is not a scalar matrix. The following formulas are easily checked:

$$\begin{aligned} (a\mathbf{I}) \cdot \mathbf{B} &= a\mathbf{B} \quad \text{for any } n \times p \text{ matrix } \mathbf{B}, \\ \mathbf{A} \cdot (a\mathbf{I}) &= a\mathbf{A} \quad \text{for any } m \times n \text{ matrix } \mathbf{A}. \end{aligned}$$

For example,

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

***Diagonal matrices.*** For any square matrix $\mathbf{A} = (a_{i,j})$ of size $n$ the entries

$$a_{1,1}, a_{2,2}, \ldots, a_{n,n}$$

are called the **diagonal entries** of $\mathbf{A}$. For example, the diagonal entries of

$$\begin{bmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{bmatrix}$$

are 3, 5, 7. A square matrix $\mathbf{D}$ is said to be a **diagonal matrix** if its only nonzero entries are on the diagonal. That is, $\mathbf{D} = (d_{i,j})$ is a diagonal matrix if and only if $d_{i,j} = 0$ for $i \neq j$.

For example, $\mathbf{I}$ and $a\mathbf{I}$ are diagonal matrices as is

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Note that the diagonal entries themselves need not be nonzero. For example

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

are also diagonal matrices.

In general, a diagonal matrix looks like

$$\mathbf{D} = \begin{bmatrix} d_{1,1} & & & \mathbf{0} \\ & d_{2,2} & & \\ & & \ddots & \\ \mathbf{0} & & & d_{n,n} \end{bmatrix},$$

where the giant 0s mean that all other entries are zero. If $\mathbf{A}$ and $\mathbf{B}$ are diagonal matrices of size $n$, then so are $\mathbf{AB}$ and $\mathbf{BA}$, and they are equal. Indeed, if

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & a_{n,n} \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} b_{1,1} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & b_{n,n} \end{bmatrix},$$

then

$$\mathbf{AB} = \begin{bmatrix} a_{1,1} \cdot b_{1,1} & & & \mathbf{0} \\ & a_{2,2} \cdot b_{2,2} & & \\ & & \ddots & \\ \mathbf{0} & & & a_{n,n} \cdot b_{n,n} \end{bmatrix} = \mathbf{BA}.$$

***Triangular matrices.*** A square matrix $\mathbf{A}$ is said to be **lower triangular** if $\mathbf{A} = (a_{i,j})$, where $a_{i,j} = 0$ if $j > i$. For example,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & -1 & -3 \end{bmatrix}$$

is a lower triangular matrix. A lower triangular matrix $\mathbf{A} = (a_{i,j})$ for which $a_{i,i} = 0$ for $i = 1, \ldots, n$ (that is, all of whose diagonal entries are also 0) is said to be **strictly lower triangular**. An example of a strictly lower triangular matrix is

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 3 & 0 \end{bmatrix}.$$

**Upper triangular matrices** are defined analogously.

***The Zero matrix.*** The **zero matrix** of size $m \times n$ is the $m \times n$ matrix $\mathbf{0}$ all of those entries are 0.

***Idempotent matrices.*** A square matrix $\mathbf{A}$ is said to be **idempotent** if $\mathbf{A}^2 = \mathbf{A}$. There are lots of idempotent matrices. Here are a few examples:

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

as may be easily checked by explicit computation.

***Nilpotent matrices.*** A square matrix $\mathbf{A}$ is said to be **nilpotent** if there is a positive integer $q$ such that $\mathbf{A}^q = \mathbf{0}$. (The smallest such integer $q$ is called the **index of nilpotence** of $\mathbf{A}$). For example, if $\mathbf{A}$ is the matrix of the shift operator on $\mathbb{R}^3$, that is,

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

then

$$\mathbf{A}^2 = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

and

$$\mathbf{A}^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so that $\mathbf{A}$ is nilpotent of index 3.

As a matter of fact, there is a very simple geometric explanation for why $\mathbf{A}^3 = \mathbf{0}$. In fact, any strictly lower triangular matrix

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{bmatrix}$$

will be nilpotent of index at most 3. Of course, we could prove this by an orgy of calculation, but that is not what we want to do. Rather, we are going to exploit the relation between matrices and linear transformations that we started to discuss in section 10.1. *Let us therefore construct a linear transformation*

$$\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

*whose matrix is* $\mathbf{B}$. To do this, we must recall that if there is such a $\mathbf{T}$, then it must satisfy

$$\begin{aligned}\mathbf{T}(1, 0, 0) &= (0, a, b),\\ \mathbf{T}(0, 1, 0) &= (0, 0, c),\\ \mathbf{T}(0, 0, 1) &= (0, 0, 0),\end{aligned}$$

if its matrix is going to be $\mathbf{B}$. (Remember how to compute the columns of $\mathbf{B}$!) Therefore,

$$\begin{aligned}\mathbf{T}(x, y, z) &= \mathbf{T}(x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1))\\ &= x\mathbf{T}(1, 0, 0) + y\mathbf{T}(0, 1, 0) + z\mathbf{T}(0, 0, 1)\\ &= x(0, a, b) + y(0, 0, c) + z(0, 0, 0)\\ &= (0, ax, bx + cy).\end{aligned}$$

That is, the formula

$$\mathbf{T}(x, y, z) = (0, ax, bx + cy)$$

defines the one and only one linear transformation

$$\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

whose matrix is $\mathbf{B}$. Let us now compute

$$\mathbf{T}^2 : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \qquad \mathbf{T}^3 : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \ldots.$$

We have

$$\begin{aligned}\mathbf{T}^2(x, y, z) &= \mathbf{T}(\mathbf{T}(x, y, z)) = \mathbf{T}(0, ax, bx + cy)\\ &= (0, 0, cax)\end{aligned}$$

and

$$\mathbf{T}^3(x, y, z) = \mathbf{T}(\mathbf{T}^2(x, y, z)) = \mathbf{T}(0, 0, cax) = (0, 0, 0).$$

Therefore ,

$$\mathbf{T}^3 = \mathbf{0} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3.$$

Hence the matrix of $\mathbf{T}^3$ relative to the standard basis of $\mathbb{R}^3$ is

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

But now remember that our definition of matrix multiplication was rigged up such that $\mathbf{B}^3$ would be the matrix of $\mathbf{T}^3$ relative to the standard basis of $\mathbb{R}^3$, and therefore we have shown that

$$\mathbf{B}^3 = \mathbf{0},$$

and hence that $\mathbf{B}$ is nilpotent of index at most 3.

Thus we see the geometric reason behind the fact that $\mathbf{B}^3 = \mathbf{0}$ is that $\mathbf{B}$ is the matrix of a linear transformation that is basically a **shift** on $\mathbb{R}^3$. (Actually, $\mathbf{T}$ is what is called a **weighted** shift, the weights being $a, b, c$.) This discussion illustrates that it is sometimes possible, and indeed advantageous, to discover properties of a matrix by examining a linear transformation that it represents. We will see more examples of this type later.

***Nonsingular matrices.*** A square matrix $\mathbf{A}$ is said to be **invertible** or **nonsingular** if there exists a matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{I} \qquad \text{and} \qquad \mathbf{BA} = \mathbf{I}.$$

If $\mathbf{A}$ is nonsingular then the matrix $\mathbf{B}$ with $\mathbf{AB} = \mathbf{I} = \mathbf{BA}$ is called the **inverse matrix** of $\mathbf{A}$ and is denoted by $\mathbf{A}^{-1}$.

It is a theorem that we will prove in the next chapter that if there exists a matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{I},$$

then also

$$\mathbf{BA} = \mathbf{I}.$$

Thus to check that $\mathbf{B} = \mathbf{A}^{-1}$ we need only calculate one of the two products $\mathbf{AB}$ and $\mathbf{BA}$ and see whether it is $\mathbf{I}$. For example, if

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

then

$$\mathbf{A}^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}.$$

For we have

$$\mathbf{AA} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I},$$

and therefore $\mathbf{A} = \mathbf{A}^{-1}$. There is actually a simple geometric explanation for why $\mathbf{A} = \mathbf{A}^{-1}$. A moment's reflection on our discussion of matrices of linear transformations shows that $\mathbf{A}$ is the matrix of the linear transformation $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by

$$\mathbf{T}(x, y, z) = (y, x, z)$$

that switches the first two coordinates, relative to the standard bases for $\mathbb{R}^3$. Clearly, switching the first two coordinates twice will change nothing, that is,

$$\mathbf{T}^2(x, y, z) = \mathbf{T}(\mathbf{T}(x, y, z)) = \mathbf{T}(y, x, z) = (x, y, z),$$

so the matrix of

$$\mathbf{T}^2 : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

relative to the standard basis of $\mathbb{R}^3$ is

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

But remember that the matrix of $\mathbf{T}^2$ is also $\mathbf{A}^2$, because that is how we defined matrix multiplication. Thus $\mathbf{A}^2 = \mathbf{I}$ for the simple geometric reason that it is the matrix of the transformation $\mathbf{T}$ that switches the first two coordinates.

The preceding discussion illustrates again that it is sometimes possible to extract information about a matrix by examining the corresponding linear transformation.

A matrix $\mathbf{A}$ with the property that $\mathbf{A} = \mathbf{A}^{-1}$ is called **involutory**, or an **involution**. The example above shows that there are nontrivial involutions.

Another example of a nonsingular matrix is

$$\mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

The inverse of $\mathbf{B}$ is the matrix

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

as the following computation shows:

$$\mathbf{BC} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}.$$

An example of a matrix that is *not* invertible is

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

and more generally we have: A nilpotent matrix is not invertible. For suppose that $\mathbf{A}$ is a nilpotent matrix that is invertible. Let $\mathbf{B}$ be an inverse for $\mathbf{A}$. Since $\mathbf{A}$ is nilpotent, there is an integer $q$ such that

$$\mathbf{A}^q = \mathbf{0}.$$

Then

$$\mathbf{0} = \mathbf{A}^q\mathbf{B} = \mathbf{A}^{q-1}\mathbf{A}\mathbf{B} = \mathbf{A}^{q-1}\mathbf{I} = \mathbf{A}^{q-1},$$

so

$$\mathbf{A}^{q-1} = \mathbf{0}.$$

We may then repeat the above trick to show

$$\mathbf{A}^{q-2} = \mathbf{0}.$$

If we repeat this trick $q - 1$ times we will get

$$\mathbf{A} = \mathbf{0}.$$

But then

$$\mathbf{I} = \mathbf{A}\mathbf{B} = \mathbf{0}\mathbf{B} = \mathbf{0},$$

which is impossible.

Similarly, we may also show: The only invertible idempotent matrix is $\mathbf{I}$. For if $\mathbf{A}$ is an idempotent matrix, then $\mathbf{A}^2 = \mathbf{A}$. If in addition $\mathbf{A}$ is invertible with inverse $\mathbf{B}$, then

$$\mathbf{A}^2 = \mathbf{A}$$

implies

$$\mathbf{A} = \mathbf{I}\mathbf{A} = \mathbf{B}\mathbf{A}\mathbf{A} = \mathbf{B}\mathbf{A}^2 = \mathbf{B}\mathbf{A} = \mathbf{I}.$$

So $\mathbf{A} = \mathbf{I}$ as claimed.

***Symmetric and skew-symmetric matrices.*** A square matrix $\mathbf{A} = (a_{i,j})$ is said to be **symmetric** if $a_{i,j} = a_{j,i}$ for $i, j = 1, \ldots, n$; it is said to be **skew-symmetric** if $a_{i,j} = -a_{j,i}$ for $i, j = 1, \ldots, n$. For example,

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 4 & 5 & 6 \\ 2 & 5 & 7 & 8 \\ 3 & 6 & 8 & 9 \end{bmatrix}$$

are symmetric matrices, and

$$\begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

are skew-symmetric matrices. Notice that the matrix

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 3 \end{bmatrix}$$

is not skew-symmetric because

$$a_{1,1} = 1 \neq -1 = -a_{1,1}.$$

That is to say, if a matrix $\mathbf{A} = (a_{i,j})$ is skew-symmetric, then the equations $a_{1,1} = -a_{1,1}$, $a_{2,2} = -a_{2,2}, \ldots, a_{n,n} = -a_{n,n}$ certainly imply that $a_{1,1} = 0$, $a_{2,2} = 0, \ldots, a_{n,n} = 0$; that is, a skew-symmetric matrix has all its diagonal entries equal to 0.

The skew-symmetric matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

is interesting because it is also nonsingular:

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus various combinations of the preceding types may occur simultaneously. By now the student must be wondering how to tell when a matrix has an inverse, and how to calculate it when it does. This is an extremely important topic[4] and we will touch on it in Chapter 17, but for the moment we will content ourselves with the $2 \times 2$ case, which is elementary.

**PROPOSITION 10.4.1:** *A $2 \times 2$ matrix*

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

*is nonsingular if and only if $ad - bc \neq 0$. If $ad - bc \neq 0$, then*

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

[4] There is a vast literature dealing with computer algorithms to find inverses.

**Proof:** Suppose that $ad - bc \neq 0$. Let

$$\mathbf{B} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Then

$$\begin{aligned}
\mathbf{BA} &= \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\
&= \frac{1}{ad-bc}\begin{bmatrix} da-bc & db-bd \\ -ca+ac & -cb+ad \end{bmatrix} \\
&= \frac{1}{ad-bc}\begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}.
\end{aligned}$$

A completely analogous computation shows that $\mathbf{AB} = \mathbf{I}$, and therefore $\mathbf{A}$ is nonsingular with

$$\mathbf{A}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Suppose conversely that $\mathbf{A}$ is nonsingular, but that $ad - bc = 0$. We will deduce a contradiction. Let

$$\mathbf{C} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Then computing as above,

$$\mathbf{CA} = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = (ad-bc)\mathbf{I} = \mathbf{0}.$$

This gives the equation

$$\mathbf{C} = \mathbf{CI} = \mathbf{C}(\mathbf{AA}^{-1}) = (\mathbf{CA})\mathbf{A}^{-1} = \mathbf{0A}^{-1} = \mathbf{0}.$$

Therefore ,

$$\mathbf{C} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

so that

$$a = 0, \qquad b = 0, \qquad c = 0, \qquad d = 0.$$

But then $\mathbf{A} = \mathbf{0}$ also, so

$$\mathbf{I} = \mathbf{AA}^{-1} = \mathbf{0A}^{-1} = \mathbf{0},$$

so

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

and hence $1 = 0$, which is impossible. □

One might wonder how the above result was originally discovered. Did somebody just make a lucky guess? Perhaps, but a more logical development may be found in Chapter 13.

## 10.5 Exercises

**1.** For each of the following linear transformations of $\mathbb{R}^3$ to $\mathbb{R}^3$ calculate the matrix relative to the standard basis:

(1) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by $\mathsf{T}(x, y, z) = (x + y + z, 0, 0)$.

(2) $\mathsf{Q} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by $\mathsf{Q}(x, y, z) = (x, x + y, x + y + z)$.

(3) $\mathsf{F} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by $\mathsf{F}(x, y, z) = (x + 2y + 3z, 2x + y, z - x)$.

(4) $\mathsf{G} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by $\mathsf{G}(x, y, z) = (y - z, x + y, z - 2x)$.

**2.** Calculate the matrices of the linear transformations

$$\mathsf{T} \cdot \mathsf{Q} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$$\mathsf{F} \cdot \mathsf{G} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$$\mathsf{Q} \cdot \mathsf{F} \cdot \mathsf{G} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

where $\mathsf{T}$, $\mathsf{Q}$, $\mathsf{F}$, and $\mathsf{G}$ are as in Exercise 1.

**3.** Suppose that the matrix of the linear transformation $\mathsf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

relative to the standard basis of $\mathbb{R}^3$. Calculate $\mathsf{S}(1, 2, 3)$.

**4.** Let $\mathsf{P} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation given by

$$\mathsf{P}(x, y, z) = (x, y, 0).$$

Calculate the matrix for $\mathsf{P}$ and for $\mathsf{P}^2$ relative to the standard basis of $\mathbb{R}^3$.

**5.** Let $\mathsf{S}, \mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformations with matrices $\mathbf{A}$ and $\mathbf{B}$ relative to the standard basis of $\mathbb{R}^3$. What is the matrix of $\mathsf{S} + \mathsf{T}$ relative to this basis?

**6.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear extension of

$$\mathsf{T}(\mathbf{E}_1) = (-1, 0, 2),$$

$$\mathsf{T}(\mathbf{E}_2) = (1, 1, -1),$$

$$\mathsf{T}(\mathbf{E}_3) = (1, -3, 4).$$

Calculate the matrix of $\mathsf{T}$ relative to the standard basis.

**7.** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear extension of

$$\mathsf{T}(1, 1) = (1, -1),$$
$$\mathsf{T}(1, -2) = (2, 2).$$

Calculate the matrix of the linear transformation $\mathsf{T}$ relative to the standard basis.

**8.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation defined by

$$\mathsf{T}(x, y, z) = (0, y, z).$$

Calculate the matrix of $\mathsf{T}$ relative to the standard basis of $\mathbb{R}^3$.

**9.** Let $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ be the linear transformation defined by

$$\mathsf{T}(x, y, z, w) = (0, x, y, z).$$

Calculate the matrix of $\mathsf{T}$ relative to the standard basis. Calculate the matrices of $\mathsf{T}^2$, $\mathsf{T}^3$, $\mathsf{T}^4$ relative to the standard basis.

**10.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix relative to the standard basis is

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 2 & -1 & 0 \end{bmatrix}.$$

(1) Calculate $\mathsf{T}(\mathbf{E}_1)$, $\mathsf{T}(\mathbf{E}_2)$, $\mathsf{T}(\mathbf{E}_3)$.
(2) Let $r$ be a real number. Calculate the matrix of $r\mathsf{T}$ relative to the standard basis.

**11.** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation whose matrix relative to the standard basis is

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

(1) Calculate $\mathsf{T}(\mathbf{E}_1)$ and $\mathsf{T}(\mathbf{E}_2)$.
(2) Find $\mathbf{A}_1, \mathbf{A}_2$ satisfying $\mathsf{T}(\mathbf{A}_i) = \mathbf{E}_i$, $i = 1, 2$.

**12.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix relative to the standard basis is

$$\begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

(1) Calculate $\mathsf{T}(\mathbf{E}_1)$, $\mathsf{T}(\mathbf{E}_2)$, $\mathsf{T}(\mathbf{E}_3)$, and $\mathsf{T}(1, 2, 3)$.
(2) Is $\mathsf{T}$ surjective? (Check whether $(1, 0, 0)$ belongs to $\mathrm{Im}(\mathsf{T})$.)

**13.** Let $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3 \in \mathbb{R}^3$ and write

$$\mathbf{A}_j = (a_{1,j}, a_{2,j}, a_{3,j}), \qquad j = 1, 2, 3.$$

Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear extension of the function $\mathsf{T}(\mathbf{E}_j) = \mathbf{A}_j$, $j = 1, 2, 3$. Show that the matrix of $\mathsf{T}$ relative to the standard basis of $\mathbb{R}^3$ is $(a_{i,j})$.

**14.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation and $\mathbf{A} = (a_{i,j})$ its matrix with respect to the standard basis of $\mathbb{R}^3$. Regard the columns of $\mathbf{A}$ as vectors in $\mathbb{R}^3$. Call these three vectors $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$. Show that $\mathrm{Im}(\mathsf{T}) = \mathcal{L}(\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3\})$.

**15.** If $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is the linear transformation with matrix

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

with respect to the standard basis of $\mathbb{R}^3$, show that $\mathsf{T}$ is an isomorphism.

**16.** Perform the following matrix computations:

(a) $2\begin{bmatrix} 3 & 1 & 4 & -1 \\ 2 & 0 & -1 & 2 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 & -1 & 7 \\ -1 & -2 & 0 & -4 \end{bmatrix}$.

(b) $2\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 4\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 8\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

**17.** Perform the following matrix multiplications:

(a) $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

(b) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$.

(c) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$.

(d) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

(e) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

**18.** Which of the following matrices are nonsingular, involutory, idempotent, nilpotent, symmetric, or skew-symmetric?

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$
$$\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$
$$\mathbf{C} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix},$$
$$\mathbf{D} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{J} = \begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix},$$
$$\mathbf{E} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{K} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Find the inverse for those that are invertible.

**19.** Show that a diagonal matrix

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & 0 & 0 \\ 0 & a_{2,2} & 0 \\ 0 & 0 & a_{3,3} \end{bmatrix}$$

is nonsingular if and only if $a_{11}, a_{22}, a_{33} \neq 0$. That is $a_{1,1} \neq 0, a_{2,2} \neq 0, a_{3,3} \neq 0$. If **A** has an inverse, what is it?

**20.** Show that a diagonal matrix

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & & & \text{\huge 0} \\ & a_{2,2} & & \\ & & \ddots & \\ \text{\huge 0} & & & a_{n,n} \end{bmatrix}$$

is nonsingular if and only if *all* its diagonal entries are nonzero. Find its inverse.

**21.** If $\mathbf{A} = (a_{i,j})$ is a matrix, we define the **transpose** of **A** to be the matrix $\mathbf{A}^{\mathrm{tr}} = (b_{i,j})$, where $b_{i,j} = a_{j,i}$. Find the transpose of each of the following matrices:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 \end{bmatrix},$$
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}.$$

Show for any matrix **A** that $(\mathbf{A}^{\mathrm{tr}})^{\mathrm{tr}} = \mathbf{A}$.

**22.** Let **A** be a square matrix, show that **A** is symmetric if and only if $\mathbf{A} = \mathbf{Atr}$, and **A** is skew-symmetric if and only if $\mathbf{A} = -\mathbf{Atr}$.

**23.** For any square matrix **A** show that $\mathbf{A} + \mathbf{A}^{tr}$ is symmetric and $\mathbf{A} - \mathbf{A}^{tr}$ is skew-symmetric.

**24.** Show every square matrix is the sum of a symmetric and a skew-symmetric matrix.

**25.** Let **A** and **B** be $3 \times 3$ matrices. Show that $\mathbf{A}^{tr}\mathbf{B}^{tr} = (\mathbf{BA})^{tr}$. (Note that the order has been reversed.)

**26.** Show that the product of two lower triangular matrices is again lower triangular. If you cannot work the general case, do the $2 \times 2$ and $3 \times 3$ cases.

**27.** Let **A** be an idempotent matrix. Show that $\mathbf{I} - \mathbf{A}$ is also idempotent.

**28.** Show that a matrix **A** is involutory if and only if $(\mathbf{I} - \mathbf{A}) \cdot (\mathbf{I} + \mathbf{A}) = \mathbf{0}$.

**29.** Let **A** be a $3 \times 3$ matrix. Compute $\mathbf{E}(1, 2) \cdot \mathbf{A}$, $\mathbf{A} \cdot \mathbf{E}(1, 2)$, $\mathbf{E}(2, 1) \cdot \mathbf{A}$, $\mathbf{A} \cdot \mathbf{E}(2, 1)$. What conclusion can you obtain in general for $\mathbf{E}(r, s) \cdot \mathbf{A}$ and $\mathbf{A} \cdot \mathbf{E}(r, s)$?

**30.** Let $\mathbf{A} = (a_{i,j})$ be a $3 \times 3$ matrix. Compute $\mathbf{A} \cdot \mathbf{E}(r, s)$ and $\mathbf{E}(r, s) \cdot \mathbf{A}$.

**31.** A square matrix **A** is said to **commute** with a matrix **B** if $\mathbf{AB} = \mathbf{BA}$. When does a $3 \times 3$ matrix **A** commute with the matrix $\mathbf{E}(r, s)$?

**32.** Show that if a $3 \times 3$ matrix **A** commutes with every $3 \times 3$ matrix **B**, then **A** is a scalar matrix. (HINT: If **A** commutes with every matrix **B**, it commutes with the 9 matrices $\mathbf{E}_{r,s}$ where $r, s = 1, 2, 3$. Use the result of the previous exercise.)

**33.** Find all $2 \times 2$ matrices that commute with

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

**34.** Construct a $3 \times 3$ matrix **A** such that $\mathbf{A}^3 = \mathbf{I}$, but $\mathbf{A}^2 \neq \mathbf{I}$. (Try to think of a simple linear transformation $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ with $\mathbf{T}^3 = \mathbf{I}$ and use its matrix relative to the standard basis.)

**35.** Let **A** be a $3 \times 3$ matrix and **D** the diagonal matrix

$$\mathbf{D} = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}.$$

(1) Compute $\mathbf{D} \cdot \mathbf{A}$.
(2) Compute $\mathbf{A} \cdot \mathbf{D}$.

**36.** If $\mathbf{A}$ is an idempotent square matrix, show that $\mathbf{I} - 2\mathbf{A}$ is invertible. (HINT: Idempotents correspond to projections. Interpret $\mathbf{I} - 2\mathbf{A}$ as a reflection. Try the $2 \times 2$ case first; then try to generalize.)

**37.** Let $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A}+\mathbf{B}$ be invertible $n \times n$ matrices. Show that $\mathbf{A}^{-1}+\mathbf{B}^{-1}$ is invertible. HINT: If you cannot think of anything else, try to show that

$$(\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1} = \mathbf{A}(\mathbf{A} + \mathbf{B})^{-1}\mathbf{B}.$$

**38.** Find all the $2 \times 2$ matrices $\mathbf{A}$ satisfying the equation $\mathbf{A}^2 + \mathbf{I} = \mathbf{0}$.

**39.** Find the inverse of the matrix

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

**40.** Show that

$$\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$$

is invertible. Find its inverse. Solve the equation

$$\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

for $x$ and $y$.

**41.** Let $A \in \mathbb{R}^n$. Write $\mathbf{A}$ for $A$ regarded as an $n \times 1$ matrix. Show that $A = 0$ if and only if the matrix product $\mathbf{A} \cdot \mathbf{A}^{\text{tr}}$ is the zero matrix.

**42.** A square matrix $\mathbf{P}$ is called a **permutation matrix** if in every column and every row there is exactly one nonzero entry, which is a 1. Show:

(1) $\mathbf{I}$ is a permutation matrix.
(2) A skew-symmetric matrix is never a permutation matrix.
(3) If $\mathbf{A}$ and $\mathbf{B}$ are permutation matrices, then so are $\mathbf{AB}$ and $\mathbf{BA}$.
(4) Permutation matrices are invertible, and the inverse is again a permutation matrix.

(If you have trouble with the general case, try the case of $3 \times 3$ matrices.)

**43.** A square matrix $\mathbf{P}$ is called a **generalized permutation matrix** if in every column and every row there is exactly one nonzero entry. Show:

(1) A diagonal matrix is a generalized permutation matrix if and only if it is nonsingular.
(2) Give an example of a skew-symmetric generalized permutation matrix.

(3) If $\mathbf{A}$ and $\mathbf{B}$ are generalized permutation matrices, then so are $\mathbf{AB}$ and $\mathbf{BA}$.
(4) Generalized permutation matrices are invertible, and the inverse is again a generalized permutation matrix.

(If you have trouble with the general case, try the case of $3 \times 3$ matrices.)

# Chapter 11
# Representing Linear Transformations by Matrices

In this chapter we return to the ideas we introduced in Chapter 10 to represent a linear transformation $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ by a $3 \times 3$ matrix. There is, of course, nothing sacred about $\mathbb{R}^3$ and its standard basis, and we will show that these ideas have far wider applications.

## 11.1 Representing a Linear Transformation by a Matrix

Let us suppose that we are given a linear transformation

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$$

between the finite-dimensional vector spaces $\mathcal{V}$ and $\mathcal{W}$. We fix an ***ordered*** basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ for $\mathcal{V}$ and an ordered basis $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ for $\mathcal{W}$. It is then possible to find unique numbers $a_{i,j}$, where $i = 1, \ldots, n, j = 1, \ldots, m$, such that

$$\begin{aligned}
\mathsf{T}(\mathbf{A}_1) &= a_{1,1}\mathbf{B}_1 + a_{2,1}\mathbf{B}_2 + \ldots + a_{m,1}\mathbf{B}_m, \\
\mathsf{T}(\mathbf{A}_2) &= a_{1,2}\mathbf{B}_1 + a_{2,2}\mathbf{B}_2 + \ldots + a_{m,2}\mathbf{B}_m, \\
&\vdots \\
\mathsf{T}(\mathbf{A}_n) &= a_{1,n}\mathbf{B}_1 + a_{2,n}\mathbf{B}_2 + \ldots + a_{m,n}\mathbf{B}_m,
\end{aligned}$$

which, we have seen may be written more compactly using the $\Sigma$-notation as

$$\mathsf{T}(\mathbf{A}_j) = \sum_{i=1}^{m} a_{i,j}\mathbf{B}_i, \qquad j = 1, 2, \ldots, n.$$

The $m \times n$ matrix $\mathsf{A} = (a_{i,j})$ is called the **matrix of T relative** to the ordered bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$, $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$. Note that just as in the case of $\mathcal{V} = \mathbb{R}^3 = \mathcal{W}$, that the **columns** of $\mathsf{A}$ are the coordinates of the vectors $\mathsf{T}(\mathbf{A}_j)$ relative to the basis $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$. Note also that in saying "**A** is the matrix of **T**" you ***must*** also specify relative to which pair of ordered bases. Here are two examples that illustrate what we mean.

**EXAMPLE 1:** Let

$$\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

be the linear transformation given by

$$\mathsf{T}(x, y) = (y, x).$$

Calculate the matrix of $\mathsf{T}$ relative to the standard basis of $\mathbb{R}^2$.

**SOLUTION:** By the "standard basis of $\mathbb{R}^2$" we mean the basis $\mathbf{E}_1 = (1, 0)$, $\mathbf{E}_2 = (0, 1)$, and that we are to use this ordered basis in both the domain and range of $\mathsf{T}$. We have

$$\mathsf{T}(1, 0) = (0, 1) \qquad \mathsf{T}(0, 1) = (1, 0),$$

so the matrix we seek is

$$\mathsf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

**EXAMPLE 2:** With $\mathsf{T}$ as in Example 1 find the matrix of $\mathsf{T}$ relative to the pair of ordered bases $\{\mathbf{E}_1, \mathbf{E}_2\}$ and $\{\mathbf{F}_1, \mathbf{F}_2\}$, where $\mathbf{F}_1 = (0, 1)$, $\mathbf{F}_2 = (1, 0)$. (Here $\{\mathbf{E}_1, \mathbf{E}_2\}$ is the basis in the domain of $\mathsf{T}$ and $\{\mathbf{F}_1, \mathbf{F}_2\}$ the basis for the range of $\mathsf{T}$.)

**SOLUTION:** We still have

$$\mathsf{T}(1, 0) = (0, 1), \qquad \mathsf{T}(0, 1) = (1, 0),$$

***but*** now we must write these equations as

$$\mathsf{T}(\mathbf{E}_1) = 1\mathbf{F}_1 + 0\mathbf{F}_2, \qquad \mathsf{T}(\mathbf{E}_2) = 0\mathbf{F}_1 + 1\mathbf{F}_2,$$

so that

$$\mathsf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is the matrix that we seek.

The lesson to be learned from the preceding example is that appearance can be deceiving! It also suggests that we might profitably inquire into when two matrices represent the same linear transformation relative to different ordered bases. Before doing so, let us see a few more examples.

**EXAMPLE 3:** Calculate the matrix of the differentiation operator

$$\mathbf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R})$$

relative to the usual basis $\left\{1, x, x^2, \ldots, x^n\right\}$ for $\mathcal{P}_n(\mathbb{R})$.

**SOLUTION:** We have for $m = 1, 2, \ldots$

$$\mathbf{D}x^m = mx^{m-1} = 0 + 0x + \cdots + mx^{m-1} + 0x^m + \cdots + 0x^n,$$

and of course

$$\mathbf{D}(1) = 0.$$

Thus the matrix that we seek is

$$\mathbf{N} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \\ 0 & 0 & 0 & & 0 \end{bmatrix}.$$

For example,

$$\mathbf{N} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{when } n = 1,$$

$$\mathbf{N} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{when } n = 2,$$

$$\mathbf{N} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{when } n = 3,$$

etc. Notice that all the nonzero entries are along the **superdiagonal**[1] of the matrix. So far, the examples have all led to square matrices, but this is an artificial restriction due to the examples having been chosen as simple as possible. For an example leading to a nonsquare matrix we can simply regard the differential operator of the previous example as a linear transformation

$$\mathbf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_{n-1}(\mathbb{R}),$$

and we can ask for its matrix relative to the standard bases of $\mathcal{P}_n(\mathbb{R})$ and $\mathcal{P}_{n-1}(\mathbb{R})$. (What size is this matrix? Answer: $n \times (n + 1)$.) Computing

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[1] The **superdiagonal** of a square matrix $\mathbf{A} = (a_{i,j})$ consists of the entries $a_{1,2}, a_{2,3}, \ldots, a_{n-1,n}$ of $\mathbf{A}$.

as before, we see, that the required matrix is

$$\mathsf{N}' = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & n \end{bmatrix}.$$

For example

$$\mathsf{N}' = \begin{bmatrix} 0 & 1 \end{bmatrix} \qquad \text{when } n = 1,$$

$$\mathsf{N}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \qquad \text{when } n = 2,$$

$$\mathsf{N}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \qquad \text{when } n = 3,$$

etc.

For a related example, try to calculate the matrix of the differential operator when we consider it as a linear transformation

$$\mathsf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_{n+1}(\mathbb{R})$$

($n + 1$ not $n - 1$). Before you write anything down decide what size it should be.

**Example 4:** Calculate the matrix of the linear transformation

$$\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathcal{P}_1(\mathbb{R})$$

given by

$$\mathsf{T}(a_1, a_2, a_3, a_4) = (a_1 + a_3) + (a_2 + a_4)x$$

relative to the ordered bases

$$\mathbf{A}_1 = (1, 1, 1, 1),$$

$$\mathbf{A}_2 = (1, 1, 1, 0),$$

$$\mathbf{A}_3 = (1, 1, 0, 0),$$

$$\mathbf{A}_4 = (1, 0, 0, 0),$$

for $\mathbb{R}^4$ and

$$\mathbf{B}_1 = 1 + x,$$

$$\mathbf{B}_2 = 1 - x,$$

for $\mathcal{P}_1(\mathbb{R})$.

**SOLUTION:** We have (note $1 = \frac{1}{2}(\mathbf{B}_1 + \mathbf{B}_2)$, $x = \frac{1}{2}(\mathbf{B}_1 - \mathbf{B}_2)$)

$$\begin{aligned}
\mathsf{T}(\mathbf{A}_1) &= \mathsf{T}(1, 1, 1, 1) = 2 + 2x = 2\mathbf{B}_1 + 0\mathbf{B}_2, \\
\mathsf{T}(\mathbf{A}_2) &= \mathsf{T}(1, 1, 1, 0) = 2 + x = \frac{3}{2}\mathbf{B}_1 + \frac{1}{2}\mathbf{B}_2, \\
\mathsf{T}(\mathbf{A}_3) &= \mathsf{T}(1, 1, 0, 0) = 1 + x = \mathbf{B}_1 + 0\mathbf{B}_2, \\
\mathsf{T}(\mathbf{A}_4) &= \mathsf{T}(1, 0, 0, 0) = 1 = \frac{1}{2}\mathbf{B}_1 + \frac{1}{2}\mathbf{B}_2,
\end{aligned}$$

so the matrix sought is

$$\begin{bmatrix} 2 & \frac{3}{2} & 1 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}.$$

**EXAMPLE 5:** Let the linear transformation

$$\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$$

be given by the formula

$$\mathsf{T}(x, y, z) = (x + y, y + z).$$

Calculate the matrix of $\mathsf{T}$ relative to:

(a) The standard bases of $\mathbb{R}^3$ and $\mathbb{R}^2$.
(b) The bases

$$\mathbf{A}_1 = (1, 0, 0), \quad \mathbf{A}_2 = (0, 0, 1), \quad \mathbf{A}_3 = (1, -1, 1)$$

and

$$\mathbf{E}_1 = (1, 0), \qquad \mathbf{E}_2 = (0, 1).$$

**SOLUTION:** We have

$$\begin{aligned}
\mathsf{T}(1, 0, 0) &= (1, 0), \\
\mathsf{T}(0, 1, 0) &= (1, 1), \\
\mathsf{T}(0, 0, 1) &= (0, 1),
\end{aligned}$$

so that

$$\mathsf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

is the matrix for part (a). On the other hand,

$$\mathsf{T}(1, -1, 1) = (0, 0),$$

so that

$$\mathsf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

is the matrix for part (b).

**EXAMPLE 6:** Let $\mathsf{T} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_5(\mathbb{R})$ be the linear transformation given by the formula

$$\mathsf{T}(p(x)) = (1 + x - x^2) \cdot p(x).$$

Find the matrix of $\mathsf{T}$ relative to the standard bases of $\mathcal{P}_3(\mathbb{R})$ and $\mathcal{P}_5(\mathbb{R})$.

**SOLUTION:** We compute

$$\begin{aligned}
\mathsf{T}(1) &= 1 + x - x^2, \\
\mathsf{T}(x) &= x + x^2 - x^3, \\
\mathsf{T}(x^2) &= x^2 + x^3 - x^4, \\
\mathsf{T}(x^3) &= x^3 + x^4 - x^5,
\end{aligned}$$

and recall that the columns of $\mathsf{T}$ are the vectors $\mathsf{T}(1), \mathsf{T}(x), \mathsf{T}(x^2)$, and $\mathsf{T}(x^3)$ expressed in terms of the basis $\left\{1, x, x^2, x^3, x^4, x^5\right\}$. So we obtain

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

as the required matrix.

As a last example consider:

**EXAMPLE 7:** Let $S = \{u, v, w\}$ and define

$$\mathsf{E} : \mathrm{Fun}_{\mathbb{R}}(S) \longrightarrow \mathbb{R}$$

by

$$\mathsf{E}(f) = f(u) + f(v) + f(w), \qquad f \in \mathrm{Fun}_{\mathbb{R}}(S).$$

Determine the matrix of $\mathsf{E}$ relative to the ordered bases

$$\begin{array}{ll} \{\chi_u, \chi_v, \chi_w\} & \text{for } \mathrm{Fun}_{\mathbb{R}}(S), \\ \{1\} & \text{for } \mathbb{R}. \end{array}$$

**SOLUTION:** We compute

$$\begin{aligned}
\mathsf{E}(\chi_u) &= \chi_u(u) + \chi_u(v) + \chi_u(w) = 1 + 0 + 0 = 1, \\
\mathsf{E}(\chi_v) &= \chi_v(u) + \chi_v(v) + \chi_v(w) = 0 + 1 + 0 = 1, \\
\mathsf{E}(\chi_w) &= \chi_w(u) + \chi_w(v) + \chi_w(w) = 0 + 0 + 1 = 1,
\end{aligned}$$

so the required matrix is

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

for $\mathsf{E}$.

**EXAMPLE 8:** Let $\mathsf{T}\mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation given by

$$\mathsf{T}(x, y) = x + 2y, 3x + 2y).$$

The matrix of $\mathsf{T}$ with respect the the standard bases is

$$\mathsf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}.$$

On the other hand,

$$\mathsf{T}(2, 3) = (8, 12) = 4(2, 3),$$
$$\mathsf{T}(1, -1) = (-1, 1) = -1(1, -1),$$

and the vectors $\mathbf{A}_1 = (2, 3), \mathbf{A}_2 = (1, -1)$ are a basis for $\mathbb{R}^2$. With respect to this basis (used twice) the matrix of $\mathsf{T}$ takes the much simpler form of the diagonal matrix

$$\mathsf{B} = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}.$$

Discovering such *hidden symetries* in a linear transformation is the subject of much of the rest of this book.

The preceding examples illustrate (I hope!) that the matrix of a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ can be exceedingly complex if some care is not exercised in the choice of the basis relative to which the matrix is computed. Indeed finding a basis relative to which the matrix is as simple as possible should clearly be one's goal if matrices are to simplify our numerical computations with linear transformations.

## 11.2 Basic Theorems

With a number of numerical examples behind us, we turn to a more careful investigation of the relation between linear transformations and matrices.

**THEOREM 11.2.1 :** *Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces. Suppose that $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ are ordered bases for $\mathcal{V}$ and $\mathcal{W}$ respectively. Then assigning to each linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ its matrix relative to these ordered bases defines an isomorphism*

$$\mathsf{M} : \mathcal{L}(\mathcal{V}, \mathcal{W}) \longrightarrow \mathrm{Mat}_{m,n}$$

*of the vector space of linear transformations from $\mathcal{V}$ to $\mathcal{W}$ with the vector space of $m \times n$ matrices.*

**PROOF:** It is clear that $\mathsf{M}$ is a linear transformation. Indeed our definition of matrix addition and scalar multiplication was rigged up so that this would be so. To show that $\mathsf{M}$ is an isomorphism we will actually construct a linear map

$$\mathsf{L} : \mathrm{Mat}_{m,n} \longrightarrow \mathcal{L}(\mathcal{V}, \mathcal{W})$$

such that

$$\mathsf{L} \cdot \mathsf{M}(\mathsf{T}) = \mathsf{T} \quad \text{for all } \mathsf{T} \text{ in } \mathcal{L}(\mathcal{V}, \mathcal{W}),$$
$$\mathsf{M} \cdot \mathsf{L}(\mathsf{A}) = \mathsf{A} \quad \text{for all } \mathsf{A} \text{ in } \mathrm{Mat}_{m,n}.$$

So suppose that $\mathsf{A} = (a_{i,j})$ is an $m \times n$ matrix. Define $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ to be the linear extension of

$$\begin{aligned}
\mathsf{T}(\mathbf{A}_1) &= a_{1,1}\mathbf{B}_1 + a_{2,1}\mathbf{B}_2 + \cdots + a_{m,1}\mathbf{B}_m,\\
\mathsf{T}(\mathbf{A}_2) &= a_{1,2}\mathbf{B}_1 + a_{2,2}\mathbf{B}_2 + \cdots + a_{m,2}\mathbf{B}_m,\\
&\vdots\\
\mathsf{T}(\mathbf{A}_n) &= a_{1,n}\mathbf{B}_1 + a_{2,n}\mathbf{B}_2 + \cdots + a_{m,n}\mathbf{B}_m.
\end{aligned}$$

Thus, if

$$\mathbf{C} = c_1\mathbf{A}_1 + c_2\mathbf{A}_2 + \cdots + c_n\mathbf{A}_n$$

is an arbitrary vector of $\mathcal{V}$, we have the horrendous formula

$$\begin{aligned}
\mathsf{T}(\mathbf{C}) &= (a_{1,1}c_1 + a_{1,2}c_2 + \cdots + a_{1,n}c_n)\mathbf{B}_1 + \cdots\\
&\quad + (a_{m,1}c_1 + a_{m,2}c_2 + \cdots + a_{m,n}c_n)\mathbf{B}_m\\
&= \sum_{i=1}^{m} \left( \sum_{j=1}^{n} a_{i,j}c_j \right) \mathbf{B}_i.
\end{aligned}$$

It is immediately clear that the assignment of the linear transformation $\mathsf{T}$ to the matrix $\mathsf{A}$ defines a linear transformation

$$\mathsf{L} : \mathrm{Mat}_{m,n} \longrightarrow \mathcal{L}(\mathcal{V}, \mathcal{W}).$$

The matrix of $\mathsf{L}(\mathsf{A})$ with respect to the basis pair $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ is by definition nothing but $\mathsf{A}$. That is,

$$\mathsf{ML}(\mathsf{A}) = \mathsf{A} \qquad \forall\, \mathsf{A} \in \mathrm{Mat}_{m,n}.$$

On the other hand, suppose that $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ is in $\mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $\mathsf{T}$ and $\mathsf{LM}(\mathsf{T})$ are given on the basis $\mathbf{A}_1, \ldots, \mathbf{A}_n$ by one and the same formula, to wit:

$$\mathsf{T}(\mathbf{A}_j) = \sum_{i=1}^{m} a_{i,j}\mathbf{B}_i = \mathsf{LM}(\mathsf{T}), \qquad j = 1, \ldots, n.$$

Since $\mathbf{A}_1, \ldots, \mathbf{A}_n$ is a basis for $\mathcal{V}$, one has for any vector a unique expression

$$\mathbf{C} = \sum_{j=1}^{n} c_j \mathbf{A}_j,$$

so we have

$$\mathsf{T}(\mathbf{C}) = \sum_{i=1}^{m} \left( \sum_{j=1}^{n} a_{i,j} c_j \right) \mathbf{B}_i = (\mathsf{LM})(\mathsf{T})(\mathbf{C}),$$

and hence $\mathsf{T} = \mathsf{LM}(\mathsf{T})$; that is,

$$\mathsf{LM}(\mathsf{T}) = \mathsf{T} \qquad \forall\, \mathsf{T} \in \mathcal{L}(\mathcal{V}, \mathcal{W}).$$

Therefore, $\mathsf{M}$ is an isomorphism with inverse $\mathsf{L}$. $\square$

Before examining several interesting consequences of Theorem 11.2.1, we should note that the proof of this theorem is of interest in its own right. Namely, given the following data,

(1) a finite-dimensional vector space $\mathcal{V}$ with basis $\mathbf{A}_1, \ldots, \mathbf{A}_n$,
(2) a finite-dimensional vector space $\mathcal{W}$ with basis $\mathbf{B}_1, \ldots, \mathbf{B}_m$,
(3) an $m \times n$ matrix $\mathsf{A} = (a_{i,j})$,

we may manufacture a linear transformation

$$\mathsf{L}(\mathsf{A}) = \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$$

by the formula

$$(*) \qquad \mathsf{T}(\mathbf{C}) = \sum_{i=1}^{m} \left( \sum_{j=1}^{n} a_{i,j} c_j \right) \mathbf{B}_i,$$

where $\mathbf{C} = \sum c_j \mathbf{A}_j$, or, what is the same thing, *by requiring that the matrix of* $\mathsf{T}$ *relative to the ordered bases* $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ *and* $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ *be* $\mathsf{A}$.

**EXAMPLE 1 :** Find the value of the linear transformation $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathcal{P}_2(\mathbb{R})$ whose matrix relative to the bases $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ and $\{1, x, x^2\}$ is

$$\mathsf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 4 & -3 \\ 3 & 0 & 2 \end{bmatrix}$$

on the vector $\mathbf{C} = (1, 1, -1)$ of $\mathbb{R}^3$.

SOLUTION: We have

$$\begin{aligned}
\mathsf{T}(\mathbf{C}) &= \mathsf{T}(1\mathbf{E}_1 + 1\mathbf{E}_2 - 1\mathbf{E}_3) \\
&= \mathsf{T}(\mathbf{E}_1) + \mathsf{T}(\mathbf{E}_2) - \mathsf{T}(\mathbf{E}_3) \\
&= (1 + 2x + 3x^2) + (4x) - (-1 - 3x + 2x^2) \\
&= 1 + 1 + 2x + 4x + 3x + 3x^2 - 2x^2 \\
&= 2 + 9x + x^2.
\end{aligned}$$

Reflection on the above example and equation (*) will show that the coordinates of $\mathsf{T}(\mathbf{C})$ relative to the ordered basis $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ appear as the entries in the ***column*** vector

$$\mathsf{A} \cdot \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

*Thus we may solve problems such as example* 1 *by matrix multiplication.*

EXAMPLE 2: Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ be the linear transformation whose matrix relative to the standard bases of $\mathbb{R}^3 \longrightarrow \mathbb{R}^4$ is

$$\mathsf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 3 & 4 & 1 \\ -1 & -5 & -1 \\ 2 & 2 & 0 \end{bmatrix}.$$

Calculate the value of $\mathsf{T}$ on $(1, 2, -4)$.

SOLUTION: We have

$$\begin{bmatrix} 0 & 1 & 0 \\ 3 & 4 & 1 \\ -1 & -5 & -1 \\ 2 & 2 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ -7 \\ 6 \end{bmatrix}.$$

Therefore, $\mathsf{T}(1, 2, 4) = (2, 7, -7, 6)$.

While it may appear strange that we calculate the value of $\mathsf{T}(\mathbf{C})$ by using the coordinates of $\mathbf{C}$ to make a column matrix and that our answer appears as a column of numbers instead of a row, let us just remark that we could very well have agreed to write vectors in $\mathbb{R}^n$ as columns of numbers instead of rows. The reason for not doing so is that English reads from left to right. The interchange of rows and columns in the mathematical formalism is an unfortunate, though not accidental, occurrence, and is tied up with the difference between covariance and contravariance in physics. In Chinese, column vectors would be perfectly natural.

**Example 3:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}$ be the linear transformation whose matrix is

$$[1, -2, 1]$$

relative to the standard bases. Find $\mathsf{T}(6, -4, 9)$.

**Solution:** We have

$$\mathsf{T}(6, -4, 9) = (1, -2, 1) \cdot \begin{bmatrix} 6 \\ -4 \\ 9 \end{bmatrix} = [6 + 8 + 9] = [23].$$

**Example 4:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ be the linear transformation with matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

relative to the standard bases of $\mathbb{R}^3$ and $\mathbb{R}^4$. Find bases for the kernel and image of $\mathsf{T}$.

**Solution:** If $(x, y, z) \in \mathbb{R}^3$, then by matrix multiplication we can find a formula for $\mathsf{T}(x, y, z)$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y + z \\ x + y \\ z \end{bmatrix}.$$

So

$$\mathsf{T}(x, y, z) = (x, y + z, x + y, z),$$

and

$$(x, y, z) \in \ker(\mathsf{T}) \quad \text{if and only if} \quad \mathbf{0} = (x, y + z, x + y, z),$$

so

$$\left.\begin{aligned} 0 &= x \\ 0 &= x + y \end{aligned}\right\} x = 0 \quad \text{and} \quad y = 0,$$

$$\left.\begin{aligned} 0 &= y + z \\ 0 &= z \end{aligned}\right\} z = 0 \quad \text{and} \quad y = 0.$$

Therefore, $\ker(\mathsf{T}) = \{\mathbf{0}\}$. To find a basis for the image of $\mathsf{T}$, note that since $\ker(\mathsf{T}) = \{\mathbf{0}\}$, the image has dimension 3. Therefore, the vectors

$$\mathsf{T}(1, 0, 0), \qquad \mathsf{T}(0, 1, 0), \qquad \mathsf{T}(0, 0, 1)$$

are a basis for the image. By the definition of the matrix of a linear transformation, the components of these vectors are the columns of the matrix, so

$$(1, 0, 1, 0), \quad (0, 1, 1, 0), \quad (0, 1, 0, 1)$$

is a basis for Im($\mathbf{T}$).

Let us return now to the consequences of Theorem 11.2.1 we hinted at previously. First we have:

**COROLLARY 11.2.2 :** *Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces. Then $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is also finite-dimensional, and moreover,* $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = \dim(\mathcal{V}) \cdot \dim(\mathcal{W})$.

**PROOF :** This is immediate from the fact that $\mathcal{L}(\mathcal{V}, \mathcal{W})$ and $\mathrm{Mat}_{m,n}$ are isomorphic, where $m = \dim(\mathcal{V})$ and $n = \dim(\mathcal{W})$. $\square$

Before we can state further corollaries, we need the following very important result.

**PROPOSITION 11.2.3:** *Let $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$ be finite-dimensional vector spaces with bases $\{\mathbf{C}_1, \ldots, \mathbf{C}_p\}$, $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$, and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$. Suppose that*

$$\mathbf{S} : \mathcal{U} \longrightarrow \mathcal{V}, \qquad \mathbf{T} : \mathcal{V} \longrightarrow \mathcal{W}$$

*are linear transformations. Let $\mathbf{B} = (b_{i,j})$ be the matrix of $\mathbf{S}$ relative to the bases $\{\mathbf{C}_1, \ldots, \mathbf{C}_p\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$, and $\mathbf{A} = (a_{j,k})$ the matrix of $\mathbf{T}$ relative to the bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$. Then the matrix of $\mathbf{T} \cdot \mathbf{S}$ relative to the bases $\{\mathbf{C}_1, \ldots, \mathbf{C}_p\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ is the matrix product $\mathbf{AB}$.*

**PROOF :** This is an immediate consequence of the definition of the product of two matrices. $\square$

**EXAMPLE 5:** Let

$$\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^4, \quad \mathbf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^2$$

be the linear transformations whose matrices relative to the standard bases are

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & -1 \\ -2 & -1 & -2 \\ 0 & 2 & 5 \\ 4 & 3 & 0 \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} -1 & 0 & 3 & -1 \\ 2 & 4 & 1 & 2 \end{bmatrix}.$$

Find the matrix of the transformation $\mathbf{T} \cdot \mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ relative to the standard bases.

**SOLUTION:** According to proposition 11.2.3 we need only calculate the matrix product **AB**, which is

$$\begin{bmatrix} -1 & 0 & 3 & -1 \\ 2 & 4 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -1 \\ -2 & -1 & -2 \\ 0 & 2 & 5 \\ 4 & 3 & 0 \end{bmatrix} = \begin{bmatrix} -5 & 3 & 16 \\ 2 & 4 & -5 \end{bmatrix},$$

and so the matrix that we seek is

$$\begin{bmatrix} -5 & 3 & 16 \\ 2 & 4 & -5 \end{bmatrix}.$$

**COROLLARY 11.2.4:** *A linear transformation* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *is an isomorphism if and only if its matrix* $\mathsf{A}$ *is invertible. (The matrix* $\mathsf{A}$ *may be computed relative to* ***any*** *pair* $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ *and* $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ *of ordered bases for* $\mathcal{V}$.*)*

**PROOF:** Suppose that $\mathsf{T}$ is an isomorphism. Let $\mathsf{S} : \mathcal{V} \longrightarrow \mathcal{V}$ be the linear transformation inverse to $\mathsf{T}$. Let $\mathsf{B}$ be the matrix of $\mathsf{S}$ relative to the basis pair $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. (N.B. We have interchanged the roles of the bases $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. Thus if $\mathsf{B} = (b_{i,j})$, then $\mathsf{S}(\mathbf{B}_j) = \sum b_{i,j}\mathbf{A}_i$.) According to Proposition 11.2.3 the matrix product $\mathsf{AB}$ is the matrix of the linear transformation $\mathsf{T} \cdot \mathsf{S} : \mathcal{V} \longrightarrow \mathcal{V}$ relative to the basis pair $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$. But $\mathsf{T} \cdot \mathsf{S}(\mathbf{C}) = \mathbf{C}$ for all $\mathbf{C} \in \mathcal{V}$, since $\mathsf{T}$ and $\mathsf{S}$ are inverse isomorphisms. In particular,

$$\mathsf{T} \cdot \mathsf{S}(\mathbf{B}_j) = 0\mathbf{B}_1 + 0\mathbf{B}_2 + \cdots + 0\mathbf{B}_{j-1} + 1\mathbf{B}_j + 0\mathbf{B}_{j+1} + \cdots + 0\mathbf{B}_n$$

and hence the matrix of $\mathsf{T} \cdot \mathsf{S}$ relative to the bases $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ is

$$\mathsf{I} = \begin{bmatrix} 1 & & 0 \\ & 1 & \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix}.$$

Therefore, $\mathsf{AB} = \mathsf{I}$. Likewise, according to Proposition 11.2.3, the matrix product $\mathsf{BA}$ is the matrix of the linear transformation $\mathsf{ST} : \mathcal{V} \longrightarrow \mathcal{V}$ relative to the basis pair $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. But $\mathsf{ST}(\mathbf{C}) = \mathbf{C}$ for all $\mathbf{C}$ in $\mathcal{V}$ because $\mathsf{S}$ and $\mathsf{T}$ are inverse isomorphisms, and hence as before, we find that $\mathsf{BA} = \mathsf{I}$. This shows that if $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is an isomorphism, then any matrix $\mathsf{A}$ representing $\mathsf{T}$ is invertible.

To prove the converse, we suppose that the matrix $\mathsf{A}$ of $\mathsf{T}$ is invertible. Let $\mathsf{B}$ be a matrix such that $\mathsf{AB} = \mathsf{I} = \mathsf{BA}$. Let $\mathsf{S} : \mathcal{V} \longrightarrow \mathcal{V}$ be the linear transformation whose matrix relative to the ordered bases $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is $\mathsf{B}$. (N.B. We have again interchanged the roles of $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$.) Then the matrix of $\mathsf{S} \cdot \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ relative to the ordered bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is

$$\mathsf{I} = \begin{bmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix}.$$

Therefore, $\mathsf{S} \cdot \mathsf{T}$ and $\mathsf{I}$ have the same matrix relative to the bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$, so that by Proposition 11.2.3 $\mathsf{S} \cdot \mathsf{T} = \mathsf{I}$; that is,

$$\mathsf{S} \cdot \mathsf{T}(\mathbf{C}) = \mathbf{C} \qquad \forall\, \mathbf{C} \in \mathcal{V}.$$

Likewise, we see that

$$\mathsf{T} \cdot \mathsf{S}(\mathbf{C}) = \mathbf{C} \qquad \forall\, \mathbf{C} \in \mathcal{V},$$

so that $\mathsf{S}$ and $\mathsf{T}$ are inverse isomorphisms. □

Note that in the proof of Corollary 11.2.4 we used the fact that the matrix of the identity transformation $\mathsf{I} : \mathcal{V} \longrightarrow \mathcal{V}$, which is defined by $\mathsf{I}(\mathbf{C}) = \mathbf{C}$ for all $\mathbf{C} \in \mathcal{V}$, relative to the bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is the identity matrix. This is not the case if we have two *different* bases (or even different orderings on the same basis) in $\mathcal{V}$.

**EXAMPLE 6:** Find the matrix of the identity linear transformation $\mathsf{I} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ relative to the ordered bases

$$\mathbf{E}_1 = (1, 0, 0),\ \mathbf{E}_2 = (0, 1, 0),\ \mathbf{E}_3 = (0, 0, 1)$$

and

$$\mathbf{F}_1 = (1, 1, 1),\ \mathbf{F}_2 = (1, 1, 0),\ \mathbf{F}_3 = (1, 0, 0)$$

of $\mathbb{R}^3$.

**SOLUTION:** We have

$$\mathsf{I}(\mathbf{E}_1) = (1, 0, 0) = (1, 0, 0) = 0\mathbf{F}_1 + 0\mathbf{F}_2 + 1\mathbf{F}_3,$$

$$\mathsf{I}(\mathbf{E}_2) = (0, 1, 0) = (0, 1, 0) = 0\mathbf{F}_1 + 1\mathbf{F}_2 + (-1)\mathbf{F}_3,$$

$$\mathsf{I}(\mathbf{E}_3) = (0, 0, 1) = (0, 0, 1) = 1\mathbf{F}_1 + (-1)\mathbf{F}_2 + 0\mathbf{F}_3.$$

So the matrix we are looking for is

$$\mathsf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix}.$$

The moral of the example is that appearances are ***really*** deceiving.

In view of Example 6 it is reasonable to expect that when we calculate with matrices of transformations $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$, we insist upon using the same ordered basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ twice to do the calculation, rather than work with distinct ordered bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$. Other reasons for insisting on using only one ordered basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ when we study a transformation $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ from the same space to itself will appear in Chapter 14.

We turn next to another consequence of Proposition 11.2.3 which substantiates a remark we made in the last chapter concerning inverses of matrices.

**COROLLARY 11.2.5:** *Suppose that* $\mathbf{A}$ *and* $\mathbf{B}$ *are square matrices of size* $n$ *and* $\mathbf{AB} = \mathbf{I}$. *Then* $\mathbf{BA} = \mathbf{I}$. *(Thus, to check that a square matrix* $\mathbf{B}$ *is the inverse of a square matrix* $\mathbf{A}$, *we need only check one of the two conditions* $\mathbf{AB} = \mathbf{I}$ *and* $\mathbf{BA} = \mathbf{I}$.*)*

**PROOF:** Let $\mathbf{T}, \mathbf{S} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be the linear transformations whose matrices with respect to the standard basis of $\mathbb{R}^n$ are $\mathbf{A}$ and $\mathbf{B}$ respectively. That is,

$$\mathbf{L}(\mathbf{A}) = \mathbf{T}, \quad \mathbf{L}(\mathbf{B}) = \mathbf{S},$$

or, what is the same thing,

$$\mathbf{M}(\mathbf{T}) = \mathbf{A}, \quad \mathbf{M}(\mathbf{S}) = \mathbf{B},$$

where $\mathbf{L}$ and $\mathbf{M}$ are as defined in Theorem 11.2.1. Then by Proposition 11.2.3,

$$\mathbf{M}(\mathbf{T} \cdot \mathbf{S}) = \mathbf{M}(\mathbf{T}) \cdot \mathbf{M}(\mathbf{S}) = \mathbf{AB} = \mathbf{I} = \mathbf{M}(\mathbf{I}).$$

Therefore, since $\mathbf{M}$ is an isomorphism,

$$\mathbf{T} \cdot \mathbf{S} = \mathbf{I}.$$

From this we may conclude that $\mathbf{S} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ is an isomorphism as follows. Suppose $\mathbf{C} \in \ker(\mathbf{S})$. Then $\mathbf{S}(\mathbf{C}) = \mathbf{0}$, so

$$\mathbf{0} = \mathbf{T}(\mathbf{0}) = \mathbf{T}(\mathbf{S}(\mathbf{C})) = (\mathbf{T} \cdot \mathbf{S})(\mathbf{C}) = \mathbf{I}(\mathbf{C}) = \mathbf{C}.$$

Therefore $\mathbf{S}$ has kernel $\{\mathbf{0}\}$ so by proposition 8.4.2 we have

$$n = \dim(\mathbb{R}^n) = \dim(\mathrm{Im}(\mathbf{S})) + \dim(\ker(\mathbf{S})) = \dim(\mathrm{Im}(\mathbf{S})).$$

This implies that $\mathrm{Im}(\mathbf{S}) = \mathbb{R}^n$ by 6.3.6 and hence $\mathbf{S}$ is an isomorphism by Proposition 8.6.1. Therefore there is a transformation

$$\overline{\mathbf{T}} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

such that

$$\mathbf{S}\overline{\mathbf{T}} = \mathbf{I}.$$

Then

$$\mathsf{T} \cdot (\mathsf{S}\overline{\mathsf{T}}) = \mathsf{T}(\mathsf{I}) = \mathsf{T}$$

$$(\mathsf{TS}) \cdot \overline{\mathsf{T}} = \mathsf{I}\overline{\mathsf{T}} = \overline{\mathsf{T}}$$

so that $\mathsf{T} = \overline{\mathsf{T}}$; that is, $\mathsf{ST} = \mathsf{I}$. Hence

$$\mathbf{I} = \mathbf{M}(\mathsf{I}) = \mathbf{M} \cdot (\mathsf{ST}) = \mathbf{M}(\mathsf{S}) \cdot \mathbf{M}(\mathsf{T}) = \mathbf{B} \cdot \mathbf{A}$$

as required. □

While the proof of Corollary 11.2.5 may seem complicated, we invite the reader to try to prove the result using only matrices. Even in the $3 \times 3$ case such a proof will be most painful! Corollaries 11.2.3 and 11.2.5 illustrate an important point, namely, that some results concerning linear transformations are best proved using matrices, and conversely, some results concerning matrices are best proved using linear transformations.

## 11.3 Change of Bases

In the study of linear transformations we will very often make use of matrix representations to prove theorems and make computations. The matrix representation for a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ depends on a choice of bases for $\mathcal{V}$ and $\mathcal{W}$. Our initial choice of these bases may be bad and unsuited to the problem at hand in that we obtain a matrix that does not convey the information we seek. It is therefore quite natural to change the bases to obtain a "better" representation for $\mathsf{T}$, because to change the bases does nothing to the linear transformation, but it does change the matrix representation. Several natural and important questions therefore arise. First of all, what is the relation (numerically, that is) between two matrix representations of $\mathsf{T}$ computed with respect to different pairs of bases in $\mathcal{V}$ and $\mathcal{W}$? Secondly, if we are given two $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, when (that is, what numerical relations must hold between them) can we conclude that they represent one and the same linear transformation $\mathsf{T}$, but relative to different bases? As it turns out, both questions may be answered simultaneously. However, before doing so, we consider one more example of the phenomena under discussion.

**EXAMPLE 1:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation given by the formula

$$\mathsf{T}(x, y, z) = (y + z, x + z, y + x).$$

Calculate the matrix of $\mathsf{T}$ relative to

(a) the standard basis of $\mathbb{R}^3$ used twice, and
(b) the basis $\{(1, 1, 1), (1, -1, 0), (1, 1, -2)\}$ used twice.

**SOLUTION:** To do part (a), we compute as follows:

$$\mathsf{T}(1, 0, 0) = (0, 1, 1),$$
$$\mathsf{T}(0, 1, 0) = (1, 0, 1),$$
$$\mathsf{T}(0, 0, 1) = (1, 1, 0).$$

Thus the desired matrix is

$$\mathsf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

To do the computations for part (b), let us set $\mathbf{F}_1 = (1, 1, 1), \mathbf{F}_2 = (1, -1, 0), \mathbf{F}_3 = (1, 1, -2)$. Then we find

$$\mathsf{T}(\mathbf{F}_1) = \mathsf{T}(1, 1, 1) = (2, 2, 2) = 2\mathbf{F}_1 + 0\mathbf{F}_2 + 0\mathbf{F}_3,$$
$$\mathsf{T}(\mathbf{F}_2) = \mathsf{T}(1, -1, 0) = (-1, 1, 0) = 0\mathbf{F}_1 - \mathbf{F}_2 + 0\mathbf{F}_3,$$
$$\mathsf{T}(\mathbf{F}_3) = \mathsf{T}(1, 1, -2) = (-1, -1, 2) = 0\mathbf{F}_1 + 0\mathbf{F}_2 - \mathbf{F}_3.$$

So the matrix for part (b) is

$$\mathsf{B} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

This illustrates that the matrix of a transformation depends heavily on the bases we use to compute it.

**THEOREM 11.3.1:** *Let* $\mathsf{A}$ *and* $\mathsf{B}$ *be* $m \times n$ *matrices,* $\mathcal{V}$ *an* $n$*-dimensional vector space, and* $\mathcal{W}$ *an* $m$*-dimensional vector space. Then* $\mathsf{A}$ *and* $\mathsf{B}$ *represent the same linear transformation* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *relative to (perhaps) different pairs of ordered bases if and only if there exist nonsingular matrices* $\mathsf{P}$ *and* $\mathsf{Q}$ *such that*

$$\mathsf{A} = \mathsf{P} \cdot \mathsf{B} \cdot \mathsf{Q}^{-1}.$$

*(Note that* $\mathsf{P}$ *is an* $m \times m$ *matrix and* $\mathsf{Q}$ *is an* $n \times n$ *matrix.)*

**PROOF:** There are two things we must prove. First, if $\mathsf{A}$ and $\mathsf{B}$ represent the same linear transformation relative to different bases of $\mathcal{V}$ and $\mathcal{W}$, we must construct invertible matrices $\mathsf{P}$ and $\mathsf{Q}$ such that

$$\mathsf{A} = \mathsf{P} \cdot \mathsf{B} \cdot \mathsf{Q}^{-1}.$$

Second, if we are given invertible matrices $\mathsf{P}$ and $\mathsf{Q}$ such that

$$\mathsf{A} = \mathsf{P} \cdot \mathsf{B} \cdot \mathsf{Q}^{-1},$$

we must construct a linear transformation

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$$

and pairs of ordered bases for $\mathcal{V}$ and $\mathcal{W}$ such that $\mathsf{A}$ represents $\mathsf{T}$ relative to one pair and $\mathsf{B}$ represents $\mathsf{T}$ relative to the other. Consider the first of these. We suppose that we are given bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ such that the matrix of $\mathsf{T}$ relative to these bases is $\mathsf{A}$, and bases $\{\mathbf{C}_1, \ldots, \mathbf{C}_n\}$ and $\{\mathbf{D}_1, \ldots, \mathbf{D}_m\}$ such that the matrix of $\mathsf{T}$ relative to these bases is $\mathsf{B}$.

Let $\mathsf{P}$ be the matrix of (remember Section 11.2 Example 6)

$$\mathsf{I} : \mathcal{W} \longrightarrow \mathcal{W}$$

relative to the bases $\{\mathbf{D}_1, \ldots, \mathbf{D}_m\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$. Then by Corollary 11.2.4, $\mathsf{P}$ is invertible. Let $\mathsf{Q}$ be the matrix of

$$\mathsf{I} : \mathcal{V} \longrightarrow \mathcal{V}$$

relative to the bases $\{\mathbf{C}_1, \ldots, \mathbf{C}_n\}$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. Then by Corollary 11.2.4, $\mathsf{Q}$ is invertible, and by Proposition 11.2.3 $\mathsf{Q}^{-1}$ represents the matrix of

$$\mathsf{I} : \mathcal{V} \longrightarrow \mathcal{V}$$

relative to the bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{C}_1, \ldots, \mathbf{C}_n\}$.

Therefore by Proposition 11.2.3 $\mathsf{P} \cdot \mathsf{B}$ is the matrix of $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ relative to the bases $\{\mathbf{C}_1, \ldots, \mathbf{C}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$. If we apply Proposition 11.2.3 again we see that $\mathsf{P} \cdot \mathsf{B} \cdot \mathsf{Q}^{-1}$ is the matrix of $\mathsf{T}$ relative to the bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$. But $\mathsf{A}$ is also the matrix of $\mathsf{T}$ relative to the bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$, so that by Theorem 11.2.1,

$$\mathsf{A} = \mathsf{P} \cdot \mathsf{B} \cdot \mathsf{Q}^{-1}$$

as required.

To prove the converse, suppose we are given invertible matrices $\mathsf{P}$ and $\mathsf{Q}$ such that $\mathsf{A} = \mathsf{P}\mathsf{B}\mathsf{Q}^{-1}$. Choose bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ for $\mathcal{V}$ and $\mathcal{W}$ respectively. Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be the linear transformation whose matrix is $\mathsf{A}$ relative to these bases. Let

$$\mathbf{C}_1 = \mathsf{P}(\mathbf{A}_1), \ldots, \mathbf{C}_n = \mathsf{P}(\mathbf{A}_n),$$

$$\mathbf{D}_1 = \mathsf{Q}(\mathbf{B}_1), \ldots, \mathbf{D}_m = \mathsf{Q}(\mathbf{B}_m).$$

Since $\mathsf{P}$ and $\mathsf{Q}$ are isomorphisms, the collections $\{\mathbf{C}_1, \ldots, \mathbf{C}_n\}$ and $\{\mathbf{D}_1, \ldots, \mathbf{D}_m\}$ are bases for $\mathcal{V}$ and $\mathcal{W}$ respectively. (See, for example, Propositions 8.5.2 and 8.5.3.) A bruteforce computation now shows that $\mathsf{B}$ is the matrix of $\mathsf{T}$ relative to the bases $\{\mathbf{C}_1, \ldots, \mathbf{C}_n\}$ and $\{\mathbf{D}_1, \ldots, \mathbf{D}_m\}$. $\square$

We will return to Theorem 11.3.1 in Chapters 16 and 17 where we will discuss in more detail when $\mathbf{A}$ and $\mathbf{B}$ represent the same linear transformation relative to distinct bases.

**EXAMPLE 2:** Let us see how Theorem 11.3.1 applies to Example 1. Recall that we are given the linear transformation $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by

$$\mathbf{T}(x, y, z) = (y + z, x + z, y + x),$$

and

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

is the matrix of $\mathbf{T}$ relative to the standard basis of $\mathbb{R}^3$, while

$$\mathbf{B} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

is the matrix of $\mathbf{T}$ relative to the basis $\{(1, 1, 1), (1, -1, 0), (1, 1, -2)\}$ of $\mathbb{R}^3$.

According to Theorem 11.3.1 there are invertible matrices $\mathbf{P}$ and $\mathbf{Q}$ such that $\mathbf{A} = \mathbf{P} \cdot \mathbf{B} \cdot \mathbf{Q}^{-1}$. Our task is to compute $\mathbf{P}$ and $\mathbf{Q}^{-1}$. The proof of Theorem 11.3.1 tells us how. Namely:

(1) $\mathbf{P}$ is the matrix of $\mathbf{I} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ relative to the basis pair $\{(1, 1, 1), (1, -1, 0), (1, 1, -2)\}$ and $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

(2) $\mathbf{Q}^{-1}$ is the matrix of $\mathbf{I} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ relative to the basis pair $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and $\{(1, 1, 1), (1, -1, 0), (1, 1, -2)\}$.

The computation of $\mathbf{P}$ is easy and gives us

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$$

The computation of $\mathbf{Q}^{-1}$ is not hard and depends on the following equations,

$$(1, 0, 0) = \frac{1}{3}(1, 1, 1) + \frac{1}{2}(1, -1, 0) + \frac{1}{6}(1, 1, -2),$$
$$(0, 1, 0) = \frac{1}{3}(1, 1, 1) - \frac{1}{2}(1, -1, 0) + \frac{1}{6}(1, 1, -2),$$
$$(0, 0, 1) = \frac{1}{3}(1, 1, 1) + 0(1, -1, 0) - \frac{1}{3}(1, 1, -2),$$

which express the standard basis in terms of the nonstandard basis. From these equations it follows that

$$\mathbf{Q}^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{6} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix}.$$

A tedious computation shows

$$\mathbf{A} = \mathbf{P} \cdot \mathbf{B} \cdot \mathbf{Q}^{-1},$$

that is,

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{6} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix}.$$

## 11.4 Exercises

**1.** Find the matrix of the following linear transformations relative to the standard bases for $\mathbb{R}^n$.

(1) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^5$ given by

$$\mathsf{T}(a_1, a_2, a_3) = (a_1, a_1 - a_3, a_2 + a_3, a_3, a_1 + a_2).$$

(2) $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ given by

$$\mathsf{T}(a_1, a_2, a_3, a_4) = (a_1, a_1 + a_2, a_1 + a_2 + a_3, a_1 + a_2 + a_3 + a_4).$$

(3) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}$ given by

$$\mathsf{T}(a_1, a_2, a_3) = a_1 + a_2 + a_3.$$

(4) $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^4$ given by

$$\mathsf{T}(a_1, a_2) = (a_1 + a_2, 2a_1, 3a_2 - a_1, 2a_1 - a_2).$$

**2.** Find the matrix of the following linear transformations relative to the usual bases of $\mathcal{P}_n(\mathbb{R})$.

(1) $\mathsf{D} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_3(\mathbb{R})$. (HINT: think about size!)
(2) $\mathsf{D} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_2(\mathbb{R})$.
(3) $\mathsf{T} : \mathcal{P}_2(\mathbb{R}) \longrightarrow \mathcal{P}_2(\mathbb{R})$ given by $\mathsf{T}(p(x)) = p(x + 1)$.
(4) $\mathsf{T} : \mathcal{P}_2(\mathbb{R}) \longrightarrow \mathcal{P}_3(\mathbb{R})$ given by $\mathsf{T}(p(x)) = xp(x)$.

**3.** Let $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^7$ be the linear transformation whose matrix relative to the standard bases is

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -1 & 4 & 1 \\ 2 & 0 & 2 & 0 \\ 0 & -1 & 3 & 1 \\ 3 & 2 & -5 & 0 \\ 0 & 0 & 7 & 1 \\ 4 & 0 & 1 & 0 \end{bmatrix}.$$

Find $\mathsf{T}(1, 2, 3, 4)$.

**4.** Let $\mathsf{S} : \mathbb{R}^7 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix relative to the standard bases is

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

and let $\mathsf{T}$ be the linear transformation of Exercise 3. Find the matrix of $\mathsf{S} \cdot \mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^3$ relative to the standard bases. What is $\mathsf{S} \cdot \mathsf{T}(1, -1, 1, -1)$?

**5.** Let $\mathsf{S}, \mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be the linear transformations with matrices

$$\begin{bmatrix} 6 & -1 & 2 \\ 2 & 4 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -5 & 0 & -7 \\ 2 & -1 & 9 \end{bmatrix}$$

respectively. What is the matrix of the linear transformation

$$3\mathsf{S} - 7\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^2?$$

Find $(3\mathsf{S} - 7\mathsf{T})(1, 2, 3)$.

**6.** Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces and $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ a linear transformation. Show that there are bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_m\}$ for $\mathcal{V}$ and $\mathcal{W}$ such that the matrix of $\mathsf{T}$ is

$$\begin{bmatrix} 1 & & & \\ & \ddots & & 0 \\ & & 1 & \\ & 0 & & 0 \end{bmatrix}.$$

(HINT: Choose a basis $\mathbf{A}'_1, \ldots, \mathbf{A}'_r$ for $\ker(\mathsf{T})$. Extend this to a basis $\mathbf{A}''_1, \ldots, \mathbf{A}''_s, A''_1, \ldots, \mathbf{A}'_r$ for $\mathcal{V}$. Then show that the vectors

$$\mathbf{B}_1 = \mathsf{T}(\mathbf{A}''_1), \ldots, \mathbf{B}_s = \mathsf{T}(\mathbf{A}''_s)$$

are linearly independent in $\mathcal{W}$. Extend them to a basis $\mathbf{B}_1, \ldots, \mathbf{B}_s, \mathbf{B}_{s+1}, \ldots, \mathbf{B}_m$ for $\mathcal{W}$. Let $\mathbf{A}_1 = \mathbf{A}_1'', \ldots, \mathbf{A}_s = \mathbf{A}_s'', \mathbf{A}_{s+1} = \mathbf{A}_1', \ldots, \mathbf{A}_n = \mathbf{A}_r'$. Calculate the matrix of $\mathsf{T}$ relative to these bases. Thus the number of 1s on the main diagonal is dim Im($\mathsf{T}$).)

**7.** Let

$$\mathsf{A} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ b & d & 0 & 0 \\ c & e & f & 0 \end{bmatrix}.$$

Show that $\mathsf{A}^4 = \mathbf{0}$. (HINT: Recall how we did an analogous problem in the $3 \times 3$ case using the linear transformation $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ whose matrix relative to the standard basis was the analogue of $\mathsf{A}$.) Can you generalize your result to $n \times n$?

**8.** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation

$$\mathsf{T}(x, y) = (x + 3y, 3x + y).$$

Find the matrix of $\mathsf{T}$ relative to :

(1) The standard basis of $\mathbb{R}^2$.
(2) The basis $\{\mathbf{F}_1 = (1, 1), \mathbf{F}_2 = (1, -1)\}$ used twice.
(3) $\{\mathbf{F}_1, \mathbf{F}_2\}$ in the domain and the standard basis in the range.

**9.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation

$$\mathsf{T}(x, y, z) = (3y + 4z, 3x, 4x).$$

Calculate the matrix of $\mathsf{T}$ relative to:

(1) The standard basis of $\mathbb{R}^3$.
(2) The basis $\{(0, 4, -3), (5, 3, 4), (5, -3, -4)\}$ used twice.

**10.** Find the matrix of the shift $\mathsf{S} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ relative to the standard basis used twice.

**11.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix relative to the standard basis is

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find the matrix of $\mathsf{T}$ relative to the basis $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\}$ used twice, where $\mathbf{F}_1 = (1, 1, 1), \mathbf{F}_2 = (1, 1, 0), \mathbf{F}_3 = (1, 0, 0)$.

**12.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix relative to the basis $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\}$ of the preceding exercise used twice is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Find the matrix of $\mathsf{T}$ relative to the standard basis.

**13.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation

$$\mathsf{T}(x, y, z) = (x + y, y + z, z + x).$$

(1) Find the matrix of $\mathsf{T}$ relative to the standard basis.
(2) Find vectors $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ satisfying $\mathsf{T}(\mathbf{A}_i) = \mathbf{E}_i$ for $i = 1, 2, 3$.
(3) Show that $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ form a set of linearly independent vectors.
(4) Show that $\mathsf{T}(\mathbf{E}_1), \mathsf{T}(\mathbf{E}_2), \mathsf{T}(\mathbf{E}_3)$ form a set of linearly independent vectors.
(5) Let $\mathsf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear extension of $\mathsf{S}(\mathbf{E}_1) = \mathbf{A}_1$, $\mathsf{S}(\mathbf{E}_2) = \mathbf{A}_2$, $\mathsf{S}(\mathbf{E}_3) = \mathbf{A}_3$, where the vectors $\mathbf{A}_i$ are those of part (b). Find the matrix of $\mathsf{S}$ relative to the standard basis.
(6) Show that the matrices obtained in (a) and (e) are inverse matrices of each other.

**14.** Let $\mathsf{T} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathbb{R}^2$ be the linear transformation defined by

$$\mathsf{T}(p(x)) = \left(p(0), \int_0^1 p(x)dx\right).$$

Find the matrix of $\mathsf{T}$ relative the ordered basis pair $\left\{1, x, x^2, x^3\right\}$ and $\{(1, 0), (0, 1)\}$.

**15.** Let $\sigma : \{a, b, c\} \longrightarrow \{a, b, c\}$ be given by

$$\sigma(a) = b, \qquad \sigma(b) = c, \qquad \sigma(c) = a.$$

Let $S = \{a, b, c\}$ and find the matrix of

$$\sigma^* : \mathrm{Fun}_{\mathbb{R}}(S) \longrightarrow \mathrm{Fun}_{\mathbb{R}}(S)$$

with respect to the basis of characteristic functions.

**16.** Determine whether the two matrices

$$\mathsf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathsf{B} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

can represent the same linear transformation $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ with respect to different pairs of ordered bases.

**17.** Find a pair of $2 \times 2$ matrices $\mathsf{A}, \mathsf{B}$ such that

$$\mathsf{AB} - \mathsf{BA} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

**18.** Show that a $3 \times 3$ matrix

$$\begin{bmatrix} a & 0 & 0 \\ 0 & b & c \\ 0 & d & e \end{bmatrix}$$

is invertible if and only if $a(be - cd) \neq 0$.

**19.** Show that the $3 \times 3$ matrix

$$\begin{bmatrix} a & 0 & c \\ 0 & b & 0 \\ c & 0 & a \end{bmatrix}$$

is invertible if and only if $b(a^2 - c^2) \neq 0$.

**20.** Let $\mathbf{A} \in \mathrm{Mat}_{m,m}$, $\mathbf{B} \in \mathrm{Mat}_{m,k}$, $\mathbf{C} \in \mathrm{Mat}_{k,k}$, and let $\mathbf{D}$ be the block matrix

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{bmatrix} \in \mathrm{Mat}_{m+k,m+k}.$$

(1) Show that $\mathbf{D}$ is invertible if and only if both $\mathbf{A}$ and $\mathbf{C}$ are invertible.
(2) Describe $\mathbf{D}^{-1}$ with the aid of $\mathbf{A}^{-1}$ and $\mathbf{C}^{-1}$.

**21.** Determine whether the matrices

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

represent the same linear transformation with respect to different ordered pairs of bases of $\mathbb{R}^3$.

**22.** Let $\mathbf{M} : \mathbb{C}^2 \longrightarrow \mathbb{C}$ be the linear transformation defined by

$$\mathbf{M}(u, v) = u \cdot v,$$

the dot denoting multiplication of complex numbers. Regard $\mathbb{C}$ and $\mathbb{C}^2$ as ***real*** vector spaces and choose $\{(1, 0), (\boldsymbol{i}, 0), (0, 1), (0, \boldsymbol{i})\}$ as bases for $\mathbb{C}^2$ and $\{1, \boldsymbol{i}\}$ for $\mathbb{C}$. Find the matrix of $\mathbf{M}$ with respect to these ordered bases.

**23.** Show that any nonzero linear combination of the matrices

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is invertible.

**24.** A vector $\mathbf{A} \in \mathcal{U}$ is called a **fixed point** of a linear transformation $\mathbf{T} : \mathcal{U} \longrightarrow \mathcal{U}$ if $\mathbf{T}(\mathbf{A}) = \mathbf{A}$. It is called an **antifixed point** if $\mathbf{T}(\mathbf{A}) = -\mathbf{A}$.

Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be represented by each of the following matrices in turn and find the fixed points and antifixed points, if any:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

**25.** Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with ordered bases $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ and $\{\mathbf{F}_1, \ldots, \mathbf{F}_m\}$. Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be a linear transformation with matrix $\mathsf{M}$ with respect to this basis pair. Let $\mathcal{V}^* = \mathcal{L}(\mathcal{V}, \mathbb{R})$ and $\mathcal{W}^* = \mathcal{L}(\mathcal{W}, \mathbb{R})$.

(1) Define $\mathsf{T}^* : \mathcal{W}^* \longrightarrow \mathcal{V}^*$ by

$$\mathsf{T}^*(f)(\mathbf{A}) = f(\mathsf{T}(\mathbf{A})), \qquad \mathbf{A} \in \mathcal{V}, \quad f \in \mathcal{W}^*.$$

Show that $\mathsf{T}$ is a linear transformation. (It is called the **dual transformation** to $\mathsf{T}$.)

(2) Define $\mathbf{E}_1^*, \ldots, \mathbf{E}_n^*$ to be the linear extension of the functions

$$\mathbf{E}_i^*(\mathbf{E}_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$$

Show that $\{\mathbf{E}_1^*, \ldots, \mathbf{E}_n^*\}$ is basis for $\mathcal{V}^*$, called the **dual basis** to $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$.

(3) Show that the matrix of $\mathsf{T}^*$ with respect to the pair of dual bases $\{\mathbf{F}_1^*, \ldots, \mathbf{F}_m^*\}$ and $\{\mathbf{E}_1^*, \ldots, \mathbf{E}_n^*\}$ is the transpose of $\mathsf{M}$.

**26.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix with respect to the standard basis of $\mathbb{R}^3$ is

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find bases for the kernel and image of $\mathsf{T}$. Use this to show that $\mathsf{T}$ is invertible. Find the matrix of $\mathsf{T}^{-1}$ relative to the standard basis.

**27.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation of the $n$-dimensional vector space $\mathcal{V}$ into itself. Show that the following two statements are equivalent:

(1) *There exists a basis $\mathbf{A}_1, \ldots, \mathbf{A}_n$ for $\mathcal{V}$ such that the matrix of $\mathsf{T}$ with respect to this basis is upper triangular.*

(2) *There exists a sequence of vector subspaces*

$$\{\mathbf{0}\} = \mathcal{U}_0 \subset \mathcal{U}_1 \subset \cdots \subset \mathcal{U}_n = \mathcal{V}$$

*with* $\dim(\mathcal{U}_i) = i$ *and* $\mathsf{T}(\mathcal{U}_i) \subseteq U_i$ *for* $i = 0, \ldots, n$.

# Chapter 12
# More on Representing Linear Transformations by Matrices

Our purpose in this chapter is to discuss some particular types of linear transformations and their matrix representations. It is to be emphasized that we are only scratching the surface of an iceberg!

## 12.1 Projections

One of the simplest types of linear transformations is a **projection**. For example, (see also Example 12.1 in Section 8.2) in $\mathbb{R}^3$ consider the plane $\mathcal{V} = \{(x, y, z) \mid x + y + z = 0\}$ (note $\mathcal{V} \subset \mathbb{R}^3$ is a 2-dimensional subspace) and the linear transformation $\mathsf{P} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ that sends a vector **A** into its projection onto the plane $\mathcal{V}$. (See Figure 12.1.1) A formula for $\mathsf{P} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is not hard to find. Notice that for any vector **A** that lies in $\mathcal{V}$ we must have $\mathsf{P}(\mathbf{A}) = \mathbf{A}$. For example,

$$\mathsf{P}(1, -1, 0) = (1, -1, 0),$$

$$\mathsf{P}(0, 1, -1) = (0, 1, -1),$$

and notice that $\{(1, -1, 0), (0, 1, -1)\}$ is a basis for $\mathcal{V}$. Notice next that for any vector **B** lying on the line through the origin perpendicular to $\mathcal{V}$ we must have $\mathsf{P}(\mathbf{B}) = \mathbf{0}$. If we make use of the fact that $\{(1, -1, 0), (0, 1, -1), (1, 1, 1)\}$ is a basis for $\mathbb{R}^3$, we can easily crank out several formulas for **P**.

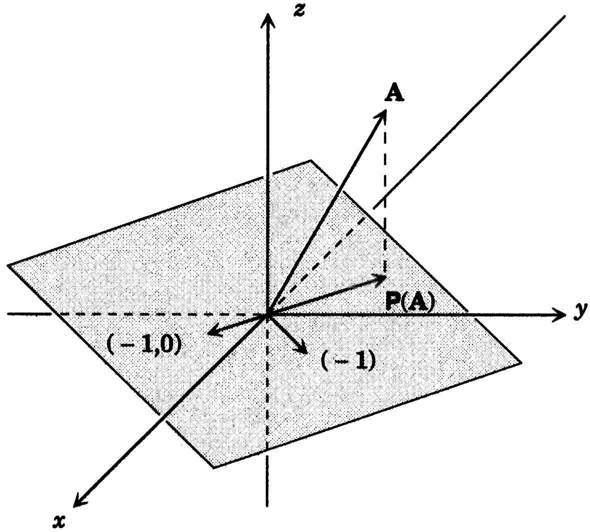


Figure 12.1.1

For example, the matrix of **P** relative to the basis $\{(1, -1, 0), (0, 1, -1), (1, 1, 1)\}$ is just

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Of course, you may object that this is cheating somewhat, and what you really want is the matrix of **P** relative to the standard basis for $\mathbb{R}^3$. This isn't hard either. Notice that

$$\begin{aligned}
(1, 0, 0) &= \frac{2}{3}(1, -1, 0) + \frac{1}{3}(0, 1, -1) + \frac{1}{3}(1, 1, 1), \\
(0, 1, 0) &= -\frac{1}{3}(1, -1, 0) + \frac{1}{3}(0, 1, -1) + \frac{1}{3}(1, 1, 1), \\
(0, 0, 1) &= -\frac{1}{3}(1, -1, 0) - \frac{2}{3}(0, 1, -1) + \frac{1}{3}(1, 1, 1).
\end{aligned}$$

Therefore, we find that

$$
\begin{aligned}
\mathbf{P}(1,0,0) &= \frac{2}{3}\mathbf{P}(1,-1,0) + \frac{1}{3}\mathbf{P}(0,1,-1) + \frac{1}{3}\mathbf{P}(1,1,1)\\
&= \frac{2}{3}(1,-1,0) + \frac{1}{3}(0,1,-1) = (\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}),\\
\mathbf{P}(0,1,0) &= -\frac{1}{3}\mathbf{P}(1,-1,0) + \frac{1}{3}\mathbf{P}(0,1,-1) + \frac{1}{3}\mathbf{P}(1,1,1),\\
&= -\frac{1}{3}(1,-1,0) + \frac{1}{3}(0,1,-1) = (-\frac{1}{3}, \frac{2}{3}, -\frac{1}{3})\\
\mathbf{P}(0,0,1) &= -\frac{1}{3}\mathbf{P}(1,-1,0) - \frac{2}{3}\mathbf{P}(0,1,-1) + \frac{1}{3}\mathbf{P}(1,1,1),\\
&= -\frac{1}{3}(1,-1,0) - \frac{2}{3}(0,1,-1) = (-\frac{1}{3}, -\frac{1}{3}, \frac{2}{3})
\end{aligned}
$$

so that the matrix of $\mathbf{P}$ relative to the standard basis is

$$
\mathbf{N} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix} = \frac{1}{3} \cdot \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix},
$$

or if one enjoys such things,

$$
\mathbf{P}(x,y,z) = (\frac{2}{3}x - \frac{1}{3}y - \frac{1}{3}z, -\frac{1}{3}x + \frac{2}{3}y - \frac{1}{3}z, -\frac{1}{3}x - \frac{1}{3}y + \frac{2}{3}z).
$$

Now the first question to ask is, Which representation of $\mathbf{P}$ tells us the most about it? I am sure you will agree that the matrix representation $\mathbf{M}$ together with the basis $\{(1,-1,0), (0,1,-1), (1,1,1)\}$ relative to which we computed $\mathbf{M}$ tells us everything we could possibly want to know about $\mathbf{P}$. For it clearly tells us that $\mathbf{P}$ is projection onto the plane spanned by the vectors $(1,-1,0)$ and $(0,1,-1)$, which is, of course, the plane $\mathcal{V} = \{(x,y,z) : x + y + z = 0\}$. On the other hand, the matrix representation $\mathbf{N}$ of $\mathbf{P}$ relative to the standard basis conveys relatively little about the simple geometric nature of $\mathbf{P}$.

This brings us to another question: How does one recognize when a linear transformation $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is a projection? Well to deal with this question we had better begin by defining what we mean by a projection. (Remember a definition is either useful or useless, not true or false!) A little experimentation shows that the following is a useful choice.

**DEFINITION :** *A linear transformation*

$$
\mathbf{P} : \mathcal{W} \longrightarrow \mathcal{W}
$$

*is called a* **projection** *if* $\mathsf{P}^2 = \mathsf{P}$, *that is, if* $\mathsf{P}(\mathsf{P}(\mathbf{A})) = \mathsf{P}(\mathbf{A})$ *for every vector* $\mathbf{A}$ *in* $\mathcal{W}$.

It pays to be warned that there are linear transformations

$$\mathsf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

that are projections in the sense of this definition that are not projections onto a plane (or line) in the sense we have spoken of so far. Let us illustrate by an example.

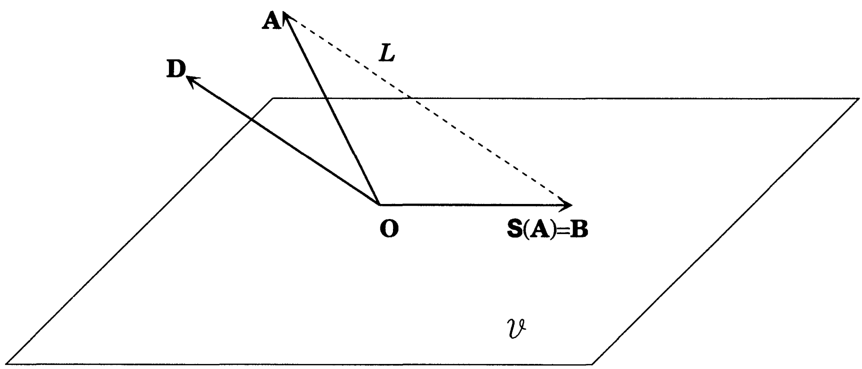


Figure 12.1.2

Let $\mathcal{V} = \left\{(x, y, z) \mid x + y + z = 0\right\}$ be the plane in $\mathbb{R}^3$ we have considered already. Let $\mathbf{D} = (1, 1, 0)$. Note that $\mathbf{D}$ is not a vector in $\mathcal{V}$. We are going to describe a sort of **skew** projection onto $\mathcal{V}$. Let $\mathbf{A}$ be a vector in $\mathbb{R}^3$. Draw a line $L$ parallel to $\mathbf{D}$ passing through the tip of the vector $\mathbf{A}$. This line will meet the plane $\mathcal{V}$ in a unique point (why?), which determines a unique vector lying in the plane $\mathcal{V}$, which we call $\mathbf{B}$. Set $\mathsf{S}(\mathbf{A}) = \mathbf{B}$ as in Figure 12.1.2. It is rather easily checked from the definition that $\mathsf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is indeed a linear transformation. Note that

$$\mathsf{S}(\mathbf{A}) = \mathbf{A} \quad \text{if } \mathbf{A} \in \mathcal{V},$$

$$\mathsf{S}(\mathbf{A}) = \mathbf{0} \quad \text{if } \mathbf{A} \in \mathcal{L}(\mathbf{D}).$$

Since $\mathbf{D}$ is not in the plane $\mathcal{V}$, if we choose a basis $\{\mathbf{F}, \mathbf{E}\}$ for $\mathcal{V}$, then $\{\mathbf{F}, \mathbf{E}, \mathbf{D}\}$ is a basis for $\mathbb{R}^3$. If $\mathbf{A}$ is any vector in $\mathbb{R}^3$, then

$$\mathbf{A} = a_1\mathbf{F} + a_2\mathbf{E} + a_3\mathbf{D},$$

so

$$\begin{aligned}\mathbf{S}(\mathbf{A}) &= a_1\mathbf{S}(\mathbf{F}) + a_2\mathbf{S}(\mathbf{E}) + a_3\mathbf{S}(\mathbf{D})\\ &= a_1\mathbf{F} + a_2\mathbf{E},\end{aligned}$$

from which it follows that $\mathbf{S}^2 = \mathbf{S}$, so $\mathbf{S}$ is a projection.

In a later chapter (Chapter 14) we are going to prove that every projection

$$\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

such that $\dim(\mathrm{Im}(\mathbf{S})) = 2$ is a skew projection of the preceding type. The more usual projections we have dealt with up until now are what are called **orthogonal** projections ($\mathbf{D}$ is orthogonal to $\mathcal{V}$) or **self adjoint** projections (see, for example, Chapter 15).

Let us return to the question that started all this, namely, how does one recognize a projection? The definition we have proposed is going to be useful if and only if we can somehow show that every projection $\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is a skew projection in the preceding sense. In fact, more is true.

**THEOREM 12.1.1:** *Let $\mathcal{W}$ be a finite-dimensional vector space of dimension $n$ and $\mathbf{S} : \mathcal{W} \longrightarrow \mathcal{W}$ a projection. Then there is a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ for $\mathcal{W}$ such that*

$$\mathbf{S}(\mathbf{A}_i) = \begin{cases} \mathbf{A}_i & \text{if } 1 \le i \le r,\\ 0 & \text{if } r+1 \le i \le n,\end{cases}$$

*where $r = \dim(\mathrm{Im}(\mathbf{S}))$, and hence the matrix of $\mathbf{S}$ relative to the basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is*

$$\begin{bmatrix} 1 & & & & & \text{\Large 0}\\ & \ddots & & & & \\ & & 1 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ \text{\Large 0} & & & & & 0\end{bmatrix}.$$

**PROOF:** Let $\{\mathbf{B}_1, \ldots, \mathbf{B}_r\}$ be a basis for $\mathrm{Im}(\mathbf{S})$ and $\{\mathbf{C}_1, \ldots, \mathbf{C}_s\}$ be a basis for $\ker(\mathbf{S})$. Notice that by the dimension formula, Theorem 8.4.2, we have $n = r + s$, so, the collection $\{\mathbf{B}_1, \ldots, \mathbf{B}_r, \mathbf{C}_1, \ldots, \mathbf{C}_s\}$ contains $n$ vectors, the correct number of vectors to be a basis for $\mathcal{W}$. Let us show that indeed it is a basis for $\mathcal{W}$. To this end notice the following two facts:

$$\begin{aligned}\mathbf{S}(\mathbf{C}_i) &= \mathbf{0} \quad \text{for } i = 1, \ldots, s,\\ \mathbf{S}(\mathbf{B}_i) &= \mathbf{B}_i \quad \text{for } i = 1, \ldots, r.\end{aligned}$$

The first of these is clear from the definition of $\{\mathbf{C}_1, \ldots, \mathbf{C}_s\}$, while the second requires some proof. (Be sensible, we still have not used the fact that $\mathbf{S}$ is a projection!) We note that since $\mathbf{B}_i$ belongs to $\operatorname{Im}(\mathbf{S})$, we may find a vector $\mathbf{D}_i$ in $\mathcal{W}$ such that $\mathbf{S}(\mathbf{D}_i) = \mathbf{B}_i$. Therefore, since $\mathbf{S}$ is a projection,

$$\mathbf{S}(\mathbf{B}_i) = \mathbf{S}(\mathbf{S}(\mathbf{D}_i)) = \mathbf{S}^2(\mathbf{D}_i) = \mathbf{S}(\mathbf{D}_i) = \mathbf{B}_i,$$

which verifies the second set of equations above.

Since there are $n$ vectors in the collection $\{\mathbf{B}_1, \ldots, \mathbf{B}_r, \mathbf{C}_1, \ldots, \mathbf{C}_s\}$ we may apply Theorem 6.3.4, and we conclude that we need only show that the set $\{\mathbf{B}_1, \ldots, \mathbf{C}_s\}$ is linearly independent for it to be a basis, that is, we must verify only one half of the basis condition.

So suppose the vectors $\{\mathbf{B}_1, \ldots, \mathbf{B}_r, \mathbf{C}_1, \ldots, \mathbf{C}_s\}$ were linearly dependent. Then we could find numbers $b_1, \ldots, b_r, c_1, \ldots, c_s$, not all zero, such that

$$(*) \qquad b_1\mathbf{B}_1 + \cdots + b_r\mathbf{B}_r + c_1\mathbf{C}_1 + \cdots + c_s\mathbf{C}_s = \mathbf{0}.$$

Apply $\mathbf{S}$ to both sides of this equation to get

$$\begin{aligned} \mathbf{0} &= \mathbf{S}(b_1\mathbf{B}_1 + \cdots + b_r\mathbf{B}_r + c_1\mathbf{C}_1 + \cdots + c_s\mathbf{C}_s) \\ &= b_1\mathbf{S}(\mathbf{B}_1) + \cdots + b_r\mathbf{S}(\mathbf{B}_r) + c_1\mathbf{S}(\mathbf{C}_1) + \cdots + c_s\mathbf{S}(\mathbf{C}_s) \\ &= b_1\mathbf{B}_1 + \cdots + b_r\mathbf{B}_r. \end{aligned}$$

Since $\mathbf{B}_1, \ldots, \mathbf{B}_r$ are a basis for $\operatorname{Im}(\mathbf{S})$, they are linearly independent, and hence $b_1 = b_2 = \cdots = b_r = 0$. Thus our original relation $(*)$ reads

$$c_1\mathbf{C}_1 + \cdots + c_s\mathbf{C}_s = \mathbf{0}.$$

The vectors $\mathbf{C}_1, \ldots, \mathbf{C}_s$ are a basis for $\ker(\mathbf{S})$, so they are linearly independent also, and hence $c_1 = c_2 = \cdots = c_s = 0$. Therefore, the collection $\{\mathbf{B}_1, \ldots, \mathbf{B}_r, \mathbf{C}_1, \ldots, \mathbf{C}_s\}$ is not linearly dependent, so it must be linearly independent. Thus, $\{\mathbf{B}_1, \ldots, \mathbf{B}_r, \mathbf{C}_1, \ldots, \mathbf{C}_s\}$ being linearly independent, is a basis for $\mathcal{W}$, as we remarked earlier. Set

$$\begin{aligned} \mathbf{A}_1 &= \mathbf{B}_1, \\ \mathbf{A}_2 &= \mathbf{B}_2, \\ \vdots &= \vdots \\ \mathbf{A}_r &= \mathbf{B}_r, \qquad \text{(remember, } r + s = n\text{)} \\ \mathbf{A}_{r+1} &= \mathbf{C}_1, \\ \vdots &= \vdots \\ \mathbf{A}_n &= \mathbf{C}_s. \end{aligned}$$

Then we will have

$$\mathbf{S}(\mathbf{A}_i) = \begin{cases} \mathbf{S}(\mathbf{B}_i) = \mathbf{B}_i = \mathbf{A}_i & \text{for } 1 \le i \le r, \\ \mathbf{S}(\mathbf{C}_{i-r}) = \mathbf{0} & \text{for } r+1 \le i \le n, \end{cases}$$

as required. $\square$

It is a rather easy consequence that every linear transformation $\mathbf{S} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ such that

$$\mathbf{S}^2 = \mathbf{S} \quad \text{and} \quad \dim(\operatorname{Im}(\mathbf{S})) = 2$$

is a skew projection as described prior to Theorem 12.1.1.

The reader should turn ahead to Proposition 14.1.1 and examine the fundamental difference between it and Theorem 12.1.1.

For a given projection $\mathbf{S} : \mathcal{W} \longrightarrow \mathcal{W}$, the actual computation of a (not ***the***!) basis in Theorem 12.1.1 may be a numerical horror, but still the proof gives you a method for doing so of the type we have successfully used before.

## 12.2 Nilpotent Transformations

A linear transformation $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is said to be **nilpotent of index $k$** if $\mathbf{T}^k = \mathbf{0}$ and $\mathbf{T}^{k-1} \neq \mathbf{0}$, that is, $\mathbf{T}^k(\mathbf{A}) = \mathbf{0}$ for all vectors $\mathbf{A}$ in $\mathcal{V}$ but there is at least one vector $\mathbf{B}$ in $\mathcal{V}$ for which $\mathbf{T}^{k-1}(\mathbf{B}) \neq \mathbf{0}$.

There is a sort of canonical example of a nilpotent transformation; namely, the differentiation operator (recall Example 1 from Section 8.3)

$$\mathbf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R})$$

is nilpotent of index $n + 1$. Another family of simple examples is provided by the shift operator

$$\mathbf{S} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

defined by

$$\mathbf{S}(a_1, a_2, \ldots, a_n) = (0, a_1, a_2, \ldots, a_{n-1}).$$

Thus if $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ denotes the usual basis for $\mathbb{R}^n$, then

$$\begin{aligned} \mathbf{S}(\mathbf{E}_1) &= \mathbf{E}_2, \\ \mathbf{S}(\mathbf{E}_2) &= \mathbf{E}_3, \\ \vdots &= \vdots \\ \mathbf{S}(\mathbf{E}_n) &= \mathbf{0}, \end{aligned}$$

so that the matrix of **S** relative to the usual basis is

$$\begin{bmatrix} 0 & 0 & & 0 & 0 \\ 1 & 0 & & 0 & 0 \\ 0 & 1 & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & 0 \end{bmatrix}.$$

In this case **S** is nilpotent of index $n$. Notice in both cases that the index of nilpotence is the dimension of the vector space on which the transformation is defined. This is a very special situation. In fact, we can determine all such nilpotent transformations. We will need the following very important fact.

**PROPOSITION 12.2.1:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation that is nilpotent of index* $k$. *Let* $\mathbf{B} \in \mathcal{V}$ *be a vector such that* $\mathsf{T}^{k-1}(\mathbf{B}) \neq \mathbf{0}$. *Then the vectors* $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \dots, \mathsf{T}^{k-1}(\mathbf{B})\right\}$ *are linearly independent. Hence* $k \leq n$.

**PROOF:** Let us suppose that

(✠) $$b_0\mathbf{B} + b_1\mathsf{T}(\mathbf{B}) + \cdots + b_{k-1}\mathsf{T}^{k-1}(\mathbf{B}) = \mathbf{0}$$

is a linear relation among $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \dots, \mathsf{T}^{k-1}(\mathbf{B})\right\}$. What we must show is that $b_0 = b_1 = \cdots = b_{k-1} = 0$. Notice that $\mathsf{T}^k(\mathbf{B}) = \mathbf{0}$, so we must have

$$\mathbf{0} = \mathsf{T}^k(\mathbf{B}) = \mathsf{T}^{k+1}(\mathbf{B}) = \cdots$$

by Proposition 8.3.1. Apply $\mathsf{T}^{k-1}$ to both sides of (✠) to get

$$\begin{aligned}\mathbf{0} &= b_0\mathsf{T}^{k-1}(\mathbf{B}) + b_1\mathsf{T}^k(\mathbf{B}) + \cdots + b_{k-1}\mathsf{T}^{2k-2}(\mathbf{B}) \\ &= b_0\mathsf{T}^{k-1}(\mathbf{B}) + b_1\mathbf{0} + \cdots + b_{k-1}\mathbf{0} \\ &= b_0\mathsf{T}^{k-1}(\mathbf{B}).\end{aligned}$$

Therefore, since $\mathsf{T}^{k-1}(\mathbf{B}) \neq \mathbf{0}$, we must have $b_0 = 0$, and our linear relation (✠) has become

(✣) $$b_1\mathsf{T}(\mathbf{B}) + \cdots + b_{k-1}\mathsf{T}^{k-1}(\mathbf{B}) = \mathbf{0}.$$

If we apply $\mathsf{T}^{k-2}$ to both sides of (✣), we get

$$\begin{aligned}\mathbf{0} &= b_1\mathsf{T}^{k-1}(\mathbf{B}) + b_2\mathsf{T}^k(\mathbf{B}) + \cdots + b_{k-1}\mathsf{T}^{2k-3}(\mathbf{B}) \\ &= b_1\mathsf{T}^{k-1}(\mathbf{B}) + b_2\mathbf{0} + \cdots + b_{k-1}\mathbf{0} \\ &= b_1\mathsf{T}^{k-1}(\mathbf{B}),\end{aligned}$$

and so $b_1 = 0$. Continuing in this way, we may show that $0 = b_2 = \cdots = b_{k-1}$. Therefore, $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \dots, \mathsf{T}^{k-1}(\mathbf{B})\right\}$ is linearly independent. □

**COROLLARY 12.2.2 :** *Suppose that $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is a nilpotent linear transformation and the index of $\mathsf{T}$ equals the dimension of $\mathcal{V}$. Call this common number $k$. Then there is a vector $\mathbf{B} \in \mathcal{V}$ such that $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \ldots, \mathsf{T}^{k-1}(\mathbf{B})\right\}$ is a basis for $\mathcal{V}$ and $\mathsf{T}^k(\mathbf{B}) = \mathbf{0}$.*

**PROOF :** The vectors $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \ldots, \mathsf{T}^{k-1}(\mathbf{B})\right\}$ are linearly independent by to Proposition 12.2.1. There are $k$ of them and since $k = \dim(\mathcal{V})$, it follows from Theorem 6.3.4 that they are basis for $\mathcal{V}$. Since $k$ is also the index of $\mathsf{T}$, we have $\mathsf{T}^k(\mathbf{B}) = \mathbf{0}$. □

Therefore, if $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is nilpotent **and** index $\mathsf{T} = k = \dim(\mathcal{V})$, then we may choose a vector $\mathbf{B}$ in $\mathcal{V}$ such that $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \ldots, \mathsf{T}^{k-1}(\mathbf{B})\right\}$ is a basis for $\mathcal{V}$ and $\mathsf{T}^k(\mathbf{B}) = \mathbf{0}$. The matrix of $\mathsf{T}$ relative to this basis is therefore

$$\begin{bmatrix} 0 & 0 & & 0 & 0 \\ 1 & 0 & & 0 & 0 \\ 0 & 1 & & 0 & 0 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & 0 \end{bmatrix},$$

and if we use this basis to identify $\mathcal{V}$ with $\mathbb{R}^n$ as in Proposition 8.5.3, we find that $\mathsf{T}$ is identified with the shift operator. Thus the structure of nilpotent transformations $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ ***of index equal to the dimension of*** $\mathcal{V}$ is completely determined. The structure of more general nilpotent transformations is more complicated and is the subject of Section 17.2.

## 12.3 Cyclic Transformations

A linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is called **cyclic** if there exists a vector $\mathbf{A}$ in $\mathcal{V}$ such that the collection $\left\{\mathbf{A}, \mathsf{T}(\mathbf{A}), \mathsf{T}^2(\mathbf{A}), \ldots\right\}$ spans $\mathcal{V}$. The vector $\mathbf{A}$ is called a **cyclic vector** for $\mathsf{T}$. Examples of cyclic transformations are plentiful. It follows from Corollary 12.2.2 that any nilpotent transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ whose index is equal to the dimension of $\mathcal{V}$ is cyclic. In dimension 2, essentially all nontrivial linear transformations are cyclic. We begin with a preliminary result of independent interest.

**PROPOSITION 12.3.1 :** *Suppose that $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is a cyclic linear transformation. Let the dimension of $\mathcal{V}$ be $n$ and let $\mathbf{A} \in \mathcal{V}$ be a cyclic vector for $\mathsf{T}$. Then the vectors $\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{n-1}(\mathbf{A})$ are a basis for $\mathcal{V}$.*

**Proof:** There are $n$ vectors in the set $\left\{\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{n-1}(\mathbf{A})\right\}$ and $\mathcal{V}$ has dimension $n$, so by Theorem 6.3.4 to show that this set is a basis we need only show that they are linearly independent. Suppose to the contrary that they are linearly dependent. By Proposition 6.2.1 there will be a positive integer $m \leq n-1$ such that

$$\mathsf{T}^m(\mathbf{A}) \in \mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{m-1}(\mathbf{A})).$$

Therefore, by applying $\mathsf{T}$,

$$\mathsf{T}^{m+1}(\mathbf{A}) \in \mathcal{L}(\mathsf{T}(\mathbf{A}), \mathsf{T}^2(\mathbf{A}), \ldots, \mathsf{T}^m(\mathbf{A})) \subseteq \mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{m-1}(\mathbf{A})),$$

since all of the vectors $\mathsf{T}(\mathbf{A}), \mathsf{T}^2(\mathbf{A}), \ldots, \mathsf{T}^m(\mathbf{A})$ belong to $\mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{m-1}(\mathbf{A}))$. Applying $\mathsf{T}$ again, we find that

$$\mathsf{T}^{m+2}(\mathbf{A}) \in \mathcal{L}(\mathsf{T}^2(\mathbf{A}), \mathsf{T}^3(\mathbf{A}), \ldots, \mathsf{T}^{m+1}(\mathbf{A})) \subseteq \mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{m-1}(\mathbf{A})),$$

and continuing in this way, we obtain that

$$\mathsf{T}^m(\mathbf{A}), \mathsf{T}^{m+1}(\mathbf{A}), \mathsf{T}^{m+2}(\mathbf{A}), \ldots \in \mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{m-1}(\mathbf{A})),$$

and therefore that

$$\mathcal{V} = \mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^m(\mathbf{A}), \mathsf{T}^{m+1}(\mathbf{A}), \ldots) \subseteq \mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{m-1}(\mathbf{A})),$$

so that $\dim(\mathcal{V}) \leq m < n$, which is a contradiction. Therefore, the vectors $\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{n-1}(\mathbf{A})$ must be linearly independent. □

**Proposition 12.3.2:** *A linear transformation* $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ *is cyclic if and only if* $\mathsf{T} \neq e\mathsf{I}$, *where* $e$ *is a number and* $\mathsf{I} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ *the identity transformation.*

**Proof:** Suppose that $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$. A vector $\mathbf{A} \in \mathbb{R}^2$ is a cyclic vector for $\mathsf{T}$ if and only if $\mathsf{T}(\mathbf{A}) \notin \mathcal{L}(\mathbf{A})$, that is, if and only if $\mathsf{T}(\mathbf{A})$ does not belong to the line through the origin spanned by $\mathbf{A}$. For then $\{\mathbf{A}, \mathsf{T}(\mathbf{A})\}$ will be a basis for $\mathbb{R}^2$, so that $\mathbb{R}^2 = \mathcal{L}(\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots)$, as required. So suppose that $\mathsf{T}$ does not have a cyclic vector. Then $\mathsf{T}(\mathbf{A}) \in \mathcal{L}(\mathbf{A})$ for every $\mathbf{A} \in \mathbb{R}^2$. In particular,

$$\mathsf{T}(1, 0) \in \mathcal{L}\left\{(1, 0)\right\} \Rightarrow \mathsf{T}(1, 0) = e_1(1, 0),$$
$$\mathsf{T}(0, 1) \in \mathcal{L}\left\{(0, 1)\right\} \Rightarrow \mathsf{T}(0, 1) = e_2(0, 1).$$

We claim that $e_1 = e_2$. For we also have

$$\mathsf{T}(1, 1) \in \mathcal{L}\left\{(1, 1)\right\} \Rightarrow \mathsf{T}(1, 1) = e(1, 1),$$

so

$$\begin{aligned} e(1,1) = \mathsf{T}(1,1) &= \mathsf{T}((1,0)+(0,1)) \\ &= \mathsf{T}(1,0)+\mathsf{T}(0,1) \\ &= e_1(1,0)+e_2(0,1), \end{aligned}$$

so

$$(e, e) = (e_1, e_2),$$

and hence $e_1 = e = e_2$. Therefore,

$$\begin{aligned} \mathsf{T}(1,0) &= (e,0), \\ \mathsf{T}(0,1) &= (0,e), \end{aligned}$$

so the matrix of $\mathsf{T}$ relative to the standard basis of $\mathbb{R}^2$ is

$$\begin{bmatrix} e & 0 \\ 0 & e \end{bmatrix} = e\mathsf{I},$$

and hence $\mathsf{T} = e\mathsf{I}$. $\square$

Suppose that $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is a cyclic transformation with cyclic vector $\mathbf{A}$. Then the matrix of $\mathsf{T}$ relative to the basis $\left\{\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{n-1}(\mathbf{A})\right\}$ is

$$\begin{bmatrix} 0 & & & 0 & a_0 \\ 1 & 0 & & \vdots & a_1 \\ 0 & \ddots & & \vdots & \vdots \\ \vdots & & 1 & 0 & \vdots \\ 0 & \cdots & 0 & 1 & a_{n-1} \end{bmatrix}$$

where $\mathsf{T}^n(\mathbf{A}) = a_0\mathbf{A} + a_1\mathsf{T}(\mathbf{A}) + \cdots + a_{n-1}\mathsf{T}^{n-1}(\mathbf{A})$. This matrix is particularly simple, and that is one reason for looking for a cyclic vector. In general, however, a linear transformation need not have any cyclic vectors at all!

## 12.4 Exercises

**1.** Find the matrix relative to the standard basis of the projection $\mathsf{P} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, onto the plane $\mathcal{V} = \left\{(x, y, z) \mid x + y = 0\right\}$.

**2.** Repeat exercise 1 for $\mathcal{V} = \left\{(x, y, z) \mid z = 0\right\}$.

**3.** Find the basis $\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3\}$ for $\mathbb{R}^3$ such that the projection $\mathsf{P}$ of Exercise 1 relative to the basis is of the form

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

**4.** Repeat Exercise 3 for the projection of Exercise 2.

**5.** Show that the differentiation operator $\mathsf{D} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_3(\mathbb{R})$ is nilpotent of index 4. Find a basis of $\mathcal{P}_3(\mathbb{R})$ of the form

$$E = \left\{\mathbf{B}, \mathsf{D}(\mathbf{B}), \mathsf{D}^2(\mathbf{B}), \mathsf{D}^3(\mathbf{B})\right\},$$

where $\mathbf{B} \in \mathcal{P}_3(\mathbb{R})$. Calculate the matrix of $\mathsf{D}$ relative to the basis $E$.

**6.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation

$$\mathsf{T}(x, y, z) = (0, x, 2y).$$

Show that $\mathsf{T}$ is nilpotent of index 3. Find a vector $\mathbf{B}$ such that

$$\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \mathsf{T}^2(\mathbf{B})\right\}$$

is a basis for $\mathbb{R}^3$.

(a) Calculate the matrix of $\mathsf{T}$ relative to this basis.
(b) Calculate the matrix of $\mathsf{T}$ relative to the standard basis.

**7.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation

$$\mathsf{T}(x, y, z) = (x + y, y + z, x).$$

Show that $\mathsf{T}$ is a cyclic linear transformation. (**Hint:** find $\mathbf{B}$ such that $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \mathsf{T}^2(\mathbf{B})\right\}$ is a set of independent vectors.)

**8.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation defined by

$$\mathsf{T}(x, y, z) = (y, x, z).$$

Show that $\mathsf{T}$ is not cyclic.

**9.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation defined by

$$\mathsf{T}(x, y, z) = (y, z, x).$$

Show that $\mathsf{T}$ is cyclic. Find a basis for $\mathbb{R}^3$ of the form $\left\{\mathbf{B}, \mathsf{T}(\mathbf{B}), \mathsf{T}^2(\mathbf{B})\right\}$ and calculate the matrix of $\mathsf{T}$ relative to this basis.

**10.** Is the product of two projections $\mathsf{P}$ and $\mathsf{Q}$ in $\mathcal{V}$ again a projection? What if $\mathsf{P}$ and $\mathsf{Q}$ commute with each other?

**11.** Is the product of two nilpotent transformations again nilpotent?

**12.** A linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is called an **involution** if $\mathsf{T}^2 = \mathsf{I} \neq \mathsf{T}$. (See also Section 10.4.) Which of the following matrices represent involutions in $\mathbb{R}^3$ with respect to the standard bases:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

**13.** Is the product of two involutions again in an involution? What if they commute with each other?

**14.** If $\mathsf{P}$ is a projection, show that $\mathsf{T} = 2\mathsf{P} - \mathsf{I}$ is an involution.

**15.** If $\mathsf{T}$ is an involution, show that $\mathsf{P} = \frac{1}{2}(\mathsf{T} + \mathsf{I})$ is a projection.

**16.** Use the previous exercises to show that with respect to a suitable basis, the matrix of an involution $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is of the form

$$\begin{bmatrix} 1 & & & & & \mathbf{0} \\ & \ddots & & & & \\ & & 1 & & & \\ & & & -1 & & \\ & & & & \ddots & \\ \mathbf{0} & & & & & -1 \end{bmatrix}.$$

**17.** Find all possible $2 \times 2$ matrices that represent nilpotent transformations $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ of index 2 with respect to some basis of $\mathbb{R}^2$.

**18.** Two transformations $\mathsf{S}, \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ are said to be **mutually annihilating** if $\mathsf{ST} = \mathsf{0}$ and $\mathsf{TS} = \mathsf{0}$. Show that the sum of two mutually annihilating projections is again a projection.

**19.** A linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ in the $n$-dimensional vector space $\mathcal{V}$ is called a **reflection** (see also Section 8.2 Example 3) if there is an $(n-1)$-dimensional subspace $\mathcal{H}$ in $\mathcal{V}$, called the **reflecting hyperplane** of $\mathsf{T}$, and a nonzero vector $\mathbf{A}$ in $\mathcal{V}$ such that $\mathbf{A}$ does not belong to $\mathcal{H}$ and

$$\mathsf{T}(\mathbf{B}) = \mathbf{B} \qquad \text{for all } \mathbf{B} \in \mathcal{H},$$

$$\mathsf{T}(\mathbf{A}) = -\mathbf{A}.$$

Show that with respect to a suitable basis the matrix of a reflection takes the form

$$\begin{bmatrix} -1 & 0 & \cdots & 0 \\ 0 & 1 & \vdots & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}.$$

**20.** Let $\mathbf{S} : \mathcal{P}_n \longrightarrow \mathcal{P}_n$ be the linear transformation defined by the equation

$$\mathbf{S}(p(x)) = p(x+1).$$

Find the matrix of $\mathbf{S}$ with respect to the standard basis $\{1, x, \ldots, x^n\}$ of $\mathcal{P}_n$.

**21.** Let $\mathbf{A}$ be the $2 \times 2$ matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

and let $\mathbf{L_A} : \mathrm{Mat}_{2,2} \longrightarrow \mathrm{Mat}_{2,2}$ be the linear transformation given by

$$\mathbf{L_A}(\mathbf{M}) = \mathbf{A} \cdot \mathbf{M}$$

for $\mathbf{M}$ in $\mathrm{Mat}_{2,2}$. Find the matrix of $\mathbf{L_A}$ with respect to the ordered basis

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

for $\mathrm{Mat}_{2,2}$.

**22.** Let $\mathbf{T} : \mathrm{Mat}_{2,2} \longrightarrow \mathrm{Mat}_{2,2}$ be the linear transformation given by

$$\mathbf{T}(\mathbf{M}) = \mathbf{M}^{\mathrm{tr}}.$$

Find the matrix of $\mathbf{T}$ with respect to the same ordered basis of $\mathrm{Mat}_{2,2}$ as in the previous exercise.

**23.** The matrix

$$\mathbf{R}_\varphi = \begin{bmatrix} \cos(\varphi) & -\sin(\varphi) \\ \sin(\varphi) & \cos(\varphi) \end{bmatrix}$$

describes a rotation in the plane $\mathbb{R}^2$ through the angle $\varphi$. Determine the values of $\varphi$ such that the corresponding linear transformation is one of the following types: *nilpotent, involutory, cyclic, nonsingular, symmetric* or a *reflection*. (There may be none.)

# Chapter 13
# Systems of Linear Equations

In the historical development of linear algebra the geometry of linear transformations and the algebra of systems of linear equations played significant and important roles. A **system of linear equations** has the form

$$(\mathcal{L}in) \qquad \begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &= b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= b_2, \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= b_m. \end{aligned}$$

Here $x_1, x_2, \ldots, x_n$ denote the unknowns, which are to be determined. The $mn$ numbers $a_{i,j}$, where $i = 1, \ldots, m, j = 1, \ldots, n$ are called the **coefficients** of the linear system $(\mathcal{L}in)$ and are, of course, fixed. A **solution** of the system $(\mathcal{L}in)$ is an ordered $n$-tuple of numbers $(s_1, \ldots, s_n)$ such that the $m$ equations

$$\begin{aligned} a_{1,1}s_1 + a_{1,2}s_2 + \cdots + a_{1,n}s_n &= b_1, \\ a_{2,1}s_1 + a_{2,2}s_2 + \cdots + a_{2,n}s_n &= b_2, \\ &\vdots \\ a_{m,1}s_1 + a_{m,2}s_2 + \cdots + a_{m,n}s_n &= b_m, \end{aligned}$$

are all true. To solve the system $(\mathcal{L}in)$ means to find *all* the solutions of $(\mathcal{L}in)$.

**EXAMPLE:** Solve the linear system

$$\begin{aligned} x_1 + x_2 + x_3 &= 3, \\ 2x_2 + x_3 &= 4, \\ x_1 - x_2 \qquad &= -1. \end{aligned}$$

**SOLUTION:** From previous experience we recall that this system may be solved as follows: First, number the equations so we can conveniently refer to them:

$$\begin{array}{lrcrcrcr} ① & x_1 & + & x_2 & + & x_3 & = & 3, \\ ② & & & 2x_2 & + & x_3 & = & 4, \\ ③ & x_1 & - & x_2 & & & = & -1, \end{array}$$

To solve, subtract ① from ③ to get a new system:

$$\begin{array}{lrcrcrcr} ❶ & x_1 & + & x_2 & + & x_3 & = & 3, \\ ❷ & & & 2x_2 & + & x_3 & = & 4, \\ ❸ & & & -2x_2 & - & x_3 & = & -4, \end{array}$$

Erase ❸, since it follows from ❷, to get

$$\begin{array}{rcrcrcr} x_1 & + & x_2 & + & x_3 & = & 3, \\ & & 2x_2 & + & x_3 & = & 4. \end{array}$$

These can be rearranged in two steps to express $x_1$ and $x_3$ in terms of $x_2$. Namely, first

$$x_1 = 3 - x_3 - x_2,$$

$$x_3 = 4 - 2x_2,$$

and then

$$x_1 = x_2 - 1,$$

$$x_3 = 4 - 2x_2.$$

Therefore the value of $x_2$ can be chosen arbitrarily and these equations can be used to find corresponding values for $x_1$ and $x_3$. Namely, if $x_2 = s$, then, $x_1 = s - 1$ and $x_3 = 4 - 2s$, so the solutions are all triples of numbers

$$(s - 1, s, 4 - 2s)$$

where $s$ is arbitrary. For example $(0, 1, 2)$, $(-1, 0, 4)$ are solutions, but $(1, 1, 2)$ is not.

On the other hand past experience should have taught us that not every system of linear equations has a solution. For example the linear system

$$\begin{array}{rcrcr} x_1 & + & x_2 & = & 1, \\ x_1 & + & x_2 & = & -1 \end{array}$$

clearly can have no solutions. Thus our first order of business in the study of linear equations should be to determine when a linear system has solutions, and only afterwards take up the discussion of actual techniques of solution.

## 13.1 Existence Theorems

Our study of matrices in the preceding chapters will come in handy here. In fact it is through the study of linear systems that matrices most frequently appear in modern scientific investigations. First let us introduce the matrices

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

The matrix $\mathbf{A}$ is called the **coefficient matrix** of the linear system ($\mathcal{Lin}$), and $\mathbf{B}$ the matrix of **constants** of the linear system. We may then write the system of linear equations ($\mathcal{Lin}$) in a more compact form as the single matrix equation

$$(\mathcal{Mat}) \qquad \mathbf{A} \cdot \mathbf{X} = \mathbf{B}.$$

Solving this matrix equation may be given a very simple interpretation in terms of linear transformations. We let

$$\mathbf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

be the linear transformation whose matrix relative to the standard bases is $\mathbf{A}$. Thus

$$\mathbf{T}(x_1, \ldots, x_n) = \left( \sum_{j=1}^{n} a_{1,j}x_j, \sum_{j=1}^{n} a_{2,j}x_j, \ldots, \sum_{j=1}^{n} a_{m,j}x_j \right).$$

Let $\mathbf{B} = (b_1, \ldots, b_m)$, which is a vector in $\mathbb{R}^m$. A **solution** of ($\mathcal{Mat}$) is a vector $(s_1, \ldots, s_n)$ in $\mathbb{R}^n$ such that

$$\mathbf{T}(s_1, s_2, \ldots, s_n) = (b_1, \ldots, b_m).$$

The following result is therefore clear.

**PROPOSITION 13.1.1:** *With the preceding notations, the linear system ($\mathcal{Lin}$) has a solution if and only if the vector $(b_1, b_2, \ldots, b_m)$ lies in the image of the transformation $\mathbf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$, that is, if and only if* $\mathbf{B} \in \mathrm{Im}(\mathbf{T})$. □

This result is all well and good from a theoretical point of view, but how does one tell whether $\mathbf{B} \in \mathrm{Im}(\mathbf{T})$ from the coefficient matrix $\mathbf{A}$? The answer is really quite simple. Let us introduce the vectors (the columns of the coefficient matrix regarded as vectors in $\mathbb{R}^m$)

$$\mathbf{A}_{(1)} = (a_{1,1}, a_{2,1}, \ldots, a_{m,1}),$$

$$\vdots$$

$$\mathbf{A}_{(n)} = (a_{1,n}, a_{2,n}, \ldots, a_{m,n})$$

in $\mathbb{R}^m$. Notice that

$$\mathbf{A}_{(1)} = \mathsf{T}(\mathbf{E}_1),$$
$$\vdots$$
$$\mathbf{A}_{(n)} = \mathsf{T}(\mathbf{E}_n).$$

Therefore by Proposition 8.3.4,

$$\operatorname{Im}(\mathsf{T}) = \mathcal{L}(\mathsf{T}(\mathbf{E}_1), \ldots, \mathsf{T}(\mathbf{E}_n)) = \mathcal{L}(\mathbf{A}_{(1)}, \ldots, \mathbf{A}_{(n)}).$$

That is to say, the image of $\mathsf{T}$ is the linear span of the vectors $\mathbf{A}_{(1)}, \ldots, \mathbf{A}_{(n)}$. This may all be summed up in a definition and a theorem.

**DEFINITION:** *If* $\mathsf{A} = (a_{i,j})$ *is an* $m \times n$ *matrix, the* **column vectors** *of* $\mathsf{A}$ *are the* $n$ *vectors*

$$\mathbf{A}_{(1)} = (a_{1,1}, a_{2,1}, \ldots, a_{m,1}),$$
$$\mathbf{A}_{(2)} = (a_{1,2}, a_{2,2}, \ldots, a_{m,2}),$$
$$\vdots$$
$$\mathbf{A}_{(n)} = (a_{1,n}, a_{2,n}, \ldots, a_{m,n})$$

*in* $\mathbb{R}^m$. *The* **column space** *of* $\mathsf{A}$ *is the linear span of its column vectors in* $\mathbb{R}^m$.

For example, if

$$\mathsf{A} = \begin{bmatrix} 1 & 0 \\ -1 & -2 \\ 0 & 2 \end{bmatrix},$$

then the column vectors of $\mathsf{A}$ are the two vectors

$$(1, -1, 0), \qquad (0, -2, 2)$$

in $\mathbb{R}^3$. The column space of $\mathsf{A}$ is the linear span in $\mathbb{R}^3$ of these two vectors. A moment's computation shows this to be the plane with the equation

$$x + y + z = 0$$

in $\mathbb{R}^3$.

From our preceding discussion we obtain:

**THEOREM 13.1.2:** *Let*

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &= b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= b_2, \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= b_m \end{aligned}$$

*be a system of linear equations. Define the matrices*

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

*Then the linear system has a solution if and only if the vector* $\mathbf{B} = (b_1, b_2, \ldots, b_m)$ *can be written as a linear combination of the column vectors of* $\mathbf{A}$.

**PROOF:** Let $\mathbf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be the linear transformation whose matrix is $\mathbf{A}$ with respect to the standard bases. Then the image of $\mathbf{T}$ is the column space of $\mathbf{A}$. Apply Proposition 13.1.1. □

**EXAMPLE 1:** Does the linear system

$$\begin{aligned} x_1 + x_2 + \ x_3 &= 1, \\ x_1 \qquad + \ x_3 &= 1, \\ 2x_1 + x_2 + 2x_3 &= 0 \end{aligned}$$

have any solutions?

**SOLUTION:** The coefficient matrix of this linear system is

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 2 \end{bmatrix},$$

and its matrix of constants is

$$\mathbf{B} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Thus the column vectors of $\mathbf{A}$ are

$$(1, 1, 2), \qquad (1, 0, 1), \qquad (1, 1, 2).$$

Thus the column space of $\mathbf{A}$ is spanned by

$$(1, 1, 2) \quad \text{and} \quad (1, 0, 1),$$

while the vector $\mathbf{B}$ is

$$(1, 1, 0).$$

Suppose

$$(1, 1, 0) = a(1, 1, 2) + b(1, 0, 1).$$

Then

$$1 = a + b,$$

$$1 = a,$$

$$0 = 2a + b,$$

which implies

$$b = 0,$$

$$a = 1,$$

$$b = -2,$$

which is impossible (compare the first and third equations). Therefore, **B** does not belong to the column space of **A**, and the system has no solutions at all.

The simplest type of linear system is one whose matrix of constants is the zero matrix, that is, where

$$\mathbf{B} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}.$$

This case merits a special name.

DEFINITION: *A system of linear equations (in matrix notation)*

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}$$

*is called* **homogeneous** *if*

$$\mathbf{B} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}.$$

**THEOREM 13.1.3:** *Let (in the matrix notation)*

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{0}$$

*be a homogeneous system of linear equations. Then the set $\mathcal{S} = \{(s_1, \ldots, s_n)\}$ of all solutions to this linear system is a linear subspace of $\mathbb{R}^n$. More precisely, if*

$$\mathbf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

*is the linear transformation whose matrix relative to the standard bases is $\mathbf{A}$, then $\mathcal{S} = \ker(\mathbf{T})$.*

**PROOF:** This theorem takes longer to state than to prove. Simply note that $(s_1, \dots, s_n)$ is a solution of $\mathbf{A} \cdot \mathbf{X} = \mathbf{0}$ if and only if $\mathbf{T}(s_1, \dots, s_n) = (0, \dots, 0)$. □

**REMARK:** You might go back and look at Examples 1 and 2 in Section 4.1 again.

Thus, in order to solve a homogeneous system

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{0}$$

we have merely to find a basis for the kernel of the linear transformation $\mathbf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ whose matrix relative to the standard basis is $\mathbf{A}$. For by Theorem 13.1.3 any solution will be a linear combination of these basis vectors.

Before we can handle the case of a general linear system, we introduce a necessary definition.

**DEFINITION:** *Let $\mathcal{V}$ be a vector space, let $\mathcal{U}$ be a linear subspace of $\mathcal{V}$, and let $\mathbf{A} \in \mathcal{V}$. Denote by $\mathcal{A} = \mathbf{A} + \mathcal{U} \subseteq \mathcal{V}$ the set of all vectors of the form $\mathbf{A} + \mathbf{X}$ where $\mathbf{X} \in \mathcal{U}$. The set $\mathbf{A} + \mathcal{U}$ is called a* **parallel translate, parallel subspace,** *or just a* **parallel** *of $\mathcal{U}$ in $\mathcal{V}$. It is said to result from* **parallel translation** *of $\mathcal{U}$ by the vector $\mathbf{A}$. A parallel of some linear subspace of $\mathcal{V}$ is called an* **affine subspace** *of $\mathcal{V}$.*

Figure 13.1.1 shows an affine subspace in $\mathbb{R}^2$. It is the result of parallel translation by the vector $\mathbf{A} = (3, 1) \in \mathbb{R}^2$ of the subspace $\mathcal{U}\ \left\{(x, y, z) \mid x + y = 0\right\}$ spanned by the vector $(1, -1) \in \mathbb{R}^2$.

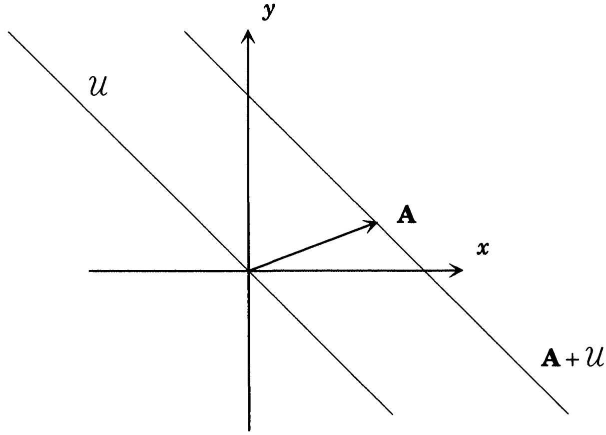


Figure 13.1.1

**PROPOSITION 13.1.4:** *If $\mathcal{U}$ is a linear subspace of a vector space $\mathcal{V}$ and $\mathbf{A} \in \mathcal{V}$, then:*

(1) $\mathbf{A} \in \mathbf{A} + \mathcal{U}$.
(2) *If* $\mathbf{B} \in \mathbf{A} + \mathcal{U}$ *then* $\mathbf{B} + \mathcal{U} = \mathbf{A} + \mathcal{U}$.
(3) *Two parallels of $\mathcal{U}$ either coincide or have no vector in common.*
(4) *If* $\mathbf{B}, \mathbf{C} \in \mathbf{A} + \mathcal{U}$ *then* $\mathbf{B} - \mathbf{C} \in \mathcal{U}$.

**PROOF:** Since $\mathbf{A} = \mathbf{A} + \mathbf{0}$, and $\mathbf{0} \in \mathcal{U}$ since $\mathcal{U}$ is a subspace, we that find $\mathbf{A} \in \mathbf{A} + \mathcal{U}$, proving (1). To prove (2) suppose that $\mathbf{B} \in \mathbf{A} + \mathcal{U}$. Then

$$\mathbf{B} = \mathbf{A} + \mathbf{X}$$

for some vector $\mathbf{X}$ in $\mathcal{U}$. Thus if $\mathbf{C} \in \mathbf{B} + \mathcal{U}$, we may find $\mathbf{Y} \in \mathcal{U}$ such that

$$\mathbf{C} = \mathbf{B} + \mathbf{Y} = (\mathbf{A} + \mathbf{X}) + \mathbf{Y} = \mathbf{A} + (\mathbf{X} + \mathbf{Y}) = \mathbf{A} + \mathbf{W}.$$

Now note that $\mathbf{W} = \mathbf{X} + \mathbf{Y} \in \mathcal{U}$ because $\mathcal{U}$ is a linear subspace of $\mathcal{V}$. Therefore, $\mathbf{C} \in \mathbf{A} + \mathcal{U}$, and hence $\mathbf{B} + \mathcal{U} \subseteq \mathbf{A} + \mathcal{U}$. Reversing the roles of $\mathbf{A}$ and $\mathbf{B}$ we may apply exactly the same argument to show that $\mathbf{A} + \mathcal{U} \subseteq \mathbf{B} + \mathcal{U}$ and hence we conclude that $\mathbf{A} + \mathcal{U} = \mathbf{B} + \mathcal{U}$, as desired.

To prove (3), we suppose that $\mathbf{A} + \mathcal{U}$ and $\mathbf{B} + \mathcal{U}$ are two parallels of $\mathcal{U}$ in $\mathcal{V}$. If they have a vector $\mathbf{C}$ in common, that is, $\mathbf{C} \in \mathbf{A} + \mathcal{U}$ and $\mathbf{C} \in \mathbf{B} + \mathcal{U}$, then by (2) we find

$$\mathbf{C} + \mathcal{U} = \mathbf{A} + \mathcal{U},$$

$$\mathbf{C} + \mathcal{U} = \mathbf{B} + \mathcal{U},$$

so that $\mathbf{A} + \mathcal{U} = \mathbf{C} + \mathcal{U} = \mathbf{B} + \mathcal{U}$ as claimed.

Finally, to prove part (4), we suppose that $\mathbf{B}, \mathbf{C} \in \mathbf{A} + \mathcal{U}$. Then we may find vectors $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{U}$ such that

$$\mathbf{B} = \mathbf{A} + \mathbf{X},$$

$$\mathbf{C} = \mathbf{A} + \mathbf{Y},$$

from which we obtain

$$\mathbf{B} - \mathbf{C} = \mathbf{X} - \mathbf{Y}.$$

But $\mathbf{X} - \mathbf{Y}$ belongs to $\mathcal{U}$, since $\mathcal{U}$ is a linear subspace of $\mathcal{V}$. □

It follows from Proposition 13.1.4 that an affine subspace $\mathcal{A}$ of $\mathcal{V}$ is completely determined by a basis for $\mathcal{U}$ and a *single* vector $\mathbf{A}$ in the affine subspace $\mathcal{A}$.

Affine subspaces may be tied in neatly with linear equations with the aid of Proposition 13.1.1 and the following:

**PROPOSITION 13.1.5:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *be a linear transformation and* $\mathbf{C} \in \mathrm{Im}(\mathsf{T})$. *Let* $\mathcal{A} = \left\{\mathbf{V} \in \mathcal{V} \mid \mathsf{T}(\mathbf{V}) = \mathbf{C}\right\}$, *that is,* $\mathcal{A}$ *is the set of vectors in* $\mathcal{V}$ *whose image under* $\mathsf{T}$ *is the vector* $\mathbf{C}$. *Then* $\mathcal{A}$ *is an affine subspace of* $\mathcal{V}$. *In fact, if* $\mathbf{A}$ *is any vector in* $\mathcal{V}$ *such that* $\mathsf{T}(\mathbf{A}) = \mathbf{C}$, *then* $\mathcal{A} = \mathbf{A} + \ker(\mathsf{T})$.

**PROOF:** Let $\mathbf{A}$ be any vector in $\mathcal{V}$ such that $\mathsf{T}(\mathbf{A}) = \mathbf{C}$. If $\mathbf{B} \in \mathbf{A} + \ker(\mathsf{T})$, then $\mathbf{B} = \mathbf{A} + \mathbf{X}$ for some $\mathbf{X} \in \ker(\mathsf{T})$. Hence

$$\mathsf{T}(\mathbf{B}) = \mathsf{T}(\mathbf{A} + \mathbf{X}) = \mathsf{T}(\mathbf{A}) + \mathsf{T}(\mathbf{X}) = \mathsf{T}(\mathbf{A}) + \mathbf{0} = \mathsf{T}(\mathbf{A}) = \mathbf{C},$$

showing that $\mathbf{B} \in \mathcal{A}$. Therefore, $\mathbf{A} + \ker(\mathsf{T}) \subseteq \mathcal{A}$. Conversely, if $\mathbf{B} \in \mathcal{A}$ then

$$\mathsf{T}(\mathbf{B} - \mathbf{A}) = \mathsf{T}(\mathbf{B}) - \mathsf{T}(\mathbf{A}) = \mathbf{C} - \mathbf{C} = \mathbf{0},$$

so that $\mathbf{B} - \mathbf{A} = \mathbf{X}$ belongs to $\ker(\mathsf{T})$. Then

$$\mathbf{B} = \mathbf{A} + \mathbf{X}$$

with $\mathbf{X} \in \ker(\mathsf{T})$, showing that $\mathbf{B} \in \mathbf{A} + \ker(\mathsf{T})$. Therefore, $\mathcal{A} \subseteq \mathbf{A} + \ker(\mathsf{T})$. Combining this with the preceding inclusion yields $\mathcal{A} = \mathbf{A} + \ker(\mathsf{T})$, as desired. $\square$

**THEOREM 13.1.6:** *Suppose (in matrix notation) that*

$$\mathsf{A} \cdot \mathsf{X} = \mathsf{B}$$

*is a system of linear equations. Then the set of all solutions* $\mathcal{S} = \left\{(s_1, \ldots, s_n)\right\}$ *to this linear system is either empty or is an affine subspace of* $\mathbb{R}^n$. *More precisely, if* $\mathsf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ *is the linear transformation whose matrix relative to the standard bases is* $\mathsf{A}$, *then* $\mathcal{S}$ *is a parallel translate of* $\ker(\mathsf{T})$ *by some vector* $\mathbf{S}$ *in* $\mathcal{S}$, *or* $\mathcal{S} = \varnothing$.

Thus we see that in order to specify all solutions of the linear system

$$\mathsf{A} \cdot \mathsf{X} = \mathsf{B}$$

we must specify a certain affine subspace of $\mathbb{R}^n$. This we may do (according to Proposition 13.1.4) by finding a basis for the solution space of the system

$$\mathsf{A} \cdot \mathsf{X} = \mathsf{0}$$

(called the **associated homogeneous** system) and finding a single particular solution to the equation (if there is one)

$$\mathsf{A} \cdot \mathsf{X} = \mathsf{B}.$$

**EXAMPLE 2:** Solve the system of linear equations

$$\begin{aligned} 2x_1 + x_2 - 2x_3 + 3x_4 &= 4, \\ 3x_1 + 2x_2 - x_3 + 2x_4 &= 6, \\ 3x_1 + 3x_2 + 3x_3 - 3x_4 &= 6. \end{aligned}$$

**SOLUTION:** The associated homogeneous system is

$$\begin{aligned} &① & 2x_1 + x_2 - 2x_3 + 3x_4 &= 0, \\ &② & 3x_1 + 2x_2 - x_3 + 2x_4 &= 0, \\ &③ & 3x_1 + 3x_2 + 3x_3 - 3x_4 &= 0, \end{aligned}$$

which may be solved as follows: Multiply the first equation by 3 and the second by 2 to get

$$\begin{aligned} &④ & 6x_1 + 3x_2 - 6x_3 + 9x_4 &= 0, \\ &⑤ & 6x_1 + 4x_2 - 2x_3 + 4x_4 &= 0. \end{aligned}$$

Subtract ④ from ⑤ to get

$$⑥ \qquad x_2 + 4x_3 - 5x_4 = 0.$$

Subtract ⑥ from ① to get

$$⑦ \qquad 2x_1 - 6x_3 + 8x_4 = 0.$$

Equations ⑦ and ⑥ yield

$$\begin{aligned} x_1 &= 3x_3 - 4x_4, \\ x_2 &= -4x_3 + 5x_4. \end{aligned}$$

A basis for the solution space to the associated homogeneous system is provided by the two vectors

$$(3, -4, 1, 0), \qquad (-4, 5, 0, 1).$$

(Note that we did *not* use equation ③. Why?) Similar manipulations show that a particular solution to the original system of equations is

$$(1, 1, 1, 1).$$

Thus the solution space of this linear system is the affine subspace

$$(1, 1, 1, 1) + \mathcal{L}((3, -4, 1, 0), (-4, 5, 0, 1))$$

of $\mathbb{R}^4$, or what is the same, all solutions to the original system are of the form

$$(1, 1, 1, 1) + a(3, -4, 1, 0) + b(-4, 5, 0, 1),$$

where $a$ and $b$ are arbitrary numbers.

## 13.2 Reduction to Echelon Form

It is time to take up in more detail explicit methods for solving systems of linear equations. The first method we will describe for solving systems of linear equations is by far the simplest of the many methods available. It will always work if one perseveres enough. However, it is usually far from the shortest method.

**DEFINITION:** *A matrix* $\mathbf{A} = (a_{i,j})$ *is said to be an* **echelon matrix** *if the first nonzero entry in any row is a 1 and it appears to the right of the first nonzero entry of the preceding row. If, in addition, the first nonzero entry in a given row is the only nonzero entry in its column we say that the matrix* **A** *is in* **reduced echelon form.**

**EXAMPLE 1:** The matrices

$$\begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

are echelon matrices, but not reduced echelon matrices. The following two matrices are reduced echelon matrices

$$\begin{bmatrix} 0 & 1 & 0 & 3 & 4 \\ 0 & 0 & 1 & 5 & 6 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Notice that an echelon matrix or reduced echelon matrix is a very special type of upper triangular matrix. If the coefficient matrix of a linear system

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}$$

is in reduced echelon form, then is possible by inspection to read off all the solutions.

**EXAMPLE 2:** Solve the linear system

$$\begin{aligned} x_1 + 0x_2 + x_3 + 0x_4 + x_5 + 0x_6 &= 1, \\ 0x_1 + x_2 + x_3 + 0x_4 + 2x_5 + x_6 &= 2, \\ 0x_1 + 0x_2 + 0x_3 + x_4 + 3x_5 + 0x_6 &= 3. \end{aligned}$$

**SOLUTION:** The coefficient matrix is in reduced echelon form. The associated homogeneous system is

$$\begin{aligned} x_1 + 0x_2 + x_3 + 0x_4 + x_5 + 0x_6 &= 0, \\ 0x_1 + x_2 + x_3 + 0x_4 + 2x_5 + x_6 &= 0, \\ 0x_1 + 0x_2 + 0x_3 + x_4 + 3x_5 + 0x_6 &= 0. \end{aligned}$$

Thus, working our way up from the bottom, we find

$$\begin{aligned} x_4 &= & & - 3x_5, & \\ x_2 &= -x_3 & & - 2x_5 & - x_6, \\ x_1 &= -x_3 & & - x_5. & \end{aligned}$$

A basis for the solution space of the associated homogeneous system is therefore seen to be

$$(-1,-1,1,0,0,0), \quad (-1,-2,0,-3,1,0), \quad (0,-1,0,0,0,1),$$

and a solution to the original system is gotten from

$$\begin{aligned} x_4 &= 3 & & - 3x_5, & \\ x_2 &= 2 - x_3 & & - 2x_5 & - x_6, \\ x_1 &= 1 - x_3 & & - x_5, & \end{aligned}$$

by setting $x_3 = x_5 = x_6 = 0$ and is seen to be

$$(1,2,0,3,0,0).$$

Thus the general solution to the original system is seen to be

$$\begin{aligned} (s_1, s_2, s_3, s_4, s_5, s_6) = (1,2,0,3,0,0) &+ a(-1,-1,1,0,0,0) \\ &+ b(-1,-2,0,-3,1,0) + c(0,-1,0,0,0,0,1). \end{aligned}$$

By the **general solution** we mean that any (and hence every) solution to the original system is obtained for a suitable choice of the three numbers (parameters) $a, b, c$.

In solving a system of linear equations

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}$$

by using echelon form, how do we proceed? Well, the idea is to exchange the given system for a new system that on the one hand has the same solution set as the original system but where the coefficient matrix has a more useful distribution of zeros. In fact, as the preceding example suggests, we would like for the new coefficient matrix to be in reduced echelon form. What kinds of operations can we perform on the original system to accomplish this? Clearly, the following three operations and any combination of them will be allowed, because they will not change the solution space, since each operation is reversible.

(1) Interchange any two equations.
(2) Multiply any equation by a ***nonzero*** number.
(3) Add one equation to another.

These operations are called **elementary row operations**. They can be combined to form more complex operations, such as subtracting twice an equation from another equation. The following theorem says that careful and perhaps lengthy application of these operations will solve any linear system.

**THEOREM 13.2.1:** *Let (in matrix notation)*

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}$$

*be a system of linear equations. Then by a finite sequence of the operations:*

(1) *interchanging two equations,*
(2) *multiplying any equation by a nonzero number, and*
(3) *adding one equation to another*

*we may obtain a system of linear equations*

$$\overline{\mathbf{A}} \cdot \overline{\mathbf{X}} = \overline{\mathbf{B}}$$

*whose coefficient matrix is in reduced echelon form.*

The proof of this theorem is an orgy of manipulation with indices and is omitted. Rather, we will illustrate the technique of the proof by working some numerical examples.

**DEFINITION:** *If* $\mathbf{A} \cdot \mathbf{X} = \mathbf{B}$ *is a system of linear equations then the matrix*

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} & b_1 \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} & b_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} & b_m \end{bmatrix}$$

*is called the* **augmented matrix** *of the linear system.*

The operations (1), (2), (3) above to be performed on the linear equations can instead be performed on the augmented matrix in order to save space. We illustrate this in the following examples.

**EXAMPLE 3:** Solve the linear system

$$\begin{aligned} x + 2y - 3z &= 4, \\ x + 3y + z &= 11, \\ 2x + 5y - 4z &= 13, \\ 2x + 6y + 2z &= 22. \end{aligned}$$

**SOLUTION:** The augmented matrix for this system is

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 1 & 3 & 1 & 11 \\ 2 & 5 & -4 & 13 \\ 2 & 6 & 2 & 22 \end{bmatrix}.$$

We proceed to apply the three basic operations to reduce this matrix to reduced echelon form. The lines of the matrix will be denoted by $L_{①}$

to $L_{\textcircled{4}}$. The operation of subtracting line 2 from line 1 will be denoted by $L_{\textcircled{1}} - L_{\textcircled{2}}$ and analogously for the other elementary row operations.

$$\begin{array}{r} \\ L_{\textcircled{2}} - L_{\textcircled{1}} \\ L_{\textcircled{3}} - 2L_{\textcircled{1}} \\ L_{\textcircled{4}} - 2L_{\textcircled{1}} \end{array} \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 1 & 4 & 7 \\ 0 & 1 & 2 & 5 \\ 0 & 2 & 8 & 14 \end{bmatrix} \Rightarrow \begin{array}{r} L_{\textcircled{1}} - 2L_{\textcircled{2}} \\ \\ L_{\textcircled{3}} - L_{\textcircled{2}} \\ L_{\textcircled{4}} - 2L_{\textcircled{2}} \end{array} \begin{bmatrix} 1 & 0 & -11 & -10 \\ 0 & 1 & 4 & 7 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{array}{r} \\ \\ -\frac{1}{2}L_{\textcircled{3}} \\ \\ \end{array} \begin{bmatrix} 1 & 0 & -11 & -10 \\ 0 & 1 & 4 & 7 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{array}{r} L_{\textcircled{1}} + 11L_{\textcircled{3}} \\ L_{\textcircled{2}} - 4L_{\textcircled{3}} \\ \\ \\ \end{array} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

which is the augmented matrix of the system

$$\begin{aligned} x_1 &= 1, \\ x_2 &= 3, \\ x_3 &= 1, \end{aligned}$$

having the same solutions as the original system. Thus the original system has the unique solution $(1, 3, 1)$.

**EXAMPLE 4:** Solve the linear system

$$\begin{aligned} 2x + y - 2z + 3w &= 1, \\ 3x + 2y - z + 2w &= 4, \\ 3x + 3y + 3z - 3w &= 5. \end{aligned}$$

**SOLUTION:** The augmented matrix for this system is

$$\begin{bmatrix} 2 & 1 & -2 & 3 & 1 \\ 3 & 2 & -1 & 2 & 4 \\ 3 & 3 & 3 & -3 & 5 \end{bmatrix},$$

which reduces to echelon form as follows:

$$\begin{array}{r} 3L_{\textcircled{1}} \\ 2L_{\textcircled{2}} \\ 2L_{\textcircled{3}} \end{array} \begin{bmatrix} 6 & 3 & -6 & 9 & 3 \\ 6 & 4 & -2 & 4 & 8 \\ 6 & 6 & 6 & -6 & 10 \end{bmatrix} \Rightarrow \begin{array}{r} \\ L_{\textcircled{2}} - L_{\textcircled{1}} \\ L_{\textcircled{3}} - L_{\textcircled{1}} \end{array} \begin{bmatrix} 6 & 3 & -6 & 9 & 3 \\ 0 & 1 & 4 & -5 & 5 \\ 0 & 3 & 12 & -15 & 7 \end{bmatrix}$$

$$\Rightarrow \begin{array}{r} \\ 3L_{\textcircled{2}} \\ \\ \end{array} \begin{bmatrix} 6 & 3 & -6 & 9 & 3 \\ 0 & 3 & 12 & -15 & 15 \\ 0 & 3 & 12 & -15 & 7 \end{bmatrix} \Rightarrow \begin{array}{r} L_{\textcircled{1}} - L_{\textcircled{2}} \\ \\ L_{\textcircled{3}} - L_{\textcircled{2}} \end{array} \begin{bmatrix} 6 & 0 & -18 & 24 & -12 \\ 0 & 3 & 12 & -15 & 15 \\ 0 & 0 & 0 & 0 & -8 \end{bmatrix}$$

$$\Rightarrow \begin{array}{r} \frac{1}{6}L_{\textcircled{1}} \\ \frac{1}{3}L_{\textcircled{2}} \\ -\frac{1}{8}L_{\textcircled{3}} \end{array} \begin{bmatrix} 1 & 0 & -3 & 4 & -2 \\ 0 & 1 & 4 & -5 & 5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

which is the augmented matrix of the linear system

$$\begin{aligned} x \quad\; - 3z + 4w &= -2, \\ y + 4z - 5w &= 5, \\ 0 &= 1. \end{aligned}$$

The impossibility of the last equation shows that this linear system has ***no*** solutions; hence the original system has no solutions. Note that the column space of the coefficient matrix for the new system is spanned by

$$(1, 0, 0), \quad (0, 1, 0), \quad (-3, 4, 0), \quad (4, -5, 0),$$

which is the $xy$-plane in $\mathbb{R}^3$, while the vector $(-2, 5, 1)$ does not lie in this plane.

As a last example of this method we work:

**EXAMPLE 5:** Solve the linear system

$$\begin{aligned} x_2 + 3x_3 + x_4 - x_5 &= 2, \\ x_1 - x_2 + 3x_3 - 4x_4 + 2x_5 &= 6, \\ x_1 + x_2 - x_3 + 2x_4 + x_5 &= 1, \\ x_1 \quad\;\; - x_3 \quad\;\; + x_5 &= 1. \end{aligned}$$

**SOLUTION:** The augmented matrix is

$$\begin{bmatrix} 0 & 1 & 3 & 1 & -1 & 2 \\ 1 & -1 & 3 & -4 & 2 & 6 \\ 1 & 1 & -1 & 2 & 1 & 1 \\ 1 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}.$$

Before reducing to echelon form we first interchange $L_{①}$ and $L_{②}$ to obtain the coefficient matrix

$$\begin{bmatrix} 1 & -1 & 3 & -4 & 2 & 6 \\ 0 & 1 & 3 & 1 & -1 & 2 \\ 1 & 1 & -1 & 2 & 1 & 1 \\ 1 & 0 & -1 & 0 & 1 & 1 \end{bmatrix},$$

which leads to

$$\begin{bmatrix} 1 & -1 & 3 & -4 & 2 & 6 \\ 0 & 1 & 3 & 1 & -1 & 2 \\ 1 & 1 & -1 & 2 & 1 & 1 \\ 1 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l} \\ \\ L_3 - L_1 \\ L_4 - L_1 \end{array} \begin{bmatrix} 1 & -1 & 3 & -4 & 2 & 6 \\ 0 & 1 & 3 & 1 & -1 & 2 \\ 0 & 2 & -4 & 6 & -1 & -5 \\ 0 & 1 & -4 & 4 & -1 & -5 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l} L_1 + L_2 \\ \\ L_3 - 2L_2 \\ L_4 - L_2 \end{array} \begin{bmatrix} 1 & 0 & 6 & -3 & 1 & 8 \\ 0 & 1 & 3 & 1 & -1 & 2 \\ 0 & 0 & -10 & 4 & 1 & -9 \\ 0 & 0 & -7 & 3 & 0 & -7 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l} \\ (-7)L_3 \\ (10)L_4 \\ \\ \end{array} \begin{bmatrix} 1 & 0 & 6 & -3 & 1 & 8 \\ 0 & 1 & 3 & 1 & -1 & 2 \\ 0 & 0 & 70 & -28 & -7 & 63 \\ 0 & 0 & -70 & 30 & 0 & -70 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l} \\ \\ \\ L_4 + L_3 \end{array} \begin{bmatrix} 1 & 0 & 6 & -3 & 1 & 8 \\ 0 & 1 & 3 & 1 & -1 & 2 \\ 0 & 0 & 70 & -28 & -7 & 63 \\ 0 & 0 & 0 & 2 & -7 & -7 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l} \\ \\ L_3/70 \\ L_4/2 \end{array} \begin{bmatrix} 1 & 0 & 6 & -3 & 1 & 8 \\ 0 & 1 & 3 & 1 & -1 & 2 \\ 0 & 0 & 1 & -\frac{28}{70} & -\frac{1}{10} & \frac{63}{70} \\ 0 & 0 & 0 & 1 & -\frac{7}{2} & -\frac{7}{2} \end{bmatrix},$$

which is the augmented matrix of the system

$$\begin{aligned} x_1 \quad + 6x_3 - 3x_4 + x_5 &= 8, \\ x_2 + 3x_3 + x_4 - x_5 &= 2, \\ x_3 - \tfrac{2}{5}x_4 - \tfrac{1}{10}x_5 &= \tfrac{9}{10}, \\ x_4 - \tfrac{7}{2}x_5 &= \tfrac{-7}{2}. \end{aligned}$$

The associated homogeneous system is

$$\begin{aligned} x_1 \quad + 6x_3 - 3x_4 + x_5 &= 0, \\ x_2 + 3x_3 + x_4 - x_5 &= 0, \\ x_3 - \tfrac{2}{5}x_4 - \tfrac{1}{10}x_5 &= 0, \\ x_4 - \tfrac{7}{2}x_5 &= 0, \end{aligned}$$

from which we obtain

$$\begin{aligned}
x_4 &= \frac{7}{2}x_5,\\
x_3 &= \frac{2}{5}x_4 + \frac{1}{10}x_5,\\
x_2 &= -3x_3 - x_4 + x_5,\\
x_1 &= -6x_3 + 3x_4 - x_5,
\end{aligned}$$

and so a basis for the solution space consists of the single vector (set $x_5 = 1$)

$$\left(\frac{1}{2}, -7, \frac{3}{2}, \frac{7}{2}, 1\right),$$

or if you don't like fractions

$$(1, -14, 3, 7, 2).$$

To find a particular solution to the nonhomogeneous system, we use the equations

$$\begin{aligned}
x_4 &= -\frac{7}{2} + \frac{7}{2}x_5,\\
x_3 &= \frac{9}{10} + \frac{2}{5}x_4 + \frac{1}{10}x_5,\\
x_2 &= 2 - 3x_3 - x_4 + x_5,\\
x_1 &= 8 - 6x_3 + 3x_4 - x_5.
\end{aligned}$$

Setting $x_5 = 0$ gives the particular solution

$$\begin{aligned}
x_5 &= 0,\\
x_4 &= -\frac{7}{2},\\
x_3 &= \frac{9}{10} + \frac{2}{5}\left(-\frac{7}{2}\right) = \frac{9}{10} - \frac{14}{10} = \frac{-5}{10} = -\frac{1}{2},\\
x_2 &= 2 - 3\left(-\frac{1}{2}\right) + \frac{7}{2} = 2 + \frac{3}{2} + \frac{7}{2} = 7,\\
x_1 &= 8 - 6\left(-\frac{1}{2}\right) + 3\left(-\frac{7}{2}\right) = 8 + 3 - \frac{21}{2} = \frac{1}{2}.
\end{aligned}$$

That is, $(\frac{1}{2}, 7, -\frac{1}{2}, -\frac{7}{2}, 0)$ solves the nonhomogeneous system. So the general solution to the original system is

$$(s_1, s_2, s_3, s_4, s_5) = (\frac{1}{2}, 7, -\frac{1}{2}, -\frac{7}{2}, 0) + a(1, -14, 3, 7, 2).$$

Setting $x_5 = 1$ gives another particular solution, namely $(1, 0, 1, 0, 1)$, so the general solution could equally well be written

$$(s_1, s_2, s_3, s_4, s_5) = (1, 0, 1, 0, 1) + a(1, -14, 3, 7, 2)$$

where $a$ is an arbitrary number.

## 13.3 The Simplex Method

The method of solving a linear system

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}$$

by reducing the coefficient matrix $\mathbf{A}$ to reduced echelon form is a system of trading off the given linear system for a new and simpler linear system

$$\overline{\mathbf{A}} \cdot \overline{\mathbf{X}} = \overline{\mathbf{B}},$$

one whose solution set is the same as the original, and from which the solutions may be read off by a purely mechanical process. There is another general approach to the solution of linear systems, which involves a more direct kind of horsetrading of *knowns* and *unknowns* To be more specific, let us examine our linear system in longhand again. It looks like

$$\begin{array}{ccccccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \cdots & + & a_{1,n}x_n & = & b_1, \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \cdots & + & a_{2,n}x_n & = & b_2, \\ & & & & & & & \vdots & \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \cdots & + & a_{m,n}x_n & = & b_m. \end{array}$$

We may regard this system as expressing the known quantities $b_1, \ldots,$ $b_m$ as linear combinations of the unknown quantities $x_1, \ldots, x_n$. To solve the system means to reverse the roles of $x_1, \ldots, x_n$ and $b_1, \ldots, b_m$, that is, to express the unknowns $x_1, \ldots, x_n$ as linear combinations of the knowns $b_1, \ldots, b_m$. For example, suppose that $a_{i,j} \neq 0$. Then we may solve the equation

$$a_{i,1}x_1 + a_{i,2}x_2 + \cdots + a_{i,j}x_j + \cdots + a_{i,n}x_n = b_i$$

for $x_j$, obtaining

$$x_j = -\frac{1}{a_{i,j}}[a_{i,1}x_1 + a_{i,2}x_2 + \cdots + a_{i,j-1}x_{j-1} - b_i + a_{i,j+1}x_{j+1} + \cdots + a_{i,n}x_n].$$

(Note that $x_j$ does ***not*** appear on the right in this formula.) We may now take this formula for $x_j$ and put it into the other equations of the system obtaining the new system

$$\begin{array}{lllll}
\left(a_{1,1}-\dfrac{a_{1,j}a_{i,1}}{a_{i,j}}\right)x_1 & +\cdots+\dfrac{a_{1,j}}{a_{i,j}}b_i+\cdots+ & \left(a_{1,n}-\dfrac{a_{1,j}a_{i,n}}{a_{i,j}}\right)x_n & = b_1, \\
 & & & \vdots \\
-\dfrac{a_{i,1}}{a_{i,j}}x_1 & -\cdots+\dfrac{1}{a_{i,j}}b_i-\cdots- & \dfrac{a_{i,n}}{a_{i,j}}x_n & = x_j, \\
 & & & \vdots \\
\left(a_{m,1}-\dfrac{a_{m,j}a_{i,1}}{a_{i,j}}\right)x_1 & +\cdots+\dfrac{a_{m,j}}{a_{i,j}}b_i+\cdots+ & \left(a_{m,n}-\dfrac{a_{m,j}a_{i,n}}{a_{i,j}}\right)x_n & = b_m.
\end{array}$$

This process of obtaining a new system of equations is called a **pivot operation** on $a_{i,j}$. Note that in this new system the unknown $x_j$ does not appear in any of the equations on the left. It has been traded for the known quantity $b_i$. By repeated application of this process we may be able to obtain a linear system in which each right hand side is an $x_j$ and each left hand side is a linear combination of the $b_j$ and those $x_k$ that do not appear on the left. In this way we will have solved the system, since each step is reversible. In order to apply this method, known as the **simplex method**, to solve systems of equations, we must have available a simple way to write down the coefficients of the system obtained by pivoting at $a_{i,j}$. To do this[1] we require a simple definition.

**DEFINITION:** *Suppose that*

$$\mathbf{C}=\begin{bmatrix} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \end{bmatrix}$$

*is a* $2\times 2$ *matrix. The* **determinant** *of* $\mathbf{C}$*, denoted by* $\det(\mathbf{C})$*, is the number* $\det(\mathbf{C}) = c_{1,1}c_{2,2} - c_{2,1}c_{1,2}$.

**EXAMPLE 1:** Find the determinant of the matrix

$$\begin{bmatrix} 1 & 4 \\ 3 & 4 \end{bmatrix}.$$

**SOLUTION:** We have

$$\det\begin{bmatrix} 1 & 4 \\ 3 & 4 \end{bmatrix} = 1(4) - 4(3) = -8.$$

**EXAMPLE 2:** Find the determinant of the matrix

$$\begin{bmatrix} 1 & -3 \\ 1 & -3 \end{bmatrix}.$$

[1] Recall from section Section 10.4 our discussion of inverting $2\times 2$ matrices.

SOLUTION: We have

$$\det\begin{bmatrix} 1 & -3 \\ 1 & -3 \end{bmatrix} = 1(-3) - (-3)1 = -3 + 3 = 0.$$

Notice that a matrix with all positive entries can have a negative determinant and a matrix with all nonzero entries a zero determinant.

Return next to our linear system

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}.$$

If we suppose that $a_{i,j} \neq 0$, then we may apply a pivot operation at $a_{i,j}$ to obtain a new linear system

$$\overline{\mathbf{A}} \cdot \overline{\mathbf{X}} = \overline{\mathbf{B}},$$

where $\overline{\mathbf{A}} = (\overline{a}_{r,s})$ and

$$\overline{a}_{r,s} = \begin{cases} \dfrac{1}{a_{i,j}} \det\begin{bmatrix} a_{r,s} & a_{r,j} \\ a_{i,s} & a_{ij} \end{bmatrix} & \text{if } r \neq i \text{ and } s \neq j, \\ \dfrac{a_{r,j}}{a_{i,j}} & \text{if } r \neq i \text{ and } s = j, \\ -\dfrac{a_{i,s}}{a_{i,j}} & \text{if } r = i \text{ and } s \neq j, \\ \dfrac{1}{a_{i,j}} & \text{if } r = i \text{ and } s = j, \end{cases}$$

and

$$\overline{\mathbf{B}} = \begin{bmatrix} b_1 \\ \vdots \\ b_{i-1} \\ x_j \\ b_{i+1} \\ \vdots \\ b_m \end{bmatrix}, \qquad \overline{\mathbf{X}} = \begin{bmatrix} x_1 \\ \vdots \\ x_{j-1} \\ b_i \\ x_{j+1} \\ \vdots \\ x_n \end{bmatrix}.$$

Note that we change not only the coefficient matrix **A** and constant matrix **B** as with the echelon form method but ***also the matrix of unknowns.*** Our notations should keep track of this fact. With these formulas[2] in mind let us run through a few examples using this method.

[2] Clearly, these formulas are ridiculously difficult to memorize. But, once they are ***programmed***, the program has no trouble remembering them, so while this method may not be well suited for people, it is very well suited for the computer, and lots faster than the echelon form method.

**EXAMPLE 3:** Solve the linear system

$$
\begin{aligned}
2x - \phantom{2}y + \phantom{2}z &= 2,\\
x + 2y - \phantom{2}z &= 3,\\
3x + \phantom{2}y + 2z &= -1.
\end{aligned}
$$

**SOLUTION:** We represent our linear system by the diagram

$$
\begin{array}{ccc|c}
x & y & z & \\
\hline
2 & -1 & 1 & 2 \\
1 & 2 & -1 & 3 \\
3 & 1 & 2 & -1
\end{array}
$$

It is always easiest to pivot on a 1 so let us pivot on the $a_{1,3}$ position where there is a 1. Our new system is then represented by the diagram

$$
\begin{array}{ccc|c}
x & y & 2 & \\
\hline
-2 & 1 & 1 & z \\
3 & 1 & -1 & 3 \\
-1 & 3 & 2 & -1
\end{array}
$$

Since we are pivoting on a 1, in the (1, 3) position, the first row changes sign and the third column changes not at all. Here are the remaining calculations:

$$
\overline{a}_{2,1} = \det \begin{bmatrix} a_{2,1} & a_{2,3} \\ a_{1,1} & a_{1,3} \end{bmatrix} = \det \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = 1 + 2 = 3,
$$

$$
\overline{a}_{2,2} = \det \begin{bmatrix} a_{2,2} & a_{2,3} \\ a_{1,2} & a_{1,3} \end{bmatrix} = \det \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = 2 - 1 = 1,
$$

$$
\overline{a}_{3,1} = \det \begin{bmatrix} a_{3,1} & a_{3,3} \\ a_{1,1} & a_{1,3} \end{bmatrix} = \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} = 3 - 4 = -1,
$$

$$
\overline{a}_{3,2} = \det \begin{bmatrix} a_{3,2} & a_{3,3} \\ a_{1,2} & a_{1,3} \end{bmatrix} = \det \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} = 1 + 2 = 3.
$$

This new system has a handy 1 in the (2, 2) spot, so let's perform a pivot there. The resulting system is represented by the diagram

$$
\begin{array}{ccc|c}
x & 3 & 2 & \\
\hline
-5 & 1 & 2 & z \\
-3 & 1 & 1 & y \\
-10 & 3 & 5 & -1
\end{array}
$$

Since $i = 2, j = 2$, the second row changes sign and the second column stays as it was. The other calculations are as follows:

$$\bar{\bar{a}}_{1,1} = \det\begin{bmatrix} \bar{a}_{1,1} & \bar{a}_{1,2} \\ \bar{a}_{2,1} & \bar{a}_{2,2} \end{bmatrix} = \det\begin{bmatrix} -2 & 1 \\ 3 & 1 \end{bmatrix} = -2 - 3 = -5,$$

$$\bar{\bar{a}}_{1,3} = \det\begin{bmatrix} \bar{a}_{1,3} & \bar{a}_{1,2} \\ \bar{a}_{2,3} & \bar{a}_{2,2} \end{bmatrix} = \det\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = 1 + 1 = 2,$$

$$\bar{\bar{a}}_{3,1} = \det\begin{bmatrix} \bar{a}_{3,1} & \bar{a}_{3,2} \\ \bar{a}_{2,1} & \bar{a}_{2,2} \end{bmatrix} = \det\begin{bmatrix} -1 & 3 \\ 3 & 1 \end{bmatrix} = -1 - 9 = -10,$$

$$\bar{\bar{a}}_{3,3} = \det\begin{bmatrix} \bar{a}_{3,3} & \bar{a}_{3,2} \\ \bar{a}_{2,3} & \bar{a}_{2,2} \end{bmatrix} = \det\begin{bmatrix} 2 & 3 \\ -1 & 1 \end{bmatrix} = 2 + 3 = 5.$$

Finally, we pivot on the (3, 1) position to obtain

| $-1$ | $3$ | $2$ | |
|---|---|---|---|
| $\frac{1}{2}$ | $-\frac{1}{2}$ | $-\frac{1}{2}$ | $z$ |
| $\frac{3}{10}$ | $\frac{1}{10}$ | $-\frac{1}{2}$ | $y$ |
| $\frac{-1}{10}$ | $\frac{3}{10}$ | $\frac{5}{10}$ | $x$ |

This time the third row changes sign and gets divided by $-10$, which effectively divides the third row by 10, and the first column (apart from the pivot position) gets divided by $-10$. The other computations go as follows:

$$\bar{\bar{\bar{a}}}_{1,2} = \frac{-1}{10}\det\begin{bmatrix} 1 & -5 \\ 3 & -10 \end{bmatrix} = \frac{-1}{10}(-10 + 15) = \frac{-5}{10} = -\frac{1}{2},$$

$$\bar{\bar{\bar{a}}}_{1,3} = \frac{-1}{10}\det\begin{bmatrix} 2 & -5 \\ 5 & -10 \end{bmatrix} = \frac{-1}{10}(-20 + 25) = -\frac{1}{2},$$

$$\bar{\bar{\bar{a}}}_{2,2} = \frac{-1}{10}\det\begin{bmatrix} 1 & -3 \\ 3 & -10 \end{bmatrix} = \frac{-1}{10}(-10 + 9) = \frac{1}{10},$$

$$\bar{\bar{\bar{a}}}_{2,3} = \frac{-1}{10}\det\begin{bmatrix} 1 & -3 \\ 5 & -10 \end{bmatrix} = \frac{-1}{10}(-10 + 15) = \frac{-5}{10} = -\frac{1}{2},$$

and thus we find

$$z = \frac{1}{2}(-1) - \frac{1}{2}(3) - \frac{1}{2}(2) = -\frac{1}{2} - \frac{3}{2} - 1 = -3,$$

$$y = \frac{3}{10}(-1) + \frac{1}{10}(3) - \frac{1}{2}(2) = -1,$$

$$x = \frac{-1}{10}(-1) + \frac{3}{10}(3) + \frac{5}{10}(2) = \frac{20}{10} = 2.$$

So there is only one solution, and it is

$$x = 2, \qquad y = -1, \qquad z = -3.$$

As a second example consider the following:

**EXAMPLE 4:** Solve the linear system

$$\begin{aligned} x - y + z &= 3, \\ 2x + y - z &= 6, \\ -x + 2y + 2z &= 1, \\ 3x - 2y - 2z &= -1. \end{aligned}$$

**SOLUTION:** We represent our system by the diagram

| $x$ | $y$ | $z$ | |
|---|---|---|---|
| 1 | –1 | 1 | 3 |
| 2 | 1 | –1 | 6 |
| –1 | 2 | 2 | 1 |
| 3 | –2 | –2 | –1 |

We first pivot on the $a_{1,1}$ position. Since we are pivoting on a 1 in position $a_{1,1}$, the first row changes sign and the first column is unchanged. The other computations look as follows:

$$\begin{aligned}
\overline{a}_{2,2} &= \det\begin{bmatrix} a_{2,2} & a_{2,1} \\ a_{1,2} & a_{1,1} \end{bmatrix} = \det\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} = 1-(-1)(2) = 3, \\
\overline{a}_{2,3} &= \det\begin{bmatrix} a_{2,3} & a_{2,1} \\ a_{1,3} & a_{1,1} \end{bmatrix} = \det\begin{bmatrix} -1 & 2 \\ 1 & 1 \end{bmatrix} = -1-2 = -3, \\
\overline{a}_{3,2} &= \det\begin{bmatrix} a_{3,2} & a_{3,1} \\ a_{1,2} & a_{1,1} \end{bmatrix} = \det\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = 2-(-1)(-1) = 1, \\
\overline{a}_{3,3} &= \det\begin{bmatrix} a_{3,3} & a_{3,1} \\ a_{1,3} & a_{1,1} \end{bmatrix} = \det\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = 2+1 = 3, \\
\overline{a}_{4,2} &= \det\begin{bmatrix} a_{4,2} & a_{4,1} \\ a_{1,2} & a_{1,1} \end{bmatrix} = \det\begin{bmatrix} -2 & 3 \\ -1 & 1 \end{bmatrix} = -2-(-1)(3) = 1, \\
\overline{a}_{4,3} &= \det\begin{bmatrix} a_{4,3} & a_{4,1} \\ a_{1,3} & a_{1,1} \end{bmatrix} = \det\begin{bmatrix} -2 & 3 \\ 1 & 1 \end{bmatrix} = -2-3 = -5.
\end{aligned}$$

Our new system is then represented by the diagram

| 3 | $y$ | $z$ | |
|---|---|---|---|
| 1 | 1 | –1 | $x$ |
| 2 | 3 | –3 | 6 |
| –1 | 1 | 3 | 1 |
| 3 | 1 | –5 | –1 |

Let us next pivot on the $a_{3,2}$ position where there is a 1 in the new matrix. This will give us the system represented by the diagram

$$
\begin{array}{ccc|c}
3 & 1 & z & \\
\hline
2 & 1 & -4 & x \\
5 & 3 & -12 & 6 \\
1 & 1 & -3 & y \\
4 & 1 & -8 & -6
\end{array}
$$

We have $i = 3, j = 2$, and

$$
\begin{aligned}
\overline{\overline{a}}_{1,1} &= \det\begin{bmatrix} \overline{a}_{1,1} & \overline{a}_{1,2} \\ \overline{a}_{3,1} & \overline{a}_{3,2} \end{bmatrix} = \det\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = 1-(-1) = 2, \\
\overline{\overline{a}}_{2,1} &= \det\begin{bmatrix} 2 & 3 \\ -1 & 1 \end{bmatrix} = 2-(-3) = 5, \\
\overline{\overline{a}}_{4,1} &= \det\begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} = 3-(-1) = 4, \\
\overline{\overline{a}}_{1,3} &= \det\begin{bmatrix} -1 & 1 \\ 3 & 1 \end{bmatrix} = -1-3 = -4, \\
\overline{\overline{a}}_{2,3} &= \det\begin{bmatrix} -3 & 3 \\ 3 & 1 \end{bmatrix} = -3-9 = -12, \\
\overline{\overline{a}}_{4,3} &= \det\begin{bmatrix} -5 & 1 \\ 3 & 1 \end{bmatrix} = -5-3 = -8.
\end{aligned}
$$

If we pivot once more, we can get rid of the $z$ and thereby solve the system. The most convenient place to pivot is on the $-8$ in the $(4, 3)$ position. This will give us

$$
\begin{array}{ccc|c}
3 & 1 & -6 & \\
\hline
 & & \frac{1}{2} & x \\
-1 & \frac{3}{2} & \frac{12}{8} & 6 \\
 & & \frac{3}{8} & y \\
\frac{1}{2} & \frac{1}{8} & -\frac{1}{8} & z
\end{array}
$$

There is no need to calculate further, as we have found the contradiction

$$-3 + \frac{3}{2} - 9 = 6,$$

which shows that the original system has no solutions.

As our final example we consider:

**EXAMPLE 5:** Solve the linear system

$$\begin{aligned} x + 2y - 3z &= 6, \\ 2x - y + 4z &= 2, \\ 4x + 3y - 2z &= 14. \end{aligned}$$

**SOLUTION:** Our system is represented by

$$\begin{array}{ccc|c} x & y & z & \\ \hline 1 & 2 & -3 & 6 \\ 2 & -1 & 4 & 2 \\ 4 & 3 & -2 & 14 \end{array}$$

Pivoting on the 1 in the (1, 1) position gives the system

$$\begin{array}{ccc|c} 6 & y & z & \\ \hline 1 & -2 & 3 & x \\ 2 & -5 & 10 & 2 \\ 4 & -5 & 10 & 14 \end{array}$$

Let us pivot next on the 10 in the (3, 3) position to obtain

$$\begin{array}{ccc|c} 6 & y & 14 & \\ \hline -\frac{1}{5} & -\frac{1}{2} & \frac{3}{10} & x \\ -2 & 0 & 1 & 2 \\ -\frac{2}{5} & \frac{1}{2} & \frac{1}{10} & z \end{array}$$

from which we find

$$x = -\frac{6}{5} - \frac{y}{2} + \frac{3(14)}{10} = \frac{42-12}{10} - \frac{y}{2} = 3 - \frac{y}{2}$$

$$-12 + 14 = 2,$$

$$z = -\frac{12}{5} + \frac{1}{2}y + \frac{14}{10} = \frac{14-24}{10} + \frac{1}{2}y = \frac{1}{2}y - 1,$$

or

$$x = 3 - \frac{y}{2},$$
$$z = -1 + \frac{y}{2},$$

so the solution space is

$$(3, 0, -1) + \mathcal{L}\left\{(-1, 2, 1)\right\}.$$

## 13.4 Exercises

**1.** Find the solution space of following systems of linear homogeneous equations.

(*a*)
$$\begin{aligned} x - y + z - w &= 0, \\ 2x + y - z + 2w &= 0, \\ 2y + 3z + w &= 0. \end{aligned}$$

(*b*)
$$\begin{aligned} 3x_1 + 2x_2 + x_3 - x_4 - x_5 &= 0, \\ x_1 - x_2 - x_3 - x_4 + 2x_5 &= 0, \\ -x_1 + 2x_2 + 3x_3 + x_4 - x_5 &= 0. \end{aligned}$$

(*c*)
$$\begin{aligned} 2x + 3y - z &= 0, \\ x - y + z &= 0, \\ x + 9y - 5z &= 0. \end{aligned}$$

(*d*)
$$\begin{aligned} 11x - 8y + 5z &= 0, \\ 3x - y - 2z &= 0, \\ -x - 12y + 13z &= 0. \end{aligned}$$

**2.** Solve the following systems of linear equations.

(*a*)
$$\begin{aligned} x - 3y + z &= -2, \\ 2x + y - z &= 6, \\ x + 2y + 2z &= 2. \end{aligned}$$

(*b*)
$$\begin{aligned} x - y + z &= 2, \\ x + y &= 1, \\ x + y + z &= 8. \end{aligned}$$

(*c*)
$$\begin{aligned} x + 2y - 3z &= 4, \\ -x + y + z &= 0, \\ 4x - 2y + z &= 9. \end{aligned}$$

(*d*)
$$\begin{aligned} 3x - 6y + z &= 7, \\ x + 2y + z &= 5, \\ -2x + 5y - 2z &= -1. \end{aligned}$$

**3.** Find a system of linear equations in three unknowns whose solution space is the line through (1, 1, 1) and the origin.

**4.** Write a computer program that reduces an $m \times n$ matrix to a reduced echelon form. If you have access to a computer, run the program on the previous problems.

**5.** Write a program to implement the simplex method on a computer If you have access to a computer, run the program on the previous problems.

**6.** Find the solution space of the following systems.

$$(a)\qquad \begin{aligned} 2x + 6y - z + w &= -3, \\ x - y + z - w &= 3, \\ -x - 3y + 3z + 2w &= 9. \end{aligned}$$

$$(b)\qquad \begin{aligned} x_2 + 2x_3 - x_4 + x_5 &= 5, \\ x_1 - x_2 + x_3 + 2x_4 - x_5 &= 6, \\ x_1 + x_2 - x_3 + x_5 &= -2, \\ x_1 + x_3 - 3x_4 - 2x_5 &= -3. \end{aligned}$$

$$(c)\qquad \begin{aligned} x - 2y + z &= 5, \\ 2x + y - 2z &= 7, \\ x - 7y + 5z &= 8. \end{aligned}$$

$$(d)\qquad \begin{aligned} 3x + y - z &= 10, \\ x - 2y - z &= -2, \\ -x + y + z &= 0, \\ 2x - y - 3z &= 7. \end{aligned}$$

**7.** Solve the linear systems:

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

**8.** For which values of $a$ does the following system of equations have a solution?

$$\begin{aligned} y + z &= 2, \\ x + y + z &= a, \\ x + y &= 2. \end{aligned}$$

**9.** For which values of $a$ does the following system of equations have a solution?

$$\begin{aligned} y + z &= 2, \\ x + ay + z &= 2, \\ x + y \phantom{{}+z} &= 2. \end{aligned}$$

**10.** Let $\mathbf{A}$ be an $n \times n$ matrix. Convert the problem of finding the inverse of $\mathbf{A}$ into a problem of solving a linear system of $n^2$ equations in $n^2$ unknowns.

**11.** Let $\overline{\mathbf{A}}$ be the result of applying an elementary row operation to the $n \times m$ matrix $\mathbf{A}$. Show that $\overline{\mathbf{A}}$ is the product of $\mathbf{A}$ with the matrix obtained by applying the same elementary row operation to the identity matrix of size $n \times n$. (Note that the order is important. What happens if you multiply in the other order?)

**12.** Let $\mathbf{A}$ be a square matrix. Suppose that there is a sequence of elementary row operations that when applied successively to $\mathbf{A}$ yield the identity matrix. Show that the same operations applied in the same order to $\mathbf{I}$ yield the inverse $\mathbf{A}^{-1}$ of $\mathbf{A}$. HINT: See the previous exercise.

# Chapter 14
# The Elements of Eigenvalue and Eigenvector Theory

One of the mysteries of a first course on the calculus and analytic geometry is sketching the graph of a quadratic curve, such as, for example,

$$x^2 + 3xy + y^2 = 11.$$

The secret to discovering the nature of this curve lies in rotating the coordinate axes $(x, y)$ to a new position $(u, v)$ so that the set of points that lie on the graph have in the new coordinate system a quadratic equation without a cross product (here the $3xy$) term. The theory of eigenvectors and eigenvalues should remove the mystery and make the solution of this, and analogous higher-dimensional problems routine.

The idea in back of eigentheory is, from among the many possible matrix representations of a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ to single out the most simple and useful ones. So we start out in this chapter by examining matrix representatives of a fixed linear transformation $\mathsf{T}$ and what they reveal about $\mathsf{T}$.

## 14.1 The Rank of an Endomorphism

Suppose that

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$$

is a linear transformation of the vector space $\mathcal{V}$ to itself. Such linear transformations have a special name (because their domain and range space are the same) they are called **endomorphisms of $\mathcal{V}$**.

**PROPOSITION 14.1.1:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the finite-dimensional vector space $\mathcal{V}$. Then there exist bases $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$*

*and* $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ *for* $\mathcal{V}$ *such that the matrix of* $\mathsf{T}$ *is*

$$\begin{array}{c}\left[\begin{array}{cccccc} 1 & & & & & \\ & \ddots & & & 0 & \\ & & 1 & & & \\ & & & 0 & & \\ & 0 & & & \ddots & \\ & & & & & 0 \end{array}\right] \begin{array}{l} \left.\vphantom{\begin{array}{c}1\\ \ddots\\ 1\end{array}}\right\} k \\ \left.\vphantom{\begin{array}{c}0\\ \ddots\\ 0\end{array}}\right\} n-k \end{array}\end{array}$$

*for some integer* $k$. *The integer* $k$ *is called the* **rank** *of the linear transformation*[1] $\mathsf{T}$, *and our proof will show that* $k = \dim(\mathrm{Im}(\mathsf{T}))$.

**PROOF:** Let $\mathbf{C}_1, \ldots, \mathbf{C}_m$ be a basis for $\ker(\mathsf{T})$. By the basis extension Theorem 6.3.2 we may find vectors $\mathbf{C}_{m+1}, \ldots, \mathbf{C}_n$ such that $\mathbf{C}_1, \ldots, \mathbf{C}_m, \mathbf{C}_{m+1}, \ldots, \mathbf{C}_n$ is a basis for $\mathcal{V}$. Thus $n = \dim(\mathcal{V})$. At this point we need:

**LEMMA 14.1.2:** *The vectors* $\mathsf{T}(\mathbf{C}_{m+1}), \mathsf{T}(\mathbf{C}_{m+2}), \ldots, \mathsf{T}(\mathbf{C}_n)$ *are linearly independent.*

**PROOF:** Suppose to the contrary that the vectors $\mathsf{T}(\mathbf{C}_{m+1}), \ldots, \mathsf{T}(\mathbf{C}_n)$ are linearly dependent. Then we may find numbers $c_{m+1}, \ldots, c_n$, not all zero, such that

$$c_{m+1}\mathsf{T}(\mathbf{C}_{m+1}) + \cdots + c_n\mathsf{T}(\mathbf{C}_n) = \mathbf{0}.$$

Therefore ,

$$\mathsf{T}(c_{m+1}\mathbf{C}_{m+1} + \cdots + c_n\mathbf{C}_n) = c_{m+1}\mathsf{T}(\mathbf{C}_{m+1}) + \cdots + c_n\mathsf{T}(\mathbf{C}_n) = \mathbf{0}.$$

Hence the vector $c_{m+1}\mathbf{C}_{m+1} + \cdots + c_n\mathbf{C}_n$ belongs to the kernel of $\mathsf{T}$. Remember that we chose the vectors $\mathbf{C}_1, \ldots, \mathbf{C}_m$ to be a basis for $\ker(\mathsf{T})$ and therefore

$$c_{m+1}\mathbf{C}_{m+1} + \cdots + c_n\mathbf{C}_n = c_1\mathbf{C}_1 + \cdots + c_m\mathbf{C}_m$$

for suitable numbers $c_1, \ldots, c_m$. But then

$$(-c_1)\mathbf{C}_1 + (-c_2)\mathbf{C}_2 + \cdots + (-c_m)\mathbf{C}_m + c_{m+1}\mathbf{C}_{m+1} + \cdots + c_n\mathbf{C}_n = \mathbf{0},$$

which (since not all of $c_1, \ldots, c_n$ are zero) means that $\mathbf{C}_1, \ldots, \mathbf{C}_n$ are linearly dependent, contrary to the fact that they are a basis for $\mathcal{V}$. Therefore, the vectors $\mathsf{T}(\mathbf{C}_{m+1}), \ldots, \mathsf{T}(\mathbf{C}_n)$ are linearly independent. $\square$

---

[1] The integer $n - k$ is often called the **corank**.

**PROOF OF PROPOSITION 14.1.1 CONTINUED :** Note that

$$\begin{aligned}\mathrm{Im}(\mathsf{T}) &= \mathcal{L}(\mathsf{T}(\mathbf{C}_1),\ldots,\mathsf{T}(\mathbf{C}_n)) = \mathcal{L}(\mathbf{0},\ldots,\mathbf{0},\mathsf{T}(\mathbf{C}_{m+1}),\ldots,\mathsf{T}(\mathbf{C}_n))\\ &= \mathcal{L}(\mathsf{T}(\mathbf{C}_{m+1}),\ldots,\mathsf{T}(\mathbf{C}_n)),\end{aligned}$$

and hence $\mathsf{T}(\mathbf{C}_{m+1}),\ldots,\mathsf{T}(\mathbf{C}_n)$ span Im(T), so by the lemma they are a basis for Im(T).

Apply the basis extension Theorem 6.3.2 again to choose vectors $\mathbf{D}_1,\ldots,\mathbf{D}_s$ such that $\{\mathbf{D}_1,\ldots,\mathbf{D}_s,\mathsf{T}(\mathbf{C}_{m+1}),\ldots,\mathsf{T}(\mathbf{C}_n)\}$ is a basis for $\mathcal{V}$. Note $s = m$ since $s + n - m = \dim(\mathcal{V}) = n$. Set

$$\begin{array}{lclclcl}
\mathbf{A}_1 & = & \mathbf{C}_{m+1}, & \quad & \mathbf{B}_1 & = & \mathsf{T}(\mathbf{C}_{m+1}),\\
\vdots & & \vdots & & \vdots & & \vdots\\
\mathbf{A}_{n-m} & = & \mathbf{C}_n, & & \mathbf{B}_{n-m} & = & \mathsf{T}(\mathbf{C}_n),\\
\mathbf{A}_{n-m+1} & = & \mathbf{C}_1, & & \mathbf{B}_{n-m+1} & = & \mathbf{D}_1,\\
\vdots & & \vdots & & \vdots & & \vdots\\
\mathbf{A}_n & = & \mathbf{C}_m, & & \mathbf{B}_n & = & \mathbf{D}_s.
\end{array}$$

Then we have

$$\begin{aligned}
\mathsf{T}(\mathbf{A}_1) &= \mathbf{B}_1,\\
\mathsf{T}(\mathbf{A}_2) &= \mathbf{B}_2,\\
&\vdots\\
\mathsf{T}(\mathbf{A}_{n-m}) &= \mathbf{B}_{n-m},\\
\mathsf{T}(\mathbf{A}_{n-m+1}) &= \mathbf{0},\\
&\vdots\\
\mathsf{T}(\mathbf{A}_n) &= \mathbf{0}.
\end{aligned}$$

If we put $k = n - m$, then the matrix of T relative to the basis pair $\{\mathbf{A}_1,\ldots,\mathbf{A}_n\}$, $\{\mathbf{B}_1,\ldots,\mathbf{B}_n\}$ is

$$\begin{array}{c}\left[\begin{array}{cccccc}1 & & & & & \\ & \ddots & & & 0 & \\ & & 1 & & & \\ & & & 0 & & \\ & 0 & & & \ddots & \\ & & & & & 0\end{array}\right]\end{array}\begin{array}{l}\left.\vphantom{\begin{array}{c}1\\ \ddots\\ 1\end{array}}\right\} k\\ \left.\vphantom{\begin{array}{c}0\\ \ddots\\ 0\end{array}}\right\} n-k\end{array}$$

as required. □

For $\mathsf{M} \in \mathrm{Mat}_{m,n}$ we define the **row rank** to be the maximal number of linearly independent rows of the matrix. Likewise, the **column rank** is the maximum number of linearly independent columns of the matrix. The previous proposition implies that these two numbers are the same, something that is not obvious directly.

**COROLLARY 14.1.3:** *Let* $\mathsf{M} \in \mathrm{Mat}_{m,n}$ *be an* $m \times n$ *matrix. Then the row rank of* $\mathsf{M}$ *is equal to the column rank of* $\mathsf{M}$.

**PROOF:** Let $\mathsf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be the linear transformation with matrix $\mathsf{M}$ with respect to the standard bases. The columns of $\mathsf{M}$ span $\mathrm{Im}(\mathsf{T})$, so the number of linearly independent columns is the rank of the linear transformation $\mathsf{T}$. Proposition 14.1.1 implies that the rank of $\mathsf{T}$ is also the number of linearly independent rows of $\mathsf{M}$. $\square$

## 14.2 Eigenvalues and Eigenvectors

It follows from Proposition 14.1.1 that when we study an endomorphism

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$$

of a finite-dimensional vector space through its matrix representatives, we will perhaps learn nothing more useful about $\mathsf{T}$ than its rank if we are free to choose different bases in the domain and range of $\mathsf{T}$. Since the domain and range of $\mathsf{T}$ are the same, it is reasonable, in seeking to force the matrix to reveal more of the structure of $\mathsf{T}$, to demand that we use the *same* basis in both the domain and range of $\mathsf{T}$. If in this way we were to obtain a *diagonal* matrix, then the structure of $\mathsf{T}$ would be completely revealed, as the following discussion shows.

Suppose that $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ is a basis for $\mathcal{V}$ such that the matrix of $\mathsf{T}$ relative to this basis (*used in both domain and range*) is diagonal, say

$$\begin{bmatrix} e_1 & & \mathbf{0} \\ & e_2 & \\ \mathbf{0} & & \ddots \\ & & & e_n \end{bmatrix}.$$

What does this mean? It means that

$$\mathsf{T}(\mathbf{E}_i) = e_i \mathbf{E}_i, \qquad i = 1, \ldots, n.$$

That is, $\mathsf{T}$ is represented by a diagonal matrix if and only if there is a basis $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ for $\mathcal{V}$ and numbers $e_1, \ldots, e_n$ such that

$$\mathsf{T}(\mathbf{E}_i) = e_i \mathbf{E}_i \qquad i = 1, \ldots, n.$$

This discussion suggests that we introduce the following definition:

**DEFINITION:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be an endomorphism of* $\mathcal{V}$. *A number* $e$ *is called an* **eigenvalue of** $\mathsf{T}$ *if there exists a* **nonzero** *vector* $\mathbf{E}$ *such that*

$$\mathsf{T}(\mathbf{E}) = e\mathbf{E}.$$

*Such a vector* $\mathbf{E}$ *is called an* **eigenvector of T associated to the eigenvalue** $e$.

Throughout our study of endomorphisms we will insist that our matrix representatives be constructed using the same basis for both the domain and range. If $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is an endomorphism and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ a basis for $\mathcal{V}$, we will use the phrase "the matrix of $\mathsf{T}$ relative to $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$" (and similar such expressions) to mean the matrix $\mathbf{A} = (a_{i,j})$ representing $\mathsf{T}$ obtained by using the basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ in both the domain and range of $\mathsf{T}$, that is,

$$\mathsf{T}(\mathbf{A}_j) = \sum_{i=1}^{n} a_{i,j}\mathbf{A}_i$$

for $j = 1, \ldots, n$. In this way if $\mathbf{A}_i$ happens to be an eigenvector of $\mathsf{T}$ corresponding to the eigenvalue $e_i$, then

$$\mathsf{T}(\mathbf{A}_i) = 0\mathbf{A}_1 + \cdots + 0\mathbf{A}_{i-1} + e_i\mathbf{A}_i + 0\mathbf{A}_{i+1} + \cdots + 0\mathbf{A}_n,$$

such that in the $i$-th column of $\mathbf{A}$ we will find

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \\ e_i \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad i\text{-th row.}$$

In fact, using this terminology, we have the following important result.

**PROPOSITION 14.2.1:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be an endomorphism of the finite-dimensional vector space* $\mathcal{V}$. *Then* $\mathsf{T}$ *is represented by a diagonal matrix using the same basis in domain and range if and only if* $\mathcal{V}$ *has a basis composed of eigenvectors of* $\mathsf{T}$.

**DEFINITION:** *An endomorphism* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *of the finitedimensional vector space* $\mathcal{V}$ *is said to be* **diagonalizable** *if there exists a basis of* $\mathcal{V}$ *with respect to which* $\mathsf{T}$ *is represented by a diagonal matrix.*

Note that in view of Proposition 14.2.1 an endomorphism $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is diagonalizable if and only if $\mathcal{V}$ has a basis composed of eigenvectors of $\mathsf{T}$. Thus, if $\mathcal{V}$ is $n$-dimensional, $\mathsf{T}$ is diagonalizable if and only if there exist $n$ linearly independent eigenvectors of $\mathsf{T}$. To **diagonalize** $\mathsf{T}$ means to find, if possible, a basis of $\mathcal{V}$ composed of eigenvectors of $\mathsf{T}$, and the matrix of $\mathsf{T}$ with respect to this basis.

Sometimes it is possible to restrict the values of possible eigenvalues, as the following proposition shows.

**PROPOSITION 14.2.2:** *Let* $\mathsf{P} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a projection, that is,* $\mathsf{P}^2 = \mathsf{P}$ *(see Section 12.1). Then the only eigenvalues of* $\mathsf{P}$ *are 0 and 1.*

**PROOF:** Suppose that $e$ is an eigenvalue of $\mathsf{P}$ and $\mathbf{E} \in \mathcal{V}$ a corresponding eigenvector, so that

$$\mathsf{P}(\mathbf{E}) = e\mathbf{E}.$$

Then, computing $\mathsf{P}^2(\mathbf{E})$ in two different ways gives

$$\mathsf{P}^2(\mathbf{E}) = \mathsf{P}(\mathbf{E}) = e\mathbf{E}$$
$$\mathsf{P}^2(\mathbf{E}) = \mathsf{P}(\mathsf{P}(\mathbf{E})) = \mathsf{P}(e\mathbf{E}) = e\mathsf{P}(\mathbf{E}) = e(e\mathbf{E}) = e^2\mathbf{E}.$$

Therefore,

$$e\mathbf{E} = e^2\mathbf{E},$$

or

$$(e^2 - e)\mathbf{E} = \mathbf{0}.$$

Since the vector $\mathbf{E} \neq \mathbf{0}$, this implies

$$0 = e^2 - e = e(e-1).$$

So $e = 0$ or $e = 1$. $\square$

In fact, in Theorem 12.1.1, we even showed that a projection has the diagonal form

$$\begin{bmatrix} \overbrace{\begin{matrix} 1 & & \\ & \ddots & \\ & & 1 \end{matrix}}^{s} & \mathbf{0} \\ \mathbf{0} & \underbrace{\begin{matrix} 0 & & \\ & \ddots & \\ & & 0 \end{matrix}}_{r} \end{bmatrix},$$

where $r = \dim(\ker(\mathsf{P}))$ and $s = \dim(\operatorname{Im}(\mathsf{P}))$.

**EXAMPLE 1:** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation given by

$$\mathsf{T}(x, y) = (x + 2y, 3x + 2y).$$

Find all the eigenvalues and eigenvectors of $\mathsf{T}$.

**SOLUTION:** Note that relative to the standard basis of $\mathbb{R}^2$, $\mathsf{T}$ is represented by the matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$$

which is hardly diagonal.

To discover the eigenvalues and eigenvectors of $\mathsf{T}$, indeed, to determine whether there are any eigenvalues at all, we must look for numbers $e$ and nonzero vectors $\mathbf{E}$ such that $\mathsf{T}(\mathbf{E}) = e\mathbf{E}$. That is, if $\mathbf{E} = (x, y)$,

$$\mathsf{T}(x, y) = (ex, ey).$$

Since by definition

$$\mathsf{T}(x, y) = (x + 2y, 3x + 2y),$$

we must determine when the system of equations

$$x + 2y = ex,$$
$$3x + 2y = ey$$

has a nontrivial solution, i.e., a solution where not both $x$ and $y$ are zero. So we must determine for which values of $e$ the homogeneous linear system

$$(1 - e)x + 2y = 0,$$
$$3x + (2 - e)y = 0$$

has a nontrivial solution. According to Theorem 13.1.3, this system has a nontrivial solution if and only if the matrix

$$\begin{bmatrix} 1-e & 2 \\ 3 & 2-e \end{bmatrix}$$

is **not** invertible. (That is, the linear transformation represented by this matrix is **not** an isomorphism.) But Proposition 10.4.1 says that a $2\times 2$ matrix $\mathbf{A}$ is not an isomorphism precisely when $\det(\mathbf{A}) = 0$. Putting all this together, we find that $e$ is an eigenvalue of $\mathsf{T}$ if and only if

$$\det\begin{bmatrix} 1-e & 2 \\ 3 & 2-e \end{bmatrix} = 0.$$

Evaluating the determinant, we find that

$$\begin{aligned} \det\begin{bmatrix} 1-e & 2 \\ 3 & 2-e \end{bmatrix} &= (1-e)(2-e) - 6 \\ &= 2 - 3e + e^2 - 6 \\ &= e^2 - 3e - 4 \\ &= (e - 4)(e + 1). \end{aligned}$$

Thus the eigenvalues of $\mathsf{T}$ are those numbers $e$ such that

$$0 = (e-4)(e+1),$$

that is,

$$e = 4, \quad e = -1$$

are the eigenvalues of $\mathsf{T}$.

Next we find the eigenvectors of $\mathsf{T}$ associated to 4 and $-1$ respectively. To do so means that we must solve the linear equations

$$\text{for } e = 4: \quad \begin{cases} -3x + 2y &= 0, \\ 3x - 2y &= 0, \end{cases}$$

$$\text{for } e = -1: \quad \begin{cases} 2x + 2y &= 0, \\ 3x + 3y &= 0. \end{cases}$$

The first system has as solution space the linear span of the vector $\mathbf{E} = (2, 3)$, and the second set has as solution space the linear span of $\mathbf{F} = (1, -1)$.

Thus we see the eigenvalues of $\mathsf{T}$ are 4 and $-1$, and all eigenvectors associated to 4 are multiples of $(2, 3)$ and all eigenvectors associated to $-1$ are multiples of $(1, -1)$.

While the preceding example is long and tedious, it is at least partly so because we really do not have enough tools yet to work such examples comfortably. In fact, the example suggests a few useful results.

**PROPOSITION 14.2.3:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the finite-dimensional vector space $\mathcal{V}$. A number $e$ is an eigenvalue of $\mathsf{T}$ if and only if the endomorphism*

$$\mathsf{T} - e\mathsf{I} : \mathcal{V} \longrightarrow \mathcal{V}$$

*is not an isomorphism.*

**PROOF:** Suppose that $e$ is an eigenvalue of $\mathsf{T}$. Let $\mathbf{E}$ be an eigenvector of $\mathsf{T}$ associated to $e$. Then

$$\mathsf{T}(\mathbf{E}) = e\mathbf{E}.$$

Therefore,

$$(\mathsf{T} - e\mathsf{I})(\mathbf{E}) = \mathsf{T}(\mathbf{E}) - e\mathbf{E} = e\mathbf{E} - e\mathbf{E} = \mathbf{0},$$

so $\mathbf{0} \neq \mathbf{E} \in \ker(\mathsf{T} - e\mathsf{I})$. Hence by Proposition 8.6.1 $\mathsf{T} - e\mathsf{I}$ is not an isomorphism. Conversely, suppose that $\mathsf{T} - e\mathsf{I}$ is not an isomorphism. By Proposition 8.6.1 again it follows that either

$$\ker(\mathsf{T} - e\mathsf{I}) \neq \{\mathbf{0}\}$$

or

$$\operatorname{Im}(\mathsf{T} - e\mathsf{I}) \neq \mathcal{V}.$$

From Theorem 8.4.2 we have the equation

$$n = \dim(\ker(\mathsf{T} - e\mathsf{I})) + \dim(\operatorname{Im}(\mathsf{T} - e\mathsf{I})).$$

Hence in either case, $\dim(\ker(\mathsf{T} - e\mathsf{I})) > 0$, that is, we must have

$$\ker(\mathsf{T} - e\mathsf{I}) \neq \{\mathbf{0}\}$$

and

$$\operatorname{Im}(\mathsf{T} - e\mathsf{I}) \neq \mathcal{V}.$$

At any rate, $\ker(\mathsf{T} - e\mathsf{I}) \neq \{\mathbf{0}\}$ and so we may select $\mathbf{E} \in \ker(\mathsf{T} - e\mathsf{I})$, $\mathbf{E} \neq \mathbf{0}$. Then

$$\mathbf{0} = (\mathsf{T} - e\mathsf{I})(\mathbf{E}) = \mathsf{T}(\mathbf{E}) - e(\mathbf{E}),$$

so

$$\mathsf{T}(\mathbf{E}) = e\mathbf{E},$$

and hence $\mathbf{E}$ is an eigenvector of $\mathsf{T}$ associated to the eigenvalue $e$. □

**DEFINITION:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the vector space $\mathcal{V}$. Suppose that $e$ is an eigenvalue of $\mathsf{T}$. Set*

$$\mathcal{V}_e = \left\{\mathbf{E} \mid \mathsf{T}(\mathbf{E}) = e\mathbf{E}\right\}.$$

*$\mathcal{V}_e$ is called the* **eigenspace** *of $\mathsf{T}$ associated to the eigenvalue $e$.*

Thus $\mathcal{V}_e$ is the set of all eigenvectors of $\mathsf{T}$ associated with the eigenvalue $e$, *together with the zero vector*. Note that by definition an eigenspace $\mathcal{V}_e$ always contains a nonzero vector, namely an eigenvector (at least one), and therefore $\dim(\mathcal{V}_e)$ is always positive.

**PROPOSITION 14.2.4:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the vector space $\mathcal{V}$. If $e$ is an eigenvalue of $\mathsf{T}$, then $\mathcal{V}_e = \ker(\mathsf{T} - e\mathsf{I})$. Hence $\mathcal{V}_e$ is a subspace of $\mathcal{V}$.*

**PROOF:** Immediate from the definitions. □

Not every linear transformation has eigenvalues, but even if a linear transformation has lots of eigenvalues, it still may be impossible to represent it by a diagonal matrix. Recall that $\ker(\mathsf{T}) = \left\{\mathbf{E} \in \mathcal{V} \mid \mathsf{T}(\mathbf{E}) = \mathbf{0}\right\}$, and so the number 0 is an eigenvalue of $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ if and only if $\ker(\mathsf{T}) \neq \{\mathbf{0}\}$, in which case $\mathcal{V}_0 = \ker(\mathsf{T})$. The following result shows that there are many nonzero endomorphisms for which 0 is the only eigenvalue.

**PROPOSITION 14.2.5:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be nilpotent. Then the only eigenvalue of* $\mathsf{T}$ *is* 0.

**PROOF:** Since $\mathsf{T}$ is nilpotent there exists an integer $k$ such that $\mathsf{T}^k = \mathbf{0}$. Let $k$ be the smallest such integer. Then $\mathsf{T}^{k-1} \neq \mathbf{0}$, so there is a vector $\mathbf{E} \in \mathcal{V}$ with $\mathsf{T}^{k-1}(\mathbf{E}) \neq \mathbf{0}$. Since

$$\mathsf{T}(\mathsf{T}^{k-1}(\mathbf{E})) = \mathsf{T}^k(\mathbf{E}) = \mathbf{0} = 0\mathbf{E}$$

it follows that $\mathbf{0} \neq \mathsf{T}^{k-1}(\mathbf{E}) \in \ker(\mathsf{T})$, so 0 is an eigenvalue of $\mathsf{T}$, and $\mathcal{V}_0 = \ker(\mathsf{T})$.

Conversely, if $e$ is any eigenvalue of $\mathsf{T}$, then there is a nonzero vector $\mathbf{F} \in \mathcal{V}$ such that

$$\mathsf{T}(\mathbf{F}) = e\mathbf{F}.$$

Then

$$\begin{aligned} \mathsf{T}^2(\mathbf{F}) &= \mathsf{T}(\mathsf{T}(\mathbf{F})) = \mathsf{T}(e\mathbf{F})) = e\mathsf{T}(\mathbf{F}) = ee\mathbf{F} = e^2\mathbf{F}, \\ \mathsf{T}^3(\mathbf{F}) &= \mathsf{T}(\mathsf{T}^2(\mathbf{F})) = e^2\mathsf{T}(\mathbf{F}) = e^3\mathbf{F}, \\ &\vdots \end{aligned}$$

and continuing in this way, we see that

$$\mathsf{T}^k(\mathbf{F}) = e^k\mathbf{F}.$$

But $\mathsf{T}^k = \mathbf{0}$, so

$$\mathsf{T}^k(\mathbf{F}) = \mathbf{0},$$

and therefore

$$e^k\mathbf{F} = \mathbf{0}.$$

Since $\mathbf{F}$ is not the zero vector, this implies that $e^k = 0$, so $e = 0$, as required. □

The only diagonalizable transformation with 0 as its only eigenvalue is the zero transformation. Therefore, the only nilpotent transformation that is diagonalizable is the zero transformation! In a very strong sense this is the reason why not all transformations can be diagonalized. In fact, if complex scalars are used, then any transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ can be written as a sum $\mathsf{D} + \mathsf{N}$, where $\mathsf{D}$ is diagonalizable with the same eigenvalues (and multiplicities; see Section 14.5) as $\mathsf{T}$, and $\mathsf{N}$ is nilpotent. This is the Jordan canonical form, which is the subject of Chapter 17.

In the following discussion it will be convenient to have a name for endomorphisms $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ that are **not** isomorphisms.

**DEFINITION:** *An endomorphism* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *is said to be* **singular** *if* $\mathsf{T}$ *is* **not** *an isomorphism.*

Note that in the course of proving Proposition 14.2.3 we also proved (look at Proposition 8.6.1 again):

**PROPOSITION 14.2.6 :** *Let $\mathcal{V}$ be a finite-dimensional vector space and $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ an endomorphism. Then $\mathsf{T}$ is singular if and only if* $\ker(\mathsf{T}) \neq \{\mathbf{0}\}$. □

**EXAMPLE 2 :** Let $\mathsf{S} : \mathcal{P}(\mathbb{R}) \longrightarrow \mathcal{P}(\mathbb{R})$ be the linear transformation given by

$$\mathsf{S}(p(x)) = xp(x).$$

Then $\mathsf{S}$ is an endomorphism of $\mathcal{P}(\mathbb{R})$. Clearly,

$$\ker(\mathsf{S}) = \{\mathbf{0}\}.$$

On the other hand, $\mathsf{S}$ is singular, since the vector 1 is not in $\mathrm{Im}(\mathsf{S})$. Thus Proposition 14.2.6 becomes false if we do not assume $\mathcal{V}$ to be finite-dimensional. (See also Exercise 10 in Chapter 9.)

**EXAMPLE 3 :** Find the eigenvalues and associated eigenspaces for the linear transformation $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ with matrix

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

relative to the standard basis.

**SOLUTION :** The eigenvalues are those for which

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} - e \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is singular. (Note that since we have insisted on using the same basis in domain and range, the matrix of $\mathsf{I}$ is

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

You should go back and look at Examples 1 and 2 of Section 11.1.) The matrix

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} - e \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-e & 2 \\ 4 & 3-e \end{bmatrix}$$

is singular if and only if its determinant vanishes. So we must have

$$0 = \det \begin{bmatrix} 1-e & 2 \\ 4 & 3-e \end{bmatrix} = (1-e)(3-e) - 8,$$

so

$$\begin{aligned} 0 &= (1-e)(3-e) - 8 = 3 - 4e + e^2 - 8 \\ &= e^2 - 4e - 5 = (e-5)(e+1). \end{aligned}$$

Hence, the eigenvalues of $\mathsf{T}$ must be $e = 5, e = -1$. To find the corresponding eigenspaces of $\mathsf{T}$, we must solve the equations

$$e = 5: \quad \begin{cases} -4x + 2y &= 0, \\ 4x - 2y &= 0, \end{cases}$$

$$e = -1: \quad \begin{cases} 2x + 2y &= 0, \\ 4x + 4y &= 0, \end{cases}$$

so

$$\mathcal{V}_5 = \mathcal{L}((1,2)) = \left\{(x, y) \,|\, 2x - y = 0\right\},$$
$$\mathcal{V}_{-1} = \mathcal{L}((1,-1)) = \left\{(x, y) \,|\, x + y = 0\right\}.$$

**EXAMPLE 4:** Find the eigenvalues and eigenvectors of the linear transformation with matrix

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

relative to the standard basis of $\mathbb{R}^2$.

**SOLUTION:** We must find when

$$\det \begin{bmatrix} -e & 1 \\ -1 & -e \end{bmatrix} = 0.$$

But

$$\det \begin{bmatrix} -e & 1 \\ -1 & -e \end{bmatrix} = e^2 + 1$$

and hence is *never* zero. Therefore, this linear transformation

$$\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

has no eigenvalues.[2]

## 14.3 Determinants

It should be clear from the preceding examples that the problem of calculating the eigenvalues and eigenvectors of a linear transformation

$$\mathsf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

[2] One can view this as a special case of a general **defect** of the real numbers: quadratic equations do not always have roots. One way to correct this defect would be to use complex numbers as scalars. See, e.g., chapter 15.

for $n > 2$ depends on having a systematic method to test when $\mathbf{T} - e\mathbf{I}$ is singular. Such a method exists, although for large $n$ it is not really simple (nor efficient), as it is enormously tedious. It involves extending the determinant to square matrices of size greater than 2. We do this by induction.

**DEFINITION:** *Let* $\mathbf{A}$ *be an* $n \times n$ *matrix. The square matrix obtained from* $\mathbf{A}$ *by deleting the* $i$*th-row and* $j$*th-column of* $\mathbf{A}$ *is called the* **minor of the element** $a_{i,j}$ *of* $\mathbf{A}$, *or the* $(i, j)$**-minor of the matrix** $\mathbf{A}$, *and is denoted by* $\mathbf{M}_{i,j}$.

**EXAMPLE 1:** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ 0 & 4 & 6 \end{bmatrix}.$$

find the minors $\mathbf{M}_{1,2}$, $\mathbf{M}_{2,3}$, $\mathbf{M}_{3,3}$ of $\mathbf{A}$.

**SOLUTION:** We have

$$\mathbf{M}_{1,2} = \begin{bmatrix} 2 & 2 \\ 0 & 6 \end{bmatrix},$$

$$\mathbf{M}_{2,3} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix},$$

$$\mathbf{M}_{3,3} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

**DEFINITION:** *Let* $\mathbf{A}$ *be a square matrix of size* $n$. *Then the* **determinant** *of* $\mathbf{A}$, *denoted by* $\det(\mathbf{A})$ *is defined by the inductive formula*

$$\det(\mathbf{A}) = a \quad \text{if } \mathbf{A} = [a] \text{ is a } 1 \times 1\text{-matrix,}$$

$$\begin{aligned} \det(\mathbf{A}) &= a_{1,1}\det(\mathbf{M}_{1,1}) - a_{1,2}\det(\mathbf{M}_{1,2}) + \cdots + (-1)^{n+1}a_{1,n}\det(\mathbf{M}_{1,n}) \\ &= \sum_{j=1}^{n}(-1)^{1+j}a_{1,j}\det(\mathbf{M}_{1,j}) \quad \text{if } \mathbf{A} \text{ is a square matrix of size } n > 1. \end{aligned}$$

**EXAMPLE 2:** Calculate the determinant of

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ 0 & 4 & 6 \end{bmatrix}.$$

**SOLUTION:** We have

$$\begin{aligned} \det(\mathbf{A}) &= 1 \cdot \det\begin{bmatrix} 1 & 2 \\ 4 & 6 \end{bmatrix} - 0 \cdot \det\begin{bmatrix} 2 & 2 \\ 0 & 6 \end{bmatrix} + 1 \cdot \det\begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix} \\ &= 6 - 8 + 0 + 8 = 6. \end{aligned}$$

**EXAMPLE 3:** Calculate the determinant of

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 2 \\ 0 & 1 & 0 & 4 \\ 1 & 0 & 2 & 1 \end{bmatrix}.$$

**SOLUTION:** We have

$$\begin{aligned}
\det(\mathbf{A}) &= 1\cdot\det\begin{bmatrix} 1 & 4 & 2 \\ 1 & 0 & 4 \\ 0 & 2 & 1 \end{bmatrix} - 2\cdot\det\begin{bmatrix} 0 & 4 & 2 \\ 0 & 0 & 4 \\ 1 & 2 & 1 \end{bmatrix} \\
&\quad -1\cdot\det\begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 4 \\ 1 & 0 & 1 \end{bmatrix} - 3\cdot\det\begin{bmatrix} 0 & 1 & 4 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix} \\
&= 1\cdot\left(\det\begin{bmatrix} 0 & 4 \\ 2 & 1 \end{bmatrix} - 4\cdot\det\begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} + 2\cdot\det\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\right) \\
&\quad -2\cdot\left(0\cdot\det\begin{bmatrix} 0 & 4 \\ 2 & 1 \end{bmatrix} - 4\cdot\det\begin{bmatrix} 0 & 4 \\ 1 & 1 \end{bmatrix} + 2\cdot\det\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}\right) \\
&\quad -\left(0\cdot\det\begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} - 1\cdot\det\begin{bmatrix} 0 & 4 \\ 1 & 1 \end{bmatrix} + 2\cdot\det\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\right) \\
&\quad -3\left(0\cdot\det\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} - \det\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} + 4\cdot\det\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\right) \\
&= (-8-4+4) - 2(0-4(-4)+0) - (0+4-2) - 3(0-0-4) \\
&= -30.
\end{aligned}$$

Hopefully, the preceding example will convince you that the evaluation of determinants of large matrices is a most painful process. However, for small values of $n$, the utility of determinants cannot be denied.

There are a number of properties of determinants that make their computation easier. The verification of these properties requires an excursion into multilinear algebra, which is undertaken in Appendix A. The basic properties of determinants are summarized in properties (1)–(8) below, of which we make free use in the sequel.

**PROPERTIES OF DETERMINANTS:** Let $\mathbf{A}$ be a matrix.

(1) If $\mathbf{A}$ has a row of zeros, then $\det(\mathbf{A}) = 0$.
(2) If $\mathbf{A}$ has two rows equal, then $\det(\mathbf{A}) = 0$.
(3) If $\mathbf{B}$ is obtained from $\mathbf{A}$ by interchanging two rows, then $\det(\mathbf{B}) = -\det(\mathbf{A})$.
(4) If $\mathbf{B}$ is obtained from $\mathbf{A}$ by adding a multiple of one row to a ***different*** row, then $\det(\mathbf{B}) = \det(\mathbf{A})$.
(5) If $\mathbf{B}$ is obtained from $\mathbf{A}$ by multiplying a row of $\mathbf{A}$ by a number $k$, then $\det(\mathbf{B}) = k\det(\mathbf{A})$.

(6) If **A** is upper triangular, then det(**A**) is the product of the diagonal entries.
(7) det(**AB**) = det(**A**) det(**B**).

The final property of the determinant that we need requires a definition.

**DEFINITION:** *The* **transpose** *of a matrix* $\mathbf{A} = (a_{i,j})$ *is the matrix* $\mathbf{A}^{\text{tr}} = (a^{\text{tr}}_{i,j})$, *where* $a^{\text{tr}}_{i,j} = a_{j,i}$.

For example when $n = 4$

$$\begin{bmatrix} 1 & 5 & 9 & 13 \\ 2 & 6 & 10 & 14 \\ 3 & 7 & 11 & 15 \\ 4 & 8 & 12 & 16 \end{bmatrix}^{\text{tr}} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}.$$

The transpose of **A** is the matrix $\mathbf{A}^{\text{tr}}$ whose rows are the columns of **A** (see Chapter 10 Exercises 21 - 25).

(8) $\det(\mathbf{A}^{\text{tr}}) = \det(\mathbf{A})$; hence everything we said about rows works equally well for columns.

Using these properties, we may do Example 3 a little less painfully.

**EXAMPLE 4** (Example 3 revisited): We have

$$\det\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 2 \\ 0 & 1 & 0 & 4 \\ 1 & 0 & 2 & 1 \end{bmatrix} = \det\begin{bmatrix} 0 & 2 & -3 & 2 \\ 0 & 1 & 4 & 2 \\ 0 & 1 & 0 & 4 \\ 1 & 0 & 2 & 1 \end{bmatrix}$$

$$= \det\begin{bmatrix} 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \\ -3 & 4 & 0 & 2 \\ 2 & 2 & 4 & 1 \end{bmatrix} = -\det\begin{bmatrix} 2 & 1 & 1 \\ -3 & 4 & 0 \\ 2 & 2 & 4 \end{bmatrix}$$

$$= -\det\begin{bmatrix} 0 & 0 & 1 \\ -3 & 4 & 0 \\ -6 & -2 & 4 \end{bmatrix} = -(-1)^{1+3}\det\begin{bmatrix} -3 & 4 \\ -6 & -2 \end{bmatrix}$$

$$= -(6 + 24) = -30.$$

**PROPOSITION 14.3.1:** *Let* **A** *be an* $n \times n$ *matrix. Then* **A** *is invertible if and only if* $\det(\mathbf{A}) \neq 0$.

**PROOF:** Suppose that **A** is invertible. Then

$$\mathbf{A} \cdot \mathbf{A}^{-1} = \mathbf{I}.$$

Taking determinants of both sides gives

$$\det(\mathbf{A} \cdot \mathbf{A}^{-1}) = \det(\mathbf{I}).$$

Since $\mathbf{I}$ is upper triangular it follows from property 6 of determinants that

$$\det(\mathbf{I}) = 1.$$

From property (7) we find that

$$\det(\mathbf{A} \cdot \mathbf{A}^{-1}) = \det(\mathbf{A})\det(\mathbf{A}^{-1}).$$

Equating gives

$$\det(\mathbf{A})\det(\mathbf{A}^{-1}) = 1.$$

Hence $\det(\mathbf{A}) \neq 0$, and in fact, $\det(\mathbf{A}^{-1}) = (\det \mathbf{A})^{-1}$. Note the use of $^{-1}$ on the left of this equation indicates the inverse of the *matrix* $\mathbf{A}$ and on the right the inverse of the *number* $\det \mathbf{A}$.

To prove the converse we will actually give a method for constructing $\mathbf{A}^{-1}$. Let us introduce the numbers

$$A_{i,j} := (-1)^{i+j} \det(\mathbf{M}_{i,j}).$$

The number $A_{i,j}$ is called the **cofactor** of $a_{i,j}$. Using this notation we have:

**LEMMA 14.3.2:** $\det(\mathbf{A}) = \sum_j a_{i,j} A_{i,j}$ *for any* $i = 1, 2, \ldots$.

**PROOF:** Let $\mathbf{B}$ be the matrix obtained from $\mathbf{A}$ by interchanging the 1-st and $i$-th rows. Then

$$\det \mathbf{A} = -\det(\mathbf{B}) = -\sum b_{1,j} B_{1,j}.$$

Next note that the minors $\mathbf{M}_{i,j}(\mathbf{B})$ of the $i$-th row of $\mathbf{B}$ differ from the minors $\mathbf{M}_{1,j}(\mathbf{A})$ of the first row of $\mathbf{A}$ in that the first row of $\mathbf{M}_{i,j}(\mathbf{B})$ is the $(i-1)$-st row of $\mathbf{M}_{1,j}(\mathbf{A})$. By interchanging this row with the one below it in $\mathbf{M}_{1,j}(\mathbf{B})$ $i-2$ times, we can put it in the $(i-1)$-st row. Therefore, for the cofactors we find that $B_{1,j} = -A_{i,j}$, and hence $\det(\mathbf{A}) = -\det(\mathbf{B}) = -\sum_j b_{1,j} B_{1,j} = \sum_j a_{i,j} A_{i,j}$. □

**REMARK:** Note that in view of the fact that $\det(\mathbf{A}) = \det(\mathbf{A}^{\mathrm{tr}})$, it follows from this lemma that $\det(\mathbf{A})$ may also be computed from the minors of any column.

**LEMMA 14.3.3:** $\sum a_{i,j} A_{k,j} = 0$ *if* $k \neq i$.

**Proof:** Let $\mathbf{B}$ be the matrix obtained from $\mathbf{A}$ by replacing the $k$-th row of $\mathbf{A}$ by the $i$-th row. Then $\mathbf{B}$ has two rows equal, so $\det(\mathbf{B}) = 0$. On the other hand the cofactors of $\mathbf{A}$ and $\mathbf{B}$ satisfy

$$B_{k,j} = A_{k,j},$$

so using the preceding lemma to evaluate $\det(\mathbf{B})$ by minors of the $k$-th row we find that

$$0 = \det(\mathbf{B}) = \sum b_{k,j} B_{k,j} = \sum a_{i,j} A_{k,j},$$

as required. $\square$

Next we define the matrix $\mathbf{A}^{\text{cof}} = (a^*_{i,j})$, called the **cofactor matrix of** $\mathbf{A}$, by $a^*_{i,j} = A_{j,i}$. (Note carefully the switch in the indices.) We then have:

**Lemma 14.3.4:** $\mathbf{A} \cdot \mathbf{A}^{\text{cof}} = \det(\mathbf{A}) \cdot \mathbf{I}$.

**Proof:** Let $\mathbf{A} \cdot \mathbf{A}^{\text{cof}} = \mathbf{C} = (c_{i,j})$. Then

$$c_{i,k} = \sum_j a_{i,j}\, a^*_{j,k} = \sum_j a_{i,j} A_{k,j},$$

so by Lemma 14.3.2

$$c_{i,k} = \begin{cases} 0 & i \neq k, \\ \det(\mathbf{A}) & i = k, \end{cases}$$

such that $\mathbf{C} = (\det \mathbf{A}) \cdot \mathbf{I}$ as required. $\square$

Returning to the proof of 14.3.1 we suppose that $\det(\mathbf{A}) \neq 0$. Let

$$\mathbf{B} = \frac{1}{\det(\mathbf{A})} \mathbf{A}^{\text{cof}}.$$

Then

$$\mathbf{A} \cdot \mathbf{B} = \frac{1}{\det(\mathbf{A})} \mathbf{A} \cdot \mathbf{A}^{\text{cof}} = \frac{1}{\det(\mathbf{A})} (\det(\mathbf{A}))\mathbf{I} = \mathbf{I},$$

i.e., $\mathbf{B} = \mathbf{A}^{-1}$, as required. $\square$

**Example 5:** Calculate the inverse of the matrix of Example 2.

**SOLUTION:** We must calculate the cofactors of $\mathbf{A}$. We find

$$A_{1,1} = (-1)^{1+1}\det\begin{bmatrix}1 & 2\\ 4 & 6\end{bmatrix} = 6-8 = -2,$$

$$A_{1,2} = (-1)^{1+2}\det\begin{bmatrix}2 & 2\\ 0 & 6\end{bmatrix} = -12,$$

$$A_{1,3} = (-1)^{1+3}\det\begin{bmatrix}2 & 1\\ 0 & 4\end{bmatrix} = 8,$$

$$A_{2,1} = (-1)^{2+1}\det\begin{bmatrix}0 & 1\\ 4 & 6\end{bmatrix} = -(0-4) = 4,$$

$$A_{2,2} = (-1)^{2+2}\det\begin{bmatrix}1 & 1\\ 0 & 6\end{bmatrix} = 6,$$

$$A_{2,3} = (-1)^{2+3}\det\begin{bmatrix}1 & 0\\ 0 & 4\end{bmatrix} = -4,$$

$$A_{3,1} = (-1)^{3+1}\det\begin{bmatrix}0 & 1\\ 1 & 2\end{bmatrix} = -1,$$

$$A_{3,2} = (-1)^{3+2}\det\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix} = 0,$$

$$A_{3,3} = (-1)^{3+3}\det\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix} = 1.$$

Hence

$$\mathbf{A}^{\text{cof}} = \begin{bmatrix}-2 & 4 & -1\\ -12 & 6 & 0\\ 8 & -4 & 1\end{bmatrix},$$

and therefore

$$\mathbf{A}^{-1} = \frac{1}{6}\begin{bmatrix}-2 & 4 & -1\\ -12 & 6 & 0\\ 8 & -4 & 1\end{bmatrix}.$$

**EXAMPLE 6** (compare Section 6.3, Example 6): Determine when the vectors $(r, 1, 1), (1, r, 1), (1, 1, r)$ are a basis for $\mathbb{R}^3$.

**SOLUTION:** Consider the linear transformation

$$\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

that is the linear extension of

$$\mathbf{T}(1, 0, 0) = (r, 1, 1),$$
$$\mathbf{T}(0, 1, 0) = (1, r, 1),$$
$$\mathbf{T}(0, 0, 1) = (1, 1, r).$$

Then $\mathbf{T}$ is an isomorphism if and only if $\{(r, 1, 1), (1, r, 1), (1, 1, r)\}$ is a basis for $\mathbb{R}^3$. The matrix of $\mathbf{T}$ relative to the standard basis of $\mathbb{R}^3$ is

$$\begin{bmatrix} r & 1 & 1 \\ 1 & r & 1 \\ 1 & 1 & r \end{bmatrix},$$

and so $\mathbf{T}$ is invertible if and only if

$$\begin{aligned} 0 \neq \det \begin{bmatrix} r & 1 & 1 \\ 1 & r & 1 \\ 1 & 1 & r \end{bmatrix} &= r \det \begin{bmatrix} r & 1 \\ 1 & r \end{bmatrix} + (-1) \det \begin{bmatrix} 1 & 1 \\ 1 & r \end{bmatrix} + 1 \det \begin{bmatrix} 1 & r \\ 1 & 1 \end{bmatrix} \\ &= r(r^2 - 1) - (r - 1) + (1 - r) \\ &= (r - 1)[r(r + 1) - 2] = (r - 1)(r^2 + r - 2) \\ &= (r - 1)^2(r + 2). \end{aligned}$$

Therefore, $\mathbf{T}$ is invertible if and only if $r \neq 1$ or $-2$, so $\{(r, 1, 1), (1, r, 1), (1, 1, r)\}$ is a basis for $\mathbb{R}^3$ if and only if $r \neq 1$ or $-2$.

As a method for computing, the above procedure is quite systematic but fraught with the perils of arithmetic error.

## 14.4 The Characteristic Polynomial

Let us return to the problem of calculating the eigenvalues of an endomorphism $\mathbf{T} : \mathcal{V} \longrightarrow V$ of the finite-dimensional vector space $\mathcal{V}$. Choose a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ for $\mathcal{V}$ and let $\mathbf{T}$ be represented by the matrix $\mathbf{A} = (a_{i,j})$ in this basis.

**Definition:** *The* **characteristic polynomial** *of* $\mathbf{T}$ *is the polynomial of degree* $n$, $\Delta(t) = \det(\mathbf{A} - t\mathbf{I})$, *where* $t$ *is a variable.*

**Remark:** It might seem as though the characteristic polynomial of $\mathbf{T}$ depends on the choice of basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. To see that this is not so suppose that $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ is another basis for $\mathcal{V}$ and that the matrix of $\mathbf{T}$ is $\mathbf{B}$ relative to this basis. Then by Theorem 11.3.1

$$\mathbf{B} = \mathbf{P}\mathbf{A}\mathbf{P}^{-1}$$

(*since we are using the same bases in domain and range*). Thus

$$\begin{aligned} \mathbf{B} - t\mathbf{I} &= \mathbf{P}\mathbf{A}\mathbf{P}^{-1} - t\mathbf{I} = \mathbf{P}\mathbf{A}\mathbf{P}^{-1} - t\mathbf{P}\mathbf{I}\mathbf{P}^{-1} \\ &= \mathbf{P}(\mathbf{A} - t\mathbf{I})\mathbf{P}^{-1}. \end{aligned}$$

Therefore,

$$\begin{aligned}
\det(\mathbf{B} - t\mathbf{I}) &= \det(\mathbf{P}(\mathbf{A} - t\mathbf{I})\mathbf{P}^{-1}) \\
&= (\det \mathbf{P})(\det(\mathbf{A} - t\mathbf{I}))(\det \mathbf{P}^{-1}) \\
&= (\det \mathbf{P})(\det \mathbf{P}^{-1})(\det(\mathbf{A} - t\mathbf{I})) \\
&= (\det \mathbf{P})(\det \mathbf{P})^{-1}(\det(\mathbf{A} - t\mathbf{I})) \\
&= \det(\mathbf{A} - t\mathbf{I}),
\end{aligned}$$

and it does not matter what basis we use to represent $\mathbf{T}$ by a matrix to compute its characteristic polynomial.

By combining propositions 14.3.1 and 14.2.3 we obtain the important:

**THEOREM 14.4.1** (Criteria for Eigenvalues): *Let* $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be an endomorphism with characteristic polynomial* $\Delta(t)$. *A number* $e$ *is an eigenvalue of* $\mathbf{T}$ *if and only if* $e$ *is a root of* $\Delta(t)$.

**PROOF:** Choose a basis for $\mathcal{V}$, say $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$, and let $\mathbf{A} = (a_{i,j})$ be the matrix of $\mathbf{T}$ relative to this basis.

Suppose that $e$ is an eigenvalue of $\mathbf{T}$. Then by Proposition 14.2.3 the transformation

$$\mathbf{T} - e\mathbf{I} : \mathcal{V} \longrightarrow \mathcal{V}$$

is singular. Its matrix relative to $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is the matrix $\mathbf{A} - e\mathbf{I}$. Since $\mathbf{T} - e\mathbf{I}$ is not an isomorphism the matrix $\mathbf{A} - e\mathbf{I}$ has no inverse (see Proposition 8.6.1). So,

$$\det(\mathbf{A} - e\mathbf{I}) = 0,$$

which says that the number $e$ is a root of the polynomial in $t$

$$\Delta(t) = \det(\mathbf{A} - t\mathbf{I}).$$

To prove the converse, suppose that $e$ is a root of $\Delta(t)$. Then

$$0 = \Delta(e) = \det(\mathbf{A} - e\mathbf{I}).$$

So by Proposition 14.2.6, $\mathbf{A} - e\mathbf{I}$ is singular, and the result follows from Proposition 14.2.3. □

**EXAMPLE 1:** Let $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation given by the formula

$$\mathbf{T}(x, y, z) = (0, x, y).$$

Find the characteristic polynomials of $\mathbf{T}$, $\mathbf{T}^2$, and $\mathbf{T}^3$.

**SOLUTION:** The matrix of $\mathbf{T}$ relative to the standard basis is

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

so those of $\mathbf{T}^2$ and $\mathbf{T}^3$ are:

$$\mathbf{A}^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \mathbf{A}^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore (since we are dealing with more than one endomorphism we will write $\Delta_{\mathbf{S}}(t)$ for the characteristic polynomial of the endomorphism $\mathbf{S}$, indicating the dependency as a subscript),

$$\begin{aligned}
\Delta_{\mathbf{T}}(t) &= \det(\mathbf{A} - t\mathbf{I}) \\
&= \det \begin{bmatrix} -t & 0 & 0 \\ 1 & -t & 0 \\ 0 & 1 & -t \end{bmatrix} = -t^3, \\
\Delta_{\mathbf{T}^2}(t) &= \det(\mathbf{A}^2 - t\mathbf{I}) \\
&= \det \begin{bmatrix} -t & 0 & 0 \\ 0 & -t & 0 \\ 1 & 0 & -t \end{bmatrix} = -t^3, \\
\Delta_{\mathbf{T}^3}(t) &= \det(\mathbf{A}^3 - t\mathbf{I}) \\
&= \det \begin{bmatrix} -t & 0 & 0 \\ 0 & -t & 0 \\ 0 & 0 & -t \end{bmatrix} = -t^3.
\end{aligned}$$

Thus $\mathbf{T}$, $\mathbf{T}^2$, and $\mathbf{T}^3$ all have the same characteristic polynomial.

**EXAMPLE 2:** Find the characteristic polynomial, the eigenvalues, and the eigenspaces of the linear transformation given by

$$\mathbf{T}(x, y, z) = (x - 2z, 0, -2x + 4z).$$

**SOLUTION:** The matrix of $\mathbf{T}$ relative to the standard basis is

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix},$$

and thus the characteristic polynomial of $\mathbf{T}$ is

$$\begin{aligned}
\det(\mathbf{A} - t\mathbf{I}) &= \det \begin{bmatrix} 1-t & 0 & -2 \\ 0 & -t & 0 \\ -2 & 0 & 4-t \end{bmatrix} \\
&= (1-t)(-t)(4-t) - (-2)(-t)(-2) = -t^3 + 5t^2.
\end{aligned}$$

So the characteristic polynomial of $\mathsf{T}$ is $\Delta(t) = -t^3 + 5t^2$. The eigenvalues are $t = 0$, $t = 0$ and $t = 5$. The corresponding eigenspaces are obtained by solving the systems of linear equations

$$t = 0: \quad \begin{cases} x \qquad\qquad -2z = 0, \\ -2x \qquad\quad +4z = 0, \end{cases}$$

$$t = 5: \quad \begin{cases} -4x \qquad\qquad\quad -2z = 0, \\ \qquad\quad -5y \qquad\qquad = 0, \\ -2x \qquad\qquad\quad -z = 0. \end{cases}$$

We find that

$$\dim(\mathcal{V}_0) = 2, \quad \text{spanned by the vectors } (0, 1, 0), (2, 0, 1)$$

$$\dim(\mathcal{V}_5) = 1, \quad \text{spanned by the vector } (1, 0, -2).$$

Thus $\mathbb{R}^3$ has as a basis the vectors

$$\mathbf{F}_1 = (2, 0, 1) \quad \mathbf{F}_2 = (0, 1, 0) \quad \mathbf{F}_3 = (1, 0, -2)$$

which are eigenvectors of $\mathsf{T}$. The matrix of $\mathsf{T}$ relative to this basis is

$$\mathsf{B} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Notice that if we were to use the matrix $\mathsf{B}$ to compute the characteristic polynomial of $\mathsf{T}$, we would get

$$\Delta(t) = \det(\mathsf{B} - t\mathsf{I}) = \det \begin{bmatrix} -t & 0 & 0 \\ 0 & -t & 0 \\ 0 & 0 & 5-t \end{bmatrix}$$

$$= (5-t)(-t)(-t) = -t^3 + 5t^2,$$

just as before.

The procedure we have gone through is called **diagonalizing** the linear transformation $\mathsf{T}$. ***Warning:*** Not every linear transformation can be diagonalized! Go back and stare at Example 4 in Section 14.2.

**EXAMPLE 3:** Diagonalize, if possible, the linear transformation

$$\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$$

given by

$$\mathsf{T}(x, y, z, w) = (x, 2x + 5y + 6z + 7w, 3x + 8z + 9w, 4x + 10w)$$

**SOLUTION :** The matrix of $\mathsf{T}$ is computed first. We have

$$\mathsf{T}(1, 0, 0, 0) = (1, 2, 3, 4),$$
$$\mathsf{T}(0, 1, 0, 0) = (0, 5, 0, 0),$$
$$\mathsf{T}(0, 0, 1, 0) = (0, 6, 8, 0),$$
$$\mathsf{T}(0, 0, 0, 1) = (0, 7, 9, 10).$$

So the matrix of $\mathsf{T}$ relative to the standard basis is

$$\mathsf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 5 & 6 & 7 \\ 3 & 0 & 8 & 9 \\ 4 & 0 & 0 & 10 \end{bmatrix}.$$

The characteristic polynomial of $\mathsf{T}$ is thus

$$\Delta(t) = \det(\mathsf{A} - t\mathsf{I}) = \det \begin{bmatrix} 1-t & 0 & 0 & 0 \\ 2 & 5-t & 6 & 7 \\ 3 & 0 & 8-t & 9 \\ 4 & 0 & 0 & 10-t \end{bmatrix}$$
$$= (1-t)\det \begin{bmatrix} 5-t & 6 & 7 \\ 0 & 8-t & 9 \\ 0 & 0 & 10-t \end{bmatrix}$$

(by property 6 of determinants)

$$= (1-t)(5-t)(8-t)(10-t).$$

Thus the eigenvalues of $\mathsf{T}$ are

$$t = 1, 5, 8, 10.$$

The eigenspaces are computed by solving systems of linear equations.

$$t = 1 : \quad \begin{cases} 2x + 4y + 6z + 7w = 0, \\ 3x + 7z + 9w = 0, \\ 4x + 9w = 0, \end{cases}$$

which may be solved as follows: the last equation gives $-4x = 9w$, which we use to eliminate $w$ from the middle equation to obtain

$$-x + 7z = 0.$$

If we use this to eliminate $w$ and $z$ from the first equation, we are left with

$$(2 + \frac{6}{7} - \frac{28}{9})x + 4y = 0,$$

from which it follows that

$$\begin{aligned} w &= -\frac{4}{9}x, \\ z &= \frac{1}{7}x, \\ y &= -\frac{4}{63}x. \end{aligned}$$

Hence setting $x = 63$, we find that

$$\mathcal{V}_1 = \operatorname{Span}\left\{(63, -4, 9, -28)\right\}.$$

For the remaining eigenvalues we have

$$t = 5 : \quad \left\{ \begin{array}{rcrcrcrcl} -4x & & & & & & & = & 0, \\ 2x & & & + & 6z & + & 7w & = & 0, \\ 3x & & & + & 3z & + & 9w & = & 0, \\ 4x & & & & & + & 5w & = & 0, \end{array} \right.$$

which has as solutions

$$x = z = w = 0, \qquad y = \text{anything},$$

so

$$\mathcal{V}_5 = \operatorname{Span}\left\{(0, 1, 0, 0)\right\},$$

and

$$t = 8 : \quad \left\{ \begin{array}{rcrcrcrcl} -7x & & & & & & & = & 0, \\ 2x & + & -3y & + & 6z & + & 7w & = & 0, \\ 3x & & & & & + & 9w & = & 0, \\ 4x & & & & & + & 2w & = & 0, \end{array} \right.$$

so

$$x = 0 = w \quad \text{and} \quad y = 2z,$$

whence

$$\mathcal{V}_8 = \operatorname{Span}\left\{(0, 2, 1, 0)\right\},$$

and finally,

$$t = 10 : \quad \left\{ \begin{array}{rcrcrcrcl} -9x & & & & & & & = & 0, \\ 2x & - & 5y & + & 6z & + & 7w & = & 0, \\ 3x & & & - & 2z & + & 9w & = & 0, \\ 4x & & & & & & & = & 0, \end{array} \right.$$

so

$$\begin{aligned} x &= 0, \\ z &= \frac{9}{2}w, \\ 5y = 7w + 6z &= 7w + 6 \cdot \frac{(9)}{2}w = 34w, \quad \text{so} \quad y = \frac{34}{5}w, \end{aligned}$$

and thus

$$\mathcal{V}_{10} = \text{Span}\left\{(0, 68, 45, 10)\right\}.$$

The vectors

$$\mathbf{F}_1 = (63, -4, 9, -28),$$
$$\mathbf{F}_2 = (0, 1, 0, 0),$$
$$\mathbf{F}_3 = (0, 2, 1, 0),$$
$$\mathbf{F}_4 = (0, 68, 45, 10)$$

may be shown to be linearly independent. (See, for example, Proposition 14.4.2 below.) Thus $\mathbb{R}^4$ has as a basis the eigenvectors $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4\}$ of $\mathsf{T}$. The matrix of $\mathsf{T}$ relative to these vectors is

$$\mathsf{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}.$$

**PROPOSITION 14.4.2:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of $\mathcal{V}$. Suppose $e_1, \ldots, e_m$ are distinct eigenvalues of $\mathsf{T}$ and $\mathbf{F}_1, \ldots, \mathbf{F}_m$ are corresponding eigenvectors. Then $\mathbf{F}_1, \ldots, \mathbf{F}_m$ are linearly independent.*

**PROOF:** Suppose to the contrary that $\mathbf{F}_1, \ldots, \mathbf{F}_m$ are linearly dependent. Applying Proposition 6.2.1 we may find an integer $k$ such that $\mathbf{F}_k$ is linearly dependent on $\mathbf{F}_1, \ldots, \mathbf{F}_{k-1}$. We may also suppose that $k$ is the smallest integer with this property and so by Proposition 6.2.1 again we may suppose that $\mathbf{F}_1, \ldots, \mathbf{F}_{k-1}$ are linearly independent. Let

① $$\mathbf{F}_k = a_1\mathbf{F}_1 + a_2\mathbf{F}_2 + \cdots + a_{k-1}\mathbf{F}_{k-1}.$$

Apply $\mathsf{T}$ to this equation to get

② $$\mathsf{T}(\mathbf{F}_k) = a_1\mathsf{T}(\mathbf{F}_1) + a_2\mathsf{T}(\mathbf{F}_2) + \cdots + a_{k-1}\mathsf{T}(\mathbf{F}_{k-1}).$$

Since the $\mathbf{F}_1, \ldots, \mathbf{F}_k$ are eigenvectors of $\mathsf{T}$ associated to the eigenvalues $e_1, \ldots, e_k$, yields, this

③ $$e_k\mathbf{F}_k = e_1a_1\mathbf{F}_1 + e_2a_2\mathbf{F}_2 + \cdots + e_{k-1}a_{k-1}\mathbf{F}_{k-1}.$$

Multiply equation ① by $e_k$ to obtain

④ $$e_k\mathbf{F}_k = e_ka_1\mathbf{F}_1 + e_ka_2\mathbf{F}_2 + \cdots + e_ka_{k-1}\mathbf{F}_{k-1}$$

and subtract equation ④ from equation ③, giving

⑤ $$\mathbf{0} = a_1(e_1 - e_k)\mathbf{F}_1 + a_2(e_2 - e_k)\mathbf{F}_2 + \cdots + a_{k-1}(e_{k-1} - e_k)\mathbf{F}_{k-1}.$$

Since the vectors $\mathbf{F}_1, \ldots, \mathbf{F}_{k-1}$ are linearly independent, it follows that

⑥
$$\begin{aligned} a_1(e_1 - e_k) &= 0, \\ a_2(e_2 - e_k) &= 0, \\ &\vdots \\ a_{k-1}(e_{k-1} - e_k) &= 0. \end{aligned}$$

Next we recall that the eigenvalues $e_1, \ldots, e_k$ are distinct, so that

⑦
$$\begin{aligned} (e_1 - e_k) &\neq 0, \\ &\vdots \\ (e_{k-1} - e_k) &\neq 0, \end{aligned}$$

and combining ⑥ and ⑦, we find that

⑧
$$\begin{aligned} a_1 &= 0, \\ &\vdots \\ a_{k-1} &= 0. \end{aligned}$$

Going back to equation ①, this says that

⑨
$$\mathbf{F}_k = \mathbf{0}.$$

But an eigenvector is by definition *nonzero*, and hence our assumption that $\{\mathbf{F}_1, \ldots, \mathbf{F}_m\}$ is linearly dependent has led to a contradiction. Therefore, $\{\mathbf{F}_1, \ldots, \mathbf{F}_m\}$ is linearly independent. $\square$

If you think that the proof of Proposition 14.4.2 is hard, compare it with the pain of proving directly that the vectors $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4$ of Example 3 are linearly independent.

Since the characteristic polynomial of an endomorphism $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ of a vector space $\mathcal{V}$ of dimension $n$ is of degree $n$, we find the following useful diagonalization result.

**COROLLARY 14.4.3:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the $n$ dimensional vector space $\mathcal{V}$. Suppose that the characteristic polynomial of $\mathsf{T}$ has $n$ distinct real roots. Then there is a basis of $\mathcal{V}$ composed of eigenvectors of $\mathsf{T}$. Relative to this basis the matrix of $\mathsf{T}$ is diagonal.*

**PROOF:** Let $e_1, \ldots, e_n$ be the eigenvalues of $\mathsf{T}$. There are $n$ of them because the characteristic polynomial has degree $n$. By assumption they are distinct. Choose eigenvectors $\mathbf{F}_1, \ldots, \mathbf{F}_n$ corresponding to $e_1, \ldots, e_n$. By Proposition 14.4.2 the vectors $\mathbf{F}_1, \ldots, \mathbf{F}_n$ are linearly independent. Therefore by Theorem 6.2.4, $\{\mathbf{E_1}, \ldots, \mathbf{E_n}\}$ is a basis for $\mathcal{V}$. $\square$

## 14.5 Diagonalization Theorems

It is natural to inquire what the situation is when the characteristic polynomial does not factor into distinct linear factors. In general, this can mean a quite complicated structure for the endomorphism $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$. There are, however, a few general results we can deduce. We begin with a definition.

**DEFINITION:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the finite-dimensional vector space $\mathcal{V}$ and $e$ an eigenvalue of $\mathsf{T}$. The* **geometric multiplicity** *of $e$ is the dimension of the eigenspace $\mathcal{V}_e$. The* **algebraic multiplicity** *of $e$ is the multiplicity of the root $e$ of the characteristic polynomial $\Delta(t)$ of $\mathsf{T}$.*

**PROPOSITION 14.5.1:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the finite-dimensional vector space $\mathcal{V}$ and $e$ an eigenvalue of $\mathsf{T}$. Then the geometric multiplicity of $e$ is less than or equal to the algebraic multiplicity of $e$.*

**PROOF:** Recall from the remark preceding Theorem 14.4.1 that we may calculate the characteristic polynomial of $\mathsf{T}$ by representing $\mathsf{T}$ with a matrix relative to any basis of $\mathcal{V}$. Let us cleverly choose such a basis for $\mathcal{V}$.

Suppose $\{\mathbf{A}_1, \ldots, \mathbf{A}_r\}$ a basis for $\mathcal{V}_e$, so that the geometric multiplicity of $e$ is equal to $r$. Let $n$ be the dimension of $\mathcal{V}$ and extend $\{\mathbf{A}_1, \ldots, \mathbf{A}_r\}$ to a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_r, \mathbf{A}_{r+1}, \ldots, \mathbf{A}_n\}$ of $\mathcal{V}$. The matrix of $\mathsf{T}$ relative to this basis is easily seen to be

$$\mathsf{A} = \begin{bmatrix} \begin{matrix} e & & 0 \\ & \ddots & \\ 0 & & e \end{matrix} & \mathsf{B} \\ \mathbf{0} & \mathsf{C} \end{bmatrix}$$

where $\mathsf{B}$ is an $r \times n - r$ matrix, $\mathsf{C}$ is an $n - r \times n - r$ matrix and $\mathbf{0}$ an $n - r \times r$ matrix of zeros.

The characteristic polynomial of $\mathbf{T}$ is therefore

$$\Delta(t) = \det(\mathbf{A} - t\mathbf{I}) = \det \begin{bmatrix} e-t & & & & \mathbf{B} \\ & e-t & & & \\ & & \ddots & & \\ & & & e-t & \\ & \mathbf{0} & & & \mathbf{C} - t\mathbf{I} \end{bmatrix},$$

and since $e - t$ is a factor of each of the first $r$ columns, we find that

$$\Delta(t) = (e-t)^r \det \begin{bmatrix} \mathbf{I} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} - t\mathbf{I} \end{bmatrix} = (e-t)^r \det(\mathbf{C} - t\mathbf{I}),$$

and therefore $e$ is a root of $\Delta(t) = 0$ with multiplicity at least $r$, which shows that the algebraic multiplicity of $e$ is at least $r$, and the result follows. $\square$

**PROPOSITION 14.5.2:** *If* $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *is a diagonalizable linear transformation, then*

$$\Delta(t) = (t - e_1)^{m_1} \cdots (t - e_k)^{m_k}$$

*and*

$$\dim(\mathcal{V}_{e_1}) = m_i, \qquad i = 1, 2, \ldots, k,$$

*where* $e_1, \ldots, e_k$ *are the distinct eigenvalues of* $\mathbf{T}$ *and* $m_1, \ldots, m_k$ *their multiplicities (either algebraic or geometric, since in this case they are equal).*

**PROOF:** Since $\mathbf{T}$ is diagonalizable, it has a matrix representation

$$\begin{array}{c@{}l} \begin{bmatrix} e_1 & & & & & & & & & & \\ & \ddots & & & & 0 & & & & & \\ & & e_1 & & & & & & & & \\ & & & e_2 & & & & & & & \\ & & & & \ddots & & & & & & \\ & & & & & e_2 & & & & & \\ & & & & & & \ddots & & & & \\ 0 & & & & & & & \ddots & & 0 & \\ & & & & & & & & \ddots & & \\ & & & & & & & & & e_k & \\ & & & 0 & & & & & & & \ddots & \\ & & & & & & & & & & & e_k \end{bmatrix} & \begin{array}{l} \left.\vphantom{\begin{matrix} a \\ a \\ a \end{matrix}}\right\} n_1 \\ \left.\vphantom{\begin{matrix} a \\ a \\ a \end{matrix}}\right\} n_2 \\ \vphantom{\begin{matrix} a \\ a \\ a \end{matrix}} \\ \left.\vphantom{\begin{matrix} a \\ a \\ a \end{matrix}}\right\} n_k \end{array} \end{array}$$

where $n_i = \dim(\mathcal{V}_{e_i})$ for $i = 1, \ldots, k$. Using this matrix representation to compute the characteristic polynomial of $\mathsf{T}$ we get

$\Delta(t) =$

$$\det \begin{bmatrix} e_1-t & & & & & & & & & \\ & \ddots & & & & \mathbf{0} & & & & \\ & & e_1-t & & & & & & & \\ & & & e_2-t & & & & & & \\ & & & & \ddots & & & & & \\ & & & & & e_2-t & & & & \\ \mathbf{0} & & & & & & \ddots & & \mathbf{0} & \\ & & & & & & & \ddots & & \\ & & & & & & & & e_k-t & \\ & & & \mathbf{0} & & & & & \ddots & \\ & & & & & & & & & e_k-t \end{bmatrix} \quad \begin{matrix} \left.\right\} n_1 \\ \left.\right\} n_2 \\ \vdots \\ \left.\right\} n_k \end{matrix}$$

$$= (e_1 - t)^{n_1} \cdots (e_k - t)^{n_k}$$

$$= (-1)^n [(t - e_1)^{n_1} \cdots (t - e_k)^{n_k}]$$

and therefore $n_1 = m_1, \ldots, n_k = m_k$, as required. □

We may summarize all this in the following:

**THEOREM 14.5.3:** *A linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ of the finite-dimensional vector space $\mathcal{V}$ is diagonalizable if and only if the following two conditions hold:*

*(i) The characteristic polynomial $\Delta(t)$ is a product of linear factors.*

*(ii) For each eigenvalue of $\mathsf{T}$ the geometric and algebraic multiplicities are the same.* □

At this point it is worthwhile to pause to reflect on the difference between the cases of real and complex scalars. (For a discussion of vector spaces with complex scalars see Chapters 2, 3 and 6.) If we look at $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ given by

$$\mathsf{T}(x, y) = (y, -x),$$

which is a rotation through 90° we find that the matrix of $\mathsf{T}$ relative to the usual basis of $\mathbb{R}^2$ is

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

so that

$$\Delta(t) = \det \begin{bmatrix} -t & 1 \\ -1 & -t \end{bmatrix} = t^2 + 1,$$

and of course as we all know

$$t^2 + 1 = 0$$

has no real roots. So this simple transformation of $\mathbb{R}^2$ has no real eigenvalues. On the other hand, it does have two distinct complex roots, namely $\pm \boldsymbol{i} = \pm\sqrt{-1} \in \mathbb{C}$. This means that the transformation

$$\mathbf{S} : \mathbb{C}^2 \longrightarrow \mathbb{C}^2$$

given by

$$\mathbf{S}(u, v) = (v, -u),$$

which has characteristic polynomial

$$\Delta(t) = t^2 + 1,$$

***can*** be diagonalized (by the complex analog of Proposition 14.4.2, since the characteristic polynomial is a product of linear factors and the eigenvalues are distinct), while the transformation $\mathbf{T}$ with the same (matrix and) characteristic polynomial does not have even a single eigenvalue. In fact, an endomorphism $\mathbf{S} : \mathcal{V} \longrightarrow \mathcal{V}$ of a (finite-dimensional) vector space over the complex number field $\mathbb{C}$ must ***always*** have an eigenvalue. Thus it has at least one eigenvector. To see this, recall the following:

**THEOREM 14.5.4** (Fundamental Theorem of Algebra): *Suppose that $p(t) = a_n t^n + \cdots + a_1 t + a_0$ is a polynomial of degree $n$ (that is, $a_n \neq 0$) with complex coefficients. Then there are complex numbers $r_1, \ldots, r_n$ (unique up to change of order) such that $p(t) = a_n(t - r_1)(t - r_2) \cdots (t - r_n)$.*

Therefore, the first condition of Theorem 14.5.3 for a linear transformation to be diagonalizable is always satisfied in the complex case.

**EXAMPLE 1:** Find the characteristic polynomial, eigenvalues, and eigenvectors of the endomorphism

$$\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

given by

$$\mathbf{T}(x, y, z) = (2x + y,\ y - z,\ 2y + 4z).$$

**SOLUTION:** We begin by calculating the matrix of $\mathbf{T}$ relative to the standard basis of $\mathbb{R}^3$. We find

$$\mathbf{T}(1, 0, 0) = (2, 0, 0),$$

$$\mathbf{T}(0, 1, 0) = (1, 1, 2),$$

$$\mathbf{T}(0, 0, 1) = (0, -1, 4),$$

and so the matrix we seek is

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix}.$$

The characteristic polynomial of $\mathbf{T}$ is therefore

$$\begin{aligned} \Delta(t) &= \det \begin{bmatrix} 2-t & 1 & 0 \\ 0 & 1-t & -1 \\ 0 & 2 & 4-t \end{bmatrix} \\ &= (2-t)\det \begin{bmatrix} 1-t & -1 \\ 2 & 4-t \end{bmatrix} \\ &= (2-t)[(1-t)(4-t)+2] = (2-t)[4-5t+t^2+2] \\ &= (2-t)[t^2-5t+6] = (2-t)(t-2)(t-3) \\ &= -(t-2)^2(t-3). \end{aligned}$$

So the eigenvalues of $\mathbf{T}$ are 2 and 3, with 2 having algebraic multiplicity 2.

To calculate the eigenspaces we must solve the linear systems corresponding to the eigenvalues.

$$e = 2: \quad \begin{cases} x + y + 0z = 0, \\ 0x - y - z = 0, \\ 0x + 2y + 2z = 0, \end{cases}$$

which yields

$$x = \text{anything}, \qquad y = 0 \qquad z = 0,$$

and therefore $\mathcal{V}_2 = \mathcal{L}(1, 0, 0)$. That is, the eigenspace associated to the eigenvalue 2 is 1-dimensional and is spanned by the vector $(1, 0, 0)$.

$$e = 3: \quad \begin{cases} -x + y + 0z = 0, \\ 0x - 2y - z = 0, \\ 0x + 2y + z = 0. \end{cases}$$

which yields

$$x - y = 0, \qquad 2y + z = 0,$$

so

$$x = y \qquad z = -2y$$

and therefore the eigenspace corresponding to 3 is spanned by the vector $(1, 1, -2)$; that is, $\mathcal{V}_3 = \mathcal{L}(1, 1, -2)$. Notice that while the eigenvalue 2 has algebraic multiplicity 2, it has geometric multiplicity only 1. It is therefore ***not*** possible to find a basis of $\mathbb{R}^3$ composed of eigenvectors of $\mathbf{T}$, and hence $\mathbf{T}$ is not diagonalizable.

**EXAMPLE 2:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be defined by

$$\mathsf{T}(x, y, z) = (0, x, y).$$

Determine whether $\mathsf{T}$ can be diagonalized.

**SOLUTION:** The matrix of $\mathsf{T}$ relative to the standard basis of $\mathbb{R}^3$ is

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

so the characteristic polynomial of $\mathsf{T}$ is

$$\Delta(t) = \det \begin{bmatrix} -t & 0 & 0 \\ 1 & -t & 0 \\ 0 & 1 & -t \end{bmatrix} = -t^3$$

and the only eigenvalue of $\mathsf{T}$ is 0, with algebraic multiplicity 3. The geometric multiplicity cannot also be 3, because $\mathcal{V}_0 = \ker(\mathsf{T})$, and $3 = \dim(\mathcal{V}_0) = \dim(\ker(\mathsf{T}))$ implies $\mathsf{T} = \mathbf{0}$. But certainly $\mathsf{T} \neq \mathbf{0}$. In fact, the kernel of $\mathsf{T}$ is the subspace spanned by $(0, 0, 1)$, so the geometric multiplicity of the eigenvalue 0 is 1.

**EXAMPLE 3:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be defined by

$$\mathsf{T}(x, y, z) = (-x, -z, y)$$

and let $\mathsf{S} : \mathbb{C}^3 \longrightarrow \mathbb{C}^3$ be defined by the same "formula" but as a linear transformation in a complex vector space, namely,

$$\mathsf{S}(u, v, w) = (-u, -w, v).$$

Determine whether $\mathsf{T}$ or $\mathsf{S}$ can be diagonalized, and if so find their diagonal forms.

**SOLUTION:** At least part of the problem can be worked simultaneously for both $\mathsf{T}$ and $\mathsf{S}$ because relative to the standard bases of $\mathbb{R}^3$ and $\mathbb{C}^3$ respectively both $\mathsf{T}$ and $\mathsf{S}$ have the same matrix

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix},$$

so their characteristic polynomials are the same, namely,

$$\Delta(t) = \det \begin{bmatrix} -1-t & 0 & 0 \\ 0 & -t & -1 \\ 0 & 1 & -t \end{bmatrix} = -(t+1)(t^2+1).$$

Now the difference between the real and complex cases appears!

$$\Delta(t) = -(t+1)(t^2+1)$$

has only one real root, so **T** does not diagonalize (though geometrically it is quite simple: reflection across the $yz$-plane followed by a 90° rotation around the $x$-axis) but **S** does, since

$$\Delta(t) = -(t+1)(t-\mathbf{i})(t+\mathbf{i})$$

has three distinct eigenvalues, $-1, \mathbf{i}, -\mathbf{i}$. The diagonal form of **S** is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \mathbf{i} & 0 \\ 0 & 0 & -\mathbf{i} \end{bmatrix}.$$

The corresponding basis of eigenvectors may be computed as follows: For $e_1 = -1$:

$$\mathbf{S}(u, v, w) = -(u, v, w) = (-u, -v, -w),$$

so

$$(-u, -w, v) = (-u, -v, -w),$$

and passing to components gives

$$-w = -v \quad \text{and} \quad v = -w,$$

so

$$w = v = 0,$$

and therefore a basis for $\mathcal{V}_{-1}$ is $(1, 0, 0)$. For $e_2 = \mathbf{i}$:

$$\mathbf{S}(u, v, w) = \mathbf{i}(u, v, w),$$

so

$$(-u, -w, v) = (\mathbf{i}u, \mathbf{i}v, \mathbf{i}w).$$

Taking components gives

$$-u = \mathbf{i}u \qquad -w = \mathbf{i}v \qquad v = \mathbf{i}w$$

so

$$u = 0, \qquad v = \mathbf{i}, \qquad w = 1.$$

Thus a basis for $\mathcal{V}_{\mathbf{i}}$ is $(0, \mathbf{i}, 1)$. For $e_3 = -\mathbf{i}$:

$$\mathbf{S}(u, v, w) = -\mathbf{i}(u, v, w),$$

so, as before passing to components gives

$$-u = -\mathbf{i}u \qquad -w = -\mathbf{i}v \qquad v = -\mathbf{i}w$$

and

$$u = 0, \qquad v = \mathbf{i}, \qquad w = -1.$$

Thus a basis for $\mathcal{V}_{-i}$ is $(0, \mathbf{i}, -1)$.

## 14.6 Exercises

**1.** Find the rank of each of the following linear transformations:

$$\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \qquad \mathsf{T}(x, y, z) = (x + y, y, z + x),$$

$$\mathsf{S} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4, \qquad \mathsf{S}(x, y, z, w) = (x - y, z - w, w + x, x - w),$$

$$\mathsf{R} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_3(\mathbb{R}), \qquad \mathsf{R}(p(x)) = x\frac{d}{dx}p(x),$$

$$\mathsf{Q} : \mathcal{P}_3(\mathbb{R}) \longrightarrow \mathcal{P}_3(\mathbb{R}), \qquad \mathsf{Q}(p(x)) = \frac{d}{dx}(xp(x)).$$

**2.** Find the cofactor of $a_{2,4}$ and $a_{4,2}$ in the matrix

$$\begin{bmatrix} 1 & 7 & 7 & 1 & 7 & 7 \\ 2 & 8 & 6 & 2 & 8 & 6 \\ 3 & 9 & 5 & 3 & 9 & 5 \\ 4 & 10 & 4 & 4 & 10 & 4 \\ 5 & 9 & 3 & 5 & 9 & 3 \\ 6 & 8 & 2 & 6 & 8 & 2 \end{bmatrix}.$$

**3.** Find the determinant of each of the following matrices

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 & 3 & 4 & 0 \\ 0 & 2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & 5 \\ 0 & 2 & 0 & 4 & 0 \\ 1 & 0 & 3 & 4 & 0 \end{bmatrix}.$$

**4.** Determine which of the following matrices are invertible. For those that are, compute the inverses.

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix},$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \\ 2 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \\ 3 & 0 & 4 & 5 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 1 & -1 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

**5.** Can you find a $2 \times 2$ matrix $\mathsf{A}$ such that $\mathsf{A} = \mathsf{A}^{\text{cof}}$?

**6.** Find the characteristic polynomial, eigenvalues, and eigenvectors, and diagonalize *if possible* each of the following linear transformations:

$(a)$ $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, $\mathsf{T}(x, y, z) = (x + 2y, 2y + x, z)$.

$(b)$ $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, $\mathsf{T}(x, y, z) = (3x - z, y - x + 2z, 4z)$.

$(c)$ $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$, $\mathsf{T}(x, y, z, w) = (x + y, x, z + 2w, 2z + w)$.

$(d)$ $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$, $\mathsf{T}(x, y, z, w) = (x + y, y, 2z, y + 2z + 2w)$.

$(e)$ $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$, $\mathsf{T}(x, y) = (y, -x)$.

$(f)$ $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$, $\mathsf{T}(x, y) = (y, -x, z)$.

**7.** Prove that if a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is nilpotent, then all its eigenvalues are zero.

**8.** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation. Prove that $\mathsf{T}$ has at least one eigenvalue.

**9.** Let $\mathsf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a linear transformation. Suppose that $\mathsf{T}^2$ has a positive eigenvalue. Show that $\mathsf{T}$ has a real eigenvalue.

**10.** Let $\mathsf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ be a linear transformation. If $\mathsf{T} - \mathsf{I}$ is invertible, show that 1 is not an eigenvalue of $\mathsf{T}$.

**11.** Suppose that $e$ is an eigenvalue of $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ with corresponding eigenvector $\mathbf{E}$. Prove that $\mathbf{E}$ is an eigenvector of $\mathsf{T}^k$ corresponding to the eigenvalue $e^k$.

**12.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in a finite-dimensional vector space and $k \in \mathbb{N}$ a positive integer. If $\mathcal{U}$ is an eigenspace of $\mathbf{T}^k$, show that $\mathbf{T}(\mathcal{U}) \subseteq \mathcal{U}$.

**13.** Suppose that $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is nonsingular. What relation is there between the eigenvalues of $\mathbf{T}$ and $\mathbf{T}^{-1}$?

**14.** Give an example of a linear transformation $\mathbf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$ without any eigenvalues. Repeat the problem for $\mathbf{T} : \mathbb{R}^{2n} \longrightarrow \mathbb{R}^{2n}$ for $n \in \mathbb{N}$.

**15.** Suppose that $\mathbf{T} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ is a linear transformation with distinct eigenvalues $e_1, \ldots, e_n$. Show that $\det(\mathbf{T}) = e_1 \cdots e_n$.

**16.** Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a diagonalizable endomorphism. Show that $\mathbf{T}^2$ is also diagonalizable. If the diagonal form of $\mathbf{T}$ is

$$\begin{bmatrix} e_1 & & & & 0 \\ & \ddots & & & \\ & & e_i & & \\ & & & \ddots & \\ 0 & & & & e_n \end{bmatrix}.$$

What is the diagonal form of $\mathbf{T}^2$?

**17.** Let $\mathbf{A} = (a_{i,j})$, $i, j = 1, 2, 3$. Then

$$\det(\mathbf{A}) = a_{1,1}\det(\mathbf{M}_{1,1}) - a_{1,2}(\det \mathbf{M}_{1,2}) + a_{1,3}(\det \mathbf{M}_{1,3}),$$

where $\mathbf{M}_{1,j}$, $j = 1, 2, 3$ are minors of the element $a_{1,j}$ of $\mathbf{A}$. Show that[3]

$$\begin{aligned} \det \mathbf{A} = {} & a_{1,1}a_{2,2}a_{3,3} + a_{1,2}a_{2,3}a_{3,1} + a_{1,3}a_{2,1}a_{3,2} - a_{1,3}a_{2,2}a_{3,1} \\ & - a_{1,2}a_{2,1}a_{3,3} - a_{1,1}a_{2,3}a_{3,2}. \end{aligned}$$

**18.** Using the above expression of $\det(\mathbf{A})$ for a $3 \times 3$ matrix, show that properties (1)-(8) of $\det(\mathbf{A})$ hold for $3 \times 3$ matrices.

**19.** Let

$$\mathbf{A} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}.$$

Show that $\det(\mathbf{A}) \neq 0$ if and only if the column vectors form a set of linearly independent vectors. (HINT: Consider $\mathbf{A}$ to be the matrix of a linear transformation with respect to the standard basis.)

[3] In Germany this is referred to the **rule of Sarrus**.

**20.** Let $(a, b, c)$, $(a', b', c')$ be two independent vectors. Suppose $(x, y, z)$ is another vector that lies in the linear span of $(a, b, c)$ and $(a', b', c')$. Then

$$\det \begin{bmatrix} x & y & z \\ a & b & c \\ a' & b' & c' \end{bmatrix} = 0,$$

i.e., $(bc' - b'c)x + (ca' - ac')y + (ab' - ba')z = 0$. (Compare to (✢✢) preceding Example 5 in Section 1.1.)

**21.** Given vectors $(a, b, c), (a', b', c') \in \mathbb{R}^3$ the vector

$$(bc' - b'c, ca' - ac', ab' - ba')$$

is called the **outer product** (or cross product) of vectors $(a, b, c)$ and $(a', b', c')$ and is denoted by $(a, b, c) \times (a', b', c')$. Show the following:

(1) $(a, b, c) \times (a', b', c') = -(a', b', c') \times (a, b, c)$.

(2) $(a, b, c) \times (a, b, c) = \mathbf{0}$.

(3) show that $|(a, b, c) \times (a', b', c')|$ is equal to the area of the parallelogram with the vectors $(a, b, c)$ and $(a', b', c')$ as two of its adjacent sides.

**22.** (Cramer's rule). Let

$$a_{i,1}x_1 + a_{i,2}x_2 + \cdots + a_{i,n}x_n = b_i, \qquad i = 1, 2, \ldots, n,$$

and suppose that the coefficient matrix $\mathbf{A} = (a_{i,j})$ of the above system of linear equations is invertible with inverse $\mathbf{A}^{-1}$. Then

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{A}^{-1}\mathbf{B}, \quad \text{where} \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Show that $x_i = \det(\mathbf{A}_i)/\det(\mathbf{A})$, $i = 1, 2, \ldots, n$, where $\mathbf{A}_i$ is the matrix obtained from $\mathbf{A}$ by replacing the $i$-th column of $\mathbf{A}$ by $\mathbf{B}$, $i = 1, 2, \ldots, n$. Verify the above statement for $n = 3$. The statement holds for any $n$ (finite!). But for all practical purposes it is useless for large $n$, since in computing $x_i$, you have to compute $n + 1$ determinants of $n \times n$ matrices.

**23.** Solve the following system of linear equations using Cramer's rule:

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 &= 5, \\ x_2 + 2x_3 - x_4 &= -3, \\ x_1 + 3x_2 + 4x_3 - x_4 &= 0, \\ 2x_1 - x_2 - x_3 + x_4 &= 4. \end{aligned}$$

**24.** Use Cramer's rule to solve each of the following systems of linear equations:

$$(a): \quad \begin{cases} x + y + z = 0, \\ 2x - y + z = 6, \\ 4x + 5y - z = 2. \end{cases}$$

$$(b): \quad \begin{cases} 3x + y - z = 1, \\ -x - y + 4z = 7, \\ 2x + y - 5z = -8. \end{cases}$$

**25.** Compute det(**H**), where

$$\mathbf{H} = \begin{bmatrix} 1 & x & x^2 & x^3 \\ 1 & y & y^2 & y^3 \\ 1 & z & z^2 & z^3 \\ 1 & w & w^2 & w^3 \end{bmatrix}.$$

**26.** Find the inverse of each of the following matrices:

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

Can you find a general pattern?

**27.** Diagonalize each of the matrices in the previous exercise.

**28.** Let $\mathbf{T} : \text{Mat}_{m,m} \longrightarrow \text{Mat}_{m,m}$ be the linear transformation given by

$$\mathbf{T}(\mathbf{M}) = \mathbf{M}^{\text{tr}}.$$

(a) Find the characteristic polynomial of **T**.
(b) Determine the eigenvalues and eigenspaces of **T**.
(c) Is **T** diagonalizable? If so, diagonalize it; if not, prove it is not.

**29.** Recall that a linear transformation $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$, where $\mathcal{V}$ is an $n$-dimensional vector, is called a **reflection** if there is an $(n-1)$-dimensional subspace $\mathcal{H}$ in $\mathcal{V}$ and a nonzero vector **X** in $\mathcal{V}$, but not in $\mathcal{H}$ such that

$$\mathbf{T}(\mathbf{H}) = \mathbf{H} \quad \text{for all} \quad \in \mathcal{H},$$

$$\mathbf{T}(\mathbf{X}) = -\mathbf{X}.$$

(a) Show that $\mathsf{T}$ is diagonalizable with diagonal form

$$\begin{bmatrix} -1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 \end{bmatrix}.$$

(b) Show that $\det(\mathsf{T}) = -1$.
(c) Let $\mathsf{T}$ and $\mathsf{S}$ be reflections. Is $\mathsf{T} \cdot \mathsf{S}$ also a reflection?

**30.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in the $n$-dimensional vector space $\mathcal{V}$ with cyclic vector $\mathbf{A}$. Show there exist unique numbers $a_0, \ldots, a_{n-1}$ such that

$$\mathsf{T}^n(\mathbf{A}) = a_0\mathbf{A} + a_1\mathsf{T}(\mathbf{A}) + \cdots + a_{n-1}\mathsf{T}^{n-1}(\mathbf{A}).$$

Show that the characteristic polynomial of $\mathsf{T}$ is:

$$\Delta(t) = (-1)^{n+1}(a_0 + a_1 t + \cdots + a_{n-1}t^{n-1} - t^n).$$

**31.** Show that the only eigenvalues of an involution are $\pm 1$.

**32.** Is $\det : \mathrm{Mat}_{m,m} \longrightarrow \mathbb{R}$ a linear transformation?

**33.** The **trace** of a square matrix $\mathbf{A}$ is defined to be the sum of the diagonal entries, and is denoted by $\mathrm{tr}(\mathbf{A})$. Is the trace $\mathrm{tr} : \mathrm{Mat}_{m,m} \longrightarrow \mathbb{R}$ a linear transformation?

# Chapter 15
# Inner Product Spaces

So far in our study of vector spaces and linear transformations we have made no use of the notions of length and angle, although these concepts play an important role in our intuition for the vector algebra of $\mathbb{R}^2$ and $\mathbb{R}^3$. In fact, the length of a vector and the angle between two vectors play very important parts in the further development of linear algebra, and it is now appropriate to introduce these ingredients into our study. There are many ways to do this, and in the approach that we will follow both length and angle will be derived from a more fundamental concept called a **scalar** or **inner product** of two vectors. Quite likely the student has encountered the scalar product in the guise of the **dot** product of two vectors in $\mathbb{R}^3$, which is usually defined by the equation

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}|\ |\mathbf{B}| \cos(\vartheta)$$

where $|\mathbf{A}|$ is the length of the vector $\mathbf{A}$, similarly for $\mathbf{B}$, and $\vartheta$ is the **angle between A** and **B**. In the study of vectors in $\mathbb{R}^3$ this is a reasonable way to introduce the scalar product, because lengths and angles are already defined and wellstudied concepts of geometry. In a more abstract study of linear algebra such as we are undertaking, this is not possible, for what is the length of a polynomial (vector) in $\mathcal{P}_4(\mathbb{R})$? Instead, we will again resort to the use of the axiomatic method. Having introduced vector spaces by axioms, it is not at all unreasonable to employ additional axioms to impose further structure on them.

## 15.1 Scalar Products

Experience has shown that the following definition[1] incorporates the essential features needed to define lengths and angles in a vector setting.

**Definition:** *A* **scalar** *(or* **inner***)* **product** *on a vector space* $\mathcal{V}$ *is a function that assigns to each pair of vectors* $\mathbf{A}, \mathbf{B}$ *in* $\mathcal{V}$ *a real number, denoted by* $<\mathbf{A}, \mathbf{B}>$*, having the following properties:*

[1] $<\mathbf{A}, \mathbf{B}> = <\mathbf{B}, \mathbf{A}>$ *for all* $\mathbf{A}, \mathbf{B}$ *in* $\mathcal{V}$.

[2] *For all vectors* $\mathbf{A}, \mathbf{B}$ *in* $\mathcal{V}$ *and numbers* $r$ *in* $\mathbb{R}$ *we have*

$$<r\mathbf{A}, \mathbf{B}> = r<\mathbf{A}, \mathbf{B}> = <\mathbf{A}, r\mathbf{B}>.$$

[3] *For all vectors* $\mathbf{A}, \mathbf{B}$, *and* $\mathbf{C}$ *in* $\mathcal{V}$ *we have*

(a) $<\mathbf{A}+\mathbf{B}, \mathbf{C}> = <\mathbf{A}, \mathbf{C}> + <\mathbf{B}, \mathbf{C}>$.

(b) $<\mathbf{A}, \mathbf{B}+\mathbf{C}> = <\mathbf{A}, \mathbf{B}> + <\mathbf{A}, \mathbf{C}>$.

[4] *For all* $\mathbf{A}$ *in* $\mathcal{V}$, $<\mathbf{A}, \mathbf{A}> \geq 0$ *and* $<\mathbf{A}, \mathbf{A}> = 0$ *if and only if* $\mathbf{A} = \mathbf{0}$.

Notice in reading the above axioms that we have used + for two different ideas, namely + for vectors, and + for numbers. This should be taken into account in reading Axioms [2], [3], and [4].

Why *scalar* product? That is, why is the concept introduced in the definition above sometimes called a **scalar** product? The answer should be apparent: because it associates to a pair of vectors a scalar (= real number). The terminology **inner** product is older, and is translated from the German "innere Produkt" (see the remark at the end of section 15.2 for one possible explanation of why) introduced by H. Grassmann (1844).

It is not difficult to verify that for the dot product

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| \cdot |\mathbf{B}| \cos(\vartheta)$$

defined in $\mathbb{R}^3$, the above axioms are satisfied. The only condition required that is not immediately clear is [3]. To verify [3(a)] consider Figure 15.1.1 where $\vartheta$ is the angle between $\mathbf{A}$ and $\mathbf{B}$ and $\vartheta_{\mathbf{X}}$ the angle between the vector $\mathbf{X}$ and $\mathbf{C}$. The last equation in the figure, which verifies [3(a)], is a consequence of the law of cosines and the preceding three equations. Thus the scalar product extends to general vector

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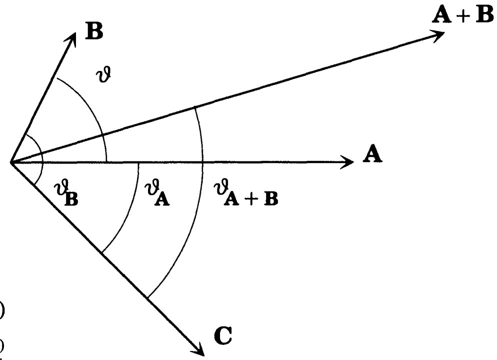
[1] Definitions in mathematics are rarely *right* or *wrong*, more often, they are either *useful* or *useless*.

spaces the dot product of everyday experience.




$\langle \mathbf{A}, \mathbf{C} \rangle = |\mathbf{A}| \cdot |\mathbf{C}| \cos(\vartheta_{\mathbf{A}})$
$\langle \mathbf{B}, \mathbf{C} \rangle = |\mathbf{B}| \cdot |\mathbf{C}| \cos(\vartheta_{\mathbf{B}})$
$\langle \mathbf{A}+\mathbf{B}, \mathbf{C} \rangle = |\mathbf{A}+\mathbf{B}| \cdot |\mathbf{C}| \cos(\vartheta_{\mathbf{A+B}})$
$|\mathbf{A}| \cos(\vartheta_{\mathbf{A}}) + |\mathbf{B}| \cos(\vartheta_{\mathbf{B}}) = (|\mathbf{A}+\mathbf{B}|)\cos(\vartheta)$

Figure 15.1.1

The notion of inner product can also be introduced into a complex vector space (see Exercise 21 at the end of this chapter), where a priori we have not a single example where length and angle have any intuitive content.

**Example 1:** The standard scalar product on $\mathbb{R}^n$.

For two vectors $\mathbf{A}, \mathbf{B}$ in $\mathbb{R}^n$ we *define*

$$\langle \mathbf{A}, \mathbf{B} \rangle = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n,$$

where $\mathbf{A} = (a_1, \ldots, a_n), \mathbf{B} = (b_1, \ldots, b_n)$. We assert that $\langle \mathbf{A}, \mathbf{B} \rangle$ is a scalar product. To verify this assertion we must check that Axioms [1]–[4] of the definition of a scalar product are satisfied.

Let's see, we have

$$\begin{aligned}\langle \mathbf{A}, \mathbf{B} \rangle &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = b_1 a_1 + b_2 a_2 + \cdots + b_n a_n \\ &= \langle \mathbf{B}, \mathbf{A} \rangle,\end{aligned}$$

so that [1] holds. Next note that

$$\begin{aligned}\langle r\mathbf{A}, \mathbf{B} \rangle &= r a_1 b_1 + r a_2 b_2 + \cdots + r a_n b_n \\ &= r(a_1 b_1 + a_2 b_2 + \cdots + a_n b_n) = r\langle \mathbf{A}, \mathbf{B} \rangle \\ &= a_1 r b_1 + a_2 r b_2 + \cdots + a_n r b_n = \langle \mathbf{A}, r\mathbf{B} \rangle,\end{aligned}$$

so that Axiom [2] also holds. To verify [3(a)] we compute as follows:

$$\begin{aligned}<\mathbf{A}+\mathbf{B},\mathbf{C}> &= (a_1+b_1)c_1+(a_2+b_2)c_2+\cdots+(a_n+b_n)c_n\\ &= a_1c_1+b_1c_1+a_2c_2+b_2c_2+\cdots+a_nc_n+b_nc_n\\ &= a_1c_1+a_2c_2+\cdots+a_nc_n+b_1c_1+b_2c_2+\cdots+b_nc_n\\ &= <\mathbf{A},\mathbf{C}>+<\mathbf{B},\mathbf{C}>.\end{aligned}$$

The verification of [3(b)] is similar. Finally, note that

$$<\mathbf{A},\mathbf{A}> = a_1a_1+a_2a_2+\cdots+a_na_n = a_1^2+a_2^2+\cdots+a_n^2,$$

and since a sum of squares is always nonnegative, being zero if and only if all the terms[2] are zero, we see that [4] is also satisfied.

Therefore,

$$<\mathbf{A},\mathbf{B}> = a_1b_1+a_2b_2+\cdots+a_nb_n$$

defines a scalar product on $\mathbb{R}^n$ called the **standard scalar product** on $\mathbb{R}^n$.

**EXAMPLE 2:** The standard scalar product in $\mathcal{V}$ with basis.

The preceding example may be generalized to any finite-dimensional vector space $\mathcal{V}$ with a ***fixed, chosen*** basis $\{\mathbf{F}_1,\ldots,\mathbf{F}_n\}$. Namely, if $\mathbf{A}$ and $\mathbf{B}$ belong to $\mathcal{V}$, then we know that

$$\begin{aligned}\mathbf{A} &= a_1\mathbf{F}_1+\cdots+a_n\mathbf{F}_n,\\ \mathbf{B} &= b_1\mathbf{F}_1+\cdots+b_n\mathbf{F}_n\end{aligned}$$

for unique $n$-tuples $(a_1,\ldots,a_n)$, $(b_1,\ldots,b_n)$, the components of $\mathbf{A}$ and $\mathbf{B}$ relative to the basis $\{\mathbf{F}_1,\ldots,\mathbf{F}_n\}$. Define

$$<\mathbf{A},\mathbf{B}> = a_1b_1+a_2b_2+\cdots+a_nb_n.$$

The same computations made in Example 1 show that $<\mathbf{A},\mathbf{B}>$ is a scalar product on $\mathcal{V}$. Notice that this scalar product depends on the choice of basis $\{\mathbf{F}_1,\ldots,\mathbf{F}_n\}$. A different choice of basis will give a different scalar product. For example, if in $\mathbb{R}^2$ we choose the standard basis and use the above definition, then we obtain the standard scalar product of $\mathbb{R}^2$. On the other hand we could use the basis $\mathbf{F}_1=(1,0)$, $\mathbf{F}_2=(1,1)$, in which case we would obtain an inner product

$$<\mathbf{A},\mathbf{B}>_{\mathbf{F}} = a_1b_1-a_1b_2-a_2b_1+2a_2b_2,$$

[2] Note that this will not work with complex numbers: some other definition of the scalar product is necessary. See Exercise 21 at the end of this chapter.

where $\mathbf{A} = (a_1, a_2) = a_1\mathbf{E}_1 + a_2\mathbf{E}_2$ and $\mathbf{B} = (b_1, b_2) = b_1\mathbf{E}_1 + b_2\mathbf{E}_2$. Thus many *different* scalar products seem possible on $\mathcal{V}$, one for each choice of basis. This difference is more apparent than real, as we shall presently see.

**EXAMPLE 3:** A scalar product on $\mathcal{P}_k(\mathbb{R})$.

For the space $\mathcal{P}_k(\mathbb{R})$ of polynomials of degree at most $k$ we may define the scalar product

$$<p(x), q(x)> = \int_{-1}^{+1} p(x)q(x)dx.$$

Certainly this definition is not at all like those of the preceding examples. In fact, except for the choice of the interval $[-1, 1]$ for the integration, it is much more natural, because it does not refer to a fixed basis. To verify the scalar product properties, we freely use facts about integration from the calculus:

[1]
$$\begin{aligned}<p(x), q(x)> &= \int_{-1}^{+1} p(x)q(x)dx \\ &= \int_{-1}^{+1} q(x)p(x)dx = <q(x), p(x)>\end{aligned}$$

[2]
$$\begin{aligned}<rp(x), q(x)> &= \int_{-1}^{+1} rp(x)q(x)dx \\ &= r\int_{-1}^{+1} p(x)q(x)dx = r<p(x), q(x)> \\ &= \int_{-1}^{+1} p(x)rq(x)dx = <p(x), rq(x)>,\end{aligned}$$

[3(a)]
$$\begin{aligned}<p(x) + q(x), r(x)> &= \int_{-1}^{+1} (p(x) + q(x))r(x)dx, \\ &= \int_{-1}^{+1} (p(x)r(x) + q(x)r(x))dx \\ &= \int_{-1}^{+1} p(x)r(x)dx + \int_{-1}^{+1} q(x)r(x)dx \\ &= <p(x), r(x)> + <q(x), r(x),\end{aligned}$$

and similarly for [3(b)].

For [4], recall that for a nonnegative continuous function $f(x)$,

$$\int_{-1}^{+1} f(x)^2dx \geq 0,$$

and

$$\int_{-1}^{+1} f(x)dx = 0 \quad \text{if and only if} \quad f(x) = 0 \quad \text{for all} \quad x \in [-1, 1].$$

Note that

$$<p(x), p(x)> = \int_{-1}^{+1} (p(x))^2 dx,$$

and that $p(x)^2$ is nonnegative. Hence

$$<p(x), p(x)> = \int_{-1}^{+1} (p(x))^2 dx \geq 0.$$

Finally, suppose that $<p(x), p(x)> = 0$. Then

$$\int_{-1}^{+1} (p(x))^2 dx = 0 \quad \text{implies} \quad p(x) = 0, \qquad x \in [-1, 1].$$

A nonzero polynomial of degree $k$ has at most $k$ roots, but $p(x) = 0$ for all $x \in [-1, 1]$, and so it has lots more than $k$ roots. Therefore, $p(x) = 0$, the zero polynomial.

Thus the axioms for a scalar product on $\mathcal{P}_k(\mathbb{R})$ are satisfied by the formula

$$<p(x), q(x)> = \int_{-1}^{+1} p(x)q(x)dx.$$

**EXAMPLE 4:** The trace scalar product on $\mathrm{Mat}_{n,n}$.

Let $\mathrm{Mat}_{n,n}$ denote the vector space of $n \times n$ matrices. Recall that $\mathbf{E}_{r,s}$ is the $n \times n$ matrix defined by

$$\mathbf{E}_{r,s} = (e_{i,j}), \qquad e_{i,j} = \begin{cases} 1 & \text{if } i = r, j = s, \\ 0 & \text{otherwise.} \end{cases}$$

The vectors $\{\mathbf{E}_{r,s}\}$ form a basis for $\mathrm{Mat}_{n,n}$ (see Proposition 10.3.2). Therefore,

$$<\mathbf{A}, \mathbf{B}> = \sum_{i,j} a_{i,j} b_{i,j}, \qquad \mathbf{A} = (a_{i,j}),\ \mathbf{B} = (b_{i,j})$$

defines a scalar product on $\mathrm{Mat}_{n,n}$ corresponding to the basis $\{\mathbf{E}_{r,s}\}$. There is another way to write this scalar product that is rather of interesting.

**DEFINITION:** *If* $\mathsf{A} = (a_{i,j})$ *is an* $n \times n$ *matrix, the* **trace** *of* $\mathsf{A}$ *is the number* $\mathrm{tr}(\mathsf{A})$ *that is the sum of the diagonal entries of* $\mathsf{A}$.

For example ($n = 3$) (compare Exercise 19 in Chapter 14)

$$\mathrm{tr}\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = 15,$$

and

$$\mathrm{tr}(\mathsf{I}) = \mathrm{tr}\begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix} = n,$$

where $\mathsf{I} \in \mathrm{Mat}_{n,n}$ is the identity matrix.

One way in which the trace and the inner product of Example 19 are connected is by the following formula[3]

$$<\mathsf{A}, \mathsf{B}> = \mathrm{tr}(\mathsf{A} \cdot \mathsf{B}^{\mathsf{tr}}).$$

To verify this we need only compute the righthand side and compare it to the definition of $<\mathsf{A}, \mathsf{B}>$. We have

$$\mathrm{tr}(\mathsf{A} \cdot \mathsf{B}^{\mathsf{tr}}) = \sum_i \sum_j a_{i,j} b^{\mathsf{tr}}_{j,i},$$

where $b^{\mathsf{tr}}_{j,i}$ denotes the entry in the $(j, i)$ position of the product matrix $\mathsf{A} \cdot \mathsf{B}^{\mathsf{tr}}$. Since $b^{\mathsf{tr}}_{j,i} = b_{i,j}$ this gives

$$\begin{aligned} \sum_i \sum_j a_{i,j} b^{\mathsf{tr}}_{j,i} &= \sum_i \sum_j a_{i,j} b_{i,j} \\ &= \sum_{i,j} a_{i,j} b_{i,j} = <\mathsf{A}, \mathsf{B}>, \end{aligned}$$

as we claimed.

The trace of a square matrix is quite a handy concept, and although we will not make use of it, it does play a rather important role in further developments of linear algebra.

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[3] Note that $\mathsf{B}^{\mathsf{tr}}$ denotes the *transpose* of the matrix $\mathsf{B}$. The font used for tr when denoting the *transpose* of a matrix is a sans serif font. The font used in tr when denoting the *trace* of a matrix is the roman font. See the chapter on Font Usage.

## 15.2 Inner Product Spaces

The following definition should come as no surprise.

**Definition:** *A vector space $\mathcal{V}$ equipped with a fixed scalar product $<—,—>$ is called an* **inner product space**.

Let us turn next to the concepts of length and angle in an inner product space.

**Definition:** *Let $(\mathcal{V}, <—,—>)$ be an inner product space and* **A** *a vector of $\mathcal{V}$. The* **length of A**, *denoted by* $|\mathbf{A}|$, *is the nonnegative number*

$$|\mathbf{A}| = \sqrt{<\mathbf{A},\mathbf{A}>}.$$

Notice by Axiom [4] of the definition of scalar product that $<\mathbf{A},\mathbf{A}> \geq 0$ for all vectors **A** in $\mathcal{V}$ and hence $<\mathbf{A},\mathbf{A}>$ has a unique nonnegative square root $\sqrt{<\mathbf{A},\mathbf{A}>}$ which we have defined to be $|\mathbf{A}|$.

**Proposition 15.2.1:** *Let $(\mathcal{V}, <—,—>)$ be an inner product space. Then the length function $|—|$ has the following properties:*

*(i) For each* **A** *in $\mathcal{V}$, $|\mathbf{A}| \geq 0$, and $|\mathbf{A}| = 0$ if and only if $\mathbf{A} = 0$.*

*(ii) For each* **A** *in $\mathcal{V}$ and a real number $r$, $|r\mathbf{A}| = |r| \cdot |\mathbf{A}|$.*

(As is becoming usual in our use of notations, we have written $|—|$ for two different concepts above. In (ii) $|r\mathbf{A}|$ is the length of the vector $r\mathbf{A}$, while $|r|$ is the absolute value of the number $r$. Actually, no confusion should arise because if we recall that $\mathbb{R} = \mathbb{R}^1$ is a vector space, then the absolute value $|r|$ is just the length of $r$ when $\mathbb{R}^1$ is given the standard scalar product.)

**Proof:** Condition (i) is just a translation of Axiom [4] for scalar products, while the computation

$$\begin{aligned}|r\mathbf{A}| &= \sqrt{<r\mathbf{A},r\mathbf{A}>} = \sqrt{r<\mathbf{A},r\mathbf{A}>}\\ &= \sqrt{r^2<\mathbf{A},\mathbf{A}>} = \sqrt{r^2}\sqrt{<\mathbf{A},\mathbf{A}>}\\ &= |r| \cdot |\mathbf{A}|\end{aligned}$$

verifies condition (ii). □

**Proposition 15.2.2** (Schwarz inequality): *Let $(\mathcal{V}, <—,—>)$ be an inner product space. Then for any pair of vectors* **A** *and* **B** *in $\mathcal{V}$,*

$$|<\mathbf{A},\mathbf{B}>| \leq |\mathbf{A}| \cdot |\mathbf{B}|.$$

**Proof:** We begin by observing that for $\mathbf{A} = \mathbf{0}$ we have

$$|\langle \mathbf{A}, \mathbf{B} \rangle| = 0 \quad \text{and} \quad |\mathbf{A}| \cdot |\mathbf{B}| = 0,$$

so that the inequality is true in this case. We may therefore assume that $\mathbf{A} \neq \mathbf{0}$, and hence that $|\mathbf{A}| \neq 0$. Let

$$\mathbf{C} = \mathbf{B} - \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\mathbf{A}.$$

Then

$$\begin{aligned}
0 \leq \langle \mathbf{C}, \mathbf{C} \rangle &= \left\langle \mathbf{B} - \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\mathbf{A}, \mathbf{B} - \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\mathbf{A} \right\rangle \\
&= \left\langle \mathbf{B} - \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\mathbf{A}, \mathbf{B} \right\rangle - \left\langle \mathbf{B} - \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\mathbf{A}, \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\mathbf{A} \right\rangle \\
&= \langle \mathbf{B}, \mathbf{B} \rangle - \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\langle \mathbf{A}, \mathbf{B} \rangle - \frac{\langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle}\langle \mathbf{B}, \mathbf{A} \rangle \\
&\quad + \frac{\langle \mathbf{A}, \mathbf{B} \rangle^2}{\langle \mathbf{A}, \mathbf{A} \rangle^2}\langle \mathbf{A}, \mathbf{A} \rangle \\
&= \langle \mathbf{B}, \mathbf{B} \rangle - \frac{\langle \mathbf{A}, \mathbf{B} \rangle^2}{\langle \mathbf{A}, \mathbf{A} \rangle} - \frac{\langle \mathbf{A}, \mathbf{B} \rangle \, \langle \mathbf{A}, \mathbf{B} \rangle}{\langle \mathbf{A}, \mathbf{A} \rangle} + \frac{\langle \mathbf{A}, \mathbf{B} \rangle^2}{\langle \mathbf{A}, \mathbf{A} \rangle} \\
&= \langle \mathbf{B}, \mathbf{B} \rangle - \frac{\langle \mathbf{A}, \mathbf{B} \rangle^2}{\langle \mathbf{A}, \mathbf{A} \rangle}.
\end{aligned}$$

That is ,

$$\langle \mathbf{B}, \mathbf{B} \rangle - \frac{\langle \mathbf{A}, \mathbf{B} \rangle^2}{\langle \mathbf{A}, \mathbf{A} \rangle} \geq 0.$$

Multiply both sides by $\langle \mathbf{A}, \mathbf{A} \rangle$, which is nonnegative and hence does not change the sense of the inequality, so we get

$$\langle \mathbf{A}, \mathbf{A} \rangle \, \langle \mathbf{B}, \mathbf{B} \rangle - \langle \mathbf{A}, \mathbf{B} \rangle^2 \geq 0,$$

which may be written as

$$|\mathbf{A}|^2 \cdot |\mathbf{B}|^2 \geq \langle \mathbf{A}, \mathbf{B} \rangle^2.$$

Taking positive square roots then gives

$$|\mathbf{A}| \cdot |\mathbf{B}| \geq |\langle \mathbf{A}, \mathbf{B} \rangle|,$$

as required. □

**PROPOSITION 15.2.3** (The Triangle Inequality): *For any vectors* **A** *and* **B** *in an inner product space* $(\mathcal{V}, \langle -, - \rangle)$ *we have*

$$|\mathbf{A}+\mathbf{B}| \leq |\mathbf{A}| + |\mathbf{B}|.$$

**PROOF:** We compute as follows:

$$\begin{aligned}
|\mathbf{A}+\mathbf{B}|^2 &= \langle \mathbf{A}+\mathbf{B}, \mathbf{A}+\mathbf{B} \rangle \\
&= \langle \mathbf{A}, \mathbf{A}+\mathbf{B} \rangle + \langle \mathbf{B}, \mathbf{A}+\mathbf{B} \rangle \\
&= \langle \mathbf{A}, \mathbf{A} \rangle + \langle \mathbf{A}, \mathbf{B} \rangle + \langle \mathbf{B}, \mathbf{A} \rangle + \langle \mathbf{B}, \mathbf{B} \rangle \\
&= |\mathbf{A}|^2 + \langle \mathbf{A}, \mathbf{B} \rangle + \langle \mathbf{A}, \mathbf{B} \rangle + |\mathbf{B}|^2 \\
&= |\mathbf{A}|^2 + 2\langle \mathbf{A}, \mathbf{B} \rangle + |\mathbf{B}|^2 \\
&\leq |\mathbf{A}|^2 + 2|\langle \mathbf{A}, \mathbf{B} \rangle| + |\mathbf{B}|^2 \\
&\leq |\mathbf{A}|^2 + 2|\mathbf{A}| \cdot |\mathbf{B}| + |\mathbf{B}|^2 = (|\mathbf{A}| + |\mathbf{B}|)^2.
\end{aligned}$$

That is,

$$|\mathbf{A}+\mathbf{B}|^2 \leq (|\mathbf{A}| + |\mathbf{B}|)^2$$

and taking positive square roots gives the desired conclusion. □

To see why Proposition 15.2.3 is called the **triangle inequality** let us define the **distance between** two vectors **A** and **B** in an inner product space $(\mathcal{V}, \langle -, - \rangle)$ by the formula

$$d(\mathbf{A}, \mathbf{B}) = |\mathbf{A} - \mathbf{B}|.$$

The distance function $d$ is easily seen to satisfy

$$d(\mathbf{A}, \mathbf{B}) \geq 0,$$

and

$$d(\mathbf{A}, \mathbf{B}) = 0 \quad \text{if and only if} \quad \mathbf{A} = \mathbf{B}.$$

In $\mathbb{R}^2$ one has the picture shown in Figure 15.2.1. Here the length of $\mathbf{A} - \mathbf{B}$ is equal to the length of $\mathbf{B} - \mathbf{A}$, which is the length of the dotted segment, since they are opposite sides of a parallelogram. This shows that $d(\mathbf{A}, \mathbf{B})$ is what you would think it was in this case.

The inequality of Proposition 15.2.3 reads

$$d(\mathbf{A}, -\mathbf{B}) \leq |\mathbf{A}| + |-\mathbf{B}|,$$

namely, the sum of the lengths of two sides of a triangle is greater than or equal to the length of the third side.

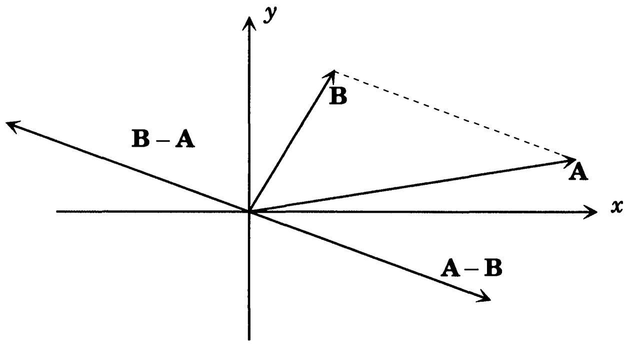


**FIGURE** 15.2.1: Distance in the plane $\mathbb{R}^2$

We have already used the inner product to define the length of a vector. We next use it to define angles.

**DEFINITION:** *Let* $(\mathcal{V}, <—,—>)$ *be an inner product space. If* **A** *and* **B** *are nonzero vectors of* $\mathcal{V}$ *the* **angle between A and B,** *denoted by* $\vartheta$, *is defined by*

$$\cos(\vartheta) = \frac{<\mathbf{A}, \mathbf{B}>}{|\mathbf{A}| \cdot |\mathbf{B}|} \leq 1$$

*and the requirement* $0 \leq \vartheta \leq \pi$.

(Note that by the Schwarz inequality (Proposition 15.2.2)

$$-1 \leq \frac{<\mathbf{A}, \mathbf{B}>}{|\mathbf{A}| \cdot |\mathbf{B}|} \leq 1,$$

so that $\vartheta$ is welldefined by the above formula.)

This definition gives the formula

$$<\mathbf{A}, \mathbf{B}> = |\mathbf{A}| \cdot |\mathbf{B}| \cos(\vartheta)$$

that we are familiar with in $\mathbb{R}^3$, so we may safely assume that the definition of $\vartheta$ is a reasonable extension of the concept of angle to an arbitrary inner product space.

**PROPOSITION 15.2.4 :** *Two vectors nonzero* **A** *and* **B** *in an inner product space* $\mathcal{V}$ *are perpendicular (or orthogonal) if and only if* $<\mathbf{A}, \mathbf{B}> = 0$.

**PROOF:** By definition, **A** and **B** are **perpendicular** if and only if the angle $\vartheta$ between them is $\pi/2$ radians = 90°. (By convention $\mathbf{0} \perp \mathbf{A}$ for any vector **A**.) For such $\vartheta$ we have $\cos(\vartheta) = 0$. Therefore,

$$\frac{<\mathbf{A}, \mathbf{B}>}{|\mathbf{A}| \cdot |\mathbf{B}|} = \cos(\vartheta) = 0,$$

so $<\mathbf{A}, \mathbf{B}> = 0$. The reverse implication, namely that $<\mathbf{A}, \mathbf{B}> = 0$ implies $\vartheta = \pi/2$, is immediate from the definition of $\vartheta$. □

Thus the inner product $<—,—>$ provides a very handy check for perpendicularity of vectors.

**NOTATION:** *I* f **A, B** are two vectors in an inner product space $\mathcal{V}, <—,—>$, then we use the notation $\mathbf{A} \perp \mathbf{B}$ to denote that **A** and **B** are perpendicular. More generally, if $\mathcal{S}, \mathcal{T} \subseteq \mathcal{V}$ we write $\mathcal{S} \perp \mathcal{T}$ to indicate that $\mathbf{A} \perp \mathbf{B}$ for all $\mathbf{A} \in \mathcal{S}$ and $\mathbf{B} \in \mathcal{B}$.

**EXAMPLE 1:** Are the vectors $(1, -1, 2), (2, 4, 1)$ in $\mathbb{R}^3$ perpendicular? (We assume that $\mathbb{R}^3$ has its standard inner product.)

**SOLUTION:** One checks that

$$<(1, -1, 2), (2, 4, 1)> = 2 - 4 + 2 = 0,$$

so they are perpendicular.

**DEFINITION:** *Let* $(\mathcal{V}, <—,—>)$ *be an inner product space. A set of vectors* $S$ *in* $\mathcal{V}$ *is said to be an* **orthogonal set of vectors** *if it does not contain the zero vector and any two distinct vectors* **A, B** *in* $S$ *are orthogonal i.e.,* $<\mathbf{A}, \mathbf{B}> = 0$ *whenever* $\mathbf{A} \neq \mathbf{B} \in S$. *An orthogonal set of vectors is said to be* **orthonormal** *if in addition* $|\mathbf{A}| = 1$ *for every* **A** *in* $S$.

**PROPOSITION 15.2.5:** *An orthogonal set of nonzero vectors* $S$ *in an inner product space* $(\mathcal{V}, <—,—>)$ *is linearly independent.*

**PROOF:** Suppose that

$$a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n = \mathbf{0}$$

is a linear relation among vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ of $S$. Then we have

$$\begin{aligned} 0 &= <\mathbf{A}_i, \mathbf{0}> = <\mathbf{A}_i, a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n> \\ &= a_1<\mathbf{A}_i, \mathbf{A}_1> + a_2<\mathbf{A}_i, \mathbf{A}_2> + \cdots + a_n<\mathbf{A}_i, \mathbf{A}_n> \\ &= a_i<\mathbf{A}_i, \mathbf{A}_i>, \end{aligned}$$

since $<\mathbf{A}_i, \mathbf{A}_j> = 0$ for $i \neq j$ because the set $S$ is orthogonal. But $\mathbf{A}_i$ is not the zero vector, and therefore $<\mathbf{A}_i, \mathbf{A}_i> \neq 0$, so the equation

$$0 = a_i <\mathbf{A}_i, \mathbf{A}_i>$$

implies $a_i = 0$. Since this is true for $i = 1, 2, \ldots, n$, there can be no nontrivial linear relation among the vectors of $S$, so $S$ is a linearly independent set. □

**EXAMPLE 2:** Which of the following sets of vectors in $\mathbb{R}^3$ are orthogonal (in the standard inner product)?

(a) $\{(0,0,0), (1,1,1)\}$.
(b) $\{(1,0,0), (0,1,0), (0,0,1)\}$.
(c) $\{(1,1,1), (1,1,0), (1,0,0)\}$.
(d) $\{(1,0,0), (0,1,0), (0,0,1), (1,1,1)\}$.

**SOLUTION:** The set (a) is not orthogonal because it contains the zero vector. The set (b) is orthogonal, but the set (c) is not, because

$$<(1,1,1), (1,0,0)> = 1 \neq 0.$$

The set (d) is not orthogonal because it contains 4 *nonzero* vectors, and if it were orthogonal, it would have to be linearly independent by Proposition 15.2.5, contrary to the fact that $\mathbb{R}^3$ is only 3-dimensional.

**EXAMPLE 3:** Is the set of vectors

$$\left\{1, x, x^2 - \frac{1}{3}\right\} \subset \mathcal{P}_k(\mathbb{R}), \qquad k \geq 2,$$

orthogonal?

**SOLUTION:** The answer is yes, and you might well ask how do I know. The *real answer* lies in the discussion of Legendre polynomials in Section 15.5. For the moment all we can do is compute. We have

$$<1, x> = \int_{-1}^{+1} x\, dx = \left.\frac{x^2}{2}\right|_{-1}^{+1} = 0,$$

$$<1, x^2 - \frac{1}{3}> = \int_{-1}^{+1} \left(x^2 - \frac{1}{3}\right) dx = \left.\frac{x^3}{3} - \frac{x}{3}\right|_{-1}^{+1} = 0,$$

$$<x, x^2 - \frac{1}{3}> = \int_{-1}^{+1} \left(x^3 - \frac{1}{3}x\right) dx = \left.\frac{x^4}{4} - \frac{1}{6}x^2\right|_{-1}^{+1} = 0.$$

The remaining scalar products are seen to be zero by symmetry of $<—,—>$ in the two variables.

Note that by Proposition 15.2.5 the vectors $\left\{1, x, x^2 - \frac{1}{3}\right\}$ are linearly independent. Since $\dim \mathcal{P}_2(\mathbb{R}) = 3$, it follows that these vectors are actually an orthogonal **basis** for $\mathcal{P}_2(\mathbb{R})$.

The construction of orthogonal sets in $\mathcal{P}_k(\mathbb{R})$ for large $k$ is a fascinating business involving many areas of mathematics including analytic number theory and differential equations. The polynomials that most often occur are named after the mathematical giants of preceding centuries. We look at one special case of this in Section 15.5.

**PROPOSITION 15.2.6:** *Suppose that $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is an orthogonal set of nonzero vectors in the inner product space $\mathcal{V}$. If*

$$\mathbf{A} \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n),$$

*then*[4]

$$\mathbf{A} = \frac{<\mathbf{A}, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1 + \cdots + \frac{<\mathbf{A}, \mathbf{A}_n>}{<\mathbf{A}_n, \mathbf{A}_n>}\mathbf{A}_n.$$

**PROOF:** The proof is very similar to that of Proposition 15.2.5. Suppose that $\mathbf{A}$ belongs to the linear span of $\mathbf{A}_1, \ldots, \mathbf{A}_n$. Then there are numbers $a_1, \ldots, a_n$ such that

$$\mathbf{A} = a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n.$$

What we are trying to prove is that the number $a_i$ must be the particular number $<\mathbf{A}, \mathbf{A}_i>/<\mathbf{A}_i, \mathbf{A}_i>$. This is easy to prove. Take the inner product of both sides of the preceding equation with $\mathbf{A}_i$, giving

$$\begin{aligned} <\mathbf{A}_i, \mathbf{A}> &= <\mathbf{A}_i, a_1\mathbf{A}_1 + \cdots + a_n\mathbf{A}_n> \\ &= a_1<\mathbf{A}_i, \mathbf{A}_1> + \cdots + a_n<\mathbf{A}_i, \mathbf{A}_n> \\ &= a_i<\mathbf{A}_i, \mathbf{A}_i>, \end{aligned}$$

and since $\mathbf{A}_i$ is nonzero, so is $<\mathbf{A}_i, \mathbf{A}_i>$, so

$$a_i = \frac{<\mathbf{A}_i, \mathbf{A}>}{<\mathbf{A}_i, \mathbf{A}_i>} = \frac{<\mathbf{A}, \mathbf{A}_i>}{<\mathbf{A}_i, \mathbf{A}_i>},$$

as required. □

---

[4] Note that since $\mathbf{A}_i \neq \mathbf{0}$, $<\mathbf{A}_i, \mathbf{A}_i> \neq 0$ also. Thus the righthand side is well-defined.

**REMARK:** The numbers $<\mathbf{A}, \mathbf{A}_i>/<\mathbf{A}_i, \mathbf{A}_i>$ are usually called the **Fourier coefficients** of $\mathbf{A}$ with respect to $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. The vector

$$\frac{<\mathbf{A}, \mathbf{A}_i>}{<\mathbf{A}_i, \mathbf{A}_i>}\mathbf{A}_i$$

is called the **component of A along** $\mathbf{A}_i$.

Since the vector

$$\frac{<\mathbf{A}, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1}\mathbf{A}_1 + \cdots + \frac{<\mathbf{A}, \mathbf{A}_n>}{<\mathbf{A}_n, \mathbf{A}_n>}\mathbf{A}_n$$

belongs to $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$ for any vector $\mathbf{A}$ in $\mathcal{V}$ we have actually proven more than is claimed in Proposition 15.2.6, namely,

$$\mathbf{A} \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n) \iff \mathbf{A} = \frac{<\mathbf{A}, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1 + \cdots + \frac{<\mathbf{A}, \mathbf{A}_n>}{<\mathbf{A}_n, \mathbf{A}_n>}\mathbf{A}_n.$$

Here are some examples to illustrate this point:

**EXAMPLE 4:** Find the Fourier coefficients of the vector

$$x^2 + x + 1$$

with respect to the orthogonal set of vectors

$$\left\{1, x, x^2 - \frac{1}{3}\right\}$$

in $\mathcal{P}_2(\mathbb{R})$.

**SOLUTION:** We must compute a number of integrals:

$$<1, x^2 + x + 1> = \int_{-1}^{+1} (x^2 + x + 1)dx = \frac{x^3}{3} + \frac{x^2}{2} + x \Big|_{-1}^{+1} = \frac{8}{3},$$

$$\begin{aligned} <x, x^2 + x + 1> &= \int_{-1}^{+1} \left(x^3 + x^2 + x\right) dx \\ &= \frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2}\Big|_{-1}^{+1} \\ &= \left(\frac{1}{4} + \frac{1}{3} + \frac{1}{2}\right) - \left(\frac{1}{4} - \frac{1}{3} + \frac{1}{2}\right) = \frac{2}{3}, \end{aligned}$$

$$\begin{aligned}
<x^2 - \frac{1}{3}, x^2 + x + 1> &= \int_{-1}^{+1} \left(x^2 - \frac{1}{3}\right)(x^2 + x + 1)dx \\
&= \int_{-1}^{+1} \left(x^4 + x^3 + x^2\right) dx - \frac{1}{3}\int_{-1}^{+1} \left(x^2 + x + 1\right) dx \\
&= \left(\frac{x^5}{5} + \frac{x^4}{4} + \frac{x^3}{3}\right)\Bigg|_{-1}^{+1} - \frac{1}{3}\left(\frac{x^3}{3} + \frac{x^2}{2} + x\right)\Bigg|_{-1}^{+1} \\
&= \frac{2}{3} + \frac{2}{5} - \frac{1}{3}\frac{8}{3} = \frac{8}{45}, \\
<1, 1> &= \int_{-1}^{+1} 1dx = 2, \\
<x, x> &= \int_{-1}^{+1} x^2\, dx = \frac{1}{3}x^3 \Bigg|_{-1}^{+1} = \frac{2}{3}, \\
<x^2 - \frac{1}{3}, x^2 - \frac{1}{3}> &= \int_{-1}^{+1} \left(x^2 - \frac{1}{3}\right)^2 dx = \int_{-1}^{+1} \left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right) dx \\
&= \frac{x^5}{5} - \frac{2}{3}\frac{x^3}{3} + \frac{1}{9}x \Bigg|_{-1}^{+1} = 2\left(\frac{1}{5} - \frac{2}{9} + \frac{1}{9}\right) = \frac{8}{45}.
\end{aligned}$$

Thus the Fourier coefficients of $1 + x + x^2$ with respect to the *ordered* basis $\left\{1, x, x^2 - \frac{1}{3}\right\}$ are $\left\{\frac{4}{3}, 1, \frac{7}{2}\right\}$.

**EXAMPLE 5:** Write the vector $\mathbf{A} = (1, -1, 1)$ as a linear combination of the vectors

$$\begin{aligned}
\mathbf{V}_1 &= (1, 1, 1), \\
\mathbf{V}_2 &= (0, 1, -1), \\
\mathbf{V}_3 &= (-2, 1, 1).
\end{aligned}$$

**SOLUTION:** Before plunging ahead and solving lists of linear equations it pays to note that $\{\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3\}$ are orthogonal and hence independent by Proposition 15.2.5 which means, since $\dim \mathbb{R}^3 = 3$, that they are a basis for $\mathbb{R}^3$. Accordingly by Proposition 15.2.6 we must have

$$\mathbf{A} = \frac{<\mathbf{A}, \mathbf{V}_1>}{<\mathbf{V}_1, \mathbf{V}_1>}\mathbf{V}_1 + \frac{<\mathbf{A}, \mathbf{V}_2>}{<\mathbf{V}_2, \mathbf{V}_2>}\mathbf{V}_2 + \frac{<\mathbf{A}, \mathbf{V}_3>}{<\mathbf{V}_3, \mathbf{V}_3>}\mathbf{V}_3,$$

and rather than solve systems of equations we need only compute inner products. We find

$$\begin{aligned}
<\mathbf{A}, \mathbf{V}_1> &= 1, & <\mathbf{V}_1, \mathbf{V}_1> &= 3, \\
<\mathbf{A}, \mathbf{V}_2> &= -2, & <\mathbf{V}_2, \mathbf{V}_2> &= 2, \\
<\mathbf{A}, \mathbf{V}_3> &= -2, & <\mathbf{V}_3, \mathbf{V}_3> &= 6,
\end{aligned}$$

and therefore

$$(1,-1,1)=\frac{1}{3}(1,1,1)-1(0,1,-1)-\frac{1}{3}(-2,1,1),$$

which is all rather easy.

**PROPOSITION 15.2.7** (Gram–Schmidt): *Let* $(\mathcal{V}, <—,—>)$ *be an inner product space and* $\mathcal{W}$ *a finite-dimensional subspace of* $\mathcal{V}$. *Then there exists in* $\mathcal{V}$ *an orthonormal set of vectors* $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ *that is a basis for* $\mathcal{W}$.

**PROOF:** The method of proof that we will employ is called the **Gram–Schmidt orthonormalization** process. It is a manufacturing process that starts with any old basis for $\mathcal{W}$ as raw material and manufactures one of the required type. We therefore begin by choosing a basis $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$ for $\mathcal{W}$. This we know we can do by Theorem 6.2.3. Define in succession vectors $\mathbf{A}_1, \ldots, \mathbf{A}_n$ by the equations

$$\begin{aligned}
\mathbf{A}_1 &= \mathbf{B}_1,\\
\mathbf{A}_2 &= \mathbf{B}_2 - \frac{<\mathbf{B}_2, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1,\\
&\;\vdots\\
\mathbf{A}_n &= \mathbf{B}_n - \frac{<\mathbf{B}_n, \mathbf{A}_{n-1}>}{<\mathbf{A}_{n-1}, \mathbf{A}_{n-1}>}\mathbf{A}_{n-1} - \cdots - \frac{<\mathbf{B}_n, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1.
\end{aligned}$$

Before proceeding further we must check that we have not committed the cardinal sin of dividing by 0 in one of the above formulas. To do so, note that $\mathbf{A}_i$ is actually a linear combination of $\mathbf{B}_1, \ldots, \mathbf{B}_i$ in disguise; that is, $\mathbf{A}_i \in \mathcal{L}(\mathbf{B}_1, \ldots, \mathbf{B}_i)$. If for some $j$, $<\mathbf{A}_j, \mathbf{A}_j> = 0$, then $\mathbf{A}_j = \mathbf{0}$ and therefore we get from the equation defining $\mathbf{A}_j$,

$$\mathbf{B}_j = \frac{<\mathbf{B}_j, \mathbf{A}_{j-1}>}{<\mathbf{A}_{j-1}, \mathbf{A}_{j-1}>}\mathbf{A}_{j-1} + \cdots + \frac{<\mathbf{B}_j, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1 \in \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_{j-1}),$$

and therefore $\mathbf{B}_j \in \mathcal{L}(\mathbf{B}_1, \ldots, \mathbf{B}_{j-1})$, contrary to the linear independence of $\{\mathbf{B}_1, \ldots, \mathbf{B}_n\}$. Therefore, $<\mathbf{A}_j, \mathbf{A}_j>$ is never zero.

Next we claim that $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_i) = \mathcal{L}(\mathbf{B}_1, \ldots, \mathbf{B}_i)$. We have already seen that $\mathbf{A}_1, \ldots, \mathbf{A}_i \in \mathcal{L}(\mathbf{B}_1, \ldots, \mathbf{B}_i)$, so that $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_i)$ is contained in $\mathcal{L}(\mathbf{B}_1, \ldots, \mathbf{B}_i)$. It will therefore suffice to show $\mathcal{L}(\mathbf{B}_1, \ldots, \mathbf{B}_i) \subseteq \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_i)$, which is easy and left to the reader.

Our next task is to show that the vectors $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ are orthogonal. We will do this by showing successively that the sets $\{\mathbf{A}_1\}$, $\{\mathbf{A}_1, \mathbf{A}_2\}$,

..., $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ are orthogonal. The first set is orthogonal because $\mathbf{A}_1 = \mathbf{B}_1 \neq \mathbf{0}$. Suppose as an induction hypothesis that we have already shown the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_i\}$ to be orthogonal. In order to prove that $\{\mathbf{A}_1, \ldots, \mathbf{A}_{i+1}\}$ is orthogonal, it is only necessary to show that $\mathbf{A}_{i+1}$ is orthogonal to $\mathbf{A}_1, \ldots, \mathbf{A}_i$. We compute for $1 \leq j \leq i$:

$$
\begin{aligned}
<\mathbf{A}_{i+1}, \mathbf{A}_j> &= <\mathbf{B}_{i+1} - \frac{<\mathbf{B}_{i+1}, \mathbf{A}_i>}{<\mathbf{A}_i, \mathbf{A}_i>}\mathbf{A}_i - \cdots - \frac{<\mathbf{B}_{i+1}, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1, \mathbf{A}_j> \\
&= <\mathbf{B}_{i+1}, \mathbf{A}_j> - \tfrac{<\mathbf{B}_{i+1}, \mathbf{A}_i>}{<\mathbf{A}_i, \mathbf{A}_i>}<\mathbf{A}_i, \mathbf{A}_j> - \cdots - \tfrac{<\mathbf{B}_{i+1}, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}<\mathbf{A}_1, \mathbf{A}_j> \\
&= <\mathbf{B}_{i+1}, \mathbf{A}_j> - \frac{<\mathbf{B}_{i+1}, \mathbf{A}_i>}{<\mathbf{A}_i, \mathbf{A}_i>} 0 - \cdots \\
&\quad - \frac{<\mathbf{B}_{i+1}, \mathbf{A}_j>}{<\mathbf{A}_j, \mathbf{A}_j>}<\mathbf{A}_j, \mathbf{A}_j> - \cdots - \frac{<\mathbf{B}_{i+1}, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{O}0 \\
&= <\mathbf{B}_{i+1}, \mathbf{A}_j> - <\mathbf{B}_{i+1}, \mathbf{A}_j> = 0,
\end{aligned}
$$

where the key step was to use the fact $<\mathbf{A}_k, \mathbf{A}_j> = 0$ if $1 \leq k \leq i$ and $k \neq j$. This computation shows that $\{\mathbf{A}_1, \ldots, \mathbf{A}_{i+1}\}$ is an orthogonal set of vectors. By induction, we may conclude that $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is an orthogonal set of nonzero vectors. By Proposition 15.2.5 it follows that the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is linearly independent. Since $\mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_n) \subseteq \mathcal{W}$ and $n = \dim \mathcal{W}$, it follows from Theorem 6.2.4 that $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is a basis for $\mathcal{W}$.

But we are not yet done! We have only constructed an orthogonal set of vectors $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ that is a basis for $\mathcal{V}$. What we want is an ortho***normal*** set. However, this is easy. We just **normalize** the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ by defining $\overline{\mathbf{A}}_i = (1/|\mathbf{A}_i|)\mathbf{A}_i$, which we may do, since $|\mathbf{A}_i| \neq 0$. We then have

$$
|\overline{\mathbf{A}}_i| = \left| \frac{1}{|\mathbf{A}_i|}\mathbf{A}_i \right| = \frac{1}{|\mathbf{A}_i|}|\mathbf{A}_i| = 1
$$

and

$$
<\overline{\mathbf{A}}_r, \overline{\mathbf{A}}_s> = \frac{1}{|\mathbf{A}_r|}\frac{1}{|\mathbf{A}_s|}<\mathbf{A}_r, \mathbf{A}_s> = 0, \qquad r \neq s,
$$

so that $\left\{\overline{\mathbf{A}}_1, \ldots, \overline{\mathbf{A}}_n\right\}$ is both a basis and an orthonormal set. □

The kind of basis that appears in the above proposition is very important and therefore comes in for a special name.

**DEFINITION :** *Let* $(\mathcal{V}, <—, —>)$ *be a finite-dimensional inner product space. A basis* $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ *for* $\mathcal{V}$ *is, called an* **orthonormal basis** *for* $\mathcal{V}$ *if* $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ *is a basis for* $\mathcal{V}$ *and an orthonormal set, that is*

$$<\mathbf{A}_i, \mathbf{A}_j> = \begin{cases} 0 & i \neq j, \\ 1 & i = j, \end{cases}$$

*or what is the same thing, the vectors are of unit length and pairwise orthogonal.*

While the Gram–Schmidt process may seem quite formidable, it really isn't. For once one has established that it works, one can apply it without worrying about the inductive check that the vectors it manufactures are orthogonal. However, one should not forget to normalize at the end if required to do so.

**EXAMPLE 6:** Apply the Gram–Schmidt orthonormalization process to find an orthogonal basis for $\mathcal{P}_2(\mathbb{R})$.

**SOLUTION :** We may start with any old basis for $\mathcal{P}_2(\mathbb{R})$ to apply the process to, so let us choose the obvious basis $\left\{1, x, x^2\right\}$. We then define vectors

$$\begin{aligned}
\mathbf{A}_1 &= 1, \\
\mathbf{A}_2 &= x - \frac{<x, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1, \\
\mathbf{A}_3 &= x^2 - \frac{<x^2, \mathbf{A}_1>}{<\mathbf{A}_1, \mathbf{A}_1>}\mathbf{A}_1 - \frac{<x^2, \mathbf{A}_2>}{<\mathbf{A}_2, \mathbf{A}_2>}\mathbf{A}_2,
\end{aligned}$$

and so we have some integrations to carry out.

Since $\mathbf{A}_1 = 1$, we get

$$<x, \mathbf{A}_1> = <x, 1> = \int_{-1}^{+1} x\, dx = x^2 \Big|_{-1}^{+1} = 0$$

$$<\mathbf{A}_1, \mathbf{A}_1> = <1, 1> = \int_{-1}^{+1} 1\, dx = 2,$$

so that $\mathbf{A}_2 = x$. Next we get for the inner product of $x^2$ with $\mathbf{A}_1$

$$<x^2, \mathbf{A}_1> = <x^2, 1> = \int_{-1}^{+1} x^2 dx = \frac{x^3}{3}\Big|_{-1}^{+1} = \frac{2}{3},$$

and for the inner product of $\mathbf{A}_2$ with $x^2$ and with itself

$$<x^2, \mathbf{A}_2> = \int_{-1}^{+1} x^2 \cdot x\,dx = \int_{-1}^{+1} x^3\,dx$$
$$= \frac{x^4}{4}\Big|_{-1}^{+1} = 0,$$
$$<\mathbf{A}_2, \mathbf{A}_2> = \int_{-1}^{+1} x^2\,dx = \frac{x^3}{3}\Big|_{-1}^{+1} = \frac{2}{3},$$

so we find that

$$\mathbf{A}_1 = 1,$$
$$\mathbf{A}_2 = x,$$
$$\mathbf{A}_3 = x^2 - \frac{\frac{2}{3}}{2} \cdot \mathbf{A}_1 - (0) \cdot \mathbf{A}_2 = x^2 - \frac{1}{3}$$

is an orthogonal basis for $\mathcal{P}_2(\mathbb{R})$. (Go back and look at Example 3 again.) The polynomials that occur here are called **Legendre polynomials.** In Section 15.5 we will examine them in more detail.

**EXAMPLE 7:** Find an orthonormal basis for the subspace $\mathcal{W}$ of $\mathbb{R}^3$ given by

$$x + 2y + 3z = 0.$$

**SOLUTION:** A basis for $\mathcal{W}$ is easily seen to be

$$\mathbf{B}_1 = (-3, 0, 1), \qquad \mathbf{B}_2 = (-2, 1, 0).$$

Applying the Gram-Schmidt process, we get the orthogonal basis

$$\mathbf{A}_1 = (-3, 0, 1),$$
$$\mathbf{A}_2 = (-2, 1, 0) - \frac{<(-3, 0, 1), (-2, 1, 0)>}{<(-3, 0, 1), (-3, 0, 1)>}(-3, 0, 1)$$
$$= (-2, 1, 0) - \frac{6}{10}(-3, 0, 1)$$
$$= (-\frac{1}{5}, 1, -\frac{3}{5}).$$

*But we are not done yet!* The vectors $\{\mathbf{A}_1, \mathbf{A}_2\}$ are only orthogonal, not ortho***normal***. We must therefore normalize them, getting

$$\mathbf{C}_1 = \frac{\mathbf{A}_1}{|\mathbf{A}_1|} = \frac{1}{\sqrt{10}}(-3, 0, 1) = \left(-\frac{3}{\sqrt{10}}, 0, \frac{1}{\sqrt{10}}\right),$$
$$\mathbf{C}_2 = \frac{\mathbf{A}_2}{|\mathbf{A}_2|} = \frac{\sqrt{5}}{\sqrt{7}}\left(-\frac{1}{5}, 1, -\frac{3}{5}\right) = \left(-\frac{1}{\sqrt{35}}, \frac{\sqrt{5}}{\sqrt{7}}, -\frac{3}{\sqrt{35}}\right),$$

which is an orthonormal basis for $\mathcal{W}$.

There is a very important consequence of Proposition 15.2.7, namely that a finite-dimensional inner product space has an orthonormal basis. We record this for future reference.

**COROLLARY 15.2.8:** *Let $(\mathcal{V}, <—,—>)$ be a finite-dimensional inner product space. Then there is an orthonormal basis for $\mathcal{V}$.* □

One reason why orthonormal bases are so useful in an inner product space is the following:

**THEOREM 15.2.9:** *Let $\{\mathbf{A}_1, \dots, \mathbf{A}_n\}$ be an orthonormal basis for the inner product space $(\mathcal{V}, <—,—>)$. If $\mathbf{A}$ belongs to $\mathcal{V}$, then*

$$\mathbf{A} = <\mathbf{A}, \mathbf{A}_1>\mathbf{A}_1 + <\mathbf{A}, \mathbf{A}_2>\mathbf{A}_2 + \cdots + <\mathbf{A}, \mathbf{A}_n>\mathbf{A}_n.$$

*If*

$$\mathbf{B} = b_1\mathbf{A}_1 + b_2\mathbf{A}_2 + \cdots + b_n\mathbf{A}_n,$$

$$\mathbf{C} = c_1\mathbf{A}_1 + c_2\mathbf{A}_2 + \cdots + c_n\mathbf{A}_n,$$

*then*

$$<\mathbf{B}, \mathbf{C}> = b_1c_1 + b_2c_2 + \cdots + b_nc_n.$$

**PROOF:** The first assertion follows directly from Proposition 15.2.6 because $<\mathbf{A}_1, \mathbf{A}_1> = <\mathbf{A}_2, \mathbf{A}_2> = \cdots = <\mathbf{A}_n, \mathbf{A}_n> = 1$. To prove the second assertion we just compute:

$$\begin{aligned} <\mathbf{B}, \mathbf{C}> &= <b_1\mathbf{A}_1 + \cdots + b_n\mathbf{A}_n, c_1\mathbf{A}_1 + \cdots + c_n\mathbf{A}_n> \\ &= \sum_{i,j} b_ic_j<\mathbf{A}_i, \mathbf{A}_j> \\ &= b_1a_1 + b_2a_2 + \cdots + b_na_n \end{aligned}$$

because

$$<\mathbf{A}_i, \mathbf{A}_j> = \begin{cases} 0 & i \neq j, \\ 1 & i = j, \end{cases}$$

since $\{\mathbf{A}_1, \dots, \mathbf{A}_n\}$ is orthonormal. □

Thus by choosing an orthonormal basis $\{\mathbf{A}_1, \dots, \mathbf{A}_n\}$ in a finite-dimensional inner product space and expressing vectors in terms of their components with respect to $\{\mathbf{A}_1, \dots, \mathbf{A}_n\}$ we obtain formulas for $<\mathbf{A}, \mathbf{B}>$ that look exactly like the formulas for $\mathbb{R}^n$ with its standard inner product. There is a good reason for this, as we will explain in the next section.

**REMARK:** If $\mathcal{V}$, $<—,—>$ is an $n$-dimensional inner product space and $\mathbf{A}, \mathbf{B} \in \mathcal{V}$, with $\mathbf{B} \neq \mathbf{0}$, then we may extend $\mathbf{B}$ to a basis for $\mathcal{V}$ and orthonormalize it to $\mathbf{B}_1 = \frac{1}{\|\mathbf{B}\|}\mathbf{B}, \mathbf{B}_2, \ldots, \mathbf{B}_n$ by the Gram-Schmidt process. If

$$\mathbf{A} = a_1\mathbf{B}_1 + a_2\mathbf{B}_2 + \cdots a_n\mathbf{B}_n$$

then $<\mathbf{A}, \mathbf{B}> = a_1$. Therefore $<\mathbf{A}, \mathbf{B}> \neq= 0$ if and only if $\mathbf{A}$ has a nonzero component *in* the direction of $\mathbf{B}$, i.e., a *component lying in the line* spanned by $\mathbf{B}$. This was apparently why Grasmann called the sacalar product an *inner product.*

## 15.3 Isometries

In studying vector spaces we found that it was important to consider linear transformations, that is, functions that preserve the basic vector operations. For vector spaces equipped with an inner product it seems reasonable to demand that the inner product also be preserved.

**DEFINITION:** *Let $\mathcal{V}$ and $\mathcal{W}$ be inner product spaces. A linear transformation* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *is called an* **isometric embedding** *if*

$$<\mathsf{T}(\mathbf{A}), \mathsf{T}(\mathbf{B})> = <\mathbf{A}, \mathbf{B}>$$

*for all vectors* $\mathbf{A}, \mathbf{B}$ *in* $\mathcal{V}$.

Isometric embeddings are a very restricted class of linear transformations. We have:

**PROPOSITION 15.3.1:** *Let $\mathcal{V}$ and $\mathcal{W}$ be inner product spaces and* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *an isometric embedding. Then* $\ker(\mathsf{T}) = \{\mathbf{0}\}$.

**PROOF:** Suppose that $\mathsf{T}(\mathbf{A}) = \mathbf{0}$. Then since $\mathsf{T}$ is an isometric embedding,

$$<\mathbf{A}, \mathbf{A}> = <\mathsf{T}(\mathbf{A}), \mathsf{T}(\mathbf{A})> = <\mathbf{0}, \mathbf{0}> = 0,$$

and therefore $\mathbf{A} = \mathbf{0}$. $\square$

**COROLLARY 15.3.2:** *Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner product spaces of the same dimension and* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ *an isometric embedding. Then* $\mathsf{T}$ *is an isomorphism.*

**PROOF:** By Propositions 15.3.1 and 8.6.1 and Theorem 6.3.4 it suffices to show that $\mathrm{Im}(\mathsf{T}) = \mathcal{W}$. By Theorem 8.4.2 we obtain $\dim(\mathcal{V}) = \dim(\mathrm{Im}(\mathsf{T}))$. Since $\dim \mathcal{V} = \dim \mathcal{W}$ by hypothesis, $\dim(\mathcal{W}) = \dim(\mathrm{Im}(\mathsf{T}))$. Therefore, since $\mathrm{Im}\ \mathsf{T}$ is a linear subspace of $\mathcal{W}$ by Proposition 8.4.1, it follows from Corollary 6.3.6 that $\mathrm{Im}(\mathsf{T}) = \mathcal{W}$. $\square$

**EXAMPLE 1:** An example of an isometric embedding of $\mathbb{R}^2$ in $\mathbb{R}^4$ is given by

$$\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^4$$

such that

$$\mathsf{T}(x, y) = \frac{1}{\sqrt{2}}(x, y, y, x).$$

To see this, we let $\mathbf{A} = (a_1, a_2)$, $\mathbf{B} = (b_1, b_2)$ and note that

$$\begin{aligned}
<\mathsf{T}(\mathbf{A}), \mathsf{T}(\mathbf{B})> &= <\frac{1}{\sqrt{2}}(a_1, a_2, a_2, a_1), \frac{1}{\sqrt{2}}(b_1, b_2, b_2, b_1)> \\
&= \frac{1}{2}(a_1b_1 + a_2b_2 + a_2b_2 + a_1b_1) \\
&= \frac{1}{2}(2a_1b_1 + 2a_2b_2) \\
&= a_1b_1 + a_2b_2 = <\mathbf{A}, \mathbf{B}>,
\end{aligned}$$

as required. $\square$

**DEFINITION:** *Let $\mathcal{V}$ and $\mathcal{W}$ be inner product spaces. A linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ is called an* **isometric isomorphism**, *an* **isometry** *for short, if $\mathsf{T}$ is an isometric embedding and there exists an isometric embedding $\mathsf{S} : \mathcal{W} \longrightarrow \mathcal{V}$ such that*

$$\mathsf{S} \cdot \mathsf{T} = \mathsf{I} : \mathcal{V} \longrightarrow \mathcal{V}$$

*and*

$$\mathsf{T} \cdot \mathsf{S} = \mathsf{I} : \mathcal{W} \longrightarrow \mathcal{W}.$$

*If there is an isometry $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$, we say that $\mathcal{V}$ and $\mathcal{W}$ are* **isometric**, *or* **isometrically isomorphic**.

**PROPOSITION 15.3.3:** *Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner product spaces of the same dimension. If $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ is an isometric embedding then $\mathsf{T}$ is an isometric isomorphism.*

**PROOF:** By Corollary 15.3.2 $\mathsf{T}$ is an isomorphism. Let $\mathsf{S} = \mathsf{T}^{-1}$. We need only show that $\mathsf{S} : \mathcal{W} \longrightarrow \mathcal{V}$ is an isometric embedding. Let $\mathbf{A}, \mathbf{B} \in \mathcal{W}$. Write $\mathbf{A} = \mathsf{T}(\mathbf{C})$, $\mathbf{B} = \mathsf{T}(\mathbf{D})$. Then

$$<\mathsf{S}(\mathbf{A}), \mathsf{S}(\mathbf{B})> = <\mathbf{C}, \mathbf{D}> = <\mathsf{T}(\mathbf{C}), \mathsf{T}(\mathbf{D})> = <\mathbf{A}, \mathbf{B}>,$$

as required. $\square$

**PROPOSITION 15.3.4:** *Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner product spaces with $\dim(\mathcal{V}) \leq \dim(\mathcal{W})$. Then there exists an isometric embedding $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$.*

**PROOF:** Choose orthonormal bases $\{\mathbf{V}_1, \ldots, \mathbf{V}_m\}$ and $\{\mathbf{W}_1, \ldots, \mathbf{W}_n\}$ for $\mathcal{V}$ and $\mathcal{W}$ respectively and let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$ be the linear extension of

$$\mathsf{T}(\mathbf{V}_i) = \mathbf{W}_i, \qquad i = 1, \ldots, m.$$

Thus if $\mathbf{A}, \mathbf{B} \in \mathcal{V}$, say

$$\mathbf{A} = \sum a_i \mathbf{V}_i, \qquad \mathbf{B} = \sum b_i \mathbf{V}_i,$$

we obtain from Theorem 15.2.9 the formula

$$<\mathbf{A}, \mathbf{B}> = \sum_i a_i \cdot b_i.$$

On the other hand, since

$$\mathsf{T}(\mathbf{A}) = \sum a_i \mathbf{W}_i, \qquad \mathsf{T}(\mathbf{B}) = \sum b_i \mathbf{W}_i,$$

we get

$$<\mathsf{T}(\mathbf{A}), \mathsf{T}(\mathbf{B})> = \sum_i a_i \cdot b_i,$$

so $\mathsf{T}$ is an isometric embedding. $\square$

Summarizing, we obtain:

**THEOREM 15.3.5:** *Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner product spaces. Then $\mathcal{V}$ and $\mathcal{W}$ are isometric if and only if* $\dim(\mathcal{V}) = \dim(\mathcal{W})$.

**PROOF:** If $\mathcal{V}$ and $\mathcal{W}$ are isometric, then they are certainly isomorphic, so $\dim \mathcal{V} = \dim \mathcal{W}$. On the other hand, if $\dim(\mathcal{V}) = \dim(\mathcal{W})$, then we may apply Proposition 15.3.3 to construct an isometric embedding $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{W}$, which by Corollary 15.3.2 will be an isometric isomorphism. $\square$

**COROLLARY 15.3.6:** *If $\mathcal{V}$ is an inner product space of finite dimension $n$ then $\mathcal{V}$, is isometric to $\mathbb{R}^n$.* $\square$

**EXAMPLE 2:** According to Theorem 15.3.5, the spaces $\mathbb{R}^3$ and $\mathcal{P}_2(\mathbb{R})$ are isometric. An explicit isometry is given by

$$\mathsf{T}(u, v, w) = \left(\frac{u}{\sqrt{2}} + \frac{(9\sqrt{5})w}{2\sqrt{2}}\right) + \frac{v\sqrt{3}}{\sqrt{2}} \cdot x + \frac{(3\sqrt{5})w}{2\sqrt{2}} \cdot x^2.$$

This isometry is constructed as follows. Note that $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is an orthogonal basis for $\mathbb{R}^3$ while $\left\{\frac{1}{\sqrt{2}}, \frac{\sqrt{3}\cdot x}{\sqrt{2}}, \frac{3\sqrt{5}}{2\sqrt{2}}(x^2 - \frac{1}{3})\right\}$ is an orthonormal[5] basis for $\mathcal{P}_2(\mathbb{R})$. Therefore the linear extension of

$$\mathbf{T}(1,0,0) = \frac{1}{\sqrt{2}},$$
$$\mathbf{T}(0,1,0) = \frac{\sqrt{3}}{\sqrt{2}} \cdot x,$$
$$\mathbf{T}(0,0,1) = \frac{3\sqrt{5}}{2\sqrt{2}}(x^2 - \frac{1}{3})$$

will be an isometry $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathcal{P}_2(\mathbb{R})$. A little algebraic manipulation brings forth the above formula.

Thus, exactly as with Theorem 8.6.3 we see that the abstraction of the axiomatics has added nothing new. That is, up to isometry there is only one inner product space of dimension $n$ (when $n < \infty$), namely $\mathbb{R}^n$. Still, the axiomatic approach is useful in that it provides a natural setting for inner products on $\mathrm{Mat}_{n,n}$ (see Section 15.1 Example 4) and the very important inner products on $\mathcal{P}_n(\mathbb{R})$. These inner products occur in nature and should not be denied.

## 15.4 The Riesz Representation Theorem

Let $<—, —>$ be a scalar product on the finite dimensional vector space $\mathcal{V}$. If $\mathbf{A} \in \mathcal{V}$ is a fixed vector, then the function

$$\mathbf{T}_{\mathbf{A}} : \mathcal{V} \longrightarrow \mathbb{R}$$

defined by

$$\mathbf{T}_{\mathbf{A}}(\mathbf{X}) = <\mathbf{X}, \mathbf{A}> \qquad \forall\, \mathbf{X} \in \mathcal{V}$$

is a linear transformation. A special case of a much more general theorem of F. Riesz says that any linear transformation

$$\mathbf{T} : \mathcal{V} \longrightarrow \mathbb{R}$$

of an inner product space $\mathcal{V}$ to the 1–dimensional space $\mathbb{R}$ may be expressed in terms of the inner product $<—, —>$. This will prove quite useful in the next chapter.

**DEFINITION :** *If $\mathcal{V}$ is a vector space, a linear transformation*

$$\mathbf{T} : \mathcal{V} \longrightarrow \mathbb{R}$$

*is called a* **linear functional**.

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[5] These are the normalizations of the Legendre polynomials from Section 15.2, Example 3.

**DEFINITION:** *Let $\mathcal{V}$ be an inner product space and $\mathcal{S}$ a subset of $\mathcal{V}$. The* **orthogonal complement** *$\mathcal{S}^\perp$ of $\mathcal{S}$ in $\mathcal{V}$ is defined by*

$$\mathcal{S}^\perp = \left\{\mathbf{A} \in \mathcal{V} \,|\, <\mathbf{A}, \mathbf{B}> = 0 \ \forall \ \mathbf{B} \in \mathcal{S}\right\}.$$

**PROPOSITION 15.4.1:** *Let $\mathcal{S} \subseteq \mathcal{V}$ be a subset of the inner product space $\mathcal{V}$. Then $\mathcal{S}^\perp$ is a linear subspace and $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$.*

**PROOF:** Let **B**, **C** belong to $\mathcal{S}^\perp$ then

$$<\mathbf{A}, \mathbf{B} + \mathbf{C}> = <\mathbf{A}, \mathbf{B}> + <\mathbf{A}, \mathbf{C}> = 0 + 0 = 0 \qquad \forall \ \mathbf{A} \in \mathcal{S},$$

so $\mathbf{B} + \mathbf{C}$ belongs to $\mathcal{S}^\perp$. If $r$ is a number, then

$$<\mathbf{A}, r\mathbf{B}> = r<\mathbf{A}, \mathbf{B}> = r \cdot 0 = 0,$$

so $r\mathbf{B} \in \mathcal{S}^\perp$. Therefore, $\mathcal{S}^\perp$ is closed under scalar products and vector addition, so it is a linear subspace of $\mathcal{V}$.

If **A** belongs to $\mathcal{S} \cap \mathcal{S}^\perp$, then $<\mathbf{A}, \mathbf{A}> = 0$ since $\mathbf{A} \in \mathcal{S}$ and $\mathbf{A} \in \mathcal{S}^\perp$. Therefore $\mathbf{A} = \mathbf{0}$. □

**REMARK:** If $\mathcal{S} \subseteq \mathcal{V}$ is a subset of the inner product space $\mathcal{V}$, then it follows from the symmetry of the scalar product (this is property [1] of the definition of a scalar product) that $\mathcal{S} \subseteq \mathcal{S}^{\perp\perp}$. Since the orthogonal complement of any sub*set* is always a sub*space*, we cannot hope to have equality unless $\mathcal{S}$ is itself a subspace. This is indeed the case if $\mathcal{V}$ is finite-dimensional, and, a proof is left to the exercises.

**EXAMPLE 1:** What is the orthogonal complement of the subspace

$$\mathcal{W} = \left\{(x, y, z) \,|\, x + 2y + 3z = 0\right\}$$

in $\mathbb{R}^3$ ?

**SOLUTION:** The orthogonal complement of the plane $\mathcal{W}$ will be a line $\mathcal{L}$. In this case the line is the subspace spanned by $(1, 2, 3)$. (These coordinates are the coefficients of the equation used to define $\mathcal{W}$. This always works.) For if $(x, y, z) \in \mathcal{W}$, then

$$<(1, 2, 3), (x, y, z)> = x + 2y + 3z = 0$$

because $(x, y, z) \in \mathcal{W}$.

**EXAMPLE 2:** What is the orthogonal complement of the subspace

$$\mathcal{W} = \left\{(x, y, z) \,|\, x + y + z = 0 \quad \text{and} \quad x - y + z = 0\right\} ?$$

**SOLUTION:** The subspace $\mathcal{W}$ is a line. Its orthogonal complement will therefore be a plane. A basis for $\mathcal{W}$ is the vector $(1, 0, -1)$, since it satisfies the equations of both planes

$$x + y + z = 0 \quad \text{and} \quad x - y + z = 0,$$

and must therefore be on their line of intersection.

The orthogonal complement of $\mathcal{W}$ must therefore be the plane $\Pi$ whose normal vector at the origin is $(1, 0, -1)$. Therefore,

$$\Pi = \left\{(x, y, z) \mid x - z = 0\right\},$$

as required.

**PROPOSITION 15.4.2:** *Let $\mathcal{V}$ be a finite-dimensional inner product space and $\mathcal{W}$ a linear subspace of $\mathcal{V}$. Then (see Chapter 2 Exercise 10)*

$$\mathcal{W} \oplus \mathcal{W}^{\perp} = \mathcal{V}.$$

**PROOF:** What we must do is show that any vector $\mathbf{A}$ in $\mathcal{V}$ may be written as a sum $\mathbf{B} + \mathbf{C}$ with $\mathbf{B}$ in $\mathcal{W}$ and $\mathbf{C}$ in $\mathcal{W}^{\perp}$. To this end apply the algorithm of Section 15.2 Example 6 to choose an orthonormal basis $\mathbf{W}_1, \ldots, \mathbf{W}_k$ for $\mathcal{W}$. Let $\mathbf{A} \in \mathcal{V}$ and define

$$\begin{aligned} \mathbf{B} &= \langle \mathbf{W}_1, \mathbf{A} \rangle \mathbf{W}_1 + \cdots + \langle \mathbf{W}_k, \mathbf{A} \rangle \mathbf{W}_k, \\ \mathbf{C} &= \mathbf{A} - \mathbf{B}. \end{aligned}$$

Clearly, $\mathbf{A} = \mathbf{B} + \mathbf{C}$, and $\mathbf{B}$ belongs to $\mathcal{W}$. So what remains to be shown is that $\mathbf{C}$ belongs to $\mathcal{W}^{\perp}$. First of all, note that

$$\mathbf{C} = \mathbf{A} - \langle \mathbf{W}_1, \mathbf{A} \rangle \mathbf{W}_1 - \cdots - \langle \mathbf{W}_k, \mathbf{A} \rangle \mathbf{W}_k.$$

Hence

$$\begin{aligned} \langle \mathbf{C}, \mathbf{W}_i \rangle &= \langle \mathbf{A}, \mathbf{W}_i \rangle - \langle \mathbf{W}_1, \mathbf{A} \rangle \langle \mathbf{W}_1, \mathbf{W}_i \rangle + \cdots + \langle \mathbf{W}_k, \mathbf{A} \rangle \langle \mathbf{W}_k, \mathbf{W}_i \rangle \\ &= \langle \mathbf{A}, \mathbf{W}_i \rangle - \langle \mathbf{W}_i, \mathbf{A} \rangle = 0, \end{aligned}$$

since

$$\langle \mathbf{W}_j, \mathbf{W}_i \rangle = \begin{cases} 0 & i \neq j, \\ 1 & i = j. \end{cases}$$

Next, suppose that $\mathbf{W}$ belongs to $\mathcal{W}$. Then

$$\mathbf{W} = w_1 \mathbf{W}_1 + \cdots + w_k \mathbf{W}_k,$$

so

$$\langle \mathbf{C}, \mathbf{W} \rangle = \sum_{i=1}^{k} w_i \langle \mathbf{C}, \mathbf{W}_i \rangle = \sum_{i=1}^{k} w_i 0 = 0,$$

and hence $\mathbf{C}$ belongs to $\mathcal{W}^{\perp}$. $\square$

**PROPOSITION 15.4.3:** *Let $\mathcal{V}$ be a finite-dimensional inner product space of dimension $n$ and $\{\mathbf{A}_1, \ldots, \mathbf{A}_k\}$ an orthonormal set of vectors in $\mathcal{V}$. Then there are vectors $\{\mathbf{A}_{k+1}, \ldots, \mathbf{A}_n\}$ such that the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_k, \mathbf{A}_{k+1}, \ldots, \mathbf{A}_n\}$ is an orthonormal basis for $\mathcal{V}$.*

**PROOF:** Let $\mathcal{W} = \text{Span}(\mathbf{A}_1, \ldots, \mathbf{A}_k)$. By Proposition 15.4.1, $\mathcal{W}^\perp$ is a linear subspace of $\mathcal{V}$, and since $\mathcal{V}$ is finite-dimensional, so is $\mathcal{W}^\perp$. Let $\{\mathbf{B}_1, \ldots, \mathbf{B}_l\}$ be an orthonormal basis for $\mathcal{W}^\perp$. One readily checks that the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_k, \mathbf{B}_1, \ldots, \mathbf{B}_l\}$ is an orthonormal set, and since none of the vectors in the set is $\mathbf{0}$, this set is linearly independent by Proposition 15.2.5. By Proposition 15.4.2 $\text{Span}(\mathbf{A}_1, \ldots, \mathbf{A}_k, \mathbf{B}_1, \ldots, \mathbf{B}_l) = \mathcal{V}$, and so setting

$$\mathbf{A}_{k+1} = \mathbf{B}_1, \ldots, \mathbf{A}_n = \mathbf{B}_l,$$

we have the required orthonormal basis. $\square$

Notice the following important consequence of the proof of Proposition 15.4.3.

**COROLLARY 15.4.4:** *Let $\mathcal{V}$ be a finite-dimensional inner product space of dimension $n$ and $\mathcal{W}$ a subspace of $\mathcal{V}$ of dimension $m$. Then* $\dim(\mathcal{W}^\perp) = n - m$. $\square$

**THEOREM 15.4.5** (F. Riesz): *Let $\mathcal{V}$ be a finite-dimensional inner product space and $\mathsf{T} : \mathcal{V} \longrightarrow \mathbb{R}$ a linear functional. Then there exists a unique vector $\mathbf{A}$ in $\mathcal{V}$ (depending on $\mathsf{T}$) such that*

$$\mathsf{T}(\mathbf{B}) = \langle \mathbf{B}, \mathbf{A} \rangle$$

*for all $\mathbf{B}$ in $\mathcal{V}$.*

**PROOF:** Let the dimension of $\mathcal{V}$ be $n$. If $\text{Im}(\mathsf{T}) = \{\mathbf{0}\}$ set $\mathbf{A} = \mathbf{0}$. Then

$$0 = \mathsf{T}(\mathbf{B}),$$

$$0 = \langle \mathbf{B}, \mathbf{0} \rangle,$$

so the theorem is true in this case. If $\text{Im}(\mathsf{T}) \neq \{\mathbf{0}\}$, then we must have $\text{Im}(\mathsf{T}) = \mathbb{R}$, since $\{\mathbf{0}\}$ and $\mathbb{R}$ are the only subspaces of $\mathbb{R}$. Therefore, by Theorem 8.4.2 the kernel of $\mathsf{T}$ has dimension $n-1$. Let $\{\mathbf{A}_1, \ldots, \mathbf{A}_{n-1}\}$ be an orthonormal basis for $\ker(\mathsf{T})$.

By Proposition 15.4.3 there is a vector $\mathbf{A}_n$ such that $\{\mathbf{A}_1, \ldots, \mathbf{A}_{n-1}, \mathbf{A}_n\}$ is a basis for $\mathcal{V}$. Let $\mathbf{B} \in \mathcal{V}$. By Proposition 15.2.6

$$\mathbf{B} = \langle \mathbf{B}, \mathbf{A}_1 \rangle \mathbf{A}_1 + \cdots + \langle \mathbf{B}, \mathbf{A}_n \rangle \mathbf{A}_n,$$

so

$$\begin{aligned} \mathsf{T}(\mathbf{B}) &= \langle \mathbf{B}, \mathbf{A}_1 \rangle \mathsf{T}(\mathbf{A}_1) + \cdots + \langle \mathbf{B}, \mathbf{A}_n \rangle \mathsf{T}(\mathbf{A}_n) \\ &= \langle \mathbf{B}, \mathbf{A}_n \rangle \mathsf{T}(\mathbf{A}_n) \\ &= \langle \mathbf{B}, \mathsf{T}(\mathbf{A}_n)\mathbf{A}_n \rangle \end{aligned}$$

(remember $\mathsf{T}(\mathbf{A}_n)$ is only a number). Thus $\mathbf{A} = \mathsf{T}(\mathbf{A}_n)\mathbf{A}_n$ is the required vector.

To see that the vector $\mathbf{A}$ is unique, suppose that $\mathbf{C}$ is another vector such that $\mathsf{T}(\mathbf{B}) = \langle \mathbf{B}, \mathbf{C} \rangle$ for all $\mathbf{B}$ in $\mathcal{V}$. Therefore,

$$\begin{aligned} \langle \mathbf{B}, \mathbf{A} - \mathbf{C} \rangle &= \langle \mathbf{B}, \mathbf{A} \rangle - \langle \mathbf{B}, \mathbf{C} \rangle \\ &= \mathsf{T}(\mathbf{B}) - \mathsf{T}(\mathbf{B}) \\ &= 0 \end{aligned}$$

for all $\mathbf{B}$ in $\mathcal{V}$. Put $\mathbf{B} = \mathbf{A} - \mathbf{C}$ to get

$$\langle \mathbf{A} - \mathbf{C}, \mathbf{A} - \mathbf{C} \rangle = 0,$$

but this implies by Axiom [4] of an inner product that $\mathbf{A} - \mathbf{C} = \mathbf{0}$, and hence $\mathbf{A} = \mathbf{C}$. □

In applying Riesz's theorem, notice that the unique vector $\mathbf{A}$, whose existence is guaranteed by the theorem, is characterized by the properties

(1) $\mathbf{A} \neq \mathbf{0}$,
(2) $\mathbf{A} \perp \ker(\mathsf{T})$,
(3) $\mathsf{T}(\mathbf{A}) = \langle \mathbf{A}, \mathbf{A} \rangle$.

If $\mathbf{E}$ is a **unit** vector, i.e., a vector of unit length, perpendicular to $\ker(\mathsf{T})$, then the proof of Riesz's theorem shows that $\mathbf{A} = (\mathsf{T}(\mathbf{E}))(\mathbf{E})$. Finally, if $\mathbf{C}$ is any nonzero vector orthogonal to $\ker(\mathsf{T})$ then $\frac{\mathbf{C}}{|\mathbf{C}|}$, is a unit vector orthogonal to $\ker(\mathsf{T})$. Combining all this leads to:

**COROLLARY 15.4.6** (F. Riesz): *Let $\mathcal{V}$ be a finite-dimensional inner product space, $\mathsf{T} : \mathcal{V} \longrightarrow \mathbb{R}$ a linear transformation and $\mathbf{C}$ a nonzero vector orthogonal to* $\ker(\mathsf{T})$. *Set*

$$\mathbf{A} = \frac{\mathsf{T}(\mathbf{C}) \cdot \mathbf{C}}{\langle \mathbf{C}, \mathbf{C} \rangle}.$$

*Then*

$$\mathsf{T}(\mathbf{B}) = \langle \mathbf{B}, \mathbf{A} \rangle$$

*for all $\mathbf{B}$ in $\mathcal{V}$.* □

**EXAMPLE 3:** Consider the linear functional

$$\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}$$

given by

$$\mathsf{T}(x, y, z) = x + y + z.$$

The kernel of $\mathsf{T}$ is thus the plane

$$\ker(\mathsf{T}) = \left\{(x, y, z) \,|\, x + y + z = 0\right\},$$

and a vector orthogonal to $\ker(\mathsf{T})$ is $(1, 1, 1)$, which normalizes to $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$. Since

$$\mathsf{T}\left(\frac{1}{\sqrt{3}}(1, 1, 1)\right) = \frac{3}{\sqrt{3}} = \sqrt{3},$$

we find that

$$\mathsf{T}(x, y, z) = \langle (x, y, z), (1, 1, 1) \rangle,$$

which can be readily checked.

In fact, more generally, if

$$\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}$$

is given by

$$\mathsf{T}(x, y, z) = ax + by + cz,$$

then the vector $\mathbf{V} = (a, b, c)$ satisfies

$$\mathsf{T}(x, y, z) = \langle (x, y, z), (a, b, c) \rangle$$

as one easily checks.

Here is a more substantial example.

**EXAMPLE 4:** Consider the linear transformation

$$\varepsilon : \mathcal{P}_2(\mathbb{R}) \longrightarrow \mathbb{R}$$

given by evaluating a polynomial at $t = 0$; i.e.,

$$\varepsilon(p(t)) = p(0).$$

The kernel of $\varepsilon$ is the subspace

$$\begin{aligned}\ker(\varepsilon) &= \left\{p(t) \,|\, p(0) = 0\right\} \\ &= \left\{a_1 t + a_2 t^2 \,|\, a_1, a_2 \in \mathbb{R}\right\}.\end{aligned}$$

We make use of Corollary 15.4.6 and first find a vector, i.e., a polynomial $h \in \mathcal{P}_2(\mathbb{R})$, orthogonal to $\ker(\varepsilon)$. This requires:

$$<t, h> = 0,$$
$$<t^2, h> = 0.$$

Set $h = h_0 + h_1 t + h_2 t^2$. Then

$$\begin{aligned} 0 = <t, h> &= \int_{-1}^{+1} (h_0 t + h_1 t^2 + h_2 t^3)\, dt \\ &= \frac{h_0 t^2}{2} + \frac{h_1 t^3}{3} + \frac{h_2 t^4}{4} \Big|_{-1}^{+1} \\ &= \left[\frac{h_0}{2} + \frac{h_1}{3} + \frac{h_2}{4}\right] - \left[\frac{h_0}{2} + \frac{-h_1}{3} + \frac{h_2}{4}\right] = \frac{2h_1}{3}, \end{aligned}$$

and hence $h_1 = 0$. Analogously,

$$\begin{aligned} 0 = <t^2, h> &= \int_{-1}^{+1} t^2\, h(t)\, dt \\ &\vdots \\ &= \frac{2h_0}{3} + \frac{2h_2}{5} \end{aligned}$$

from which we obtain $0 = \frac{2h_0}{3} + \frac{2h_2}{5}$, and therefore

$$h = 1 - \frac{5}{3} t^2,$$

which is orthogonal to $\ker(\varepsilon)$. By Corollary 15.4.6, the quadratic polynomial

$$q(t) = \frac{\varepsilon(h)}{<h, h>} h = \frac{1}{\frac{8}{9}} h = \frac{9}{8} \left(1 - \frac{5}{3} t^2\right)$$

satisfies the condition of Riesz's theorem, so

$$f(0) = \varepsilon(f) = \frac{9}{8} \int_{-1}^{+1} f(t) \left(1 - \frac{5}{3} t^2\right) dt$$

for all quadratic polynomials $f$.

There is nothing sacred about the evaluation of polynomials at 0 nor on the restriction to quadratic polynomials. We could equally well consider the linear transformation

$$\varepsilon_a : \mathcal{P}_k(\mathbb{R}) \longrightarrow \mathbb{R},$$

defined by $\varepsilon_a(f) = f(a)$ for some fixed $a \in \mathbb{R}$. The theorem of Riesz implies the existence of a polynomial $h_a$ of degree at most $k$ such that

$$f(a) = \varepsilon_a(f) = \int_{-1}^{+1} f(t) \cdot h_a(t)\, dt$$

for all $f \in \mathcal{P}_k(\mathbb{R})$. If we replace $a$ by a variable $x$ and write $h_x(t)$ as $H(t, x)$, then the preceding equation reads:

$$f(x) = \int_{-1}^{+1} f(t) \cdot H(t, x)\, dt$$

The function of two variables $H(t, x)$ is called a **reproducing kernel** for $\mathcal{P}_k(\mathbb{R})$ because by integrating a polynomial against it we recover the polynomial. Reproducing kernels play an important role in many parts of functional analysis and partial differential equations, where they occur in more general spaces of functions than $\mathcal{P}(\mathbb{R})$.

## 15.5 Legendre Polynomials

In Section 15.2, Example 3, we examined the application of the Gram-Schmidt orthonormalization construction to the polynomials $1, t, t^2 \in \mathcal{P}_k(\mathbb{R})$. In this section we take a brief look at the general case, namely $1, t, \ldots, t^k$. We will concentrate on finding an orthogonal basis for $\mathcal{P}_k(\mathbb{R})$ and not worry about the normalization constants until the very end.

Let $L_0, L_1, \ldots, L_k$ be the result of applying the Gram–Schmidt orthogonalization process to the polynomials $1, t, , \ldots, t^k$. Then it is clear from the construction that

$$\operatorname{Span}\left\{1, \ldots, t^m\right\} = \left\{L_0, \ldots, L_m\right\} \qquad \forall\, m \mid 0 \le m \le k,$$

and therefore that $L_n$ is characterized up to a nonzero constant by the properties:

(1) $L_n$ is a nonzero polynomial of degree $n$.

(2) $L_n \perp \operatorname{Span}\left\{1, \ldots, t^{n-1}\right\}$.

To determine $L_n$ we make use of a formula (known as Rodrigues's formula) for the Legendre polynomials. To this end we set

$$R_n(t) = \frac{d^n}{dt^n}\left((t^2 - 1)^n\right), \qquad n = 0, 1, \ldots,$$

so $R_n(t)$ is a polynomial of degree $n$ in $t$. We call it the *n-th* **Rodrigues polynomial**.

**LEMMA 15.5.1:** *For each integer $n \geq 0$ the Rodrigues polynomials satisfy*

$$\text{Span}\{1, t, \ldots, t^n\} = \{R_0, R_1, \ldots, R_n\}.$$

**PROOF:** Let$k$ be a nonnegative integer. By direct computation, using the definition of $R_n$, we see that the coefficient of $t^k$ in $R_n$ is nonzero if $k = n$ and the coefficient of $t^k$ in $R_n$ is zero if $k > n$. Therefore, $\deg(R_n) = n$ for all $k \in \mathbb{N}_0$ and $\text{Span}\{R_0, R_1, \ldots, R_n\} \subseteq \text{Span}\{1, t, \ldots, t^n\}$. Moreover,

$$\begin{aligned}
R_1 &= r_{0,0} = 1, \\
R_1 &= r_{1,0} + r_{1,1}t, \\
R_2 &= r_{2,0} + r_{2,1}t + r_{2,2}t^2, \\
&\vdots \\
R_n &= r_{n,0} + r_{n,1}t + \cdots + r_{n,n}t^n.
\end{aligned}$$

The matrix $\mathbf{R} = [r_{i,j}]$ is strictly lower triangular and has all nonzero entries on the diagonal, and hence its determinant is nonzero. Therefore, $1, t, \ldots, t^n$ may be expressed (Cramer's rule) as linear combinations of $R_0, R_1, \ldots, R_n$ and hence $\text{Span}\{1, t, \ldots, t^n\} \subseteq \text{Span}\{R_0, R_1, \ldots, R_n\}$ for $n = 0, 1, \ldots$, from which the result follows. □

It is convenient to introduce the following notations:

(*a*) $p_n = (t^2 - 1)^n$.

(*b*) $f'$ is the derivative of the function $f$, and more generally.

(*c*) $f^{(k)}$ is the $k$-th derivative of the function $f$ if $k > 0$.

(*d*) $f^{(0)} = f$.

**LEMMA 15.5.2:** *Let $f$ and $h$ be polynomials, $s \in \mathbb{R}$, and suppose*
*(i) $h(s) \neq 0$, and*
*(ii) $f(t) = (t - s)^n h(t)$.*

*Then, if $n > 0$ we have*

$$f(s) = 0, \quad f'(s) = 0, \quad f^{n-1}(s) = 0, \quad f^n(s) \neq 0.$$

**PROOF:** If $n = 0$ there is nothing to prove, so we may assume $n > 0$. By direct computation we have

$$f'(t) = (t - s)^{n-1} h_1(t)$$

where

$$h_1(t) = nh(t) + (t - s)h'(t).$$

In particular,

$$h_1(s) = nh(s) \neq 0,$$

since $n > 0$. Therefore, the pair of functions $f'$ and $h_1$ satisfy the hypothesis of the lemma with $n$ replaced by $n-1$. If $n-1=0$ we are done, otherwise by induction

$$f'(s) = 0, \quad (f')'(s) = 0, \quad (f')^{(n-2)}(s) = 0, \quad (f')^{(n-1)}(s) \neq 0.$$

Since in general $(f')^{(k)} = f^{(k+1)}$, the result is established. □

**PROPOSITION 15.5.3 :** *For each positive integer $n$ the Rodrigues polynomial $R_n$ is orthogonal to the linear subspace* Span $\left\{1, t, \ldots, t^{n-1}\right\}$ $\subset \mathcal{P}(\mathbb{R})$ *for all $n$.*

**PROOF :** By definition

$$R_n = \frac{d^n}{dt^n}\left((t^2-1)^n\right).$$

The polynomial $p_n(t) = (t^2-1)^n$ may be factored into $(t-1)^n(t+1)^n$, and therefore, satisfies the preceding lemma for $s=-1$ and $s=1$, from which it follows that

$$\begin{aligned}
&p_n^{(i)}(1) = 0 \quad \text{for} \quad i = 1, \ldots, n-1,\\
&p_n^{(n)}(1) \neq 0,\\
&p_n^{(i)}(-1) = 0 \quad \text{for} \quad i = 1, \ldots, n-1,\\
&p_n^{(n)}(-1) \neq 0.
\end{aligned}$$

To compute

$$<t^i, R_n> = \int_{-1}^{+1} t^i R_n \, dt$$

we apply the rule for integration by parts and obtain

$$\begin{aligned}
<t^i, R_n> &= \int_{-1}^{+1} t^i p_n^{(n)}(t)\, dt = \int_{-1}^{+1} t^i (p_n^{(n-1)}(t))' \, dt\\
&= t^i p_n^{(n-1)}(t)\Big|_{-1}^{+1} - i\int_{-1}^{+1} t^{i-1} p_n^{(n-1)}(t)\, dt.
\end{aligned}$$

The first term in the last line above vanishes by Lemma 15.5.2. If we apply integration by parts to the second term, we obtain apart from a factor of $-i$,

$$t^{i-1}p_n^{(n-2)}(t)\Big|_{-1}^{+1} - (i-1)\int_{-1}^{+1} t^{i-2} p_n^{(n-2)}(t)\, dt,$$

and again the first term vanishes by Lemma 15.5.2. Hence by repeated integration by parts we obtain

$$
\begin{aligned}
<t^i, R_n> &= \int_{-1}^{+1} t^i\, p_n^{(n)}(t)\, dt = \int_{-1}^{+1} t^i\, (p_n^{(n-1)}(t))'\, dt \\
&= t^i p_n^{(n-1)}(t)\Big|_{-1}^{+1} - i \int_{-1}^{+1} t^{i-1}\, p_n^{(n-1)}(t)\, dt \\
&= -i \left[ t^{i-1} p_n^{(n-2)}(t)\Big|_{-1}^{+1} - (i-1) \int_{-1}^{+1} t^{i-2}\, p_n^{(n-2)}(t)\, dt \right] \\
&\ \ \vdots \\
&= 0,
\end{aligned}
$$

as required. $\square$

It therefore follows that the Legendre polynomial $L_n$ is equal to the Rodrigues polynomial $R_n(t) = \frac{d^n}{dt^n}\left[(t^2-1)^n\right]$ up to a nonzero constant factor. In fact, the exact formula is

$$
L_n(t) = \frac{1}{2^n n!}\frac{d^n}{dt^n}\left[(t^2-1)^n\right] = \sum_{m=0}^{n} \frac{(-1)^m (m+n)!}{2^m (m!)^2 (n-m)!}(1-t)^m
$$

for $n = 0, 1, \ldots$.

The Legendre polynomial $L_n$ satisfies the differential equation

$$
(1-t^2)\frac{d^2 y}{dt^2} - 2t\frac{dy}{dt} + n(n+1)y = 0,
$$

which is known as **Legendre's differential equation** of order $n$. This equation has been well studied, and the order need not be restricted to the positive integers; any real number can serve as the order. For integral values of the order, the equation has $L_n$ as one solution, and a power series with radius of convergence 1 as another. Together these span the space of all solutions in the sense that in a small neighborhood of 1 any solution is a linear combination of these two.

## 15.6 Exercises

**1.** Compute the following inner products in $\mathbb{R}^3$ relative to the standard inner product. Find the cosine of the angle between the indicated pairs of vectors:

(a) $<(1,2,3),(4,5,6)>$.

(b) $<(1,2,3),(3,2,1)>$.

(c) $<(1,0,1),(0,4,0)>$.

(d) $<(1,1,1),(1,-2,1)>$.

**2.** Compute the following inner products in $\mathcal{P}_2(\mathbb{R})$. Find the cosine of the angle between the indicated pair of vectors:

(a) $<1 + 2x + 3x^2, 4 + 5x + 6x^2>$.
(b) $<1 + 2x + 3x^2, 3 + 2x + x^2>$.
(c) $<1 + x^2, 4x>$.
(d) $<1 + x + x^2, 1 - 2x + x^2>$.

**3.** Compute the following inner products in $\mathrm{Mat}_{3,3}$ using the trace inner product:

(a) $\left\langle \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 0 \\ 0 & 1 & 3 \end{bmatrix}, \begin{bmatrix} 4 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & 4 \end{bmatrix} \right\rangle$.

(b) $\left\langle \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \right\rangle$.

**4.** Let $<—,—> : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by

$$<(x, y), (u, v)> = 5xu - 2(xv + yu) + yv.$$

(a) Show that $<—,—>$ is an inner product.
(b) Determine the lengths of the vectors

$$(1, 0), \quad (0, 1), \quad (1, 3), \quad (-1, 2)$$

using this inner product.
(c) Draw a picture of the circle of radius of length 1 ***in this inner product*** around the origin .

**5.** Show that the formula

$$<(a_i), (b_i)> = \sum_{i=0}^{\infty} \frac{a_i b_i}{i^2}$$

defines an inner product on the vector space $\ell_\infty$ of all bounded sequences of real numbers. Find a ***nonzero***, ***proper*** subspace $\mathcal{U}$, i.e., $\{\mathbf{0}\} \subsetneq \mathcal{U} \subsetneq \ell_\infty$, such that $\mathcal{U}^\perp = \{\mathbf{0}\}$.

**6.** Let $w(x)$ be a nonnegative-valued function on the interval $[-1, 1]$, and for polynomials $p(x), q(x)$ define

$$<p(x), q(x)> = \int_{-1}^{1} p(x)q(x)w(x)dx.$$

Show that $<,>$ is an inner product on $\mathcal{P}_n(\mathbb{R})$.

**7.** Let $(\mathcal{V}, <—,—>)$ be an inner product space and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ an orthogonal set of vectors in $\mathcal{V}$. Establish the following:

**Bessel's inequality:** $$\sum_{i=1}^{n} \frac{|<\mathbf{A}, \mathbf{B}_i>|^2}{|\mathbf{A}_i|^2} \le |\mathbf{B}|^2.$$

**Parseval's identity:** $$<\mathbf{B}, \mathbf{C}> = \sum <\mathbf{B}, \mathbf{A}_i> \cdot <\mathbf{A}_i, \mathbf{C}>.$$

**8.** Let $\mathcal{V}$ be an inner product space. Prove the following (draw pictures in $\mathbb{R}^2$ to see what is going on!)

**Law of cosines:** $$|\mathbf{A} - \mathbf{B}|^2 = |\mathbf{A}|^2 + |\mathbf{B}|^2 - 2|\mathbf{A}|\,|\mathbf{B}| \cos(\vartheta),$$

where $\vartheta$ is the angle between $\mathbf{A}$ and $\mathbf{B}$.

**9.** Which of the following sets of vectors in $\mathbb{R}^3$ are orthogonal?

(a) $\{(0,0,0), (1,1,1), (0,1,-1)\}$.
(b) $\{(1,1,1), (1,-2,1), (-2,1,-2)\}$.
(c) $\{(1,0,0), (0,1,0), (0,0,1), (1,1,1)\}$.
(d) $\{(1,-1,2), (2,0,-1), (0,1,1), (0,0,1)\}$.

**10.** Which of the following sets of vectors in $\mathcal{P}_3(\mathbb{R})$ are orthogonal? (Use the inner product of Section 15.1 Example 3.)

(a) $\left\{1, x - \frac{1}{2}\right\}$.
(b) $\{1, x - 1\}$.
(c) $\left\{x, x^2 - 1\right\}$.
(d) $\left\{x, 3x^2 - 2, 2x^3\right\}$.

**11.** Which of the following sets of vectors in $\mathrm{Mat}_{3,3}$ are orthogonal? (Use the trace scalar product of Section 15.1 example 4

(a) $$\left\{ \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 3 & 4 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \right\}.$$

(b) $$\left\{ \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}, \begin{bmatrix} -3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \right\}.$$

**12.** Find an orthogonal basis for each of the indicated subspaces:

(a) $$\left\{(x, y, z) \in \mathbb{R}^3 \mid x + y - z = 0\right\}.$$

(b) $$\left\{p(x) \in \mathcal{P}_3(\mathbb{R}) \mid x \frac{d}{dx} p(x) = p(x)\right\}.$$

(c) $$\left\{\mathbf{A} \in \mathrm{Mat}_{3,3} \mid \mathrm{tr}(\mathbf{A}) = 0\right\}.$$

**13.** Find the orthogonal complement of each of the subspaces in the preceding exercise.

**14.** Which of the following linear transformations are isometric embeddings?

(a) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^4 \quad \mathsf{T}(x, y, z) = (x + y, y + z, z + x, x + y + z)$.

(b) $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^3 \quad \mathsf{T}(x, y) = (x, 0, y)$.

(c) $\mathsf{T} : \mathcal{P}_2(\mathbb{R}) \longrightarrow \mathcal{P}_3(\mathbb{R}) \quad \mathsf{T}(p(x)) = xp(x)$.

(d) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathrm{Mat}_{3,3} \quad \mathsf{T}(x, y, z) = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$.

(e) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathrm{Mat}_{3,3} \quad \mathsf{T}(x, y, z) = \begin{bmatrix} x & 0 & 0 \\ y & 0 & 0 \\ z & 0 & 0 \end{bmatrix}$.

**15.** Let $\mathcal{V}$ be a finite-dimensional inner product space and $\mathcal{S} \subseteq \mathcal{V}$ a subset. Show that $\mathcal{S} = \mathcal{S}^{\perp\perp}$ if and only if $\mathcal{S}$ is a vector subspace of $\mathcal{V}$. What happens if $\mathcal{V}$ is *not* finite-dimensional?

**16.** A linear transformation $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ is called a **rotation** if $\mathsf{T}$ is an isometry and $\det(\mathsf{T}) > 0$. (See also Exercise 23 in Chapter 12.) Show that the matrix of $\mathsf{T}$ relative to the standard basis must be

$$\begin{bmatrix} \cos(\vartheta) & -\sin(\vartheta) \\ \sin(\vartheta) & \cos(\vartheta) \end{bmatrix}$$

for a unique angle $\vartheta$. In fact, $\cos(\vartheta) = \langle \mathbf{E}_1, \mathsf{T}(\mathbf{E}_1) \rangle$. (Draw a picture.)

**17.** If $\mathsf{T}_1, \mathsf{T}_2 : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ are rotations, show that $\mathsf{T}_1\mathsf{T}_2$ is also a rotation. Show that $\mathsf{T}_1\mathsf{T}_2 = \mathsf{T}_2\mathsf{T}_1$.

**18.** Suppose $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ is an orthonormal basis for $\mathcal{V}$. If $\mathbf{A}$ is a vector in $\mathcal{V}$ such that

$$\langle \mathbf{A}, \mathbf{A}_i \rangle = 0, \qquad i = 1, 2, \ldots, n,$$

show that $\mathbf{A} = \mathbf{0}$.

**19.** Verify that the trace of a square matrix has the following properties for any two square matrices $\mathsf{A}$ and $\mathsf{B}$ and numbers $a$:

(a) $\mathrm{tr}(\mathsf{A}) = \mathrm{tr}(\mathsf{A}^{\mathrm{tr}})$.
(b) $\mathrm{tr}(\mathsf{AB}) = \mathrm{tr}(\mathsf{BA})$.
(c) $\mathrm{tr}(\mathsf{A} + \mathsf{B}) = \mathrm{tr}(\mathsf{A}) + \mathrm{tr}(\mathsf{B})$.
(d) $\mathrm{tr}(a\mathsf{A}) = a\,\mathrm{tr}(\mathsf{A})$.
(e) If $\mathsf{A}$ is invertible then $\mathrm{tr}(\mathsf{ABA}^{-1}) = \mathrm{tr}(\mathsf{B})$.

**20.** Show that there do not exist square matrices $\mathsf{A}$ and $\mathsf{B}$ such that $\mathsf{AB} - \mathsf{BA} = \mathsf{I}$. (HINT: see the previous exercise.)

The notion of inner product can also be introduced in complex vector spaces by a slight modification (remember: definitions are useful or useless, ***not*** true or false) of the axioms we have used in the real case. To this end recall that for a complex number $z = a + b\boldsymbol{i}$ ($a$, $b$ real numbers) the **complex conjugate** of $z$ is the complex number

$$\overline{z} = a - b\boldsymbol{i}.$$

It is very important to note that

$$z\overline{z} = (a + b\boldsymbol{i})(a - b\boldsymbol{i}) = a^2 - b^2 i^2 = a^2 + b^2$$

is a nonnegative real number. It is zero if and only if $z = 0$.

**21.** Let $\mathsf{T} : \mathbb{C} \longrightarrow \mathbb{R}^2$ be given by $\mathsf{T}(a + bi) = (a, b)$. Show that $\mathsf{T}$ is an isomorphism of real vector spaces and $|\mathsf{T}(z)|^2 = z\overline{z}$.

DEFINITION: *If $\mathcal{V}$ is a complex vector space, a* **Hermitian inner product** *on $\mathcal{V}$ is a function that assigns to each pair* $\mathbf{A}, \mathbf{B}$ *of vectors in $\mathcal{V}$ a complex number* $<\mathbf{A}, \mathbf{B}>$ *such that*

(i) $<\mathbf{A}, \mathbf{B}> = \overline{<\mathbf{B}, \mathbf{A}>}$ *for all* $\mathbf{A}, \mathbf{B} \in \mathcal{V}$.

(ii) *For all* $\mathbf{A}, \mathbf{B} \in \mathcal{V}$ *and* $z \in \mathbb{C}$ *we have*

$$<z\mathbf{A}, \mathbf{B}> = z<\mathbf{A}, \mathbf{B}>,$$

$$<\mathbf{A}, z\mathbf{B}> = \overline{z}<\mathbf{A}, \mathbf{B}>.$$

(iii) *For all vectors* $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathcal{V}$ *we have*

$$<\mathbf{A}, \mathbf{B} + \mathbf{C}> = <\mathbf{A}, \mathbf{B}> + <\mathbf{A}, \mathbf{C}>,$$

$$<\mathbf{A} + \mathbf{B}, \mathbf{C}> = <\mathbf{A}, \mathbf{C}> + <\mathbf{B}, \mathbf{C}>.$$

(iv) *For all* $\mathbf{A}$ *in* $\mathcal{V}$, $<\mathbf{A}, \mathbf{A}> \geq 0$ *and* $<\mathbf{A}, \mathbf{A}> = 0$ *if and only if* $\mathbf{A} = \mathbf{0}$.

**22.** On $\mathbb{C}^n$ define

$$<\mathbf{Z}, \mathbf{W}> = \sum_{i=1}^{n} z_i \overline{w}_i,$$

where $\mathbf{Z} = (z_1, z_2, \ldots, z_n)$, $\mathbf{W} = (w_1, w_2, \ldots, w_n)$. Verify that $<—, —>$ is an Hermitian inner product on $\mathbb{C}^n$.

**23.** Compute the following inner products in $\mathbb{C}^3$:

$$<(3, \boldsymbol{i}, 2 - \boldsymbol{i}), (5, -\boldsymbol{i}, 1)>.$$
$$<(\boldsymbol{i}, \boldsymbol{i}, \boldsymbol{i}), (1, 1, 1)>.$$
$$<(\boldsymbol{i}, 1, \boldsymbol{i}), (1, \boldsymbol{i}, 1)>.$$

**24.** The **length** of a vector $\mathbf{Z}$ in a complex inner product space is by definition

$$|\mathbf{Z}| = <\mathbf{Z}, \mathbf{Z}>^{1/2}.$$

Verify the triangle inequality in the complex case.

**25.** A basis for a complex inner product space is **orthonormal** if it consists of orthogonal vectors of length 1, just as in the real case. Which of the following are orthonormal bases for $\mathbb{C}^3$?

(1) $(\boldsymbol{i}, 0, 0), (0, \boldsymbol{i}, 0), (0, 0, \boldsymbol{i})$.
(2) $(1, 0, 0), (0, \boldsymbol{i}, 0), (0, 0, 1)$.
(3) $\sqrt{\frac{1}{2}}(1 + \boldsymbol{i}, 1 - \boldsymbol{i}, 0), \sqrt{\frac{1}{2}}(0, 1, 1), (0, 0, 1)$.
(4) $(1, \boldsymbol{i}, 0), (-1, \boldsymbol{i}, 0), (0, 0, 1)$.

**26.** What changes, if any, are needed to prove the theorems of this chapter in the complex case. In particular, show that two complex inner product spaces of finite dimensions are isometric if and only if they have the same dimension. In particular, a complex inner product space of dimension $n$ is isometric to $\mathbb{C}^n$.

**27.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in an $n$-dimensional inner product space $\mathcal{V}$. $\mathsf{T}$ is called an **orthogonal reflection** if there exists an $(n-1)$-dimensional subspace $\mathcal{H}$ in $\mathcal{V}$ and a (nonzero) vector of unit length $\mathbf{V}$ orthogonal to $\mathcal{H}$ (orthogonality of vectors, or sets of vectors, is defined as in the real case) such that

$$\mathsf{T}(\mathbf{H}) = \mathbf{H} \quad \text{for all} \quad \mathbf{H} \in \mathcal{H},$$
$$\mathsf{T}(\mathbf{V}) = -\mathbf{V}.$$

Show that

$$\mathsf{T}(\mathbf{W}) = \mathbf{W} - 2<\mathbf{W}, \mathbf{V}> \cdot \mathbf{V} \qquad \forall\, \mathbf{W} \in \mathcal{W}.$$

**28.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a reflection. Show that with respect to a properly chosen inner product, $\mathsf{T}$ is orthogonal.

**29.** Let $S$ be a finite set, Fun($S$) the vector space of real-valued functions on $S$, and $\varphi : S \longrightarrow S$ a function. Define

$$\varphi^* : \mathrm{Fun}(S) \longrightarrow \mathrm{Fun}(S)$$

by

$$\varphi^*(f)(s) = f(\varphi(s)).$$

By Chapter 8 Exercise 28, $\varphi^*$ is a linear transformation. Show that the trace of $\varphi^*$ is the number of fixed points of $\varphi$. (A point $x \in S$ is called a **fixed point** of $\varphi$ if and only if $\varphi(x) = x$.)

# Chapter 16
# The Spectral Theorem and Quadratic Forms

So far in our study of linear transformations we have concentrated on trying to find conditions that assure that the matrix of the transformation has a particular form. We have not asked the related question what are the properties of those transformations whose matrices are *assumed* to have a particularly simple form. There is, in fact, a good reason for this, and it is tied up with our work of the last chapter. For example we might propose to study those linear transformations whose matrices are symmetric. We would therefore like to introduce the following:

**PROPOSED DEFINITION:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation. We say that* $\mathsf{T}$ *is* **symmetric** *if there is a basis* $\{\mathbf{A}_1, \dots, \mathbf{A}_n\}$ *for* $\mathcal{V}$ *such that the matrix of* $\mathsf{T}$ *relative to this basis is symmetric.*

One reason we have indicated that this is only a proposed definition, and not one we intend to work with, is because it is not really a useful concept. Remember, a definition is not right or wrong, only useful or useless. The preceding one is pretty much useless. The following example will help illustrate why.

**EXAMPLE:** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation given by

$$\mathsf{T}(x,\ y) = (x + 2y,\ 2x + y).$$

Let us calculate the matrix of $\mathsf{T}$ relative to the standard basis of $\mathbb{R}^2$. We find

$$\mathsf{T}(1, 0) = (1, 2), \qquad \mathsf{T}(0, 1) = (2, 1)$$

so the matrix we seek is

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix},$$

which is a symmetric matrix. On the other hand, we may compute the matrix of $\mathsf{T}$ relative to the basis $\{\mathbf{F}_1 = (1,1), \mathbf{F}_2 = (1,0)\}$. We get

$$\begin{aligned} \mathsf{T}(1,1) &= (3,3) = 3\mathbf{F}_1 + 0\mathbf{F}_2, \\ \mathsf{T}(1,0) &= (1,2) = 2\mathbf{F}_1 - \mathbf{F}_2, \end{aligned}$$

so the required matrix is

$$\mathsf{B} = \begin{bmatrix} 3 & 2 \\ 0 & -1 \end{bmatrix},$$

which is not symmetric.

The preceding example illustrates a crucial drawback to using the proposed definition of symmetric transformation. ***The concept is not a property of the linear transformation that is invariant under change of basis.*** This has several consequences. For one, as defined, symmetry is a property of the coordinate system in which the matrix of the transformation is computed. Thus we will have to specify this basis and work only with it if we wish to take advantage of the symmetry property. Secondly, and more importantly, a linear transformation $\mathsf{T}$ might be symmetric in terms of the proposed definition and we would never know it because we are unable to discover a basis in which $\mathsf{T}$ has a symmetric matrix!

## 16.1 Self-Adjoint Transformations

The aforementioned difficulties can be overcome by introducing a slightly different concept, that depends only on the linear transformation and not on the choice of basis. However, we must assume that the vector space in question is equipped with a *fixed* inner product. The particular definition we are going to give was arrived at through a process of trial, error, and experimentation; the basis of all modern science.

We begin with an important construction.

LEMMA 16.1.1: *Let $\mathcal{V}$ be a finite-dimensional inner product space and $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ a linear endomorphism. For each vector $\mathbf{A} \in \mathcal{V}$ define the function*

$$\mathsf{T}_{\mathbf{A}} : \mathcal{V} \longrightarrow \mathbb{R} \quad \text{by} \quad \mathsf{T}_{\mathbf{A}}(\mathbf{B}) = \langle \mathsf{T}(\mathbf{B}), \mathbf{A} \rangle,$$

*where $\langle -, - \rangle$ is the inner product in $\mathcal{V}$. Then the function $\mathsf{T}_{\mathbf{A}} : \mathcal{V} \longrightarrow \mathbb{R}$ is a linear transformation.*

**PROOF:** Suppose $\mathbf{B}_1 + \mathbf{B}_2$ belong to $\mathcal{V}$. Then

$$\begin{aligned} \mathsf{T}_{\mathbf{A}}(\mathbf{B}_1 + \mathbf{B}_2) &= \langle \mathsf{T}(\mathbf{B}_1 + \mathbf{B}_2), \mathbf{A} \rangle \\ &= \langle \mathsf{T}(\mathbf{B}_1) + \mathsf{T}(\mathbf{B}_2), \mathbf{A} \rangle \\ &= \langle \mathsf{T}(\mathbf{B}_1), \mathbf{A} \rangle + \langle \mathsf{T}(\mathbf{B}_2), \mathbf{A} \rangle \\ &= \mathsf{T}_{\mathbf{A}}(\mathbf{B}_1) + \mathsf{T}_{\mathbf{A}}(\mathbf{B}_2). \end{aligned}$$

A similar computation shows that $\mathsf{T}_{\mathbf{A}}$ preserves scalar multiples, and hence $\mathsf{T}_{\mathbf{A}}$ is a linear transformation. $\square$

Suppose again that $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is a linear transformation in the finite-dimensional inner product space $\mathcal{V}$. For each $\mathbf{A}$ in $\mathcal{V}$ there is by the lemma the linear transformation

$$\mathsf{T}_{\mathbf{A}} : \mathcal{V} \longrightarrow \mathbb{R} \quad \text{given by} \quad \mathsf{T}_{\mathbf{A}}(\mathbf{B}) = \langle \mathsf{T}(\mathbf{B}), \mathbf{A} \rangle.$$

According to Theorem 15.4.5 there exists a unique vector $\mathbf{A}^*$ in $\mathcal{V}$ such that

$$\mathsf{T}_{\mathbf{A}}(\mathbf{B}) = \langle \mathbf{B}, \mathbf{A}^* \rangle$$

for all $\mathbf{B}$ in $\mathcal{V}$. Define a function

$$\mathsf{T}^* : \mathcal{V} \longrightarrow \mathcal{V} \quad \text{by} \quad \mathsf{T}^*(\mathbf{A}) = \mathbf{A}^*.$$

The function $\mathsf{T}^*$ is called the **adjoint** of $\mathsf{T}$.

**PROPOSITION 16.1.2:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in the finite-dimensional inner product space $\mathcal{V}$. Then the adjoint of $\mathsf{T}$,*

$$\mathsf{T}^* : \mathcal{V} \longrightarrow \mathcal{V},$$

*is also a linear transformation.*

**PROOF:** We are going to repeatedly use the ***uniqueness*** clause of the Riesz representation Theorem 15.4.5.

The adjoint of $\mathsf{T}$ is characterized by the equations

$$\langle \mathsf{T}(\mathbf{B}), \mathbf{A} \rangle = \mathsf{T}_{\mathbf{A}}(\mathbf{B}) = \langle \mathbf{B}, \mathsf{T}^*(\mathbf{A}) \rangle$$

for all vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$. So suppose that $\mathbf{A}_1, \mathbf{A}_2$ belong to $\mathcal{V}$. Then for any $\mathbf{B}$ in $\mathcal{V}$

$$\begin{aligned} \langle \mathbf{B}, \mathsf{T}^*(\mathbf{A}_1 + \mathbf{A}_2) \rangle &= \langle \mathsf{T}(\mathbf{B}), \mathbf{A}_1 + \mathbf{A}_2 \rangle \\ &= \langle \mathsf{T}(\mathbf{B}), \mathbf{A}_1 \rangle + \langle \mathsf{T}(\mathbf{B}), \mathbf{A}_2 \rangle \\ &= \langle \mathbf{B}, \mathsf{T}^*(\mathbf{A}_1) \rangle + \langle \mathbf{B}, \mathsf{T}^*(\mathbf{A}_2) \rangle \\ &= \langle \mathbf{B}, \mathsf{T}^*(\mathbf{A}_1) + \mathsf{T}^*(\mathbf{A}_2) \rangle. \end{aligned}$$

If we set $\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2$, then the above equations show that

$$<\mathbf{B}, \mathsf{T}^*(\mathbf{A}_1 + \mathbf{A}_2)> = \mathsf{T}_{\mathbf{A}}(\mathbf{B}) = <\mathbf{B}, \mathsf{T}^*(\mathbf{A}_1) + \mathsf{T}^*(\mathbf{A}_2)>$$

for all $\mathbf{B}$ in $\mathcal{V}$. By the ***uniqueness*** part of Theorem 15.4.5 we therefore find

$$\mathsf{T}^*(\mathbf{A}_1 + \mathbf{A}_2) = \mathsf{T}^*(\mathbf{A}_1) + \mathsf{T}^*(\mathbf{A}_2)$$

so $\mathsf{T}^*$ preserves vector sums.

To show that $\mathsf{T}^*$ preserves scalar products we take $\mathbf{A}$ in $\mathcal{V}$ and $r$ a number. Then for all $\mathbf{B}$ in $\mathcal{V}$

$$\begin{aligned} <\mathbf{B}, \mathsf{T}^*(r\mathbf{A})> &= <\mathsf{T}(\mathbf{B}), r\mathbf{A}> \\ &= r<\mathsf{T}(\mathbf{B}), \mathbf{A}> \\ &= r<\mathbf{B}, \mathsf{T}^*(\mathbf{A})> \\ &= <\mathbf{B}, r\mathsf{T}^*(\mathbf{A})>, \end{aligned}$$

and therefore

$$<\mathbf{B}, \mathsf{T}^*(r\mathbf{A})> = \mathsf{T}_{r\mathbf{A}}(\mathbf{B}) = <\mathbf{B}, r\mathsf{T}^*(\mathbf{A})>$$

and the uniqueness part of Theorem 15.4.5 yields

$$\mathsf{T}^*(r\mathbf{A}) = r\mathsf{T}^*(\mathbf{A})$$

so that $\mathsf{T}^*$ preserves scalar products. □

**PROPOSITION 16.1.3:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in the finite-dimensional inner product space $\mathcal{V}$. Then*

$$<\mathsf{T}(\mathbf{B}), \mathbf{A}> = <\mathbf{B}, \mathsf{T}^*(\mathbf{A})>$$

*for all $\mathbf{A}, \mathbf{B} \in \mathcal{V}$.*

**PROOF:** This follows instantly from the formulas

$$\mathsf{T}_{\mathbf{A}}(\mathbf{B}) = <\mathsf{T}(\mathbf{B}), \mathbf{A}>,$$
$$\mathsf{T}_{\mathbf{A}}(\mathbf{B}) = <\mathbf{B}, \mathsf{T}^*(\mathbf{A})>,$$

which have already found application in the proof of Proposition 16.1.2. □

Let us work a few numerical examples.

**EXAMPLE 1:** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation given by

$$\mathsf{T}(x, y) = (x + 2y, 2x + y).$$

Calculate $\mathsf{T}^* : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$.

**SOLUTION:** To compute $\mathsf{T}^*(u, v)$ we will use the equation

$$<(x, y), \mathsf{T}^*(u, v)> = <\mathsf{T}(x, y), (u, v)>$$

for all $(x, y), (u, v)$ in $\mathbb{R}^2$. We find that

$$\begin{aligned} <(x, y), \mathsf{T}^*(u, v)> &= <\mathsf{T}(x, y), (u, v)> \\ &= <(x + 2y, y + 2x), (u, v)> \\ &= xu + 2yu + yv + 2xv \\ &= xu + 2xv + yv + 2yu \\ &= x(u + 2v) + y(v + 2u) \\ &= <(x, y), (u + 2v, v + 2u)>, \end{aligned}$$

and therefore

$$\mathsf{T}^*(u, v) = (u + 2v, v + 2u),$$

or if you prefer $x$ and $y$ to $u$ and $v$;

$$\mathsf{T}^*(x, y) = (x + 2y, y + 2x).$$

Therefore, $\mathsf{T}^* = \mathsf{T}$.

**EXAMPLE 2:** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation given by

$$\mathsf{T}(x, y) = (x + 3y, 2x + y).$$

Calculate $\mathsf{T}^* : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$.

**SOLUTION:** We have

$$\begin{aligned} <(x, y), \mathsf{T}^*(u, v)> &= <\mathsf{T}(x, y), (u, v)> \\ &= <(x + 3y, 2x + y), (u, v)> \\ &= xu + 3yu + 2xv + yv \\ &= xu + 2xv + 3yu + yv \\ &= x(u + 2v) + y(3u + v) \\ &= <(x, y), (u + 2v, 3u + v)>, \end{aligned}$$

and therefore

$$\mathsf{T}^*(u, v) = (u + 2v, 3u + v),$$

and so $\mathsf{T} \neq \mathsf{T}^*$.

**Definition:** *Let $\mathcal{V}$ be a finite-dimensional inner product space. A linear transformation* $\mathsf{T} : \mathcal{V} \longrightarrow V$ *is called* **self-adjoint** *if* $\mathsf{T} = \mathsf{T}^*$.

Notice that if $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is self-adjoint, then since $\mathsf{T} = \mathsf{T}^*$

$$<\mathsf{T}(\mathbf{B}), \mathbf{A}> = <\mathbf{B}, \mathsf{T}(\mathbf{A})>$$

for all vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$. On the other hand, if $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is a linear transformation such that

$$<\mathsf{T}(\mathbf{B}), \mathbf{A}> = <\mathbf{B}, \mathsf{T}(\mathbf{A})>$$

for all vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$, then $\mathsf{T}$ is self-adjoint. To see this, recall that we always have the equation

$$<\mathsf{T}(\mathbf{B}), \mathbf{A}> = <\mathbf{B}, \mathsf{T}^*(\mathbf{A})>$$

for all vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$. Hence for our hypothesized $\mathsf{T}$ we will have

$$<\mathbf{B}, \mathsf{T}(\mathbf{A})> = <\mathbf{B}, \mathsf{T}^*(\mathbf{A})>$$

for all vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$. This may be rewritten as

$$<\mathbf{B}, \mathsf{T}(\mathbf{A}) - \mathsf{T}^*(\mathbf{A})> = 0$$

for all vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$. In particular, putting $\mathbf{B} = \mathsf{T}(\mathbf{A}) - \mathsf{T}^*(\mathbf{A})$, we find that

$$<\mathsf{T}(\mathbf{A}) - \mathsf{T}^*(\mathbf{A}), \mathsf{T}(\mathbf{A}) - \mathsf{T}^*(\mathbf{A})> = 0$$

for all vectors $\mathbf{A}$ in $\mathcal{V}$. Therefore, $\mathsf{T}(\mathbf{A}) - \mathsf{T}^*(\mathbf{A}) = \mathbf{0}$ for all vectors $\mathbf{A}$ in $\mathcal{V}$, or what is the same thing,

$$\mathsf{T}(\mathbf{A}) = \mathsf{T}^*(\mathbf{A})$$

for all vectors $\mathbf{A}$ in $\mathcal{V}$. Hence $\mathsf{T} = \mathsf{T}^*$, so that $\mathsf{T}$ is self-adjoint, as we claimed.

As an example of how we may apply the above fact to check whether a transformation is self-adjoint consider the linear transformation

$$\mathsf{T} : \mathrm{Mat}_{n,n} \longrightarrow \mathrm{Mat}_{n,n}$$

given by

$$\mathsf{T}(\mathbf{A}) = \mathbf{A}^{\mathrm{tr}}$$

where $\mathbf{A}^{\mathrm{tr}}$ is the transpose of the $n \times n$ matrix $\mathbf{A}$. Recall (see Section 15.1 Example 4) that the vector space $\mathrm{Mat}_{n,n}$ has an inner product defined by

$$<\mathbf{A}, \mathbf{B}> = \mathrm{tr}(\mathbf{A}\mathbf{B}^{\mathrm{tr}})$$

and with respect to this inner product we have

$$\begin{aligned} <\mathbf{T}(\mathbf{B}^{tr}), \mathbf{A}> &= \mathrm{tr}(\mathbf{T}(\mathbf{B}^{tr})\mathbf{A}^{tr}) \\ &= \mathrm{tr}((\mathbf{B}^{tr})^{tr}\mathbf{A}^{tr}) \\ &= \mathrm{tr}(\mathbf{B}\mathbf{A}^{tr}) \end{aligned}$$

and

$$\begin{aligned} <\mathbf{B}^{tr}, \mathbf{T}(\mathbf{A})> &= \mathrm{tr}(\mathbf{B}^{tr}(\mathbf{T}(\mathbf{A}))^{tr}) \\ &= \mathrm{tr}(\mathbf{B}^{tr}(\mathbf{A}^{tr})^{tr}) \\ &= \mathrm{tr}(\mathbf{B}^{tr}\mathbf{A}). \end{aligned}$$

Using the definition of the trace, it is very easy to verify (see Chapter 15 Exercise 19) that

$$\mathrm{tr}(\mathbf{C}^{tr}) = \mathrm{tr}(\mathbf{C}),$$

$$\mathrm{tr}(\mathbf{CD}) = \mathrm{tr}(\mathbf{DC}).$$

Recall also that

$$(\mathbf{CD})^{tr} = \mathbf{D}^{tr}\mathbf{C}^{tr}.$$

We therefore may compute as follows:

$$<\mathbf{T}(\mathbf{B}^{tr}), \mathbf{A}> = \mathrm{tr}(\mathbf{B}\mathbf{A}^{tr}) = \mathrm{tr}(\mathbf{B}^{tr}\mathbf{A}) = <\mathbf{B}^{tr}, \mathbf{T}(\mathbf{A})>,$$

and hence $\mathbf{T} : \mathrm{Mat}_{n,n} \longrightarrow \mathrm{Mat}_{n,n}$ must be self-adjoint.

Of course there are other methods to check whether a linear transformation is self-adjoint. One such method is to examine the matrix relative to an orthonormal basis.

**THEOREM 16.1.4 :** *Let $\mathcal{V}$ be a finite-dimensional inner product space, $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ a linear transformation, and $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ an orthonormal basis for $\mathcal{V}$. Let $\mathbf{M}$ be the matrix of $\mathbf{T}$ relative to the basis $\{\mathbf{A}_1, \cdots, \mathbf{A}_n\}$ and $\mathbf{N}$ the matrix of $\mathbf{T}^*$ relative to the same basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. Then $\mathbf{N} = \mathbf{M}^{tr}$.*

**PROOF:** What we need is Theorem 15.2.9 and the definitions of adjoint and transpose. Let $(\mathbf{M}) = (m_{i,j})$ and $\mathbf{N} = (n_{i,j})$. Then

$$\mathbf{T}(\mathbf{A}_j) = \sum m_{i,j}\mathbf{A}_i,$$

$$\mathbf{T}^*(\mathbf{A}_s) = \sum n_{r,s}\mathbf{A}_r.$$

According to Theorem 15.2.9

$$m_{i,j} = <\mathbf{T}(\mathbf{A}_j), \mathbf{A}_i>$$

and

$$n_{r,s} = <\mathbf{T}^*(\mathbf{A}_s), \mathbf{A}_r>.$$

But

$$\langle \mathsf{T}(\mathbf{A}_j), \mathbf{A}_i \rangle = \langle \mathbf{A}_j, \mathsf{T}^*(\mathbf{A}_i) \rangle = \langle \mathsf{T}^*(\mathbf{A}_i), \mathbf{A}_j \rangle,$$

so that

$$m_{i,j} = n_{j,i}.$$

But this just says that $\mathsf{N} = \mathsf{M}^{\mathrm{tr}}$ as required. □

**THEOREM 16.1.5 :** *Let $\mathcal{V}$ be a finite-dimensional inner produce space and $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ a self-adjoint linear transformation. If $\{\mathbf{A}_1, \dots, \mathbf{A}_n\}$ is an orthonormal basis for $\mathcal{V}$ then the matrix of $\mathsf{T}$ relative to this basis is symmetric.*

*For this reason self-adjoint transformations are often called* **symmetric linear transformations** *or* **symmetric linear operators**.

**PROOF:** Let $\mathsf{M}$ be the matrix of $\mathsf{T}$ and $\mathsf{N}$ the matrix of $\mathsf{T}^*$, both relative to the basis $\{\mathbf{A}_1, \dots, \mathbf{A}_n\}$. Then $\mathsf{N} = \mathsf{M}^{\mathrm{tr}}$ by Theorem 16.1.4. But since $\mathsf{T}$ is self-adjoint $\mathsf{T} = \mathsf{T}^*$, so $\mathsf{M} = \mathsf{N}$. Thus $\mathsf{M} = \mathsf{N} = \mathsf{M}^{\mathrm{tr}}$. That is, $\mathsf{M} = \mathsf{M}^{\mathrm{tr}}$ so $\mathsf{M}$, is symmetric. □

**EXAMPLE 3:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation given by

$$\mathsf{T}(x, y, z) = (x - y + z, x + y, z - x).$$

Determine whether if $\mathsf{T}$ is self-adjoint.

**SOLUTION:** The matrix of $\mathsf{T}$ relative to the standard orthonormal basis may be easily computed. We get

$$\mathsf{T}(1, 0, 0) = (1, 1, -1),$$
$$\mathsf{T}(0, 1, 0) = (-1, 1, 0),$$
$$\mathsf{T}(0, 0, 1) = (1, 0, 1),$$

so the required matrix is

$$\mathsf{M} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

Then

$$\mathsf{M}^{\mathrm{tr}} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

so that $\mathsf{M} \neq \mathsf{M}^{\mathrm{tr}}$; that is, $\mathsf{M}$ is not symmetric. By Theorem 16.1.5 it follows that $\mathsf{T}$ is not self-adjoint.

Actually we have enough information to compute the adjoint $\mathsf{T}^*: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$. In view of Theorem 16.1.4 we need only calculate

$$\mathsf{M}^{tr}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y-z \\ y-x \\ x+z \end{bmatrix},$$

from which we get the formula

$$\mathsf{T}^*(x, y, z) = (x + y - z, y - x, x + z).$$

Of course, we could also calculate $\mathsf{T}^*$ using the method of Examples 2 and 3. You ought to try this.

**THEOREM 16.1.6:** *Let $\mathsf{T}: \mathcal{V} \longrightarrow \mathcal{V}$ be an endomorphism of the finite-dimensional inner product space $\mathcal{V}$. Let $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ be an orthonormal basis for $\mathcal{V}$ and suppose that the matrix of $\mathsf{T}$ is symmetric with respect to this basis. Then $\mathsf{T}$ is self-adjoint.*

**PROOF:** Let the matrix of $\mathsf{T}$ relative to the basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ be $\mathsf{M}$. By Theorem 16.1.4 the matrix of $\mathsf{T}^*$ relative to this basis is also $\mathsf{M}$, since $\mathsf{M}$ is symmetric. Therefore, by Theorem 11.2.1, $\mathsf{T} = \mathsf{T}^*$. □

**ALTERNATIVE PROOF:** Let $\mathsf{M}$ be the matrix of $\mathsf{T}$ relative to the basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. Let $\mathbf{A}$ and $\mathbf{B}$ be vectors in $\mathcal{V}$ with

$$\begin{aligned} \mathbf{A} &= a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + \cdots + a_n\mathbf{A}_n, \\ \mathbf{B} &= b_1\mathbf{A}_1 + b_2\mathbf{A}_2 + \cdots + b_n\mathbf{A}_n. \end{aligned}$$

Then if $\mathsf{M} = (m_{i,j})$,

$$\mathsf{T}(\mathbf{A}) = \sum_{i,j} m_{i,j}a_j\mathbf{A}_i,$$

$$\mathsf{T}(\mathbf{B}) = \sum_{i,j} m_{i,j}b_j\mathbf{A}_i.$$

Applying Theorem 15.4.5 we get

$$<\mathsf{T}(\mathbf{A}), \mathbf{B}> = \sum_{i,j} m_{i,j}a_jb_i = \sum_{r,s} a_rb_sm_{s,r},$$

$$<\mathbf{A}, \mathsf{T}(\mathbf{B})> = \sum_{i,j} a_im_{i,j}b_j = \sum_{r,s} a_rb_sm_{r,s}.$$

Since $\mathsf{M}$ is symmetric, $m_{r,s} = m_{s,r}$, so we get

$$<\mathsf{T}(\mathbf{A}), \mathbf{B}> = \sum_{r,s} a_rb_sm_{s,r} = \sum_{r,s} a_rb_sm_{r,s} = <\mathbf{A}, \mathsf{T}(\mathbf{B})>$$

for all **A**, **B** in $\mathcal{V}$. But according to the definition of $\mathsf{T}^*$,

$$<\mathsf{T}(\mathbf{B}), \mathbf{A}> = <\mathbf{B}, \mathsf{T}^*(\mathbf{A})>$$

for all **A**, **B** in $\mathcal{V}$. Therefore, we get

$$<\mathbf{B}, \mathsf{T}(\mathbf{A})> = <\mathsf{T}(\mathbf{A}), \mathbf{B}> = <\mathbf{A}, \mathsf{T}(\mathbf{B})> = <\mathsf{T}(\mathbf{B}), \mathbf{A}> = <\mathbf{B}, \mathsf{T}^*(\mathbf{A})>,$$

so that

$$<\mathbf{B}, \mathsf{T}(\mathbf{A})> = <\mathbf{B}, \mathsf{T}^*(\mathbf{A})>$$

for all **A**, **B** in $\mathcal{V}$. The definition of the adjoint now gives $\mathsf{T}(\mathbf{A}) = \mathsf{T}^*(\mathbf{A})$ for all **A** in $\mathcal{V}$. Thus $\mathsf{T} = \mathsf{T}^*$. □

Notice that Theorem 16.1.6 provides us with a painless method for constructing loads of self-adjoint linear transformations. All we have to do is take a symmetric matrix **S** of size $n$ and an $n$-dimensional inner product space $\mathcal{V}$ with an an orthonormal basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$. The linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ whose matrix is **S** with respect to the basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ will then be self-adjoint.

Notice also that Theorem 16.1.6 suggests ways to modify our proposed definition of a symmetric linear transformation so that we get a useful concept. We will not pursue this idea here, but relegate it to a flock of exercises.

## 16.2 The Spectral Theorem

We are ready to state the main result of this chapter (and indeed of this book so far).

**THEOREM 16.2.1** (Spectral Theorem): *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a self-adjoint linear transformation in the finite-dimensional inner product space $\mathcal{V}$. Then there exists an orthonormal basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ and numbers $\lambda_1, \ldots, \lambda_n$ such that*

$$\mathsf{T}(\mathbf{A}_i) = \lambda_i \mathbf{A}_i, \qquad i = 1, 2, \ldots, n.$$

*Therefore, the matrix of* $\mathsf{T}$ *relative to this basis is*

$$\begin{bmatrix} \lambda_1 & & & \mathbf{0} \\ & \lambda_2 & & \\ & & \ddots & \\ \mathbf{0} & & & \lambda_n \end{bmatrix}.$$

Note that it is not asserted that the numbers $\lambda_1, \ldots, \lambda_n$ are distinct; they need not be. They are of course the eigenvalues of **T**. What

Theorem[1] 16.2.1 says is that the characteristic polynomial of $\mathsf{T}$ factors completely and the geometric multiplicity of any eigenvalue coincides with its algebraic multiplicity.

The proof of Theorem 16.2.1 is an induction argument. We will give the proof for $n = 2$ and 3, and deal with the general case in Section 16.4. An alternative proof of the general case is also sketched out in the exercises. We need two preliminary results.

**PROPOSITION 16.2.2:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a self-adjoint linear transformation in the finite-dimensional inner product space $\mathcal{V}$. Suppose that $\mathcal{W}$ is a linear subspace of $\mathcal{V}$ with the property that $\mathsf{T}(\mathcal{W}) \subseteq \mathcal{W}$. Then $\mathcal{W}^{\perp}$ has the same property, namely, $\mathsf{T}(\mathcal{W}^{\perp}) \subseteq \mathcal{W}^{\perp}$.*

**PROOF:** Let $\mathbf{A} \in \mathcal{W}^{\perp}$. We must show that

$$\langle \mathsf{T}(\mathbf{A}), \mathbf{B} \rangle = 0$$

for all $\mathbf{B}$ in $\mathcal{W}$. We have

$$\begin{aligned} \langle \mathsf{T}(\mathbf{A}), \mathbf{B} \rangle &= \langle \mathbf{A}, \mathsf{T}^*(\mathbf{B}) \rangle \\ &= \langle \mathbf{A}, \mathsf{T}(\mathbf{B}) \rangle \quad (\text{since})\mathsf{T} = \mathsf{T}^* \\ &= 0 \quad (\text{since})\ \mathsf{T}(\mathbf{B}) \in \mathcal{W} \text{and } \mathbf{A} \in \mathcal{W}^{\perp}) \end{aligned}$$

as required. $\square$

**PROPOSITION 16.2.3:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a self-adjoint linear transformation in the finite-dimensional inner product space $\mathcal{V}$. Suppose that $e'$ and $e''$ are distinct eigenvalues of $\mathsf{T}$ with corresponding eigenvectors $\mathbf{E}'$ and $\mathbf{E}''$. Then $\langle \mathbf{E}', \mathbf{E}'' \rangle = 0$, that is, $\mathbf{E}'$ and $\mathbf{E}''$ are orthogonal.*

**PROOF:** By hypothesis we have

$$\mathsf{T}(\mathbf{E}') = e'\mathbf{E}', \qquad \mathsf{T}(\mathbf{E}'') = e''\mathbf{E}'',$$

so since $\mathsf{T}$ is self-adjoint, we get

$$\begin{aligned} e' \langle \mathbf{E}', \mathbf{E}'' \rangle = \langle e'\mathbf{E}', \mathbf{E}'' \rangle &= \langle \mathsf{T}(\mathbf{E}'), \mathbf{E}'' \rangle \\ &= \langle \mathbf{E}', \mathsf{T}^*(\mathbf{E}'') \rangle = \langle \mathbf{E}', \mathsf{T}(\mathbf{E}'') \rangle \\ &= \langle \mathbf{E}', e''\mathbf{E}'' \rangle = e'' \langle \mathbf{E}', \mathbf{E}'' \rangle, \end{aligned}$$

which means that

$$e' \langle \mathbf{E}', \mathbf{E}'' \rangle = e'' \langle \mathbf{E}', \mathbf{E}'' \rangle.$$

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[1] This result is sometimes called the **principal axis theorem**. We reserve this term for Theorem 16.3.1.

Since $e' \neq e''$ by hypothesis, we must have $<\mathbf{E}', \mathbf{E}''> = 0$, as required. $\square$

**Proof of The Spectral Theorem for $n = 2$:** Let $\{\mathbf{E}_1, \mathbf{E}_2\}$ be an orthonormal basis for $\mathcal{V}$. By Theorem 16.1.4 the matrix $\mathbf{M}$ of $\mathbf{T}$ relative to this basis is symmetric, say

$$\mathbf{M} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}.$$

If $b = 0$, there is nothing to prove, so we will assume that $b \neq 0$. The characteristic polynomial of $\mathbf{T}$ is given by

$$\begin{aligned} \Delta(t) = \det(\mathbf{M} - t\mathbf{I}) &= \det \begin{bmatrix} a - t & b \\ b & c - t \end{bmatrix} \\ &= (a - t)(c - t) - b^2 \\ &= t^2 - (a + c)t + ac - b^2. \end{aligned}$$

By the quadratic formula, the roots of $\Delta(t)$ are given by

$$\begin{aligned} r_{\pm} &= \frac{a + c \pm \sqrt{(a + c)^2 - 4(ac - b^2)}}{2} \\ &= \frac{a + c \pm \sqrt{a^2 + 2ac + c^2 - 4ac + 4b^2}}{2} \\ &= \frac{a + c \pm \sqrt{4b^2 + (a - c)^2}}{2}. \end{aligned}$$

Notice that since $b \neq 0$,

$$4b^2 + (a - c)^2 > 0$$

so that the characteristic roots are distinct. Let $\{\mathbf{F}_+, \mathbf{F}_-\}$ be the corresponding eigenvectors. By Proposition 16.2.3 $\mathbf{F}_+$ and $\mathbf{F}_-$ are orthogonal. Once we normalize $\mathbf{F}_+$ and $\mathbf{F}_-$ are an orthonormal basis of eigenvectors, completing the proof. $\square$

**Proof of The Spectral Theorem for $n = 3$:** Let $\Delta(t)$ be the characteristic polynomial of $\mathbf{T}$. Then $\Delta(t)$ is of degree 3 and hence must have at least one *real* root, say $e$. Then $e$ is an eigenvalue of $\mathbf{T}$. Let $\mathbf{E}$ be a corresponding eigenvector of unit length and $\mathcal{W} = \mathrm{Span}(\mathbf{E}) \subset \mathcal{V}$. Of course $\mathbf{T}(\mathcal{W}) \subseteq \mathcal{W}$ and therefore by 16.2.2 $\mathbf{T}(\mathcal{W}^\perp) \subseteq \mathcal{W}^\perp$. Let $\mathbf{F}_1, \mathbf{F}_2$ be an orthonormal basis for $\mathcal{W}^\perp$. Then $\{\mathbf{E}, \mathbf{F}_1, \mathbf{F}_2\}$ is an orthonormal basis for $\mathcal{V}$ (see the proof of Proposition 15.4.3). By Theorem 16.1.4 the matrix $\mathsf{M}$ of $\mathsf{T}$ relative to the basis $\{\mathbf{E}, \mathbf{F}_1, \mathbf{F}_2\}$ will be symmetric. Since $e$ is an eigenvalue, $\mathsf{M}$ must look like

$$\mathsf{M} = \begin{bmatrix} e & 0 & 0 \\ 0 & a & b \\ 0 & b & c \end{bmatrix}.$$

The characteristic polynomial of $\mathsf{T}$ is therefore

$$\Delta(t) = (e - t)(t^2 - (a + c)t + ac - b^2).$$

If $b = 0$ there is nothing to prove. If $b \neq 0$ then as in the case $n = 2$ there are two distinct eigenvalues, $e_1, e_2$ the zeros of the quadratic

$$t^2 - (a + c)t + ac - b^2.$$

If $\mathbf{E}_1$ and $\mathbf{E}_2$ are corresponding eigenvectors, then $\mathbf{E}_1, \mathbf{E}_2$ are orthogonal by Proposition 16.2.3, while they both are orthogonal to $\mathbf{E}$, since they belong to $\mathcal{W}^\perp$. Therefore, $\{\mathbf{E}, \mathbf{E}_1, \mathbf{E}_2\}$ is an orthonormal basis for $\mathcal{V}$, and the matrix of $\mathsf{T}$ relative to this basis is

$$\mathsf{M} = \begin{bmatrix} e & 0 & 0 \\ 0 & e_1 & 0 \\ 0 & 0 & e_2 \end{bmatrix}.$$

(Note that it is possible that $e = e_1$ or $e = e_2$ even if $b \neq 0$. If $b = 0$ it is even possible for $e_1 = e = e_2$.) $\square$

A proof of the spectral theorem for arbitrary $n$ proceeds along the lines of the proof for $n = 3$: the key point is to show that $\mathsf{T}$ has an eigenvalue, i.e., that the characteristic polynomial of $\mathsf{T}$ has at least one real root. There are many ways to organize such a proof, and we present the details of one such proof in Section 16.4

**EXAMPLE 1:** Let $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation given by

$$\mathsf{T}(x, y) = (x + 2y, 2y + x).$$

Find an orthonormal basis for $\mathbb{R}^2$ of eigenvectors of $\mathsf{T}$.

**SOLUTION:** We already have seen that $\mathsf{T}$ is self-adjoint. Therefore, by applying Theorem 16.2.1 we know that it is possible to find an orthonormal basis for $\mathbb{R}^2$ composed of eigenvectors of $\mathsf{T}$. As a matter of fact, the orthogonality will take care of itself because of Proposition 16.2.3. This will even provide a numerical check on our work.

The matrix of $\mathsf{T}$ relative to the standard orthonormal basis of $\mathbb{R}^2$ was computed to be

$$\mathsf{M} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

in Section 16.1 Example 1. The characteristic polynomial of $\mathsf{T}$ is therefore

$$\begin{aligned}\Delta(\mathsf{T}) &= \det(\mathsf{M} - t\mathsf{I}) = \det\begin{bmatrix} 1-t & 2 \\ 2 & 1-t \end{bmatrix} \\ &= (1-t)^2 - 4 = t^2 - 2t + 1 - 4 \\ &= (t^2 - 2t - 3) = (t-3)(t+1).\end{aligned}$$

The eigenvalues of $\mathsf{T}$ are therefore 3 and $-1$. Corresponding eigenvectors are easily computed.

$$e = 3: \qquad \begin{cases} -2x + 2y = 0, \\ 2x - 2y = 0, \end{cases}$$

which means that $x = y$, and so $(1, 1)$ is an eigenvector and $(\sqrt{2}/2, \sqrt{2}/2)$ an eigenvector of unit length.

$$e = -1: \qquad \begin{cases} 2x + 2y = 0, \\ 2x + 2y = 0, \end{cases}$$

so $x = -y$, and $(1, -1)$ is an eigenvector and $(\sqrt{2}/2, -\sqrt{2}/2)$ an eigenvector of unit length. Therefore, $\mathbf{F}_1 = (\sqrt{2}/2, \sqrt{2}/2), \mathbf{F}_2 = (\sqrt{2}/2, -\sqrt{2}/2)$ is, by Proposition 16.2.3, an orthonormal basis of eigenvectors of $\mathsf{T}$ for $\mathbb{R}^2$. We can check the orthogonality easily enough:

$$<\mathbf{F}_1, \mathbf{F}_2> = \frac{2}{4} - \frac{2}{4} = 0,$$

as required.

**EXAMPLE 2:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be given by

$$\mathsf{T}(x, y, z) = (3x - z, 2y, -x + 3z).$$

Verify that $\mathsf{T}$ is self-adjoint and put it into diagonal form.

**SOLUTION:** By direct calculation,

$$\begin{aligned}\mathsf{T}(1, 0, 0) &= (3, 0, -1), \\ \mathsf{T}(0, 1, 0) &= (0, 2, 0), \\ \mathsf{T}(0, 0, 1) &= (-1, 0, 3).\end{aligned}$$

Therefore, the matrix of $\mathsf{T}$ relative to the standard basis of $\mathbb{R}^3$ is

$$\begin{bmatrix} 3 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix},$$

which is symmetric, so $\mathsf{T}$ is self-adjoint. The characteristic polynomial of $\mathsf{T}$ is

$$\begin{aligned}
\Delta(t) &= \det \begin{bmatrix} 3-t & 0 & -1 \\ 0 & 2-t & 0 \\ -1 & 0 & 3-t \end{bmatrix} \\
&= (2-t)\det \begin{bmatrix} 3-t & -1 \\ -1 & 3-t \end{bmatrix} \\
&= (2-t)((3-t)^2-1) = (2-t)(t^2-6t+8) \\
&= (2-t)(t-4)(t-2).
\end{aligned}$$

Therefore, the eigenvalues of $\mathsf{T}$ are $t = 2, t = 4$ with multiplicities 2 and 1, respectively. A basis for the eigenspace corresponding to the eigenvalue 2 is obtained by solving the linear system

$$\mathsf{T}(x, y, z) = 2(x, y, z),$$

that is,

$$\begin{aligned}
3x - z &= 2x, \\
2y &= 2y, \\
-x + 3z &= 2z,
\end{aligned}$$

or

$$\begin{aligned}
x - z &= 0, \\
-x + z &= 0,
\end{aligned}$$

so a basis for the eigenspace corresponding to eigenvalue 2 consists of the vectors $(1, 0, 1), (0, 1, 0)$.

To find an eigenvector corresponding to eigenvalue 4, we solve

$$\mathsf{T}(x, y, z) = 4(x, y, z),$$

that is,

$$\begin{aligned}
3x - z &= 4x, \\
2y &= 4y, \\
-x + 3z &= 4z,
\end{aligned}$$

or

$$\begin{aligned}
-z &= x, \\
-2y &= 0, \\
-x &= z,
\end{aligned}$$

which has $(-1, 0, 1)$ as a solution. Therefore, the matrix of $\mathsf{T}$ relative to the ordered basis $\mathbf{F}_1 = (1, 0, 1)$, $\mathbf{F}_2 = (0, 1, 0)$, $\mathbf{F}_3 = (-1, 0, 1)$ is

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

Notice that $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\}$ is an orthogonal basis for $\mathbb{R}^3$.

**EXAMPLE 3:** Put $\mathsf{T} : \mathrm{Mat}_{2,2} \longrightarrow \mathrm{Mat}_{2,2}$ given by $\mathsf{T}(\mathbf{A}) = \mathbf{A}^{\mathrm{tr}}$ into diagonal form.

**SOLUTION:** We showed that $\mathsf{T}$ is self-adjoint, so by the spectral theorem $\mathsf{T}$ is diagonalizable. As an orthonormal basis for $\mathcal{V} = \mathrm{Mat}_{2,2}$ we choose

$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{M}_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$
$$\mathbf{M}_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{M}_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

So

$$\mathsf{T}(\mathbf{M}_1) = \mathbf{M}_1, \qquad \mathsf{T}(\mathbf{M}_2) = \mathbf{M}_3, \qquad \mathsf{T}(\mathbf{M}_3) = \mathbf{M}_2, \qquad \mathsf{T}(\mathbf{M}_4) = \mathbf{M}_4,$$

and the matrix of $\mathsf{T}$ relative to this basis is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

**NOTE:** From $\mathsf{T}(\mathbf{M}_1) = \mathbf{M}_1$, and $\mathsf{T}(\mathbf{M}_4) = \mathbf{M}_4$, we see immediately that $\mathbf{M}_1$ and $\mathbf{M}_4$ are eigenvectors of $\mathsf{T}$ corresponding to the eigenvalue 1.

The characteristic polynomial of $\mathsf{T}$ is

$$\begin{aligned} \Delta(t) &= \det \begin{bmatrix} 1-t & 0 & 0 & 0 \\ 0 & -t & 1 & 0 \\ 0 & 1 & -t & 0 \\ 0 & 0 & 0 & 1-t \end{bmatrix} \\ &= (1-t)\det \begin{bmatrix} -t & 1 & 0 \\ 1 & -t & 0 \\ 0 & 0 & 1-t \end{bmatrix} \\ &= (1-t)(-1)^{3+3}(1-t)\det \begin{bmatrix} -t & 1 \\ 1 & -t \end{bmatrix} \\ &= (1-t)^2(t^2-1) = (t-1)^3(t+1), \end{aligned}$$

so the eigenvalues of $\mathbf{T}$ are $+1$ and $-1$ with multiplicities 3 and 1, respectively. By the spectral theorem, $\dim \mathcal{V}_1 = 3$, $\dim \mathcal{V}_{-1} = 1$. We can find three orthonormal eigenvectors of $\mathbf{T}$ corresponding to eigenvalue $+1$ by solving the system

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix},$$

which is

$$x = x, \qquad z = y, \qquad y = z, \qquad w = w.$$

So

$$(1, 0, 0, 0), \qquad \sqrt{\frac{1}{2}}(0, 1, 1, 0), \qquad (0, 0, 0, 1)$$

are the components relative to $\{\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3, \mathbf{M}_4\}$ of an orthonormal basis for $\mathcal{V}_1$. The corresponding vectors of $\mathrm{Mat}_{2,2}$ are

$$\mathbf{N}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{N}_2 = \sqrt{\frac{1}{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{N}_3 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

To find a basis for the eigenspace $\mathcal{V}_{-1}$ requires that we solve the system

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = - \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix},$$

which is

$$x = -x, \qquad z = -y, \qquad y = -z, \qquad w = -w.$$

Thus $x = 0$, $w = 0$ and $y = 1$, $z = -1$. Normalizing this, we have

$$\sqrt{\frac{1}{2}}(0, 1, -1, 0)$$

as a basis for the solution space relative to the basis $\{\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3, \mathbf{M}_4\}$, so the vector

$$\mathbf{N}_4 = \sqrt{\frac{1}{2}} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

is an orthonormal basis for $\mathcal{V}_{-1}$. The matrix of $\mathbf{T}$ relative to the basis $\{\mathbf{N}_1, \mathbf{N}_2, \mathbf{N}_3, \mathbf{N}_4\}$ is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

## 16.3 The Principal Axis Theorem for Quadratic Forms

As an application of the spectral theorem we take up the so-called *principal axis theorem* for quadratic forms. In the calculus one spends considerable time in studying the graphs of quadratic equations, whose graphs are the conic sections, ellipse, hyperbola, parabola, and various degenerate cases. There is a foolproof procedure for drawing the graph of, say,

$$2x^2 + 3y^2 = 7,$$

but for

$$x^2 + 4xy + y^2 = 7$$

there is a difficulty caused by the cross-product term $4xy$. Of course there is a formula for rotating the coordinate system so as to eliminate the cross-product term, and it is just this formula that we wish to discuss in the light of our work with self-adjoint linear transformations! Rather than introduce the final, sophisticated version of the theory, let us consider an example first.

**EXAMPLE 1:** Sketch the graph of

$$x^2 + 4xy + y^2 = 7.$$

**SOLUTION:** We make the following observation:

$$x^2 + 4xy + y^2 = [x, y]\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}.$$

What we wish to discover is a new coordinate system for $\mathbb{R}^2$ in which the quadratic form has no cross-product term. Let

$$\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

be the linear transformation whose matrix in the standard coordinate system is

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

That is,

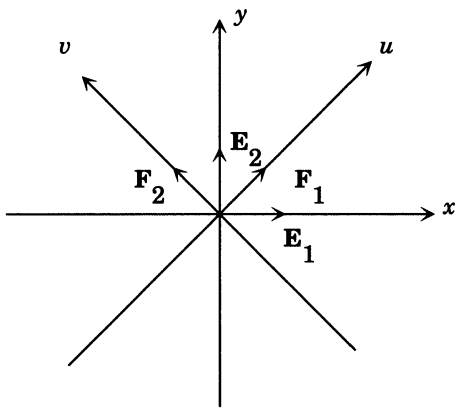
$$\mathsf{T}(x, y) = (x + 2y, y + 2x).$$

Then we see that

$$[x, y]\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \langle (x, y), \mathsf{T}(x, y)\rangle.$$

That is, for a vector $\mathbf{A}$ in $\mathbb{R}^2$ with coordinates $(x, y)$ we have

$$x^2 + 4xy + y^2 = \langle \mathbf{A}, \mathsf{T}(\mathbf{A})\rangle.$$




16.3.1: Eigenvectors for **T**

The number $<\mathbf{A}, \mathsf{T}(\mathbf{A})>$ depends not on the coordinate system in which we express **A**, but only on **T**. Thus we could equally well use the coordinate system determined by the unit eigenvectors of **T**, namely

$$\mathbf{F}_1 = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \qquad \mathbf{F}_2 = \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$

See Figure 16.3.1. Let us write $(u, v)$ for the coordinates of a vector in the coordinate system determined by the vectors $\mathbf{F}_1, \mathbf{F}_2$. Then

$$<\mathbf{A}, \mathsf{T}(\mathbf{A})> = [u, v]\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = 3u^2 - v^2,$$

since the matrix of **T** relative to the basis $\{\mathbf{F}_1, \mathbf{F}_2\}$ is

$$\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}.$$

Thus our original quadratic form

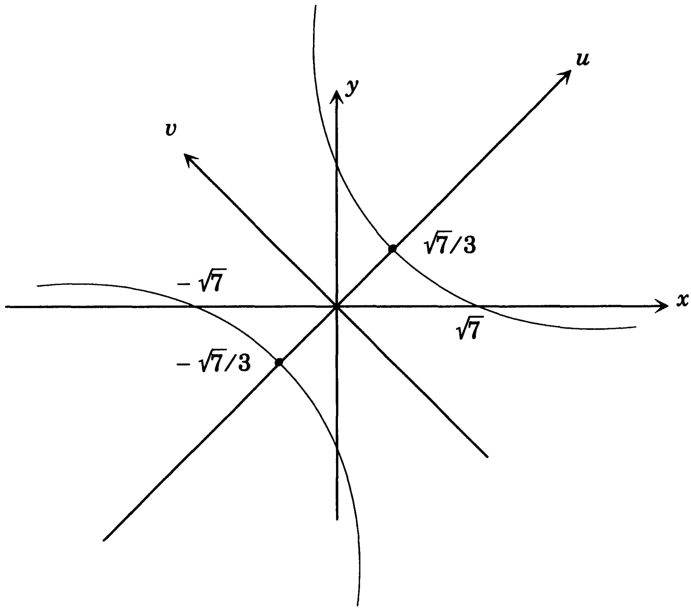
$$7 = x^2 + 4xy + y^2$$

may be written (changing coordinate systems has not changed the value of $<\mathbf{A}, \mathsf{T}(\mathbf{A})>$)

$$7 = 3u^2 - v^2$$

in the $(u, v)$ coordinate system. One readily checks that this is a hyperbola and the graph is easily sketched as in Figure 16.3.2. The vertices of the hyperbola are at the points $(u, v) = (\pm\sqrt{7/3}, 0)$. The $(x, y)$ coordinates of these points are easily computed:

$$(x, y) = \pm\sqrt{\frac{7}{3}}\mathbf{F}_1 + 0\mathbf{F}_2 = \pm\sqrt{\frac{7}{3}}\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$




16.3.2: The rotated hyperbola

The relation between the $(x, y)$ and $(u, v)$ coordinates of a point $\mathbf{A}$ in $\mathbb{R}^2$ is computed thus:

$$\begin{aligned}
x\mathbf{E}_1 + y\mathbf{E}_2 = \mathbf{A} &= u\mathbf{F}_1 + v\mathbf{F}_2 \\
&= u\left(\frac{\sqrt{2}}{2}\mathbf{E}_1 + \frac{\sqrt{2}}{2}\mathbf{E}_2\right) + v\left(-\frac{\sqrt{2}}{2}\mathbf{E}_1 + \frac{\sqrt{2}}{2}\mathbf{E}_2\right) \\
&= \frac{\sqrt{2}}{2}(u - v)\mathbf{E}_1 + \frac{\sqrt{2}}{2}(u + v)\mathbf{E}_2,
\end{aligned}$$

whence

$$\begin{aligned}
x &= \frac{\sqrt{2}}{2}u - \frac{\sqrt{2}}{2}v, \\
y &= \frac{\sqrt{2}}{2}u + \frac{\sqrt{2}}{2}v,
\end{aligned}$$

or, equivalently,

$$\begin{aligned} u\mathbf{F}_1 + v\mathbf{F}_2 &= \mathbf{A} = x\mathbf{E}_1 + y\mathbf{E}_2 \\ &= x\left(\frac{\sqrt{2}}{2}\mathbf{F}_1 - \frac{\sqrt{2}}{2}\mathbf{F}_2\right) + y\left(\frac{\sqrt{2}}{2}\mathbf{F}_1 + \frac{\sqrt{2}}{2}\mathbf{F}_2\right) \\ &= \frac{\sqrt{2}}{2}(x+y)\mathbf{F}_1 + \frac{\sqrt{2}}{2}(y-x)\mathbf{F}_2, \end{aligned}$$

whence

$$\begin{aligned} u &= \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y, \\ v &= -\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y. \end{aligned}$$

The equations

$$\sqrt{3}u \pm v = \sqrt{7}$$

for the asymptotes of the hyperbola may thus be rewritten

$$\sqrt{3}\left(\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y\right) \pm \left(-\frac{\sqrt{2}}{2}x + \frac{\sqrt{2}}{2}y\right) = \sqrt{7},$$

or

$$\frac{\sqrt{6} \mp \sqrt{2}}{2}x + \frac{\sqrt{6} \pm \sqrt{2}}{2}y = \sqrt{7}.$$

Notice that the angle $\vartheta$ between the $(u, v)$ and $(x, y)$ coordinate systems (that is, the angle between $\mathbf{E}_1$ and $\mathbf{F}_1$) may be computed from the formula

$$\cos(\vartheta) = \langle \mathbf{E}_1, \mathbf{F}_1 \rangle = \frac{\sqrt{2}}{2}.$$

It is time to reveal the basic theory behind the preceding example. This will require a definition.

**DEFINITION:** *Let $\mathcal{V}$ be a finite-dimensional inner product space. A function $\varphi : \mathcal{V} \longrightarrow \mathbb{R}$ is called* **quadratic** *if there exists a self-adjoint linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ such that*

$$\varphi(\mathbf{A}) = \langle \mathbf{A}, \mathsf{T}(\mathbf{A}) \rangle$$

*for all vectors $\mathbf{A}$ in $\mathcal{V}$.*

**EXAMPLE 2:** The function

$$\varphi(x, y) = x^2 + 4xy - y^2$$

is quadratic. For if

$$\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

is the linear transformation with matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

in the standard orthonormal coordinate system, then $\mathsf{T}$ is self-adjoint by Theorem 16.1.4. Moreover,

$$<(x, y), \mathsf{T}(x, y)> = (x, y)\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = x^2 + 4xy - y^2 = \varphi(x, y),$$

and so $\varphi$ is quadratic. More generally,

$$\varphi(x, y) = ax^2 + 2bxy + cy^2$$

is quadratic. We let

$$\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

have matrix

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

in the standard orthonormal basis, so that $\mathsf{T}$ is self-adjoint, and one checks that

$$\varphi(x, y) = <(x, y), \mathsf{T}(x, y)>.$$

Thus $\varphi$ is quadratic.

Strictly speaking, we should call a function $\varphi : \mathcal{V} \longrightarrow \mathbb{R}$ given by

$$\varphi(\mathbf{A}) = <\mathbf{A}, \mathsf{T}(\mathbf{A})>$$

a **homogeneous** quadratic function, or **quadratic form**, because it has no linear terms.

**THEOREM 16.3.1** (Principal Axis Theorem): *Suppose that*

$$\varphi : \mathcal{V} \longrightarrow \mathbb{R}$$

*is a homogeneous quadratic function on the finite-dimensional inner product space $\mathcal{V}$. Then there exists a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_n\}$ for $\mathcal{V}$ and numbers $\lambda_1, \ldots, \lambda_n$ such that for a vector $\mathbf{X} = x_1\mathbf{A}_1 + \cdots + x_n\mathbf{A}_n$ in $\mathcal{V}$ the value of $\varphi$ is given by the formula*

$$\varphi(\mathbf{X}) = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \cdots + \lambda_n x_n^2.$$

**PROOF:** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a self-adjoint transformation such that

$$\varphi(\mathbf{X}) = \langle \mathbf{X}, \mathsf{T}(\mathbf{X}) \rangle$$

for all $\mathbf{X}$ in $\mathcal{V}$. By the spectral theorem we may find a basis $\mathbf{A}_1, \ldots, \mathbf{A}_n$ for $\mathcal{V}$ composed of eigenvectors of $\mathsf{T}$. Let $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues. If $\mathbf{X} = x_1\mathbf{A}_1 + \cdots + x_n\mathbf{A}_n$, then

$$\begin{aligned}
\varphi(\mathbf{X}) &= \langle \mathbf{X}, \mathsf{T}(\mathbf{X}) \rangle \\
&= \langle x_1\mathbf{A}_1 + \cdots + x_n\mathbf{A}_n, \mathsf{T}(x_1\mathbf{A}_1 + \cdots + x_n\mathbf{A}_n) \rangle \\
&= \langle x_1\mathbf{A}_1 + \cdots + x_n\mathbf{A}_n, x_1\mathsf{T}(\mathbf{A}_1) + \cdots + x_n\mathsf{T}(\mathbf{A}_n) \rangle \\
&= \langle x_1\mathbf{A}_1 + \cdots + x_n\mathbf{A}_n, x_1\lambda_1\mathbf{A}_1 + \cdots + x_n\lambda_n\mathbf{A}_n \rangle \\
&= x_1\lambda_1x_1 + x_2\lambda_2x_2 + \cdots + x_n\lambda_nx_n \\
&= \lambda_1x_1^2 + \lambda_2x_2^2 + \cdots + \lambda_nx_n^2,
\end{aligned}$$

as required. □

Note that the proof of Theorem 16.3.1 is itself important, as it provides a method for finding the basis $\mathbf{A}_1, \ldots, \mathbf{A}_n$, whose elements are called the **principal axes** of the quadratic form $\varphi$.

**EXAMPLE 3:** Find the principal axes of the quadratic function

$$\varphi : \mathbb{R}^3 \longrightarrow \mathbb{R}$$

given by

$$\varphi(x, y, z) = 3x^2 + 2y^2 + 3z^2 - 2xy - 2yz.$$

**SOLUTION:** A little experimentation will show that

$$\varphi(x, y, z) = \langle (x, y, z), \mathsf{T}(x, y, z) \rangle,$$

where $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is the self-adjoint linear transformation whose matrix is (do you see the gimmick?)

$$\begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

relative to the standard orthonormal basis of $\mathbb{R}^3$. The characteristic polynomial of T is

$$\begin{aligned}
\Delta(t) &= \det\begin{bmatrix} 3-t & -1 & 0 \\ -1 & 2-t & -1 \\ 0 & -1 & 3-t \end{bmatrix} \\
&= (3-t)\det\begin{bmatrix} 2-t & -1 \\ -1 & 3-t \end{bmatrix} + \det\begin{bmatrix} -1 & -1 \\ 0 & 3-t \end{bmatrix} \\
&= (3-t)[(2-t)(3-t)-1]-(3-t) \\
&= (3-t)[6-5t+t^2-1-1] \\
&= (3-t)(t^2-5t+4) \\
&= -(t-3)(t-4)(t-1),
\end{aligned}$$

so the eigenvalues of T are 1, 3, and 4. Corresponding eigenvectors may be readily found. We get

$$t=1 \quad \left\{\begin{array}{rrrcl} 2x & -y & & = & 0, \\ -x & +y & -z & = & 0, \\ & -y & +2z & = & 0, \end{array}\right. .$$

which yields

$$\begin{array}{rrrl} & y & & = 2x, \\ x & & -z & = 0, \\ -2x & & +2z & = 0, \end{array}$$

so

$$\begin{aligned} y &= 2x, \\ z &= x, \end{aligned}$$

which has a nontrivial solution,

$$x=1, \quad y=2, \quad z=1,$$

and a unit eigenvector is thus $\mathbf{A}_1 = (1/\sqrt{6}, 2/\sqrt{6}, 1/\sqrt{6})$.

Likewise, we find that $\mathbf{A}_2 = (1/\sqrt{2}, 0, -1/\sqrt{2})$ is a unit eigenvector corresponding to $t=3$, and $\mathbf{A}_3 = (1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3})$ is an eigenvector corresponding to $t=4$.

Thus the principal axes of $\varphi$ are

$$\begin{aligned}
\mathbf{A}_1 &= \left(\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{6}\right), \\
\mathbf{A}_2 &= \left(\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}\right), \\
\mathbf{A}_3 &= \left(\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right),
\end{aligned}$$

and if $(u, v, w)$ denotes the coordinates of a vector $\mathbf{X}$ with respect to the principal axes, then

$$\varphi(\mathbf{X}) = u^2 + 3v^2 + 4w^2.$$

**EXAMPLE 4:** Sketch the graph of the quadratic equation

$$2x^2 + 3xy - 2y^2 = 10.$$

**SOLUTION:** Consider the function

$$\varphi(x, y) = 2x^2 + 3xy - 2y^2.$$

Of course, $\varphi : \mathbb{R}^2 \longrightarrow \mathbb{R}$ is a quadratic function and

$$\varphi(x, y) = \langle (x, y), \mathsf{T}(x, y) \rangle,$$

where $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ is the self-adjoint transformation whose matrix in the standard orthonormal basis of $\mathbb{R}^2$ is

$$\begin{bmatrix} 2 & \frac{3}{2} \\ \frac{3}{2} & -2 \end{bmatrix}.$$

The characteristic polynomial of $\mathsf{T}$ is

$$\begin{aligned} \Delta(t) &= \det \begin{bmatrix} 2-t & \frac{3}{2} \\ \frac{3}{2} & -2-t \end{bmatrix} = (2-t)(-2-t) - \frac{9}{4} \\ &= t^2 - 4 - \frac{9}{4} \\ &= t^2 - \frac{25}{4}. \end{aligned}$$

The eigenvalues of $\mathsf{T}$ are therefore $\pm 5/2$. Corresponding unit eigenvectors are easily found to be

$$\begin{aligned} \left(\frac{3\sqrt{10}}{10}, \frac{\sqrt{10}}{10}\right) &= \mathbf{F}_+ \quad \text{corresponding to } \tfrac{5}{2}, \\ \left(\frac{-\sqrt{10}}{10}, \frac{3\sqrt{10}}{10}\right) &= \mathbf{F}_- \quad \text{corresponding to } -\tfrac{5}{2}. \end{aligned}$$

If we write $(u, v)$ for the coordinates of a vector $\mathbf{A}$ in $\mathbb{R}^2$ with respect to the orthonormal basis $\{\mathbf{F}_+, \mathbf{F}_-\}$, we find

$$\varphi(\mathbf{A}) = \frac{5u^2}{2} - \frac{5v^2}{2}.$$

We seek the graph of the equation

$$\varphi(\mathbf{A}) = 10.$$

But if we employ the $(u, v)$ coordinates determined by the principal axes of $\mathbf{T}$, then this is just the graph of the equation

$$\frac{5u^2}{2} - \frac{5v^2}{2} = 10,$$

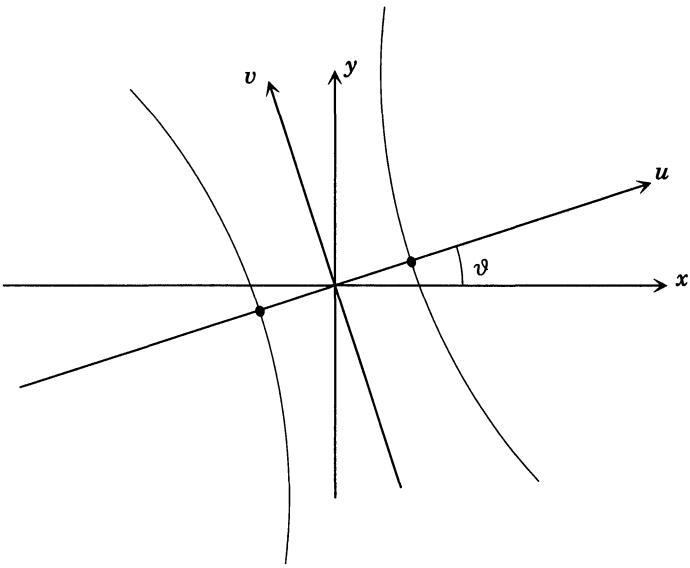
or

$$u^2 - v^2 = 4.$$

One recognizes this straight off as a hyperbola as in Figure 16.3.3. The angle $\vartheta$ between the principal axes of $\vartheta$ and the usual axes is given by

$$\cos(\vartheta) = \langle \mathbf{E}_1, \mathbf{F}_+ \rangle = \frac{3\sqrt{10}}{10},$$

which is about 18.5°.




16.3.3: Another rotated hyperbola

The coordinates of the vertices of the hyperbola are

$$(u, v) = (\pm 2, 0),$$

or

$$(x, y) = \pm\left(\frac{3\sqrt{10}}{5}, \frac{\sqrt{10}}{5}\right).$$

As a final example, we wish to indicate that the application of the principal axis theorem to the study of conic sections is not limited to the homogeneous case.

**EXAMPLE 5:** Sketch the graph of the equation

$$16x^2 - 24xy + 9y^2 - 30x - 40y = 0.$$

**SOLUTION:** Let us first find the principal axes of the homogeneous quadratic function $\varphi : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ given by

$$\varphi(x, y) = 16x^2 - 24xy + 9y^2.$$

We begin by noting that

$$\varphi(x, y) = \langle (x, y), \mathsf{T}(x, y) \rangle,$$

where $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ is the self-adjoint transformation with matrix

$$\begin{bmatrix} 16 & -12 \\ -12 & 9 \end{bmatrix}$$

relative to the standard orthonormal basis of $\mathbb{R}^2$. The characteristic polynomial of $\mathsf{T}$ is

$$\begin{aligned} \Delta(t) &= \det \begin{bmatrix} 16-t & -12 \\ -12 & 9-t \end{bmatrix} = (9-t)(16-t) - 144 \\ &= t^2 - 25t + 144 - 144 \\ &= t^2 - 25t. \end{aligned}$$

The eigenvalues of $\mathsf{T}$ are therefore $t = 0$, $t = 25$. Corresponding eigenvectors are $\mathbf{F}_1 = (\frac{3}{5}, \frac{4}{5})$ and $\mathbf{F}_2 = (-\frac{4}{5}, \frac{3}{5})$. If we let $(u, v)$ denote the coordinates of a vector in the $\{\mathbf{F}_1, \mathbf{F}_2\}$ basis, then

$$\begin{aligned} x\mathbf{E}_1 + y\mathbf{E}_2 = \mathbf{A} = u\mathbf{F}_1 + v\mathbf{F}_2 &= u\left(\frac{3}{5}\mathbf{E}_1 + \frac{4}{5}\mathbf{E}_2\right) + v\left(\frac{-4}{5}\mathbf{E}_1 + \frac{3}{5}\mathbf{E}_2\right) \\ &= \left(\frac{3u}{5} - \frac{4v}{5}\right)\mathbf{E}_1 + \left(\frac{4u}{5} + \frac{3v}{5}\right)\mathbf{E}_2. \end{aligned}$$

That is, the relation between the $(x, y)$ and $(u, v)$ coordinate systems is

$$\begin{aligned} x &= \frac{3u}{5} - \frac{4v}{5}, \\ y &= \frac{4u}{5} + \frac{3v}{5}. \end{aligned}$$

To graph the equation

$$16x^2 - 24xy - 9y^2 - 30x - 40y = 0,$$

is therefore the same as to graph the equation

$$\varphi(x, y) - 30x - 40y = 0$$

which is the same as

$$\varphi(u, v) - 30\left(\frac{3u}{5} - \frac{4v}{5}\right) - 40\left(\frac{4u}{5} + \frac{3v}{5}\right) = 0,$$

which is the same as

$$0u^2 + 25v^2 - 18u + 24v - 32u - 24v = 0,$$

or

$$25v^2 = 50u,$$

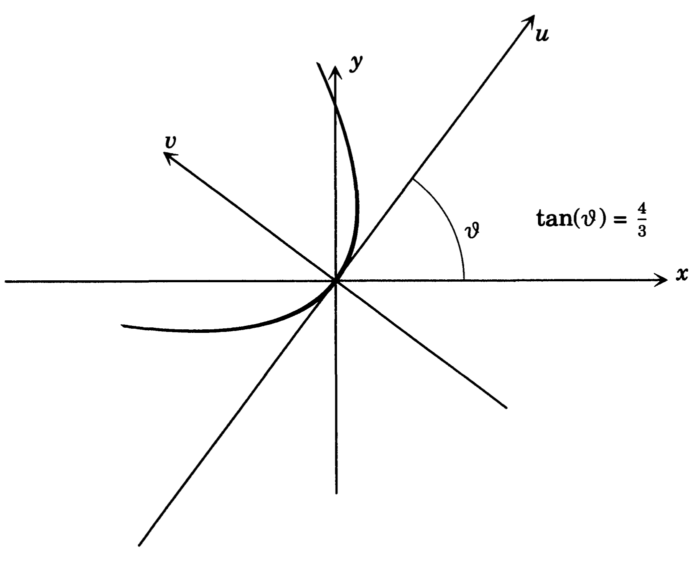
or

$$v^2 = 2u,$$

which is instantly seen to be the parabola depicted in Figure 16.3.4. The angle $\vartheta$ between the $(u, v)$ and $(x, y)$ coordinate systems is given by

$$\cos(\vartheta) = \langle \mathbf{E}_1, \mathbf{F}_1 \rangle = \frac{3}{5},$$

which is about $53.1^\circ$.




16.3.4: A parabola

## 16.4 A Proof of the Spectral Theorem in the General Case

In this section we prove the spectral theorem along the lines of the proof for $n = 3$ from Section 16.2. The key point is to show that a self–adjoint linear transformation has an eigenvalue, i.e., that the characteristic polynomial of a self-adjoint transformation has a real zero.

The idea for the proof is to reverse the procedure used in Section 16.3 and show that an eigenvector of a self-adjoint linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ may be found by minimizing the quadratic function

$$\varphi(\mathbf{X}) = <\mathsf{T}(\mathbf{X}), \mathbf{X}>$$

restricted to the unit sphere of $\mathcal{V}$. The position vector $\mathbf{V}$ of a point achieving this minimum is an eigenvector of $\mathsf{T}$. Let us fix some notations.

**NOTATION:** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a self-adjoint linear transformation in the finite-dimensional inner product space $(\mathcal{V}, <—, —>)$. We assume that $\mathcal{V}$ has dimension 2 at least. Denote by $\mathbf{S}(\mathcal{V}) \subset \mathcal{V}$ the unit sphere of $\mathcal{V}$, i.e.,

$$\mathbf{S}(\mathcal{V}) = \left\{ z \in \mathcal{V} \,\middle|\, \|z\| = 1 \right\}.$$

Define the function

$$f : \mathbf{S}(\mathcal{V}) \longrightarrow \mathbb{R} \quad \text{by} \quad f(\mathbf{X}) = <\mathsf{T}(\mathbf{X}), \mathbf{X}>.$$

The function $f$ is a differentiable function on $\mathbf{S}(\mathcal{V})$. The set $\mathbf{S}(\mathcal{V}) \subset \mathcal{V}$ is closed and bounded, and so by the Heine–Borel theorem, the function $f$ attains a minimum on $\mathbf{S}(\mathcal{V})$ at some point $\mathbf{V} \in \mathbf{S}(\mathcal{V})$.

**LEMMA 16.4.1:** *With the preceding notations, choose a unit vector* $\mathbf{Y}$ *orthogonal to* $\mathbf{V}$, *i.e.,* $\mathbf{Y} \in \mathbf{S}(\mathcal{V}) \cap \operatorname{Span}\{\mathbf{V}\}^{\perp}$. *(Such a point exists because* $\mathcal{V}$ *has at least dimension 2.) Then for any* $t \in \mathbb{R}$ *the vector* $\dfrac{\mathbf{V} + t\mathbf{Y}}{\sqrt{1 + t^2}}$ *has unit length.*

**PROOF:** This follows by expanding the inner product, viz.,

$$\begin{aligned} <\frac{\mathbf{V} + t\mathbf{Y}}{\sqrt{1+t^2}}, \frac{\mathbf{V} + t\mathbf{Y}}{\sqrt{1+t^2}}> &= \frac{1}{1+t^2} <\mathbf{V} + t\mathbf{Y}, \mathbf{V} + t\mathbf{Y}> \\ &= \frac{1}{1+t^2} \left[ <\mathbf{V}, \mathbf{V}> + t<\mathbf{V}, \mathbf{Y}> + t<\mathbf{Y}, \mathbf{V}> + t^2<\mathbf{Y}, \mathbf{Y}> \right] \\ &= \frac{1}{1+t^2} \left[ 1 + 0 + 0 + t^2 \right] = 1, \end{aligned}$$

where we have used that $<\mathbf{V}, \mathbf{Y}> = 0 = <\mathbf{Y}, \mathbf{V}>$. $\square$

From Lemma 16.4.1 it follows that the function

$$\omega : \mathbb{R} \longrightarrow \mathcal{V} \quad \text{defined by} \quad \omega(t) = \frac{\mathbf{V} + t\mathbf{Y}}{\sqrt{1+t^2}}$$

defines a path in $\boldsymbol{S}(\mathcal{V})$. This path goes through the point $\mathbf{V}$ when $t = 0$, where $f$ takes on its minimum value on $\boldsymbol{S}(\mathcal{V})$. In addition, since $f$ is differentiable, the point $t = 0$ must be a critical point of the composite function $g : \mathbb{R} \longrightarrow \mathbb{R}$ defined by $g(t) := f(\omega(t))$. The function $g$ is also differentiable. Its derivative may be computed as follows:

$$\begin{aligned} g(t) &= \left\langle \mathsf{T}\left(\frac{\mathbf{V} + t\mathbf{Y}}{\sqrt{1+t^2}}\right), \frac{\mathbf{V} + t\mathbf{Y}}{\sqrt{1+t^2}} \right\rangle \\ &= \frac{1}{1+t^2} \langle \mathsf{T}(\mathbf{V} + t\mathbf{Y}), \mathbf{V} + t\mathbf{Y} \rangle. \end{aligned}$$

Set

$$h(t) = \langle \mathsf{T}(\mathbf{V} + t\mathbf{Y}), \mathbf{V} + t\mathbf{Y} \rangle,$$

so

$$g(t) = \frac{1}{1+t^2} h(t),$$

and therefore

$$\begin{aligned} \frac{dg}{dt} &= \frac{1}{1+t^2}\frac{dh}{dt} + \frac{-2t}{(1+t^2)^2} h(t) \\ &= \frac{1}{1+t^2}\left[2\langle \mathsf{T}(\mathbf{V}), \mathbf{Y} \rangle + 2t \langle \mathsf{T}(\mathbf{Y}, \mathbf{Y} \rangle\right] - \frac{2t}{(1+t^2)^2}\left[\langle \mathsf{T}(\mathbf{V} + t\mathbf{Y}), \mathbf{V} + t\mathbf{Y} \rangle\right] \\ &= \frac{1}{(1+t^2)^2}\left[(2 - 2t^2)\langle \mathsf{T}(\mathbf{V}), \mathbf{Y} \rangle + 2t\langle \mathsf{T}(\mathbf{Y}), \mathbf{Y} \rangle - 2t \langle \mathsf{T}(\mathbf{V}), \mathbf{V} \rangle\right], \end{aligned}$$

and in particular,

$$\left.\frac{dg}{dt}\right|_{t=0} = 2\langle \mathsf{T}(\mathbf{V}), \mathbf{Y} \rangle = 0.$$

This has the following important consequence:

**PROPOSITION 16.4.2:** *With the preceding notations, the vector* $\mathbf{V}$ *is an eigenvector of* $\mathsf{T}$.

**PROOF:** Let $\mathcal{L} = \operatorname{Span}\{\mathbf{V}\}^{\perp}$. From the preceding discussion it follows that

$$0 = \left.\frac{dg}{dt}\right|_{t=0} = 2\langle \mathsf{T}(\mathbf{V}), \mathbf{Y} \rangle$$

for every unit vectors $\mathbf{Y}$ in $\mathcal{L}$. Therefore, $\mathsf{T}(\mathbf{V}) \perp \mathcal{L}$. But $\mathcal{L}^{\perp} = \operatorname{Span}\{\mathbf{V}\}$, and therefore $\mathsf{T}(\mathbf{V}) \in \operatorname{Span}\{\mathbf{V}\}$, so $\mathsf{T}(\mathbf{V}) = \lambda\mathbf{V}$ for some $\lambda \in \mathbb{R}$, which says that $\mathbf{V}$ is an eigenvector of $\mathsf{T}$ corresponding to the eigenvalue $\lambda$. $\square$

**PROPOSITION 16.4.3:** *With the preceding notations, let* $\mathcal{L} = \text{Span}\,\{\mathbf{V}\}^{\perp}$. *Then* $\mathsf{T}(\mathcal{L}) \subseteq \mathcal{L}$.

**PROOF:** By definition, $\mathbf{Z} \in \mathcal{L}$ if and only if $\mathbf{Z} \perp \mathbf{V}$, i.e., $0 = \langle \mathbf{Z}, \mathbf{V} \rangle$. In this case,

$$\begin{aligned} \langle \mathsf{T}(\mathbf{Z}), \mathbf{V} \rangle &= \langle \mathbf{Z}, \mathsf{T}^*(\mathbf{V}) \rangle = \langle \mathbf{Z}, \mathsf{T}(\mathbf{V}) \rangle \\ &= \langle \mathbf{Z}, \lambda\mathbf{V} \rangle = \lambda \langle \mathbf{Z}, \mathbf{V} \rangle = 0, \end{aligned}$$

and therefore $\mathsf{T}(\mathbf{Z}) \perp \mathbf{V}$, so $\mathsf{T}$ maps $\mathcal{L}$ into itself. □

**PROOF OF THE SPECTRAL THEOREM:** We proceed by induction on $n$. The case $n = 1$ is trivial, and the cases $n = 2$ and 3 were done in Section 16.2. We may therefore assume that $n$ is at least 2 and that the theorem has been proven for all self-adjoint transformations in inner product spaces of dimension $< n$.

From Propositions 16.4.2 and 16.4.3 we may choose a unit eigenvector $\mathbf{V}$ such that $\mathsf{T}(\text{Span}\,\{\mathbf{V}\}^{\perp}) \subseteq \text{Span}\,\{\mathbf{V}\}^{\perp}$. Choose an orthonormal basis $\mathbf{V}_2, \ldots, \mathbf{V}_n$ for $\mathcal{L} := \text{Span}\,\{\mathbf{V}\}^{\perp}$. Then the vectors $\mathbf{V} = \mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_n$ form orthonormal basis for $\mathcal{V}$. The matrix of $\mathsf{T}$ with respect to this basis is of the form

$$\mathsf{A} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & \mathsf{B} & \\ 0 & & & \end{bmatrix},$$

where $\lambda_1$ is the eigenvalue corresponding to the eigenvector $\mathbf{V}$, and by Theorem 16.1.4 $\mathsf{A}$, and hence also $\mathsf{B}$, is symmetric. The matrix $\mathsf{B}$ is therefore the matrix of a self-adjoint linear transformation

$$\mathsf{T}\,|_{\mathcal{L}} : \mathcal{L} \longrightarrow \mathcal{L}$$

The vector space $\mathcal{L}$ has dimension $n - 1$, and so by the induction assumption there exists an orthonormal basis $\mathbf{W}_2, \ldots, \mathbf{W}_n$ for $\mathcal{L}$ composed of eigenvectors of $\mathsf{T}\,|_{\mathcal{L}}$. The matrix of $\mathsf{T}$ with respect to the basis $\mathbf{V} = \mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_n$ therefore has the form

$$\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ 0 & \cdots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix},$$

where $\lambda_i$ is the eigenvalue corresponding to the eigenvector $\mathbf{W}_i$ for $i = 2, \ldots, n$. Therefore, $\mathsf{T}$ is diagonalizable, completing the induction step. □

Since the eigenvalues of a linear transformation in an inner product space are per definition *real* numbers we obtain:

**COROLLARY 16.4.4:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a self-adjoint linear transformation in the* $n$*-dimensional inner product space* $\mathcal{V}$. *Then the characteristic polynomial of* $\mathsf{T}$ *has* $n$ *real roots counting multiplicities.* □

## 16.5 Exercises

**1.** Compute the adjoint of each of the following:
(a) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, $\mathsf{T}(x, y, z) = (x + y + z, x + 2y + 2z, x + 2y + 3z)$.
(b) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, $\mathsf{T}(x, y, z) = (x + y - z, -x + 2y + 2z, x + 2y + 3z)$.
(c) $\mathsf{T} : \mathcal{P}_2(\mathbb{R}) \longrightarrow \mathcal{P}_2(\mathbb{R})$, $\mathsf{T}(p(x)) = x\frac{d}{dx}p(x) - \frac{d}{dx}(xp(x))$.
(d) $\mathsf{T} : \mathbb{R}^4 \longrightarrow \mathbb{R}^4$, $\mathsf{T}(x, y, z, w) = (x + y, y + z, z + w, w + x)$.
(e) $\mathsf{T} : \mathrm{Mat}_{3,3} \longrightarrow \mathrm{Mat}_{3,3}$, $\mathsf{T}(\mathbf{A}) = \mathbf{A}^{\mathrm{tr}} + \mathbf{A}$.
(f) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, $\mathsf{T}(x, y, z) = (-y + z, -x + 2z, x + 2y)$.

**2.** Let $\mathsf{S}, \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$. Prove:
(a) $(\mathsf{S} + \mathsf{T})^* = \mathsf{S}^* + \mathsf{T}^*$.
(b) $(a\mathsf{S})^* = a\mathsf{S}^*$.
(c) $(\mathsf{S}\mathsf{T})^* = \mathsf{T}^*\mathsf{S}^*$.
(d) $\mathsf{T} + \mathsf{T}^*$ is self-adjoint.

**3.** Diagonalize each of the following self-adjoint transformations:
(a) $\mathsf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$, $\mathsf{T}(x, y) = (2x + y, 2y + x)$.
(b) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, $\mathsf{T}(x, y, z) = (2x + y + z, x + 2y + 2z, x + y + 2z)$.
(c) $\mathsf{T} : \mathbb{R}^5 \longrightarrow \mathbb{R}^5$, $\mathsf{T}(x, y, z, u, v) = (2x + y, 2y + x, 2z + u + v, z + 2u + v, z + u + 2v)$.
(d) $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, $\mathsf{T}(x, y, z) = (y + z, x + z, x + y)$.
(e) $\mathsf{T} : \mathrm{Mat}_{3,3} \longrightarrow \mathrm{Mat}_{3,3}$, $\mathsf{T}(\mathbf{A}) = \mathbf{A}^{\mathrm{tr}}$.
(f) $\mathsf{T} : \mathrm{Mat}_{3,3} \longrightarrow \mathrm{Mat}_{3,3}$, $\mathsf{T}(\mathbf{A}) = \mathbf{A} - \mathbf{A}^{\mathrm{tr}}$.

**4.** Sketch the graph of each of the following quadratic forms:
(a) $8x^2 - 12xy + 17y^2 = 80$.
(b) $3x^2 + 2xy + 3y^2 = 4$.
(c) $5x^2 - 8xy + 5y^2 = 9$.
(d) $11x^2 - 24xy + 4y^2 + 6x + 8y = -15$.
(e) $16x^2 - 24xy + 9y^2 - 30x + 40y = 5$.

**5.** Suppose that $\varphi : \mathcal{V} \longrightarrow \mathbb{R}$ is a quadratic form. Show that for any two vectors $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$,

$$<\mathbf{A}, \mathsf{T}(\mathbf{B})> = \frac{1}{2}(\varphi(\mathbf{A} + \mathbf{B}) - \varphi(\mathbf{A}) - \varphi(\mathbf{B})).$$

**6.** Suppose that $\varphi : \mathcal{V} \longrightarrow \mathbb{R}$ is a quadratic form and

$$\varphi(A) = <\mathbf{A}, \mathsf{T}(\mathbf{A})>,$$
$$\varphi(A) = <\mathbf{A}, \mathsf{S}(\mathbf{A})>$$

for two self-adjoint transformations $\mathsf{S}, \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$. Show that $\mathsf{S} = \mathsf{T}$. (**HINT:** Use Exercise 5 to show that $<\mathbf{B}, \mathsf{T}(\mathbf{A})> = <\mathbf{B}, \mathsf{S}(\mathbf{A})>$ for all $\mathbf{B}$ and $\mathbf{A}$ in $\mathcal{V}$. Then recall that $<\mathsf{S}(\mathbf{B}), (\mathbf{A})> = <\mathbf{B}, \mathsf{S}(\mathbf{A})>$ for all $\mathbf{A}, \mathbf{B}$ in $\mathcal{V}$. Use this to show that $\mathsf{T} = \mathsf{S}^*$. Appeal to self-adjointness.)

**7.** Let $\mathcal{V}, <—,—>$ be an inner product space and denote by $^* : \mathcal{L}(\mathcal{V}, \mathcal{V}) \longrightarrow \mathcal{L}(\mathcal{V}, \mathcal{V})$ the mapping that associates to a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ its adjoint $\mathsf{T}^* : \mathcal{V} \longrightarrow \mathcal{V}$. Show that $^*$ is a linear transformation. Introduce an inner product in $\mathcal{L}(\mathcal{V}, \mathcal{V})$ with respect to which $^*$ is self adjoint. (**HINT:** Reexamine Example 4 in Section 15.1.)

The exercises that follow are intended to lead to a proof of the spectral theorem for arbitrary dimensions $n$. There are two basic steps in all the proofs that I know of. The first of these is to show that the transformation always has at least one eigenvector. In the next exercise we use complex numbers and self-adjointness to show that a **real** eigenvalue exists. The second step is to show that the subspace orthogonal to an eigenvector is invariant. The proof can then be completed by induction. This is the purpose of the following exercise.

**8.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a self-adjoint linear transformation on the finite-dimensional inner product space $\mathcal{V}$. Let $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ be an orthonormal basis for $\mathcal{V}$ and $\mathsf{A}$ the matrix of $\mathsf{T}$ relative to $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$. Introduce the linear transformation $\mathsf{T}_{\mathbb{C}} : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ whose matrix with respect to the standard basis is $\mathbf{A}$.

[1] Show that the characteristic polynomial of $\mathsf{T}_{\mathbb{C}}$ is equal to the characteristic polynomial of $\mathsf{T}$.

The fundamental theorem of algebra tells us that $\Delta_{\mathsf{T}_{\mathbb{C}}}(x)$ must have a root, which may, however, be complex. We want to show that it is real so that $\Delta_{\mathsf{T}}(X)$ has real roots, and hence $\mathsf{T}$ has an eigenvalue. To do this, let $\lambda$ be a root of $\Delta_{\mathsf{T}_{\mathbb{C}}}(x)$, $\mathbf{W} \in \mathbb{C}^n$ a corresponding eigenvector, and $\overline{\mathbf{W}}$ the vector whose coordinates are the complex conjugates of those of $\mathbf{W}$.

[2] Show that $\mathsf{T}_{\mathbb{C}}(\overline{\mathbf{W}}) = \overline{\lambda}\overline{\mathbf{W}}$, where $\overline{\lambda}$ is the complex conjugate of $\lambda$. (**HINT:** Compute the complex conjugate of the matrix product $\mathsf{A}(\mathbf{W})$ and use the fact that the entries of $\mathsf{A}$ are real numbers.)

[3] Show that the number $\overline{\mathbf{W}} \cdot \mathbf{W}^{\mathrm{tr}}$ is nonzero ($\cdot$ denotes the matrix product of the row vector $\overline{\mathbf{W}}$ with the column vector $\mathbf{W}^{\mathrm{tr}}$).

[4] Show that $\overline{|w|}\mathsf{A}(\mathbf{W}^{\mathrm{tr}}) = \lambda(\overline{\mathbf{W}} \cdot \mathbf{W}^{\mathrm{tr}})$, $\mathbf{W}\mathsf{A}\overline{\mathbf{W}}^{\mathrm{tr}} = \overline{\lambda}(\mathbf{W} \cdot \overline{\mathbf{W}}^{\mathrm{tr}})$.

[5] Show that $\lambda$ is real. (**Hint:** Use part [4] and take the transposes to show that $\overline{\mathbf{W}}\mathbf{A}(\mathbf{W})^{tr} = \mathbf{W}\mathbf{A}\overline{\mathbf{W}}^{tr}$ because $\mathbf{A}$ is symmetric.)

[6] Show that the characteristic polynomial of $\mathbf{T}$ has a real root.

[7] Show that all the roots of the characteristic polynomial of $\mathbf{T}$ are real.

**9.** Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a self-adjoint linear transformation on the finite-dimensional inner product space $\mathcal{V}$. Using either Proposition 16.4.2 or the previous exercise, we know that $\mathbf{T}$ has at least one eigenvector. Let $\mathbf{V}$ be such an eigenvector with corresponding eigenvalue $\lambda$.

[1] If $\mathbf{Y} \in \mathcal{V}$, is orthogonal to $\mathbf{V}$ then $\mathbf{T}(\mathbf{Y})$ is also orthogonal to $\mathbf{V}$. (**Hint:** Use $<\mathbf{T}(\mathbf{Y}), \mathbf{V}> = <\mathbf{Y}, \mathbf{T}(\mathbf{V})>$.)

[2] Let $\mathcal{W}$ be the subspace of $\mathcal{V}$ orthogonal to Span $\{\mathbf{V}\}$. Show that $\mathbf{T}(\mathcal{W}) \subseteq \mathcal{W}$.

[3] Show that $\mathbf{T}|_{\mathcal{W}}$ is self-adjoint.

[4] Show that there is an orthonormal basis $\{\mathbf{V}, \mathbf{W}_1, \ldots, \mathbf{W}_n\}$ for $\mathcal{V}$ with respect to which the matrix of $\mathbf{T}$ is

$$\begin{bmatrix} \lambda & 0 & \ldots & 0 \\ 0 & & & \\ \vdots & & \mathbf{B} & \\ 0 & & & \end{bmatrix},$$

where $\mathbf{B}$ is symmetric.

[5] Use Exercises 8 [5], 9 [4] and induction to prove the spectral theorem.

**10.** Let $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix with respect to the standard orthonormal basis of $\mathbb{R}^3$ is

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Diagonalize $\mathbf{T}$.

**11.** Let $\mathbf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation whose matrix with respect to the standard orthonormal basis of $\mathbb{R}^3$ is

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix}.$$

Diagonalize $\mathbf{T}$.

**12.** Same problem for each of the following matrices

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}. \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}. \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}.$$

**13.** Let $\mathsf{T} : \mathbb{C}^2 \longrightarrow \mathbb{C}^2$ be the linear transformation given by

$$\mathsf{T}(a, b) = (2ia + b, a),$$

where $a, b$ are complex numbers.

(a) Find the matrix of $\mathsf{T}$ with respect to the standard basis of $\mathbb{C}^2$.
(b) Show that the matrix of $\mathsf{T}$ is symmetric.
(c) Find the eigenvalues of $\mathsf{T}$.
(d) Show that $\mathsf{T}$ is *not* diagonalizable.
(e) What do you conclude about the validity of the following statement? Symmetric matrices are diagonalizable. (**HINT:** Show that this is true if the entries of the matrix are real, and can be false if the entries are allowed to be complex.)

**14.** Suppose $\mathsf{A} \in \mathrm{Mat}_{n,n}$ and that the columns of $\mathsf{A}$ regarded as vectors in $\mathbb{R}^n$ are orthogonal. Show that regarded as vectors in $\mathbb{R}^n$ the rows are also orthogonal. (**HINT:** Think about the relation between matrices and linear transformations.)

# Chapter 17
# Jordan Canonical Form

This chapter is of a distinctly more difficult nature than the preceding ones. In it will treat the problem of finding a particular simple *canonical*, or *normal*, matrix representative of a linear transformation. In the preceding chapter we treated this problem for self-adjoint linear transformations in a finite-dimensional inner product space. For a finite-dimensional inner product space $\mathcal{V}$ and a self-adjoint linear transformation

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$$

we saw that we could always find an orthonormal basis for $\mathcal{V}$ such that the corresponding matrix of $\mathsf{T}$ was diagonal. The required basis would be composed of the normed eigenvectors of $\mathsf{T}$, and the diagonal entries of the corresponding matrix are the eigenvalues of $\mathsf{T}$. The problem to be considered here is Can we achieve a similar **normal form** in general? One answer to this question leads to the **Jordan canonical form**, also called the **Jordan normal form**. Before we arrive at this goal we will require quite a few preparatory steps.

To begin, let us consider where we might encounter difficulties in the general case if we were to proceed by analogy with self-adjoint linear transformations. What we achieved in the self-adjoint case is a decomposition

$$\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_q$$

of the vector space $\mathcal{V}$ into a direct sum[1] of eigenspaces $\mathcal{U}_i$ for $1 \leq i \leq q$,

---

[1] A direct sum decomposition of a vector space with respect to subspaces is a special case of the sum of subspaces, and will be defined, explained, and illustrated in what follows.

of $\mathsf{T}$. For a general linear transformation

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$$

there are two problems that prevent us from obtaining an analogous result.

(1) $\mathsf{T}$ may not have any eigenvalues at all.

(2) $\mathsf{T}$ may have eigenvalues, but the sum of the distinct eigenspaces $\mathcal{U}_1 + \cdots + \mathcal{U}_q$ may not be all of $\mathcal{V}$.

The first problem is not so serious. We can overcome it by working with vector spaces over the **complex numbers**. (The advantage so gained has already been indicated in Chapter 15 and the exercises to Chapter 14.) On the other hand, the second problem is more serious. It reflects more a defect in the linear transformation $\mathsf{T}$ rather than in the scalars. Here are some examples to illustrate this point.

**EXAMPLE:** We will consider three linear transformations from $\mathbb{R}^3$ to itself,

$$\mathsf{T}_i : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \qquad i = 1, 2, 3;$$

each one of which has the real number $e$ as eigenvalue with multiplicity 3, but, whose corresponding eigenspace $\mathcal{V}_e$ will have dimension 1, 2, and 3, respectively. The first, $\mathsf{T}_1 : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is the linear transformation represented by the matrix

$$\mathbf{A}_1 = \begin{bmatrix} e & 0 & 0 \\ 0 & e & 0 \\ 0 & 0 & e \end{bmatrix}$$

with respect to the standard basis of $\mathbb{R}^3$. Then $e$ is an eigenvalue of $\mathsf{T}_1$ with the geometric multiplicity 3, and the standard basis vectors are eigenvectors of $\mathsf{T}_1$.

$\mathsf{T}_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ is the linear transformation represented by the matrix

$$\mathbf{A}_2 = \begin{bmatrix} e & 1 & 0 \\ 0 & e & 0 \\ 0 & 0 & e \end{bmatrix}$$

with respect to the standard basis of $\mathbb{R}^3$. The characteristic polynomial of $\mathsf{T}_2$ is

$$p(t) = (t - e)^3,$$

so $e$ is an eigenvalue of algebraic multiplicity 3. But there are only two linearly independent eigenvectors belonging to this eigenvalue. To see this, recall that the eigenvectors $\mathbf{E}$ are the nonzero solutions of

$$(*) \qquad (\mathsf{T}_2 - e\mathsf{I})(\mathbf{E}) = \mathbf{0}.$$

The vectors $\mathbf{E} = (x, y, z) \in \mathbb{R}^3$ that satisfy $(*)$ are the solutions of the linear system

$$y = 0.$$

So $(1, 0, 0), (0, 0, 1)$ is a basis for $\mathcal{V}_e$, and $e$ is of geometric multiplicity 2.

The third transformation in this example, $\mathsf{T}_3 : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$, is the linear transformation represented by the matrix

$$\mathsf{A}_3 = \begin{bmatrix} e & 1 & 0 \\ 0 & e & 1 \\ 0 & 0 & e \end{bmatrix}$$

with respect to the standard basis of $\mathbb{R}^3$. The characteristic polynomial is again

$$p(t) = (t - e)^3,$$

so $e$ is again an eigenvalue with algebraic multiplicity 3. To find the geometric dimension we must find a basis for

$$\ker(\mathsf{T}_3 - e\mathsf{I}).$$

This leads to the system of linear equations

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

namely,

$$y = 0,$$
$$z = 0,$$

so that $\mathcal{V}_e = \mathcal{L}((1, 0, 0))$, and $e$ has geometric multiplicity only 1. From this example we learn that there are linear transformations that simply do not have enough ***eigenvectors*** (a lack of eigen***values*** is not the problem) to be representable by a diagonal matrix. In order to represent such a linear transformation by as simple a matrix as possible, we must attempt to cleverly enlarge the eigenspaces whose dimensions are smaller than the algebraic multiplicity of the corresponding eigenvalue. The enlargement should be made in such a way that the enlarged space admits a basis with respect to which the matrix of $\mathsf{T}$ is almost diagonal.

## 17.1 Invariant Subspaces

To achieve our goal we are going to need a number of new ideas. The first, that of an **invariant subspace**, is central to all that follows.

**DEFINITION :** $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation. A vector subspace* $\mathcal{U}$ *in* $\mathcal{V}$ *is called* **$\mathsf{T}$-invariant** *if* $\mathsf{T}(\mathcal{U}) \subseteq \mathcal{U}$.

**Example 1:** If $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ has an eigenvalue $e$ then the eigenspace $\mathcal{V}_e$ is $\mathsf{T}$-invariant. To see this, note that $\mathbf{E} \in \mathcal{V}_e$, then $\mathsf{T}(\mathbf{E}) = e\mathbf{E}$ is a multiple of $\mathbf{E}$ and so belongs to $\mathcal{V}_e$, since $\mathcal{V}_e$ is a subspace.

**Example 2:** Let $\mathsf{T} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be defined by the matrix (see the example above)

$$\mathsf{A} = \begin{bmatrix} e & 1 & 0 \\ 0 & e & 0 \\ 0 & 0 & e \end{bmatrix}.$$

Then $\mathcal{U} = \mathcal{L}((1, 0, 0), (0, 1, 0))$ is an invariant subspace. To see this, note that

$$\mathsf{T}(1, 0, 0) = (e, 0, 0) = e(1, 0, 0) + 0(0, 1, 0),$$

$$\mathsf{T}(0, 1, 0) = (1, e, 0) = 1(1, 0, 0) + e(0, 1, 0),$$

so $\mathsf{T}(1, 0, 0), \mathsf{T}(0, 1, 0) \in \mathcal{U}$. By linearity of $\mathsf{T}$, this implies that $\mathsf{T}(\mathcal{U}) \subsetneq \mathcal{U}$, so $\mathcal{U}$ is $\mathsf{T}$-invariant. Since $(0, 1, 0)$ is ***not*** an eigenvector of $\mathsf{T}$, the subspace $\mathcal{U}$ is not contained in an eigenspace of $\mathsf{T}$, because if it were, then since $(1, 0, 0) \in \mathcal{U}$ it would follow that $\mathcal{U} \subseteq \mathcal{V}_e$, which is definitely false.

When a linear transformation has an invariant subspace, we can find a simple matrix representation for it.

**Proposition 17.1.1:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in the $n$-dimensional vector space $\mathcal{V}$. Suppose $\mathcal{U}$ is an $m$-dimensional $\mathsf{T}$-invariant subspace of $\mathcal{V}$. Then with respect to a suitable basis for $\mathcal{V}$, $\mathsf{T}$ can be represented by a matrix in block form*

$$\begin{bmatrix} \mathsf{A} & \mathsf{C} \\ \mathsf{0} & \mathsf{B} \end{bmatrix}.$$

*The matrix $\mathsf{A}$ is $m \times m$, the matrix $\mathsf{B}$ is $(n-m)\times(n-m)$, the matrix $\mathsf{C}$ is $m \times (n-m)$, and the zero matrix $\mathsf{0}$ has size $(n-m)\times m$.*

**Proof:** Choose a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_m\}$ for $\mathcal{U}$ and extend this to a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1, \ldots, \mathbf{B}_{n-m}\}$ for $\mathcal{V}$. Since $\mathcal{U}$ is $\mathsf{T}$-invariant, $\mathsf{T}(\mathbf{A}_i) \in \mathcal{U} = \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_m)$, so we have

$$\mathsf{T}(\mathbf{A}_i) = \sum_{j=1}^{m} a_{j,i}\mathbf{A}_j,$$

and the first $m$ columns of the matrix of $\mathsf{T}$ with respect to this basis have no entries in rows $m+1$ through $n$. This says, the matrix is of the above block form. $\square$

**DEFINITION:** *Let $\mathcal{V}$ be a vector space and let $\mathcal{S}$, $\mathcal{T}$ be subspaces of $\mathcal{V}$. We say that $\mathcal{V}$ is the* **direct sum** *of $\mathcal{S}$ and $\mathcal{T}$ if*

$$\mathcal{V} = \mathcal{S} + \mathcal{T} \qquad \text{and} \qquad \mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}.$$

*In this case we write $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$.*

**EXAMPLE 3:** In $\mathbb{R}^3$ consider the subspaces

$$\mathcal{S} = \mathcal{L}((1, -1, 0), (0, 1, -1)), \text{ and } \mathcal{T} = \mathcal{L}((1, 1, 1)).$$

Then it is not hard to verify that

$$\mathbb{R}^3 = \mathcal{S} \oplus \mathcal{T}.$$

**PROPOSITION 17.1.2:** *Let $\mathcal{V}$ be a vector space and let $\mathcal{S}$, $\mathcal{T}$ be subspaces of $\mathcal{V}$. Then $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ if and only if every vector in $\mathcal{V}$ can be written uniquely in the form $\mathbf{S} + \mathbf{T}$, where $\mathbf{S} \in \mathcal{S}$ and $\mathbf{T} \in \mathcal{T}$.*

**PROOF:** Suppose $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$. Let $\mathbf{A} \in \mathcal{V}$. Since $\mathcal{V} = \mathcal{S} + \mathcal{T}$, there exist vectors $\mathbf{S} \in \mathcal{S}$, $\mathbf{T} \in \mathcal{T}$ such that

$$(*) \qquad \mathbf{A} = \mathbf{S} + \mathbf{T}.$$

If $\overline{\mathbf{S}} \in \mathcal{S}$, $\overline{\mathbf{T}} \in \mathcal{T}$ are any vectors such that

$$\mathbf{A} = \overline{\mathbf{S}} + \overline{\mathbf{T}},$$

then

$$\mathbf{S} + \mathbf{T} = \mathbf{A} = \overline{\mathbf{S}} + \overline{\mathbf{T}},$$

which implies

$$\mathbf{S} - \overline{\mathbf{S}} = \mathbf{T} - \overline{\mathbf{T}}.$$

So $\mathbf{S} - \overline{\mathbf{S}} \in \mathcal{T}$. But $\mathbf{S} - \overline{\mathbf{S}} \in \mathcal{S}$, since $\mathcal{S}$ is a subspace. Therefore,

$$\mathbf{S} - \overline{\mathbf{S}} \in \mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\},$$

so $\mathbf{S} - \overline{\mathbf{S}} = \mathbf{0}$, or $\mathbf{S} = \overline{\mathbf{S}}$. A similar argument shows that $\mathbf{T} - \overline{\mathbf{T}} = \mathbf{0}$, or $\mathbf{T} = \overline{\mathbf{T}}$, so the representation of $\mathbf{A}$ in $(*)$ is unique.

The converse is immediate. $\square$

**PROPOSITION 17.1.3:** *Let $\mathcal{V}$ be a finite-dimensional vector space and let $\mathcal{S}$, $\mathcal{T}$ be subspaces of $\mathcal{V}$. If $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$, then*

$$\dim(\mathcal{V}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$$

*Moreover, if $\{\mathbf{A}_1, \ldots, \mathbf{A}_s\}$ is a basis for $\mathcal{S}$ and $\{\mathbf{B}_1, \ldots, \mathbf{B}_t\}$ a basis for $\mathcal{T}$, then $\{\mathbf{A}_1, \ldots, \mathbf{A}_s, \mathbf{B}_1, \ldots, \mathbf{B}_t\}$ is a basis for $\mathcal{V}$.*

**PROOF:** The proof is similar to that of Theorem 8.4.2. □

**EXAMPLE 4:** In $\mathbb{R}^3$ consider the subspaces

$$\mathcal{S} = \mathcal{L}((1, -1, 0), (0, 1, -1)), \text{ and } \mathcal{R} = \mathcal{L}((1, 1, 1), (1, -2, 1)).$$

Then

$$\mathbb{R}^3 = \mathcal{R} + \mathcal{S}$$

but

$$\mathbb{R}^3 = \mathcal{S} \oplus \mathcal{T}.$$

because $\dim \mathcal{R} + \dim \mathcal{S} = 4 \neq 3 = \dim \mathbb{R}^3$. In fact,

$$1 \cdot (1, -1, 0) + (-1)(0, 1, -1) = (1, -2, 1)$$

shows that

$$(1, -2, 1) \in \mathcal{R} \cap \mathcal{S}.$$

**PROPOSITION 17.1.4:** *Let $\mathcal{V}$ be a finite-dimensional vector space, and let $\mathcal{S}, \mathcal{T}$ be subspaces of $\mathcal{V}$ such that $\mathcal{V} = \mathcal{S} + \mathcal{T}$. Then $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ if and only if* $\dim(\mathcal{V}) = \dim \mathcal{S} + \dim \mathcal{T}$.

**PROOF:** Do Exercise 10 in Chapter 6. □

**COROLLARY 17.1.5:** *Let $(\mathcal{V}, \langle —, — \rangle)$ be an inner product space and let $\mathcal{W}$ be a subspace of $\mathcal{V}$. Then $\mathcal{V} = \mathcal{W} \oplus \mathcal{W}^{\perp}$.*

**PROOF:** This follows from Propositions 17.1.4 and 15.4.1. □

**DEFINITION:** *Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation. A $\mathbf{T}$-invariant subspace $\mathcal{U}$ in $\mathcal{V}$ is called* **completely invariant,** *or* **fully invariant,** *if there exists a $\mathbf{T}$-invariant subspace $\mathcal{W}$ in $\mathcal{V}$ such that $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$. The subspace $\mathcal{W}$ is called a* **T-invariant complement** *for $\mathcal{U}$, and together $\mathcal{U}$ and $\mathcal{W}$ are called a* **T-invariant complementary pair of subspaces.**

**COROLLARY 17.1.6:** *Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in the $n$-dimensional vector space $\mathcal{V}$. Suppose $\mathcal{U}$ is a $\mathbf{T}$-completely invariant subspace of dimension $m$ in $\mathcal{V}$. Then with respect to a suitable basis for $\mathcal{V}$, $\mathbf{T}$ can be represented in block matrix form*

$$\begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix},$$

*where $\mathbf{A}$ is an $m \times m$ matrix and $\mathbf{B}$ is an $(n - m) \times (n - m)$ matrix.*

**PROOF:** Let $\mathcal{W}$ be a $\mathsf{T}$-invariant complement for $\mathcal{U}$. Let $\mathbf{A}_1, \ldots, \mathbf{A}_m$ be a basis for $\mathcal{U}$ and $\mathbf{B}_1, \ldots, \mathbf{B}_{n-m}$ a basis for $\mathcal{W}$. Then $\mathbf{A}_1, \ldots, \mathbf{A}_m, \mathbf{B}_1, \ldots, \mathbf{B}_{n-m}$ is a basis for $\mathcal{V}$. The matrix of $\mathsf{T}$ with respect to this basis is of the required form. □

This corollary shows that one way to simplify a matrix representative for $\mathsf{T}$ is to find a complementary pair of $\mathsf{T}$-invariant subspaces. But how does one find such pairs? The following results start us on the right track.

**PROPOSITION 17.1.7:** *Suppose we are given linear transformations*

$$\mathsf{S}, \mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$$

*such that*

$$\mathsf{S} \cdot \mathsf{T} = \mathsf{T} \cdot \mathsf{S}$$

*(that is,* $\mathsf{S}$ *and* $\mathsf{T}$ *commute with each other). Then* $\ker(\mathsf{S})$ *and* $\operatorname{Im}(\mathsf{S})$ *are* $\mathsf{T}$*-invariant subspaces.*

**PROOF:** If $\mathbf{A} \in \ker(\mathsf{S})$, then the computation

$$\mathsf{S} \cdot \mathsf{T}(\mathbf{A}) = \mathsf{T} \cdot \mathsf{S}(\mathbf{A}) = \mathsf{T}(\mathbf{0}) = \mathbf{0}$$

shows that $\mathsf{T}(\mathbf{A}) \in \ker(\mathsf{S})$; so $\ker(\mathsf{S})$ is a $\mathsf{T}$-invariant subspace. Similarly, if $\mathbf{B} \in \operatorname{Im}(\mathsf{S})$, then $\mathbf{B} = \mathsf{S}(\mathbf{C})$ for some $\mathbf{C} \in \mathcal{V}$, and therefore

$$\mathsf{T}(\mathbf{B}) = \mathsf{T} \cdot \mathsf{S}(\mathbf{C}) = \mathsf{S} \cdot \mathsf{T}(\mathbf{C}) = \mathsf{S}\big(\mathsf{T}(\mathbf{C})\big)$$

since $\mathsf{S}$ and $\mathsf{T}$ commute with each other, so $\mathsf{T}(\mathbf{B}) \in \operatorname{Im}(\mathsf{S})$, and so $\operatorname{Im}(\mathsf{S})$ is a $\mathsf{T}$-invariant subspace. □

In general, it can be very difficult to determine the set $\mathsf{S}$ of linear transformations that commute with a fixed linear transformation $\mathsf{T}$. However, certain elements in this set are obvious.

**COROLLARY 17.1.8:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation and*

$$\mathsf{S} = a_0\mathsf{I} + a_1\mathsf{T} + \cdots + a_m\mathsf{T}^m : \mathcal{V} \longrightarrow \mathcal{V},$$

*where* $a_0, \ldots, a_m$ *are scalars. Then* $\ker(\mathsf{S})$ *and* $\operatorname{Im}(\mathsf{S})$ *are* $\mathsf{T}$*-invariant subspaces.*

**PROOF:** Clearly $\mathsf{S}$ commutes with $\mathsf{T}$ so the result follows from Proposition 17.1.7. □

## 17.2 Nilpotent Transformations

It is impossible to distinguish the nilpotent linear transformations in $\mathbb{R}^4$ whose matrices with respect to the standard basis are

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

by examining their characteristic polynomials: Both have characteristic polynomial $t^4$. Nor will examining their eigenspaces distinguish them: Both have 0 as a quadruple eigenvalue, with corresponding eigenspace, which is the kernel, of dimension 2. Since the square of the first matrix is zero and the second nonzero, they are certainly different in the sense that no change of basis can turn one into the other.

The next step in our program, therefore, and the key that unlocks the door to the Jordan canonical form, is the generalization of the structure theorem 12.2.1 in Section 12.2 of nilpotent transformations with a cyclic vector (i.e., nilpotent transformations whose nilpotence index is the dimension of the vector space in which they are defined) to completely general nilpotent transformations.

**THEOREM 17.2.1** (Nilpotent Structure Theorem): *Let $\mathcal{V}$ be a finite-dimensional vector space and suppose that $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is a nilpotent linear transformation. Then there exists a direct sum decomposition*

$$\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_t$$

*such that for $i = 1, \ldots, t$ $\mathcal{U}_i$ is $\mathbf{T}$-invariant and the linear transformation $\mathbf{T} : \mathcal{U}_i \longrightarrow \mathcal{U}_i$ induced by $\mathbf{T}$ has a cyclic vector.*

**PROOF:** Let $p$ be the index of nilpotence (see Section 12.2) of $\mathbf{T}$ that is $p$ is the smallest integer such that $\mathbf{T}^p = \mathbf{0}$. Define

$$\mathcal{K}_i = \ker(\mathbf{T}^i) \qquad \text{for} \quad i = 1, \ldots, p.$$

Then the $\mathcal{K}_i$ are vector subspaces of $\mathcal{V}$. Notice that $\mathcal{K}_0 = \{\mathbf{0}\}$, since $\mathbf{T}^0 = \mathbf{I}$ by definition, and $\mathcal{K}_p = \mathcal{V}$. Moreover, if $0 \leq i \leq p$, then we have

$$\mathbf{A} \in \mathcal{K}_i \iff \mathbf{T}^i(\mathbf{A}) = \mathbf{0} \Rightarrow \mathbf{T}^{i+1}(\mathbf{A}) = \mathbf{T}(\mathbf{T}^i(\mathbf{A})) = \mathbf{T}(\mathbf{0}) = \mathbf{0},$$

so $\mathbf{A} \in \mathcal{K}_{i+1}$. Thus we have an **ascending chain of subspaces**[2]

$$\{\mathbf{0}\} = \mathcal{K}_0 \subseteq \mathcal{K}_1 \subseteq \cdots \subseteq \mathcal{K}_p = \mathcal{V},$$

and $\mathbf{T}(\mathcal{K}_i) \subseteq \mathcal{K}_{i-1}$.

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[2] By an **ascending chain of subspaces** we mean a sequence of subspaces that are included one in the next.

**REMARK:** In the case that $\mathsf{T}$ has a cyclic vector $\mathbf{A}$, then

$$\mathcal{K}_i = \mathcal{L}\left\{\mathsf{T}^{p-i}(\mathbf{A}), \ldots, \mathsf{T}^{p-1}(\mathbf{A})\right\}$$

for $i = 1, \ldots, p$, and $\mathcal{K}_0 = \{\mathbf{0}\}$.

We proceed to construct the invariant subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_t$ of $\mathcal{V}$. To do this we need a new idea.

**DEFINITION:** *Let $\mathcal{V}$ be a vector space and $\mathcal{W}$ a subspace. A set of vectors $E$ is said to be* **linearly dependent over** $\mathcal{W}$, *or* **modulo** $\mathcal{W}$, *if there are distinct vectors $\mathbf{A}_1, \ldots, \mathbf{A}_k$ in $E$ and numbers $a_1, \ldots, a_k$, not all zero, such that*

$$a_1\mathbf{A}_1 + \cdots + a_k\mathbf{A}_k \in \mathcal{W}.$$

*A set of vectors $E$ that is not linearly dependent over $\mathcal{W}$ is said to be* **linearly independent over** $\mathcal{W}$.

**PROPOSITION 17.2.2:** *Let $\mathcal{V}$ be a vector space and $\mathcal{W}$ a finite-dimensional subspace of $\mathcal{V}$. A set of vectors $E$ is linearly independent over $\mathcal{W}$ if and only if for every basis $\{\mathbf{B}_1, \ldots, \mathbf{B}_s\}$ of $\mathcal{W}$, the set of vectors $E \cup \{\mathbf{B}_1, \ldots, \mathbf{B}_s\}$ is linearly independent in $\mathcal{V}$.*

**PROOF:** Suppose that $E$ is linearly independent over $\mathcal{W}$ and that $\{\mathbf{B}_1, \ldots, \mathbf{B}_s\}$ is a basis for $\mathcal{W}$. Let $\{\mathbf{A}_1, \ldots, \mathbf{A}_k\}$ be vectors of $E$, and $a_1, \ldots, a_k$ and $b_1, \ldots, b_s$ numbers such that

$$a_1\mathbf{A}_1 + \cdots + a_k\mathbf{A}_k + b_1\mathbf{B}_1 + \cdots + b_s\mathbf{B}_s = \mathbf{0}.$$

Then, of course,

$$a_1\mathbf{A}_1 + \cdots + a_k\mathbf{A}_k = -(b_1\mathbf{B}_1 + \cdots + b_s\mathbf{B}_s) \in \mathcal{W},$$

so $a_1 = \cdots = a_k = 0$. But then

$$b_1\mathbf{B}_1 + \cdots + b_s\mathbf{B}_s = \mathbf{0},$$

so $b_1 = \cdots = b_s = 0$ also. The converse is just as easy. $\square$

**DEFINITION:** *Let $\mathcal{V}$ be a vector space and $\mathcal{W}$ a subspace of $\mathcal{V}$. A set of vectors $E$ is said to* **span** $\mathcal{V}$ **over** $\mathcal{W}$, *or* **modulo** $\mathcal{W}$, *if $\mathcal{V} = \mathcal{L}(E) + \mathcal{W}$. If, in addition, $E$ is linearly independent over $\mathcal{W}$, then we say that $E$ is a* **basis** *for $\mathcal{V}$* **over** $\mathcal{W}$.

**PROPOSITION 17.2.3:** *If $\mathcal{V}$ is a finite-dimensional vector space and $\mathcal{W}$ is a subspace, then there exists a basis for $\mathcal{V}$ over $\mathcal{W}$. The number of vectors in a basis for $\mathcal{V}$ over $\mathcal{W}$ is* $\dim(\mathcal{V}) - \dim(\mathcal{W})$.

**Proof:** To prove the existence of a basis for $\mathcal{V}$ over $\mathcal{W}$ is easy. We choose a basis $\{\mathbf{B}_1, \ldots, \mathbf{B}_s\}$ for $\mathcal{W}$ and extend this to a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_k, \mathbf{B}_1, \ldots, \mathbf{B}_s\}$ for $\mathcal{V}$. Then by Proposition 17.2.2 the set $\{\mathbf{A}_1, \ldots, \mathbf{A}_k\}$ is linearly independent over $\mathcal{W}$. Since

$$\begin{aligned}\mathcal{V} &= \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k, \mathbf{B}_1, \ldots, \mathbf{B}_s)\\ &= \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k) + \mathcal{L}(\mathbf{B}_1, \ldots, \mathbf{B}_s) = \mathcal{L}(\mathbf{A}_1, \ldots, \mathbf{A}_k) + \mathcal{W},\end{aligned}$$

it follows that $\{\mathbf{A}_1, \ldots, \mathbf{A}_k\}$ is a basis for $\mathcal{V}$ over $\mathcal{W}$. Since $s + k = \dim(\mathcal{V})$, it follows that $k = \dim(\mathcal{V}) - \dim(\mathcal{W})$. $\square$

We return to the proof of Theorem 17.2.1. We have the nested chain of subspaces

$$\{\mathbf{0}\} = \mathcal{K}_0 \subseteq \mathcal{K}_1 \subseteq \cdots \subseteq \mathcal{K}_p = \mathcal{V}.$$

Choose a basis $\{\mathbf{A}_1, \ldots, \mathbf{A}_r\}$ for $\mathcal{V} = \mathcal{K}_p$ over $\mathcal{K}_{p-1}$. Then

$$\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_r) \in \mathcal{K}_{p-1}.$$

We claim that $\{\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_r)\}$ are linearly independent over $\mathcal{K}_{p-2}$. For suppose that there are numbers $a_1, \ldots, a_r$ such that

$$a_1\mathsf{T}(\mathbf{A}_1) + \cdots + a_r\mathsf{T}(\mathbf{A}_r) \in \mathcal{K}_{p-2}.$$

Apply $\mathsf{T}^{p-2}$ to this equation. We get, by definition of $\mathcal{K}_{p-2}$,

$$\mathbf{0} = \mathsf{T}^{p-2}(a_1\mathsf{T}(\mathbf{A}_1) + \cdots + a_r\mathsf{T}(\mathbf{A}_r)) = \mathsf{T}^{p-1}(a_1\mathbf{A}_1 + \cdots + a_r\mathbf{A}_r).$$

Therefore,

$$a_1\mathbf{A}_1 + \cdots + a_r\mathbf{A}_r \in \mathcal{K}_{p-1}.$$

But $\mathbf{A}_1, \ldots, \mathbf{A}_r$ is a basis for $\mathcal{K}_p$ over $\mathcal{K}_{p-1}$, and so $a_1 = \cdots = a_r = 0$, establishing the claim.

Extend $\{\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_r)\}$ to a basis $\{\mathsf{T}(\mathbf{A}_1), \ldots, \mathsf{T}(\mathbf{A}_r), \mathbf{A}_{r+1}, \ldots, \mathbf{A}_q\}$ for $\mathcal{K}_{p-1}$ over $\mathcal{K}_{p-2}$. Then

$$\mathsf{T}^2(\mathbf{A}_1), \ldots, \mathsf{T}^2(\mathbf{A}_r), \mathsf{T}(\mathbf{A}_{r+1}), \ldots, \mathsf{T}(\mathbf{A}_q) \in \mathcal{K}_{p-2},$$

and the argument we just used may be applied again to show that $\left\{\mathsf{T}^2(\mathbf{A}_1), \ldots, \mathsf{T}^2(\mathbf{A}_r), \mathsf{T}(\mathbf{A}_{r+1}), \ldots, \mathsf{T}(\mathbf{A}_q)\right\}$ are linearly independent over $\mathcal{K}_{p-3}$.

If we continue in this way, we obtain after $p$ steps a basis for $\mathcal{V}$ arranged as in the following scheme:

$$\begin{array}{llll}
\mathbf{A}_1, \ldots\ldots\ldots, \mathbf{A}_r, & & & \\
\mathsf{T}(\mathbf{A}_1), \ldots\ldots, \mathsf{T}(\mathbf{A}_r), & \mathbf{A}_{r+1}, \ldots\ldots\ldots, \mathbf{A}_q, & & \\
\ldots\ldots\ldots\ldots & \ldots\ldots\ldots\ldots & \ldots & \ldots \\
\mathsf{T}^{p-1}(\mathbf{A}_1), \ldots, \mathsf{T}^{p-1}(\mathbf{A}_r), & \mathsf{T}^{p-2}(\mathbf{A}_{r+1}), \ldots, \mathsf{T}^{p-2}(\mathbf{A}_q), & \ldots, \mathbf{A}_s, & \ldots, \mathbf{A}_m.
\end{array}$$

Notice that the vectors in each column span an invariant subspace for which the vector at the top of the column is a cyclic vector. $\square$

**REMARK:** Notice that the dimension of $\mathcal{K}_i$ over $\mathcal{K}_{i-1}$ is the number of elements in the $i$-th row, which is never zero, so the inclusions in the chain

$$\{\mathbf{0}\} = \mathcal{K}_0 \subset \mathcal{K}_1 \subset \cdots \subset \mathcal{K}_p$$

are all proper, i.e., no two succesive subspaces are the same.

Here is an example to help clarify what is going on in Theorem 17.2.1. It will be convenient in this and succeeding examples to introduce one more definition.

**DEFINITION:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a nilpotent linear transformation of index* $p$. *If* $\mathbf{A} \neq \mathbf{0} \in \mathcal{V}$, *there is a smallest integer* $r$ *such that* $\mathsf{T}^r(\mathbf{A}) = \mathbf{0}$ *(note* $r \leq p$*). The set of vectors*

$$\left\{\mathbf{A}, \mathsf{T}(\mathbf{A}), \ldots, \mathsf{T}^{r-1}(\mathbf{A})\right\}$$

*are called the* **T-orbit of A** *(or the* **orbit of A under T***). The integer* $r$ *is called the* **length** *of the orbit.*

**EXAMPLE 1:** Let $\mathsf{T} : \mathbb{R}^5 \longrightarrow \mathbb{R}^5$ be the linear transformation given by

$$\mathsf{T}(x, y, u, v, w) = (0, u + v, 0, u, x + v).$$

Then

$$\mathsf{T}^2(x, y, u, v, w) = (0, u, 0, 0, u)$$
$$\mathsf{T}^3(x, y, u, v, w) = (0, 0, 0, 0, 0)$$

so $\mathsf{T}$ is nilpotent of index 3. The matrix of $\mathsf{T}$ with respect to the standard basis of $\mathbb{R}^5$ is

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix},$$

which despite the large number of zero entries is not very informative. The proof of Theorem 17.2.1 provides a more suitable basis for $\mathbb{R}^5$.

We first need the chain of subspaces

$$\{\mathbf{0}\} = \mathcal{K}_0 \subset \mathcal{K}_1 \subset \mathcal{K}_2 \subset \mathcal{K}_3 = \mathbb{R}^5.$$

The vector space $\mathcal{K}_1$ is the kernel of $\mathsf{T}$ and so consists of all vectors in $\mathbb{R}^5$ satisfying the equations

$$\left.\begin{array}{l}\left.\begin{array}{rcrl} u & + & v & = 0 \\ & & u & = 0 \end{array}\right] \Rightarrow \quad v = 0 \\ \begin{array}{rcrl} x & + & v & = 0 \end{array}\end{array}\right] \Rightarrow \quad x = 0,$$

so $\mathcal{K}_1$ is a 2-dimensional vector space. The vector space $\mathcal{K}_2$ is the kernel of $\mathsf{T}^2$ and so consists of all the vectors in $\mathbb{R}^5$ satisfying

$$u = 0,$$

so $\mathcal{K}_2$ is 4-dimensional.

The next step is to find a basis for $\mathcal{K}_3 = \mathbb{R}^5$ over $\mathcal{K}_2$. Since $\mathcal{K}_2$ is 4-dimensional and $\mathcal{K}_5 = \mathbb{R}^5$ is 5-dimensional, such a basis consists of exactly one vector, which has a $\mathsf{T}$-orbit of length 3. The vector

$$\mathbf{A}_1 = (0, 0, 1, 0, 0)$$

fills the criterion, since

$$\mathsf{T}(\mathbf{A}_1) = (0, 1, 0, 1, 0),$$
$$\mathsf{T}^2(\mathbf{A}_1) = (0, 1, 0, 0, 1).$$

The number of vectors in a basis for $\mathcal{K}_2$ over $\mathcal{K}_1$ is 2 by Proposition 17.2.3. One such vector is $\mathsf{T}(\mathbf{A}_1)$. The elements of $\mathcal{K}_2$ all satisfy $u = 0$, and those elements in $\mathcal{K}_1$ also satisfy

$$y = 0 = v.$$

We need a vector in $\mathcal{K}_2$, but not in Span $\{\mathcal{K}_1, \mathsf{T}(\mathbf{A}_1)\}$ to extend $\mathsf{T}(\mathbf{A}_1)$ to a basis for $\mathcal{K}_2$ over $\mathcal{K}_1$. One possible choice is the vector

$$\mathbf{B}_1 = (1, 0, 0, 0, 0),$$

since

$$\mathsf{T}(\mathbf{B}_1) = (0, 0, 0, 0, 1) \neq \mathbf{0}$$

but

$$\mathsf{T}^2(\mathbf{B}_2) = \mathbf{0}.$$

Therefore,

$$\mathsf{T}(\mathbf{A}_1) = (0, 1, 0, 1, 0),$$
$$\mathbf{B}_1 = (1, 0, 0, 0, 0)$$

is a basis for $\mathcal{K}_2$ over $\mathcal{K}_1$. Finally note that the vectors

$$\mathsf{T}^2(\mathbf{A}_1) = (0, 1, 0, 0, 1)$$
$$\mathsf{T}(\mathbf{B}_1) = (0, 0, 0, 0, 1)$$

are a basis for $\mathcal{K}_1 = \ker(\mathsf{T})$. Arranging these vectors in a schema as in the proof of Theorem 17.2.1 gives

$$\begin{array}{llll} \mathbf{A}_1 & & & \text{basis for } \mathcal{K}_3 = \mathbb{R}^5 \text{ over } \mathcal{K}_2 \\ \mathsf{T}(\mathbf{A}_1) & \mathbf{B}_1 & & \text{basis for } \mathcal{K}_2 \text{ over } \mathcal{K}_1 \\ \mathsf{T}^2(\mathbf{A}_1) & \mathsf{T}(\mathbf{B}_1) & & \text{basis for } \mathcal{K}_1 = \ker(\mathsf{T}). \end{array}$$

The vectors $\left\{\mathbf{A}_1, \mathsf{T}(\mathbf{A}_1), \mathsf{T}^2(\mathbf{A}_1)\right\}$ span a $\mathsf{T}$-invariant subspace $\mathcal{U}_1$ with cyclic vector $\mathbf{A}_1$, and the vectors $\{\mathbf{B}_1, \mathsf{T}(\mathbf{B}_1)\}$ span another $\mathsf{T}$-invariant subspace $\mathcal{U}_2$. Altogether, $\mathbb{R}^5 = \mathcal{U}_1 \oplus \mathcal{U}_2$.

The matrix interpretation of Theorem 17.2.1 is most important in applications and is contained in the following corollary.

**COROLLARY 17.2.4:** *Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a nilpotent linear transformation of index $p$ in the finite-dimensional vector space $\mathcal{V}$. Then there exists a basis for $\mathcal{V}$ such that the matrix of $\mathsf{T}$ is of the form*

$$\begin{bmatrix} \mathbf{J}_p & & & & & & & & & & \\ & \mathbf{J}_p & & & & & & & & & \\ & & \ddots & & & & & & & & \\ & & & \mathbf{J}_p & & & & & & & \\ & & & & \mathbf{J}_{p-1} & & & & & & \\ & & & & & \ddots & & & & \mathbf{0} & \\ & & & & & & \mathbf{J}_{p-1} & & & & \\ & & & & & & & \ddots & & & \\ & & & & & & & & \mathbf{J}_{p-1} & & \\ & & & & & & & & & \ddots & \\ & & & & & & & & & & \mathbf{J}_1 \\ & \mathbf{0} & & & & & & & & & & \ddots & \\ & & & & & & & & & & & & \mathbf{J}_1 \end{bmatrix}.$$

*The matrix $\mathbf{J}_i$ has the form*

$$\mathbf{J}_i = \begin{bmatrix} 0 & & & & & \\ 1 & 0 & & & & \\ 0 & 1 & 0 & & \mathbf{0} & \\ 0 & 0 & \ddots & & & \\ \vdots & \vdots & \vdots & \ddots & & \\ 0 & 0 & 0 & & 1 & 0 \end{bmatrix}.$$

*If $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_p$ is the decomposition of Theorem 17.2.1 then the matrix $\mathbf{J}_i$ occurs $n_i = \dim(\mathcal{U}_i) - \dim(\mathcal{U}_{i-1})$ times.*

**PROOF:** By Theorem 17.2.1 we can find a direct sum decomposition

$$\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_p$$

of $\mathcal{V}$ where each $\mathcal{U}_i$ is $\mathsf{T}$-invariant and the induced linear transformation

$$\mathsf{T} : \mathcal{U}_i \longrightarrow \mathcal{U}_i$$

has a cyclic vector $\mathbf{A}_i$. If we choose a basis for $\mathcal{U}_i$ composed of $\mathbf{A}_i, \mathsf{T}(\mathbf{A}_i), \ldots, \mathsf{T}^{p_i-1}(\mathbf{A}_i)$, where $\mathsf{T}^{p_i}(\mathbf{A}_i) = \mathbf{0}$, then the matrix of

$$\mathsf{T} : \mathcal{U}_i \longrightarrow \mathcal{U}_i$$

with respect to this basis is the $p_i \times p_i$ matrix

$$\begin{bmatrix} 0 & 0 & & & \\ 1 & 0 & & \mathbf{0} & \\ & \ddots & 0 & & \\ & & \ddots & & \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Assembling these blocks, we obtain a matrix of the stated form representing $\mathsf{T}$. $\square$

**EXAMPLE 2 :** If $\mathsf{T} : \mathbb{R}^5 \longrightarrow \mathbb{R}^5$ is the linear transformation of Example 1, then in the notations of that example, the vectors

$$\mathbf{A}_1, \mathsf{T}(\mathbf{A}_1), \mathsf{T}^2(\mathbf{A}_1), \mathbf{B}_1, \mathsf{T}(\mathbf{B}_1)$$

form a basis for $\mathbb{R}^5$ of the type specified in Corollary 17.2.4, and the matrix of $\mathsf{T}$ with respect to this basis is

$$\begin{bmatrix} 0 & 0 & 0 & & \\ 1 & 0 & 0 & \multicolumn{2}{c}{\mathbf{0}} \\ 0 & 1 & 0 & & \\ & & & 0 & 0 \\ \multicolumn{3}{c}{\mathbf{0}} & 1 & 0 \end{bmatrix},$$

from which we see that $\mathsf{T}$ is composed of two shift operators: one acting on the 3-dimensional space spanned by $\left\{\mathbf{A}_1, \mathsf{T}(\mathbf{A}_1), \mathsf{T}^2(\mathbf{A}_1)\right\}$ giving a nilpotent operator of nilpotence index 3 and cyclic vector $\mathbf{A}_1$; and a second acting on the 2-dimensional subspace spanned by $\{\mathbf{B}_1, \mathsf{T}(\mathbf{B}_1)\}$, which is nilpotent of nilpotence index 2 and has $\mathbf{B}_1$ as a cyclic vector.

## 17.3 The Jordan Normal Form

The **Jordan canonical form** provides a decomposition analogous to that of Theorem 17.2.1 for an arbitrary linear transformation in a finite-dimensional vector space, provided that the characteristic polynomial splits into linear factors. As we saw in Chapter 14, we can always assume that this is the case if we choose to use the complex numbers $\mathbb{C}$ as scalars.

A central fact we will need is that in a finite-dimensional vector space $\mathcal{V}$ an ascending chain of subspaces

$$\{\mathbf{0}\} = \mathcal{U}_0 \subseteq \mathcal{U}_1 \subseteq \cdots \subseteq \mathcal{U}_n \subseteq \cdots \subseteq \mathcal{V}$$

must eventually stabilize; that is, there must be an integer $m$ such that

$$\mathcal{U}_m = \mathcal{U}_{m+1} = \cdots.$$

This is because

$$0 = \dim(\mathcal{U}_0) \leq \dim(\mathcal{U}_1) \leq \cdots \leq \dim(\mathcal{U}_n) \leq \cdots \leq \dim(\mathcal{V}),$$

and a bounded sequence of *integers* must eventually stabilize, whence the chain of vector spaces must stabilize by Corollary 6.3.6 applied to the pair $\mathcal{U}_i \subseteq \mathcal{U}_{i+1}$ whenever $i \geq m$.

Similarly, a **descending chain of subspaces**

$$\mathcal{V} = \mathcal{W}_0 \supseteq \mathcal{W}_1 \supseteq \cdots \supseteq \mathcal{W}_n \supseteq \cdots$$

must also stabilize at a subspace $\mathcal{W}_k$ such that

$$W_k = W_{k+1} = \cdots,$$

as easily follows from the inequalities

$$\dim(\mathcal{V}) = \dim(\mathcal{W}_0) \geq \dim(\mathcal{W}_1) \geq \cdots \geq \dim(\mathcal{W}_k) \geq \cdots \geq 0.$$

We hope to obtain a direct sum decomposition of $\mathcal{V}$ into $\mathsf{T}$-invariant subspaces for an arbitrary linear transformation

$$\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V},$$

where $\mathcal{V}$ is finite-dimensional, by cleverly *enlarging* the eigenspaces of $\mathsf{T}$. But how? Here is the idea. Suppose that $e$ is an eigenvalue of $\mathsf{T}$. Then

$$\mathcal{V}_e = \ker(\mathsf{T} - e\mathsf{I}).$$

Let us look at the sequence of subspaces

$$\mathcal{V}_{e,i} = \ker((\mathsf{T} - e\mathsf{I})^i) \qquad \text{for } i = 1, 2 \ldots.$$

Notice that these subspaces are nested one within the other; that is,

$$(\maltese) \qquad \mathcal{V}_e = \mathcal{V}_{e,1} \subseteq \mathcal{V}_{e,2} \subseteq \mathcal{V}_{e,3} \subseteq \cdots \subseteq \mathcal{V}.$$

If $\mathcal{V}$ is finite-dimensional, then as we just noticed, this chain of subspaces cannot go on getting bigger forever. Therefore, there must be an integer $n$, depending of course on $\mathsf{T}$ and $e$, such that

$$\mathcal{V}_{e,j} = \mathcal{V}_{e,n} \quad \forall\, j \geq n.$$

The vector space $\mathcal{V}_{e,n} = \ker((\mathsf{T} - e\mathsf{I})^n)$ at which the chain $(\maltese)$ stabilizes is $\mathsf{T}$-invariant, since

$$(\mathsf{T} - e\mathsf{I})^n = \mathsf{T}^n - e\mathsf{T}^{n-1} + \cdots + (-1)^n e^n \mathsf{I}$$

is a polynomial in $\mathsf{T}$, and we may therefore apply Proposition 17.1.7. Let us introduce a name for $\mathcal{V}_{e,n}$.

**DEFINITION:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation with eigenvalue* $e$. *Let* $\mathcal{V}_{e,i} = \ker((\mathsf{T} - e\mathsf{I})^i)$ *for* $i = 1, 2, \ldots,$ *and let* $n$ *be the smallest integer such that* $\mathcal{V}_{e,n} = \mathcal{V}_{e,n+1} = \cdots$. *The subspace* $\mathcal{V}_{e,n}$ *is called the* **closure** *of the eigenspace* $\mathcal{V}_e$, *and is denoted by* $\overline{\mathcal{V}_e}$.

**REMARK:** Note that

$$\mathcal{V}_e = \mathcal{V}_{e,1} \subset \mathcal{V}_{e,2} \subset \cdots \subset \mathcal{V}_{e,n} = \overline{\mathcal{V}_e}$$

is a chain of subspaces of $\mathcal{V}$ with no two adjacent subspaces equal, i.e., where all the inclusions are proper. To see this, suppose that for some $1 \leq i \leq n$ $\mathcal{V}_{e,i} = \mathcal{V}_{e,i+1}$. By definition, $\mathcal{V}_{e,i+2} = \ker((\mathsf{T} - e\mathsf{I})^{i+2})$. Suppose $\mathbf{A} \in \mathcal{V}_{e,i+2}$. Then

$$(\mathsf{T} - e\mathsf{I})(\mathbf{A}) \in \mathcal{V}_{e,i+1} = \mathcal{V}_{e,i},$$

so

$$(\mathsf{T} - e\mathsf{I})^{i+1}(\mathbf{A}) = (\mathsf{T} - e\mathsf{I})^i((\mathsf{T} - e\mathsf{I})(\mathbf{A})) = \mathbf{0},$$

since $\mathcal{V}_{e,i} = \ker((\mathsf{T} - e\mathsf{I})^i)$. But this says that $\mathbf{A} \in \mathcal{V}_{e,i+1}$. Therefore, $\mathcal{V}_{e,i+2} \subseteq \mathcal{V}_{e,i+1}$. Since the reverse inclusion also holds, we must have equality, so $\mathcal{V}_{e,i+2} = \mathcal{V}_{e,i+1}$. Continuing in this way we see that $\mathcal{V}_{e,i} = \mathcal{V}_{e,i+1} = \mathcal{V}_{e,i+2} = \cdots = \overline{\mathcal{V}_e}$ contrary to the way we chose $n$.

The discussion preceding the definition of the closure of an eigenspace may be summarized as follows.

**PROPOSITION 17.3.1:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation in the finite-dimensional vector space* $\mathcal{V}$ *and* $e$ *an eigenvalue of* $\mathsf{T}$. *Then,* $\overline{\mathcal{V}_e}$, *the closure of the eigenspace corresponding to* $e$, *is* $\mathsf{T}$*-invariant.* $\square$

How does $\mathsf{T}$ act on the closure of the eigenspace? The following proposition tells us.

**PROPOSITION 17.3.2:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation in the finite-dimensional vector space* $\mathcal{V}$ *with eigenvalue* $e$. *Let* $\overline{\mathcal{V}_e}$ *be the closure of the corresponding eigenspace* $\mathcal{V}_e$. *Then the induced linear transformation*

$$\mathsf{T} - e\mathsf{I} : \overline{\mathcal{V}_e} \longrightarrow \overline{\mathcal{V}_e}$$

*is nilpotent. If* $n$ *is the index of nilpotence, then*

$$\mathcal{V}_{e,n} = \mathcal{V}_{e,n+1} = \cdots = \overline{\mathcal{V}_e}.$$

**PROOF:** Recall the chain of subspaces

$$(\maltese) \qquad \mathcal{V}_e = \mathcal{V}_{e,1} \subseteq \mathcal{V}_{e,2} \subseteq \mathcal{V}_{e,3} \subseteq \cdots \subseteq \mathcal{V}.$$

We have $\overline{\mathcal{V}_e} = \mathcal{V}_{e,n}$, where $n$ is the smallest integer such that

$$\mathcal{V}_{e,n} = \mathcal{V}_{e,n+1}.$$

From the definitions, $\overline{\mathcal{V}_e} = \ker((\mathsf{T} - e\mathsf{I})^n)$, so

$$(\mathsf{T} - e\mathsf{I})^n = \mathbf{0} : \overline{\mathcal{V}_e} \longrightarrow \overline{\mathcal{V}_e}$$

but since $\mathcal{V}_{e,n-1} \subsetneqq \mathcal{V}_{e,n}$ there is a vector in $\mathcal{V}_{e,n} = \overline{\mathcal{V}_e}$ that is not in the kernel of $(\mathsf{T} - e\mathsf{I})^{n-1}$, and therefore

$$(\mathsf{T} - e\mathsf{I}) : \overline{\mathcal{V}_e} \longrightarrow \overline{\mathcal{V}_e}$$

is nilpotent of index $n$. $\square$

We are now in a position to state the main result of this chapter.

**THEOREM 17.3.3:** *Let* $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation in the finite-dimensional complex vector space* $\mathcal{V}$. *Let*

$$p(t) = (t - e_1)^{d_1} \cdots (t - e_q)^{d_q}$$

*be the characteristic polynomial of* $\mathsf{T}$. *Then:*

(1) *The closure of each eigenspace* $\overline{\mathcal{V}_{e_i}}$ *is fully* $\mathsf{T}$*-invariant.*
(2) $\dim(\overline{\mathcal{V}_{e_i}}) = d_i$.
(3) *The characteristic polynomial of the induced linear transformation* $\mathsf{T} : \overline{\mathcal{V}_{e_i}} \longrightarrow \overline{\mathcal{V}_{e_i}}$ *is* $(t - e_i)^{d_i}$.
(4) $\mathcal{V} = \overline{\mathcal{V}_{e_1}} \oplus \cdots \oplus \overline{\mathcal{V}_{e_q}}$.

We will need a number of preliminary steps before undertaking a proof of this major result. However, before embarking, notice that combining Theorem 17.3.3 with the canonical matrix representation for nilpotent transformations of Theorem 17.2.1 will give us a matrix canonical form for any linear transformation in a finite-dimensional complex vector space,[3] namely:

**COROLLARY 17.3.4:** *Let* $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ *be a linear transformation in the finite-dimensional complex vector space* $\mathcal{V}$*, with distinct eigenvalues* $e_1, \ldots, e_m$ *occuring*[4] *with multiplicities* $d_1, \ldots, d_m$*. Then there exists a basis for* $\mathcal{V}$ *with respect to which* $\mathbf{T}$ *has the matrix representation*

(❶) $$\begin{bmatrix} \mathbf{M}_1 & & 0 \\ & \mathbf{M}_2 & \\ & & \ddots & \\ 0 & & & \mathbf{M}_m \end{bmatrix},$$

*where the matrix* $\mathbf{M}_j$ *has the block format*

(❷) $$\begin{bmatrix} \mathbf{M}_{j,1} & & & \\ & \mathbf{M}_{j,2} & 0 & \\ & & \ddots & \\ & 0 & & \\ & & & \mathbf{M}_{j,d_j} \end{bmatrix}$$

*where in turn, the matrix* $\mathbf{M}_{j,k}$ *has the block format*

(❸) $$\begin{bmatrix} e_j & & & \\ 1 & e_j & 0 & \\ & 1 & \ddots & \\ & 0 & & \\ & & 1 & e_j \end{bmatrix}.$$

Thus, taking these properties all together, Theorem 17.3.4 says that $\mathbf{T}$ has a matrix of the form with the eigenvalues on the diagonal, 1 or 0

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3 Or more generally for any linear transformation whose characteristic polynomial splits into a product of linear factors.

4 We say that the eigenvalue $e$ occurs with multiplicity $d$ when $(t - e)^d$ is a factor of the characteristic polynomial, but $(t - e)^{d+1}$ is not.

on the subdiagonal, and 0 otherwise, i.e., of the form

$$\begin{bmatrix}
e_1 &&&&&&&&&& \\
1 & e_1 &&&&&&&&& \\
& \ddots & \ddots &&&&&&&& \\
&& 1 & e_1 &&&&&&& \\
&&& 0 & e_2 &&&& \text{\Large 0} && \\
&&&& 1 & \ddots &&&&& \\
&&&&& 1 & e_2 &&&& \\
&&&&&& \ddots &&&& \\
&&&&&&& 0 & e_m && \\
&&&&&&&& \ddots & \ddots & \\
& \text{\Large 0} &&&&&&&& 1 & e_m
\end{bmatrix},$$

where all the nonzero entries are either on the diagonal or the subdiagonal.

**Proof:** Let

$$\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$$

be the decomposition of $\mathcal{V}$ into a direct sum of $\mathsf{T}$-invariant subspaces guaranteed by Theorem 17.3.3. By repeated use of Corollary 17.1.6 we can choose bases for each $\mathcal{U}_j$ individually, such that the matrix of $\mathsf{T}$ with respect to this basis will have the block format (❶).

For the block $\mathcal{U}_j$ the linear transformation $\mathsf{T} - e_j\mathsf{I} : \mathcal{U}_j \longrightarrow \mathcal{U}_j$ is nilpotent of index $d_j$ by Theorem 17.3.3. Hence we may apply Corollary 17.2.4 to conclude that there is a basis for $\mathcal{U}_j$ with respect to which $\mathsf{T}$ has a matrix representation of the form (❷) and the subblocks have the form (❸). □

The matrix representation in Corollary 17.3.4 is referred to as the **Jordan normal form**, or **Jordan canonical form**, of the linear transformation $\mathsf{T}$. The submatrices (❸) are called the **Jordan blocks** of $\mathsf{T}$. This matrix has the eigenvalues of $\mathsf{T}$ on the diagonal, and a pattern of 0s and 1s on the subdiagonal, the pattern being determined by the structure Theorem 17.2.1 for the nilpotent operators

$$\mathsf{T} - e_j\mathsf{I} : \overline{\mathcal{V}_{e_j}} \longrightarrow \overline{\mathcal{V}_{e_j}},$$

where $e_1, \ldots, e_m$ are the distinct eigenvalues of $\mathsf{T}$. All in all the Jordan form looks like

$$\begin{bmatrix} \left.\begin{matrix} e_1 & & & \\ * & e_1 & & \\ & * & \ddots & \\ & & 0 & e_1 \end{matrix}\right\} d_1 & & & \\ & \left.\begin{matrix} e_2 & & & \\ * & e_2 & & \\ & * & \ddots & \\ & & 0 & e_2 \end{matrix}\right\} d_2 & & \mathbf{0} \\ & & \ddots & \\ & & \ddots & \\ \mathbf{0} & & & \left.\begin{matrix} e_m & & & \\ * & e_m & & \\ & * & \ddots & \\ & & 0 & e_m \end{matrix}\right\} d_m \end{bmatrix},$$

where $* = 0$ or 1, the 0s separate the Jordan blocks. Let us illustrate this with an example.

**Example 1:** $\mathsf{T} : \mathbb{C}^3 \longrightarrow \mathbb{C}^3$ be given by

$$\mathsf{T}(x, y, z) = (5x + 4y + 3z, -x - 3z, x - 2y + z).$$

With respect to the standard basis for $\mathbb{C}^3$ the matrix representation of $\mathsf{T}$ is

$$\mathbf{A} = \begin{bmatrix} 5 & 4 & 3 \\ -1 & 0 & -3 \\ 1 & -2 & 1 \end{bmatrix}.$$

The characteristic polynomial of $\mathsf{T}$ is

$$\Delta(t) = \det(\mathbf{A} - t\mathbf{I}) = \det \begin{bmatrix} 5-t & 4 & 3 \\ -1 & -t & -3 \\ 1 & -2 & 1-t \end{bmatrix}$$

$$= (5-t)[-t(1-t)-6)] - 4[-1(1-t)+3] + 3[2-(-t)]$$

$$= (5-t)[t^2 - t - 6] - 4[t+2] + 3[t+2]$$

$$= (5-t)(t-3)(t+2) - (t+2) = -(t+2)[(t-5)(t-3)+1]$$

$$= -(t+2)(t-4)^2,$$

so there are two eigenvalues,

$$e_1 = 4 \text{ of algebraic multiplicity } 2,$$

$$e_2 = -2 \text{ of algebraic multiplicity } 1.$$

Next we find the eigenspace corresponding to the eigenvalue 4. To do so we must solve the linear system

$$\begin{bmatrix} 1 & 4 & 3 \\ -1 & -4 & -3 \\ 1 & -2 & -3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

This leads to the equivalent systems

$$\begin{bmatrix} 1 & 4 & 3 \\ 0 & 0 & 0 \\ 1 & -2 & -3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

$$\begin{bmatrix} 2 & 2 & 0 \\ 0 & 0 & 0 \\ 1 & -2 & -3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

so the solutions are given by the conditions

$$x = -y = z,$$

and the eigenspace is 1-dimensional with basis $(1, -1, 1) = \mathbf{A}_1$.

To find a basis for the eigenspace corresponding to the eigenvalue $-2$, we need to solve the linear system

$$\begin{bmatrix} 7 & 4 & 3 \\ -1 & 2 & -3 \\ 1 & -2 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

which leads to the solution

$$x = -y = -z.$$

Therefore, the eigenspace corresponding to the eigenvalue $-2$ has dimension 1 also, with basis $\mathbf{A}_2 = (1, -1, -1)$.

The linear transformation $\mathsf{T}$ ***cannot*** be put into diagonal form because the eigenvalue 4 has algebraic multiplicity 2, but the geometric multiplicity is only 1.

The Jordan canonical form of $\mathsf{T}$ is

$$\begin{bmatrix} 4 & 0 & 0 \\ \delta & 4 & 0 \\ 0 & 0 & -2 \end{bmatrix},$$

where $\delta = 0, 1$. Note that $\delta$ cannot be 0 because $\mathsf{T}$ is not diagonalizable. Thus we already know that

$$\begin{bmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

is the Jordan canonical form for $\mathsf{T}$ ***although we have not yet found the basis with respect to which*** $\mathsf{T}$ ***is represented by this matrix.***

To find a basis corresponding to the Jordan form for $\mathsf{T}$, we must find a basis for $\overline{\mathcal{V}_4}$. Since we know that $\dim(\overline{\mathcal{V}_4}) = 2$ and $\mathcal{V}_4 \subset \overline{\mathcal{V}_4}$ we look for a vector $\mathbf{A}_3$ such that

$$(\mathsf{T} - 4\mathsf{I})^2(\mathbf{A}_3) = \mathbf{0},$$

but

$$(\mathsf{T} - 4\mathsf{I})(\mathbf{A}_3) \neq \mathbf{0}.$$

To this end, notice that

$$(\mathsf{T} - 4\mathsf{I})^2 = \begin{bmatrix} 1 & 4 & 3 \\ -1 & -4 & -3 \\ 1 & -2 & -3 \end{bmatrix}^2 = \begin{bmatrix} 0 & -18 & -18 \\ 0 & 18 & 18 \\ 0 & 18 & 18 \end{bmatrix}.$$

So a basis for $\ker((\mathsf{T} - 4\mathsf{I})^2$ is the solution space of the linear system

$$y + z = 0.$$

One solution is the eigenvector $\mathbf{A}_1 = (1, -1, 1)$, and a second, *linearly independent* solution is $\mathbf{A}_3 = (1, 0, 0)$, and $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ is a basis for $\mathbb{R}^3$. Since

$$(\mathsf{T} - 4\mathsf{I})(\mathbf{A}_3) = (1, -1, 1) = \mathbf{A}_1,$$

we see that by putting the basis in the order $\mathbf{A}_3, \mathbf{A}_1, \mathbf{A}_2$ we have a basis in which the matrix of $\mathsf{T}$ has Jordan form.

For the sake of clarity here is another example.

**EXAMPLE 2:** Let $\mathsf{T} : \mathbb{C}^3 \longrightarrow \mathbb{C}^3$ be the linear transformation represented by the matrix

$$\begin{bmatrix} 2 & 2 & -1 \\ -1 & -1 & 1 \\ -1 & -2 & 2 \end{bmatrix}$$

with respect to the standard basis of $\mathbb{C}^3$.

The characteristic polynomial of $\mathsf{T}$ is

$$\Delta(t) = \det \begin{bmatrix} 2-t & 2 & -1 \\ -1 & -1-t & 1 \\ -1 & -2 & 2-t \end{bmatrix} = \cdots = -(t-1)^3.$$

Therefore, $t = 1$ is an eigenvalue of algebraic multiplicity 3. The matrix is therefore certainly not diagonalizable, because if it were, the diagonal form would have to be

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix},$$

which would say $\mathsf{T} = -\mathsf{I}$ which certainly is not the case: For example, $\mathsf{T}(1, 0, 0) = (2, -1, -1)$. To determine the eigenspace for $e = 1$, we solve the linear system

$$\begin{bmatrix} 1 & 2 & -1 \\ -1 & -2 & 1 \\ -1 & -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

This leads to the equation

$$x + 2y - z = 0,$$

which defines a plane in $\mathbb{C}^3$ with a basis $\mathbf{A}_1 = (1, 0, 1), \mathbf{A}_2 = (0, 1, 2)$. Since the eigenspace $\mathcal{V}_1$ is 2-dimensional and 1 is the only eigenvalue, we have $\overline{\mathcal{V}_1} = \mathbb{C}^3$, and so $\overline{\mathcal{V}_1}$ has dimension 1 over $\mathcal{V}_1$. To find a basis for $\overline{\mathcal{V}_1}$ over $\mathcal{V}_1$ we may choose any vector in $\mathbb{C}^3$ not in $\mathcal{V}_1$, since it will automatically be in $\ker(\mathsf{T} - \mathsf{I})^2$. (Of course $(\mathsf{T} - \mathsf{I})^2 = \mathbf{0}$ in this case.)

One such vector is $\mathbf{B}_1 = (1, 0, 0)$. We now follow the prescription of Theorem 17.2.1:

$$(\mathsf{T} - \mathsf{I})(\mathbf{B}_1) = \begin{bmatrix} 1 & 2 & -1 \\ -1 & -2 & 1 \\ -1 & -2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix},$$

and $\mathbf{B}_2 = (1, -1, -1)$ belongs to $\mathcal{V}_1$. If we extend $\mathbf{B}_2$ to a basis for $\mathcal{V}_1$ by adjoining, say, $\mathbf{B}_3 = (0, 1, 2)$, we find that the matrix of $\mathsf{T}$ with respect to the basis $\{\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3\}$ is in the Jordan normal form

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We turn next to the proof of Theorem 17.3.3 and separate out the key step in the proof as a lemma.

**LEMMA 17.3.5:** *Suppose $\mathsf{S} : \mathcal{V} \longrightarrow \mathcal{V}$ is a linear transformation in the finite-dimensional vector space $\mathcal{V}$. Then there exists an integer $k$ such that*

$$\operatorname{Im}(\mathsf{S}^j) = \operatorname{Im}(\mathsf{S}^k),$$

$$\ker(\mathsf{S}^j) = \ker(\mathsf{S}^k),$$

*whenever $j \geq k$, and moreover, $\ker(\mathsf{S}^k)$ and $\operatorname{Im}(\mathsf{S}^k)$ are a complementary pair of $\mathsf{S}$-invariant subspaces of $\mathcal{V}$.*

**Proof:** The nested chains of subspaces

$$
\begin{array}{ccccccccccccc}
\{\mathbf{0}\} & = & \ker(\mathbf{S}^0) & \subseteq & \ker(\mathbf{S}^1) & \subseteq & \cdots & \subseteq & \ker(\mathbf{S}^i) & \subseteq & \cdots & \subseteq & \mathcal{V}, \\
\mathcal{V} & = & \mathrm{Im}(\mathbf{S}^0) & \supseteq & \mathrm{Im}(\mathbf{S}^1) & \supseteq & \cdots & \supseteq & \mathrm{Im}(\mathbf{S}^j) & \supseteq & \cdots & \supseteq & \{\mathbf{0}\}
\end{array}
$$

must eventually both stabilize because $\mathcal{V}$ is finite-dimensional. Say,

$$
\begin{array}{rcll}
\mathrm{Im}(\mathbf{S}^{j'}), & = & \mathrm{Im}(\mathbf{S}^{k'}), & j' \geq k' \\
\ker(\mathbf{S}^{j''}), & = & \ker(\mathbf{S}^{k''}) & j'' \geq k''.
\end{array}
$$

If we choose $k = \max\{k', k''\}$, then both sequences are stable beginning at the $k$-th term. We claim that with this choice of $k$

$$\mathrm{Im}(\mathbf{S}^k) \cap \ker(\mathbf{S}^k) = \{\mathbf{0}\}.$$

For if $\mathbf{A} \in \mathrm{Im}(\mathbf{S}^k) \cap \ker(\mathbf{S}^k)$, then there is a vector $\overline{\mathbf{A}} \in \mathcal{V}$ such that

$$\mathbf{A} = \mathbf{S}^k(\overline{\mathbf{A}}).$$

Then of course,

$$\mathbf{0} = \mathbf{S}^k(\mathbf{A}) = \mathbf{S}^{2k}(\overline{\mathbf{A}}),$$

so $\overline{\mathbf{A}} \in \ker(\mathbf{S}^{2k})$. But by the choice of $k$,

$$\ker(\mathbf{S}^k) = \ker(\mathbf{S}^{k+1}) = \cdots = \ker(\mathbf{S}^{2k}),$$

so $\overline{\mathbf{A}} \in \ker(\mathbf{S}^k)$. Then

$$\mathbf{A} = \mathbf{S}^k(\overline{\mathbf{A}}) = \mathbf{0},$$

as claimed.

For the linear transformation

$$\mathbf{S}^k : \mathcal{V} \longrightarrow \mathcal{V},$$

we have by the dimension formula (Theorem 8.4.2)

$$\dim \mathcal{V} = \dim(\mathrm{Im}(\mathbf{S}^k)) + \dim(\ker(\mathbf{S}^k)).$$

On the other hand, since

$$\mathrm{Im}(\mathbf{S}^k) \cap \ker(\mathbf{S}^k) = \{\mathbf{0}\},$$

we have

$$\dim\left(\mathrm{Im}(\mathbf{S}^k) + \ker(\mathbf{S}^k)\right) = \dim(\mathrm{Im}(\mathbf{S}^k)) + \dim(\ker(\mathbf{S}^k)) = \dim(\mathcal{V}),$$

so $\mathrm{Im}(\mathbf{S}^k) + \ker(\mathbf{S}^k) = \mathcal{V}$ by Corollary 6.3.6. Hence by Proposition 17.1.4

$$\mathcal{V} = \mathrm{Im}(\mathbf{S}^k) \oplus \ker(\mathbf{S}^k),$$

as required. $\square$

**PROOF OF THEOREM 17.3.3 :** We begin by applying Lemma 17.3.5 to the linear transformation

$$\mathbf{S}_1 = \mathbf{T} - e_1\mathbf{I} : \mathcal{V} \longrightarrow \mathcal{V}.$$

We conclude that for a suitable integer $k$,

$$\mathcal{V} = \ker(\mathbf{T} - e_1\mathbf{I})^k \oplus \operatorname{Im}(\mathbf{T} - e_1\mathbf{I})^k,$$

where by the definition of $\overline{\mathcal{V}_{e_1}}$ we have

$$\overline{\mathcal{V}_{e_1}} = \ker(\mathbf{T} - e_1\mathbf{I})^k.$$

By Proposition 17.1.7,

$$\operatorname{Im}(\mathbf{T} - e_1\mathbf{I})^k, \qquad \ker(\mathbf{T} - e_1\mathbf{I})^k$$

are $\mathbf{T}$-invariant complementary subspaces, so by Proposition 17.1.1, $\mathbf{T}$ has a matrix representation

$$\mathbf{M} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix},$$

where $\mathbf{A}$ is a matrix representation for

$$\mathbf{T} : \overline{\mathcal{V}_{e_1}} \longrightarrow \overline{\mathcal{V}_{e_1}}$$

and $\mathbf{B}$ is a matrix representation for

$$\mathbf{T} : \operatorname{Im}(\mathbf{T} - e_1\mathbf{I})^k \longrightarrow \operatorname{Im}(\mathbf{T} - e_1\mathbf{I})^k.$$

From the matrix $\mathbf{M}$ we can compute the characteristic polynomial of $\mathbf{T}$, and we get

$$\Delta(t) = \Delta_1(t) \cdot \Delta_2(t),$$

where

$$\Delta_1(t) = \det(\mathbf{A} - t\mathbf{I}),$$
$$\Delta_2(t) = \det(\mathbf{B} - t\mathbf{I})$$

are the characteristic polynomials of $\mathbf{T}$ on $\overline{\mathcal{V}_{e_1}}$ and $\widehat{\mathcal{V}_{e_1}} = \operatorname{Im}(\mathbf{T} - e_1\mathbf{I})^k$, respectively. Since $\overline{\mathcal{V}_{e_1}}$ contains the eigenspace $\mathcal{V}_{e_1}$, it follows that

$$\Delta_1(t) = (t - e_1)^{d_1}\hat{\Delta}(t),$$

and $\Delta_2(t)$ cannot contain $(t - e_1)$ as a factor. We claim that $\hat{\Delta}(t) = +1$. To this end; recall that

$$\mathbf{T} - e_1\mathbf{I} : \overline{\mathcal{V}_{e_1}} \longrightarrow \overline{\mathcal{V}_{e_1}}$$

is nilpotent of order $d_1$, so it has a matrix representation in the form

$$\begin{bmatrix} 0 & & & & \\ * & 0 & & \mathbf{0} & \\ 0 & * & 0 & & \\ & & \ddots & \ddots & \\ & \mathbf{0} & & * & 0 \end{bmatrix} \qquad * = 0 \text{ or } 1.$$

Therefore, $\mathsf{T} : \overline{\mathcal{V}_{e_1}} \longrightarrow \overline{\mathcal{V}_{e_1}}$ has a matrix representation

$$\begin{bmatrix} e_1 & & & & \\ * & e_1 & & & \\ 0 & * & \ddots & & \\ 0 & 0 & \ddots & & \\ 0 & & & * & e_1 \end{bmatrix},$$

again where $* = 0$ or $1$, so

$$\Delta_1(t) = \det \begin{bmatrix} e_1 - t & & & & \\ * & e_1 - t & & \mathbf{0} & \\ 0 & * & \ddots & & \\ & & \ddots & \ddots & \\ & \mathbf{0} & & * & e_1 - t \end{bmatrix} = (e_1 - t)^{\dim(\overline{\mathcal{V}_{e_1}})}.$$

Therefore $\hat{\Delta}(t) = +1$, because the factorization of a polynomial into linear factors is unique. Summarizing, we have $\Delta_1(t) = (t - e_1)^{d_1}$ and $e_1$ is not a root of $\Delta_2(t)$. If we now repeat the preceding argument for the eigenvalues $e_2$, $e_3$, etc. of $\mathsf{T}$, we arrive at the conclusion that $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$ and that the characteristic polynomial of $\mathsf{T}$ regarded as a linear transformation $\mathsf{T} : \mathcal{U}_i \longrightarrow \mathcal{U}_i$ is $\Delta_i(t) = (t - e_i)^{d_i}$ with

$$\Delta(t) = \Delta_1(t) \cdots \Delta_m(t)$$

the characteristic polynomial of $\mathsf{T}$, as required. $\square$

**EXAMPLE 3:** As a last illustrative example we consider the linear transformation represented by the $4 \times 4$ matrix

$$\mathsf{A} = \begin{bmatrix} 7 & 1 & 2 & 2 \\ 1 & 4 & -1 & -1 \\ -2 & 1 & 5 & -1 \\ 1 & 1 & 2 & 8 \end{bmatrix}.$$

We proceed step by step to obtain the Jordan form using a slightly different method than in the two preceding examples. First we compute the characteristic polynomial (this takes some work!)

$$\Delta(t) = \det\begin{bmatrix} 7-t & 1 & 2 & 2 \\ 1 & 4-t & -1 & -1 \\ -2 & 1 & 5-t & -1 \\ 1 & 1 & 2 & 8-t \end{bmatrix} = (t-6)^4.$$

So there is exactly one eigenvalue, namely $t = 6$, with algebraic multiplicity 4. To find a basis for the eigenspace $\mathcal{V}_6$ we must solve the linear system

$$\mathbf{0} = (\mathbf{A} - 6\mathbf{I})\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & -2 & -1 & -1 \\ -2 & 1 & -1 & -1 \\ 1 & 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}.$$

Since the first and last row are identical and the sum of the second and third rows is the negative of the first row, we can work with the equivalent system

$$\begin{aligned} x + y + 2z + 2w &= 0, \\ x - 2y - z - w &= 0, \end{aligned}$$

which easily leads to yet another equivalent system

$$\begin{aligned} x + z + w &= 0, \\ y + z + w &= 0, \end{aligned}$$

and the basis

$$\mathbf{A}_1 = (-1, -1, 1, 0), \qquad \mathbf{A}_2 = (-1, -1, 0, 1)$$

for the solution space. Therefore,

$$2 = \dim(\mathcal{V}_6) < \dim(\overline{\mathcal{V}_6}) = 4.$$

In the next step we need to extend $\mathbf{A}_1, \mathbf{A}_2$ to a basis for

$$\mathcal{V}_{6,2} = \ker(\mathbf{A} - 6\mathbf{I})^2.$$

So we look for vectors $\mathbf{F}$ such that

$$\mathbf{0} \neq (\mathbf{A} - 6\mathbf{I})(\mathbf{F}) \in \mathcal{L}(\mathbf{A}_1, \mathbf{A}_2) = \mathcal{V}_{6,1}.$$

This means that we are looking for solutions to

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & -2 & -1 & -1 \\ -2 & 1 & -1 & -1 \\ 1 & 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = a\mathbf{A}_1 + b\mathbf{A}_2 = \begin{bmatrix} -a-b \\ -a-b \\ a \\ b \end{bmatrix},$$

where $a$ and $b$ must first be chosen such that solutions exist and not both of them are zero. By Theorem 13.1.2 these equations have solutions, i.e., are consistent, if and only if the vector $(-a - b, -a - b, a, b)$ lies in the column space of the coefficient matrix

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & -2 & -1 & -1 \\ -2 & 1 & -1 & -1 \\ 1 & 1 & 2 & 2 \end{bmatrix}.$$

From the first and fourth equation we see that $b = -a - b$, so $2b = -a$. If we set $b = 1$ and $a = -2$ then $(-a - b, -a - b, a, b) = (1, 1, -2, 1)$ is the first column of the coefficient matrix, so after simplifying we obtain the consistent system

$$\begin{aligned} x - 2y - z - w &= 1 \\ -2x + y - z - w &= -2. \end{aligned}$$

(Note that the first and last equations of the previous set are consequences of these.) From this pair of equations we obtain

$$y = x - 1 = -(z + w),$$

so a solution is

$$\mathbf{A}_3 = (1, 0, 0, 0).$$

In addition to $\mathbf{A}_1, \mathbf{A}_2$ and $\mathbf{A}_3$ there are, in fact, no further linearly independent solutions to the system

$$(\mathbf{A} - 6\mathbf{I})^2 \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

because any such solution must be of the form

$$c\mathbf{A}_3 + \mathbf{B}, \qquad \mathbf{B} \in \mathcal{L}(\mathbf{A}_1, \mathbf{A}_2).$$

Hence $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ is a basis for $\mathcal{V}_{6,2}$. Finally, we must extend these three vectors to a basis for

$$\overline{\mathcal{V}_6} = \mathcal{V}_{6,3} = \ker(\mathbf{A} - 6\mathbf{I})^3.$$

This is equivalent to finding a solution to the system

$$(\mathbf{A} - 6\mathbf{I}) \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + a_3\mathbf{A}_3$$

with $a_1\mathbf{A}_1 + a_2\mathbf{A}_2 + a_3\mathbf{A}_3 \neq \mathbf{0}$. Thus we must find values of $a_1$, $a_2$ and $a_3$ such that the system

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & -2 & -1 & -1 \\ -2 & 1 & -1 & -1 \\ 1 & 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -a_1 - a_2 + a_3 \\ -a_1 - a_2 \\ a_1 \\ a_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

is consistent, and find a solution. We again apply Theorem 13.1.2. From the first and fourth rows we get

$$-a_1 - a_2 + a_3 = a_2.$$

The choices

$$a_1 = -1, \qquad a_2 = 2, \qquad a_3 = 3$$

lead to a consistent system, since in this case

$$\begin{bmatrix} -a_1 - a_2 + a_3 \\ -a_1 - a_2 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -1 \\ 2 \end{bmatrix}$$

coincides with the last column of the coefficient matrix, which gives as solution

$$\mathbf{A}_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

As basis for $\mathbb{C}^4$ we have

$$\mathbf{B}_1 = \mathbf{A}_4,$$

$$\mathbf{B}_2 = (\mathbf{A} - 6\mathbf{I})(\mathbf{B}_1) = \begin{bmatrix} 2 \\ -1 \\ -1 \\ 2 \end{bmatrix},$$

$$\mathbf{B}_3 = (\mathbf{A} - 6\mathbf{I})(\mathbf{B}_2) = \begin{bmatrix} 3 \\ 3 \\ -6 \\ 3 \end{bmatrix},$$

$$\mathbf{B}_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -2 \end{bmatrix},$$

where $\mathbf{B}_4$ is an eigenvector linearly independent from the eigenvector $\mathbf{B}_3$. With respect to this basis the matrix of $\mathsf{T}$ is

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 1 & 6 & 0 & 0 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}.$$

## 17.4 Square Roots

In the next chapter we will undertake a major application of the Jordan normal form (namely to the solution of linear systems of differential equations). In the remaining sections of this chapter we present some more elementary applications, the first of which is a proof of the following remarkable theorem.

**THEOREM 17.4.1:** *Let $\mathsf{T} : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ be a nonsingular linear transformation. Then there exists a linear transformation $\mathsf{S} : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ such that $\mathsf{T} = \mathsf{S}^2$.*

A transformation $\mathsf{S}$ whose existence is guaranteed by the theorem is called a **square root** of $\mathsf{T}$. A square root of $\mathsf{S}$ is not unique and, indeed, there may be more than two. The matrix interpretation of this theorem runs as follows:

**THEOREM 17.4.2:** *Let $\mathsf{A}$ be an $n \times n$ matrix with complex entries and nonzero determinant. Then there exists a matrix $\mathsf{B}$ such that $\mathsf{A} = \mathsf{B}^2$.*

Needless to say, the matrix $\mathsf{B}$ of this theorem is called a **square root** of $\mathsf{A}$, and the two results are equivalent, one following directly from the other. So we will discuss them simultaneously.

To see how the Jordan form is relevant to these theorems, suppose that $\mathsf{T} : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ is nonsingular and that

$$\mathsf{T} = \begin{bmatrix} \mathsf{J}_1 & & & \\ & \mathsf{J}_2 & & \\ & & \ddots & \\ & & & \mathsf{J}_k \end{bmatrix}$$

is the Jordan normal form of $\mathsf{T}$, so each matrix $\mathsf{J}_i$ has the form

$$\mathsf{J}_i = \begin{bmatrix} e_i & & & \\ 1 & e_i & \mathbf{0} & \\ & \ddots & \ddots & \\ & & 1 & e_i \end{bmatrix}$$

where $e_i$ is an eigenvalue of $\mathsf{T}$. Since

$$\det(\mathsf{T}) = e_1^{\mu_1} \cdots e_k^{\mu_k}$$

is the product of the eigenvalues, each one occurring as often as its algebraic multiplicity, none of the eigenvalues is zero. The block multiplication rules for matrices show that it will therefore be enough to prove that a matrix of the form

$$\mathsf{J} = \begin{bmatrix} e & & & \\ 1 & e & \text{\Large 0} & \\ & \ddots & \ddots & \\ \text{\Large 0} & & 1 & e \end{bmatrix},$$

with $e \neq 0$ has a square root. Note that

$$\mathsf{J} = e\mathsf{I} + \mathsf{N},$$

where $\mathsf{N}$ is a nilpotent matrix. Since $e \neq 0$ and nonzero complex numbers have square roots, we can divide by $e$, and thus we need to show that a matrix of the form

$$\mathsf{J} = \mathsf{I} + \mathsf{N},$$

where $\mathsf{N}$ is nilpotent, has a square root.

To this end, recall the power series expansion

$$\sqrt{1+x} = \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} x^k,$$

where

$$\binom{\frac{1}{2}}{k} = \frac{\frac{1}{2}(\frac{1}{2}-1)\cdots(\frac{1}{2}-k+1)}{1\cdot 2 \cdot \cdots \cdot k}.$$

Let the index of nilpotence of $\mathsf{N}$ be $m$. Then substituting $\mathsf{N}$ for $x$ in this formula leads to the polynomial in $\mathsf{N}$

$$\sqrt{\mathsf{I}+\mathsf{N}} = \sum_{k=0}^{m-1} \binom{\frac{1}{2}}{k} \mathsf{N}^k,$$

which upon squaring gives $\mathsf{I} + \mathsf{N}$. To summarize, we have shown:

**PROPOSITION 17.4.3 :** *Let $\mathsf{N}$ be a nilpotent matrix of nilpotence index $m$. Then the matrix*

$$\mathsf{S} = \sum_{k=0}^{m-1} \binom{\frac{1}{2}}{k} \mathsf{N}^k$$

*satisfies $\mathsf{S}^2 = \mathsf{I} + \mathsf{N}$.* □

**Proof of Theorems 17.4.1 and 17.4.2:** Both theorems follow immediately from Proposition 17.4.3. □

Here is a simple example to illustrate all this.

**Example 1:** Consider the linear transformation of Example 2 in Section 17.3. The Jordan normal form of this linear transformation is

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which we may write in the form

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{I} + \mathbf{N},$$

where $\mathbf{N}$ is nilpotent of index 2. (Note that there are two Jordan blocks in this case, but only one eigenvalue.) Resorting to the formula in Proposition 17.4.3, we find that

$$\sqrt{\mathbf{T}} = \mathbf{I} + \frac{1}{2}\mathbf{N} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

## 17.5 The Hamilton-Cayley Theorem

In Section 10.3 we showed (Proposition 10.3.2) that $\mathrm{Mat}_{n,n}$ is an $n^2$-dimensional vector space. If $\mathbf{A}$ is an $n \times n$ matrix, then the $n^2 + 1$ matrices

$$\mathbf{A}^{n^2}, \mathbf{A}^{n^2-1}, \ldots, \mathbf{A}, \mathbf{I}$$

must be linearly dependent. By Proposition 5.2.3 it therefore follows there is a smallest integer $m$ such that $\mathbf{A}^m$ can be written as a linear combination of the matrices

$$\mathbf{I}, \mathbf{A}, \ldots, \mathbf{A}^{m-1},$$

and we may choose to write this relation in the form

$$\mathbf{A}^m + a_{m-1}\mathbf{A}^{m-1} + \cdots + a_1\mathbf{A} + a_0\mathbf{I} = \mathbf{0}.$$

The polynomial

$$x^m + a_{m-1}x^{m-1} + \cdots + a_1 x + a_0$$

is called the **minimal polynomial** of $\mathbf{A}$. A priori, the degree of this polynomial could be as much as $n^2$, but in fact it is at most $n$, and indeed more is true, namely:

**THEOREM 17.5.1** (Hamilton–Cayley): *If* $\mathbf{A}$ *is an* $n \times n$ *matrix, then the minimal polynomial of* $\mathbf{A}$ *divides the characteristic polynomial of* $\mathbf{A}$. *Hence the degree of the minimal polynomial is at most* $n$.

**PROOF:** Let $\Delta(t) = \det(\mathbf{A} - t\mathbf{I})$ be the characteristic polynomial of $\mathbf{A}$ and $m(t)$ the minimal polynomial. Let $\mathbf{T} : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ be the linear transformation with matrix $\mathbf{A}$ with respect to the standard basis. Since $m(\mathbf{A})$ is the zero matrix, $m(\mathbf{T})$ must be the zero transformation. The characteristic polynomial of $\mathbf{A}$ is the same as the characteristic polynomial of $\mathbf{T}$, and is independent of the basis used to represent $\mathbf{T}$ by a matrix. If $\mathbf{B}$ is a matrix representing $\mathbf{T}$, then since $m(\mathbf{T}) = \mathbf{0}$, it follows that $m(\mathbf{B}) = \mathbf{0}$ is the zero matrix. The roles of $\mathbf{A}$ and $\mathbf{B}$ can be reversed, and so we conclude that $m(t)$ is the minimal polynomial of any matrix representing $\mathbf{T}$.

Choose a basis for $\mathbb{C}^n$ in which $\mathbf{T}$ has Jordan normal form. The rules for block matrix multiplication and addition show that it is sufficient to prove the theorem for a single Jordan block

$$\mathbf{J} = \begin{bmatrix} e & & & \\ 1 & e & \mathbf{0} & \\ & \ddots & \ddots & \\ \mathbf{0} & & 1 & e \end{bmatrix}.$$

If the size of this Jordan block is $r \times r$, then the powers of $\mathbf{J}$ for $i = 1, \ldots, r-1$ are of the form

$$\mathbf{J}^i = \begin{bmatrix} e^i & & & \\ * & e^i & \mathbf{0} & \\ \vdots & \ddots & \ddots & \\ * & * & * & e^i \end{bmatrix},$$

These matrices are linearly independent in $\mathrm{Mat}_{r \times r}$ and since $(\mathbf{J} - e\mathbf{I})^r = \mathbf{0}$, the minimal polynomial of $\mathbf{J}$ is $(t-e)^r$. But this is also the characteristic polynomial of $\mathbf{J}$, and hence the minimal polynomial certainly divides the characteristic polynomial in this case, completing the proof. □

If you are wondering whether the minimal polynomial is always equal to the characteristic polynomial, the answer is no. Simply consider the case

$$\mathbf{A} = \begin{bmatrix} \mathbf{N} & \mathbf{0} \\ \mathbf{0} & \mathbf{N} \end{bmatrix},$$

where, say, $\mathbf{N}$ is a nilpotent matrix of nilpotence index 2 and of size $n \times n$. Then the characteristic polynomial of $\mathbf{A}$ is $t^{2n}$, but the minimal polynomial is $t^2$, so chosing $n > 1$, the minimal polynomial divides, but is not equal to, the characteristic polynomial.

## 17.6 Inverses

The Hamilton–Cayley theorem has a surprising application to computing inverses of matrices. For suppose $\mathbf{A}$ is a matrix with $\det(\mathbf{A}) \neq 0$. Consider the characteristic polynomial

$$\Delta(t) = \det(\mathbf{A}) = (-1)^n t^n + (-1)^{n-1} c_{n-1} t^{n-1} + \cdots - c_1 t + c_0$$

of $\mathbf{A}$. The coefficient $c_0$ of the characteristic polynomial cannot be zero, because if it were, $t$ would be a factor of $\Delta(t)$ and hence 0 an eigenvalue, which is equivalent to $\mathbf{A}$ being noninvertible, contrary to the assumption that $\det(\mathbf{A}) \neq 0$. In fact $c_0 = \det(\mathbf{A})$, as follows by putting $t = 1$ in the formula for the characteristic polynomial. From the Hamilton-Cayley theorem it follows that the matrix

$$\Delta(\mathbf{A} - t\mathbf{I}) = (-1)^n \mathbf{A}^n + (-1)^{n-1} c_{n-1} \mathbf{A}^{n-1} + \cdots - c_1 \mathbf{A} + c_0 \mathbf{I}$$

is the zero matrix, and hence we can solve the equation

$$(-1)^n \mathbf{A}^n + (-1)^{n-1} c_{n-1} \mathbf{A}^{n-1} + \cdots - c_1 \mathbf{A} + c_0 \mathbf{I} = \mathbf{0}$$

for $c_0\mathbf{I}$ to obtain

$$-c_0 \mathbf{I} = (-1)^n \mathbf{A}^n + (-1)^{n-1} c_{n-1} \mathbf{A}^{n-1} + \cdots - c_1 \mathbf{A},$$

which may be rearranged as follows:

$$\mathbf{I} = \frac{-1}{c_0}\left[\mathbf{A} \cdot \left((-1)^n \mathbf{A}^{n-1} + (-1)^{n-1} c_{n-1} \mathbf{A}^{n-2} + \cdots - c_1 \mathbf{I}\right)\right].$$

A moment's thought shows that we have proved:

**PROPOSITION 17.6.1:** *Let $\mathbf{A}$ be an $n \times n$ matrix with nonzero determinant and characteristic polynomial*

$$\Delta(t) = (-1)^n t^n + (-1)^{n-1} c_{n-1} t^{n-1} + \cdots - c_1 t + c_0.$$

*Then*

$$\mathbf{A}^{-1} = \frac{-1}{c_0}\left[(-1)^n \mathbf{A}^{n-1} + (-1)^{n-1} c_{n-1} \mathbf{A}^{n-2} + \cdots - c_1 \mathbf{I}\right]. \quad \square$$

**EXAMPLE 1:** Consider the matrix

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ 0 & 4 & 6 \end{bmatrix}$$

of Example 2 in Section 14.3. The determinant of this matrix is 6, and the characteristic polynomial can be computed with the use of some elementary row and column operations:

$$\begin{aligned}
\Delta(t) &= \det \begin{bmatrix} 1-t & 0 & 1 \\ 2 & 1-t & 2 \\ 0 & 4 & 6-t \end{bmatrix} \\
&= \frac{1}{1-t} \det \begin{bmatrix} 1-t & 0 & 1-t \\ 2 & 1-t & 2(1-t) \\ 0 & 4 & (6-t)(1-t) \end{bmatrix} \\
&= \frac{1}{1-t} \det \begin{bmatrix} 1-t & 0 & 0 \\ 2 & 1-t & 2(1-t)-2 \\ 0 & 4 & (6-t)(1-t) \end{bmatrix} \\
&= \frac{1-t}{1-t} \det \begin{bmatrix} 1-t & 2(1-t)-2 \\ 4 & (6-t)(1-t) \end{bmatrix} \\
&= (1-t)^2(6-t) + 8t = -t^3 + 8t^2 - 5t + 6.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathsf{T}^{-1} &= \frac{-1}{6}\left(-\mathsf{T}^2 + 8\mathsf{T} - 5\mathsf{I}\right) \\
&= \frac{-1}{6}\left(-\begin{bmatrix} 1 & 4 & 7 \\ 4 & 9 & 16 \\ 8 & 28 & 44 \end{bmatrix} + 8\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ 0 & 4 & 6 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\right) \\
&= -\frac{1}{6}\begin{bmatrix} 2 & -4 & 1 \\ 12 & -6 & 0 \\ -8 & 4 & -1 \end{bmatrix}.
\end{aligned}$$

## 17.7 Exercises

**1.** Give an example of a linear transformation in a finite-dimensional ***real*** vector space with no nontrivial invariant subspaces.

**2.** Let $\mathsf{T} : \mathbb{C}^2 \longrightarrow \mathbb{C}^2$ be the linear transformation given by the matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

with respect to some basis. What are the invariant subspaces of $\mathsf{T}$?

**3.** Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation with invariant subspaces $\mathcal{U}, \mathcal{W}$. Show that $\mathcal{U} + \mathcal{W}$ and $\mathcal{U} \cap \mathcal{W}$ are also invariant subspaces of $\mathbf{T}$.

**4.** Let $\mathbf{D} : \mathcal{P}_n(\mathbb{R}) \longrightarrow \mathcal{P}_n(\mathbb{R})$ be the differentiation operator. Show that $\mathcal{P}_i(\mathbb{R})$ for $i = 1, \ldots, n$ are all invariant subspaces of $D$. Does any of them have an invariant complement?

**5.** Which of the following subspaces of $\mathbb{C}^3$ are complementary to

$$\mathcal{U} = \left\{(x, y, z) \mid x + y + z = 0\right\}?$$

(1) $\left\{(x, y, z) \mid x + 2y + z = 0\right\}$.
(2) $\left\{(x, y, z) \mid x - y + z = 0\right\}$.
(3) $\left\{(x, y, z) \mid x + 2y + z = 0 \text{ and } x - y + z = 0\right\}$.
(4) $\mathcal{L}((1, -1, 1))$.
(5) $\mathcal{L}((1, -2, 1))$.

**6.** Define two subspaces of $\mathcal{P}_n(\mathbb{R})$ by

$$\mathcal{S} = \left\{p \in \mathcal{P}_n(\mathbb{R}) \mid p(x) = p(-x)\right\},$$
$$\mathcal{A} = \left\{p \in \mathcal{P}_n(\mathbb{R}) \mid p(x) = -p(-x)\right\}.$$

Show that $\mathcal{S}$ and $\mathcal{A}$ are complementary subspaces. Are they invariant under the differentiation operator $\mathbf{D}$?

**7.** Verify that each of the following transformations is nilpotent and find their Jordan normal forms:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} 1 & 2 & -1 \\ -1 & -2 & 1 \\ -1 & -2 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

**8.** Let $\mathbf{T} : \mathbb{C}^2 \longrightarrow \mathbb{C}^2$ with just one eigenvalue $e$. Show that $\mathbf{T} - e\mathbf{I}$ is a nilpotent linear transformation.

**9.** Let $\mathbf{N} : \mathcal{V} \longrightarrow \mathcal{V}$ be a nilpotent transformation. Show that for any linear transformation $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$, $\mathbf{N} \cdot \mathbf{T} - \mathbf{T} \cdot \mathbf{N} = \mathbf{S}$ is a nilpotent transformation.

**10.** Let $\mathbf{N} : \mathcal{V} \longrightarrow \mathcal{V}$ be a nilpotent transformation in an $n$-dimensional vector space $\mathcal{V}$. Show that the characteristic polynomial $\Delta_{\mathbf{N}}(t)$ is $(-1)^n t^n$.

**11.** Show that the trace of a nilpotent transformation is zero. Is the converse true?

**12.** Let $\mathbf{T} : \mathbb{C}^3 \longrightarrow \mathbb{C}^3$ be represented with respect to the standard basis by each of the following matrices in turn. Find the Jordan normal form of $\mathbf{T}$.

$$\mathbf{A} = \begin{bmatrix} 5 & 4 & 3 \\ -1 & 0 & -3 \\ 1 & -2 & 1 \end{bmatrix}. \quad \mathbf{D} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

$$\mathbf{B} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \quad \mathbf{E} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

$$\mathbf{C} = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}. \quad \mathbf{F} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}.$$

**13.** Let $\mathbf{T} : \mathbb{C}^6 \longrightarrow \mathbb{C}^6$ be represented by the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

with respect to the standard basis of $\mathbb{C}^6$. Find the Jordan normal form of $\mathbf{T}$.

**14.** Let $\mathbf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a linear transformation in a finite-dimensional complex vector space $\mathcal{V}$. Show that $\mathbf{T} = \mathbf{A} + \mathbf{N}$, where $\mathbf{A} : \mathcal{V} \longrightarrow \mathcal{V}$ is an isomorphism and $\mathbf{N} : \mathcal{V} \longrightarrow \mathcal{V}$ is nilpotent.

**15.** Find a square root of each of the following matrices.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \quad \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

**16.** Find the inverse of the matrices in the preceding exercise.

**17.** Let $\mathcal{V}$ be an inner product space and let $\mathbf{T} : \mathcal{V} \longrightarrow V$ be a linear transformation with invariant subspace $\mathcal{W}$. Show that $\mathcal{W}^{\perp}$ is invariant for the adjoint $\mathbf{T}^* : \mathcal{V} \longrightarrow \mathcal{V}$.

**18.** A linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ in an inner product space is called **normal** if it commutes with its adjoint. Suppose that $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is diagonalizable with respect to an orthonormal basis in the inner product space $\mathcal{V}, <—,—>$. Show that $\mathsf{T}$ is normal. Is it enough to assume that $\mathsf{T}$ be diagonalizable with respect to some, perhaps nonorthonormal basis?

**19.** Let $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ be a normal transformation in a finite-dimensional complex Hermitian inner product space (see Exercise 21 in Section 15.6).

(a) Show that $\mathsf{T}$ has a nonzero invariant subspace. (HINT Since the scalars are complex there must be eigenvalues.)

(b) Let $\mathcal{W} \subseteq \mathcal{V}$ be a $\mathsf{T}$-invariant subspace. Show that it is completely invariant, i.e., $\mathcal{W}$ has an invariant complement. (HINT: Do Exercise 17 and use Proposition 17.1.7.)

(c) Show that $\mathsf{T}$ is diagonalizable. **HINT:** Use induction on the dimension of $\mathcal{V}$.

# Chapter 18 Application to Differential Equations

Differential equations have played an important part in the development of mathematics since the time of Newton. They have also been central to many applications of mathematics to the physical sciences and technology. Often the equations relevant to practical applications are so difficult to solve explicitly that they can only be handled with approximation techniques on large computer systems. In this chapter we will be concerned with a simple form of differential equation, and systems thereof, namely, linear differential equations with constant coefficients. These systems have a great deal in common with systems of linear equations, and we are in a position to apply the hardwon knowledge about Jordan canonical forms to solve such systems.

## 18.1 Linear Differential Systems: Basic Definitions

Differential equations come in all sizes. Here is a relatively simple example.

**EXAMPLE 1:** Find all differentiable functions $y : \mathbb{R} \longrightarrow \mathbb{R}$ with the property that the derivative $dy/dt$ of $y$ satisfies the equation

$$\frac{dy}{dt}(t) = r \cdot y(t), \qquad \forall\, t \in \mathbb{R},$$

where $r$ is some fixed number.

**SOLUTION:** $y = ce^{rt}$, where $c$ is a constant, as follows from the fundamental theorem[1] of the calculus.

We begin by defining some basic concepts.

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[1] By the **fundamental theorem of the calculus** we understand the following

**Definition:** *Let $n$ be a positive integer, $B \subseteq \mathbb{R}^{n+2}$ a nonempty subset, and*

$$F : B \longrightarrow \mathbb{R}$$

*a function. A function*

$$y : J \longrightarrow \mathbb{R}$$

*defined on the interval $J = [a, b] \subseteq \mathbb{R}$ (i.e., $J = \{t \in \mathbb{R}, a \le t \le b\}$) is called a* **solution to the differential equation**

$$F\left(t, y, \frac{dy}{dt}, \dots, \frac{d^n y}{dt^n}\right) = 0$$

*on the interval $J$ if the following conditions are fulfilled:*

(1) *$y$ is an $n$-times differentiable function on $J$.*

(2) *$\forall\, t \in J$ the $(n+2)$-tuple*

$$\left(t, y(t), \frac{dy}{dt}(t), \dots, \frac{d^n y}{dt^n}(t)\right)$$

*belongs to $B$.*

(3) *$\forall\, t \in J$ we have*

$$F\left(t, y(t), \frac{dy}{dt}(t), \dots, \frac{d^n y}{dt^n}(t)\right) = 0.$$

A class of differential equations of particular interest are those of the form

$$\frac{d^n y}{dt^n} - f\left(t, y, \dots, \frac{d^{n-1} y}{dt^{n-1}}\right) = 0,$$

or, equivalently,

$$\frac{d^n y}{dt^n} = f\left(t, y, \dots, \frac{d^{n-1} y}{dt^{n-1}}\right).$$

Such an equation is said to be of $n$-th **order**.

We will also consider ***systems*** of differential equations. They are analogous to systems of simultaneous equations.

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theorem:

**Theorem 18.1.1** (Fundamental Theorem of the Calculus): *If $f(t)$ is a continuous function on the interval $[a, b]$ and*

$$F(x) = \int_a^x f(t)dt \qquad \forall\, x \in [a, b]$$

*then $F'(x) = f(x)$ for all $x \in (a, b)$. If $H(x)$ is any other function with $H'(x) = f(x)\ \forall\, x \in (a, b)$ then $H(x) = F(x) + C$ for some constant $C \in \mathbb{R}$.*

**DEFINITION:** *A* **solution** *to the* **first-order system**

$$\frac{dy_1}{dt} = f_1(t, y_1, \dots, y_n),$$
$$\frac{dy_2}{dt} = f_2(t, y_1, \dots, y_n),$$
$$\vdots \quad \vdots$$
$$\frac{dy_n}{dt} = f_n(t, y_1, \dots, y_n),$$

*where $f_i : B \longrightarrow \mathbb{R}$, for $i = 1, \dots, n$, are functions defined on an open nonempty set $B \subseteq \mathbb{R}^{n+1}$, consists of an interval $J \subseteq \mathbb{R}$ and functions*

$$y_i : J \longrightarrow \mathbb{R}, \qquad i = 1, \dots, n,$$

*such that:*

(1) *$y_1, \dots, y_n$ are differentiable on $J$.*
(2) *$\forall\ t \in J$ the $(n+1)$-tuple $(x, y_1(t), \dots, y_n(t))$ belongs to $B$.*
(3) *$\forall\ t \in J$ and each $1 \le i \le n$*

$$\frac{dy_i}{dt}(t) = f_i(t, y_1(t), \dots, y_n(t)).$$

*If each $f_i$, for $1 \le i \le n$, is a linear function, that is,*

$$f_i(t_1, \dots, t_{n+1}) = \sum_{j=1}^{n+1} a_{i,j} t_j, \qquad 1 \le i \le n,$$

*for $a_{i,j} \in \mathbb{R}$, then the above system is called a* **linear differential system of the first order.** *If furthermore, $a_{i,1} = 0$ for $i = 1, \dots, n$ (i.e., "$t = t_1$ does not appear on the right-hand side") then the system is called* **homogeneous.**

**REMARK:** Do not let these definitions scare you! We will only be concerned with linear differential systems of the first order, and that only in concrete cases.

**NOTE:** For linear systems, the functions $f_i$ are defined on all of $\mathbb{R}^{n+1}$.

We can write a linear differential system in the matrix form

$$\begin{bmatrix} \frac{dy_1}{dt}(t) \\ \vdots \\ \frac{dy_n}{dt}(t) \end{bmatrix} = t\mathbf{B} + \mathbf{A} \cdot \begin{bmatrix} y_1(t) \\ \vdots \\ y_n(t) \end{bmatrix},$$

where

$$\mathbf{B} = \begin{bmatrix} a_{1,1} \\ \vdots \\ a_{n,1} \end{bmatrix} \in \mathbb{R}^n, \qquad \mathbf{A} = \begin{bmatrix} a_{1,2} & \cdots & a_{1,n+1} \\ & \ddots & \\ a_{n,2} & \cdots & a_{n,n+1} \end{bmatrix} \in \mathrm{Mat}_{n,n}.$$

The system is homogeneous when $\mathbf{B} = \mathbf{0} \in \mathbb{R}^n$.

Suppose that $f : B \longrightarrow \mathbb{R}$, $B \subseteq \mathbb{R}^{n+1}$, and that we are looking for a solution $y : J \longrightarrow \mathbb{R}$ to the $n$-th order differential equation

$$(*) \qquad \frac{d^n y}{dt^n}(t) = f\left(t, y, \frac{dy}{dt}, \ldots, \frac{d^{n-1}y}{dt^{n-1}}\right).$$

If we set

$$\begin{aligned} y_1 &= y, \\ y_2 &= \frac{dy}{dt} = \frac{dy_1}{dt}, \\ &\vdots \\ y_n &= \frac{d^{n-1}y}{dt^{n-1}} = \frac{dy_{n-1}}{dt}, \end{aligned}$$

then $y$ is a solution to $(*)$ if and only if $(y_1, \ldots, y_n)$ is a solution to the rather special first order system of differential equations

$$\begin{aligned} \frac{dy_1}{dt} &= y_2, \\ &\vdots \\ \frac{dy_{n-2}}{dt} &= y_{n-1}, \\ \frac{dy_{n-1}}{dt} &= y_n, \\ \frac{dy_n}{dt} &= f(t, y_1, \ldots, y_{n-1}). \end{aligned}$$

In this way, solving a single differential equation of order $n$ is equivalent to solving a first order system of differential equations.

Before we take up some examples, let us consider the structure of the set of all solutions to a linear differential system (compare Theorem 13.1.3), which justifies the use of the word *linear*.

**PROPOSITION 18.1.2:** *Suppose*

$$(*) \qquad \begin{bmatrix} \frac{dy_1}{dt} \\ \vdots \\ \frac{dy_n}{dt} \end{bmatrix} = t\mathbf{B} + \mathbf{A} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix},$$

$\mathbf{A} \in \mathrm{Mat}_{n,n}$, $\mathbf{B} \in \mathbb{R}^n$, *is a linear differential system of the first order. Let* $\mathcal{S}$ *be the set of all solutions on some interval* $J = [a, b]$ *to the associated homogeneous system*

$$(**) \qquad \frac{dy}{dt} = \mathbf{A}y, \qquad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

*Then*

(1) $\mathcal{S}$ *is a vector subspace of* $C(a, b) \oplus \cdots \oplus C(a, b)$ *(Example* 5 *in Section* 3.2.*)*

(2) *If* $z = (z_1, \ldots, z_n)$ *is a solution to* $(*)$, *then every solution to* $(*)$ *is of the form* $z + y$ *for some* $y \in \mathcal{S}$.

**PROOF:** First of all $\mathbf{0} \in \mathcal{S}$ because if $\mathbf{0} = (0, \cdots, 0)$, then

$$\frac{d\mathbf{0}}{dt} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{A} \cdot \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}.$$

Next, if $\varphi, \psi \in \mathcal{S}$ and $a, b \in \mathbb{R}$, then using the linearity of differentiation, we find

$$\begin{aligned} \frac{d}{dt}(a\varphi + b\psi) &= a\frac{d\varphi}{dt} + b\frac{d\psi}{dt} \\ &= a\mathbf{A}\varphi + b\mathbf{A}\psi \\ &= \mathbf{A}(a\varphi + b\psi), \end{aligned}$$

which says that $a\varphi + b\psi$ solves the equation $(**)$ and so $a\varphi + b\psi \in \mathcal{S}$.

To prove (2), suppose $z$ is a fixed solution to $(*)$ and $w$ any other solution. Then again using the linearity of differentiation, we obtain

$$\begin{aligned} \frac{d}{dt}(z - w) &= \frac{dz}{dt} - \frac{dw}{dt} \\ &= (\mathbf{B}t + \mathbf{A}z) - (\mathbf{B}t + \mathbf{A}w) \\ &= \mathbf{A}(z - w), \end{aligned}$$

so $z - w$ solves $(**)$, and hence $z - w = y \in \mathcal{S}$. $\square$

## 18.2 Diagonalizable Systems

A large family of examples are covered by the definition and methods that follow in this section, namely the diagonalizable systems. Here is the fundamental definition.

**DEFINITION:** *A linear homogeneous differential system of the first order*

$$\frac{dy}{dt} = \mathbf{A}y, \qquad \mathbf{A} \in \mathrm{Mat}_{n,n},$$

*is said to be* **diagonalizable** *if the matrix* **A** *is diagonalizable.*

Suppose that

$$(\maltese) \qquad \frac{dy}{dt} = \mathbf{A}y$$

is a diagonalizable system. Then there exists a diagonal matrix **D** and an invertible matrix $\mathbf{P} \in \mathrm{Mat}_{n,n}$ such that

$$\mathbf{A} = \mathbf{P} \cdot \mathbf{D} \cdot \mathbf{P}^{-1}.$$

Thus we can write ($\maltese$) in the form

$$\frac{dy}{dt} = \mathbf{P} \cdot \mathbf{D} \cdot \mathbf{P}^{-1}y.$$

If we multiply both sides of this equation on the left by $\mathbf{P}^{-1}$, we obtain the equivalent equation

$$\mathbf{P}^{-1}\frac{dy}{dt} = \mathbf{D} \cdot \mathbf{P}^{-1}y.$$

Since $\mathbf{P}^{-1}$ is a matrix of constants, i.e., does not depend on the variable $t$, we can write further

$$\mathbf{P}^{-1}\frac{dy}{dt} = \frac{d}{dt}(\mathbf{P}^{-1}y).$$

Now comes the crucial step. Set

$$w = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \mathbf{P}^{-1}\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

and combine the two preceding equations in

$$\frac{dw}{dt} = \mathbf{D}w.$$

The matrix $\mathbf{D}$ being a diagonal matrix, this latter system is just

$$\begin{aligned} \frac{dw_1}{dt} &= d_1 w_1, \\ \vdots \quad & \quad \vdots \\ \frac{dw_n}{dt} &= d_n w_n, \end{aligned}$$

and so the fundamnetal theorem of the calculus tells us that the solutions are

$$\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} c_1 e^{d_1 t} \\ \vdots \\ c_n e^{d_n t} \end{bmatrix}.$$

Finally, we can solve the original system (✲) by reversing the steps. Since

$$\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \mathbf{P}^{-1} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix},$$

we have

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \mathbf{P} \cdot \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \mathbf{P} \cdot \begin{bmatrix} c_1 e^{d_1 t} \\ \vdots \\ c_n e^{d_n t} \end{bmatrix},$$

and thus we have proven the following theorem:

**THEOREM 18.2.1:** *Let*

$$\frac{dy}{dt} = \mathbf{A} y, \qquad \mathbf{A} \in \mathrm{Mat}_{n,n},$$

*be a diagonalizable linear system, with*

$$\mathbf{A} = \mathbf{P} \cdot \mathbf{D} \cdot \mathbf{P}^{-1}$$

*for a diagonal matrix* $\mathbf{D}$ *and an invertible matrix* $\mathbf{P}$. *Then the solutions are of the form*

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \mathbf{P} \cdot \begin{bmatrix} c_1 e^{d_1 t} \\ \vdots \\ c_n e^{d_n t} \end{bmatrix},$$

*where* $d_1, \ldots, d_n$ *are the diagonal entries of* $\mathbf{D}$ *and* $c_1, \ldots, c_n$ *are arbitrary constants.* □

**EXAMPLE 1:** Solve the linear differential system

$$\frac{dy_1}{dt} = y_2,$$
$$\frac{dy_2}{dt} = y_1.$$

**SOLUTION:** In matrix form this is the linear differential system

$$\frac{d\mathbf{y}}{dt} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

The coefficient matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is symmetric, and therefore diagonalizable. To diagonalize $\mathbf{A}$ we first compute the characteristic polynomial

$$\Delta(t) = \det \begin{bmatrix} -t & 1 \\ 1 & -t \end{bmatrix} = (t^2 - 1) = (t-1)(t+1).$$

There are two eigenvalues, $t = \pm 1$, and corresponding eigenvectors can be easily found:

$$t = 1: \quad \left\{ \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right. \quad \text{Solution: } \mathbf{A}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$
$$t = -1: \quad \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right. \quad \text{Solution: } \mathbf{A}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

We can therefore write

$$\mathbf{A} = \mathbf{P} \cdot \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{P}^{-1},$$

where

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The associated diagonalized system is therefore

$$\frac{dw_1}{dt} = w_1,$$
$$\frac{dw_2}{dt} = -w_2,$$

and has as solutions the functions

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} c_1 e^{t} \\ c_2 e^{-t} \end{bmatrix}.$$

Thus the solutions to the equations of this example are

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{P} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} c_1 e^t \\ c_2 e^{-t} \end{bmatrix}$$
$$= \begin{bmatrix} c_1 e^t + c_2 e^{-t} \\ c_1 e^t - c_2 e^{-t} \end{bmatrix}.$$

**EXAMPLE 2:** Solve the differential equation

$$\frac{d^3 z}{dt^3} + 4\frac{d^2 z}{dt^2} + \frac{dz}{dt} - 6z = 0.$$

**SOLUTION:** To do so, we first convert this single equation to an equivalent first order system by setting

$$y_1 = z,$$
$$y_2 = \frac{dz}{dt},$$
$$y_3 = \frac{d^2 z}{dt^2}.$$

We then have

$$\frac{dy_1}{dt} = y_2,$$
$$\frac{dy_2}{dt} = y_3,$$

and the original equation reads

$$\frac{dy_3}{dt} = -4y_3 - y_2 + 6y_1,$$

so we have the system

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -1 & -4 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

The characteristic polynomial of the coefficient matrix is

$$\Delta(t) = \det \begin{bmatrix} -t & 1 & 0 \\ 0 & -t & 1 \\ 6 & -1 & -4-t \end{bmatrix}$$
$$= -t(t(t+4)+1) + 6(1)$$
$$= -t^3 - 4t^2 - t + 6$$
$$= -(t-1)(t+2)(t+3).$$

Therefore, the coefficient matrix has three distinct eigenvalues and can be diagonalized. The computation of eigenvectors proceeds as usual. We find

$$t_1 = 1: \quad \left\{ \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 6 & -1 & -5 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right. \quad \text{Solution: } \mathbf{A}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix},$$

$$t_2 = -2: \quad \left\{ \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 6 & -1 & -2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right. \quad \text{Solution: } \mathbf{A}_2 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix},$$

$$t_3 = -3: \quad \left\{ \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 6 & -1 & -1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right. \quad \text{Solution: } \mathbf{A}_3 = \begin{bmatrix} 1 \\ -3 \\ 9 \end{bmatrix}.$$

The matrix of the basis change is

$$\mathsf{P} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix},$$

that is,

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -1 & -4 \end{bmatrix} = \mathsf{P} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \cdot \mathsf{P}^{-1}.$$

Set

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \mathsf{P}^{-1} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

The associated diagonal system is

$$\frac{d}{dt} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix},$$

and it has the solutions

$$w_1 = c_1 e^{t},$$

$$w_2 = c_2 e^{-2t},$$

$$w_3 = c_3 e^{-3t}.$$

If we make the change of variables

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \mathsf{P}^{-1} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},$$

then we find that

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \mathbf{P} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix},$$

and thus

$$y_1 = c_1 e^t + c_2 e^{-2t} + c_3 e^{-3t},$$

$$y_2 = c_1 e^t - 2c_2 e^{-2t} - 3c_3 e^{-3t},$$

$$y_3 = c_1 e^t + 4c_2 e^{-2t} + 9c_3 e^{-3t}.$$

Since $z = y_1$, we obtain for the general solution to the original third order linear differential equation,

$$z = c_1 e^t + c_2 e^{-2t} + c_3 e^{-3t}.$$

This example illustrates that the $n$-th order differential equation

$$\frac{d^n z}{dt^n} + a_{n-1}\frac{d^{n-1}z}{dt} + \cdots + a_1 \frac{dz}{dt} + a_0 z = 0$$

is equivalent to the linear system

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & \ddots & \cdots & \vdots \\ 0 & \cdots & 0 & 1 \\ -a_0 & \cdots & & -a_{n-1} \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

To apply the preceding method of solution, we need to know when a matrix of the form,

$$\begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & \ddots & \cdots & \vdots \\ 0 & \cdots & 0 & 1 \\ -a_0 & \cdots & & -a_{n-1} \end{bmatrix},$$

is diagonalizable. Matrices of the precedding form, i.e. with 1s on the superdiagonal and 0s except for the last row otherwise, are called **companion matrices.** The following proposition provides us with a method to decide when such a companion matrix is diagonalizable.

**PROPOSITION 18.2.2:** *The characteristic polynomial of the companion matrix*

$$\begin{bmatrix} 0 & 1 & & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & & 0 & 1 \\ b_0 & & \cdots & b_{n-1} \end{bmatrix}$$

*is* $\Delta(t) = (-1)^n(t^n - b_{n-1}t^{n-1} - \cdots - b_1 t - b_0)$.

**PROOF:** For $n = 2$ we have

$$\Delta(t) = \det\begin{bmatrix} -t & 1 \\ b_0 & b_1 - t \end{bmatrix} = -t(b_1 - t) - b_0$$
$$= t^2 - b_1 t - b_0,$$

so the proposition is true for $n = 2$. Suppose that we knew that the proposition were true for all companion matrices of size $(n-1)\times(n-1)$ and that $\mathbf{B}$ is a companion matrix of size $n \times n$. Then using Lagrange expansion with respect to the first column gives

$$\Delta(t) = \det(\mathbf{B} - t\mathbf{I}) = \det\begin{bmatrix} -t & 1 & 0 & & 0 \\ 0 & -t & 1 & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -t & 1 \\ b_0 \cdots & \cdots & \cdots & \cdots & b_{n-1} - t \end{bmatrix}$$

$$= -t \det\begin{bmatrix} -t & 1 & 0 & & 0 \\ 0 & -t & 1 & & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -t & 1 \\ b_1 & b_2 & \cdots & & b_{n-1} - t \end{bmatrix} + (-1)^{n+1} b_0 \det\begin{bmatrix} 1 & 0 & & & 0 \\ -t & \ddots & & & \\ 0 & \ddots & \ddots & & \\ \vdots & & & & \\ 0 & \cdots & 0 & -t & 1 \end{bmatrix}.$$

Notice that

$$\det\begin{bmatrix} -t & 1 & 0 & \cdots & 0 \\ 0 & -t & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & & \vdots \\ 0 & 0 & & -t & 1 \\ b_1 & \cdots & & b_{n-2} & b_{n-1} - t \end{bmatrix}$$

is the characteristic polynomial of the companion matrix

$$\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & & \vdots \\ 0 & 0 & & 0 & 1 \\ b_1 & \cdots & & b_{n-2} & b_{n-1} \end{bmatrix}$$

of size $(n-1)\times(n-1)$. By our inductive assumption we know its characteristic polynomial, and substituting this, and also that the determinant of the lower triangular matrix is 1, in the preceding equation gives

$$\Delta(t) = -t((-1)^{n-1}(t^{n-1} - b_{n-1}t^{n-2} - \cdots - b_1)) + (-1)^{n+1} b_0$$
$$= (-1)^n [t^n - b_{n-1}t^{n-1} - \cdots - b_1 t - b_0],$$

establishing the proposition by induction. □

**THEOREM 18.2.3:** *Consider the differential equation*

$$\frac{d^n z}{dt^n} + a_{n-1}\frac{d^{n-1}z}{dt^{n-1}} + \cdots + a_1\frac{dz}{dt} + a_0 z = 0.$$

*If the polynomial*

$$\Delta(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0$$

*has n distinct real roots, $d_1, \ldots, d_n$, then the general solution of this differential equation is*

$$z = c_1 e^{d_1 t} + \cdots + c_n e^{d_n t}.$$

**PROOF:** Set

$$\begin{aligned} y_1 &= z, \\ y_2 &= \frac{dz}{dt}, \\ \vdots\;\; & \;\;\vdots \\ y_n &= \frac{d^n z}{dt^n}. \end{aligned}$$

Then the original $n$-th order differential equation is equivalent to the linear differential system

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & & 0 \\ \vdots & \ddots & & \\ 0 & \cdots & 0 & 1 \\ -a_0 & \cdots & & -a_{n-1} \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \mathbf{A} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

The characteristic polynomial of the coefficient matrix is (by Proposition 18.2.2)

$$\Delta(t) = (-1)^n t^n + a_{n-1}t^{n-1} + \cdots + a_0.$$

By hypothesis this polynomial has $n$ distinct roots, $d_1, \ldots, d_n$, so the coefficient matrix $\mathbf{A}$ is diagonalizable, say,

$$\mathbf{A} = \mathbf{P}\begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}\mathbf{P}^{-1}.$$

The corresponding diagonal system

$$\frac{d}{dt}\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} d_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & d_n \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$$

has as solution

$$\begin{aligned} w_1 &= c_1 e^{d_1 t}, \\ \vdots \quad & \quad \vdots \\ w_n &= c_n e^{d_n t}, \end{aligned}$$

where $c_1, \ldots, c_n$ are arbitrary constants and

$$\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \mathbf{P}^{-1} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

So, in particular,

$$z(t) = y_1(t) = p_{1,1} c_1 e^{d_1 t} + \cdots + p_{1,n} c_n e^{d_n d t}.$$

It only remains to show that none of the entries $p_{1,j}$ is zero, so that none of the constants $p_{1,j} c_j$ vanish, and thus remain arbitrary. So suppose $p_{1,j} = 0$. Then from the equation

$$y_2 = \frac{dy_1}{dt}$$

we get by substitution

$$\begin{aligned} & p_{2,1} c_1 e^{d_1 t} + \cdots + p_{2,j} c_j e^{d_j t} + \cdots + p_{2,n} c_n e^{d_n t} \\ & \quad = p_{1,1} c_1 d_1 e^{d_1 t} + \cdots + p_{1,j} c_j d_j e^{d_j t} + \cdots + p_{1,n} c_n d_n e^{d_n t}, \end{aligned}$$

for all values of the constants $c_1, \ldots, c_n$, in particular, for $c_1 = \cdots = c_{j-1} = c_{j+1} = \cdots = c_n = 0$ and $c_j = 1$. This gives

$$p_{2,j} e^{d_j t} = 0,$$

so $p_{2,j} = 0$. Continuing in this way we obtain in succession $p_{3,j} = 0, \ldots, p_{n,j} = 0$. But this says that $\mathbf{P}$ has a column of zeros, and hence the determinant of $\mathbf{P}$ is zero, contrary to the fact that $\mathbf{P}$ is invertible. $\square$

Let us try another example using all that we have learned up to here.

**EXAMPLE 3:** Solve the differential equation

$$\frac{d^3 z}{dt^3} - \frac{dz}{dt} = 0.$$

**SOLUTION :** The associated polynomial is

$$p(t) = t^3 - t = t(t^2 - 1) = t(t-1)(t+1).$$

So by Theorem 18.2.3 the general solution is (remember $e^0 = 1$)

$$z = c_1 e^{0t} + c_2 e^t + c_3 e^{-t} = c_1 + c_2 e^t + c_3 e^{-t}.$$

## 18.3 Application of Jordan Form

Up until this point we have considered only linear differential systems

$$\frac{dy}{dt} = \mathbf{A}y,$$

where the coefficient matrix $\mathbf{A}$ is diagonalizable. In this case Theorem 18.2.3 provides an almost mechanical[2] means to find all the solutions. We know, of course, that not all matrices are diagonalizable. What happens then? Here is where the Jordan canonical form comes into play! Let us begin by looking at a simple example.

**EXAMPLE 1:** Consider the differential system

$$\begin{bmatrix} \frac{dy_1}{dt} \\ \frac{dy_2}{dt} \end{bmatrix} = \begin{bmatrix} a & 0 \\ 1 & a \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \end{bmatrix},$$

where $a$ is a fixed real number. The characteristic polynomial of the coefficient matrix is

$$\Delta(t) = (a - t)^2,$$

so $a$ is a double root, and the coefficient matrix cannot be diagonalized. (Think this through!) On the other hand, we can easily solve this differential system. We have

$$\frac{dy_1}{dt} = a y_1 \quad \Rightarrow \quad y_1 = c_1 e^{at}.$$

If we put this into the second equation, we find

$$\frac{dy_2}{dt} = c_1 e^{at} + a y_2.$$

[2] Finding the roots of the characteristic polynomial can be very hard, and is in general not possible with a ***mechanical*** method. But for polynomials up to degree four there are simple algebraic formulas for their roots.

Consider $y_2e^{-at}$, and compute its derivative by the product rule. We find

$$\begin{aligned}\frac{d}{dt}(y_2e^{-at}) &= -a\,y_2e^{-at} + (c_1e^{at} + a\,y_2)e^{-at}\\ &= -a\,y_2e^{-at} + c_1 + a\,y_2e^{-at}\\ &= c_1.\end{aligned}$$

That is, $y_2e^{-at}$ has a constant derivative. Therefore, by the fundamental theorem of the calculus,

$$y_2e^{-at} = c_1t + c_2,$$

or

$$y_2 = c_1te^{at} + c_2e^{at},$$

where $c_2$ is a second arbitrary constant. So the general solution is

$$\begin{aligned}y_1 &= c_1e^{at},\\ y_2 &= c_1te^{at} + c_2e^{at}.\end{aligned}$$

This is hardly any more complicated than our initial analysis of the diagonalizable case. Here is another example.

**EXAMPLE 2:** Solve the differential equation

$$\frac{d^2z}{dt^2} + \frac{2dz}{dt} + z = 0.$$

**SOLUTION:** The equivalent linear differential system is

$$\begin{bmatrix}\frac{dy_1}{dt}\\ \frac{dy_2}{dt}\end{bmatrix} = \begin{bmatrix}0 & 1\\ -1 & -2\end{bmatrix}\cdot\begin{bmatrix}y_1\\ y_2\end{bmatrix}.$$

The characteristic polynomial of the coefficient matrix $\mathbf{A}$ is

$$\Delta(t) = t^2 + 2t + 1 = (t+1)^2,$$

which has $t = -1$ as a double root. The matrix $\mathbf{A}$ is therefore not diagonalizable. (Because if it were, the diagonal form would be scalar, and since scalar matrices commute with every matrix, the equation $\mathbf{A} = \mathbf{P}\cdot\mathbf{D}\cdot\mathbf{P}^{-1}$ would yield $\mathbf{A} = \mathbf{D}$. But $\mathbf{A}$ is not itself diagonal.) On the other hand $\mathbf{A}$, has Jordan canonical form

$$\mathbf{J} = \begin{bmatrix}-1 & 0\\ 1 & -1\end{bmatrix},$$

and we saw how to solve

$$\frac{d}{dt}\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \mathbf{J}\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

in Example 1. This gives us:

$$w_1 = c_1 e^{-t}$$

$$w_2 = c_1 t e^{-t} + c_2 e^{-t}.$$

What we want, however, is not a formula for $(w_1, w_2)$, but for $(y_1, y_2)$. This is also not hard: what we need is the transformation matrix passing between $\mathbf{A}$ and its Jordan form, that is, a matrix $\mathbf{Q}$ such that

$$\mathbf{A} = \mathbf{Q} \cdot \mathbf{J} \cdot \mathbf{Q}^{-1}.$$

This proceeds as in Chapter 17 by finding a Jordan basis. Let $\mathbf{T} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear transformation represented by the matrix $\mathbf{A}$ with respect to the standard basis of $\mathbb{R}^2$. Let $\mathbf{F}_1 = (u, v) \in \mathbb{R}^2$ be a vector such that

$$\mathbf{F}_2 = \mathbf{A}\begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} u \\ v \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Such a vector exists because $\mathbf{A}$ is not diagonalizable. Furthermore, since $-1$ is the only eigenvalue of $\mathbf{A}$, it follows that $\mathbf{F}_2 \notin \mathcal{L}(\mathbf{F}_1)$. Therefore, $\{\mathbf{F}_1, \mathbf{F}_2\}$ is a basis for $\mathbb{R}^2$. Moreover,

$$\begin{aligned} \mathbf{T}(\mathbf{F}_1) &= -\mathbf{F}_1 + \mathbf{F}_2, \\ \mathbf{T}(\mathbf{F}_2) &= (\mathbf{T} + \mathbf{I} - \mathbf{I})(\mathbf{T} + \mathbf{I})(\mathbf{F}_1) \\ &= (\mathbf{T} + \mathbf{I})^2(\mathbf{F}_1) - (\mathbf{T} + \mathbf{I})(\mathbf{F}_1) \\ &= \mathbf{0} - \mathbf{F}_2, \end{aligned}$$

because $\mathbf{T} + \mathbf{I}$ is nilpotent of index 2. Thus $\mathbf{J}$ is the matrix of $\mathbf{T}$ with respect to the basis $\{\mathbf{F}_1, \mathbf{F}_2\}$ for $\mathbb{R}^2$. We need only find a vector $\mathbf{F}_1$. A possible choice for $\mathbf{F}_1$ is (0, 1), whence $\mathbf{F}_2 = (1, -1)$. Therefore, setting

$$\mathbf{Q} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix},$$

we have

$$\mathbf{A} = \mathbf{Q}\begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}\mathbf{Q}^{-1}.$$

We can complete the solution as in the diagonalizable case and set

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \mathbf{Q}^{-1}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix},$$

so

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{Q}\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} c_1 e^{-t} \\ c_1 t e^{-t} + c_2 e^{-t} \end{bmatrix}$$

$$= \begin{bmatrix} c_1 t e^{-t} + c_2 e^{-t} \\ c_1 e^{-t} - c_1 t e^{-t} - c_2 e^{-t} \end{bmatrix},$$

whence we find that the general solution to the original differential equation is

$$z = c_1 t e^{-t} + c_2 e^{-t}.$$

The step from this example to the general solution of the differential system

$$\frac{d}{dt}\mathbf{Y} = \mathbf{J}_n(a)\mathbf{Y},$$

where

$$\mathbf{J}_n(a) = \begin{bmatrix} a & & & \\ 1 & a & \mathbf{0} & \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & 1 & a \end{bmatrix}$$

is an $n \times n$ Jordan block with eigenvalue $a \in \mathbb{R}$, is easy. We simply recycle the argument of Example 1 as often as possible. We start with

$$\frac{dy_1}{dt} = a y_1,$$

so that

$$y_1 = c_1 e^{at},$$

$$\frac{dy_2}{dt} = y_1 + a y_2,$$

and by the product rule (see Example 1 again)

$$\frac{d}{dt}(e^{-at} y_2) = c_1.$$

This says that the function $y_2 e^{-at}$ is a constant. Therefore,

$$y_2 e^{-at} = c_1 t + c_2,$$

and

$$y_2 = c_1 t e^{at} + c_2 e^{at} = (c_1 t + c_2)e^{at}.$$

Next we deal with $y_3$:

$$\frac{dy_3}{dt} = y_2 + a y_3,$$

so by the product rule

$$\frac{d}{dt}(y_3 e^{-at}) = y_2 e^{-at}$$
$$= c_1 t + c_2.$$

As before, the fundamental theorem of the calculus gives

$$y_3 e^{-at} = \frac{c_1}{2}t^2 + c_2 t + c_3,$$

where $c_3$ is yet another constant. This says that

$$y_3 = \left(\frac{c_1}{2}t^2 + c_2 t + c_3\right)e^{at}.$$

Continuing in this way gives[3]

$$y_k = \left(\frac{c_1}{(k-1)!}t^{k-1} + \frac{c_1}{(k-2)!}t^{k-2} + \cdots + c_{k-1}t + c_k\right)e^{at}$$

for $k = 1, 2, \ldots, n$.

We may summarize this in a theorem.

**THEOREM 18.3.1:** *The general solution to the linear system*

$$\frac{d}{dt}\mathbf{Y} = \mathbf{J}_n(a)\mathbf{Y}$$

*is given by*

$$y_k = \left(\frac{c_1}{(k-1)!}t^{k-1} + \cdots + c_{k-1}t + c_k\right)e^{at}, \qquad k = 1, \ldots, n,$$

*where $c_1, \ldots, c_n$ are arbitrary constants.* □

The step from a single Jordan block to a general linear system with constant coefficients is clear: find the Jordan form of the coefficient matrix, and the coordinate change relating the standard basis to the Jordan basis, etc. Here is an example to illlustrate this.

**EXAMPLE 3:** Solve the differential system

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 & 2 & -1 \\ -1 & -1 & 1 \\ -1 & -2 & 2 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

[3] If $m$ is a positive integer then $m!$ denotes the product $1 \cdot 2 \cdots m$ of all the integers up to and including $m$. $0!$, the empty product, is defined to be 1.

**SOLUTION :** The characteristic polynomial of the coefficient matrix **A** is

$$\Delta(t) = \det \begin{bmatrix} 2-t & 2 & -1 \\ -1 & -1-t & 1 \\ -1 & -2 & 2-t \end{bmatrix} = \cdots = -(t-1)^3.$$

There is therefore just one eigenvalue, 1, of multiplicity 3. The Jordan normal form of **A** is (see Example 2 in Section 17.3)

$$\mathbf{J} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

and a corresponding Jordan basis for $\mathbb{R}^3$ is

$$\mathbf{F}_1 = (1, 0, 0), \qquad \mathbf{F}_2 = (1, -1, 1), \qquad \mathbf{F}_3 = (0, 1, 2).$$

So

$$\mathbf{A} = \mathbf{Q} \cdot \mathbf{J} \cdot \mathbf{Q}^{-1},$$

where

$$\mathbf{Q} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

If we set

$$\mathbf{W} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \mathbf{Q}^{-1} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},$$

then we obtain for **W** the differential system

$$\frac{d}{dt}\mathbf{W} = \mathbf{J}\mathbf{W} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{W} = \begin{bmatrix} \mathbf{J}_2(1) & 0 \\ 0 & 1 \end{bmatrix} \mathbf{W}$$

and working block by block, we find that

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} c_1 e^t \\ (c_1 t + c_2)e^t \end{bmatrix}$$

and

$$w_3 = c_3 e^t,$$

where $c_1$, $c_2$, $c_3$ are arbitrary constants. Thus

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \mathbf{Q} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \cdots = \begin{bmatrix} c_1 e^t + c_1 t e^t + c_2 e^t \\ -c_1 t e^t - c_2 e^t + c_3 e^t \\ c_1 t e^t + c_2 e^t + 2c_3 e^t \end{bmatrix}.$$

As a last example we consider:

**EXAMPLE 4:** Solve the differential equation

$$\frac{d^3y}{dt^3} - \frac{d^2y}{dt^2} - \frac{dy}{dt} + y = 0.$$

**SOLUTION:** The coefficient matrix of the associated linear system is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}.$$

The characteristic polynomial is (by Proposition 18.2.2)

$$\begin{aligned} \Delta(t) &= (-1)^3(t^3 - t^2 - t + 1) \\ &= -(t-1)^2(t+1), \end{aligned}$$

so **A** has two distinct eigenvalues, $d_1 = 1$ and $d_2 = -1$, with respective multiplicities 2 and 1. Following the procedure of Chapter 17 the Jordan form of **A** works out to be

$$\mathbf{J} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

and a corresponding Jordan basis for $\mathbb{R}^3$ is

$$\mathbf{F}_1 = (0, 1, 2), \qquad \mathbf{F}_2 = (1, 1, 1), \qquad \mathbf{F}_3 = (1, -1, 1).$$

So if we set

$$\mathbf{Q} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix},$$

then $\mathbf{A} = \mathbf{Q} \cdot \mathbf{J} \cdot \mathbf{Q}^{-1}$. Set

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \mathbf{Q}^{-1} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},$$

where as usual

$$\begin{aligned} y_1 &= y, \\ y_2 &= \frac{dy}{dt}, \\ y_3 &= \frac{d^2y}{dt^2}. \end{aligned}$$

Our original differential equation is equivalent to the system

$$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \mathbf{A} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},$$

which in turn is equivalent to

$$\frac{d}{dt}\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \mathbf{J}\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}.$$

By Theorem 18.3.1 the general solution of the linear system

$$\frac{d}{dt}\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

is given by (work block by block)

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} c_1e^t \\ (c_1t + c_2)e^t \\ c_3e^{-t} \end{bmatrix}.$$

From the definition of $(w_1, w_2, w_3)$ we thus find that

$$\begin{aligned}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \mathbf{P}\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_1e^t \\ (c_1t + c_2)e^t \\ c_3e^{-t} \end{bmatrix} \\ &= \begin{bmatrix} (c_1t + c_2)e^t + c_3e^{-t} \\ c_1e^t + (c_1t + c_2)e^t - c_3e^{-t} \\ 2c_1e^t + (c_1t + c_2)e^t + c_3e^{-t} \end{bmatrix},\end{aligned}$$

so we conclude that

$$y = y_1 = (c_1t + c_2)e^t + c_3e^{-t}$$

is the general solution to the original equation.

## 18.4 Exercises

**1.** Among all the solutions of the differential system

$$\frac{dy_1}{dt} = -y_2$$
$$\frac{dy_2}{dt} = -y_1$$

find a solution that satisfies the ***initial conditions***

$$y_1(0) = 2,$$
$$y_2(0) = 0.$$

(**HINT:** First find all solutions. Then determine for which values of the constants the initial conditions are satisfied.)

**2.** Solve the differential system

$$\frac{dy_1}{dt} = y_2 + y_3,$$
$$\frac{dy_2}{dt} = y_1 + y_3,$$
$$\frac{dy_3}{dt} = y_1 + y_2.$$

Find a solution satisfying

$$y_1(0) = 1,$$
$$y_2(0) = 0,$$
$$y_3(0) = -1.$$

**3.** Solve each of the following differential equations:

(a) $$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0.$$

(b) $$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 0.$$

(c) $$\frac{d^2y}{dt^2} - y = 0.$$

(d) $$\frac{d^2y}{dt^2} + y = 0.$$

**4.** Solve each of the following linear systems

(a) $$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

(b) $$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

(c) $$\frac{d}{dt}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

**5.** Solve each of the following differential equations:

(a) $$\frac{d^3y}{dt^3} + \frac{d^2y}{dt^2} - \frac{dy}{dt} - y = 0.$$

(b) $$\frac{d^3y}{dt^3} + 3\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + y = 0.$$

(c) $$\frac{d^3y}{dt^3} - 6\frac{d^2y}{dt^2} - 7\frac{dy}{dt} - 6y = 0.$$

In each case determine a solution satisfying the initial conditions

$$y(0) = 1,$$
$$\frac{dy}{dt} = 0,$$
$$\frac{d^2y}{dt^2} = 1.$$

# Chapter 19
# The Similarity Problem

The last two chapters of this book have made extensive use of the Jordan normal form. In working through the examples and exercises it must have become apparent that there is no simple method for finding the Jordan form. By contrast, finding the diagonal form of a symmetric matrix reduces to factoring the characteristic polynomial.[1] Why is this? Why is the Jordan form so much more difficult to find than the diagonal or row echelon form? The reason is that linear algebra alone is not sufficient to solve this problem: One needs other methods and techniques, in particular not just algebra. It is the purpose of this chapter to make this more precise, and to provide a proof in the case of $2 \times 2$ matrices that no such simple method for finding the Jordan form can exist.

## 19.1 The Fundamental Problem of Linear Algebra

The Jordan normal form is essential to many of the most important theorems and applications of linear algebra, and it is natural to adopt the viewpoint that the fundamental problem of linear algebra is to compute the Jordan normal form of a linear transformation $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ in a finite-dimensional vector space. What should a solution to this problem look like?

Starting with $\mathsf{T}$, we would choose some arbitrary basis for $\mathcal{V}$ and represent $\mathsf{T}$ by a square matrix $\mathbf{A}$. The hope would be to be able to compute the Jordan form of $\mathsf{T}$ from the coefficients of $\mathbf{A}$. Denote the Jordan form

[1] Be warned: there is no *formula* for the roots of polynomials of degree greater than four.

of **T** by $\mathbf{J} = (f_{i,j})$. What we would like is a formula to express $f_{i,j}$ in terms of the entries $a_{r,s}$, $r, s = 1, \ldots, n$, of the matrix **A**. In other words, $f_{i,j}$ should be a function of $n^2$ variables, $n = \dim(\mathcal{V})$, which when evaluated on the entries of **A** gives the entry in the $i$-th row and $j$-th column of **J**, the Jordan normal form of **T**.

Suppose that such a formula were to exist. What could we say about the functions $f_{i,j}$ as functions of the entries of **A**? Remember that we started with an ***arbitrary*** matrix representative of **T**, so whatever these formulas looked like, if we chose another matrix representative **B** for **T**, then the function $f_{i,j}$ evaluated on the entries of **B** would have to give the same result as when evaluated on the entries of **A**, because the Jordan normal form of **T** is unique apart from the order of the blocks.

The relation between the two matrices, **A**, **B** that represent one and the same linear transformation, here **T**, with respect to two different bases was described in Theorem 11.3.1; namely, there is an invertible matrix $\mathbf{P} \in \mathrm{Mat}_{n,n}$ such that

$$\mathbf{B} = \mathbf{P} \cdot \mathbf{A} \cdot \mathbf{P}^{-1}. \tag{$*$}$$

Therefore, if two matrices $\mathbf{A}, \mathbf{B} \in \mathrm{Mat}_{n,n}$ are related as in the previous equation then the functions $f_{i,j}$ must take the same values on the entries of **A** and **B**. Two functions that obviously fulfill this condition are the determinant and the trace: Both of them are, in fact, polynomial functions of the $n^2$ matrix entries, and if two matrices are related by equation $(*)$, then the determinant and the trace take the same values on both these matrices.

Given these preliminaries, we can formulate the fundamental problem of linear algebra as follows:

> **PROBLEM:** For each positive integer $n$ find $n^2$ functions
>
> $$f_{i,j} : \mathrm{Mat}_{n,n} \longrightarrow \mathbb{C}, \qquad i, j = 1, \ldots, n,$$

such that for any matrix $\mathbf{A} \in \mathrm{Mat}_{n,n}$ the matrix $\big(f_{i,j}(\mathbf{A})\big)$ is the Jordan canonical form of **A**.

In the preceding discussion we saw that if such functions $f_{i,j}$ were to exist, then $f_{i,j}(a_{r,s}) = f(b_{r,s})$ whenever the matrices $(a_{r,s})$ and $(b_{r,s})$ are related by the equation $(*)$. Let us call two matrices **similar** if they are related by equation $(*)$.

## 19.2 A Bit of Invariant Theory

In order to formulate a positive result that implies a negative solution to the main problem of linear algebra we introduce some notations. The vector space $\mathrm{Mat}_{n,n}$ of $n \times n$ matrices is of dimension $n^2$, so an element of $\mathrm{Mat}_{n,n}$ has this many coordinates whenever we choose a basis for $\mathrm{Mat}_{n,n}$. We will say that a function

$$f : \mathrm{Mat}_{n,n} \longrightarrow \mathbb{C}$$

is a **polynomial function** if it can be written as a polynomial in the $n^2$ entries of the matrices in $\mathrm{Mat}_{n,n}$. We denote by $\mathbb{C}[\mathrm{Mat}_{n,n}]$ the set of all polynomial functions from $\mathrm{Mat}_{n,n}$ to $\mathbb{C}$. Polynomial functions can be added and multiplied amongst themselves as well as multiplied by complex numbers by using the pointwise operations

$$(f + h)(\mathbf{A}) = f(\mathbf{A}) + h(\mathbf{A}),$$

$$(f \cdot h)(\mathbf{A}) = f(\mathbf{A}) \cdot h(\mathbf{A}),$$

$$(a \cdot f)(\mathbf{A}) = a \cdot f(\mathbf{A}),$$

where $\mathbf{A} \in \mathrm{Mat}_{n,n}$, $f, h \in \mathbb{C}[\mathrm{Mat}_{n,n}]$, and $a \in \mathbb{C}$. Therefore, the polynomial functions form an algebra.

If a polynomial function $f \in \mathbb{C}[\mathrm{Mat}_{n,n}]$ takes the same value on any two polynomials that satisfy the condition $(*)$ of the previous section, i.e., if

$$f(\mathbf{A}) = f(\mathbf{P} \cdot \mathbf{A} \cdot \mathbf{P}^{-1})$$

whenever $\mathbf{A}, \mathbf{P} \in \mathrm{Mat}_{n,n}$ and $\mathbf{P}$ is invertible, then we say that $f$ is **invariant**. These polynomials also form an algebra, in that the sum and product of an invariant polynomials, as well as the product of an invariant polynomial by a complex number are again invariant. The set of invariant polynomials is therefore a subalgebra of $\mathbb{C}[\mathrm{Mat}_{n,n}]$, which we denote by $\mathbb{C}[\mathrm{Mat}_{n,n}]^{\mathrm{GL}(n,\mathbb{C})}$. Although this notation may seem overly complicated, it is logical and standard: $\mathrm{GL}(n, \mathbb{C}) \subset \mathrm{Mat}_{n,n}$ is one standard notation for the subset of invertible matrices, and the $^{\mathrm{GL}(n,\mathbb{C})}$ indicates that we are interested in polynomial functions that take the same value on two matrices that are related by the equation $(*)$ of the preceding section.

To show that the problem formulated at the end of the last section must be answered in the negative, we will compute all the invariant polynomial functions and show that there is no way they can distinguish Jordan normal forms. Specifically, formulated in a positive fashion, if $f \in \mathbb{C}[\mathrm{Mat}_{n,n}]^{\mathrm{GL}(n,\mathbb{C})}$ is an invariant polynomial function and $\mathbf{A}, \mathbf{B}$ are two matrices with the same eigenvalues (counting multiplicity), then $f(\mathbf{A}) = f(\mathbf{B})$. We will present a proof for the special case $n = 2$. To do so we prove:

**PROPOSITION 19.2.1:** *If $f \in \mathbb{C}[\mathrm{Mat}_2(\mathbb{C})]^{\mathrm{GL}(2,\mathbb{C})}$ is an invariant polynomial function, then $f$ can be written as a polynomial in the two invariant polynomials* det *and* tr.

**PROOF:** Every matrix $\mathsf{T} \in \mathrm{Mat}_2(\mathbb{C})$ has a Jordan normal form, which is one of the two standard forms

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, \qquad \begin{bmatrix} c & 0 \\ 1 & c \end{bmatrix},$$

for if a matrix has distinct eigenvalues, then it is diagonalizable, and a necessary condition that $\mathsf{T}$ ***not*** be diagonalizable is that it have a double eigenvalue. This happens if and only if the discriminant of its characteristic polynomial vanishes; i.e., if

$$\Delta_{\mathsf{T}}(t) = t^2 + c_1 t + c_2$$

is the characteristic polynomial of $\mathsf{T}$, then

$$c_1^2 - 4c_2 = 0.$$

If a matrix has a characteristic polynomial with vanishing discriminant, then a very small change in its entries will yield a matrix with a nonzero discriminant. Therefore, if $\Delta_{\mathsf{T}}$ has vanishing discriminant, then every small open neighborhood of $\mathsf{T}$ in $\mathrm{Mat}_{n,n}$ contains matrices with characteristic polynomial with nonvanishing discriminant. Hence the set of matrices with characteristic polynomial with nonvanishing discriminant is dense in $\mathrm{Mat}_{2,2}$.

If $\mathsf{A} \in \mathrm{Mat}_{2,2}$ is any matrix, then among all the matrices related to $\mathsf{A}$ by the equation $(*)$ of the last section, there is exactly one in Jordan normal form. This means if $f : \mathrm{Mat}_{2,2} \longrightarrow \mathbb{C}$ is an invariant polynomial then $f$ is determined as soon as we know the value of $f$ on all the Jordan normal forms. Furthermore, since the matrices with distinct eigenvalues are dense in the set of matrices it suffices to know the value of $f$ on matrices of the form

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}.$$

So $f$ must be a polynomial function of the two eigenvalues.

Among all the invertible matrices $GL(2, \mathbb{C})$ there is the involution

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

which when applied to the diagonal matrix

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

by forming the similar matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & 0 \\ 0 & a \end{bmatrix}$$

exchanges the diagonal entries. If $f$ is an invariant polynomial, it must have the same value on these two matrices, so $f(a, b) = f(b, a)$, i.e. $f$ is a symmetric function of the two variables and hence is a polynomial in the two elementary symmetric functions[2]

$$a + b = \mathrm{tr}(\mathbf{T}), \qquad a \cdot b = \det(\mathbf{T})$$

as required. □

Notice that the two matrices

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

have the same eigenvalues but are not similar. Since both matrices have trace 2 and determinant 1, it follows from Proposition 19.2.1 that ***any*** invariant polynomial $f \in \mathbb{C}[\mathrm{Mat}_{2,2}]^{GL(2,\mathbb{C})}$ must take exactly the same value on these two matrices. Hence there can be no possible way to express the Jordan form of these matrices in the form

$$\begin{bmatrix} f_{1,1} & f_{2,1} \\ f_{2,1} & f_{2,2} \end{bmatrix},$$

where $f_{i,j}$ are invariant polynomial functions of the entries.

COROLLARY 19.2.2 : *The fundamental problem of linear algebra cannot be solved by linear algebra alone. In other words, it is not possible to distinguish similar matrices by polynomial functions of the matrix entries.* □

## 19.3 Exercises

**1.** Let $p(\mathbf{A})$ be the function that assigns to a square matrix $\mathbf{A}$ its characteristic polynomial, i.e.,

$$p(\mathbf{A}) = (-1)^n t^n + (-1)^{n-1} c_1 t^{n-1} + \cdots + (-1)^1 c_{n-1} t + c_n.$$

The coefficients of this polynomial are also functions of $\mathbf{A}$, and hence of the $n^2$ entries of $\mathbf{A}$.

(1) Show that $c_1$ is the trace of $\mathbf{A}$.
(2) Show that $c_n$ is the determinant of $\mathbf{A}$.
(3) Show that $c_i$ is a polynomial function of the entries of $\mathbf{A}$ for $i = 1, \ldots, n$.

Show that the polynomial functions

$$c_i : \mathrm{Mat}_{n,n} \longrightarrow \mathbb{C} \qquad i = 1, \ldots, n$$

[2] You may need to consult an algebra book for a proof of this important fact.

are ***invariant*** in the sense of Section 19.2.

**2.** Let $f_i : \mathrm{Mat}_{n,n} \longrightarrow \mathbb{C}$ be the function defined by

$$f_i(\mathbf{A}) = \mathrm{tr}\left(\mathbf{A}^i\right).$$

Show that $f_i$ is an invariant polynomial in the sense of section Section 19.2.

**3.** Let $\mathbf{A}$ be a $3 \times 3$ matrix. Find a formula for the coefficients of the characteristic polynomial of $\mathbf{A}$ in terms of the eigenvalues of $\mathbf{A}$.

**4.** Let $f(x, y)$ be a polynomial in the two variables $x$ and $y$. Suppose that $f(x, y) = f(y, x)$ for all $x$ and $y$. Show that $f(x, y)$ can be written as a polynomial in the two **elementary symmetric polynomials**

$$e_1 = x + y,$$

$$e_2 = xy.$$

**5.** Find all the polynomial functions $f(x, y)$ of two real variables $x$ and $y$ that satisfy the identities

$$f\left(\tfrac{x+(q-1)y}{\sqrt{q}}, \tfrac{x-y}{\sqrt{q}}\right) = f(x, y)$$
$$f(-x, -y) = f(x, y)$$

where $q$ is a fixed positive real number. Note that we are demanding that the polynomial be *invariant* under the linear change of variables given by the matrices

$$\mathbf{A} = \frac{1}{\sqrt{q}}\begin{bmatrix} 1 & q-1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

# Appendix A Multilinear Algebra and Determinants

In this appendix we collect some ideas not central to our development of the theory of a single linear transformation and give the proofs of the basic properties of determinants used beginning with Chapter 14.

## A.1 Multilinear Forms

Let $n$ be a positive integer, and let $\mathcal{V}_1, \ldots, \mathcal{V}_n, \mathcal{W}$ be vector spaces. A function

$$f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W}$$

is called a **multilinear form** if for each integer $i$, $1 \leq i \leq n$, and each fixed $(n-1)$-tuple $(v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n)$ the function

$$f_i : \mathcal{V}_i \longrightarrow \mathcal{W}$$

defined by

$$f_i(v) = f(v_1, \ldots, v_{i-1}, v, v_{v+1}, \ldots, v_n)$$

is a linear transformation. For $n = 1$ a multilinear form is simply a linear transformation. For $n = 2$ we speak of a bilinear form. Often we just say **form** for a multilinear form.

**EXAMPLE 1:** The determinant of $2 \times 2$ matrices defines a bilinear form

$$\mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}$$

by the formula

$$\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} c \\ d \end{bmatrix} \longmapsto \det \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

**EXAMPLE 2:** Let $\mathcal{V}_1, \ldots, \mathcal{V}_n$ be vector spaces then the map

$$f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{V}_1 \oplus \cdots \oplus \mathcal{V}_n$$

defined by

$$f(v_1, \ldots, v_n) = (v_1, v_2, \ldots, v_n)$$

is a multilinear form.

**EXAMPLE 3:** Let $a, b$ be positive integers. The multiplication of polynomials defines a bilinear form

$$\mu : \mathcal{P}_a(\mathbb{R}) \times \mathcal{P}_b(\mathbb{R}) \longrightarrow \mathcal{P}_{a+b}(\mathbb{R}).$$

**EXAMPLE 4:** Let $a, b$ be positive integers and define

$$f : \mathbb{R}^{a+1} \times \mathbb{R}^{b+1} \longrightarrow \mathbb{R}^{a+b+1}$$

by (this formula is often referred to as the **Cauchy product**: note that the indexing starts with a zero)

$$f(x, y) = (z_0, \ldots, z_{a+b}),$$

where

$$x = (x_0, \ldots, x_a), \qquad y = (y_0, \ldots, y_b),$$

and

$$z_c = \sum_{i+j=c} x_i y_j, \qquad c = 0, \ldots, a+b.$$

The function $f$ is a multilinear form.

**DEFINITION:** *Two multilinear forms*

$$f' : \mathcal{V}_1' \times \cdots \times \mathcal{V}_n' \longrightarrow \mathcal{W}',$$
$$f'' : \mathcal{V}_1'' \times \cdots \times \mathcal{V}_n'' \longrightarrow \mathcal{W}'',$$

*are said to be* **equivalent** *if there exist vector-space isomorphisms*

$$\varphi_i : \mathcal{V}_i' \longrightarrow \mathcal{V}_{\sigma(i)}'', \qquad i = 1, \ldots, n,$$
$$\psi : \mathcal{W}' \longrightarrow \mathcal{W}'',$$

*where* $(\sigma(1), \ldots, \sigma(n))$ *is a rearrangement (i.e.,* **permutation**, *see Section* 8.1 *Example* 3*) of the integers* $(1, \ldots, n)$ *such that*

$$\psi f'(v_1', \ldots, v_n') = f''(\varphi_1(v_1'), \ldots, \varphi_n(v_n')) \qquad \forall\ v_i' \in \mathcal{V}_i, \quad i = 1, \ldots, n,$$

*where*

$$v_j'' = \varphi_j(v_{\tau(j)}'), \qquad j = 1, \ldots, n,$$

*and* $\tau$ *is the* **inverse rearrangement** *of* $\sigma$*; that is,*

$$\tau(\sigma(j)) = j, \qquad j = 1, \ldots, n.$$

**THEOREM A.1.1** (Linear Extension Construction for Forms): *Let $n$ be a positive integer and let $\mathcal{V}_1, \ldots, \mathcal{V}_n, \mathcal{W}$ be vector spaces. Assume that $\dim(\mathcal{V}_i) = d_i$ is finite, and let $\mathbf{E}_{i,1}, \ldots, \mathbf{E}_{i,d_i}$ be a basis for $\mathcal{V}_i$, for $i = 1, \ldots, n$. For each index $(j_1, \ldots, j_n)$ with $1 \leq j_s \leq d_s$ let $w_{(j_1,\ldots,j_n)} \in \mathcal{W}$. Then there is one and only one form*

$$f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W}$$

*such that*

$$f(\mathbf{E}_{1,j_1}, \ldots, \mathbf{E}_{n,j_n}) = w_{(j_1,\ldots,j_n)}.$$

**PROOF:** This is analogous to Theorem 8.5.1, namely, the linear extension construction. To construct $f$, we suppose that $v_i \in \mathcal{V}_i$, $i = 1, \ldots, n$. We can then write

$$v_i = a_{i,1}\mathbf{E}_{i,1} + \cdots + a_{i,d_i}\mathbf{E}_{i,d_i}$$

for unique scalars $a_{i,1}, \ldots, a_{i,d_i}$. We set (this definition is forced by the desired multilinearity)

$$f(v_1, \ldots, v_n) = \sum_{\substack{(j_1,\ldots,j_n) \\ 1 \leq j_i \leq d_i}} a_{(1,j_1)} a_{(2,j_2)} \cdots a_{(n,j_n)} w_{(j_1,\ldots,j_n)}.$$

A simple check of the definition shows that $f$ is indeed a multilinear form. Furthermore, if

$$g : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W}$$

is any multilinear form such that

$$g(\mathbf{E}_{1,j_1}, \ldots, \mathbf{E}_{n,j_n}) = w_{(j_1,\ldots,j_n)}, \qquad 1 \leq j_i \leq d_i,$$

then by multilinearity

$$g(v_1, \ldots, v_n) = \sum a_{(1,j_1)} \cdots a_{(n,j_n)} w_{(j_1,\ldots,j_n)};$$

that is, $f = g$. $\square$

If $a$ is a scalar and

$$f, g : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W}$$

are multilinear forms, then so are

$$af : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W},$$

where

$$(af)(v_1, \ldots, v_n) = af(v_1, \ldots, v_n),$$

and

$$f + g : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W},$$

where

$$(f + g)(v_1, \ldots, v_n) = f(v_1, \ldots, v_n) + g(v_1, \ldots, v_n)\,.$$

Here is a partial verification of the first assertion for $n = 2$.

$$\begin{aligned}(f + g)(x' + x'', y) &= f(x' + x'', y) + g(x' + x'', y)\\ &= f(x', y) + f(x'', y) + g(x', y) + g(x'', y)\\ &= f(x', y) + g(x', y) + f(x'', y) + g(x'', y)\\ &= (f + g)(x', y) + (f + g)(x'', y).\end{aligned}$$

The first equality follows from the definitions, the second by bilinearity, and the remaining equalities by rearrangement. The verification in general is completely analogous. We record this as our first result.

**Proposition A.1.2:** *The set of all multilinear forms*

$$f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W}$$

*is a vector space with respect to the addition of forms and multiplication of a form by a scalar as defined above.* □

**Proposition A.1.3:** *The dimension of the vector space of multilinear forms*

$$f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W}$$

*is $d \cdot d_1 \cdots d_n$, where $d_i = \dim \mathcal{V}_i$, $i = 1, \ldots, n$ and $d = \dim \mathcal{W}$.*

**Proof:** This follows from Theorem **A**.1.1. □

The study of multilinear forms (multilinear algebra) leads to many interesting, and in some cases totally unsolved problems. To illustrate this we need to introduce one new concept.

**Definition:** *A form*

$$f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W}$$

*is called* **nonsingular** *if whenever*

$$f(v_1, v_2, \ldots, v_n) = 0,$$

*at least one of the vectors $v_1, \ldots, v_n$ is the zero vector.*

**EXAMPLE 5:** The form $\mu : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$\mu : (x, y) = (x_1 y_1 - x_2 y_2, x_1 y_2 + x_2 y_1)$$

is a nonsingular bilinear form.

It is possible to construct nonsingular bilinear forms

$$\mathbb{R}^4 \times \mathbb{R}^4 \longrightarrow \mathbb{R}^4,$$
$$\mathbb{R}^8 \times \mathbb{R}^8 \longrightarrow \mathbb{R}^8.$$

Just as $\mu$ corresponds to multiplication of complex numbers, these forms derive from multiplication of quaternions and Cayley numbers. These facts were known in the nineteenth century. That the only possible nonsingular forms

$$\mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

are, in fact, the forms where $n = 1, 2, 4$, or $8$ was first proven in 1960 by J.F. Adams.

**OPEN PROBLEM:** Let $n, k$ be positve integers. Find the largest integer $b(k, n)$ such that there exists a nonsingular bilinear form

$$\mathbb{R}^n \times \mathbb{R}^{b(k,n)} \longrightarrow \mathbb{R}^k.$$

## A.2 Determinants

Our purpose in introducing multilinear forms is to define and establish the basic properties of the determinant of $n \times n$ matrices. We turn to this next. We will be concerned with multilinear forms

$$f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \longrightarrow \mathcal{W},$$

where $\mathcal{V}_1 = \cdots = \mathcal{V}_n$ that is, multilinear forms of the type

$$f : \mathcal{V} \times \cdots \times \mathcal{V} \longrightarrow \mathcal{W}.$$

**DEFINITION:** *A form*

$$f : \mathcal{V} \times \cdots \times \mathcal{V} \longrightarrow \mathcal{W}$$

*is called* **sign-symmetric** *if for every pair of integers $i, j$, with $1 \le i \ne j \le n$, and $v_1, \ldots, v_n \in \mathcal{V}$,*

$$f(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_n) = -f(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_n).$$

A sign-symmetric form is also referred to as **alternating**. Note that whenever $v_i = v_j$ for some $1 \le i \neq j \le n$, then interchanging $v_i$ and $v_j$ does not change the value of $f(v_1, \ldots, v_n)$. On the other hand, if $f$ is sign-symmetric, the sign of the value must change. This is only possible if

$$f(v_1, \ldots, v_n) = 0$$

whenever there is a pair of integers $i, j$ such that $1 \le i \neq j \le n$ and $v_i = v_j$.

**EXAMPLE 1:** Let

$$\Delta : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}$$

be defined by

$$\Delta\big((a, b), (c, d)\big) = ad - bc.$$

Then $\Delta$ is bilinear and sign-symmetric. The verification, while tedious, is straightforward and left to the reader.

**THEOREM A.2.1:** *Let $\mathcal{V}$ be an $n$-dimensional vector space. Then the sign-symmetric forms (Note: $n = \dim \mathcal{V}$)*

$$f : \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\longleftarrow n \longrightarrow} \longrightarrow \mathbb{R}$$

*form a one-dimensional subspace in the space of all forms.*

**PROOF:** Let $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ be a basis for $\mathcal{V}$. By the linear extension construction for forms,

$$f : \mathcal{V} \times \cdots \times \mathcal{V} \longrightarrow \mathbb{R}$$

is completely determined when we know the values $f(\mathbf{E}_{i_1}, \ldots, \mathbf{E}_{i_n}) \in \mathbb{R}$, $1 \le i_j \le n$. If the form is alternating then

$$f(\mathbf{E}_{i_1}, \ldots, \mathbf{E}_{i_n}) = 0$$

whenever $\mathbf{E}_{i_a} = \mathbf{E}_{i_b}$ for a pair of indices $i_a \neq i_b$. This means that an alternating form

$$f : \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\leftarrow n = \dim(\mathcal{V}) \rightarrow} \longrightarrow \mathbb{R}$$

is determined by the values

$$f(\mathbf{E}_{i_1}, \ldots, \mathbf{E}_{i_n}),$$

where the indices are all distinct. If the form is sign-symmetric and $(\mathbf{E}_{i_1}, \ldots, \mathbf{E}_{i_n})$ has distinct indices, then

$$f(\mathbf{E}_{i_1}, \ldots, \mathbf{E}_{i_n}) = (-1)^s f(\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_n),$$

where $s$ is the number of pairs of indices that have to be switched to rearrange the $n$-tuple $(i_1, \ldots, i_n)$ to give $(1, \ldots, n)$. Thus a sign-symmetric (or alternating) form

$$f: \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\leftarrow n = \dim(\mathcal{V}) \rightarrow} \longrightarrow \mathbb{R}$$

is completely determined by sign-symmetry and

$$f(\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_n) \in \mathbb{R}.$$

Let

$$g: \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\leftarrow n = \dim(\mathcal{V}) \rightarrow} \longrightarrow \mathbb{R}$$

be the unique sign-symmetric form defined by

$$g(\mathbf{E}_1, \ldots, \mathbf{E}_n) = 1.$$

Let

$$f : \mathcal{V} \times \cdots \times \mathcal{V} \longrightarrow \mathbb{R}, \quad n = \dim \mathcal{V},$$

be a sign-symmetric form. Then

$$h := f - f(\mathbf{E}_1, \ldots, \mathbf{E}_n) \cdot g, \qquad h : \mathcal{V} \times \cdots \times \mathcal{V} \longrightarrow \mathbb{R},$$

is also sign-symmetric and alternating. Moreover

$$h(\mathbf{E}_1, \ldots, \mathbf{E}_n) = 0.$$

Therefore, $h$ is identically 0., Thus $f = f(\mathbf{E}_1, \ldots, \mathbf{E}_n) \cdot g$ and $g$ spans the space of sign-symmetric forms

$$f : \mathcal{V} \times \cdots \times \mathcal{V} \longrightarrow \mathbb{R}, \quad n = \dim \mathcal{V},$$

so this space is one-dimensional. $\square$

Suppose $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ is a linear transformation and

$$g: \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\leftarrow n = \dim(\mathcal{V}) \rightarrow} \longrightarrow \mathbb{R}$$

is a nonzero sign-symmetric form. Then

$$g_{\mathsf{T}} : \mathcal{V} \times \cdots \times \mathcal{V} \longrightarrow \mathbb{R}$$

defined by

$$g_{\mathsf{T}}(v_1, \ldots, v_n) = g(\mathsf{T}v_1, \ldots, \mathsf{T}v_n) \in \mathbb{R}$$

is also a sign-symmetric form. Since the space of such forms is one-dimensional, and $g \neq 0$, it follows that

$$g_{\mathsf{T}} = d_g(\mathsf{T}) \cdot g$$

for a unique scalar $d_g(\mathsf{T}) \in \mathbb{R}$. Thus we are led to:

**DEFINITION:** *Let* $\mathsf{M} \in \mathrm{Mat}_{n,n}$ *be an* $n \times n$ *matrix. Let*

$$g : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \longrightarrow \mathbb{R}$$

*be the sign-symmetric form defined by*

$$g(\mathbf{E}_1, \ldots, \mathbf{E}_n) = 1,$$

*where* $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ *is the standard basis for* $\mathbb{R}^n$. *The form g is called the* **fundamental sign-symmetric** *or* **alternating** *form. Let*

$$\mathsf{M} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

*be the linear transformation represented by the matrix* $\mathsf{M}$ *with respect to the standard basis. The* **determinant** *of* $\mathsf{M}$ *is the unique real number* $\det(\mathsf{M})$ *such that* $g_{\mathsf{M}} = \det(\mathsf{M}) \cdot g$.

**THEOREM A.2.2:** *The function*

$$\det : \mathrm{Mat}_{n,n} \longrightarrow \mathbb{R}$$

*has the following properties:*

(i) *If* $\mathsf{A}$ *has a column of zeros, then* $\det(\mathsf{A}) = 0$.
(ii) *If* $\mathsf{A}$ *has two columns equal, then* $\det(\mathsf{A}) = 0$.
(iii) *If* $\mathsf{B}$ *is obtained from* $\mathsf{A}$ *by interchanging two columns, then* $\det(\mathsf{B}) = -\det(\mathsf{A})$.
(iv) *If* $\mathsf{B}$ *is obtained from* $\mathsf{A}$ *by adding a multiple of one column to a different column, then* $\det(\mathsf{B}) = \det(\mathsf{A})$.
(v) *If* $\mathsf{B}$ *is obtained by multiplying a column of* $\mathsf{A}$ *by a number* $k$, *then* $\det(\mathsf{B}) = k\det(\mathsf{A})$.
(vi) *If* $\mathsf{A}$ *is upper triangular, then* $\det(\mathsf{A})$ *is the product of the diagonal entries.*
(vii) $\det(\mathsf{A}^{\mathrm{tr}}) = \det(\mathsf{A})$. *Thus everything said about columns in (i)–(vi) holds also for rows.*
(viii) $\det(\mathsf{A} \cdot \mathsf{B}) = \det(\mathsf{A})\det(\mathsf{B})$.
(ix) $\mathsf{A}$ *is invertible if and only if* $\det(\mathsf{A}) \neq 0$.

Before taking up the proof proper we require some preliminary maneuvering. The proof of Theorem **A**.2.1 shows that a sign-symmetric form

$$f : \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\leftarrow n = \dim(\mathcal{V}) \rightarrow} \longrightarrow \mathbb{R}$$

is completely determined by the value

$$f(\mathbf{E}_1, \ldots, \mathbf{E}_n) \in \mathbb{R},$$

where $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ is a basis for $\mathcal{V}$. In fact, if

$$g : \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\leftarrow n = \dim(\mathcal{V}) \rightarrow} \longrightarrow \mathbb{R}$$

is the unique sign-symmetric form such that

$$g(\mathbf{E}_1, \ldots, \mathbf{E}_n) = 1,$$

then $f = f(\mathbf{E}_1, \ldots, \mathbf{E}_n) \cdot g$ as forms. Thus we have shown:

**PROPOSITION A.2.3:** *Two sign-symmetric forms*

$$f', f'' : \underbrace{\mathcal{V} \times \cdots \times \mathcal{V}}_{\leftarrow n = \dim(\mathcal{V}) \rightarrow} \longrightarrow \mathbb{R}$$

*are equal if and only if for some basis* $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ *of* $\mathcal{V}$

$$f'(\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_n) = f''(\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_n). \quad \square$$

Let $\mathsf{A} \in \mathrm{Mat}_{n,n}$ and write[1]

$$\mathsf{A} : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

for the linear transformation corresponding to the matrix $\mathsf{A}$ with respect to the standard basis $\{\mathbf{E}_1, \ldots, \mathbf{E}_n\}$ for $\mathbb{R}^n$. Let

$$\mathsf{A} = [\mathsf{A}_1, \ldots, \mathsf{A}_n],$$

where

$$\mathsf{A}_{(i)} = \begin{bmatrix} a_{i,1} \\ \vdots \\ a_{i,n} \end{bmatrix}, \qquad i = 1, \ldots, n,$$

is the $i$-th column of $\mathsf{A}$. Then

$$\mathsf{A}(\mathbf{E}_i) = \mathsf{A}^{\mathrm{tr}}_{(i)}, \qquad i = 1, \ldots, n,$$

where $\mathsf{A}^{\mathrm{tr}}_{(i)}$ denotes the transpose of $\mathsf{A}_{(i)}$.

Let

$$g : \underbrace{\mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{\longleftarrow n \longrightarrow} \longrightarrow \mathbb{R}$$

be the unique sign-symmetric form defined by

$$g(\mathbf{E}_1, \ldots, \mathbf{E}_n) = 1.$$

To compute the determinant of $\mathsf{A}$ we introduce the form

$$g_{\mathsf{A}} : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \longrightarrow \mathbb{R}$$

---

[1] The use of $\mathsf{A}$ for both the linear transformation and its matrix with respect to the standard basis should cause no confusion, as only the standard basis of $\mathbb{R}^n$ appears in the discussion.

defined by

$$g_{\mathbf{A}}(v_1, \ldots, v_n) = g(\mathbf{A}v_1, \ldots, \mathbf{A}v_n).$$

Then $\det(\mathbf{A})$ is the unique number such that

$$g_{\mathbf{A}} = (\det(\mathbf{A})) \cdot g : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \longrightarrow \mathbb{R}.$$

The remarks preceding Proposition A.2.3 thus show that

$$\begin{aligned} g_{\mathbf{A}} &= g_{\mathbf{A}}(\mathbf{E}_1, \ldots, \mathbf{E}_n) \cdot g = g(\mathbf{A}(\mathbf{E}_1, \ldots, \mathbf{A}(\mathbf{E}_n)) \\ &= g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}), \end{aligned}$$

and therefore we conclude:

**LEMMA A.2.4:** *With the preceding notations we have*

$$\det(\mathbf{A}) = g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)})$$

*for* $\mathbf{A} \in \mathrm{Mat}_{n,n}$. □

**PROOF OF THEOREM A.2.2:** In permuted order here are the proofs.

(i) Suppose $\mathbf{A}$ has a column of zeros, say $\mathbf{A}_i$. Then $\mathbf{A}^{\mathrm{tr}}_{(i)} = (0, \ldots, 0)$, so $0 = g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) = \det(\mathbf{A})$ by multilinearity.

(ii) Suppose $\mathbf{A}$ has two columns equal, say $\mathbf{A}_{(i)} = \mathbf{A}_{(j)}$, $i \neq j$. Then $\mathbf{A}^{\mathrm{tr}}_{(i)} = \mathbf{A}^{\mathrm{tr}}_{j}$, so $0 = g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(j)}, \ldots, \mathbf{A}_{(n)}) = \det(\mathbf{A})$ because $g$ is alternating.

(iii) If $\mathbf{B}$ is obtained from $\mathbf{A}$ by interchanging the two columns $\mathbf{A}_{(i)}$ and $\mathbf{A}_{(j)}$, then if we arrange the incices so that $i < j$; we see that

$$\begin{aligned} \det(\mathbf{B}) &= g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(j)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= -g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(j)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) = -\det(\mathbf{A}), \end{aligned}$$

because $g$ is sign-symmetric.

(iv) Suppose $\mathbf{B}$ is obtained from $\mathbf{A}$ by adding $a \cdot \mathbf{A}^{\mathrm{tr}}_{(i)}$ to $\mathbf{A}^{\mathrm{tr}}_{(j)}$, where $i \neq j$. Then

$$\begin{aligned} \det(\mathbf{B}) &= g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{j} + a\mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{j}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &\quad + g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, 0, \ldots, a\mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= \det(\mathbf{A}) + g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, 0, \ldots, \mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= \det(\mathbf{A}) + 0 = \det(\mathbf{A}), \end{aligned}$$

by property $(i)$.

(v) Suppose $\mathbf{B}$ is obtained from $\mathbf{A}$ by multiplying $\mathbf{A}^{\mathrm{tr}}_{(i)}$ by $a$. Then

$$\begin{aligned}\det(\mathbf{B}) &= g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, a\mathbf{A}^{\mathrm{tr}}_{(i)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= ag(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) = a\det(\mathbf{A}),\end{aligned}$$

by multilinearity.

(vi) Suppose that $\mathbf{A}$ is an upper triangular matrix, that is,

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & & \vdots \\ & & \ddots & \\ \mathbf{0} & & & a_{n,n} \end{bmatrix}$$

Then

$$\begin{aligned}\mathbf{A}^{\mathrm{tr}}_{(1)} &= a_{1,1}\mathbf{E}_1, \\ \mathbf{A}^{\mathrm{tr}}_{(2)} &= a_{1,2}\mathbf{E}_1 + a_{2,2}\mathbf{E}_2, \\ &\vdots \\ \mathbf{A}^{\mathrm{tr}}_{(n)} &= a_{1,n}\mathbf{E}_1 + \cdots + a_{n,n}\mathbf{E}_n,\end{aligned}$$

as vectors in $\mathbb{R}^n$. Therefore, by using the linearity of $g$ in the successive columns and dropping terms that vanish because $g$ is also alternating, we obtain

$$\begin{aligned}\det(\mathbf{A}) &= g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= g(a_{1,1}\mathbf{E}_1, \mathbf{A}^{\mathrm{tr}}_{(2)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= a_{1,1}g(\mathbf{E}_1, a_{1,2}\mathbf{E}_1 + a_{2,2}\mathbf{E}_2, \mathbf{A}^{\mathrm{tr}}_{(3)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= a_{1,1}a_{1,2}g(\mathbf{E}_1, \mathbf{E}_1, \mathbf{A}^{\mathrm{tr}}_{(3)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &\quad + a_{1,1}a_{2,2}(\mathbf{E}_1, \mathbf{E}_2, \mathbf{A}^{\mathrm{tr}}_{(3)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= a_{1,1}a_{1,2} \cdot 0 + a_{1,1}a_{2,2}g(\mathbf{E}_1, \mathbf{E}_2, \mathbf{A}^{\mathrm{tr}}_{(3)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= a_{1,1}a_{2,2}a_{1,3}g(\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_1, \mathbf{A}^{\mathrm{tr}}_{(4)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &\quad + a_{1,1}a_{2,2}a_{2,3}g(\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_2, \mathbf{A}^{\mathrm{tr}}_{(4)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &\quad + a_{1,1}a_{2,2}a_{3,3}g(\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \mathbf{A}_4, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) \\ &= a_{1,1}a_{2,2}a_{1,3} \cdot 0 + a_{1,1}a_{2,2}a_{2,3} \cdot 0 \\ &\quad + a_{1,1}a_{2,2}a_{2,3}g(\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \mathbf{A}^{\mathrm{tr}}_{(4)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)})\end{aligned}$$

$$\vdots$$

$$= a_{1,1}a_{2,2}\cdots a_{1,n}\cdot 0 + \cdots + a_{1,1}a_{2,2}\cdots a_{1,n-1}\cdot 0$$
$$+ a_{1,1}\cdots a_{n,n}g(\mathbf{E}_1,\ldots,\mathbf{E}_n)$$
$$= a_{1,1}\cdots a_{n,n},$$

as required.
It is convenient, before proving (viii), to prove (ix).

(ix) Let $\mathsf{A} \in \mathrm{Mat}_{n,n}$. Suppose that $\mathsf{A}$ is invertible. Then the vectors $\{\mathsf{A}(\mathbf{E}_1),\ldots,\mathsf{A}(\mathbf{E}_n)\}$ form a basis for $\mathbb{R}^n$. Suppose

$$g\big(\mathsf{A}(\mathbf{E}_1),\ldots,\mathsf{A}(\mathbf{E}_n)\big) = 0.$$

By Proposition **A**.2.3 this implies that

$$g = 0 : \underbrace{\mathbb{R}^n \times\cdots\times \mathbb{R}^n}_{n} \longrightarrow \mathbb{R}$$

as a multilinear form, since of course,

$$0\big(\mathsf{A}(\mathbf{E}_1),\ldots,\mathsf{A}(\mathbf{E}_n)\big) = 0,$$

and an alternating form is completely determined by its value on ***any*** ordered basis, in particular, on the ordered basis $\mathsf{A}(\mathbf{E}_1),\ldots,\mathsf{A}(\mathbf{E}_n)$. Therefore,

$$\det(\mathsf{A}) = \det(\mathsf{A})\cdot g = g_{\mathsf{A}} \neq 0,$$

as required.

(viii) Suppose $\mathsf{B}$ is singular. Then so is $\mathsf{A}\cdot\mathsf{B}$ so by (ix)

$$\det(\mathsf{AB}) = 0 = \det(\mathsf{A})\cdot 0 = \det(\mathsf{A})\cdot\det\mathsf{B}.$$

If $\mathsf{B}$ is nonsingular, then the form

$$g_{\mathsf{B}} : \mathbb{R}^n \times\cdots\times \mathbb{R}^n \longrightarrow \mathbb{R}$$

is nonzero, since by (ix),

$$g_{\mathsf{B}} = \det(\mathsf{B})\cdot g \neq 0.$$

Therefore, $g_{\mathsf{B}}$ is a basis for the alternating forms, since the space of such forms is one-dimensional. Thus there exists $d \in \mathbb{R}$ such that

$$g_{\mathsf{AB}} = d g_{\mathsf{B}} : \underbrace{\mathbb{R}^n \times\cdots\times \mathbb{R}^n}_{n} \longrightarrow \mathbb{R}.$$

Therefore, we obtain from the definitions

$$g_{\mathsf{AB}} = d g_{\mathsf{B}} = d\det(\mathsf{B})g,$$

and

$$g_{\mathsf{AB}} = \det(\mathsf{A} \cdot \mathsf{B})g.$$

So it remains to show that $d = \det(\mathsf{A})$. To this end, note that since $\mathsf{B}$ is nonsingular, we can consider the point

$$\left(\mathsf{B}^{-1}(\mathbf{E}_1), \ldots, \mathsf{B}^{-1}(\mathbf{E}_n)\right) \in \underbrace{\mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{n}.$$

So evaluating the two sides of the equation $g_{\mathsf{AB}} = dg_{\mathsf{B}}$ on

$$\left(\mathsf{B}^{-1}(\mathbf{E}_1), \ldots, \mathsf{B}^{-1}(\mathbf{E}_n)\right) \in \mathbb{R}^n \times \cdots \times \mathbb{R}^n$$

gives, on the one hand,

$$\begin{aligned} g_{\mathsf{AB}}\left(\mathsf{B}^{-1}(\mathbf{E}_1), \ldots, \mathsf{B}^{-1}(\mathbf{E}_n)\right) &= g\left(\mathsf{ABB}^{-1}(\mathbf{E}_1), \ldots, \mathsf{ABB}^{-1}(\mathbf{E}_n)\right) \\ &= g\left(\mathsf{A}(\mathbf{E}_1), \ldots, \mathsf{A}(\mathbf{E}_n)\right) \\ &= \det(\mathsf{A})g(\mathbf{E}_1, \ldots, \mathbf{E}_n) = \det(\mathsf{A}), \end{aligned}$$

and, on the other hand,

$$\begin{aligned} dg_{\mathsf{B}}\left(\mathsf{B}^{-1}(\mathbf{E}_1), \ldots, \mathsf{B}^{-1}(\mathbf{E}_n)\right) &= dg\left(\mathbf{BB}^{-1}(\mathbf{E}_1), \ldots, \mathbf{BB}^{-1}(\mathbf{E}_n)\right) \\ &= dg(\mathbf{E}_1, \ldots, \mathbf{E}_n) = d, \end{aligned}$$

and so equating yields $d = \det(\mathsf{A})$, as required.

Before turning to the proof of (vi) we record a lemma.

**LEMMA A.2.5:** *Let $\mathsf{H}_i : \mathbb{R}^{n-1} \longrightarrow \mathbb{R}^n$ be defined by the formula*

$$\mathsf{H}_i(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{i-1}, 0, x_i, \ldots, x_{n-1}),$$

*and let*

$$g_n : \underbrace{\mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{n} \longrightarrow \mathbb{R},$$

*and*

$$g_{n-1} : \overbrace{\mathbb{R}^{n-1} \times \cdots \times \mathbb{R}^{n-1}}^{n-1} \longrightarrow \mathbb{R}$$

*be the fundamental sign-symmetric forms. Define*

$$g^i_{n-1} : \underbrace{\mathbb{R}^{n-1} \times \cdots \times \mathbb{R}^{n-1}}_{n-1} \longrightarrow \mathbb{R}$$

*by*

$$g^i_{n-1}(\mathbf{A}_{(1)}, \ldots, \mathbf{A}_{n-1}) = g_n\left(\mathsf{H}_i(\mathbf{A}_1), \ldots, \mathsf{H}_i(\mathbf{A}_{n-1}), \mathbf{E}_i\right).$$

*Then*

$$g^i_{n-1} = (-1)^{n-i} g_{n-1}.$$

**Proof:** It is enough to show that

$$g^i_{n-1}(\mathbf{E}_1, \dots, \mathbf{E}_{n-1}) = (-1)^{n-i}.$$

Note that

$$\mathsf{H}_i(\mathbf{E}_j) = \begin{cases} \mathbf{E}_j, & j \le i-1, \\ \mathbf{E}_{j+1}, & j \ge i, \end{cases}$$

so by definition

$$g^i_{n-1}(\mathbf{E}_1, \dots, \mathbf{E}_{n-1}) = g_n(\mathbf{E}_1, \dots, \mathbf{E}_{i-1}, \mathbf{E}_{i+1}, \dots, \mathbf{E}_n, \mathbf{E}_i)$$

$$n - i \text{ interchanges} \quad \begin{cases} = -g_n(\mathbf{E}_1, \dots, \mathbf{E}_{i-1}, \mathbf{E}_{i+1}, \dots, \mathbf{E}_i, \mathbf{E}_n) \\ \vdots \\ = (-1)^{n-i-1} g_n(\mathbf{E}_1, \dots, \mathbf{E}_n) = (-1)^{n-i-1}, \end{cases}$$

as required. □

Let $\mathsf{A} = [\mathsf{A}_{(1)}, \dots, \mathsf{A}_{(n)}] \in \mathrm{Mat}_{n,n}$, where

$$\mathsf{A}_{(i)} = \begin{bmatrix} a_{1,i} \\ a_{2,i} \\ \vdots \\ a_{n,i} \end{bmatrix}, \qquad i = 1, \dots, n,$$

are the columns of $\mathsf{A}$. Then,

$$\mathsf{A}^{\mathrm{tr}}_{(i)} = \sum_{k=1}^{n} a_{k,i}\mathbf{E}_k = \mathbf{B}_{(i)} + a_{n,i}\mathbf{E}_n$$

which defines $\mathbf{B}_{(1)}, \dots, \mathbf{B}_{(n)} \in \mathrm{Span}\{\mathbf{E}_1, \dots, \mathbf{E}_{n-1}\}$. By definition we have

$$\begin{aligned} \det(\mathsf{A}) &= g(\mathsf{A}^{\mathrm{tr}}_{(1)}, \dots, \mathsf{A}^{\mathrm{tr}}_{(n)}) \\ &= g(\mathbf{B}_{(1)} + a_{n,1}\mathbf{E}_n, \dots, \mathbf{B}_{(n)} + a_{n,n}\mathbf{E}_n). \end{aligned}$$

We are going to expand this last expression using the multilinearity of $g(—, \dots, —)$. Among the terms of this expansion will be

$$g(\mathbf{B}_{(1)}, \dots, \mathbf{B}_{(n)}).$$

This term is, however, zero. To see this, notice that

$$g(\mathbf{B}_{(1)}, \dots, \mathbf{B}_{(n)}) = \det \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \cdots & \vdots \\ a_{n-1,1} & \cdots & a_{n-1,n} \\ 0 & \cdots & 0 \end{bmatrix}.$$

The matrix on the right is obviously singular because it has a row of zeros. By property (viii) we conclude that the determinant of this matrix is zero, so

$$g(\mathbf{B}_{(1)}, \ldots, \mathbf{B}_{(n)}) = 0.$$

Another typical term of the expansion is

$$g(\mathbf{B}_{(1)}, \ldots, \mathbf{B}_{(i-1)}, a_{n,i}\mathbf{E}_n, \mathbf{B}_{(i+1)}, \ldots, a_{n,j}\mathbf{E}_n, \mathbf{B}_{(j+1)}, \ldots, \mathbf{B}_{(n)}).$$

This term also vanishes because by multilinearity and sign-symmetry it is equal to

$$\begin{aligned} &a_{n,i}a_{n,j}g(\mathbf{B}_{(1)}, \ldots, \mathbf{B}_{(i-1)}, \mathbf{E}_n, \mathbf{B}_{(i+1)}, \ldots, \mathbf{B}_{(j-1)}, \mathbf{E}_n, \mathbf{B}_{(j+1)}, \ldots, \mathbf{B}_n) \\ &= a_{n,1}a_{n,j} \cdot 0 = 0. \end{aligned}$$

These considerations then lead to the following formula

$$\begin{aligned} \det(\mathbf{A}) = g(a_{n,1}\mathbf{E}_n, \mathbf{B}_{(2)}, \ldots, \mathbf{B}_{(n)}) + g(\mathbf{B}_1, a_{n,2}\mathbf{E}_n, \mathbf{B}_{(3)}, \ldots, \mathbf{B}_{(n)}) \\ + \cdots + g(\mathbf{B}_{(1)}, \ldots, \mathbf{B}_{(n-1)}, a_{n,n}\mathbf{E}_n). \end{aligned}$$

So by linearity and the preceding lemma we get

$$\begin{aligned} \det(\mathbf{A}) &= a_{n,1}g(\mathbf{E}_n, \mathbf{B}_{(2)}, \ldots, \mathbf{B}_{(n)}) + \cdots + a_{n,n}g_{n-1}(\mathbf{B}_{(1)}, \ldots, \mathbf{B}_{(n-1)}, \mathbf{E}_{(n)}) \\ &= \sum_{i=1}^{n}(-1)^{n-i}a_{n,i}g(\mathbf{B}_{(1)}, \ldots, \mathbf{B}_{(i-1)}, \mathbf{B}_{(i+1)}, \ldots, \mathbf{B}_{(n)}) \\ &= \sum_{i=1}^{n}(-1)^{n-i}a_{n,i}\det(\mathbf{M}_{n,i}), \end{aligned}$$

by the definition of the $(n, i)$ minor of $\mathbf{A}$ and the sign-symmetry of $g$. This formula is the **Lagrange expansion formula** with respect to the $n$-th row. There is a similar formula for $\det(\mathbf{A})$ that employs the $n$-th column of $\mathbf{A}$. To obtain this we write

$$\begin{aligned} \det(\mathbf{A}) = g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n)}) = g\left(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n-1)}, \sum_{k=1}^{n} a_{k,n}\mathbf{E}_k\right) \\ = \sum_{k=1}^{n} a_{k,n}g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \ldots, \mathbf{A}^{\mathrm{tr}}_{(n-1)}, \mathbf{E}_k). \end{aligned}$$

Let us now analyze the individual terms of this sum. For $1 \leq k \leq n$ we write

$$\mathbf{A}^{\mathrm{tr}}_{(i)} = \mathbf{C}_{k,i} + a_{k,i}\mathbf{E}_k.$$

Then reasoning as above, and using the fact that

$$g(\mathbf{C}_{k,1}, \ldots, \mathbf{C}_{k,n}) = 0,$$

since it may be interpreted as the determinant of a matrix with a column of zeros, and

$$g(\mathbf{C}_{k,1}, \ldots, a_{k,i}\mathbf{E}_k, \mathbf{C}_{k,i+1}, \ldots, a_{k,j}\mathbf{E}_k, \mathbf{C}_{k,j+1}, \ldots, \mathbf{C}_{k,n})$$

vanishes by sign-symmetry, we get the formula

$$\begin{aligned}\det(\mathbf{A}) &= \sum_{k=1}^{n} a_{k,n} g(\mathbf{C}_{k,1}, \ldots, \mathbf{C}_{k,n-1}, \mathbf{E}_k) \\ &= \sum_{k=1}^{n} (-1)^{n-k} a_{k,n} g(\mathbf{C}_{k,1}, \ldots, \mathbf{C}_{k,n-1}) \\ &= \sum_{k=1}^{n} (-1)^{n-k} a_{k,n} \det(\mathbf{M}_{k,n}),\end{aligned}$$

which is the **Lagrange expansion formula** with respect to the $n$-th column.

Finally we come to:

(vii) We may prove (vii) by observing that the Lagrange expansion with respect to the $n$-th column of $\mathbf{A}^{\mathrm{tr}}$ is the same as the Lagrange expansion of $\mathbf{A}$ with respect to the $n$-th row. But the Lagrange expansions compute det, so

$$\begin{aligned}\det(\mathbf{A}) &= \text{Lagrange expansion of } \mathbf{A} \text{ with respect to the } n\text{-th row} \\ &= \text{Lagrange expansion of } \mathbf{A}^{\mathrm{tr}} \text{ with respect to the } n\text{-th column} \\ &= \det(\mathbf{A}^{\mathrm{tr}}),\end{aligned}$$

as required. □

Let us derive an explicit form for the determinant of a $2 \times 2$ matrix. Let

$$\mathbf{A} = (\mathbf{A}_{(1)}, \mathbf{A}_2),$$

where

$$\mathbf{A}_{(i)} = \begin{bmatrix} a_{1,i} \\ a_{2,i} \end{bmatrix}, \qquad i = 1, 2,$$

are the columns of $\mathbf{A}$. As in the discussion preceding the proof of Lemma

A.2.4 we see that

$$\begin{aligned}\det(\mathbf{A}) &= g(\mathbf{A}^{\mathrm{tr}}_{(1)}, \mathbf{A}^{\mathrm{tr}}_{(2)}) \\ &= g(a_{1,1}\mathbf{E}_1 + a_{2,1}\mathbf{E}_2, a_{1,2}\mathbf{E}_1 + a_{2,2}\mathbf{E}_2) \\ &= a_{1,1}g(\mathbf{E}_1, a_{1,2}\mathbf{E}_1 + a_{2,2}\mathbf{E}_2) + a_{2,1}g(\mathbf{E}_2, a_{1,2}\mathbf{E}_1 + a_{2,2}\mathbf{E}_2) \\ &= a_{1,1}(a_{1,2}g(\mathbf{E}_1, \mathbf{E}_1) + a_{2,2}(\mathbf{E}_1, \mathbf{E}_2)) \\ &\quad + a_{2,1}(a_{1,2}g(\mathbf{E}_2, \mathbf{E}_1) + a_{2,2}g(\mathbf{E}_2, \mathbf{E}_2)) \\ &= a_{1,1}(0 + a_{2,2}) + a_{2,1}(-a_{1,2} + 0) \\ &= a_{1,1}a_{2,2} - a_{2,1}a_{1,2},\end{aligned}$$

because

$$g(\mathbf{E}_1, \mathbf{E}_1) = 0 = g(\mathbf{E}_2, \mathbf{E}_2),$$
$$g(\mathbf{E}_1, \mathbf{E}_2) = 1 = -g(\mathbf{E}_2, \mathbf{E}_1).$$

Thus

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc,$$

exactly as we had already defined the determinant in Chapter 13 for $2 \times 2$ matrices.

It should be apparent that one can employ the preceding procedure to arrive at a formula for $\det(\mathbf{A})$, where $\mathbf{A}$ is an $n \times n$ matrix. Such a formula would look like

$$(*) \qquad \det(\mathbf{A}) = \sum (-1)^{\sigma(j_1,\ldots,j_n)} a_{j_1,1} \cdots a_{j_n,n},$$

where each term in the sum

$$a_{j_1,1} \cdots a_{j_n,n}$$

is a product of matrix elements, one element chosen out of each column such that no two elements lie in the same row, and $\sigma(j_1, \ldots, j_n)$ is the number of pairs of indices that must be interchanged to bring $(\mathbf{E}_{j_1}, \ldots, \mathbf{E}_{j_n})$ to the form $(\mathbf{E}_1, \ldots, \mathbf{E}_n)$.

**NOTE:** There are as many terms in the sum $(*)$ as there are possible rearrangements of $(\mathbf{E}_1, \ldots, \mathbf{E}_n)$, or, what is the same thing, $(1, \ldots, n)$. An elementary counting argument shows that the number of such arrangements is

$$n! = n(n-1) \cdots (2)(1).$$

So for a $5 \times 5$ matrix the preceding sum contains

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

terms. Thus the formula (*) is not particularly efficacious for computations.

We close our discussion of determinants with the Lagrange expansion formula.

**PROPOSITION A.2.6:** *If* **A** *is a square matrix of size* $n$*, then*

$$\det(\mathbf{A}) = \sum_{j=1}^{n} (-1)^{i+j} a_{j,i} \det \mathbf{M}_{j,i}.$$

**PROOF:** This follows from the Lagrange expansion formula with respect to the $n$-th column. □

## A.3 Exercises

**1.** Find a nonsingular bilinear form

$$f : \mathbb{R}^4 \times \mathbb{R}^4 \longrightarrow \mathbb{R}^4.$$

**2.** Show that the bilinear forms of Examples 3 and 4 are isomorphic.

**3.** Let $b(n, k)$ be defined as in the open problem at the end of Section A.1

(a) Show that $b(k, n) \geq 1$.
(b) If $k$ and $n$ are even show that $b(k, n) \geq 2$.

**4.** Compute the determinant of each of the following matrices and determine which, if any, are invertible.

$$\begin{bmatrix} 0 & -1 & -1 & -1 \\ -1 & 0 & -1 & -1 \\ -1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix},$$

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 4 & -1 & -1 \\ -1 & 4 & -1 \\ 4 & -1 & -1 \end{bmatrix}.$$

**5.** (The Lewis Carroll Rule[2]) Suppose **A, B, C, D** $\in \mathrm{Mat}_{n,n}$ and let **M** be the block matrix

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}.$$

[2] I learned of this rule from R. Howe.

Prove that

$$\det \mathbf{M} = \det(\mathbf{A})\det(\mathbf{D}) - \det(\mathbf{B})\det(\mathbf{C}).$$

**HINT:** Show that the function $f : \mathrm{Mat}_{2n,2n} \longrightarrow \mathbb{R}$ that associates to a $2n \times 2n$ matrix $\mathbf{M}$ the number $\det \mathbf{M} = \det(\mathbf{A})\det(\mathbf{D}) - \det(\mathbf{B})\det(\mathbf{C})$, where $\mathbf{M}$ has been partitioned into a block matrix as above, is a multilinear function of the columns of $\mathbf{M}$, and $f(\mathbf{I}) = 1$. Then use the uniqueness of the standard sign-symmetric form and the definition of the determinant as given in Section A.2.

**6.** Let

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}.$$

The **diagonals** of $\mathbf{A}$ are

$$\begin{matrix} a_{1,1} & & & a_{1,2} & & & a_{1,3} & & \\ & a_{2,2} & & & a_{2,3} & & & a_{2,1} & \\ & & a_{3,3} & & & a_{3,1} & & & a_{3,2} \end{matrix}$$

and the **antidiagonals** are

$$\begin{matrix} & & a_{1,3} & & & a_{1,2} & & & a_{1,1} \\ & a_{2,2} & & & a_{2,1} & & & a_{2,3} & \\ a_{3,1} & & & a_{3,3} & & & a_{3,2} & & \end{matrix} .$$

(a) Show that the determinant of $\mathbf{A}$ is the sum of the products of the diagonal elements minus the sum of the products of the antidiagonal elements. (This is Exercise 17 in Chapter 14.) In Germany this is referred to the **rule of Sarrus**.

(b) Show by an example that the analogous rule is false for $4 \times 4$ matrices.

**7.** Let $a_i \in \mathbb{R}$ be numbers, $i = 1, 2, \ldots,$ , and define the square matrices

$$\mathbf{A}(n) = \begin{bmatrix} a_1 & 1 & 0 & 0 \\ -1 & \ddots & \ddots & 0 \\ 0 & \ddots & \ddots & 1 \\ 0 & & -1 & a_n \end{bmatrix}, \qquad n = 1, 2, \ldots.$$

Define $d_n = \det(\mathbf{A}(n))$ for $n \in \mathbb{N}$ and set $d_0 = 1$. Show that the recursive formula

$$d_n = a_n d_{n-1} + d_{n-2}$$

holds for all $n > 2$.

**8.** Let $\mathbf{A} = (a_1, a_2)$, $\mathbf{B} = (b_1, b_2) \in \mathbb{R}^2$, and suppose that $\mathbf{A} \neq \mathbf{B}$. Show that the equation of the line through $\mathbf{A}$ and $\mathbf{B}$ is given by

$$\det \begin{bmatrix} x & y & 1 \\ a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \end{bmatrix} = 0.$$

**9.** Show that the area of the triangle in $\mathbb{R}^2$ with vertices $(a_1, b_1)$, $(a_2, b_2)$, $(a_3, b_3)$ is given by

$$\det \begin{bmatrix} a_1 & b_1 & 1 \\ a_2 & b_2 & 1 \\ a_3 & b_3 & 1 \end{bmatrix}.$$

**10.** If $\mathbf{A} \in \mathrm{Mat}_{n,n}$ is skew-symmetric and $n$ is odd, show that $\det(\mathbf{A}) = 0$.

**11.** Let $\sigma$ be a **permutation** of $(1, \ldots, n)$, that is, a rearrangement $(\sigma(1), \ldots, \sigma(n))$ of $(1, \ldots, n)$ in perhaps some other order. Define sgn $(\sigma)$ by

$$\prod_{i<j} (x_{\sigma(i)} - x_{\sigma(j)}) = \mathrm{sgn}(\sigma)\Delta(x),$$

where

$$\Delta(x) = \prod_{i<j} (x_i - x_j).$$

(a) Show that $\mathrm{sgn}(\sigma) = -1$ if $\sigma$ is the permutation where

$$\sigma(k) = \begin{cases} k, & k \neq i, j, \\ j, & k = i, \\ i, & k = j. \end{cases}$$

Such a permutation is simply the interchange of $i$ and $j$.

(b) Show that any permutation can be accomplished by a succession of interchanges.

(c) Show that $\mathrm{sgn}(\sigma) = (-1)^s$ where $s$ is the minimum number of interchanges required to accomplish $\sigma$.

(d) Let $\sigma$ be a permutation and $\mathbf{L}_\sigma : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ by taking the linear extension of

$$\mathbf{L}_\sigma(\mathbf{E}_i) = \mathbf{E}_{\sigma(i)}, \qquad i = 1, \ldots, n.$$

The matrix of $\mathbf{L}_\sigma$ with respect to the standard basis of $\mathbb{R}^n$ is called a **permutation matrix**. Write down all the permutation matrices of size 3 and compute their determinants.

(e) Show that $\det \mathbf{L}_\sigma = \mathrm{sgn}(\sigma)$.

**12.** Let $\mathbf{A} \in \mathrm{Mat}_{n,n}$. Show that

$$\det(\mathbf{A}) = \sum (-1)^{\mathrm{sgn}(\sigma)} a_{\sigma(1)} \cdots a_{\sigma(n)},$$

where the sum extends over all permutations of $(1, \ldots, n)$.

**13.** Evaluate

$$\det \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix}.$$

**14.** Show that

$$\det \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & & x_n^{n-1} \end{bmatrix} = \prod_{i<j} (x_i - x_j).$$

**15.** Let $\mathbf{A} \in \mathrm{Mat}_{m,n}$, $\mathbf{B} \in \mathrm{Mat}_{n,m}$ where $m > n$. Show that $\det(\mathbf{AB}) = 0$. (N.B. $\mathbf{A} \cdot \mathbf{B}$ is square of size $m \times m$.)

# Appendix B
# Complex Numbers

Beginning with Chapter 14 we were often confronted with the problem of whether the characteristic polynomial of $\mathsf{T} : \mathcal{V} \longrightarrow \mathcal{V}$ has any roots. If a solution to this problem is understood to mean ***real*** roots, i.e., numbers in $\mathbb{R}$, then the answer was often no. Although the real numbers are adequate for describing many of the phenomena encountered in mathematics, the physical sciences, and everyday life, the fact that polynomials do not of necessity have roots can be seen as a defect. In this chapter we describe one way to remedy this defect, and that is to introduce the **complex numbers**. There is, of course, a price to be paid for this, and we will indicate what it is once the basic definitions and elementary properties of the complex numbers have been explained. We assume that the reader is familiar with matrix algebra as explained in Chapter 10.

## B.1 The Complex Numbers

Before giving a definition of complex numbers, let us decide what it is we desire. We are hoping to define a ***number system*** that is an enlargement $\mathbb{C}$ of $\mathbb{R}$ in which we can perform the usual arithmetic operations, addition and multiplication, which agree with the operations when applied to real numbers and enjoy as many of the properties of these rules as possible, e.g., existence of 0, 1, negatives, and inverses for nonzero numbers,. In addition the complex number system should have the property that polynomials with coefficients in $\mathbb{C}$ always have roots in $\mathbb{C}$. Here is one way to proceed.

DEFINITION: *The* **complex numbers**, *denoted by* $\mathbb{C}$, *are defined* *to be the set of all $2 \times 2$ matrices of the form*

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix},$$

*where $a, b \in \mathbb{R}$ are arbitrary.*

**LEMMA B.1.1:** *If $\mathbb{C} \subset \mathrm{Mat}_{2,2}$ is the set of complex numbers and $\mathbf{A}', \mathbf{A}'' \in \mathbb{C}$, then $\mathbf{A}' + \mathbf{A}''$ and $\mathbf{A}'\mathbf{A}''$ also belong to $\mathbb{C}$. In addition, $\mathbf{A}'\mathbf{A}'' = \mathbf{A}''\mathbf{A}'$.*

**PROOF:** This is elementary. □

**LEMMA B.1.2:** *The set of complex numbers $\mathbb{C}$ is a linear subspace of $\mathrm{Mat}_{2,2}$ of dimension 2. The matrices*

$$(*) \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

*form a basis for $\mathbb{C}$ as vector space over $\mathbb{R}$.*

**PROOF:** If $\mathbf{A} \in \mathbb{C}$, then

$$\mathbf{A} = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$

and therefore the two matrices $(*)$ span $\mathbb{C}$ as a vector space over $\mathbb{R}$. Since these matrices are also linearly independent, they form a basis. □

**LEMMA B.1.3:** *If $\mathbf{A} \in \mathbb{C}$ is a nonzero complex number, then it has an inverse.*

**PROOF:** If

$$\mathbf{A} = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \in \mathbb{C} \subset \mathrm{Mat}_{2,2},$$

then $\det(\mathbf{A}) = a^2 + b^2$, and since $\mathbf{A}$ is not zero, neither is $\det(\mathbf{A})$. Hence by Proposition 10.4.1, $\mathbf{A}$ is invertible, and

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} a & -b \\ b & a \end{bmatrix}. \qquad \square$$

**NOTATION:** It is convenient when working with complex numbers to introduce the notation

$$1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{i} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

for the basis vectors of $\mathbb{C}$ as real vector space. With this notation we can write each complex number in the form $a \cdot 1 + b \cdot \boldsymbol{i}$ or in the abbreviated forms $a + b\boldsymbol{i} = a + \boldsymbol{i}b$ in exactly one way.

A priori, using this notation in abridged form could lead to an ambiguity; namely, 7 could be interpreted as either a complex number or a real number. The following lemma tells us that this is not important, because the real numbers are part of the complex numbers, and the operations of + and $\cdot$ for complex numbers applied to real numbers yield the usual answers.

**LEMMA B.1.4:** *The set of all complex numbers of the form*

$$\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix},$$

*or in the previous notations $a + 0\boldsymbol{i}$, or $a$, can be identified with the real numbers $\mathbb{R}$ via the correspondence*

$$\mathbb{R} \ni a \overset{\varphi}{\longmapsto} \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \in \mathbb{C}.$$

*This correspondence is compatible with + and $\cdot$ , i.e.,*

$$\varphi(a' + a'') = \varphi(a') + \varphi(a'') \quad \text{and} \quad \varphi(a' \cdot a'') = \varphi(a') \cdot \varphi(a'').$$

*Moreover, $\varphi(1) = 1$ and $\varphi(0) = 0$.*

**PROOF:** This follows from the fact that scalar matrices add and multiply to give scalar matrices in such a simple way. □

Using the previous notations for complex numbers leads to the (perhaps already familiar) multiplication rule

$$(a + b\boldsymbol{i})(c + d\boldsymbol{i}) = (ac - bd) + (ad + bc)\boldsymbol{i}$$

and in particular

$$\boldsymbol{i}^2 = -1.$$

From this it follows for $a + b\boldsymbol{i} \neq 0$ that

$$(a + b\boldsymbol{i})^{-1} = \frac{1}{a^2 + b^2}(a - b\boldsymbol{i}).$$

It is also easy to see that any nonzero real number $a$ has exactly two complex square roots, viz., $\pm\sqrt{a}$ if $a > 0$ and $\pm\boldsymbol{i}\sqrt{|a|}$ if $a < 0$. Indeed, more is true.

**LEMMA B.1.5:** *Every nonzero complex number has two complex square roots.*

**PROOF:** Let $a + b\boldsymbol{i} \in \mathbb{C}$. Since $-1$ has a square root (indeed two, $\pm\boldsymbol{i}$) in $\mathbb{C}$, we may, by multiplying by $-1$ if need be, assume that $b \geq 0$. We are looking for complex numbers $x + \boldsymbol{i}y$ that satisfy

$$(x + \boldsymbol{i}y)^2 = a + b\boldsymbol{i},$$

or, what is the same thing,

$$(x^2 - y^2) + (2xy)\boldsymbol{i} = a + b\boldsymbol{i}.$$

Since $x$, $y$, $a$, and $b$ are real numbers, and $1, \boldsymbol{i}$ a basis for $\mathbb{C}$ as a real vector space, we may equate coefficients to obtain

$$(\maltese) \qquad a = x^2 - y^2 \quad \text{and} \quad b = 2xy.$$

So we need to solve the equations $(\maltese)$ for $x$ and $y$.

The three numbers $\sqrt{a^2 + b^2}$, $b$, and $|a|$ are a Pythagorean triple, so there is a right triangle with sides of length $\sqrt{a^2 + b^2}$, $b$, $|a|$ (Figure B.1.1).

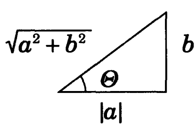


Figure 19.1.1

The function $\cos(\vartheta)$ is a continuous, strictly decreasing function on the interval $0 \leq \vartheta \leq \pi$ and has $+1$ as a maximum and $-1$ as a minimum. The number $\frac{a}{\sqrt{a^2+b^2}}$ lies between $-1$ and $+1$, and therefore by the intermediate value theorem there is a unique value $\Theta_0 \in [0, \pi]$ with $\cos(\Theta_0) = \frac{a}{\sqrt{a^2+b^2}}$.

The corresponding value of $\sin(\Theta_0)$ is then $\sin(\Theta_0) = \frac{b}{\sqrt{a^2+b^2}}$.

Let $\Theta_0 = 2\vartheta_0$. Recall the addition formulas from elementary trigonometry

$$\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta),$$

$$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta).$$

In particular, there are the double angle formulas

$$\sin(2\vartheta) = 2\sin(\vartheta)\cos(\vartheta),$$

$$\cos(2\vartheta) = \cos^2(\vartheta) - \sin^2(\vartheta),$$

Let $x_0 = \left(\sqrt[4]{a^2+b^2}\cdot\cos(\vartheta_0)\right)$ and $y_0 = \left(\sqrt[4]{a^2+b^2}\cdot\sin(\vartheta_0)\right)$. Then

$$\begin{aligned}
(x_0 + y_0\mathbf{i})^2 &= \left(\left(\sqrt[4]{a^2+b^2}\cdot\cos(\vartheta_0)\right) + \left(\sqrt[4]{a^2+b^2}\cdot\sin(\vartheta_0)\mathbf{i}\right)\right)^2 \\
&= \sqrt{a^2+b^2}\left(\cos^2(\vartheta_0) - \sin^2(\vartheta_0)\right) + \sqrt{a^2+b^2}\left(2\sin(\vartheta_0)\cos(\vartheta_0)\right)\mathbf{i} \\
&= \sqrt{a^2+b^2}\left(\cos(2\vartheta_0) + \mathbf{i}\sin(2\vartheta_0)\right) \\
&= \sqrt{a^2+b^2}\left(\cos(\Theta_0) + \mathbf{i}\sin(\Theta_0)\right) \\
&= \sqrt{a^2+b^2}\left(\frac{a}{\sqrt{a^2+b^2}} + \frac{b}{\sqrt{a^2+b^2}}\mathbf{i}\right) \\
&= a + b\mathbf{i}.
\end{aligned}$$

Therefore, the complex numbers $\pm(x_0 + y_0\mathbf{i})$ are square roots of $a + b\mathbf{i}$. $\square$

Finally, this leads easily to our first real result.

**PROPOSITION B.1.6:** *If $Q(x) = rx^2 + px + q$ is a quadratic polynomial with complex coefficients, and $r \neq 0$, then counting multiplicity $Q(x)$ has two complex roots.*

**PROOF:** By dividing by $r$ we may assume that $r = 1$, so $Q(x) = x^2 + px + q$. We next complete the square to obtain

$$\begin{aligned}
Q(x) = x^2 + px + q &= \left(x^2 + px + \frac{p^2}{4}\right) + \left(q - \frac{p^2}{4}\right) \\
&= (x+p)^2 + \left(q - \frac{p^2}{4}\right),
\end{aligned}$$

which upon solving for $(x + p)$ leads to

$$x + p = \pm\sqrt{\frac{p^2 - 4q}{4}},$$

or

$$x = -p \pm \sqrt{\frac{p^2 - 4q}{4}},$$

the familiar **quadratic formula**. $\square$

The complex numbers are a 2-dimensional vector space over $\mathbb{R}$, so we can represent a complex number by a vector in the plane (Figure B.1.2).

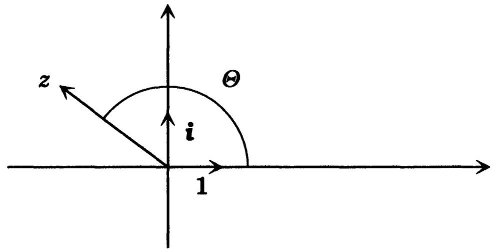


**FIGURE** B.1.2: Geometric interpretation of complex numbers

As in the proof of Lemma **B**.1.5, if $b \le 0$, we may find a unique $\Theta \in [\pi, 2\pi)$ such that $a + b\boldsymbol{i} = \sqrt{a^2 + b^2}(\cos(\Theta) + \boldsymbol{i}\sin(\Theta))$, and combined with the representation for $b \ge 0$ from Lemma **B**.1.5 we find that every complex number $z = a + b\boldsymbol{i}$ has a unique representation in the form $z = r\big(\cos(\Theta) + \boldsymbol{i}\sin(\Theta)\big)$, where $\Theta \in [0, 2\pi)$ and[1] $r = \sqrt{a^2 + b^2}$. This is often referred to as **de Moivre's representation**, and we introduce the notations $|z| = \sqrt{a^2 + b^2}$ and $\arg(z) = \Theta$ for this representation of a complex number. The nonnegative real number $|z|$ is called the **norm** of the complex number $z$ and $\arg(z)$ the **argument** of $z$. Note that the norm of a complex number $z$ is the norm of $z$ regarded as a vector in $\mathbb{R}^2$ with its usual inner product. The following two properties of the norm and argument are elementary consequences of the addition formulas for the trigonometric functions.

**LEMMA B.1.7:** *If $z', z'' \in C$ are complex numbers, then*

$$|z'z''| = |z'| \cdot |z''|,$$

$$\arg(z'z'') = \arg(z') + \arg(z'').$$

**PROOF:** Let

$$z' = |z'|(\cos(\Theta') + \boldsymbol{i}\sin(\Theta')),$$

$$z'' = |z''|(\cos(\Theta'') + \boldsymbol{i}\sin(\Theta'')).$$

Then direct computation gives

$$z'z'' = |z'| \cdot |z''| \big((\cos(\Theta')\cos(\Theta'') - \sin(\Theta')\sin(\Theta'')) + \boldsymbol{i}(\cos(\Theta')\sin(\Theta'') + \sin(\Theta')\cos(\Theta''))\big)$$

and the result follows from the addition formulas for sin and cos. □

[1] Note that $2\pi$ is excluded to obtain uniqueness of $\Theta$.

We can next prove the **Fundamental Theorem of Algebra**, which tells us that polynomials with complex coefficients always have complex roots, justifying our use of complex numbers in the development of the Jordan normal form. As we will see, the proof[2] uses essentially *nonalgebraic* ideas: in this case a *topological* idea.

THEOREM **B.1.8** (Fundamental Theorem of Algebra): *Every polynomial of positive degree with complex coefficients has a complex root.*

PROOF: Let $p(z) = a_m z^m + \cdots + a_1 z + a_0$ be a polynomial with complex coefficients of degree $m$, i.e., $a_m \neq 0$. By dividing by $a_m$ we obtain a new polynomial $q(z) = z^m + c_{m-1}z^{m-1} + \cdots + c_1 z + c_0$, also of degree $m$, with the same roots as $p(z)$ (if there are any at all!) and leading coefficient 1.

Let $C_r \subset \mathbb{C} = \mathbb{R}^2$ be the circle centered at the origin of radius $r \geq 0$. Then the image of $C_r$ under the polynomial map $q : \mathbb{C} \longrightarrow \mathbb{C}$ is a smooth curve in the plane. Let us assume that $q(z)$ has no roots. Then none of the curves $q(C_r)$, $r \in \mathbb{R}$, pass through the origin. Therefore, we can consider the number of times $\omega(r)$, the curve $q(C_r)$, **winds** around the origin.

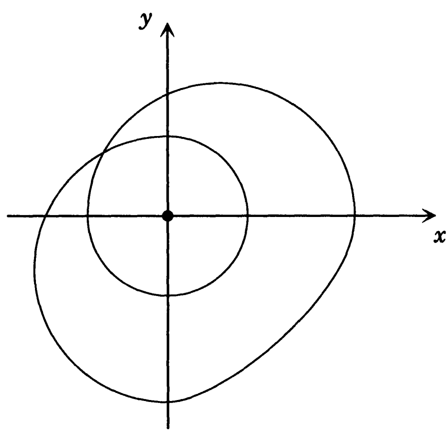


Figure 19.1.3

This can be defined as follows: let the points of $C_r$ be parameterized by their argument $0 \leq \arg(z) < 2\pi$. Starting at $\Theta = 0$ we draw the vector from the origin to the point $q(r(\cos(\Theta) + \sin(\Theta))$ on the curve $q(C_r)$, and record the angle between this vector and the positive $x$-axis. This is just the argument of the point on the curve regarded as a complex number. As $\Theta$ varies from 0 to $2\pi$ this angle varies, and the number of

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[2] As do all proofs I know.

times it passes through $2\pi$ (to become 0 again) is the **winding number** $\omega(r)$. In Figure B.1.3 the curve winds 2 times around the origin.

Clearly, as $r$ varies, the winding number of $q(C_r)$ remains constant, so long as the curves do not pass through the origin, as is our case. Again, so long as the curves do not pass through the origin, the winding number $\omega_r$ depends continuously on $r$. This says that for our family of curves $q(C_r)$, the winding number is a constant.

For $r = 0$ the curve $q(C_0)$ is just the point $q(0) = c_0$, and hence the winding number is 0. On the other hand, we may write $q(z)$ in the form

$$q(z) = z^m \left(1 + \sum_{k=1}^{m} \frac{c_{m-k}}{z^k}\right),$$

and so by de Moivre's representation and the preceding lemma

$$\arg(q(z)) = m \arg(z) + \arg\left(1 + \sum_{k=1}^{m} \frac{c_{m-k}}{z^m}\right).$$

Hence as we move around the curve, letting $\Theta$ vary from 0 to $2\pi$, we find that the net change in the value of $\arg(q(z))$ is $m(2\pi)$ plus the net change of $\arg\left(1 + \sum_{k=1}^{m} \frac{c_{m-k}}{z^m}\right)$. If we choose $r$ large enough, then

$$\left|\sum_{k=1}^{m} \frac{c_{m-k}}{z^m}\right| < \frac{1}{2},$$

and hence all the points

$$1 + \sum_{k=1}^{m} \frac{c_{m-k}}{z^m}$$

lie inside a circle of radius $\frac{1}{2}$ centered at $z = 1$, so the curve traced by these points as $\Theta$ varies from 0 to $2\pi$ goes around the origin no times at all, and the winding number $\omega_r$ for $r$ very large is therefore $m$. Since $m \neq 0$, we have a contradiction. Therefore, at least one of the curves $q(C_r)$ must pass through the origin; i.e., there is a point $z_{r_0}$ on the circle $C_{r_0}$ for which $q(z_{r_0}) = 0$. $\square$

Although the complex numbers fulfill the wish that polynomials always have roots, we have paid a price for this. The price is that the complex numbers cannot be ordered the way the real numbers can. In other words, there is no way to define a notion of *less than*, say $\prec$ for complex numbers, so that the usual properties of $<$ hold.

## B.2 Exercises

**1.** Write out the complex number $\frac{5+3i}{\sqrt{7}}$ in the long matrix form. Find its square roots.

**2.** If $p(z)$ is a polynomial with complex coefficients, show that a complex number $a$ is a zero of $p(z)$ if and only if the linear polynomial $z - a$ divides $p(z)$, i.e., $p(z) = (z - a)q(z)$ for some polynomial $q(z)$. **Hint:** Use polynomial division.

**3.** Use Exercise 2 to define the **multiplicity** of a root of a complex polynomial $p(z)$ as follows: $a \in \mathbb{C}$ is a root of $p(z)$ of multiplicity $k$ if $(z - a)^k$ divides $p(z)$ but $(z - a)^{k+1}$ does not. Show that with this definition each polynomial of degree $m$ has exactly $m$ roots, counting each root as often as its multiplicity.

**4.** Let $k \in \mathbb{N}$ be a positive integer. Does every complex number have a $k$-th root?

**5.** Let the complex numbers $\mathbb{C}$ be defined as in the text as a subset of $\mathrm{Mat}_{2,2}$. Show that $\mathbf{A} \longmapsto \sqrt{\det(\mathbf{A})}$ defines a norm on $\mathbb{C}$ regarded as a real vector space.

**6.** In this exercise we try to repeat the construction of the complex numbers that started with the real numbers, but this time *starting with the complex numbers*. The result is the **algebra of quaternions**, originally discovered by W.R. Hamilton in his study of rotations in $\mathbb{R}^3$. Let $\mathbb{H}$ denote the set of all $2 \times 2$ matrices with complex entries of the form

$$\mathbf{A} = \begin{bmatrix} \bar{u} & w \\ -\bar{w} & u \end{bmatrix}.$$

The elements of $\mathbb{H}$ are called **quaternions.**

(a) Show that the sum and product of any two matrices in $\mathbb{H}$ is again in $\mathbb{H}$.

(b) Show that whenever $0 \neq \mathbf{A} \in \mathbb{H}$, then $-\mathbf{A}$ and $\mathbf{A}^{-1}$ also belong to $\mathbb{H}$.

(c) Show by example that the multiplication in $\mathbb{H}$ is ***not*** commutative.

(d) Show that $\mathbb{H}$ is a 4-dimensional vector space over $\mathbb{R}$ with basis the four quaternions

$$\mathbf{1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{i} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \boldsymbol{j} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, \quad \boldsymbol{k} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}.$$

(e) Show $\boldsymbol{i}^2 = \boldsymbol{j}^2 = \boldsymbol{k}^2 = -1$.

(f) Show that $\boldsymbol{ij} = -\boldsymbol{ji}$ and the corresponding results for the products of $\boldsymbol{i}$ and $\boldsymbol{k}$, and $\boldsymbol{j}$ and $\boldsymbol{k}$.

(g) Show that the polynomial $w^2 + 1$ when regarded as having quaternion coefficients has infinitely many quaternion roots.

**7.** The naive way to try do the previous exercise would have been to consider the $2 \times 2$ matrices of the form

$$\begin{bmatrix} u & w \\ -w & u \end{bmatrix}, \qquad u, w \in \mathbb{C}.$$

Why doesn't this work?

**8.** Let $\mathcal{V}$ be the 3-dimensional real vector subspace of $\mathbb{H}$ spanned by the three quaternions $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$.

(a) Show that for any nonzero quaternion $\mathbf{h} \in \mathbb{H}$ and any vector $\mathbf{A} \in \mathcal{V}$, that the product of the three quaternions $\mathbf{hAh}^{-1}$ lies in $\mathcal{V}$.

(b) Let $\mathbf{h}$ be a fixed nonzero quaternion and define the function $\mathsf{T}_{\mathbf{h}} : \mathcal{V} \longrightarrow \mathcal{V}$ by $\mathsf{T}_{\mathbf{h}}(\mathbf{A}) = \mathbf{h} \cdot \mathbf{A} \cdot \mathbf{h}^{-1}$, where as in the previous exercise the product is as quaternions. Show $\mathsf{T}_{\mathbf{h}}$ is a linear transformation.

(c) Describe the linear transformations $\mathsf{T}_{\boldsymbol{i}}$, $\mathsf{T}_{\boldsymbol{j}}$, and $\mathsf{T}_{\boldsymbol{k}}$. HINT: Use the basis $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ for $\mathcal{V} = \mathbb{R}^3$.

# Font Usage

The following conventions for font usage are adhered to in the text.

| | |
|---|---|
| *a, b,* ........ | denote scalar quantities, i.e. numbers. |
| *A, B,* ....... | denote sets (of numbers, vectors, functions etc.) |
| **A**, **B**,........ | are used to denote vector quantities |
| $\mathcal{V}$, $\mathcal{U}$,....... | are used to denote vector spaces |
| **T**, **M**,........ | denote linear transformations and matrices, |
| **a, b,** ........ | is used for defining terms, |
| *a, b,*......... | is used for emphasis, |
| ***a, b,*** ........ | is used for extra emphasis. |

The naming conventions used for these fonts in *LS* TeX are:

| | |
|---|---|
| scalars,sets | `\mit` |
| vectors, defining terms | `\bf` |
| vectorspaces | `\zapf` |
| linear transformations, matrices | `\sansbold` |
| emphasis | `\it` |
| extra emphasis | `\bit` |

The font `\mit` is based on the font[1] *New Century School Book Italic* together with TeX and POSTSCRIPT hackery, and \bf is **New Century Schoolbook Bold**. The font `\sansbold` is **Helvetica bold** and `\zapf` is the font designed by Hermann Zapf called *Zapf Chancery medium italic*. The lowercase Greek letters are taken from the Greek character set created by Dr. A.V. Hershey while working at the U. S. National Bureau of Standards. The ***Helvetica bold italic*** font makes a brief apperance in Chapter 16 to denote the unit sphere in an inner product space. The running text is printed in New Century Schoolbook Roman.

[1] In the Adobe POSTSCRIPT naming conventions NewCenturySchlbk-Italic.

# Notations

$\Rightarrow$ = implies
$\in$ = is an element of
$\forall$ = for all
$\exists$ = there exists
$\emptyset$ = the empty set
$\subseteq$ = contained in
$\subset$ = properly contained in
$\subsetneqq$ = properly contained in
$\supseteq$ = contains
$\supset$ = properly contains
$\supsetneqq$ = properly contains
$|A|$ = number of elements in the set $A$
$\delta_{i,j} = \begin{cases} 1 & i = j, \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{N}$ = positive integers = $\{1, 2, \ldots\}$
$\mathbb{N}_0$ = natural numbers = $\{0, 1, \ldots\}$
$\mathbb{Z}$ = integers = $\{0, \pm 1, \ldots\}$
$\mathbb{Q}$ = rational numbers
$\mathbb{R}$ = real numbers
$\mathbb{C}$ = complex numbers
$\boldsymbol{i} = \sqrt{-1}$
$\mathbb{R}^k$ = $k$-dimensional real Cartesian space
$\mathbb{C}^k$ = $k$-dimensional complex Cartesian space
$\mathcal{P}_n(\mathbb{R})$ = polynomials of degree $\leq n$ with real coefficients
$\mathcal{P}_k(\mathbb{C})$ = polynomials of degree $\leq k$ with complex coefficients
$\mathcal{P}(\mathbb{R})$ = all polynomials with real coefficients
$\mathcal{C}(a, b)$ = continuous functions on the interval $(a, b)$
$\mathrm{Fun}_{\mathbb{R}}(S)$ = realvalued functions on the nonempty set $S$
$\mathrm{Fun}_{\mathbb{R}}(S, T)$ = realvalued functions on the nonempty set $S$ that are 0 on $T$
$\mathrm{Fun}(S, \mathcal{V})$ = functions on the nonempty set $S$ with values in $\mathcal{V}$
$\mathrm{Fun}_{\mathbb{C}}(S)$ = complexvalued functions on the nonempty set $S$
$\mathcal{L}(E)$ = linear span of the set $E$
$\mathrm{Span}\{E\}$ = linear span of the set $E$
$\ell_0$ = vector space of sequences of real numbers
$\ell_\infty$ = vector space of bounded sequences of real numbers
$\mathcal{V} + \mathcal{W}$ = sum of the vector sub-

spaces $\mathcal{V}$ and $\mathcal{W}$

$\mathcal{V} \oplus \mathcal{W}$ = direct sum of the vector spaces $\mathcal{V}$ and $\mathcal{W}$

$\langle \mathbf{A}, \mathbf{B} \rangle$ = inner product of the two vectors **A** and **B**

$\mathrm{Mat}_{m,n}$ = vector space of $m \times n$ matrices

$\mathrm{GL}(n, \mathbb{R})$ = invertible $n \times n$ matrices with real entries

$\mathrm{GL}(n, \mathbb{C})$ = invertible $n \times n$ matrices with complex entries

$\mathrm{tr}\mathbf{A}$ = trace of the matrix **A**

$\mathbf{A}^{\mathrm{tr}}$ = transpose of the matrix **A**

$\mathbf{A}^{\mathrm{cof}}$ = cofactor matrix

$E(r, s)$ = elementary matrix with entries $e_{i,j} = \delta_{i,r}\delta_{s,j}$

$\mathbf{A} \perp \mathbf{B}$ = **A** is perpendicular to **B**

$\mathcal{S} \perp \mathcal{T} = \mathbf{A} \perp \mathbf{B}\ \forall\ \mathbf{A} \in \mathcal{S}, \mathbf{B} \in \mathcal{T}$

$\mathcal{S}^{\perp}$ = orthogonal compliment of $\mathcal{S}$

$(a, b)$ = open interval of real numbers $x$ such that $a < x < b$

$[a, b]$ = closed interval of real numbers $x$ such that $a \leq x \leq b$

$(a, b]$ = half open interval of real numbers $x$ such that $a < x \leq b$

$[a, b)$ = half open interval of real numbers $x$ such that $a \leq x < b$

$\mathbf{S}(\mathcal{V})$ = unit sphere in the inner product space $\mathcal{V}, \langle -, - \rangle$

# Index

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**Troutman:** Variational Calculus and Optimal Control. Second edition.
**Valenza:** Linear Algebra: An Introduction to Abstract Mathematics.
**Whyburn/Duda:** Dynamic Topology.
**Wilson:** Much Ado About Calculus.